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Fluctuations, Inflation and Growth in
Closed and Open Economies

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Fernando de Holanda Barbosa
FGV EPGE Escola Brasileira de Economia e Finanças
Rio de Janeiro, Brazil

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*“IT DOES REQUIRE MATURITY TO
REALIZE THAT MODELS ARE TO BE USED
BUT NOT TO BE BELIEVED” [Theil (1971),
p. VI].*

*“THE PROOF OF THE PUDDING IS IN
THE EATING.”*

*“ANY POLICY-MAKER OR ADVISER WHO
THINKS HE IS NOT USING A MODEL IS
KIDDING BOTH HIMSELF AND US”
[Tobin, James, quoted by Lombra and Moran
(1980), p. 41].*

*“... IN THE DYNAMIC FIELD OF
SCIENCE THE MOST IMPORTANT GOAL
IS TO BE SEMINAL AND PATHBREAKING,
TO LOOK FORWARD BOLDLY EVEN IF
IMPERFECTLY” [Samuelson (1971),
p X-XI].*

*“IT IS MUCH EASIER TO DEMONSTRATE
TECHNICAL VIRTUOSITY THAN TO MAKE
A CONTRIBUTION TO KNOWLEDGE.
UNFORTUNATELY IT IS ALSO MUCH
LESS USEFUL” [Summers (1991) p. 18].*

“GENTLEMEN, IT IS A DISAGREEABLE CUSTOM TO WHICH ONE IS TOO EASILY LED BY THE HARSHNESS OF THE DISCUSSIONS, TO ASSUME EVIL INTENTIONS. IT IS NECESSARY TO BE GRACIOUS AS TO INTENTIONS; ONE SHOULD BELIEVE THEM GOOD, AND APPARENTLY THEY ARE; BUT WE DO NOT HAVE TO BE GRACIOUS AT ALL TO INCONSISTENT LOGIC OR TO ABSURD REASONING. BAD LOGICIANS HAVE COMMITTED MORE INVOLUNTARY CRIMES THAN BAD MEN HAVE DONE INTENTIONALLY” [Pierre S. du Pont, quoted in Friedman (1994), p. 265].

Preface

This book grew out of the macroeconomics graduate courses I have taught at the Production Engineering Department, Fluminense Federal University and at FGV EPGE Brazilian School of Economics and Finance over more than three decades. During this period, macroeconomics has evolved greatly, as I describe succinctly in the Introduction, giving up the Keynesian Agenda with its behavioral equations and entering the Lucas Agenda, with its equations based on microeconomic foundations.

I have tried very hard not to be idiosyncratic, presenting models that are considered to be the core of mainstream macroeconomics, using both microeconomic-based and ad hoc models. There are at least two reasons that justify the inclusion of ad hoc models in a macroeconomics textbook: (i) pedagogical and (ii) empirical evidence. From a pedagogical point of view, nobody disputes that the simple and elegant Solow model is the best way to introduce economic growth models. From an empirical point of view, the traditional Keynesian models of fluctuations, either for closed or for small open economies (the Mundell-Fleming-Dornbusch model) have not been rejected by the empirical evidence.

I like to say that a teacher is a storyteller and the classroom a stage. I try to make each story as simple as possible without loss of generality. What lessons do these stories teach? The moral of each one is explicitly conveyed. Models are falsifiable representations of certain phenomena. Thus, I strive to present, for each model, empirically testable falsifiable predictions. The structure of almost every section is the same, consisting of: (i) model specification; (ii) algebra to transform the model into a dynamical system; (iii) system equilibrium and stability analysis; and (iv) comparative dynamics experiments. The book is self-contained. There is a mathematical appendix that simply and succinctly provides the required tools to understand the material.

In a market of monopolistic competition, producers need to differentiate their products. This book differs from other macroeconomics textbooks for its emphasis on open economies. My experience in teaching macroeconomics in Brazil has convinced me that a great number of Brazilian economists analyze the Brazilian economy as if it were a closed economy. For example, it is not unusual for someone

to adopt the same technique as the FED, the US Central Bank, to estimate the natural rate of interest. You can find this type of ‘vice’ all over the world. One just has to search for international works that estimate the natural rate of interest in small open economies. My hypothesis is that this behavior stems from studying macroeconomics textbooks that follow the English and American tradition of analyzing their economies as if they were closed economies. Those books do not provide enough scope for the analysis of open economies.

The representative agent model became the workhorse of modern macroeconomics. Chapter 2 might seem to be an idiosyncratic chapter, but this is not the case. It presents several hypotheses that allow for the use of the representative agent model for open economies. However, these hypotheses are either ad hoc, contrary to common sense, or they are rejected by empirical evidence. A model with heterogeneous agents, such as the overlapping generations model, presented in Chap. 3, is well suited to model open economies.

Latin America, as well as many other countries all over the world, has vast experience with economic crises. These crises can be understood by studying four pathologies: (i) public debt default; (ii) chronic inflation; (iii) hyperinflation; and (iv) foreign debt default. Chapter 10 of this book deals with the government budget constraint, the framework that allows for analysis of the sustainability of public debt and hyperinflation. Foreign debt default can be analyzed as a simple extension of public debt default, as Chap. 4, Section 3 shows. The sixth section of Chap. 7 presents a model of chronic inflation that requires a minor change to the Keynesian models, namely, that of changing the monetary policy rule. Instead of an interest-rate rule, the monetary-policy rule states that the Central Bank issues money to finance the public deficit.

This book can be used as basic material for four different graduate courses: (i) Macroeconomics; (ii) Open Economy Macroeconomics; (iii) Monetary Theory; and (iv) Economic Growth. The Macroeconomics course should include Chaps. 1, 3, 6, 7, and 10. The Open Economy Macroeconomics course would cover Chaps. 1, 2, 3, 8, and 9. The Monetary Theory course should include Chaps. 1, 6, 7, 10, and 11. The Economic Growth course would cover Chaps. 1, 3, 4, and 5. Each course should be supplemented by a reading list of papers at the frontier of each field. This book can also be used in an advanced undergraduate Macroeconomics course.

The book was written in Portuguese. Allan Vidigal translated it into English and I revised it to make sure that we are using accurate economic terms. I thank Allan for his excellent job. I am responsible for any errors and imperfections. I thank my students, from different cohorts, whose feedback has helped me to improve my ability as a Macroeconomics ‘storyteller.’ I hope that they will approve of the outcome of this long endeavor. I thank Mariana Biojone Brandão at Springer for her support, which has made publication of this book possible. My thanks also go to Regina Helena Luz for the wonderful job and hard work in drawing up most of the phase diagrams for this book. Vera Lúcia de Abreu went on to complete this task and I thank her for helping me to finalize this book.

Introduction

Macroeconomics

Macroeconomics is a branch of economics that applies economic theory to study growth, business cycles, and price-level determination. Macroeconomics takes account of stylized facts observed in the real world and builds theoretical frameworks to explain these facts. Generally speaking, these frameworks include two types of mechanism: impulse and propagation. The impulse mechanisms, or shocks, are the cause of changes in the model's variables. Propagation mechanisms, as the name indicates, transmit impulses over time and are responsible for the model's dynamics.

Economic growth has been a stylized fact of market economies since England's 19th-century industrial revolution. Until then, poverty was a common good for humanity. Economic growth consists of the persistent, smooth, and sustained increase of income per-capita. Why does one country have a higher level of income per-capita than another? What forces cause one country to grow faster than another? What are the roles of the market and the state in the growth process? This question is part of any economic discussion. In most cases, the answer is based on value judgments. This book is about positive economics only, and does not address any issue from the perspective of normative economics.

Any market economy shows fluctuations in output and other variables, with periods of expanding and contracting economic activity. This phenomenon is called the business cycle. The cycle's main characteristics are as follows: (i) brief economic contraction phase and lengthy economic expansion phase, and (ii) variable duration. What causes the business cycle? What kinds of nominal and (or) real shocks cause the economic activity to fluctuate? What are the roles of the market and the state in the cycle?

Determination of the price level, or the value of money, is a fascinating macroeconomic issue. The value of money is its purchasing power measured in terms of a basket of goods and services. Therefore, purchasing power equals the inverse of the

price level. Why does a financial asset such as paper currency, with no intrinsic value whatsoever, and dominated by another interest-bearing asset, have value? Does the value of money affect the business cycle? Does the value of money affect growth? Under what circumstances does the value of money go to zero, as is the case with hyperinflation?

A stylized fact observed in market economies is the short run non-neutrality of money. A nominal interest rate reduction by the Central Bank causes output to expand. An interest rate increase causes a contraction of the economy's output. Successful stabilization programs end hyperinflation without bringing about recession. How can one reconcile the non-neutrality of money over the cycle with its neutrality at the end of hyperinflation episodes?

In addition to issuing money by the Central Bank, the government, through the Treasury, issues interest-bearing securities with varying characteristics. Does issuing securities affect the economy's real and (or) nominal variables? Under what circumstances does public debt become unsustainable?

Economies are not autarchic, closed to the rest of world, but rather live in permanent contact with other economies in an increasingly globalized world. Each country (or group of countries) has its own currency, goods and services, capital, and flow across countries. The mobility of labor is generally restricted by immigration policies. Does the foreign exchange rate regime affect the functioning of the economy? A stylized fact of the flexible exchange rate regime, since its adoption in the first half of the 1970s by the world's leading economies, is a positive correlation between nominal and real exchange rates. How does one explain this non-neutrality of money? Are growth and the business cycle affected by the economy's degree of openness?

Keynes Agenda

Macroeconomics began with the General Theory. This innovative book by Keynes (1936) was motivated by the great depression that began in 1929 and extended through the 1930s. The adjustment of the market economy, under conditions of unemployment, should be done through the price system. Real wages and real interest rates would bring the economy to full employment. The mechanism, according to Keynes, was not working. The purpose of the General Theory was not just to explain what was going on, but also to propose economic policies to address the problem.

The General Theory set the research agenda for almost half a century. The Keynes agenda led Hicks (1937), Modigliani (1944), Phillips (1958), Mundell (1963), Fleming (1962), and Friedman (1968) to design the architecture of the macroeconomic model of the late 1960s and to write undergraduate textbooks in the latter half of the 1970s. The agenda's basic short-term model for a closed economy consists of the combination of Hicks's *IS/LL* model (1937), which Hansen (1949) calls *IS/LM*,

Modigliani's price and (or) wage rigidity (1944) and a Phillips (1958) curve that is vertical in the long run, in Friedman's (1968) version.

In an open economy, the mechanism that determines the price of the money of a country (or group of countries) in relation to the currencies of other countries and the mobility of capitals across countries are crucial to the functioning of the economy. Two foreign exchange regimes exist as polar cases: the fixed and flexible exchange rate regimes. Under a fixed exchange rate regime, the price of a foreign currency is determined by the Central Bank. Under a flexible foreign exchange regime, the market sets the price. In practice, no fixed exchange rate regime exists that is eternal, nor does a flexible exchange regime free from government intervention exist. Mundell (1963) and Fleming (1962) extended the Keynes short-term model framework for a closed economy to an open economy by introducing the relationship between domestic and foreign interest rates arising from the mobility of capital and the arbitrage that this movement produces. In addition, they analyzed the behavior of the economy under the foreign exchange regime in place.

On the economic growth front, the Keynes agenda began with the breakthrough papers of Harrod (1939) and Domar (1946). This growth model produced a razor's edge on which the economy should travel. There was no salvation outside of the razor's edge, as no other mechanism existed leading the economy to full utilization of labor and capital. Solow (1956) showed that the razor's edge did not exist in reality. The price system would take care of the allocation of resources through changes in the capital-output ratio. Solow's model then became the agenda's basic economic growth model.

In the early 1970s, the Keynes agenda came to an end with two contributions from Lucas (1972, 1976). The former, referred to as rational expectations, allows for the building of consistent models in which agents' expectations regarding future events play a crucial role. Until then, agents had one prediction and the model produced another entirely different from agents' expectations. After a while, rational expectations were entirely absorbed by the Keynes agenda's models, in a choice for rigor, coherence, and empirical evidence.

Lucas Agenda

Lucas's second contribution is known as the Lucas critique. The Lucas critique is devastating for econometric models developed under the Keynes agenda. Why? Because it argues that people change behaviors when the rules of the game change. The explanation for this is so simple that, once you get it, you'll ask yourself, "Why didn't I think of this before?" Assume that you play soccer with your friends twice a week, and that there is always a team waiting off-side for its turn to play. On one day of the week, the game is organized as follows: the winning team stays on for the next match, and the losing team leaves the field. On the other day of the week, the organization differs. For the first match, the winner criterion prevails. From the second match on, each team, winner or loser, plays only two matches. Does the

behavior of a player who plays on both days of the week stay the same? The answer is no: everyone dances according to the music.

The Lucas agenda used two types of models that had been developed earlier, but were not part of the training of macroeconomists until the mid-1970s. The representative agent model of Ramsey (1928), Cass (1965), Koopmans (1965), and Samuelson's (1958) overlapping generations model. In the early 1980s, Kydland and Prescott (1982) built a model based on the representative agent framework to explain the business cycle, which was known as the real business cycle because it was caused by technology shocks instead of the nominal shocks of the Keynes agenda's business cycle models. This model influenced an entire generation of economists for two reasons. Firstly, because it did not rely on any ad hoc hypotheses, such as the price rigidity hypothesis of Keynesian models. Secondly, because a general equilibrium model in the tradition of Arrow/Debreu was capable of producing the business cycle phenomenon. Still, a sizeable share of the profession remained unconvinced that technology shocks would have the necessary magnitude to produce cycles, or that nominal monetary policy shocks might be irrelevant to the business cycle.

In the Keynes agenda's economic growth model – the Solow-Swan model – the rate of technological progress, which determines the growth rate of income per-capita in the long run, is an exogenous variable. In the short run, different income per-capita growth rates may be explained by the dynamics of the transition path to the economy's long run equilibrium. This, however, is not a satisfactory solution to understanding observed differences between countries in the growth rates of their income per-capita.

On the economic growth front, the Lucas agenda brought about the rebirth of this field with two works that originated in the endogenous growth models, one by Lucas himself (1988) and another by Romer (1986), that aim to make the long run growth rate of income per-capita endogenous. These models may be divided into four groups according to the mechanisms that produce endogenous growth rates. These are: (i) externalities from capital and knowledge accumulation; (ii) human capital accumulation; (iii) production of an increasing variety of intermediate goods used in the production process, an outcome of firms' research-and-development investment; and (iv) technological innovations leading to machinery and equipment more efficient than those in existence, in a creative destruction process, since innovations render older machinery and equipment obsolete.

Model Fundamentals

The Keynes agenda's models are based on behavioral decisions, which take the limits of human rationality into account. The Lucas agenda's models are optimization models. This statement needs qualification to prevent misinterpretation. The Keynes agenda produced a large number of important works that built models for decisions on consumption [Friedman (1957), Modigliani & Brumberg (1954)],

investment [Jorgenson (1963), Tobin (1969)], and demand for money [Friedman (1956), Baumol (1952), Tobin (1958)] based on the neo-classical theory. But short-term macroeconomic models such as the Klein and Goldberger (1955) model were built specifying equation by equation, without a shared theoretical framework to determine each equation's specifications.

Models based on behavioral decisions do not show how these decisions might have emerged in a process of choice with duly explicit options and constraints. The models are built to simulate economic policies, which are the rules of the game for economic agents, consumers, workers, and businessmen. Models in which agents' decisions are unchangeable with regard to economic policies must be taken with a grain of salt.

Models derived from the solution of optimization problems assume that the players make decisions while knowing the rules by which they play, and the literature calls them 'microfounded models.' The prefix 'micro' comes from micro-economics. In these models, agents maximize their objective function conditioned by the constraints to which they are subject and to the economic environment in which they live.

To what extent should one dismiss models based on behavioral decisions and use only microfounded ones? If the only model-selection criteria were theoretical structure and their foundation on the basic principles of neoclassical theory, behavioral models should be discarded. However, the models' ability to explain facts observed in the real world overcomes theoretical soundness. Empirical evidence does not yet enable a definitive answer to this question. For as long as that is the case, then, the two types of models should be part of macroeconomics training.

New Keynesian (New Neoclassical) Synthesis

Short-term models have adopted the monetary policy rule that Taylor (1993) proposed for the interest rate in the interbank reserve market, which Central Banks control. The rule's success is due to the fact that, in countries that adopt a flexible exchange-rate regime, Central Banks implement monetary policy by setting the interbank reserve market's interest rate, rather than the amount of banking reserves, as the LM curve implicitly assumed.

In the latter half of the 1990s, the Lucas agenda faced the challenge of redressing the Keynes agenda's business cycle models, embracing the price rigidity hypothesis, but looking to microeconomics for fundamentals for the IS and Phillips curves. Some [Clarida, Gali, & Gertler (1999)] call this new synthesis new Keynesian, others refer to it as new neoclassical [Goodfriend (2004)].

Short-term models for a closed economy, whether they may exist under the Keynes agenda or the new Keynesian (or new neoclassical) synthesis therefore consist of three equations: an IS curve, a Phillips curve, and a Taylor rule. The LM curve is not an explicit part of the models because money became endogenous.

Integrating the Two Agendas

Undergraduate macroeconomics textbooks essentially contain the Keynes agenda's macroeconomic models, whereas those for graduate programs feature the Lucas agenda's. This book does not embrace this split, and includes both types of models expressed in a single mathematical language. The approach enables not only a better understanding of the business cycle and economic growth models, but also a comparison of the predictions made by each model. Ultimately, models must be judged by their predictive capacity. Those that the data reject must be abandoned and kept as historic relics.

Open-Economy Macroeconomics

The tradition of macroeconomics, which developed in England in the first half of the 20th century and in the United States in the latter half, is to model a closed economy. The tradition may be due to the fact that the economies of those two countries were so big that they might be seen as the world economy itself. Over time, the rest of the world grew and the British and American economies ceased to be prevalent. Still, the force of habit endures and many macroeconomics textbooks and manuals do not devote enough space to open-economy issues.

Not so this book. Chapter 2, which addresses the representative agent model in an open economy, shows that this model is not appropriate for a small open economy, unless one is willing to accept ad hoc hypotheses.

Chapter 3, which covers the overlapping generations model, shows that this kind of model may be applied to a small open economy. A small open economy model requires heterogeneous agents. The representative agent model is inappropriate because it either leads to absurd conclusions or requires ad hoc hypotheses. Therefore, policy conclusions derived from this model do not in general apply to a small open economy. The metrics of optimum policy, the maximization of the representative agent's welfare, do not apply in the overlapping generations model. According to this model, an optimum policy depends on the weights assigned to the welfare of the various generations, including those yet unborn.

Chapter 8 addresses the basics of open-economy macroeconomics and Chap. 9 is entirely concerned with open-economy economic fluctuation models. In addition to the Mundell-Fleming-Dornbusch model, Chap. 9 presents the overlapping generations new Keynesian model for both fixed and flexible exchange-rate regimes. The chapters that deal with economic growth models analyze two open-economy models: the Solow model in a small open economy under perfect capital mobility, in Chap. 4, and Ventura and Acemoglu's AK model in an open economy with no capital mobility, in Chap. 5.

Mathematical Tool: Dynamical Systems

Economic models use three types of language: (i) verbal, (ii) graphical, and (iii) mathematical. Verbal language has the advantage of being more accessible, but sometimes at the cost of logical rigor. Graphical language offers the benefit of visual understanding, but at the potential cost of allowing the hand to draw graphs that fail to abide by the model's properties. Mathematical language has the benefit of logical rigor, but the cost of learning mathematical technique is not always to be disregarded.

The mathematical tool for the models in this book is one that allows for the analysis of dynamical systems. Such systems may be built with discrete or continuous variables. This book uses dynamical systems with continuous variables that allow graphical representation on phase diagrams. The continuous variables dynamical system is a system of differential equations:

$$\dot{x} = F(x, \alpha),$$

where $\dot{x} = dx/dt$, x is a vector of endogenous variables, and α is a vector of the model's exogenous variables and (or) parameters.

The model as described by such a dynamical system must be analyzed in order to establish properties concerning: (i) equilibrium, (ii) stability, and (iii) comparative dynamics. Equilibrium analysis checks whether an equilibrium exists and whether it is unique or not. That is, is there a vector $\bar{x}$ such that $\dot{x} = 0$ in the system of differential equations? If so, the vector is obtained by solving the equations system:

$$F(\bar{x}, \alpha) = 0.$$

Assuming that a solution exists, the equilibrium value is a function of the model's exogenous variables and (or) parameters. That is:

$$\bar{x} = \bar{x}(\alpha).$$

Vector α is the model's driving force, or impulse mechanism. The dynamical system may be linearized around equilibrium point $\bar{x}$, according to:

$$\dot{x} = F_{\bar{x}}(x - \bar{x}),$$

where $F_{\bar{x}}$ is a matrix of partial derivatives of F relative to the x variables, valued at the stationary equilibrium point. The purpose of (local) stability analysis is to determine what happens to a dynamical system when variable x is different from its equilibrium value $\bar{x}$. When the system is stable, the economy converges to equilibrium. When unstable, the economy does not converge to the stationary equilibrium.

Stability analysis also allows for the checking for bubbles in the economy. A bubble occurs when the endogenous variables change without a change in the model's fundamentals, e.g., exogenous variables and (or) parameters.

A model is a falsifiable representation of a phenomenon. Comparative dynamics experiments, together with model stability analysis, enable obtaining the model's propositions that may be falsifiable and, therefore, empirically refuted. The purpose of comparative dynamics is to analyze the response of endogenous variables to changes in the model's exogenous variables and (or) parameters. Comparative dynamics shows not only what happens to the new equilibrium, but also the path of the economy following a change in the exogenous variables and parameters of the model.

The basic difference between models built on behavioral decisions and microfounded ones is that, in the latter, the differential equations system is obtained from the solution of an optimization problem, derived from a Hamiltonian. No such Hamiltonian exists in models built based on behavioral decisions. Furthermore, in microfounded models, the solution to the optimization problem must satisfy a transversality condition that selects, out of several possible paths that meet the first-order conditions, the path that maximizes the problem's objective function.

Organization of the Book

This book is organized in three parts and three appendices. Part I deals with flexible-price models. Part II introduces sticky-price models. Part III presents monetary- and fiscal-policy models. The mathematical appendices succinctly introduce the mathematical techniques needed to understand the book. Each chapter includes a list of exercises. Many of these exercises are based on the literature listed in the References section, although the sources may not be documented.

Part I, for flexible-price models, has five chapters. Chapter 1 introduces the representative agent model. Chapter 2 extends the representative agent model to the open economy. Chapter 3 presents the overlapping generations model. Chapter 4 covers the Solow growth model, and Chap. 5 deals with endogenous savings and growth models.

Chapter 1 introduces the representative agent model, which has become, since the 1980s, the workhorse of macroeconomics. The representative agent model is extended to an economy endowed with a government and money. Analysis of the latter includes the issue of money neutrality with two monetary policy rules. The Central Bank controls money stock in one and the nominal interest rate in the other. This chapter also introduces a model in which the agent faces the consumption versus leisure choice, assuming a production function subject to technology shocks. Combined, the two hypotheses give rise to the real business cycle model.

The Achilles heel of the representative agent model is the small open economy. As Chap. 2 shows, in a small open economy the representative agent model, with its constant rate of time preference, lacks a stationary equilibrium unless two parameters are equal by pure chance. This stationary equilibrium exists in three cases that the chapter analyses: (i) variable rate of time preference; (ii) risk premium on the foreign interest rate; and (iii) complete financial markets. In the first case, however, the

wealthy individual must be impatient, an assumption that goes against common sense. In the second case, the risk premium's specification is not microfounded and, therefore, subject to the Lucas critique. In the final case, empirical evidence rejects the hypothesis of complete financial markets.

Chapter 3 addresses overlapping generations models and presents two models, one with infinite and another with finite life. The infinite-life overlapping generations model, in which at every moment in time a generation is born with no financial assets and, therefore, disconnected from the existing generations, applies to an economy endowed with a government as well as to a small open economy. This chapter shows that the overlapping generations model, unlike the representative agent model, may be applied to a small open economy without the need for any ad hoc hypotheses. The finite-life overlapping generations model employs the simplifying assumption that an individual's probability of death is independent of their age.

Chapter 4 introduces the Solow economic growth model, the purpose of which is to explain the causes that determine the level and growth rate of labor productivity. The model is generalized by the inclusion of human capital as a factor of production. One of the chapter's sections analyzes application of the Solow model in a small open economy with perfect capital mobility. The chapter also introduces the theoretical framework of growth accounting.

In the Solow growth model, consumption is not deduced from intertemporal resource allocation. Chapter 5 introduces the Ramsey-Cass-Koopmans and overlapping generations (OLG) models, in which savings is an endogenous variable. The chapter analyzes and discusses endogenous growth models, the AK model, the Lucas human capital model, the Romer varieties model, and the Aghion-Howitt Schumpeterian model. It also introduces Ventura and Acemoglu's AK model for an open economy with no capital mobility, but which takes part in international trade.

Part II covers rigid-price models and includes four chapters. Chapter 6 addresses the specification of the equations in the Keynesian and new Keynesian models. Chapter 7 analyzes economic fluctuation and stabilization in a closed economy according to the two types of model. Chapter 8 introduces the basic issues of open-economy macroeconomics. Chapter 9 introduces economic fluctuation models for the small open-economy.

Chapter 6 covers the specification of three equations for short-term macroeconomic models: (i) the relationship between the real interest rate and real output, the IS curve; (ii) the relationship between the nominal interest rate and quantity of money, the LM curve; (iii) the relationship between the unemployment rate (or output gap) and the inflation rate, the Phillips curve. Each equation is specified from two approaches. From the traditional Keynesian point of view, the equations are motivated by behavioral decisions, and not founded on optimization models. From the new Keynesian approach, specifications are based on microeconomics. The two approaches produce not only distinct specifications, but also different and empirically testable predictions.

Chapter 7 presents equilibrium and dynamics under six sticky price models. The first one has an IS curve, a Phillips curve, a Taylor monetary policy rule, and inertial inflation rate. The second model has the same equations as the former, but no inertial

inflation rate. The third one is the new Keynesian model, in which the IS curve is indeed the Euler equation. The fourth model is an encompassing specification of Keynesian models, where each model is obtained as a particular case depending on parameter values and the initial state of the model's variables. The fifth model is the Friedman model, in which the Central Bank controls the rate of growth of the stock of money. In the sixth model, the Central Bank does not control either the money stock or the rate of interest. It provides resources to finance the fiscal deficit of the government. Inertia affects both the price level and the inflation rate. This chronic inflation model has an IS curve, a Phillips curve and the only difference is the monetary policy rule, which states that the Central Bank issues money to finance a given fiscal deficit.

Chapter 8 is concerned with open-economy macroeconomics. This chapter introduces arbitrage pricing models for goods and services that are the subject of international trade and for the domestic and foreign interest rates on the assets involved in capital flows between countries. It then deduces the Marshall-Lerner condition that sets the restrictions for a positive correlation between the current account of the balance of payments and the real exchange rate. The specifications of the open-economy IS curve, which relates real output, real interest rate, and the real exchange rate, are provided for the traditional and the microfounded models. The chapter analyzes the determination of the long run equilibrium real exchange rate, the natural exchange rate, showing that the determination of the natural interest rate is completely different from its determination in a closed economy. This chapter also addresses the specification of the open-economy Phillips curve, whose arguments include changes in the real exchange rate.

Chapter 9 analyzes economic fluctuation and stabilization in an open economy's models under both the fixed and flexible exchange rate regimes. We first analyze the Mundell-Fleming-Dornbusch model. The model is then expanded with the introduction of wealth in the consumption function. The chapter also includes the analysis of the overlapping generations new Keynesian small open economy model.

Part III is made up of two chapters covering monetary and fiscal policy models. Chapter 10 analyzes the government budget constraint. Chapter 11 introduces a few monetary theory and policy models.

Chapter 10 addresses the government budget constraint and several issues that can be analyzed based on this accounting framework. The government budget constraint derives from the consolidated Treasury and Central Bank accounts. It allows for the setting of the conditions for public debt to be sustainable. The inflation rate may be regarded as the inflation tax rate. The chapter therefore introduces different calculations of the social cost of this tax. The pathology of hyperinflation is scrutinized by means of a review of the various models that attempt to explain this phenomenon. The conditions for Ricardian equivalence are duly analyzed in this chapter, which also covers the fiscal theory of the price level and the necessary conditions for the sustainability of a monetary regime.

Chapter 11 introduces several monetary theory and policy issues. It first analyzes the determination of the price of money as the price of a financial asset, with both of its components: fundamentals and bubbles. It then demonstrates the possibility of multiple equilibria in a monetary economy, in which money has no value at one

equilibrium. The chapter then proceeds to address the issue of the indeterminacy of the price of money when the Central Bank adopts a rigid monetary policy rule, setting a nominal interest rate irrespective of the economy's prevailing conditions. The optimum quantity of money in a flexible price economy is a classic subject in the literature that no macroeconomics book can dismiss, despite its irrelevance for monetary policy practice. This chapter analyzes the liquidity trap in its modern version, with zero-limit nominal interest rate. It also covers dynamic inconsistency, when incentives exist for not carrying out decisions made in the present with the future in mind. Dynamic inconsistency is part of human behavior and has monetary policy implications. The smoothing of the interest rate by central bankers who prefer not to change it abruptly, but rather gradually, leads to some inertia in the behavior of interbank interest rates, which is a stylized fact of monetary policy. The consequences of this behavior are duly analyzed according to the Keynesian and new Keynesian models. This chapter shows how inflation targeting, a system adopted by several Central Banks around the world after the Central Bank of New Zealand first invented it, may be incorporated into the framework of short-term macroeconomic models. One of the chapter's sections analyzes the operational procedures of monetary policy in the interbank reserve market, where the Central Bank plays a dominant role. The chapter also shows how the term structure of interest rates can be introduced into short-term macroeconomic models. The framework allows for the analysis of the effect of Central Bank announcements regarding future short-term interest rates on the economy's activity level and inflation rate.

The mathematical appendix is comprised of three chapters. Appendix A deals with differential equations. Appendix B addresses the theory of optimal control. Appendix C introduces stability analysis for finite-difference equation models. Although this book uses continuous dynamic models, this appendix has been included because Keynesian and new Keynesian models are usually presented with discrete variables. Readers will find here a summary of the main properties needed to analyze such models' equilibria and dynamics.

Appendix A presents basic results of first- and second-order linear differential equations, as well as of first-order linear differential equation systems, which are widely used throughout the text.

Appendix B succinctly introduces the theory of optimal control, addressing the basic optimal control problem, the Hamiltonian, and the transversality condition. The appendix also discusses discount-rate and infinite-horizon optimal control, linear optimal control, and analyzes the comparative dynamics optimal control problem's solution.

Appendix C analyzes first-order finite-difference equation models, with rational expectations, forward and backward variables, and hybrid models. The tools are then used to analyze the Keynesian, new Keynesian, encompassing (which includes the former two as particular cases), and hybrid new Keynesian models.

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Part I

Flexible Price Models

Chapter 1

The Representative Agent Model



The representative agent model has been the workhorse of macroeconomics since the 1980s. This chapter deals with this model. Section 1.1 presents the basic model. Section 1.2 introduces a government into the economy. Section 1.3 addresses the model of a monetary economy with two monetary policy rules. The central bank controls the money stock under one policy rule and the nominal interest rate under the other policy rule. Section 1.4 introduces the consumption and leisure choice and assumes that the production function is subject to technology shocks. Combined, these two assumptions give rise to the real business cycle model.

1.1 Basic Model

The representative agent model maximizes the present value of the consumption instantaneous utility flow, $u(c)$, throughout its infinite life, discounted at the rate of time preference ρ . The population grows at a continuous rate equal to n . At the initial moment, population is normalized as unity ($L_0 = 1$). The representative agent's problem, therefore, consists of maximizing:

$$\int_0^{\infty} e^{-\rho t} u(c) L_0 e^{nt} dt = \int_0^{\infty} e^{-(\rho-n)t} u(c) dt, \rho > n, \quad (1.1)$$

subject to the constraints:

$$\dot{k} = f(k) - c - (\delta + n)k, \quad (1.2)$$

$$k(0) = k_0, \text{ given.} \quad (1.3)$$

The rate of time preference must be greater than the population growth rate; otherwise, the integral would not converge. The first constraint is the capital accumulation equation, in which the economy's output is absorbed as consumption

or investment. The production function assumes constant returns to scale on the capital and labor factors of production. The production function is written in intensive form—that is, output per capita depends on the capital-labor ratio k : $Y = F(K, L) = Lf(k)$. The capital depreciation rate is constant and equal to δ . The model's second constraint states that the model's initial capital-labor ratio is given.

The problem's current value Hamiltonian H is:

$$H = u(c) + \lambda[f(k) - c - (\delta + n)k],$$

where λ is the costate variable. The first-order conditions are:

$$\begin{aligned}\frac{\partial H}{\partial c} &= u'(c) - \lambda = 0, \\ \dot{\lambda} &= (\rho - n)\lambda - \frac{\partial H}{\partial k} = (\rho - n)\lambda - \lambda[f'(k) - (\delta + n)], \\ \frac{\partial H}{\partial \lambda} &= f(k) - c - (\delta + n)k = \dot{k}.\end{aligned}$$

The optimal solution must satisfy the transversality condition:

$$\lim_{t \rightarrow \infty} \lambda k e^{-(\rho - n)t} = 0.$$

This condition states that the optimal path must be such that the present value of capital, in terms of utility, discounted at the rate of time preference, net population growth rate, must equal zero when time goes to infinity. Otherwise, agents might increase their welfare by reducing capital accumulation and consuming their resources.

This model's representative agent must decide, at each moment in time, whether to consume or to save and invest in capital accumulation. The second first-order condition equation is precisely this decision's arbitrage condition. Should the agent choose to save and invest one dollar more, their return will equal the net marginal product of capital: $f'(k) - \delta$. On the other hand, should the agent decide to consume this dollar, their return will equal the rate of return on consumption, the rate of time preference ρ plus capital gain, which is a loss in this case because the price of consumption (the marginal utility of consumption) decreases as consumption increases. The equilibrium arbitrage condition is:

$$f'(k) - \delta = \rho - \frac{\dot{\lambda}}{\lambda}.$$

The first among the first-order condition equations establishes that the marginal utility of consumption must equal the costate variable. The derivative of this equation with respect to time is:

$$u''(c)\dot{c} = \dot{\lambda} \Rightarrow \frac{\dot{\lambda}}{\lambda} = \frac{u''(c)}{u'(c)}\dot{c}.$$

Dynamic System

Substituting the arbitrage equation's $\dot{\lambda}$ expression in the previous equation yields the differential equation of consumption. The model's second differential equation corresponds to the third first-order condition, which reproduces the problem's constraint. The model's dynamical system has two differential equations. The former is the celebrated Keynes-Ramsey rule, according to which the growth rate of consumption depends on the difference between the rate of time preference and the interest rate. The latter equation shows the evolution of the economy's capital stock, that is:

$$\dot{c} = \frac{u'(c)}{u''(c)}[\rho + \delta - f'(k)], \quad (1.4)$$

$$\dot{k} = f(k) - c - (\delta + n)k. \quad (1.5)$$

The system's Jacobian matrix at the steady state equilibrium is:

$$J = \begin{bmatrix} \frac{\partial \dot{c}}{\partial c} & \frac{\partial \dot{c}}{\partial k} \\ \frac{\partial \dot{k}}{\partial c} & \frac{\partial \dot{k}}{\partial k} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{u'(\bar{c})}{u''(\bar{c})}f''(\bar{k}) \\ -1 & f'(\bar{k}) - \delta - n \end{bmatrix}.$$

The matrix's determinant is negative because the marginal utility of consumption and the marginal product of capital are decreasing, that is:

$$|J| = -\frac{u'(\bar{c})}{u''(\bar{c})}f''(\bar{k}) < 0.$$

Therefore, the steady-state equilibrium is a saddle point.

Figure 1.1 shows the phase diagrams for the dynamical system's two differential equations. At the points where consumption does not change ($\dot{c} = 0$), the net marginal product of capital equals the representative agent's rate of time preference:

$$f'(k) - \delta = \rho.$$

The capital-labor ratio remains constant ($\dot{k} = 0$) when consumption is:

$$c = f(k) - (\delta + n)k.$$

The point of maximum consumption occurs when the interest rate, the net marginal product of capital, equals the population growth rate:

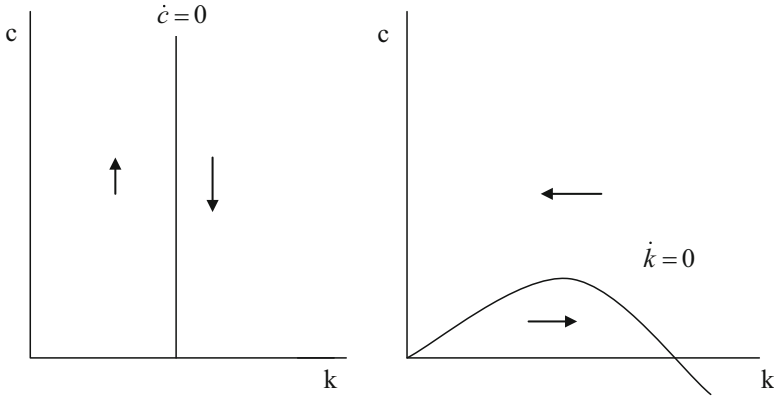


Fig. 1.1 The dynamics of c and k

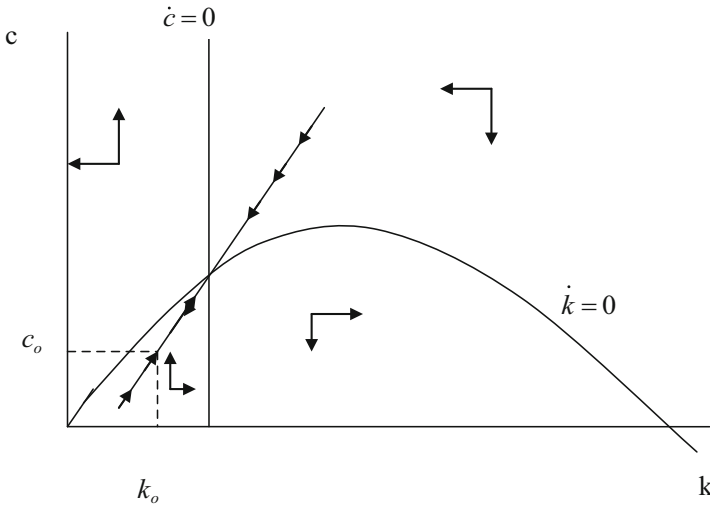


Fig. 1.2 The phase diagram for the c and k system

$$f'(k_G) - \delta = n. \quad (1.6)$$

The quantity of capital (k_G) corresponds to the golden rule level of consumption. The arrows in Fig. 1.1 show the variables' movement when out of equilibrium

Figure 1.2 shows the model's phase diagram. For a given initial capital-labor ratio, the only path that satisfies the transversality condition is the *SS* saddle path. At steady-state equilibrium, the interest rate equals the rate of time preference, which is greater than the population growth rate by hypothesis. Therefore, the representative agent model does not give rise to dynamical inefficiency, that is, the capital-labor ratio will always be lower than the corresponding golden-rule value.

Experiment

Figure 1.3 shows an unanticipated, permanent change in the rate of time preference. The rate of time preference drops from ρ_0 to ρ_1 . The economy's long-term interest rate drops and the quantity of capital rises, as Fig. 1.4 illustrates. Consumption shows an instant drop at first, and the economy describes a path on the model's new saddle. The level of consumption will be higher than before the change in the rate of time preference at the long-term equilibrium. The same will be true for the capital stock.

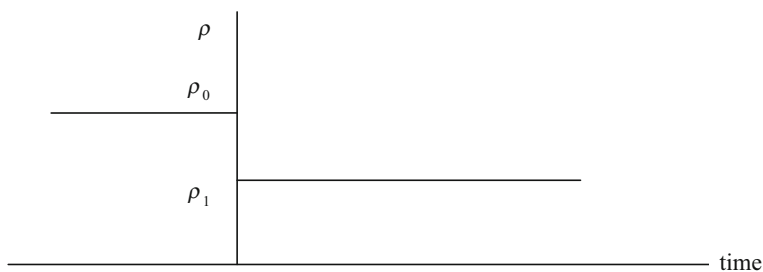


Fig. 1.3 An unanticipated permanent decrease in the rate of time preference

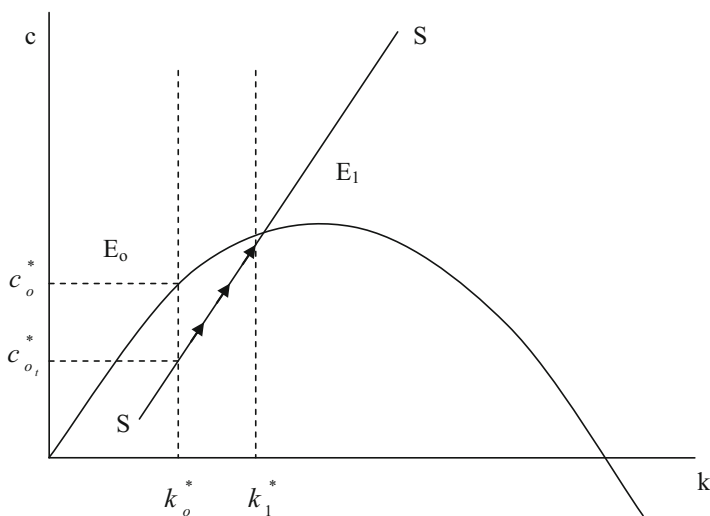


Fig. 1.4 The effects of an unanticipated permanent decrease in the rate of time preference

1.2 Economy with a Government

The representative agent model with a government has Ricardian equivalence, that is, public debt does not affect the agent's decision to consume and save. To demonstrate this statement, let us begin with the agent's and the government's budget constraints in terms of flows:

$$\begin{aligned}\frac{\dot{A}}{L} &= ra + y - c - \tau, \\ \frac{\dot{B}}{L} &= rb + g - \tau,\end{aligned}$$

where $\dot{A}$ is the change in the agent's wealth, L is population, r is the interest rate, a is per-capita wealth, y is per-capita income, c is per-capita consumption, τ is per-capita tax. The change in public debt is represented by $\dot{B}$, b is per-capita public debt, and g is per-capita public spending. Since:

$$\frac{\dot{A}}{L} = \dot{a} + na; \frac{\dot{B}}{L} = \dot{b} + nb,$$

the two budget constraints become:

$$\begin{aligned}\dot{a} &= (r - n)a + y - c - \tau, \\ \dot{b} &= (r - n)b + g - \tau.\end{aligned}$$

The solutions to these two first-order differential equations, assuming no Ponzi scheme, result in the private and public sectors' intertemporal constraints:

$$\begin{aligned}a(t) &= \int_t^\infty e^{-(r-n)(v-t)} (c + \tau - y) dv, \\ b(t) &= \int_t^\infty e^{-(r-n)(v-t)} (\tau - g) dv.\end{aligned}$$

Subtracting the latter constraint from the former yields:

$$a(t) - b(t) + \int_t^\infty e^{-(r-n)(v-t)} (y - g) dv = \int_t^\infty e^{-(r-n)(v-t)} c dv.$$

We conclude that the path of taxes is irrelevant to the agent's consumption decision. Public debt is not regarded as wealth for this economy's agents because public debt stock b is subtracted from total wealth a .

The representative agent model with a government assumes that the government's budget is always balanced: $g = \tau$, where g is the government's spending and τ is a lump-sum tax, which does not distort the agent's decisions. The assumption is not restrictive because there is Ricardian equivalence in this environment. Assume also

that the agent's welfare is not affected by government spending. The representative agent's problem, therefore, consists of maximizing:

$$\int_0^{\infty} e^{-(\rho-n)t} u(c) dt, \quad (1.7)$$

subject to the constraints:

$$\dot{k} = f(k) - c - g - (\delta + n)k, \quad (1.8)$$

$$k(0) = k_o, \text{ given.} \quad (1.9)$$

The solution to this model is the same as the previous section's, except that the capital accumulation equation includes government spending. Therefore, the dynamic system has two differential equations:

$$\dot{c} = \frac{u'(c)}{u''(c)} [\rho + \delta - f'(k)], \quad (1.10)$$

$$\dot{k} = f(k) - c - g - (\delta + n)k. \quad (1.11)$$

The system's Jacobian matrix is the same as the previous section's. The equilibrium point is a saddle point. The only change occurs in the equation for $\dot{k}$, since the one for $\dot{c}$ remains the same. Figure 1.5 illustrates the model's phase diagram. Given the initial quantity of capital, the SS saddle is the equilibrium path that satisfies the transversality condition.

Experiments

An interesting question associated with this model is finding out what happens with the interest rate when government spending changes. Figure 1.6 addresses an unanticipated, permanent increase in government spending from g_0 to g_1 .

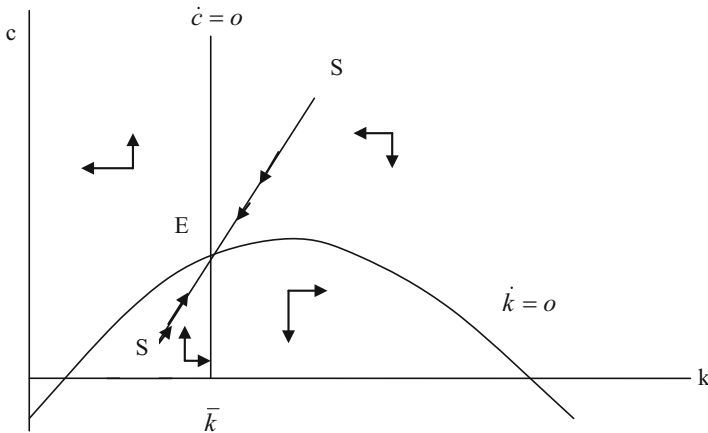


Fig. 1.5 The dynamics of c and k in a model with government

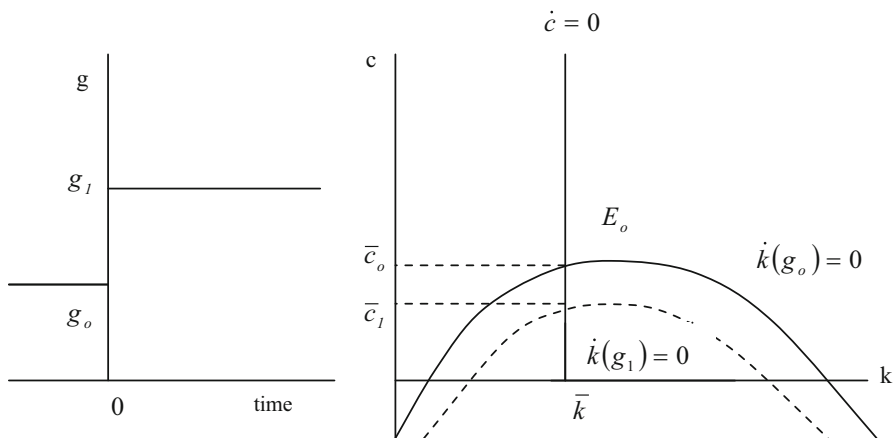


Fig. 1.6 An unanticipated permanent increase in government spending and its effects

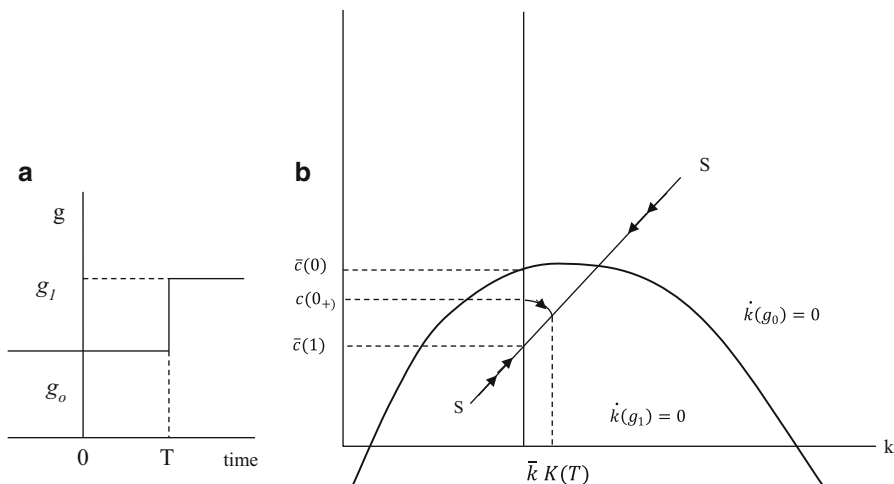


Fig. 1.7 (a, b) An anticipated permanent increase in government spending and its effects

Consumption drops instantly and the capital stock remains unchanged. The interest rate, therefore, remains the same as it was before government spending increased.

Figure 1.7 analyzes the case of an anticipated, permanent increase in government spending. At the time of the announcement consumption decreases, savings increase, and the capital stock increases. The interest rate, therefore, decreases. However, at the instant of the change in spending, the economy must be at the new saddle, otherwise the transversality condition would not be satisfied and the economy would not converge to its new stationary equilibrium. At this new equilibrium, the quantity of capital is precisely the same as at the time of the

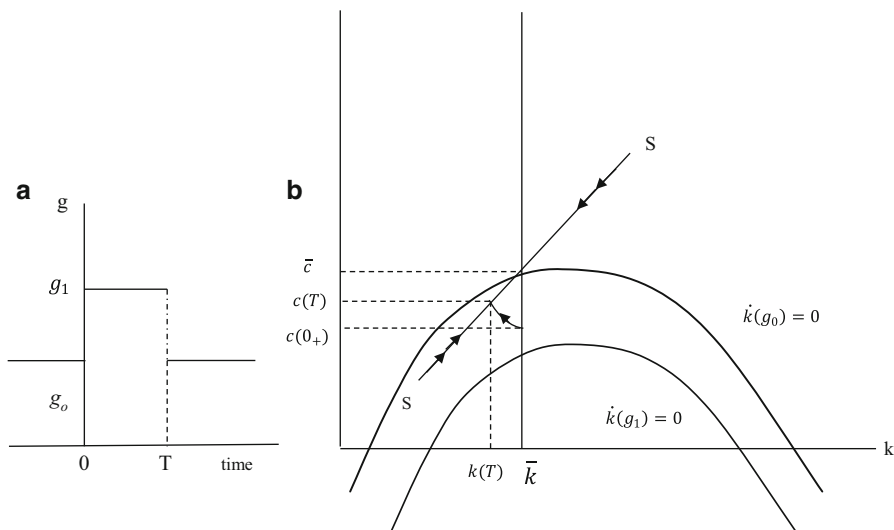


Fig. 1.8 (a, b) An unanticipated transitory increase in government spending and its effects

announcement of the increased spending policy. The conclusion is that, in this case, the drop in the interest rate is only transitory.

Figure 1.8 illustrates an unanticipated, transitory increase in government spending. At the time of the announcement, consumption drops, but by not as much as the increase in spending. Private-sector savings drop and so does capital accumulation. The reduction in capital stock makes the interest rate rise. When government spending returns to its original level at time T , the economy must be at the model's previous saddle. Otherwise, the transversality condition would not be satisfied. The economy then converges to its previous steady state equilibrium. At this point, the interest rate is the same as before the policy change. The conclusion is that the increase in government spending affects the interest rate, but the interest rate increase is just transitory.

Figure 1.9 illustrates an anticipated, transitory increase in government spending. At the time of the announcement, consumption decreases, savings increase, and the capital stock increases as well. The interest rate therefore decreases. When the increase in government spending occurs, savings decrease and the capital stock starts to decline. The process continues until the end of the increased government spending policy. At this moment, the economy must be on the model's previous saddle path. Otherwise, the economy would not converge to the steady-state equilibrium. In this case, the interest rate initially decreases, then increases and returns to its initial level. The conclusion is that the interest rate change is merely transitory.

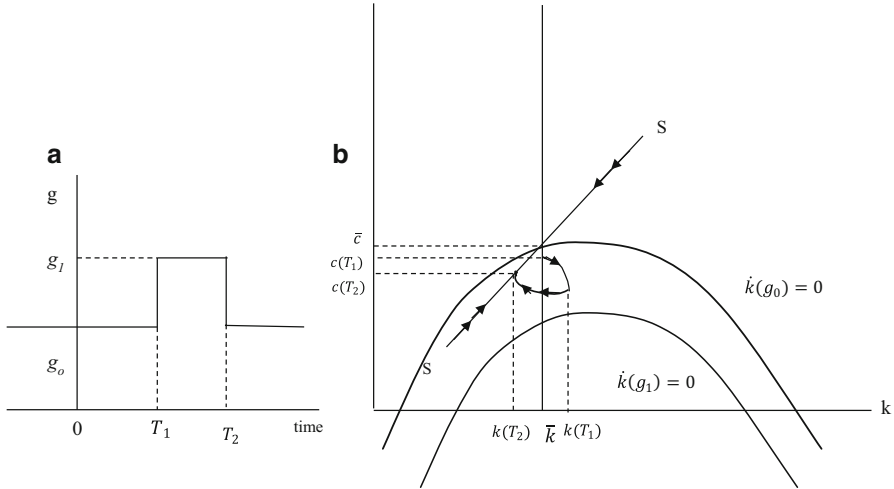


Fig. 1.9 (a, b) An anticipated transitory increase in government spending and its effects

1.3 Monetary Economy

The purpose of this section is to analyze how money affects real output, capital stock, and consumption. In Sect. 3.1, the central bank follows a monetary policy rule of money stock control. In Sect. 3.2, the model's monetary policy rule has the central bank controlling the nominal interest rate.

1.3.1 Monetary Policy Rule: Money Stock Control

Money is introduced into the models from different approaches. One assumes that money, unlike other financial assets, produces services because it is used as a means of payment. Assume, then, that the representative agent's instantaneous utility depends on consumption (c) and money services (m) according to the separable function:

$$U(c, m) = u(c) + v(m).$$

Functions $u(c)$ and $v(m)$ have traditional properties, i.e.: positive, decreasing marginal utilities; positive marginal utilities when the variables tend to zero; and close to zero when the variables tend to infinity.

The agent's budget constraint, in terms of flows, states that revenues from income (y) and government transfers (τ) finance their consumption (c), investment (i), and increase in money stock ($\dot{M}/P$). In other words:

$$y + \tau = c + i + \frac{\dot{M}}{P}.$$

Given that gross investment (i) equals the increase in capital ($\dot{k}$) plus the depreciation of capital (δk), $i = \dot{k} + \delta k$ and $(\dot{M}/P) = \dot{m} + m\pi$, the budget constraint may be written as:

$$y + \tau = c + \dot{k} + \delta k + \dot{m} + m\pi.$$

The agent's wealth (a) equals the sum of their capital and money stocks: $a = k + m$. Therefore: $\dot{a} = \dot{k} + \dot{m}$. The agent's budget constraint, in terms of flows, shows how their wealth changes:

$$\dot{a} = y + \tau - c - \delta k - \pi m.$$

In this economy, production occurs by combining capital and labor, in a production function with constant returns to scale. The intensive-form production function is expressed by $f(k)$. The budget constraint, therefore, is:

$$\dot{a} = f(k) + \tau - c - \delta k - \pi m.$$

The agent's problem consists of maximizing:

$$\int_0^{\infty} e^{-\rho t} [u(c) + v(m)] dt, \quad (1.12)$$

subject to constraints:

$$\dot{a} = f(k) + \tau - c - \delta k - \pi m, \quad (1.13)$$

$$a = k + m, \quad (1.14)$$

$$a(0) = a_o, \text{ given.} \quad (1.15)$$

The constraint $a = k + m$ will be used to substitute capital stock, $k = a - m$, in the state variable transition equation $\dot{a}$. The current-value Hamiltonian is:

$$H = u(c) + v(m) + \lambda [f(a - m) + \tau - c - \delta(a - m) - \pi m].$$

The first-order conditions are:

$$\frac{\partial H}{\partial c} = u'(c) - \lambda = 0,$$

$$\frac{\partial H}{\partial m} = v'(m) + \lambda [f'(k)(-1) + \delta - \pi] = 0,$$

$$\dot{\lambda} = \rho\lambda - \frac{\partial H}{\partial a} = \rho\lambda - \lambda[f'(k) - \delta],$$

$$\frac{\partial H}{\partial \lambda} = f(a - m) + \tau - c - \delta(a - m) - \pi m = \dot{a}.$$

The problem's optimal solution must satisfy the transversality condition:

$$\lim_{t \rightarrow \infty} \lambda a e^{-\rho t} = 0.$$

According to the first first-order condition equation, marginal utility of consumption is equal to the costate variable $u'(c) = \lambda$. Combining the first two conditions yields the traditional property according to which the marginal rate of substitution between consumption and money equals the nominal interest rate:

$$\frac{v'(m)}{u'(c)} = f'(k) - \delta + \pi. \quad (1.16)$$

Government

This economy's government issues money and transfers the seigniorage (τ) to society. The government budget constraint is:

$$\tau = \frac{\dot{M}}{P}. \quad (1.17)$$

The monetary policy consists of expanding the nominal stock of money at a constant rate equal to μ . The expansion rate of the real quantity of money equals the difference between the expansion rate of the nominal stock of money and the inflation rate:

$$\dot{m} = m(\mu - \pi).$$

Goods and Services Market Equilibrium

The goods and services market is at equilibrium when production equals expenditure:

$$y = f(k) = c + \dot{k} + \delta k. \quad (1.18)$$

Expenditure has two components: consumption and gross investment.

Dynamical System

The model's dynamical system has three differential equations; for consumption, capital stock, and the real quantity of money. That is:

$$\dot{c} = \frac{u'(c)}{u''(c)} [\rho + \delta - f'(k)], \quad (1.19)$$

$$\dot{k} = f(k) - c - \delta k, \quad (1.20)$$

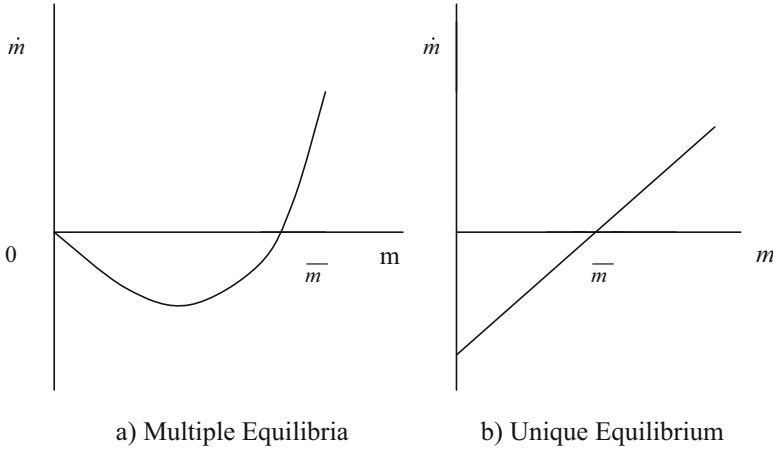


Fig. 1.10 (a) Multiple equilibria, (b) unique equilibrium

$$\dot{m} = (\mu + \rho)m - \frac{mv'(m)}{u'(c)}. \quad (1.21)$$

This dynamical system is separable, that is, the first two differential equations can be solved independently from the third. In the steady state, $\dot{c} = 0$, $\dot{k} = 0$, capital and consumption do not depend on the inflation rate:

$$f'(\bar{k}) - \delta = \rho; \bar{c} = f(\bar{k}) - \delta\bar{k}. \quad (1.22)$$

In the steady state, money, in addition to being neutral, is also superneutral. Money is neutral when a change in the nominal stock of money does not affect the economy's real output. Money is superneutral when a change in the expansion rate of the money stock does not affect the economy's real output.

The solution of the differential equation for the real stock of money can have a unique steady-state equilibrium, or two equilibria, as Fig. 1.10 illustrates. Whether or not the equilibrium is unique depends on the limit:

$$\lim_{m \rightarrow 0} mv'(m).$$

When the limit is positive, a unique equilibrium exists, as Fig. 1.10b shows. When the limit equals zero, $m = 0$ is a steady-state equilibrium for the model, that is, the economy includes the possibility of money having no value. Some authors refer to this situation as a bubble. It is not, however, a proper bubble because the value of the money services approaches zero as the real stock of money tends to zero. This proposition is easily understood by solving the differential equation for the real stock of money:

$$m(t) = \int_t^\infty e^{-(\mu+\rho)(x-t)} \frac{m v'(m)}{u'(c)} dx + C e^{(\mu+\rho)t}. \quad (1.23)$$

The expression's latter component is the bubble solution; the former is the fundamentals solution. Clearly, if $C = 0$ and $\lim_{m \rightarrow 0} m v'(m) = 0$, $m(t) = 0$ is not a bubble solution, but rather a fundamentals solution. In this model, the initial price level is an endogenous variable and its value may change instantaneously, so that the real quantity of money is the equilibrium quantity when the nominal stock of money is a given for the problem.

1.3.2 Monetary Policy Rule: Nominal Interest Rate Control

The representative agent's instantaneous utility function depends on consumption (c) and the real stock of money (m):

$$u(c, m), u_c > 0, u_m \geq 0.$$

The marginal utilities of consumption and money are positive. The utility function is concave, and its Hessian matrix is:

$$H = \begin{bmatrix} u_{cc} & u_{cm} \\ u_{mc} & u_{mm} \end{bmatrix}$$

Consumption and money are complementary, that is, the marginal utility of consumption increases when the real stock of money increases. Because the function is concave, the marginal utilities of consumption and money are decreasing:

$$u_{cm} = u_{mc}, u_{cm} \geq 0, u_{cc} < 0, u_{mm} \leq 0.$$

The utility function's concavity implies a positive determinant of the Hessian matrix:

$$|H| = u_{cc}u_{mm} - u_{cm}^2 \geq 0$$

The representative agent's problem therefore consists of maximizing:

$$\int_0^\infty e^{-\rho t} u(c, m) dt, \quad (1.24)$$

subject to the constraints:

$$\dot{a} = f(k) + \tau - c - \delta k - \pi m, \quad (1.25)$$

$$a = k + m, \quad (1.26)$$

$$a(0) = a_0 \text{ given.} \quad (1.27)$$

The current-value Hamiltonian is:

$$H = u(c, m) + \lambda[f'(a - m) + \tau - c - \delta(a - m) - \pi m],$$

where the constraint $a = k + m$ is used to eliminate the variable k . The first-order conditions are:

$$\frac{\partial H}{\partial c} = u_c - \lambda = 0,$$

$$\frac{\partial H}{\partial m} = u_m + \lambda[f'_k(-1) + \delta - \pi] = 0,$$

$$\dot{\lambda} = \rho\lambda - \frac{\partial H}{\partial a} = \rho\lambda - \lambda[f'_k(k) - \delta],$$

$$\frac{\partial H}{\partial \lambda} = f(a - m) + \tau - c - \delta(a - m) - \pi m = \dot{a}.$$

The problem's optimal solution satisfies the transversality condition:

$$\lim_{t \rightarrow \infty} \lambda a e^{-\rho t} = 0.$$

Monetary Policy Rule

The central bank sets the nominal interest rate (R) by adding the real interest rate to the inflation rate:

$$R = f_k - \delta + \pi. \quad (1.28)$$

Government Budget Constraint

The government issues money and transfers the funds to society:

$$\tau = \frac{\dot{M}}{P} = \frac{\dot{M}}{M} \frac{M}{P} = \mu m. \quad (1.29)$$

Algebra

Combining the first two first-order conditions, we reach the conclusion that the rate of substitution between money and consumption is equal to the nominal interest rate:

$$\frac{u_m}{u_c} = R.$$

This equation implicitly defines the money demand equation: $m = L(c, R)$. This equation's differential is obtained by differentiating the previous equation:

$$u_{mm}dm + u_{mc}dc = u_c dR + Ru_{cm}dm + Ru_{cc}dc,$$

which can be rewritten, after rearranging terms, as:

$$(u_{mm} - Ru_{cm})dm = u_c dR + (Ru_{cc} - u_{mc})dc.$$

Therefore, the differential money demand equation, in percentage terms, is:

$$\frac{dm}{m} = \frac{u_c R}{(u_{mm} - Ru_{cm})m} \frac{dR}{R} + \left(\frac{Ru_{cc} - u_{mc}}{u_{mm} - Ru_{cm}} \right) \frac{c}{m} \frac{dc}{c}.$$

The coefficients of dR/R and dc/c are the elasticities of the real quantity of money with respect to the interest rate, $\varepsilon_{m,R}$, and consumption, $\varepsilon_{m,c}$:

$$\varepsilon_{m,R} = \frac{u_m}{(u_{mm} - Ru_{cm})m},$$

$$\varepsilon_{m,c} = \left(\frac{Ru_{cc} - u_{mc}}{u_{mm} - Ru_{cm}} \right) \frac{c}{m}.$$

The expression for the elasticity of the real quantity of money relative to the interest rate takes into account that $u_c R = u_m$. This elasticity's absolute value equals ξ . That is:

$$|\varepsilon_{m,R}| = \frac{u_m}{(u_{mm} - Ru_{cm})m} = \xi.$$

According to the first-order condition equation, marginal utility of consumption equals the costate variable λ :

$$u_c = \lambda.$$

The derivative of this expression with respect to time yields:

$$u_{cc}\dot{c} + u_{cm}\dot{m} = \dot{\lambda}.$$

Therefore, the derivative of the real quantity of money relative to time equals:

$$\dot{m} = \frac{\dot{\lambda} - u_{cc}\dot{c}}{u_{cm}}.$$

The derivative of the equation that implicitly defines money demand, $u_m = Ru_c$, with respect to time yields:

$$u_{mc}\dot{c} + u_{mm}\dot{m} = \dot{R}u_c + Ru_{cc}\dot{c} + Ru_{cm}\dot{m}.$$

Rearranging the expression's terms, we obtain:

$$(u_{mc} - Ru_{cc})\dot{c} + (u_{mm} - Ru_{cm})\dot{m} = u_c \dot{R}.$$

Substituting the value of the derivative of the real quantity of money relative to time, as obtained previously, yields:

$$(u_{mc} - Ru_{cc})\dot{c} + (u_{mm} - Ru_{cm})\left(\frac{\dot{\lambda} - u_{cc}\dot{c}}{u_{cm}}\right) = u_c\dot{R}.$$

This equation, after rearranging the terms that multiply $\dot{c}$, becomes:

$$\left(u_{mc} - \frac{u_{mm}u_{cc}}{u_{cm}}\right)\dot{c} + \left(\frac{u_{mm} - Ru_{cm}}{u_{cm}}\right)\dot{\lambda} = u_c\dot{R}.$$

Multiplying both sides by u_{cm} yields:

$$(u_{mc}u_{cm} - u_{mm}u_{cc})\dot{c} = (Ru_{cm} - u_{mm})\dot{\lambda} + u_cu_{cm}\dot{R}.$$

The conclusion is that the growth rate of consumption depends on the growth rates of the costate variable and the nominal interest rate:

$$\frac{\dot{c}}{c} = \frac{(Ru_{cm} - u_{mm})u_c}{c(u_{cm}^2 - u_{mm}u_{cc})} \frac{\dot{\lambda}}{\lambda} + \frac{1}{c} \left(\frac{u_{cm}u_m}{u_{cm}^2 - u_{mm}u_{cc}} \right) \frac{\dot{R}}{R}.$$

Separating the coefficient of $\dot{\lambda}/\lambda$ yields:

$$\frac{\dot{c}}{c} = \frac{(Ru_{cm} - u_{mm})u_c}{c(u_{cm}^2 - u_{mm}u_{cc})} \left[\frac{\dot{\lambda}}{\lambda} + \frac{u_{cm}u_m}{(Ru_{cm} - u_{mm})u_c} \frac{\dot{R}}{R} \right].$$

Given that:

$$\frac{\dot{\lambda}}{\lambda} = \rho - f_k + \delta, \text{ and } \frac{u_m}{Ru_{cm} - u_{mm}} = m\xi,$$

the growth rate of consumption equals:

$$\frac{\dot{c}}{c} = \frac{(Ru_{cm} - u_{mm})u_c}{c(u_{cm}^2 - u_{mm}u_{cc})} \left[\rho - f_k + \delta + \xi \frac{u_{cm}m}{u_c} \frac{\dot{R}}{R} \right].$$

The elasticity of the marginal utility of consumption with respect to money is defined by:

$$\phi = \frac{u_{cm}m}{u_c} = \frac{\partial u_c}{\partial m} \frac{m}{c}.$$

Therefore, the growth rate of consumption may be written as:

$$\frac{\dot{c}}{c} = \frac{(Ru_{cm} - u_{mm})u_c}{c(u_{mm}u_{cc} - u_{cm}^2)} \left[f_k - \delta - \rho - \xi\phi \frac{\dot{R}}{R} \right].$$

To simplify the notation, the parameter $\sigma(c, m)$ is defined by:

$$\sigma(c, m) = \frac{(Ru_{cm} - u_{mm})u_c}{c(u_{mm}u_{cc} - u_{cm}^2)}.$$

Dynamical System

The model's dynamical system has two differential equations, one for consumption and the other for the capital stock:

$$\frac{\dot{c}}{c} = \sigma(c, m) \left[f_k - \delta - \rho - \xi \phi \frac{\dot{R}}{R} \right], \quad (1.30)$$

$$\dot{k} = f(k) - \delta k - c. \quad (1.31)$$

In this dynamical system's steady state, the real interest rate equals:

$$f_k - \delta = \rho + \xi \phi \frac{\dot{R}}{R}.$$

The parameter γ is defined by:

$$\xi \phi \frac{\dot{R}}{R} = -\gamma,$$

and the nominal interest rate's growth rate equals:

$$\frac{\dot{R}}{R} = -\frac{\gamma}{\xi \phi}.$$

Therefore, the economy's real interest rate is:

$$f_k - \delta = \rho - \gamma. \quad (1.32)$$

Monetary policy may choose the parameter γ and affect the economy's equilibrium capital stock. The model's optimal solution satisfies the transversality condition: $\lim_{t \rightarrow \infty} \lambda a e^{-\rho t} = 0$. According to this condition, wealth (a) valued in terms of utility (λa) may grow at a lower rate than the rate of time preference (ρ). The growth rate of costate variable (λ) grows at a rate equal to γ . That is:

$$\frac{\dot{\lambda}}{\lambda} = \rho - (f_k - \delta) = \rho - (\rho - \gamma) = \gamma.$$

Therefore, the parameter γ must satisfy the following inequality:

$$\gamma < \rho.$$

The conclusion is that the monetary policy is not superneutral. The central bank may select a path for the nominal interest rate and affect the economy's steady state capital stock and output.

1.4 Real Business Cycle

The real business cycle model admits that changes in real output and employment along the economic cycle are not caused by rigidity in the price system or by nominal monetary policy shocks. Cycles are caused by technology shocks. Hence the term “real” in the model's name. Its main characteristics include: (i) the decisions of economic agents – consumers, workers, firms – are based on intertemporal optimization with rational expectations; (ii) the economy's general equilibrium is specified; and (iii) the model's qualitative and quantitative properties are analyzed and studied.

Production function:

$$Y = AF(K, L),$$

is first-degree homogeneous for the factors of production, capital (K), and labor (L). The letter A represents technology shocks.

The representative agent's utility function is separable into consumption (C) and leisure (ℓ):

$$u(C) + v(\ell),$$

with the functions' traditional properties.

The agent allocates their available time ($\bar{t}$) between labor (L) and leisure (ℓ):

$$L + \ell = \bar{t}.$$

The agent's income (Y) is spent on consumption (C) and investment, $I = \dot{K} + \delta K$, to increase the capital stock ($\dot{K}$) and replace the capital spent (δK) in the production process, where δ is the depreciation rate:

$$Y = C + I = C + \dot{K} + \delta K.$$

The representative agent solves the following problem:

$$\max \int_0^{\infty} e^{-\rho t} [u(C) + v(\ell)] dt, \quad (1.33)$$

subject to the constraints:

$$\dot{K} = AF(K, L) - C - \delta K, \quad (1.34)$$

$$L + \ell = \bar{t}, \quad (1.35)$$

$$K(0) = K_o, \quad \text{given.} \quad (1.36)$$

The time constraint may be used to substitute for leisure ($\ell = \bar{t} - L$) in the instantaneous utility function. The current-value Hamiltonian is:

$$H = u(C) + v(\bar{t} - L) + \lambda[AF(K, L) - C - \delta K].$$

The first-order conditions are:

$$\begin{aligned} \frac{\partial H}{\partial C} &= u_c - \lambda = 0, \\ \frac{\partial H}{\partial L} &= -v_\ell + \lambda AF_L = 0, \\ \dot{\lambda} &= \rho\lambda - \frac{\partial H}{\partial K} = \rho\lambda - \lambda(AF_K - \delta), \\ \frac{\partial H}{\partial \lambda} &= AF(K, L) - C - \delta K = \dot{K}. \end{aligned}$$

According to the first condition, the costate variable must equal the marginal utility of consumption. At each point in time, the agent must allocate their time between leisure and work. The second first-order condition equation addresses this choice: the marginal product of labor must equal the marginal rate of substitution between leisure and consumption. The third first-order condition equation is concerned with the choice between consumption and savings: the rate of time preference must equal the net marginal product of capital plus gains from reduced consumption. The final first-order condition equation reproduces the agent's budget constraint.

The transversality condition establishes that the present value of capital, in terms of utility, must equal zero in the distant future:

$$\lim_{t \rightarrow \infty} \lambda K e^{-\rho t} = 0.$$

The solution to this problem is given by the equations system:

$$\begin{cases} u_c = \lambda \\ AF_L = \frac{v_\ell}{\lambda} \\ \dot{\lambda} = \lambda[\rho - AF_K + \delta] \\ \dot{K} = AF(K, L) - C - \delta K \\ L + \ell = \bar{t} \\ \lim_{t \rightarrow \infty} \lambda K e^{-\rho t} = 0 \end{cases}.$$

The first equation, for the equality between the marginal utility of consumption and the costate variable, may be plugged into the equations system. This produces the following system:

$$\begin{cases} AF_L = \frac{v_\ell}{u_c} \\ \dot{C} = \frac{u_c}{u_{cc}}[\rho - AF_K + \delta] \\ \dot{K} = AF(K, L) - C - \delta K \\ L + \ell = \bar{l} \\ \lim_{t \rightarrow \infty} u_c K e^{-\rho t} = 0 \end{cases} \quad (1.37)$$

The system will be solved for the particular case in which the production, utility of consumption, and utility of leisure functions have the following functional forms: $F(K, L) = K^\alpha L^{1-\alpha}$; $U(C) = \log C$; and $v(\ell) = \beta \ell$. These functions' partial derivatives are: $F_L = (1 - \alpha)K^\alpha L^{-\alpha} = (1 - \alpha)(K/L)^\alpha$, $F_K = \alpha K^{\alpha-1} L^{1-\alpha} = \alpha(K/L)^{\alpha-1}$, $u_c = 1/C$, and $v_\ell = \beta$. The marginal product of labor equals the marginal rate of substitution between leisure and consumption:

$$AF_L = \frac{v_\ell}{u_c} \Rightarrow A(1 - \alpha)k^\alpha = \frac{\beta}{1/C},$$

where k is the K/L ratio. It follows that consumption depends on the k ratio according to:

$$C = \frac{A(1 - \alpha)}{\beta} k^\alpha.$$

The growth rate of consumption, the derivative of the log of consumption relative to time, is proportional to the growth rate of the capital-labor ratio k :

$$\frac{\dot{C}}{C} = \alpha \frac{\dot{k}}{k}.$$

The problem's Euler equation is:

$$\dot{C} = \frac{1/C}{-1/C^2} [\rho - AF_K + \delta].$$

Therefore, combining the last two equations yields the following differential equation for k :

$$\frac{\dot{k}}{k} = \frac{1}{\alpha} [A\alpha k^{\alpha-1} - \rho - \delta].$$

Dynamical System

The model's dynamical system has two differential equations:

$$\begin{cases} \dot{k} = Ak^\alpha - \frac{(\rho + \delta)}{\alpha} k & (1.38) \\ \dot{K} = AKk^{\alpha-1} - \frac{A(1-\alpha)}{\beta} k^\alpha - \delta K & (1.39) \end{cases}$$

The system's steady state equilibrium solution, $\dot{k} = \dot{K} = 0$, is:

$$\bar{k} = \left(\frac{\alpha A}{\rho + \delta} \right)^{\frac{1}{1-\alpha}}; \bar{K} = \frac{\alpha(1-\alpha)A}{\beta[\rho + (1-\alpha)\delta]} \left(\frac{\alpha A}{\rho + \delta} \right)^{\frac{\alpha}{1-\alpha}}.$$

The steady-state equilibrium of the number of working hours, $\bar{L} = \bar{K}/\bar{k}$, is independent of the parameter A that represents technological progress:

$$\bar{L} = \frac{(1-\alpha)(\rho + \delta)}{\beta(\rho + (1-\alpha)\delta)}.$$

The dynamical system's Jacobian matrix equals:

$$J = \begin{bmatrix} \frac{\partial \dot{k}}{\partial k} & \frac{\partial \dot{k}}{\partial K} \\ \frac{\partial \dot{K}}{\partial k} & \frac{\partial \dot{K}}{\partial K} \end{bmatrix} = \begin{bmatrix} \alpha Ak^{\alpha-1} - \frac{\rho + \delta}{\alpha} & 0 \\ (\alpha - 1)AKk^{\alpha-2} - \frac{A\alpha(1-\alpha)}{\beta} & Ak^{\alpha-1} - \delta \end{bmatrix}.$$

The matrix's determinant at the steady state equilibrium point is:

$$|J| = \left(\alpha Ak^{\alpha-1} - \frac{\rho + \delta}{\alpha} \right) (Ak^{\alpha-1} - \delta) = -\frac{(1-\alpha)}{\alpha^2} (\rho + \delta)(\rho + (1-\alpha)\delta) < 0.$$

We therefore conclude that the model's steady-state equilibrium point is a saddle point.

Figure 1.11 provides the phase diagram for the k differential equation. When $\dot{k} = 0$, $k = \bar{k}$. For k values greater than $\bar{k}$, k decreases, and the opposite occurs for lower values, as the arrows show in Fig. 1.11.

Figure 1.12 shows the phase diagram for the K differential equation. When $\dot{K} = 0$,

$$K = \frac{A(1-\alpha)k^\alpha}{\beta[Ak^{\alpha-1} - \delta]}.$$

The arrows in Fig. 1.12 show that for points above this curve K increases, and the opposite takes places for points below the curve.

Figure 1.13 shows the model's phase diagram. The SS saddle path shows that, for a given initial capital stock, the economy converges to the steady state.

Experiments

In the real business cycle model, the economic cycle is generated by technological progress. A positive technology shock increases the parameter A . A negative

Fig. 1.11 The dynamics of k

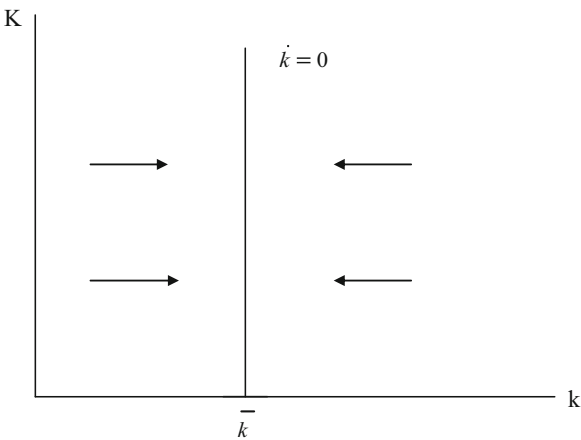


Fig. 1.12 The dynamics of K

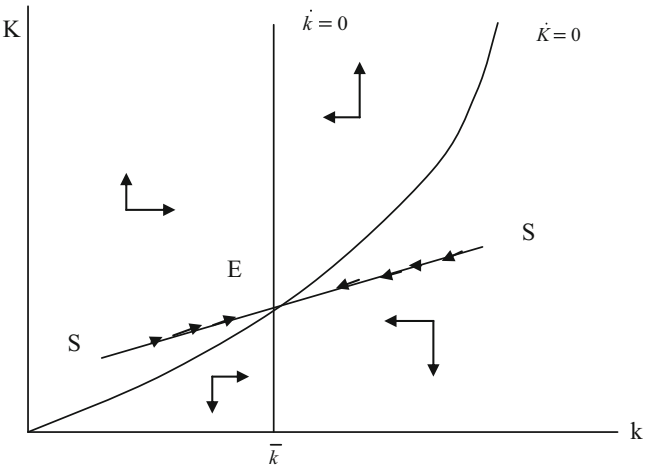
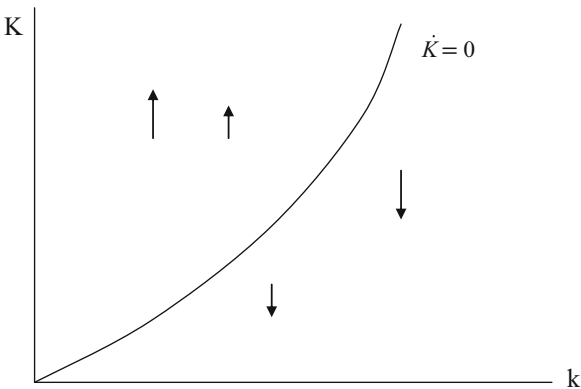


Fig. 1.13 The phase diagram for the K and k system

technology shock decreases the parameter A . A positive shock is due to technology advances that increase the productivities of the factors of production. Negative shocks must be understood as constraints, due largely to policy-related reasons that prevent the efficient use of resources, as it makes no sense at all to assume that society may have unlearned what it once knew. To analyze the consequences of changes in technology parameter A , one may carry out comparative dynamics exercises with permanent or transitory, anticipated or unanticipated changes. Figure 1.14 illustrates an unanticipated, permanent increase in parameter A .

Figure 1.15 describes the comparative dynamics exercise. The system's initial equilibrium was at point E_0 . The graph does not show the curves that went through that point for the sake of simplicity. When the technology change occurs, the $\dot{k} = 0$

Fig. 1.14 An unanticipated permanent increase in A

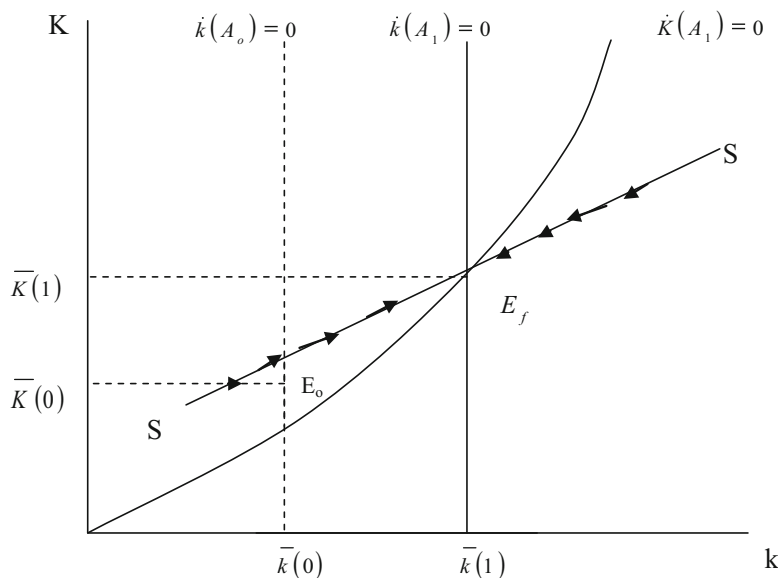
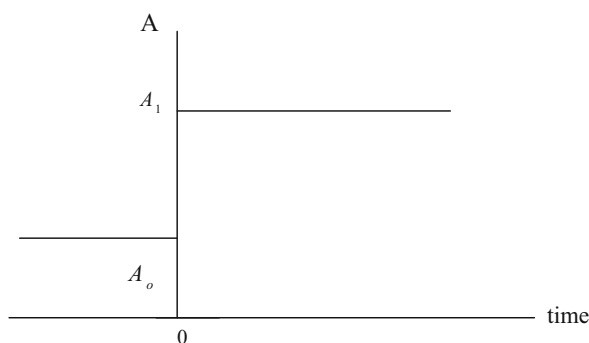
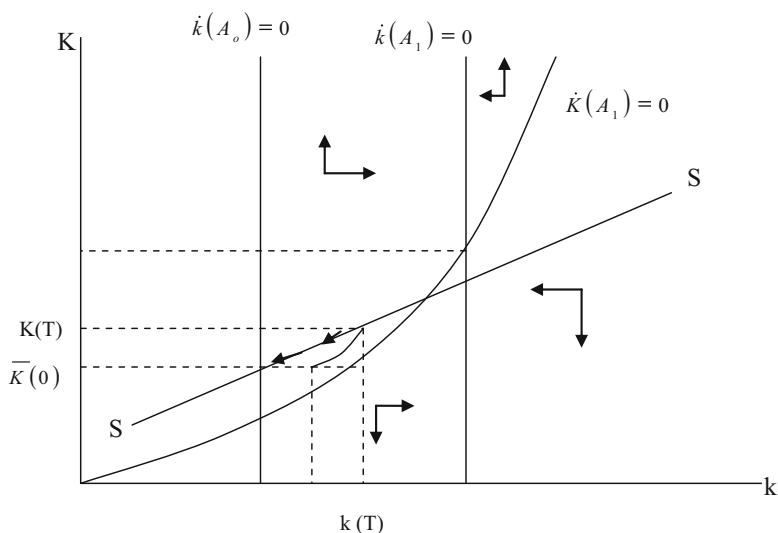
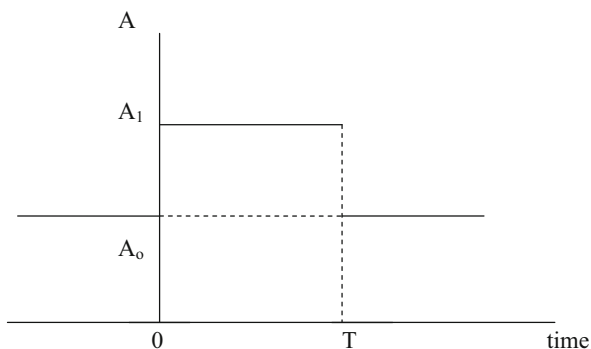


Fig. 1.15 The effects of an unanticipated permanent increase in A

Fig. 1.16 An unanticipated transitory increase in A **Fig. 1.17** The effects of an unanticipated transitory increase in A

and $\dot{K} = 0$ curves shift to go through point E_f . The initial capital was at the initial equilibrium point. Saddle path SS leads the economy to its new steady-state equilibrium, as only this path satisfies the transversality condition. In this new equilibrium, both the capital stock and the capital-labor ratio increase. However, at the new equilibrium, the number of working hours remains the same as under the previous equilibrium. That is, Fig. 1.15 must have $\bar{K}(1)/k(1) = \bar{K}(0)/k(0) = \bar{L}$. During the adjustment process, the number of working hours does not remain constant. It initially increases as the k ratio decreases. Over time, the number of working hours decreases until it returns to the steady-state equilibrium.

Figure 1.16 illustrates a transitory change in the technological progress parameter A . After a period T , the parameter A returns to its previous value. Figure 1.17 shows the dynamics of the system as a result of the shock. For simplicity, Fig. 1.17 does not include curve $\dot{K}(A_0) = 0$. At instant T , the economy must be at the SS saddle;

otherwise, it will not converge to steady state equilibrium. The k ratio initially increases, that is, the number of working hours decreases, and savings and the capital stock increase. From instant T onward, the capital stock and the capital-labor ratio start to decrease. In the long term, the economy returns to its former steady-state equilibrium.

Real Business Cycle in an Economy with a Government

When the government is introduced into the real business cycle model, the only change in the dynamical system is the variable that stands for government spending (g). It enters the differential equation for capital stock K , as in this case expenditures include government spending in addition to consumption and investment. The dynamical system has two differential equations:

$$\begin{cases} \dot{k} = Ak^\alpha - \frac{(\rho + \delta)}{\alpha} k & (1.40) \\ \dot{K} = AKk^{\alpha-1} - \frac{A(1-\alpha)}{\beta} k^\alpha - g - \delta K & (1.41) \end{cases}$$

In this model, a shift in government spending changes the economy's equilibrium. The changes may be permanent or transitory, announced or unanticipated. The economy's cycles may be caused not only by technology, but also by fiscal policy.

1.5 Exercises

1. The representative agent maximizes:

$$\int_0^\infty \beta(t) u(c) dt,$$

subject to the constraints:

$$\begin{aligned} \dot{a} &= ra + y - c \\ a(0) &= a_0 \quad \text{given.} \end{aligned}$$

where $\beta(t)$ is the agent's discount factor. Assume that for $s > 0$, the agent chooses to maximize:

$$\int_s^\infty \beta(t-s) u(c) dt,$$

subject to the constraints:

$$\begin{aligned}\dot{a} &= ra + y - c \\ a(s) &= a_s \quad \text{given.}\end{aligned}$$

What condition must the discount factor $\beta(\tau)$ satisfy so that the solution to the two problems is the same?

2. The representative agent maximizes:

$$\int_0^\infty e^{-\rho \tau} [u(c) + v(m)] dt,$$

subject to the constraints:

$$\begin{aligned}y + \tau &= c + \frac{\dot{M}}{P}, \\ M(0) &\text{ given,}\end{aligned}$$

where the symbols have the following meanings: c = consumption; $m = M/P$ = real money supply; y = income; τ = government transfers; M = nominal stock of money; $\dot{M} = dM/dt$; P = price level.

- (a) Derive the problem's first-order conditions.
 - (b) Analyze the model's equilibrium and dynamics with the costate variable (λ) on the vertical axis and the real quantity of money on the horizontal axis.
 - (c) Repeat the previous item (b), with consumption on the vertical axis instead of the costate variable, and the real quantity of money on the horizontal axis.
3. Add the following assumptions to the previous question's model:

Government: $\tau = \frac{\dot{M}}{P},$

Goods and Services Market Equilibrium: $c = y.$

- (a) Monetary policy holds the growth rate of money stock constant: $\mu = \dot{M}/M$. Can there be hyperinflation? Can there be hyperdeflation?
- (b) Suppose that the monetary and fiscal policy is such that:

$$\frac{\dot{M}}{P} = \text{constant}$$

Can there be hyperinflation in this model? Can there be hyperdeflation?

4. Assume that the cash in advance constraint (CIA) is:

$$M(t) \geq F(\theta) = \int_t^{t+\theta} C(s) ds.$$

(a) Show that a Taylor expansion of function $F(\theta)$ can be expressed by:

$$F(\theta) = F(0) + F'(0)\theta + \frac{F''(0)}{2}\theta^2 + \dots = C(t)\theta + \frac{\dot{C}(t)}{2}\theta^2 + \dots$$

(b) Show that the cash in advance constraint is approximately expressed by:

$$M(t) \geq \theta C(t).$$

5. The representative agent maximizes:

$$\int_0^\infty e^{-\rho t} u(c) dt,$$

subject to constraints:

$$\begin{aligned} \dot{a} &= f(k) + \tau - c - \delta k - \pi m, \\ m &\geq c, \\ k(0) \text{ and } M(0) &\text{ given.} \end{aligned}$$

(a) Derive the problem's first-order conditions.

(b) Is money superneutral in this model?

6. The representative agent maximizes:

$$\int_0^\infty e^{-\rho t} u(c) dt,$$

subject to the constraints:

$$\begin{aligned} f(k) + \tau &= c + \dot{k} + \delta k + \frac{\dot{M}}{P}, \\ m &\geq c + \dot{k} + \delta k, \\ k(0) \text{ and } M(0) &\text{ given.} \end{aligned}$$

The model's second constraint – the cash in advance constraint – assumes that there is a need for money to buy both consumer goods and capital goods.

(a) Derive the problem's first-order conditions.

(b) Is money neutral?

(c) Is money superneutral?

7. The representative agent maximizes:

$$\int_0^\infty e^{-\rho t} [u(c) + v(m)] dt,$$

subject to constraints:

$$(1 - \tau)(rb + y) = c + \frac{\dot{B}}{P} + \frac{\dot{M}}{P},$$

$m(0)$ and $b(0)$ given.

At the goods and services market equilibrium, output equals expenditure:

$$y = c + g.$$

The government budget constraint is:

$$g + rb - \tau(rb + y) = \frac{\dot{M}}{P} + \frac{\dot{B}}{P}.$$

The central bank increases the nominal stock of money at a constant rate and the differential equation of the real stock of money is:

$$\dot{m} = m(\mu - \pi), \mu = \bar{\mu} = \text{constant}.$$

- (a) On a phase diagram with $b(=B/P)$ on the vertical axis and $m(=M/P)$ on the horizontal axis, show the model's equilibrium and dynamics.
 - (b) On the previous item's phase diagram, analyze the following experiment: at instant zero, the central bank reduces the monetary expansion rate from μ_0 to μ_1 . After a period of time T , when public debt reaches its ceiling (b^s), the central bank must change the monetary policy to accommodate the public deficit. What happens to the inflation rate at instant zero?
8. In a representative agent model with government bonds, the following transversality condition must be satisfied:

$$\lim_{T \rightarrow \infty} \lambda b e^{-\rho T} = 0,$$

where λ is the costate variable and b is the real stock of public debt.

- (a) Suppose that $\dot{b} = f + \rho b$, where f is the primary deficit. Show that:

$$b(T) = b(0)e^{\rho T} + \left[\int_0^T f e^{-\rho s} ds \right] e^{\rho T}.$$

- (b) Assume a constant f . Show that:

$$\lim_{T \rightarrow \infty} \lambda b e^{-\rho T} \neq 0.$$

- (c) Suppose that:

$$g - \tau + rb = \dot{b} = f = \text{constant},$$

(d) where f is now the real deficit. Show that $b(T) - b(t) = fT$ and that,

$$\lim_{T \rightarrow \infty} \lambda b e^{-\rho T} = 0.$$

(e) What conclusion can you draw from items (b) and (c)?

9. Consider the real business cycle model with a government. The model's dynamical system is:

$$\begin{aligned}\dot{k} &= Ak^\alpha - \frac{(\rho + \delta)}{\alpha} k, \\ \dot{K} &= AKk^{\alpha-1} - \frac{A(1-\alpha)}{\beta} k^\alpha - g - \delta K.\end{aligned}$$

- (a) Analyze an unanticipated, permanent change in government spending.
- (b) Analyze an anticipated, permanent change in government spending.
- (c) Analyze an unanticipated, transitory change in government spending.
- (d) Analyze an anticipated, transitory change in government spending.

10. The representative agent maximizes:

$$\int_0^\infty e^{-\rho t} u(c) dt,$$

subject to the constraint:

$$\dot{k} = f(k) - (n + \delta)k - c.$$

Population grows at a rate n and, for the sake of simplicity, the initial population is normalized at unity.

- (a) Does the model's net marginal product equal the rate of time preference?
- (b) Why was the previous item's result obtained?

Chapter 2

The Open-Economy Representative Agent Model



This chapter addresses the Achilles' heel of the representative agent model: the open economy. Section 2.1 discusses the issue of goods aggregation in an open economy. Section 2.2 shows that, in a small open economy, the representative agent model requires ad hoc assumptions for a stationary equilibrium. Section 2.3 analyzes the model when the rate of time preference is variable. Section 2.4 presents the model in which the interest rate carries a risk premium. Section 2.5 shows the calculation of the IS curve for a small open economy from a representative agent model with complete financial markets.

2.1 Goods Aggregation

An open economy assumes that goods and services are traded among countries. The open economy's models must, therefore, specify the kinds of goods and services that characterize the economy. The literature offers at least two types of structures: one Keynesian-inspired, and the other according to the Australian dependent economy model. Table 2.1 describes the two structures. Two goods (or products), X and Y , exist. Under the Keynesian model, one product is domestic and the other is imported. The domestic product is also exported. According to the dependent economy model, one product is tradable and the other is not tradable because its transportation cost is prohibitive. The first question to analyze in connection with open-economy models is how to aggregate these goods.

Assume two goods, X and Y . Consumption of these goods is represented by C_x and C_y , respectively. Aggregate consumption C is calculated using the aggregator function $F(C_x, C_y) = C$. Price level P must be set so that the budget constraint $PC = P_x C_x + P_y C_y$ is satisfied. To determine the price level, the following problem must be solved. Minimize:

Table 2.1 Structure of Goods Aggregation

Models	Good X	Good Y
Keynesian	Domestic	Imported
Dependent economy	Tradable	Non-tradable

$$E = P_x C_x + P_y C_y, \quad (2.1)$$

subject to the constraint:

$$F(C_x, C_y) = C. \quad (2.2)$$

The problem's Lagrangian is:

$$L = P_x C_x + P_y C_y + \lambda [C - F(C_x, C_y)]$$

where λ is the Lagrange multiplier. The first-order conditions for the minimum are:

$$\begin{aligned} \frac{\partial L}{\partial C_x} &= P_x - \lambda \frac{\partial F}{\partial C_x} = 0, \\ \frac{\partial L}{\partial C_y} &= P_y - \lambda \frac{\partial F}{\partial C_y} = 0. \end{aligned}$$

Dividing one equation by the other to eliminate the Lagrange multiplier, the relative price equals the two goods' marginal rate of substitution. This equation, together with the aggregator equation, forms a system with two equations and two unknowns:

$$\begin{cases} \frac{P_x}{P_y} = \frac{\frac{\partial F}{\partial C_x}}{\frac{\partial F}{\partial C_y}} \\ C = F(C_x, C_y) \end{cases},$$

To resolve the system, consumption of the two goods depends on their relative price and on aggregate consumption:

$$\begin{cases} C_x = C_x(P_x/P_y, C) \\ C_y = C_y(P_x/P_y, C) \end{cases}.$$

Substituting the two equations in the objective function yields the expenditure function:

$$E = P_x C_x(P_x/P_y, C) + P_y C_y(P_x/P_y, C).$$

Total expenditure equals the product of aggregate consumption times price level: $E = PC$, when $C = 1 \Rightarrow P = E$. Therefore, the price level is obtained by setting $C = 1$ in the expenditure equation:

$$P = P_x C_x (P_x / P_y, 1) + P_y C_y (P_x / P_y, 1). \quad (2.3)$$

Example: The Cobb-Douglas Function

Assume that the aggregator function is a Cobb-Douglas function:

$$C = C_x^\alpha C_y^\beta. \quad (2.4)$$

The relative price is equal to the marginal rate of substitution:

$$\frac{P_x}{P_y} = \frac{\frac{\partial F}{\partial C_x}}{\frac{\partial F}{\partial C_y}} = \frac{\alpha C_x^{\alpha-1} C_y^\beta}{\beta C_x^\alpha C_y^{\beta-1}} = \frac{\alpha C_y}{\beta C_x}.$$

This equation, together with the aggregator equation, yields each product's consumption equation:

$$C_x = \left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha+\beta}} \left(\frac{P_x}{P_y} \right)^{-\frac{\beta}{\alpha+\beta}} C^{\frac{1}{\alpha+\beta}},$$

$$C_y = \left(\frac{\beta}{\alpha} \right)^{\frac{\alpha}{\alpha+\beta}} \left(\frac{P_x}{P_y} \right)^{-\frac{\alpha}{\alpha+\beta}} C^{\frac{1}{\alpha+\beta}}.$$

Substituting these two equations in the expenditure equation and setting $C = 1$ yields the price level associated with the Cobb-Douglas function:

$$P = E(C = 1) = \left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha+\beta}} P_x \left(\frac{P_x}{P_y} \right)^{-\frac{\beta}{\alpha+\beta}} + \left(\frac{\beta}{\alpha} \right)^{\frac{\alpha}{\alpha+\beta}} P_y \left(\frac{P_x}{P_y} \right)^{\frac{\alpha}{\alpha+\beta}}.$$

The price level is also a Cobb-Douglas geometric index:

$$P = \gamma P_x^{\frac{\alpha}{\alpha+\beta}} P_y^{\frac{\beta}{\alpha+\beta}} \dots \gamma = \left(\frac{\alpha + \beta}{\alpha} \right) \left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha+\beta}}. \quad (2.5)$$

Utility of the Representative Agent

In the representative agent model, the representative agent's welfare is measured by an instantaneous utility function that depends on aggregate consumption:

$$u = u(C).$$

Aggregate consumption, in its turn, is obtained by a function that aggregates the two goods:

$$C = F(C_X, C_Y).$$

This aggregator function is associated with a price level such that the following relationship is valid:

$$PC = P_X C_X + P_Y C_Y.$$

2.2 Constant Rate of Time Preference

In a small open economy, the representative agent maximizes the present value of the instantaneous utility flow,

$$\int_0^{\infty} e^{-\rho t} u(c) dt, \quad (2.6)$$

subject to budget constraint:

$$\dot{a} = ra + y - c. \quad (2.7)$$

Revenue from financial investments (ra) plus income from other sources (y) minus consumption spending (c) equals the agent's wealth change ($\dot{a}$). The initial value of the agent's wealth, which may be either positive or negative, is a given. That is:

$$a(0) = a_0 \quad \text{given.} \quad (2.8)$$

A small open economy's interest rate is determined by international financial markets and the country cannot change it. The country may be either a net creditor or a net debtor.

The current-value Hamiltonian is:

$$H = u(c) + \lambda(ra + y - c).$$

The problem's first-order conditions are:

$$\begin{aligned} \frac{\partial H}{\partial c} &= \frac{\partial u}{\partial c} - \lambda = 0, \\ \dot{\lambda} &= \rho\lambda - \frac{\partial H}{\partial a} = \rho\lambda - \lambda r, \\ \frac{\partial H}{\partial \lambda} &= ra + y - c = \dot{a}, \end{aligned}$$

and the following transversality condition must be satisfied:

$$\lim_{t \rightarrow \infty} \lambda a e^{-rt} = 0.$$

Algebra

The derivative of $u'(c) = \lambda$ with respect to time yields $u''(c)\dot{c} = \dot{\lambda}$. Dividing this equation by the former yields:

$$\frac{u''(c)}{u'(c)}\dot{c} = \frac{\dot{\lambda}}{\lambda} = \rho - r.$$

Assuming, for simplicity, that the utility function has constant elasticity of substitution, the differential equation for consumption is:

$$\dot{c} = \sigma c(r - \rho).$$

Dynamical System

This model's two differential equations form the following dynamical system:

$$\begin{cases} \dot{c} = \sigma c(r - \rho) \\ \dot{a} = ra + y - c \end{cases} \quad (2.9)$$

The Jacobian matrix, evaluated at steady state, is:

$$J = \begin{bmatrix} \frac{\partial \dot{c}}{\partial c} & \frac{\partial \dot{c}}{\partial a} \\ \frac{\partial \dot{a}}{\partial c} & \frac{\partial \dot{a}}{\partial a} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & r \end{bmatrix}.$$

The matrix's determinant equals zero:

$$|J| = 0.$$

This means that one of the roots equals zero, producing what literature calls a zero root. This dynamical system only has a steady-state equilibrium if, by chance, the representative agent's rate of time preference is equal to the international interest rate. When they differ, two possibilities exist. If the international interest rate is lower than the rate of time preference, the agents borrow from the international market to finance consumption. This behavior, however, is not sustainable in the long term because debt will grow indefinitely. In the second possibility, where the international interest rate is greater than the rate of time preference, the country will prefer to delay consumption, save, and invest its funds in the international market, becoming a creditor country. However, no mechanism exists to prevent this country from becoming the owner of the world, and eventually ceasing to be a small country. The conclusion is that the representative agent model cannot be applied to a small open economy, unless one is willing to introduce an assumption, such as a rate of

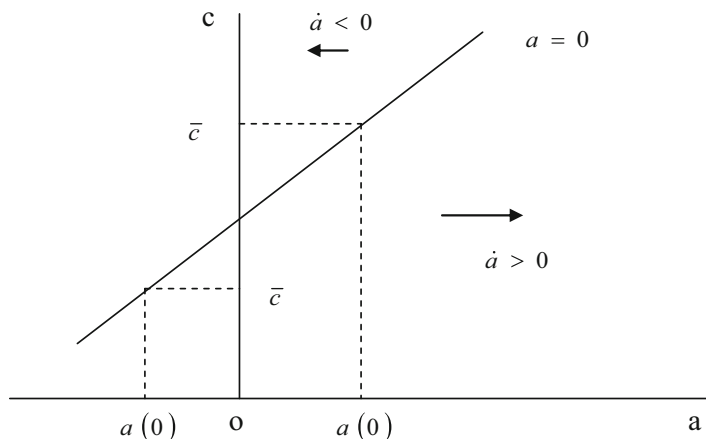
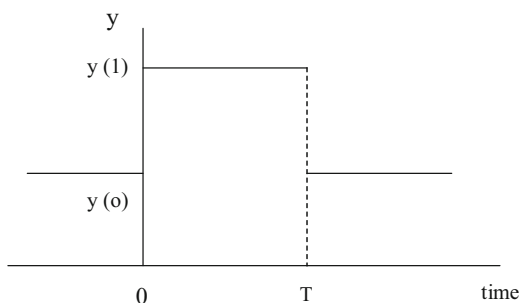


Fig. 2.1 The phase diagram for the c and a system

Fig. 2.2 Unanticipated transitory increase in the agent's income



time preference that depends on consumption, or an interest rate premium that depends on the stock of foreign debt.

Figure 2.1 shows the phase diagram for the differential equation of financial wealth. The arrows show what happens to the variable's movement when it is not in equilibrium. If, by chance, the international interest rate is equal to the rate of time preference, consumption will be constant. Figure 2.1 shows two cases. The country is an international creditor in one and a debtor in the other.

Experiment

Figure 2.2 illustrates a comparative dynamics exercise, assuming a transitory, unanticipated increase in the representative agent's income. Consumption changes instantly at the time of the income change and remains at this new level indefinitely. Some of the transitory income increase will be saved and the agent will show a permanent increase in wealth. Figure 2.3 describes the entire adjustment dynamics.

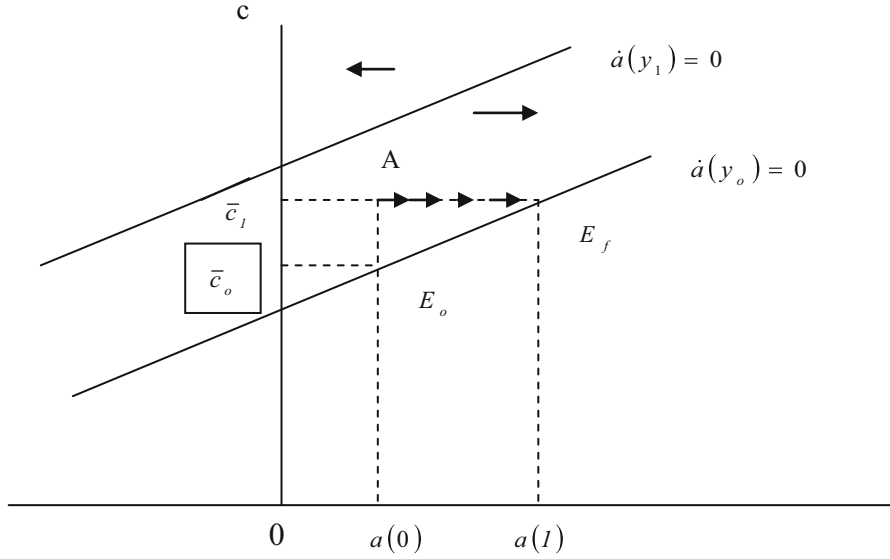


Fig. 2.3 The effects of an unanticipated transitory increase in the agent's income

2.3 Variable Rate of Time Preference

The representative agent model for a small open economy only has a steady-state equilibrium assuming the implausible assumption that the rate of time preference is equal to the international interest rate. A possible solution to the problem is to assume that the rate of time preference changes with some variable. Consumption is a natural candidate for such a variable. The representative agent would then maximize the present value of the instantaneous utility flow discounted at a variable rate $\rho(c)$:

$$\int_0^{\infty} e^{-\int_0^t \rho(c) ds} u(c) dt, \quad (2.10)$$

subject to the following constraints:

$$\dot{a} = ra + y - c, \quad (2.11)$$

$$a(0) = a_0 \quad \text{given.} \quad (2.12)$$

Solving the representative agent's optimal control problem becomes easier with the introduction of a new state variable defined as:

$$S = \int_0^t [\rho(c) - r] ds.$$

The derivative of S relative to time equals:

$$\dot{S} = \rho(c) - r.$$

The present value of the utility flow may be rewritten as:

$$\int_0^\infty e^{-rt} e^{-\left[\int_0^t [\rho(c)-r] ds\right]} u(c) dt = \int_0^\infty e^{-rt} e^{-S} u(c) dt.$$

Therefore, the representative agent's problem is to maximize:

$$\int_0^\infty e^{-rt} e^{-S} u(c) dt, \quad (2.13)$$

subject to the constraints:

$$\dot{a} = ra + y - c \quad (2.14)$$

$$\dot{S} = \rho(c) - r \quad (2.15)$$

$$a(0) = a_0 \quad \text{given} \quad (2.16)$$

$$S(0) = 0 \quad \text{given} \quad (2.17)$$

The current-value Hamiltonian's expression is:

$$H = e^{-S} u(c) + \lambda(ra + y - c) + \mu(\rho(c) - r),$$

where λ and μ are costate variables. The first-order conditions are:

$$\begin{aligned} \frac{\partial H}{\partial c} &= e^{-S} u_c - \lambda + \mu \rho_c = 0, \\ \dot{\lambda} &= r\lambda - \frac{\partial H}{\partial a} = r\lambda - \lambda r = 0, \\ \dot{\mu} &= r\mu - \frac{\partial H}{\partial S} = r\mu + e^{-S} u(c), \\ \frac{\partial H}{\partial \lambda} &= \dot{a} = ra + y - c, \\ \frac{\partial H}{\partial \mu} &= \dot{S} = \rho(c) - r. \end{aligned}$$

The $u(c) > 0$ hypothesis implies a negative costate variable μ . The second derivative of the $\rho(c)$ function with respect to consumption ρ_{cc} must be positive for the Hamiltonian to be maximum, relative to consumption. The solution to this equation system may be simplified with the introduction of two variables that allow for elimination of the variable S from the equation system. They are defined as:

$$\begin{aligned} \Gamma &= \lambda e^S, \\ M &= \mu e^S. \end{aligned}$$

Algebra

The derivative of $\Gamma = \lambda e^S$ and $M = \mu e^S$ with respect to time yields:

$$\begin{aligned}\dot{\Gamma} &= \dot{\lambda} e^S + \lambda e^S \dot{S} = \Gamma[\rho(c) - r], \\ \dot{M} &= \dot{\mu} e^S + \mu e^S \dot{S} = u(c) + M\rho(c).\end{aligned}$$

The first first-order condition equation may be written as:

$$\Gamma = u_c + M\rho_c.$$

The derivative of this expression with respect to time yields:

$$\dot{\Gamma} = u_{cc}\dot{c} + \dot{M}\rho_c + M\rho_{cc}\dot{c}$$

Substituting the values of $\dot{\Gamma}$, $\dot{M}$ and M in the expression yields, after simplification, the following differential equation for consumption:

$$\dot{c} = \alpha(c, \Gamma)[\rho(c, \Gamma) - r],$$

where:

$$\begin{aligned}\alpha(c, \Gamma) &= \frac{\Gamma}{u_{cc} + \frac{\Gamma - u_c}{\rho_c} \rho_{cc}}, \\ \rho(c, \Gamma) &= \rho(c) \left(\frac{u_c - \rho_c u(c) / \rho(c)}{\Gamma} \right).\end{aligned}$$

The coefficient $\alpha(c, \Gamma)$ is negative and so is the partial derivative of $\rho(c, \Gamma)$ with respect to consumption. When discount rate ρ is constant, the traditional result is found: $\alpha(c, \Gamma) = u_c / u_{cc}$ and $\rho(c, \Gamma) = \rho$.

Dynamical System

The representative agent model's dynamical system with a variable rate of time preference has three differential equations:

$$\dot{\Gamma} = \Gamma[\rho(c) - r], \quad (2.18)$$

$$\dot{c} = \alpha(c, \Gamma)[\rho(c, \Gamma) - r], \quad (2.19)$$

$$\dot{a} = ra + y - c. \quad (2.20)$$

The first equation shows that any discrepancy between the rate of time preference and the interest rate produces a change in the marginal utility of wealth (Γ). The second equation states that changes in consumption depend on the difference between the $\rho(c, \Gamma)$ rate and the interest rate, and not on a comparison of the rate of time preference and the interest rate. At a steady-state equilibrium, the $\rho(c, \Gamma)$ rate equals the rate of time preference. The third equation is the agent's budget constraint in terms of flows. The first two equations do not depend on state variable a and

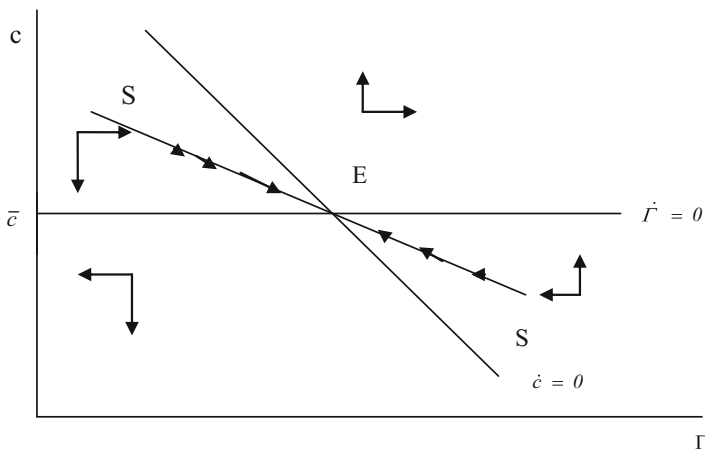


Fig. 2.4 The phase diagram for the c and the marginal utility of wealth (Γ) system

therefore form a system of two differential equations. This model's stationary equilibrium point is given by:

$$\begin{aligned} \rho(\bar{c}) &= r, \\ \bar{\Gamma} &= u_c(\bar{c}) - \frac{u(\bar{c})\rho_c(\bar{c})}{r}. \end{aligned}$$

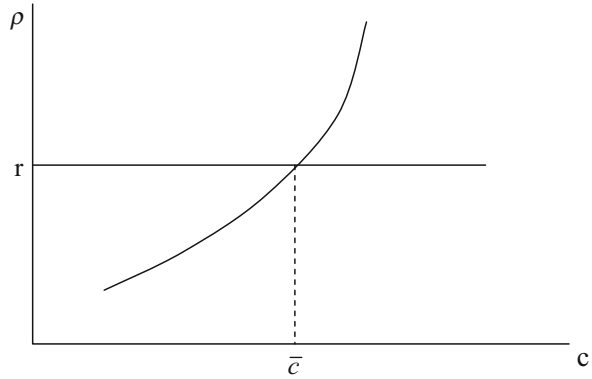
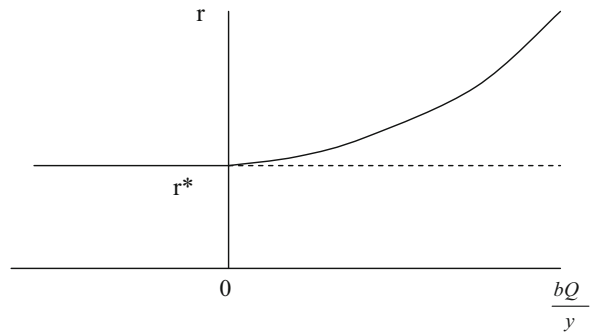
The system's Jacobian matrix determinant at the stationary equilibrium point equals:

$$|J| = \frac{r\Gamma\rho_c}{u_{cc} - \frac{u\rho_{cc}}{r}}.$$

Assuming that $\rho_{cc} > 0$, $u > 0$, $u_{cc} < 0$, it follows that $|J| < 0$ if $\rho_c > 0$. Therefore, the stationary equilibrium point is a saddle point.

Figure 2.4 shows the model's phase diagram, where the vertical axis measures consumption and the horizontal axis measures variable Γ . The $\dot{\Gamma} = 0$ curve is horizontal because $\rho(c) = r$, when $\dot{c} = 0$, $\rho(c, \Gamma) = r$. It is easily seen that, in this case, $\partial c / \partial \Gamma < 0$. The SS saddle has a negative slope.

Figure 2.5 shows the rate of time preference curve, assuming $\rho_c > 0$ and $\rho_{cc} > 0$. Given the international interest rate, this curve determines the consumption that corresponds to a stationary equilibrium. For the model to have a saddle point, the rate of time preference must increase as consumption increases. This means that a rich agent must be more impatient than a poor one. This assumption runs counter to common sense, because one imagines that poor people must be more impatient than wealthier ones. Thus, we assume a variable rate of time preference cannot save the representative agent model in a small open economy.

Fig. 2.5 The rate of time preference curve**Fig. 2.6** The risk premium curve

2.4 Interest Rate Risk Premium

The previous section showed that a variable rate of time preference that depends on consumption is not a satisfactory solution for the representative agent model in a small open economy. An alternative solution is to assume a premium on the foreign interest rate and that this premium depends on some variable in the model. Let r^* be the real international interest rate, b the stock of foreign debt denominated in the international currency (dollars, for example), P^* the international price level, P the domestic price level, and S the foreign exchange rate. Domestic interest rate r equals the foreign interest rate plus a risk premium dependent, for example, on the foreign debt-to-gross domestic product ratio, according to:

$$r = r^* + p_r \left(\frac{SP^*b}{Py} \right), b > 0, p'_r > 0, p''_r > 0. \quad (2.21)$$

Figure 2.6 shows the risk premium curve, taking into account that the real exchange rate is defined by: $Q = SP^*/P$. When debt is zero or negative (the economy accumulates international assets), no risk premium exists.

The representative agent's budget constraint in the domestic currency shows that the change in foreign debt equals the difference between the representative agent's spending and income. That is:

$$\dot{B} = RB + Pc_d + SP^*c_m - Y,$$

where R is the nominal interest rate, B is the debt stock in the domestic currency, c_d is the consumption of the domestic product, P is the price of the domestic product, c_m is the consumption of the imported product, S is the exchange rate, P^* is the price of the imported product, and Y is the nominal income.

Foreign debt in the international currency equals:

$$b = \frac{B}{SP^*}.$$

The derivative of this expression with respect to time yields:

$$\frac{\dot{B}}{SP^*} = \dot{b} + b\left(\frac{\dot{S}}{S} + \frac{\dot{P}^*}{P^*}\right) = \dot{b} + b(\dot{s} + \pi^*),$$

where $\dot{s} = d(\log S)/dt$ and $\pi^* = d(\log P^*)/dt$. Dividing both sides of the budget constraint by SP^* yields, after some simplification, the following budget constraint:

$$\dot{b} = rb + \frac{c_d + c_m Q - y}{Q},$$

where:

$$r = R - \pi^* - \dot{s}.$$

The representative agent maximizes the present value of the instantaneous utility flow,

$$\int_0^\infty e^{-\rho t} u(c) dt.$$

Consumption c is an aggregate index of the consumption of the domestic and imported goods,

$$c = F(c_d, c_m).$$

This consumption index corresponds to a price index P_c that satisfies the constraint:

$$P_c c = Pc_d + SP^*c_m$$

Let $p_c = P_c/P$. It follows that expenditure, at domestic prices, is:

$$p_c c = c_d + Qc_m.$$

The representative agent's budget constraint may be written as:

$$\dot{b} = rb + \frac{p_c c - y}{Q}.$$

Therefore, the representative agent maximizes:

$$\int_0^{\infty} e^{-\rho t} u(c) dt, \quad (2.22)$$

subject to the constraints:

$$\dot{b} = rb + \frac{p_c c - y}{Q}, \quad (2.23)$$

$$r = r^* + p_r \left(\frac{bQ}{y} \right), \quad (2.24)$$

$$b(0) = b_0 \quad \text{given.} \quad (2.25)$$

This problem's current-value Hamiltonian is:

$$H = u(c) - \lambda \left[rb + \frac{p_c c - y}{Q} \right].$$

The first-order conditions are:

$$\begin{aligned} \frac{\partial H}{\partial c} &= u_c - \lambda \frac{p_c}{Q} = 0, \\ \dot{\lambda} &= \rho \lambda - \frac{\partial H}{\partial b} = \rho \lambda - \lambda \left(r + b \frac{\partial r}{\partial b} \right), \\ \frac{\partial H}{\partial \lambda} &= -\dot{b} = - \left[\frac{p_c c - y}{Q} + rb \right]. \end{aligned}$$

The second first-order condition equation may be written as:

$$\frac{\dot{\lambda}}{\lambda} = \rho - r - b \frac{\partial r}{\partial b}.$$

Assuming, for simplicity, that the real exchange rate does not vary over time, the first first-order condition equation derivative with respect to time yields:

$$\frac{\dot{\lambda}}{\lambda} = \frac{u_{cc}}{u} \dot{c} = \rho - r - b \frac{\partial r}{\partial b}.$$

Therefore, the differential equation for consumption is:

$$\dot{c} = \frac{u_c}{u_{cc}} \left[\rho - r - b \frac{\partial r}{\partial b} \right].$$

Dynamical System

This model's dynamical system has two differential equations:

$$\dot{c} = \frac{u_c}{u_{cc}} \left[\rho - r - b \frac{\partial r}{\partial b} \right], \quad (2.26)$$

$$\dot{b} = \frac{p_c c - y}{Q} = r b. \quad (2.27)$$

This system's steady-state equilibrium $(\bar{c}, \bar{b})$ is determined by the solution of the system:

$$\rho = r^* + p_r \left(\frac{\bar{b} Q}{y} \right) + \bar{b} \frac{\partial r}{\partial b}(\bar{b}),$$

$$\frac{y - p_c \bar{c}}{Q} = \left(\rho^* + p_r \left(\frac{\bar{b} Q}{y} \right) \right) \bar{b}.$$

In this model, the rate of time preference is greater than the foreign interest rate and the country is an international debtor. For the representative agent model to be used in a small open economy and to have a steady-state solution – without equality between the rate of time preference and the foreign interest rate – one must introduce an ad hoc assumption for the interest rate risk premium. It is worth pointing out that the assumed interest rate risk premium does not solve the problem of the representative agent model in a small open economy when the country is an international net creditor. A positive or negative risk premium would make no sense under such circumstances.

2.5 The New Keynesian IS Curve

This section presents the IS curve model for the small open economy. The model assumes complete financial markets, an unrealistic assumption that is not observed in the real world and will be duly analyzed.

The economy includes countless consumers denominated by $j \in [0, 1]$. These consumers are producers at the same time. They produce differentiated – that is, non-homogeneous – goods, have the same preferences, and allocate time between leisure and the labor used to produce the products. The representative agent then maximizes the expected value of the objective function that depends on aggregate consumption (C) and the time dedicated to labor (N). The utility function of labor carries a negative sign because the agent would rather use their time for leisure than for labor. The objective function is given by:

$$U_t(j) = \sum_{k=0}^{\infty} \beta^k [u(C_{t+k}(j)) - v(N_{t+k}(j))]. \quad (2.28)$$

Aggregate consumption depends on the consumption of domestic (H) and imported (F) goods, according to a CES function:

$$C_t = \left[(1 - \gamma)^{\frac{1}{\eta}} (C_{H,t})^{\frac{\eta-1}{\eta}} + \gamma^{\frac{1}{\eta}} (C_{F,t})^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \eta > 0. \quad (2.29)$$

The Greek letter η is the elasticity of substitution between H and F . Consumption of domestic and foreign goods is also aggregated by means of a CES function:

$$C_{H,t} = \left[\int_0^1 C_{H,t}(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}}, \theta > 1, C_{F,t} = \left[\int_0^1 C_{F,t}(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}}. \quad (2.30)$$

Elasticity of substitution is represented by θ . The economy's price index depends on the prices of domestic and imported goods according to:

$$P_t = [(1 - \gamma)(P_{H,t})^{1-\eta} + \gamma(P_{F,t})^{1-\eta}]^{\frac{1}{1-\eta}}. \quad (2.31)$$

This price index, like the others we will define below, is built in such a way that total expenditure is equal to the sum of the expenditures of the goods included in the consumer's goods basket. In this case: $PC = P_{H,t} C_{H,t} + P_{F,t} C_{F,t}$. Demands for domestic and foreign goods depend on total consumption and on the relative prices according to:

$$C_{H,t} = (1 - \gamma) \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} C_t, C_{F,t} = \gamma \left(\frac{P_{F,t}}{P_t} \right)^{-\eta} C_t. \quad (2.32)$$

The price indices of domestic and imported goods are:

$$P_{H,t} = \left[\int_0^1 P_{H,t}(j)^{1-\theta} dj \right]^{\frac{1}{1-\theta}}, P_{F,t} = \left[\int_0^1 P_{F,t}(j)^{1-\theta} dj \right]^{\frac{1}{1-\theta}}. \quad (2.33)$$

The demand equations for each product – domestic and imported – are specified by:

$$C_{H,t}(j) = \left[\frac{P_{H,t}(j)}{P_{H,t}} \right]^{-\theta} C_{H,t}, C_{F,t}(j) = \left[\frac{P_{F,t}(j)}{P_{F,t}} \right]^{-\theta} C_{F,t}. \quad (2.34)$$

The log approximation of the economy's price index equals the geometric average of the prices of the domestic and imported goods:

$$p_t = (1 - \gamma)p_{H,t} + \gamma p_{F,t}.$$

Let $P_t^* = P_{F,t}^*$ be the price index of the imported goods. The price of the imported product in the domestic currency equals price in dollars multiplied by the exchange rate:

$$P_{F,t} = S_t P_{F,t}^* = S_t P_t^*.$$

The real exchange rate equals the ratio of the price indices valued in the same currency:

$$Q_t = \frac{S_t P_t^*}{P_t}.$$

Domestic Goods Market Equilibrium

The market for each product is at equilibrium when production equals the sum of domestic and foreign demands. That is:

$$Y_t(j) = C_{H,t}(j) + C_{H,t}^*(j).$$

Substituting (2.34) in this expression yields:

$$Y_t(j) = \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{-\theta} C_{H,t} + \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{-\theta} C_{H,t}^*.$$

Substituting (2.32) in the preceding equation yields:

$$Y_t(j) = \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{-\theta} (1 - \gamma) \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} C_t + \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{-\theta} \gamma^* \left(\frac{P_{H,t}}{S_t P_t^*} \right)^{-\eta} C_t^*.$$

This expression may be rewritten as:

$$Y_t(j) = \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{-\theta} \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} \left[(1 - \gamma) C_t + \gamma^* \left(\frac{P_t}{S_t P_t^*} \right)^{-\eta} C_t^* \right].$$

By taking into account the definition of the real exchange rate, this equation becomes:

$$Y_t(j) = \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{-\theta} \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} [(1 - \gamma) C_t + \gamma^* Q_t^\eta C_t^*]. \quad (2.35)$$

The economy's real output is obtained by an aggregate CES function. That is:

$$Y_t = \left[\int_0^1 Y_t(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}}.$$

Substituting (2.35) in this aggregate equation for real output yields:

$$Y_t = \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} [(1 - \gamma)C_t + \gamma^* Q_t^\eta C_t^*] \left[\int_0^1 \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{-\theta \left(\frac{\theta-1}{\theta} \right)} dj \right]^{\frac{\theta}{\theta-1}}.$$

The definition of the domestic goods price index implies that:

$$\left[\int_0^1 \left(\frac{P_{H,t}(j)}{P_{H,t}} \right)^{1-\theta} dj \right]^{\frac{1}{1-\theta}} = 1.$$

By taking this into account, it follows that the real output equation becomes:

$$Y_t = \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} [(1 - \gamma)C_t + \gamma^* Q_t^\eta C_t^*]. \quad (2.36)$$

Complete Financial Markets

The financial markets in this economy are complete, that is, claims contingent on every possible state of the world are traded on the financial market. Let $V_{t,t+1}$ be the price, in period t and in the domestic currency, of a one-period claim paying one unit of the domestic currency in period $t + 1$ if the state of the world for ξ , and paying nothing otherwise. The conditional probability of the state of the world ξ occurring in period $t + 1$ is equal to $\pi_t(\xi_{t+1})$. The equilibrium condition for the representative agent upon purchasing the claim is that the cost in terms of welfare of the amount spent on the purchase must equal the expected value of the benefit should state of the world ξ occur:

$$\frac{V_{t,t+1}}{P_t} u'(C_t) = \beta \pi_t(\xi_{t+1}) \frac{1}{P_{t+1}} u'(C_{t+1}).$$

Abroad, the price of a claim similar to the domestic one equals $V_{t,t+1}^*$. Therefore, the equilibrium condition for an agent living in the foreign country is:

$$\frac{V_{t,t+1}^*}{P_t^*} u'(C_t^*) = \beta \pi_t(\xi_{t+1}) \frac{1}{P_{t+1}^*} u'(C_{t+1}^*).$$

International arbitrage implies that the prices of the claims are related according to:

$$V_{t,t+1} = V_{t,t+1}^* \frac{S_t}{S_{t+1}}. \quad (2.37)$$

where S is the exchange rate. Dividing the domestic agent's equilibrium condition equation by the foreign agent's and taking the arbitrage condition into account yields:

$$\frac{u'(C_t)}{u'(C_t^*)} Q_t = \frac{u'(C_{t+1})}{u'(C_{t+1}^*)} Q_{t+1},$$

where Q is the real exchange rate. It therefore follows that:

$$\frac{u'(C_t)}{u'(C_t^*)} Q_t = \text{constant}.$$

Assume that the instantaneous utility function is:

$$u(C) = \frac{C^{1-\frac{1}{\sigma}} - 1}{1 - \frac{1}{\sigma}}. \quad (2.38)$$

With this utility function, domestic consumption is proportional to foreign consumption for a given real exchange rate, according to:

$$C_t = \kappa C_t^* Q_t^\sigma.$$

For a small open economy, the consumption of the rest of the world equals the real-world output:

$$C_t^* = Y_t^*.$$

Substituting this expression in the consumption equation yields:

$$C_t = \kappa Y_t^* Q_t^\sigma. \quad (2.39)$$

Domestic consumption is proportional to world output, and the proportionality coefficient depends on the real exchange rate.

Equilibrium Real Output

Substituting the domestic consumption expression (2.39) in (2.36), and bearing in mind that the rest of the world's consumption equals the world output, yields:

$$Y_t = \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} Y_t^* Q_t^\eta [(1-\gamma)\kappa Q_t^{\sigma-\eta} + \gamma\kappa]. \quad (2.40)$$

We assume $\gamma^* = \kappa\gamma$, a simplifying assumption that will be justified at the end of this section. The log of both sides of the equation yields:

$$\log Y_t = -\eta \log \left(\frac{P_{H,t}}{P_t} \right) + \log Y_t^* + \eta \log Q_t + \log \kappa + \log [(1-\gamma)Q_t^{\sigma-\eta} + \gamma].$$

The log of the relative price that appears in this expression equals:

$$\log \frac{P_{H,t}}{P_t} \cong -\frac{\gamma}{1-\gamma} \log Q_t = -\frac{\gamma}{1-\gamma} q_t. \quad (2.41)$$

Obtaining the preceding result took into account the fact that the price index equals: $p_t = (1 - \gamma)p_{H,t} + \gamma p_{F,t} = (1 - \gamma)p_{H,t} + \gamma(s_t + p_t^*)$. The final term in the log of real output equation may be approximated as:

$$\log[(1 - \gamma)Q_t^{\sigma - \eta} + \gamma] \cong (1 - \gamma)(\sigma - \eta)q_t.$$

with the appropriate substitutions and a little algebra, we arrive at the following equilibrium real output equation:

$$y_t = y_t^* + \frac{\sigma\omega}{1 - \gamma}q_t. \quad (2.42)$$

The lower-case variables represent the logs of the respective upper-case variables, and the parameter ω equals:

$$\omega = 1 + \gamma(2 - \gamma)\left(\frac{\eta}{\sigma} - 1\right).$$

The parameter ω depends on the economy's openness degree (γ), on the intertemporal elasticity of substitution for consumption (σ), and the intratemporal elasticity of consumption (η). The parameter ω is equal to unity in two cases: (i) when the economy is closed ($\gamma = 0$) and (ii) when the intertemporal and intratemporal elasticities of substitution are the same ($\sigma = \eta$).

The Euler Equation

The representative agent's equilibrium condition in the complete market's economy may be written as:

$$\frac{V_{t,t+1}}{\pi_t(\xi_{t+1})} = D_{t,t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{P_t}{P_{t+1}}.$$

$D_{t,t+1}$ is the discount rate for world state ξ . Taking the expression's mathematical expectation conditioned by the information available in period t yields the model's Euler equation:

$$D_t = \beta E_t \frac{u'(C_{t+1})}{u'(C_t)} \frac{P_t}{P_{t+1}}.$$

D_t is the price of a one-period claim paying one unit of the domestic currency for any state of the world in period $t + 1$:

$$D_t = E_t D_{t,t+1} = \frac{1}{1 + i_t}.$$

Therefore, the Euler equation for the utility function (2.38) is:

$$D_t = \beta E_t \left(\frac{C_{t+1}}{C_t} \right)^{-\frac{1}{\sigma}} \frac{P_t}{P_{t+1}}.$$

By taking the log of both sides of the expression, using linear approximation, and the notation according to which a lower-case letter represents the log of the corresponding upper-case variable, the Euler equation has the following functional form:

$$c_t = E_t c_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \rho), \quad (2.43)$$

where: $i_t = -\log D_t$, $\rho = -\log \beta$, $\pi_{t+1} = p_{t+1} - p_t$.

The IS Curve

This model's IS curve is obtained by combining the equations for the real output in goods and services market equilibrium, for consumption in complete markets, the Euler equation for intertemporal consumption allocation, and the equation that relates consumer inflation to domestic goods inflation, all of which are repeated below for convenience:

$$y_t = y_t^* + \frac{\sigma\omega}{1-\gamma} q_t, \quad (2.44)$$

$$c_t = y_t^* + \sigma q_t, \quad (2.45)$$

$$c_t = E_t c_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \rho), \quad (2.46)$$

$$\pi_t = \pi_{H,t} + \frac{\gamma}{1-\gamma} \Delta q_t. \quad (2.47)$$

Equations (2.44) and (2.46) correspond to Eqs. (2.42) and (2.43). Equation (2.45) is obtained by taking the logs of both sides of Eq. (2.39) and disregarding the constant. Equation (2.47) is obtained from Eq. (2.41) and the definition of the consumer and domestic prices inflation rates.

Obtaining this model's IS curve requires a little algebra: (i) Eq. (2.45) is used to eliminate y_t^* from Eq. (2.44); (ii) consumption c_t is then replaced with its expression of the Euler equation, eliminating period $t+1$'s consumption via the output equation for the same period; (iii) Eq. (2.47) substitutes for consumer inflation; and (iv) the real exchange's variation rate is eliminated with help from Eq. (2.44). The IS curve is specified as follows:

$$x_t = E_t x_{t+1} - \omega \sigma(i_t - E_t \pi_{H,t+1} - \bar{r}_t). \quad (2.48)$$

The coefficient of the real interest rate gap depends on parameter ω and on elasticity of substitution σ . This specification of the IS curve for a small open economy is similar to its closed-economy specification. The difference lies only in the parameter ω . When this parameter equals unity, there will be no difference between the two economies' IS curves.

The IS curve's natural interest rate in this small open economy model depends on the rate of time preference, on the growth rate of the economy's potential output, and on the growth rate of the world economy's output, according to:

$$\bar{r}_t = \rho + \frac{1}{\sigma\omega} E_t \Delta \bar{y}_{t+1} - \frac{(1-\omega)}{\sigma\omega} E_t \Delta y_{t+1}^*. \quad (2.49)$$

When parameter ω equals unity, this small open economy's natural interest rate is equal to the closed economy's natural interest rate.

Uncovered Real Interest Rate Parity

The prices of domestic and foreign contingent assets are related as follows:

$$\frac{V_{t,t+1}}{\pi_t(\xi_{t+1})} = \frac{V_{t,t+1}^*}{\pi_t(\xi_{t+1}^*)} \frac{S_t}{S_{t+1}},$$

This equation is obtained by dividing both sides of Eq. (2.37) by the conditional probability of the state of the world in period $t+1$. It therefore follows that the discount rates, defined as the ratio of the asset price to the conditional probability, are:

$$D_{t,t+1} = D_{t,t+1}^* \frac{S_t}{S_{t+1}}.$$

A foreign claim paying one unit of the foreign currency regardless of the state of the world is priced at:

$$D_t^* = E_t D_{t,t+1}^* = \frac{1}{1 + i_t^*}.$$

Taking the previous expression into account, this price can be written as:

$$E_t D_{t,t+1} \frac{S_{t+1}}{S_t} = \frac{1}{1 + i_t^*}.$$

Combining this expression with the one for the domestic asset, we have:

$$E_t \left\{ D_{t,t+1} \left[(1 + i_t) - (1 + i_t^*) \frac{S_{t+1}}{S_t} \right] \right\} = 0.$$

This equation may be written as:

$$E_t \{ D_{t,t+1} [\exp i_t - \exp(i_t^* + \Delta s_{t+1})] \} = 0.$$

A linear approximation around the perfect foresight stationary equilibrium point yields:

$$E_t \{ D_{t,t+1} [i_t - (i_t^* + \Delta s_{t+1})] \} = 0.$$

The covariance of variables X and Y is defined by: $\text{cov}(X, Y) = EXY - EXEY$. Denoting the discount rate as X and the gap between domestic and foreign rates of return as Y , and taking into account that $EXY = 0$ yields the uncovered nominal interest rate parity. According to this parity, the nominal interest rate equals the sum of the foreign interest rate with the anticipated change in the nominal exchange rate and risk premium:

$$i_t = i_t^* + E_t \Delta s_{t+1} + \text{pr}.$$

The risk premium (pr) equals the ratio of the covariance of the discount rate to the domestic asset's excess return relative to the foreign asset and the mathematical expectation for the discount rate, multiplied by minus one. Subtracting the respective expected inflation rates from the two nominal interest rates yields:

$$i_t - E_t \pi_{H,t+1} = i_t^* - E_t \pi_{t+1}^* + E_t \Delta s_{t+1} + E_t \pi_{t+1}^* - E_t \pi_{H,t+1}.$$

This expression disregards the risk premium. The left side of the equation is the domestic real interest rate. The right side is the sum of the real foreign interest rate and a multiple of the expected value of the real exchange rate variation. That is:

$$r_t = r_t^* + \frac{1}{1 - \gamma} E_t \Delta q_{t+1}.$$

Natural Interest Rate and International Interest Rate

In the long-run equilibrium, when the economy is growing, the real exchange rate is not constant. Therefore, the natural interest rate is not equal to the real foreign interest rate:

$$\bar{r} \neq r^*. \quad (2.50)$$

We will show next that under the new Keynesian small open economy model, the natural interest rate equals a weighted average of the natural interest rate that would occur if the small economy were closed, and the real international interest rate.

The natural world rate is given by:

$$\bar{r}_t^* = \rho + \frac{1}{\sigma} E_t \Delta \bar{y}_{t+1}^*,$$

and the natural rate in the small open economy if it were a closed economy would be given by:

$$\bar{r}_t = \rho + \frac{1}{\sigma} E_t \Delta \bar{y}_{t+1}.$$

The world economy's IS curve is given by:

$$E_t(\Delta y_{t+1}^* - \Delta \bar{y}_{t+1}^*) = \sigma(r_t^* - \bar{r}_t^*).$$

Substituting this expression in Eq. (2.17) yields:

$$\bar{r}_t = \rho + \frac{1}{\sigma\omega} E_t \Delta \bar{y}_{t+1} - \frac{(1-\omega)}{\sigma\omega} (E_t \Delta y_{t+1}^* + \sigma(r_t^* - \bar{r}_t^*)).$$

Using the two natural interest rate definitions – for the small closed economy and the world economy – we find that the natural interest rate in the new Keynesian small open economy model is given by the weighted average of the small open economy's natural interest rate if it were a closed economy and of the real international interest rate. That is:

$$\bar{r}_t = \frac{1}{\omega} \bar{r}_t^{CE} + \frac{\omega-1}{\omega} r_t^*. \quad (2.51)$$

When ω equals one, the economy is closed and the natural interest rate corresponds to that of a closed economy. When $\omega \rightarrow \infty$ ($\eta \rightarrow \infty$ or $\sigma \rightarrow 0$), the small open economy's natural interest rate will equal the real international interest rate. It is worth pointing out that, even in this particular case, the small open economy's natural interest rate is not equal to the world natural interest rate. In the small open economy, the natural interest rate varies according to the real international interest rate.

Perfect Foresight Stationary Equilibrium

Under a perfect foresight situation, the stationary equilibrium condition in the goods and services market is given by Eq. (2.40), assuming that the variables are not time-dependent. That is:

$$Y = \left(\frac{P_H}{P}\right)^{-\eta} Q^\eta [(1-\gamma)\kappa Q^{\sigma-\eta} + \gamma\kappa] Y^*.$$

The relative price that appears in the expression equals:

$$\frac{P_H}{P} = Q^{-\frac{\gamma}{1-\gamma}}.$$

Substituting this expression in the goods and services market equilibrium condition yields:

$$Y = Q^{\frac{\eta}{1-\gamma}} [(1-\gamma)\kappa Q^{\sigma-\eta} + \gamma\kappa] Y^*. \quad (2.52)$$

In the labor market, the equilibrium condition implies that marginal disutility of labor equals the marginal utility of consumption obtained from the purchase of goods and services that the real wage enables:

$$N^{\frac{1}{\varphi}} = \frac{W}{P} C^{-\frac{1}{\sigma}}.$$

The expression assumes that the utility function of labor has the following functional form:

$$v(N) = \frac{N^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}}.$$

Producers of domestic goods operate in a monopolistic competition market and the price of their goods is obtained by adding a margin to the marginal cost of production:

$$P_H = \frac{\theta}{\theta - 1} Cmg.$$

The margin depends on the price elasticity of demand (θ) of the domestically produced goods, as specified in Eq. (2.34).

The production function is linear for the quantity of labor:

$$Y = AN,$$

where A is the technical coefficient of production. The marginal cost of production is therefore proportional to wages:

$$Cmg = \frac{W}{A}.$$

The price of the domestic product at perfect foresight stationary equilibrium is given by:

$$P_H = \frac{\theta}{\theta - 1} \frac{W}{A}.$$

The value of the wage in this expression, when substituted in the labor market equilibrium equation, yields:

$$N^{\frac{1}{\varphi}} C^{\frac{1}{\sigma}} = \frac{A(\theta - 1)}{\theta} \frac{P_H}{P}.$$

Substituting N , C , and relative price P_H/P in this equation yields:

$$Y^{\frac{1}{\varphi}} (\kappa Y^* Q^{\sigma})^{\frac{1}{\sigma}} = A^{\frac{1+\varphi}{\varphi}} \frac{\theta - 1}{\theta} Q^{-\frac{\gamma}{1-\gamma}}.$$

In this equation, real output may be written as:

$$Y = A^{1+\varphi} \left(\frac{(\theta - 1)Q^{-\frac{1}{1-\gamma}}}{(\kappa Y^*)^{\frac{1}{\sigma}} \theta} \right)^{\varphi}. \quad (2.53)$$

This equation, together with Eqs. (2.51) and (2.39), determines real output, consumption, and the real exchange rate at perfect foresight stationary equilibrium. It is easily seen that $Q = 1$, $C = Y = \kappa Y^*$ are solutions for the model and that real output at stationary equilibrium is:

$$Y = A^{\frac{\sigma(1+\varphi)}{\sigma+\varphi}} \left(\frac{\theta - 1}{\theta} \right)^{\frac{\sigma\varphi}{\sigma+\varphi}}.$$

In this perfect foresight stationary equilibrium, purchasing power parity applies ($Q = 1$); since consumption equals output, net exports are zero, a finding that emerges from the previous assumption that $\gamma^* = \kappa\gamma$.

2.6 Exercises

1. Price index P_t is defined by:

$$P_t = \left[(1 - \gamma)(P_{H,t})^{1-\eta} + \gamma(P_{F,t})^{1-\eta} \right]^{\frac{1}{1-\eta}},$$

(a) Show that this expression may be written as:

$$p_t = p_{H,t} + \frac{1}{1-\eta} \log \left[(1 - \gamma) + \gamma \left(\frac{P_{F,t}}{P_{H,t}} \right)^{1-\eta} \right],$$

where $p_t = \log P_t$ and $p_{H,t} = \log P_{H,t}$.

(b) Show that:

$$\log \left[(1 - \gamma) + \gamma \left(\frac{P_{F,t}}{P_{H,t}} \right)^{1-\eta} \right] \cong \gamma(1 - \eta)(p_{F,t} - p_{H,t}),$$

where $p_{F,t} = \log P_{F,t}$.

(c) Combine the results of the two preceding items to show that:

$$p_t = (1 - \gamma)p_{H,t} + \gamma p_{F,t}$$

2. The price of imported good (F) is given by $P_{F,t} = S_t P_t^*$. Therefore:

$$p_t = (1 - \gamma)p_{H,t} + \gamma(s_t + p_t^*),$$

where $s_t = \log S_t$ and $p_t^* = \log P_t^*$. The real exchange rate is defined by: $q_t = s_t + p_t^* - p_t$. Use this definition to show that:

$$p_t = p_{H,t} + \frac{\gamma}{1-\gamma} q_t.$$

3. Show how consumption indices:

$$\text{I. } C_t = \left[(1-\gamma)^{\frac{1}{\eta}} (C_{H,t})^{\frac{\eta-1}{\eta}} + \gamma^{\frac{1}{\eta}} (C_{F,t})^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \eta > 0,$$

$$\text{II. } C_{H,t} = \left[\int_0^1 C_{H,t}(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}}, \theta > 1,$$

$$\text{III. } C_{F,t} = \left[\int_0^1 C_{F,t}(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}},$$

can be used to obtain:

(a) The corresponding price indices:

$$\text{I. } P_t = \left[(1-\gamma)(P_{H,t})^{1-\eta} + \gamma(P_{F,t})^{1-\eta} \right]^{\frac{1}{1-\eta}},$$

$$\text{II. } P_{H,t} = \left[\int_0^1 P_{H,t}(j)^{1-\theta} dj \right]^{\frac{1}{1-\theta}},$$

$$\text{III. } P_{F,t} = \left[\int_0^1 P_{F,t}(j)^{1-\theta} dj \right]^{\frac{1}{1-\theta}}.$$

(b) Each good's demand equations:

$$C_{H,t} = (1-\gamma) \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} C_t, \quad C_{F,t} = \gamma \left(\frac{P_{F,t}}{P_t} \right)^{-\eta} C_t,$$

$$C_{H,t}(j) = \left[\frac{P_{H,t}(j)}{P_{H,t}} \right]^{-\theta} C_{H,t}, \quad C_{F,t}(j) = \left[\frac{P_{F,t}(j)}{P_{F,t}} \right]^{-\theta} C_{F,t},$$

4. The representative agent maximizes the functional:

$$\int_0^\infty e^{-\int_0^t \rho(s) ds} u(c) dt,$$

subject to the constraints:

$$\dot{a} = ra + y - c,$$

$$a(0) = a_0 \quad \text{given.}$$

Define:

$$\delta(t) = \int_0^t \rho(s) ds.$$

- (a) Show that $\dot{\delta} = \rho(t)$.
 - (b) What are the first-order conditions for the solution to this optimal control problem?
 - (c) What is this model's dynamical system for the consumption and wealth (a) variables?
 - (d) Is the system autonomous?
 - (e) Is there stationary equilibrium?
5. The dynamical system for the representative agent model with variable rate of time preference is:

$$\begin{aligned}\dot{\Gamma} &= \Gamma[\rho(c) - r], \\ \dot{c} &= \alpha(c, \Gamma)[\rho(c, \Gamma) - r], \\ \dot{a} &= ra + y - c,\end{aligned}$$

where:

$$\begin{aligned}\alpha(c, \Gamma) &= \frac{\Gamma}{u_{cc} + \frac{\Gamma - u_c}{\rho_c} \rho_{cc}}, \\ \rho(c, \Gamma) &= \rho(c) \left(\frac{u_c - \rho_c u(c)/\rho(c)}{\Gamma} \right).\end{aligned}$$

- (a) Show that:
- $$\alpha(c, \Gamma) < 0, \quad \frac{\partial \rho(c, \Gamma)}{\partial c} < 0.$$
- (b) Calculate the determinant of the Jacobian matrix of the system made up of the two first differential equations.
 - (c) What happens in this economy when the foreign interest rate rises?
6. The dynamical system for the representative agent model with interest rate risk premium is:

$$\begin{aligned}\dot{c} &= \frac{u_c}{u_{cc}} \left[\rho - r - b \frac{\partial r}{\partial b} \right], \\ \dot{b} &= \frac{p_c c - y}{Q} + rb, \\ r &= r^* + p_r \left(\frac{Qb}{y} \right), \quad b > 0, \quad p'_r > 0, \quad p''_r > 0.\end{aligned}$$

- (a) Analyze this model's dynamics on a phase diagram with consumption (c) on the vertical axis and the debt stock (b) on the horizontal axis.

- (b) Analyze what happens in this economy when an anticipated permanent change occurs in real exchange rate Q .
- (c) Analyze what happens in this economy when an anticipated transitory change occurs in real exchange rate Q .

7. The functional U is defined by:

$$U = \int_t^\infty u(c) e^{-\int_t^v \rho(c) ds} dv.$$

- (a) Show that:

$$\dot{U} = \rho(c)U - u(c).$$

- (b) What is the economic interpretation of this differential equation?

8. The representative agent maximizes:

$$\int_0^\infty e^{-\left[\int_0^t \rho(c) ds - nt\right]} u(c) dt,$$

subject to the constraints:

$$\begin{aligned} \dot{k} &= f(k) - (n + \delta)k - c, \\ k(0) &= k_0 \text{ given.} \end{aligned}$$

- (a) Define $S = \int_0^t \rho(c) ds - nt$. Show that $\dot{S} = \rho(c) - n$.
- (b) Solve the representative agent's problem using the new state variable S .
- (c) Analyze the model's equilibrium and dynamics on a phase diagram with consumption (c) on the vertical axis and capital (k) on the horizontal axis.
- (d) What happens in this model when the population growth rate (n) decreases?
- (e) What happens in this model when depreciation rate δ increases?

9. The representative agent maximizes the functional:

$$U = \int_0^\infty e^{-\rho t} u(c, z) dt,$$

where utility function $u(c, z)$ depends on consumption (c) and on a past consumption index z , according to:

$$z(t) = \int_{-\infty}^t \beta e^{-\beta(t-\tau)} c(\tau) d\tau.$$

(a) Show that:

$$\dot{z} = \beta(c - z).$$

(b) Establish the first-order conditions of the following problem:

$$\max \int_0^{\infty} e^{-\rho t} u(c, z) dt,$$

subject to the constraints:

$$\begin{aligned} \dot{a} &= ra + y - c, \\ \dot{z} &= \beta(c - z), \\ a(0) &= a_0 \quad \text{given}, \\ z(0) &= z_0 \quad \text{given}. \end{aligned}$$

(c) In a small open economy, what is the condition needed for a stationary equilibrium for consumption to exist?

10. The representative agent maximizes:

$$\int_0^{\infty} e^{-\rho t} u(c) dt,$$

subject to the constraints:

$$\begin{aligned} \dot{a} &= ra + y - c, \\ a(0) &= a_0 \quad \text{given}. \end{aligned}$$

Rate r is the foreign interest rate.

(a) Assume that $r = \rho$. Show that:

$$\dot{a} = y - y^p,$$

where $y^p = \int_0^{\infty} e^{-rt} y dt$. How would you interpret this result?

(b) Assume that $r \neq \rho$ and $u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}$. Show that:

$$\dot{a} = y - y^p + \sigma(r - \rho)W,$$

where $W = \int_0^{\infty} ye^{-rt} dt + a_0$. How would you interpret this result?

Chapter 3

Overlapping Generations



This chapter presents the overlapping generations (OLG) model. Section 1 introduces an infinite-life OLG model where, at each moment, a generation is born with no financial assets and is, therefore, disconnected from existing generations. Section 2 introduces the government into this model and analyzes the question of Ricardian equivalence. Section 3 shows that the OLG model, unlike the representative agent model, can be applied to a small open economy without the need for any ad hoc assumptions. Section 4 derives the OLG model's new Keynesian IS curve. Section 5 shows a finite-life OLG model. This model uses the simplifying assumption that an individual's probability of death is independent of their age.

3.1 Infinite-Life Overlapping Generations Model

The overlapping generations (OLG) model with infinite-life assumes that at each point in time a new generation is born that is not connected to existing generations. Genetically, the assumption presupposes parthenogenesis—that is, the existence of an embryo with no fertilization. It intends to show that a large share of the population has no financial assets.

The growth rate of new generations equals the population growth rate. Each generation is born with no financial assets, but with the same human capital as other generations. At instant t , each agent maximizes the utility flow throughout their life, discounted at the rate of time preference ρ , which is the same for all agents. The agent therefore maximizes the integral:

$$\int_t^{\infty} e^{-\rho(v-t)} u[c(s, v)] dv, \quad (3.1)$$

subject to the following constraints:

$$\dot{a}(s, v) = r a(s, v) + \omega(v) - c(s, v), \quad (3.2)$$

$$a(s, t) \text{ given.} \quad (3.3)$$

The notation is as follows: $c(s, v)$ is the generation's consumption at instant v , indexed by s ; $a(s, v)$ is generation s 's total financial assets on date v ; $\omega(v)$ is the agent's wages; and r is the interest rate on the financial investment.

This problem's current-value Hamiltonian is:

$$H = u[c(s, v)] + \lambda[r a(s, v) + \omega(v) - c(s, v)].$$

According to the first-order conditions: (i) the marginal utility of consumption is equal to the costate variable λ ; (ii) the agent chooses between consuming and investing at every moment, arbitraging the return of the two choices; and (iii) the budget constraint must be met. That is:

$$\frac{\partial H}{\partial c} = u'(c(s, v)) - \lambda = 0,$$

$$\dot{\lambda} = \rho\lambda - \frac{\partial H}{\partial a} = \rho\lambda - r\lambda,$$

$$\frac{\partial H}{\partial \lambda} = \omega - c + ra = \dot{a}.$$

This optimal solution must satisfy the transversality condition:

$$\lim_{t \rightarrow \infty} \lambda a e^{-\rho t} = 0.$$

Consumption Function

The derivative of both sides of $u'(c) = \lambda$ with respect to time yields: $u''(c)\dot{c} = \dot{\lambda}$. Dividing both sides of this expression by λ yields:

$$\frac{\dot{c}}{c} = \frac{u'(c)}{c u''(c)} \frac{\dot{\lambda}}{\lambda} = \frac{u'(c)}{c u''(c)} (\rho - r).$$

The agent's instantaneous utility function has a constant elasticity of substitution σ :

$$u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}$$

Calculating the marginal utility of consumption and the derivative of this marginal utility, and substituting in the growth rate of consumption, the Euler equation is given by the Keynes-Ramsey rule:

$$\frac{\dot{c}(s, v)}{c(s, v)} = \sigma(r - \rho). \quad (3.4)$$

The agent's budget constraint, in terms of flows, states that the change in their wealth must equal the difference between revenues, interest on the financial investment, and wages on the one hand, and expenses with the purchase of consumer goods and services on the other:

$$\dot{a}(s, v) = ra(s, v) + \omega(v) - c(s, v).$$

The solution to this differential equation is the agent's intertemporal constraint, assuming no Ponzi game. The present value of consumption must equal the present value of wages, plus the value of the financial assets held. That is:

$$a(s, t) + \int_t^\infty e^{-r(v-t)} \omega(v) dv = \int_t^\infty e^{-r(v-t)} c(s, v) dv.$$

The present value of wages is the agent's human capital h :

$$h(t) = \int_t^\infty e^{-r(v-t)} \omega(v) dv. \quad (3.5)$$

Using this notation, generation s 's intertemporal constraint may be written as:

$$a(s, t) + h(t) = \int_t^\infty e^{-r(v-t)} c(s, v) dv.$$

According to the Euler equation, generation s 's consumption must grow at a rate equal to $\sigma(r - \rho)$:

$$c(s, v) = c(s, t) e^{\sigma(r-\rho)(v-t)}.$$

Substituting this expression in the integral of present value of consumption yields:

$$\begin{aligned} \int_t^\infty e^{-r(v-t)} c(s, v) dv &= \int_t^\infty e^{-r(v-t)} c(s, t) e^{\sigma(r-\rho)(v-t)} dv \\ &= c(s, t) \int_t^\infty e^{-[r-\sigma(r-\rho)](v-t)} dv. \end{aligned}$$

Therefore, the present value of consumption equals:

$$\int_t^\infty e^{-r(v-t)} c(s, v) dv = \frac{c(s, t)}{r - \sigma(r - \rho)}.$$

The consumption of a generation s agent is, thus, proportional to their wealth, with the proportionality coefficient given by $\theta = r - \sigma(r - \rho)$. That is:

Fig. 3.1 Generation born at past instant s

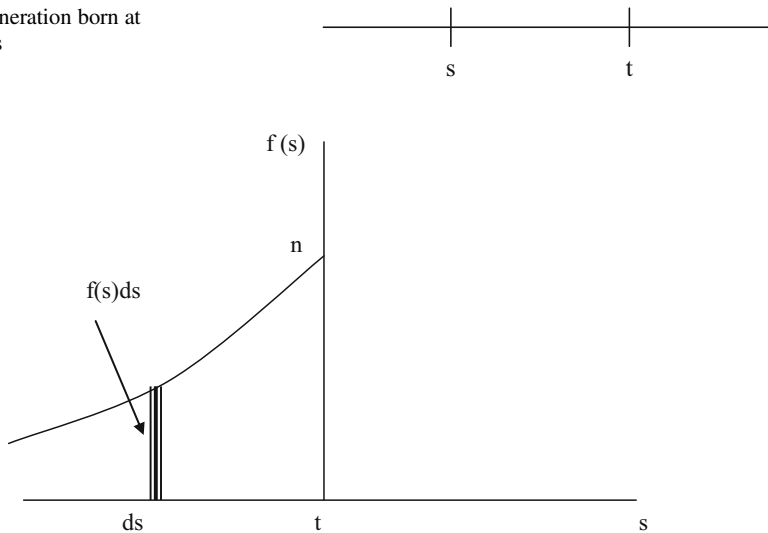


Fig. 3.2 Weights assigned to each generation: $f(s) = ne^{n(s-t)}$

$$c(s, t) = \theta[a(s, t) + h(t)]. \quad (3.6)$$

When the elasticity of substitution is equal to one, the proportionality coefficient equals the rate of time preference: $\theta = \rho$.

Aggregation

According to the OLG model, different generations exist at instant t . Figure 3.1 shows that a generation was born at a past instant s . What is this generation's size? To simplify the calculations without loss of generality, assume that the population at instant t is equal to unity. With this simplification, each variable's aggregate value is its per-capita value.

Because the population at instant t is equal to unity, $P(t) = 1$, population at s was equal to: $P(s) = e^{-n(t-s)}$. The number of individuals in generation s was, therefore, equal to $nP(s)$. The aggregate value of any given variable $x(s, t)$ is defined by:

$$x(t) = \int_{-\infty}^t nP(s)x(s, t)ds = \int_{-\infty}^t ne^{-n(t-s)}x(s, t)ds. \quad (3.7)$$

Figure 3.2 shows the weights assigned to each generation in the aggregation process. Older generations have a lower weight, while younger generations contribute more value because of each generation's size.

Consumption per capita is then obtained by means of the aggregation:

$$c(t) = \int_{-\infty}^t ne^{n(s-t)}c(s, t)ds.$$

Because consumption is proportional to wealth, it follows that:

$$c(t) = \int_{-\infty}^t n e^{n(s-t)} \theta[a(s, t) + h(t)] ds.$$

Using the same aggregation formula for human capital and financial assets, we conclude that consumption per capita is proportional to wealth per capita:

$$c(t) = \theta[a(t) + h(t)]. \quad (3.8)$$

Aggregate Consumption Change

To obtain the rate of change of aggregate consumption, we apply the Leibnitz rule [see exercise 15, Chap. 10] to the consumption aggregation formula. The result is:

$$\frac{dc(t)}{dt} = n e^{n(t-t)} c(t, t) + n \int_{-\infty}^t \left[(-n) e^{n(s-t)} c(s, t) + e^{n(s-t)} \dot{c}(s, t) \right] ds.$$

Substituting the previously obtained $\dot{c}(s, t)$ equation in the expression's last integral yields:

$$\frac{dc(t)}{dt} = nc(t, t) - nc(t) + \int_{-\infty}^t n e^{n(s-t)} \sigma(r - \rho) c(s, t) ds,$$

which is equivalent to:

$$\frac{dc(t)}{dt} = n[c(t, t) - c(t)] + \sigma(r - \rho)c(t).$$

Since $c(t, t) = \theta[a(t, t) + h(t)]$ and $c(t) = \theta[a(t) + h(t)]$, it follows that:

$$c(t, t) - c(t) = \theta[a(t, t) + h(t) - a(t) - h(t)].$$

Substituting the assumption that each generation has no financial assets upon birth, $a(t, t) = 0$, in the preceding equation yields:

$$c(t, t) - c(t) = -\theta a(t).$$

Hence:

$$\frac{dc(t)}{dt} = \dot{c} = \sigma(r - \rho)c(t) - n\theta a(t). \quad (3.9)$$

The rate of change of consumption has two components. The first one is common to all agents. The second one takes into account the fact that new agents have no financial assets upon entering the economy. The aggregate Euler equation is different from the individual Euler equation.

Financial Wealth Change

The economy's per-capita assets at moment t is given by the integral:

$$a(t) = \int_{-\infty}^t n e^{n(s-t)} a(s, t) ds.$$

The derivative of this integral with respect to time, using the Leibnitz rule [see exercise 15, Chap. 10] yields:

$$\frac{da(t)}{dt} = n e^{n(t-t)} a(t, t) + n \int_{-\infty}^t \left[(-n) e^{n(s-t)} a(s, t) + e^{n(s-t)} \dot{a}(s, t) \right] ds.$$

Taking into account the budget constraint, in terms of flows, and the assumption that each generation is born with no financial assets, $a(t, t) = 0$, the rate of change of the society's stock of financial assets is:

$$\frac{da(t)}{dt} = \dot{a} = (r - n)a(t) + \omega(t) - c(t).$$

Output

In this economy, the only existing asset is capital, and k is its quantity per capita. Therefore, $a = k$ and $\dot{a} = \dot{k}$. Interest rate r equals the marginal product of capital:

$$r = f'(k) - \delta. \quad (3.10)$$

Wages equals the marginal product of labor:

$$\omega = f(k) - f'(k)k, \quad (3.11)$$

where $f(k)$ is the intensive form of the production function with constant returns to scale. Now substituting these expressions for the rate of interest and the wage rate in the equation for $\dot{a}$ yields:

$$\dot{k} = (f'(k) - \delta - n)k + f(k) - f'(k)k - c,$$

which may be simplified and rewritten as:

$$\dot{k} = f(k) - (\delta + n)k - c. \quad (3.12)$$

The equation of the rate of change of consumption per capita, after substitution of the interest rate for the marginal product of capital, transforms into:

$$\dot{c} = \sigma(f'(k) - \delta - \rho)c - n((1 - \sigma)(f'(k) - \delta) + \sigma\rho)k.$$

Dynamical System

This model's dynamical system has two differential equations:

$$\dot{c} = \sigma(f'(k) - \delta - \rho)c - n((1 - \sigma)(f'(k) - \delta + \sigma\rho)k, \quad (3.13)$$

$$\dot{k} = f(k) - (\delta + n)k - c. \quad (3.14)$$

This system's Jacobian matrix, when the elasticity of substitution is equal to one ($\sigma = 1$), is:

$$J = \begin{bmatrix} \frac{\partial \dot{c}}{\partial c} & \frac{\partial \dot{c}}{\partial k} \\ \frac{\partial \dot{k}}{\partial c} & \frac{\partial \dot{k}}{\partial k} \end{bmatrix} = \begin{bmatrix} (f'(k) - \delta - \rho) & f''(k)c - n\rho \\ -1 & f'(k) - \delta - n \end{bmatrix}.$$

This matrix's determinant, at the stationary equilibrium point, is:

$$|J| = \bar{r}(\bar{r} - \rho - n) + f''(\bar{k})\bar{c} < 0.$$

When the elasticity of substitution is equal to one, the interest rate, at stationary equilibrium, will be equal to the rate of time preference plus a component that depends on the capital-consumption ratio according to:

$$\bar{r} = f'(k) - \delta = \rho + n\rho \frac{\bar{k}}{\bar{c}}.$$

It is straightforward to show that the equilibrium interest rate has the following bounds:

$$\rho < \bar{r} < \rho + n.$$

The second inequality is a result of the fact that $\rho(\bar{k}/\bar{c}) < 1$ because consumption is proportional to total wealth, $\bar{c} = \rho(\bar{h} + \bar{k})$. Therefore, the Jacobian determinant is negative and the stationary equilibrium is a saddle point.

In the infinite-life OLG model, the rate of time preference does not have to be greater than the population growth rate, as in the representative agent model. Therefore, dynamic inefficiency is possible. For it to occur, the rate of time preference must be lower than the population growth rate, as Fig. 3.3 illustrates.

3.2 Economy with a Government

The government budget constraint, in terms of flows, implies that the increase in public debt (B) is equal to the difference between expenditures, including interest (rB) and government spending, and the tax revenue (T). That is:

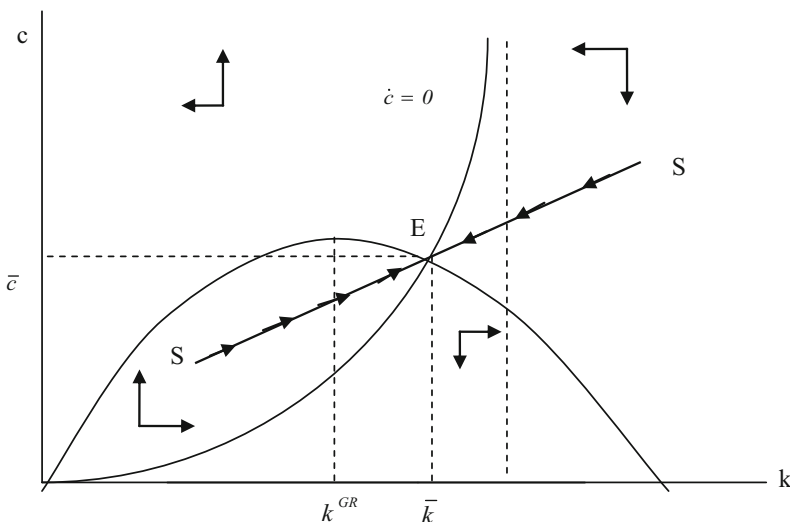


Fig. 3.3 The phase diagram for the c and k system

$$\dot{B} = rB + G - T.$$

Denoting the per-capita values of public debt, government spending, and taxes as $b = B/L$, $g = G/L$, $\tau = T/L$, respectively, the government budget constraint transforms into:

$$\dot{b} = (r - n)b + g - \tau.$$

The solution to this differential equation, assuming no Ponzi game ($r > n$), is:

$$b(t) = \int_t^\infty e^{-(r-n)(x-t)} (\tau - g) dx.$$

Per-capita public debt at instant t equals the present value of future primary surpluses discounted at the interest rate minus the population growth rate.

The private sector's wealth in an economy with OLG, assuming that government bonds are the only financial assets, has two components: government bonds and human capital. That is:

$$a(t) = b(t) + h(t)$$

The value of human capital equals the present value of wages net of taxes:

$$h(t) = \int_t^\infty e^{-r(x-t)} (\omega - \tau) dx.$$

This equation may be rewritten as follows:

$$h(t) = \int_t^\infty e^{-r(x-t)}(\omega - g)dx - \int_t^\infty e^{-r(x-t)}(\tau - g)dx.$$

Combining this equation with the expression for the government budget constraint allows for the description of total wealth with two components: one for human capital and the other for government bonds:

$$a(t) = \int_t^\infty e^{-r(x-t)}(\omega - g)dx + \int_t^\infty e^{-r(x-t)}\left(e^{n(x-t)} - 1\right)(\tau - g)dx.$$

Therefore, Ricardian equivalence does not exist in the OLG model. It would only be present if the population growth rate was equal to zero. In this case, however, the representative agent model is a particular case of the infinite-life OLG model.

Assuming a constant primary surplus, $\tau - g$, the government-bonds component of wealth is:

$$\frac{n}{r}b(t). \quad (3.15)$$

In this particular case, this component depends on the relationship between the population growth rate and the interest rate. The conclusion is that Ricardian equivalence does not exist in the OLG model because society regards part of the public debt as wealth.

3.3 Open Economy

In an open economy, individuals of the various generations may invest their funds in a foreign asset paying an interest rate, in the domestic currency, equal to r . The country is small and the foreign interest rate is given. The budget constraint, in terms of flows, equals:

$$\dot{a}(s, v) = ra(s, v) + \omega(v) - c(s, v).$$

Each generation maximizes the present value of the utility flow subject to this constraint and the model's initial condition. The optimal control problem has already been solved in Sect. 1 of this chapter. The Euler equation for each generation is:

$$\frac{\dot{c}(s, v)}{c(s, v)} = \sigma(r - \rho).$$

In an economy with a single agent, consumption would only be constant in the unusual case where the rate of time preference is exactly equal to the economy's interest rate. This would only happen by chance. In an economy with heterogeneous agents and OLG, per-capita consumption may be stationary, even if consumption is not constant for each agent, as we will see in the following section.

Dynamical System

This model has two differential equations:

$$\dot{c} = \sigma(r - \rho)c - n\theta a, \quad (3.16)$$

$$\dot{a} = (r - n)a + \omega - c. \quad (3.17)$$

This system's Jacobian matrix, evaluated at the equilibrium point, is:

$$J = \begin{bmatrix} \frac{\partial \dot{c}}{\partial c} & \frac{\partial \dot{c}}{\partial a} \\ \frac{\partial \dot{a}}{\partial c} & \frac{\partial \dot{a}}{\partial a} \end{bmatrix} = \begin{bmatrix} \sigma(r - \rho) & -n\theta \\ -1 & r - n \end{bmatrix}.$$

This matrix's determinant is equal to:

$$|J| = r[\sigma(r - \rho) - n].$$

For the determinant to be negative, the following inequality must hold:

$$r < \rho + \frac{n}{\sigma} \quad (3.18)$$

In this event, the dynamical system has a saddle point. Two cases are possible, as Figs. 3.4 and 3.5 show. In Fig. 3.4 the interest rate is greater than the rate of time preference. The opposite occurs in Fig. 3.5.

When the international interest rate is greater than the rate of time preference, as in Fig. 3.4, the country will be a creditor and accumulate wealth equal to $\bar{a}$ in the stationary state. When the international interest rate is lower than the rate of time

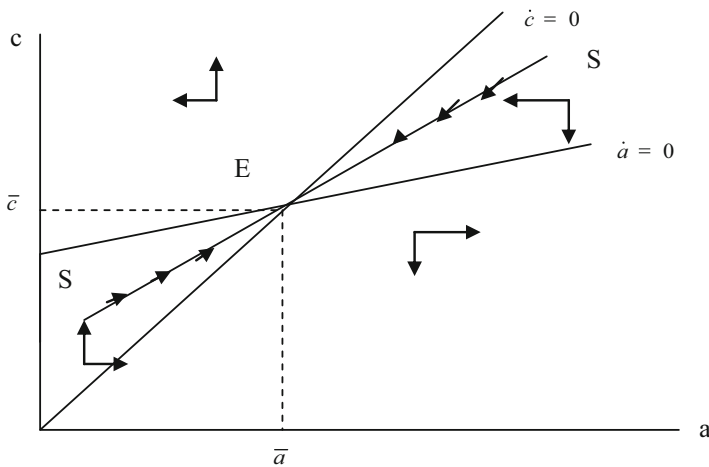


Fig. 3.4 The phase diagram for a creditor country: $r > \rho$

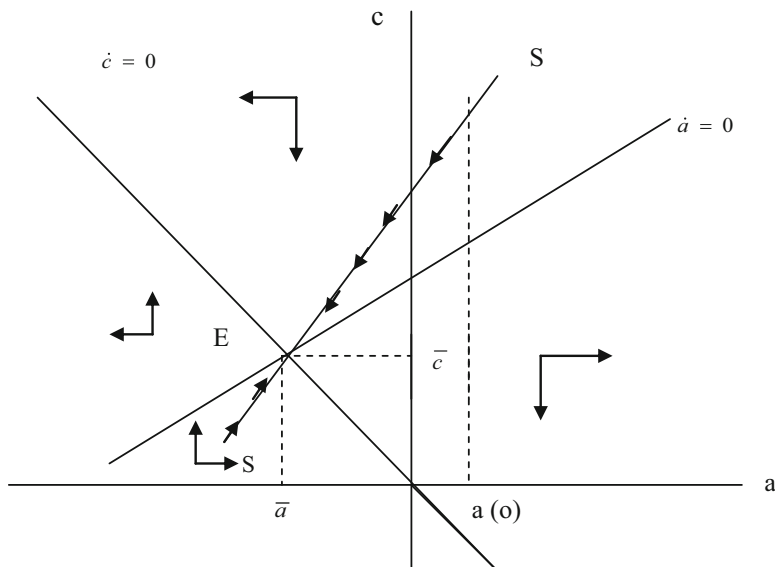
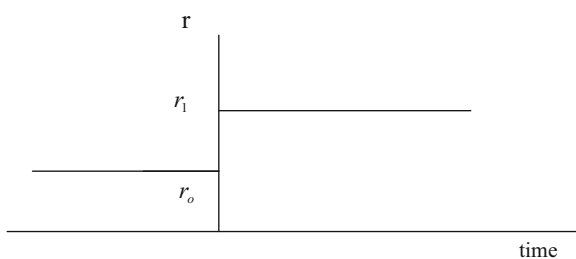


Fig. 3.5 The phase diagram for a debtor country: $r < \rho$

Fig. 3.6 An unanticipated permanent increase in the foreign interest rate



preference, the country will be a debtor, as seen in Fig. 3.5. In the stationary state, foreign debt will equal $\bar{a}$.

Experiment

Figure 3.6 assumes an experiment in which the real foreign interest rate undergoes an unanticipated, permanent increase from r_0 to r_1 . Figure 3.7 shows what happens in this economy assuming that the real interest rate is greater than the rate of time preference. The country is a net creditor. When the real interest rate increases, the curves corresponding to $\dot{c} = 0$ and $\dot{a} = 0$ shift. The former curve is down and to the right, the latter curve is up and to the left. At the new stationary equilibrium point, consumption and the stock of wealth will be greater than those associated with the previous stationary equilibrium.

When the foreign interest rate changes, consumption instantly changes as well. In Fig. 3.7, consumption decreases at first. This figure assumes that the substitution

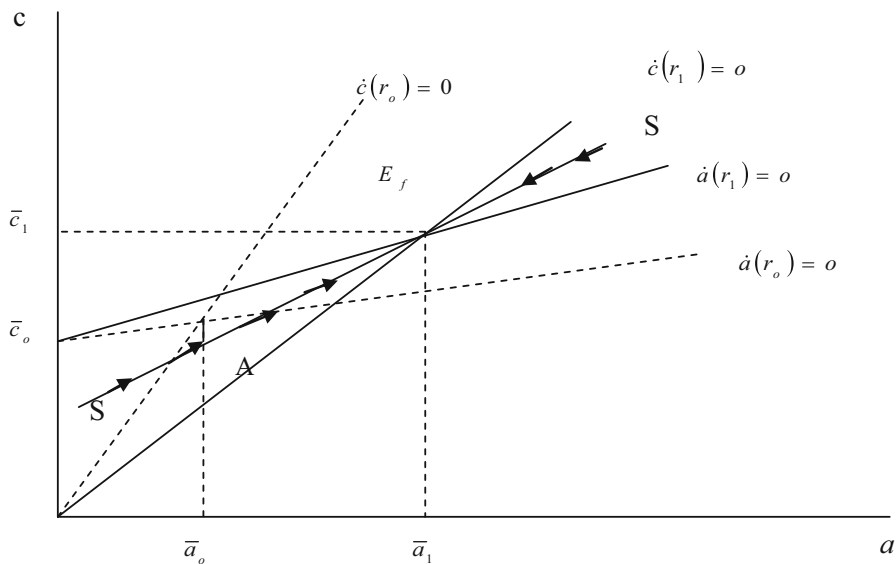


Fig. 3.7 The effects of an unanticipated permanent increase in the foreign interest rate

effect is greater than the income effect. When the two effects cancel one another, the elasticity of substitution is equal to one, and initial consumption does not change. On the other hand, if the elasticity of substitution is lower than one, initial consumption increases. As consumption changes instantly, the economy then enters the saddle path and converges to the new stationary equilibrium.

3.4 Open Economy New Keynesian IS Curve

This section specifies a new Keynesian IS curve in a small open economy. The agents are heterogeneous, with OLG born with no financial wealth.

The infinite-life OLG model's differential equations system for aggregate consumption and financial wealth is:

$$\begin{cases} \dot{C} = \sigma(r - \rho)C - \eta\theta A \\ \dot{A} = (r - n)A + y - C \end{cases} \quad (3.19)$$

The upper-case letter C represents consumption and the letter A stands for financial wealth, while y is the economy's output.

A small open economy with perfect capital mobility may have a stationary equilibrium even if the rate of time preference is different from the international interest rate, as the phase diagrams in Figs. 3.4 and 3.5 show. In Fig. 3.4 the country is a net creditor, while in Fig. 3.5 the country is an international net debtor.

The ratio of financial wealth to consumption is represented by letter a : $a = A/C$, and $\bar{a}$ represents this ratio's stationary equilibrium value. The first equation in the differential equations system may be rewritten as:

$$\dot{c} = \frac{\dot{C}}{C} = \sigma(r - \rho) - \eta\theta a.$$

Adding and subtracting $\eta\theta\bar{a}$ from the right side of the equation yields:

$$\dot{c} = \sigma(r - \bar{r}) - \eta\theta(a - \bar{a}),$$

where the natural interest rate $\bar{r}$ is equal to the rate of time preference plus a component that depends on wealth at the stationary state and on model parameters. That is:

$$\bar{r} = \rho + \frac{\eta\theta}{\sigma} \bar{a}.$$

In this model, in the long term, the natural interest rate will be equal to the international interest rate. The long-term equality between the domestic rate of interest and the international interest rate occurs as a result of the change in the ratio of financial wealth to consumption, which may be positive or negative, depending on the economy's degree of impatience compared with the international interest rate:

$$\bar{r} = \rho + \frac{n\theta}{\sigma} \frac{\bar{a}}{\bar{c}_y} = r^*. \quad (3.20)$$

In this expression, the notation has been changed and the variables are normalized by potential output: $\bar{a} = \bar{A}/\bar{y}$ and $\bar{c}_y = \bar{C}/\bar{y}$.

Algebra

The new Keynesian IS curve for the OLG model can be easily derived, but it requires a little algebra. The first-order expansion of $f(A/C) = A/C$ is:

$$\frac{A}{C} = \frac{\bar{A}}{\bar{C}} + \frac{1}{\bar{C}}(A - \bar{A}) - \frac{1}{\bar{C}^2}\bar{A}(C - \bar{C}).$$

This expression may be rewritten as:

$$\frac{A}{C} = \frac{\bar{A}}{\bar{C}} + \frac{\bar{y}}{\bar{C}} \left(\frac{A}{\bar{y}} - \frac{\bar{A}}{\bar{y}} \right) - \frac{\bar{A}/\bar{y}}{\bar{C}/\bar{y}} \left(\frac{C - \bar{C}}{\bar{C}} \right).$$

Therefore, the gap of the wealth-to-consumption ratio depends on the wealth and consumption gaps according to:

$$\frac{A}{C} - \frac{\bar{A}}{\bar{C}} = \frac{1}{\bar{c}_y} (a - \bar{a}) - \frac{\bar{a}}{\bar{c}_y} \log \frac{C}{\bar{C}}.$$

Substituting this expression in the equation for the growth rate of consumption ($\dot{c}$), where the lower-case c represents the log of consumption, $c = \log C$, yields:

$$\dot{c} = \sigma(r - \bar{r}) - \frac{n\theta}{\bar{c}_y} (a - \bar{a}) + \frac{n\theta\bar{a}}{\bar{c}_y} (c - \bar{c}).$$

The IS curve for a small open economy can be obtained using the goods and services market equilibrium equation for the equality of aggregate demand and the output from the representative agent model's IS curve [Chap. 2, Sect. 5, Eq. (2.35)]. Output is determined by aggregate demand and has two components: domestic consumption and foreign consumption, according to:

$$Y = (1 - \gamma)S^\eta C + \gamma^* S^\eta C^*. \quad (3.21)$$

The symbols have the following meanings: S represents the terms of trade, η is the elasticity of substitution between domestic and foreign goods, γ measures the domestic preference for foreign goods, and a variable with an asterisk $*$ represents a foreign variable.

Log-linear Approximation

A log-linear approximation of the expression:

$$Y = \gamma X^\alpha + \delta Z^\beta,$$

around point $\bar{X} = e^{\bar{x}}$, $\bar{Z} = e^{\bar{z}}$ is given by:

$$y - \bar{y} = \alpha\omega(x - \bar{x}) + \beta(1 - \omega)(z - \bar{z}),$$

where $\omega = \gamma\bar{X}^\alpha / \bar{Y}$.

Application of this linear approximation to the log of Eq. ((3.21)) around the stationary equilibrium point results in:

$$x = y - \bar{y} = \omega(c - \bar{c}) + \eta[1 - \omega(1 - \gamma)](s - \bar{s}) + (1 - \omega)(y^* - \bar{y}^*).$$

The symbol ω is the ratio of the corresponding variable at the stationary equilibrium. Therefore, the output gap in a small open economy has three components: the consumption gap, the terms of trade gap, and the foreign output gap.

IS Curve

The derivative of the output gap equation with respect to time yields:

$$\dot{x} = \omega_1 \dot{c} + \omega_2 \dot{s},$$

where $\omega_1 = \omega$, $\omega_2 = \eta[1 - \omega(1 - \gamma)]$ and assuming, for simplicity, that the foreign output gap equals zero. Combining the equations for $\dot{c}$, $\dot{x}$, and x , and the interest rate's uncovered parity condition, yields the IS curve:

$$\dot{x} = \beta_x x + \beta_r (r - \bar{r}) - \beta_a (a - \bar{a}) - \beta_s (s - \bar{s}). \quad (3.22)$$

In this equation, the rate of change of the output gap depends on its own level, and on the gaps of the interest rate, wealth, and the terms of trade. Therefore, total foreign liquid assets play an important role in the IS curve. This curve's parameters are related to the structural parameters according to:

$$\begin{aligned} \beta_x &= \frac{n\theta\bar{a}}{\bar{c}_y}; & \beta_r &= \omega_1\sigma, \\ \beta_a &= \frac{\omega_1 n\theta}{\bar{c}_y}; & \beta_s &= \frac{n\theta\bar{a}\omega_2\xi}{\bar{c}_y}. \end{aligned}$$

The coefficients β_r and β_a are positive. On the other hand, the coefficients β_x and β_s may be positive or negative. They are positive if the country is a net creditor ($\bar{a} > 0$) and negative if it is a net debtor ($\bar{a} < 0$). When $n = 0$, the particular case of the representative agent model's IS curve occurs because $\beta_x = \beta_a = \beta_s = 0$.

In the representative agent model, the rate of time preference and the international interest rate would only be equal by chance. When the two rates are different, opportunities arise for arbitrage and the representative agent model produces paradoxical conclusions: the small open economy becomes the owner of the world, or the small country becomes so indebted that it has to use its entire income to repay its foreign debt. These paradoxical conclusions do not exist in the infinite-life OLG model because the natural interest rate equals the international interest rate in the long term. The ratio of wealth (debt) to consumption adjusts to cover the difference between the rate of time preference and the international interest rate.

3.5 Finite-Life Overlapping Generations Model

The infinite-life OLG model assumes that individuals are immortal. The assumption simplifies the algebra, but flies in the face of facts. One certainty in life is death: life is finite. Its duration, however, is uncertain and no one knows when they will die. Death has two causes: (i) purely random factors and (ii) natural factors brought about by the aging of the human machine. The former is independent of one's age, but the latter depends on age. This section's model assumes that an individual's probability of death at each instant in time is constant and not age-dependent. The cause of death in this model is therefore random. This simplification allows a more elegant solution for the model, which would not be the case were we to introduce the more realistic assumption that the probability of death increases with age.

Because life is finite and its duration is unpredictable, let X be a random variable representing the duration of life with a probability density function,

$$f(x) = \begin{cases} \phi e^{-\phi(x-t)}, & x \geq t \\ 0, & x < t \end{cases} \quad (3.23)$$

where ϕ is a positive parameter whose interpretation will be provided below. The distribution function of random variable X , which is defined by the probability of the individual's life duration being lower than or equal to ν , is calculated by means of the integral:

$$F(\nu) = P(X \leq \nu) = \int_t^\nu \phi e^{-\phi(x-t)} dx = 1 - e^{-\phi(\nu-t)}.$$

Therefore, the probability of an individual living longer than a time interval ν equals: $P(X \geq \nu) = e^{-\phi(\nu-t)}$. The probability of event A , the individual dying within the interval $(\nu, \nu + d\nu)$, conditioned by event B , that they have not died until date ν , is called the hazard rate. That is:

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{f(\nu)}{1 - F(\nu)}.$$

The probability of the intersection of events, $P(A \cap B)$, is equal to $f(\nu)$, and the probability of event B , $P(B)$, is equal to $1 - F(\nu)$. For the density function of random variable X , this probability is:

$$P(A/B) = \frac{f(\nu)}{1 - F(\nu)} = \frac{\phi e^{\phi(\nu-t)}}{e^{-\phi(\nu-t)}} = \phi, \quad (3.24)$$

where the parameter ϕ is, therefore, the instant probability of death in interval $(\nu, \nu + d\nu)$, which is independent of the individual's age. The assumption is quite unrealistic, but enables easier calculations in the aggregation process. Probability ϕ may be regarded as the mortality rate of this economy's population.

Actuarial Notes

This model with finite-life individuals has a wealth-accumulation problem. At the time of death, wealth cannot be either positive or negative, that is, the individual cannot die with assets, as they have no heirs, or die with outstanding loans, because there would be no one to answer for their obligations. The solution to this problem is to assume an insurance company that can issue actuarial notes, selling them to, or buying them from, individuals.

Actuarial notes are claims that enable individuals to obtain better returns than the interest rate on their wealth, but with the condition of transferring their wealth to the insurance company upon death. When an individual issues an actuarial note, they borrow funds at a higher interest rate because the debt ceases to exist upon death.

What premium would the insurer be willing to charge from individuals under such circumstances? Assume that the insurance market is competitive and no entry barriers exist against business firms in this industry. The company's profit is zero. Therefore, the premium equals the economy's mortality rate, because the insurer's revenue will be equal to the mortality rate times the value of the assets it collects from individuals upon their death, ϕa , and its expenses will be an equal amount transferred to living individuals.

Consumer

The individuals forming each generation in this economy maximize the expected value of the instantaneous utility flow discounted at the rate of time preference. The expected value is obtained by multiplying the discounted value of the utility by the individual's probability of being alive. That is:

$$\int_t^\infty e^{-\phi(v-t)} e^{-\rho(v-t)} u[c(s, v)] dv = \int_t^\infty e^{-(\phi+\rho)(v-t)} u[c(s, v)] dv$$

Interpretation of the above integral is straightforward: for the individual, the discount rate equals the rate of time preference plus the mortality rate. The fact that the individual knows that they will die causes them to discount the future at a higher rate that takes the probability of death into account. Individuals in each generation maximize:

$$\int_t^\infty e^{-(\phi+\rho)(v-t)} u[c(s, v)] dv, \quad (3.25)$$

subject to the constraints:

$$\dot{a}(s, v) = (\phi + r)a(s, v) + \omega(v) - c(s, v), \quad (3.26)$$

$$a(s, t) \quad \text{given.} \quad (3.27)$$

For the individual, the rate of return on assets in this economy equals the interest rate plus the premium on the actuarial claim, the individual's probability of death.

This problem's current-value Hamiltonian is:

$$H = u[c(s, v)] + \lambda[(\phi + r)a(s, v) + \omega(v) - c(s, v)].$$

The first-order conditions are:

$$\begin{aligned} \frac{\partial H}{\partial C} &= u'(c(s, v)) - \lambda = 0, \\ \dot{\lambda} &= (\phi + \rho)\lambda - \frac{\partial H}{\partial A} = (\rho + \phi)\lambda - (\phi + r)\lambda, \end{aligned}$$

$$\frac{\partial H}{\partial \lambda} = (\phi + r)a(s, v) + \omega(v) - c(s, v) = \dot{a}(s, v).$$

According to the first condition, the marginal utility of consumption is equal to the costate variable. The second is the arbitrage condition between present consumption and future consumption. The third one reproduces the problem's constraint.

$$\begin{cases} u'(c(s, v)) = \lambda \\ \frac{\dot{\lambda}}{\lambda} = \rho - r \\ u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} \end{cases}$$

The first two first-order conditions, reproduced above, plus the assumption that the utility function has constant elasticity of substitution, imply a constant growth rate of consumption:

$$\frac{\dot{c}(s, v)}{c(s, v)} = \sigma(r - \rho).$$

Solving the differential equation for the budget constraint yields the intertemporal budget constraint, assuming no Ponzi game. The present value of consumption must equal the present value of income plus the initial stock of wealth:

$$a(s, t) + \int_t^\infty e^{-(\phi+r)(v-t)} w(v) dv = \int_t^\infty e^{-(\phi+r)(v-t)} c(s, v) dv,$$

where the discount rate is equal to the sum of the interest rate and the insurance premium. The present value of labor income will be the individual's human capital:

$$h(t) = \int_t^\infty e^{-(\phi+r)(v-t)} w(v) dv.$$

The intertemporal budget constraint may then be written as:

$$a(s, t) + h(t) = \int_t^\infty e^{-(\phi+r)(v-t)} c(s, v) dv.$$

Individual consumption grows at a rate equal to $\sigma(r - \rho)$. Therefore:

$$c(s, v) = c(s, t) e^{\sigma(r-\rho)(v-t)}.$$

Substituting this expression in the intertemporal budget constraint yields:

$$a(s, t) + h(t) = \int_t^\infty e^{-[(\phi+r)-\sigma(r-\rho)](v-t)} c(s, t) dv.$$

It then follows that consumption is proportional to individual's total wealth:

$$c(s, t) = \theta[a(s, t) + h(t)],$$

where the proportionality coefficient, assuming that it is positive, is:

$$\theta = \phi + r - \sigma(r - \rho) = \phi + (1 - \sigma)r + \sigma\rho > 0.$$

When the elasticity of substitution is equal to one, the proportionality coefficient equals the sum of the rate of time preference and the risk premium.

Aggregation

Assume that the population is constant. The birth rate is therefore equal to the mortality rate. The number of individuals from generation s in existence at moment t equals:

$$P(s, t) = \phi e^{-\phi(t-s)}. \quad (3.28)$$

The aggregation of any variable, say z , in this model takes into account the number of individuals in each generation. Therefore, for variable $z(s, t)$ of generation s at time t , the aggregate value is obtained by adding together all currently existing generations:

$$z(t) = \int_{-\infty}^t \phi e^{-\phi(t-s)} z(s, t) ds.$$

Aggregate Consumption

Aggregate consumption is then obtained by:

$$c(t) = \int_{-\infty}^t \phi e^{\phi(s-t)} c(s, t) ds.$$

Because individual consumption is proportional to wealth, it follows that aggregate consumption is also proportional to the society's total wealth:

$$c(t) = \theta[a(t) + h(t)]. \quad (3.29)$$

The rate of change of aggregate consumption is obtained by applying the Leibnitz rule [see exercise 15, Chap. 10] to the consumption aggregation formula. That is:

$$\frac{dc(t)}{dt} = \phi e^{\phi(t-t)} c(t, t) + \int_{-\infty}^t \phi(-\phi) e^{\phi(s-t)} c(s, t) ds + \int_{-\infty}^t \phi e^{\phi(s-t)} \dot{c}(s, t) ds.$$

Simplifying this expression and taking into account the growth rate of individual consumption yields:

$$\frac{dc(t)}{dt} = \phi[c(t, t) - c(t)] + \int_{-\infty}^t \phi e^{\phi(s-t)} \sigma(r - \rho) c(s, t) ds.$$

The integral featured in this equation is proportional to aggregate consumption. Therefore:

$$\frac{dc(t)}{dt} = \phi(c(t, t) - c(t)) + \sigma(r - \rho)c$$

The difference between the consumption of individuals in generation t and aggregate consumption, both in period t , is obtained by assuming that the financial wealth of individuals at birth equals zero, $a(t, t) = 0$. Therefore:

$$c(t, t) - c(t) = \theta[a(t, t) + h(t)] - \theta[a(t) + h(t)] = -\theta a(t).$$

The rate of change of consumption depends on the level of consumption and on aggregate wealth, according to:

$$\dot{c} = \sigma(r - \rho)c - \phi\theta a. \quad (3.30)$$

Aggregate Wealth

Aggregate wealth is defined by:

$$a(t) = \int_{-\infty}^t \phi e^{\phi(s-t)} a(s, t) ds$$

The rate of change of aggregate wealth is obtained by applying the Leibnitz rule [see exercise 15, Chap. 10] to the above expression:

$$\frac{da(t)}{dt} = \phi e^{\phi(t-t)} a(t, t) + \int_{-\infty}^t \phi(-\phi) e^{\phi(s-t)} a(s, t) ds + \int_{-\infty}^t \phi e^{\phi(s-t)} \dot{a}(s, t) ds.$$

Simplifying this equation and taking into account the change in the individual's wealth yields:

$$\frac{da(t)}{dt} = \phi a(t, t) - \phi a(t) + \int_{-\infty}^t \phi e^{\phi(s-t)} [(\phi + r)a(s, v) + \omega(v) - c(s, v)] dv.$$

Because of the assumption $a(t, t) = 0$, it follows that:

$$\frac{da(t)}{dt} = -\phi a + (\phi + r)a + \omega - c.$$

Therefore, the rate of change of aggregate wealth is:

$$\dot{a} = ra + \omega - c. \quad (3.31)$$

The relevant interest rate for the change of aggregate wealth is the economy's interest rate, and not the one that each individual takes into account when making decisions. The interest rate for each individual is the economy's interest rate plus the risk premium arising from the fact that life is finite. In aggregate terms, the risk premium only ensures a transfer of income among individuals.

Dynamical System

The finite-life OLG model's dynamical system has two differential equations:

$$\begin{cases} \dot{c} = \sigma(r - \rho)c - \phi\theta a \\ \dot{a} = ra + \omega - c \end{cases}. \quad (3.32)$$

This system's Jacobian matrix, evaluated at the equilibrium point, is:

$$J = \begin{bmatrix} \frac{\partial \dot{c}}{\partial c} & \frac{\partial \dot{c}}{\partial a} \\ \frac{\partial \dot{a}}{\partial c} & \frac{\partial \dot{a}}{\partial a} \end{bmatrix} = \begin{bmatrix} \sigma(r - \rho) & -\phi\theta \\ -1 & r \end{bmatrix}$$

This matrix's determinant, at the stationary equilibrium point, is negative:

$$|J| = \sigma(r - \rho)r - \phi\theta = \phi\theta \left(\frac{\bar{a}r}{\bar{c}} - 1 \right) = -\frac{\omega}{\bar{c}}.$$

The dynamical system's equilibrium point is a saddle point. Given the initial wealth, the economy converges to the stationary equilibrium point on the saddle path.

Economy with Production

In the model of a finite-life OLG economy with production, financial wealth corresponds to the capital stock. Therefore, $a = k$, and the interest rate equals the net marginal product of capital:

$$r = f'(k) - \delta. \quad (3.33)$$

The sum of the compensation of capital and labor, which are paid according to their marginal products, is precisely equal to output, assuming that the production function has constant returns to scale. That is:

$$ra + \omega = (f'(k) - \delta)k + f(k) - kf'(k) = f(k) - \delta k.$$

This model's dynamical system has two differential equations:

$$\begin{cases} \dot{c} = \sigma[f'(k) - \delta - \rho]c - \phi\theta k \\ \dot{k} = f(k) - \delta k - c \end{cases}. \quad (3.34)$$

Analysis of this system is similar to that for the infinite-life OLG model.

Open Economy

The model of a small open economy with finite-life OLG assumes that financial wealth corresponds to internationally traded claims at an interest rate that the small country is unable to affect. This model's dynamical system has two differential equations:

$$\begin{cases} \dot{c} = \sigma(r - \rho)c - \phi\theta a \\ \dot{a} = r^*a + \omega - c \end{cases} \quad (3.35)$$

where the international interest rate is equal to r^* . At stationary equilibrium, the international interest rate and the rate of time preference are represented by:

$$r^* = \rho + \frac{\phi\theta}{\sigma} \frac{\bar{a}}{\bar{c}}. \quad (3.36)$$

When the country is a creditor, the international interest rate is greater than the rate of time preference, $r^* > \rho$, since $\bar{a} > 0$. In the case of a debtor country, the international interest rate is lower than the rate of time preference, $r^* < \rho$, since $\bar{a} < 0$. Unlike the representative agent model, in the finite-life OLG model the presence of the stationary state for consumption does not depend on equality between the international interest rate and the rate of time preference. The model's adjustment variable is the current account on the balance of payments, generating deficits or surpluses until the ratio of the stock of assets to consumption reaches a certain proportion.

Economy with a Government

In an economy with a government, the public deficit is financed issuing government securities. The rate of change of public debt is:

$$\dot{b} = rb + g - \tau.$$

This equation is the government budget constraint in terms of flows. The solution to this differential equation, assuming no Ponzi game, is the government's intertemporal budget constraint:

$$b(t) = \int_t^\infty e^{-r(x-t)} (\tau - g) dx.$$

According to this constraint, the present value of future primary surpluses must equal the value of the public debt held by the market.

The private sector's total wealth is the sum of financial wealth, government bonds, plus the value of human capital:

$$b(t) + h(t) = b(t) + \int_t^\infty e^{-(\phi+r)(v-t)} (w - \tau) dv.$$

The value of human capital is equal to the present value of wages net of taxes, and can be written as follows:

$$\begin{aligned} h(t) &= \int_t^\infty e^{-(\phi+r)(v-t)} (w - \tau) dv \\ &= \int_t^\infty e^{-(\phi+r)(v-t)} (\omega - g) d\nu - \int_t^\infty e^{-(\phi+r)(v-t)} (\tau - g) d\nu. \end{aligned}$$

Substituting this expression in the previous one yields:

$$b(t) + h(t) = b(t) + \int_t^\infty e^{-(\phi+r)(\nu-t)} (\omega - g) d\nu - \int_t^\infty e^{-(\phi+r)(\nu-t)} (\tau - g) d\nu.$$

This society's total wealth includes a share of the value of the government bonds. This occurs because the interest rate at which individuals discount primary surpluses is different to the rate at which the public sector discounts them. That is:

$$\begin{aligned} b(t) - \int_t^\infty e^{-(\phi+r)(\nu-t)} (\tau - g) d\nu &= \int_t^\infty e^{-r(\nu-t)} (\tau - g) d\nu \\ &\quad - \int_t^\infty e^{-(\phi+r)(\nu-t)} (\tau - g) d\nu. \end{aligned}$$

Where parameter ϕ is equal to zero, that is, life is infinite, public debt will not be part of society's wealth. Therefore, Ricardian equivalence is not valid in the finite-life OLG model.

3.6 Exercises

1. Individuals have two periods of life. In the former – youth – they work. In the latter – old age – they retire and live off the savings from their youth and its interest. Consumption in the former period is represented by $c_{1,t}$ and by $c_{2,t+1}$ in the latter. The individual's budget constraint is:

$$\begin{aligned} c_{1,t} &= \omega_t - s_t, \\ c_{2,t+1} &= (1 + r_{t+1})s_t, \end{aligned}$$

where ω_t is wages, s_t is savings, and r_{t+1} is the interest rate.

The individual's preferences are represented by the utility function:

$$u(c_{1,t}) + \frac{1}{1+\rho} u(c_{2,t+1}), \quad u' > 0, \quad u'' < 0,$$

where ρ is the rate of time preference.

- (a) Deduce the optimization problem's first-order condition.
 (b) Show that savings is a function of wages and the interest rate:

$$s_t = s(\omega_t, r_{t+1}).$$

(c) Show that: $\frac{\partial s_t}{\partial \omega_t} > 0$,

(d) Show that: $\frac{\partial s_t}{\partial r_{t+1}} \geq 0$.

(e) Assume that the utility function is:

$$u(c) = \begin{cases} \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}, & \sigma \neq 1 \\ \log c, & \sigma = 1 \end{cases}.$$

Deduce the savings function in each of the two cases and show that if $\sigma > 1$,

$$\frac{\partial s_t}{\partial r_{t+1}} > 0.$$

2. In the previous question's OLG economy, the production function has constant returns to scale,

$$Y = F(K, L),$$

where Y is the output, K is the capital stock, and L is the working population, whose growth rate is equal to n : $L_t = (1+n)L_{t-1}$. Business firms maximize profit, wages, and the interest rate:

$$\begin{aligned} \omega_t &= f(k_t) - kf'(k), \\ r_t &= f'(k_t) - \delta, \end{aligned}$$

where $k = K/L$, $f(k) = F(\frac{K}{L}, 1)$ and δ is the depreciation rate.

Savings equals investment:

$$K_{t+1} - K_t + \delta K_t = L_t s_t.$$

- (a) Show that:

$$k_m = \frac{(1-\delta)k_t + s_t(\omega_t, r_m)}{1+n},$$

where:

$$\omega_t = f(k_t) - kf'(k),$$

$$r_{t+1} = f'(k_{t+1}) - \delta.$$

- (b) Analyze the model's equilibrium and dynamics.
 (c) In this model, can the economy show capital over-accumulation and be dynamically inefficient? That is, is the equilibrium Pareto-efficient?
3. Answer the previous problem's three questions in the particular case where the utility and output functions are respectively specified by:

$$u(c) = \frac{C^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}},$$

$$f(k) = Ak^\alpha.$$

4. Consider the infinite-life OLG model,

$$\dot{c} = \sigma(r - \rho)c - n\theta a,$$

$$\dot{a} = (r - n)a + \omega - c.$$

Assume that the international interest rate is lower than the rate of time preference. Show what happens in this economy when the international interest rate decreases under the following conditions:

- (a) unanticipated permanent change;
 (b) anticipated permanent change;
 (c) unanticipated transitory change;
 (d) anticipated transitory change.
5. Consider the following model of a finite-life OLG economy with production:

$$\begin{cases} \dot{c} = \sigma[f'(k) - \delta - \rho]c - \phi\theta k \\ \dot{k} = f(k) - \delta k - c \end{cases}.$$

- (a) Analyze the model's equilibrium and dynamics on a phase diagram with consumption on the vertical axis and capital on the horizontal axis.
 (b) Is there dynamic inefficiency in this model?
 (c) Analyze the effects of an unanticipated change in the mortality rate on consumption and capital.

Chapter 4

The Solow Growth Model



The aim of the economic growth theory is to explain the causes that determine the level and growth rate of labor productivity. This theory must be able to explain Kaldor's stylized facts: (i) the productivity of labor has been growing systematically; (ii) the capital to labor ratio has been growing over time; (iii) the rate of return on capital has been reasonably constant; (iv) the capital to output ratio has not changed over time; (v) the shares of labor and capital in output have not been showing upwards or downwards trends; (vi) the growth rate of the productivity of labor has been varying from one country to another. The Solow model introduced in this chapter attempts to explain these facts.

Section 4.1 introduces the Solow economic growth model. Section 4.2 includes human capital as a factor of production in the Solow model. Section 4.3 presents the Solow model in a small open economy with perfect capital mobility. Section 4.4 deals with economic growth accounting.

4.1 The Solow Model

The economy's real output (Y) equals the average productivity of labor (Y/L) multiplied by labor (L). That is:

$$Y = \frac{Y}{L}L.$$

The growth rate of real output is, therefore, equal to the sum of the growth rate of labor productivity and the growth rate of the labor force:

$$\hat{Y}_n = \left(\frac{\hat{Y}}{L} \right) + \hat{L},$$

where the symbol $\hat{X}$ represents the growth rate of variable X : $\hat{X} = \dot{X}/X$. Assuming constant rates of growth of labor productivity and population,

$$\left(\frac{\hat{Y}}{\hat{L}}\right) = g; \quad \hat{L} = n,$$

the growth rate of real output, which we call the natural growth rate ($\hat{Y}_n$), is:

$$\hat{Y}_n = g + n. \quad (4.1)$$

An alternative way to compute the growth rate of real output is obtained from the growth rate of capital stock:

$$\hat{Y}_\omega = \frac{\dot{K}}{K} = \frac{\dot{K}}{Y} \frac{Y}{K},$$

where the expression is multiplied and divided – after the second equal sign – by real output. The growth rate of real output therefore depends on the proportion of the increase in capital stock relative to output and the output to capital ratio. The change in capital stock equals gross capital investment minus depreciation:

$$\dot{K} = I - \delta K.$$

Substituting this expression in the equation for the growth rate of real output yields:

$$\hat{Y}_\omega = \frac{I - \delta K}{Y} \frac{Y}{K} = \frac{I/Y}{K/Y} - \delta.$$

Assuming a constant rate of investment and capital to output ratio,

$$\frac{I}{Y} = s; \quad \frac{K}{Y} = v.$$

The growth rate of real output, which we call the warranted rate, is:

$$\hat{Y}_\omega = \frac{s}{v} - \delta. \quad (4.2)$$

The natural and warranted rates of growth of real output would only be equal by chance. Economics literature refers to this fact as the razor's edge. The two rates are usually different:

$$\hat{Y}_n = g + n \neq \frac{s}{v} - \delta = \hat{Y}_\omega. \quad (4.3)$$

The natural growth rate of real output depends on technology while the warranted rate corresponds to full utilization of capital, as it is a consequence of the condition that savings must equal investment for the goods and services market to be at

equilibrium. If the natural rate is greater than the warranted rate ($Y_n > Y_w$), then either the unemployment rate will increase indefinitely, or the utilization rate of installed capacity will increase continuously. On the other hand, if the natural rate is lower than the warranted rate ($Y_n < Y_w$), either there is excess demand for labor, or the utilization of installed capacity will drop systematically. The Solow model assumes that the prices of the factors of production, capital, and labor adjust to solve this problem.

Three equations specify the Solow model: (i) a production function with constant returns to scale and labor-saving technological progress; (ii) a savings and investment equilibrium condition in the goods and services market; and (iii) a behavioral assumption that consumption is proportional to income. That is:

$$Y = F(K, AL), \quad (4.4)$$

$$Y = C + I = C + \dot{K} + \delta K, \quad (4.5)$$

$$S = Y - C = Y - (1 - s)Y = sY, \quad (4.6)$$

where A is a parameter representing technological progress, C is consumption, S is savings, $1-s$ is average (= marginal) propensity to consume, and s is the average propensity to save. The Solow model further assumes a constant rate of technological progress (g) and population growth rate (n):

$$\frac{\dot{A}}{A} = g, \quad \frac{\dot{L}}{L} = n. \quad (4.7)$$

Algebra

When the factors of production are multiplied by any parameter in the production function with constant returns to scale, the quantity of output increases by the same magnitude:

$$F(\lambda K, \lambda AL) = \lambda F(K, AL).$$

Parameter λ may be selected arbitrarily and is conveniently defined as:

$$\lambda = \frac{1}{AL}.$$

Substituting this expression in the production function yields:

$$y = \frac{Y}{AL} = F\left(\frac{K}{AL}, \frac{AL}{AL}\right) = F\left(\frac{K}{AL}, 1\right) = f(k),$$

where $f(k)$ is the intensive form of the production function. Real output and the quantity of capital, in labor efficiency units, are defined by:

$$y = \frac{Y}{AL} \quad k = \frac{K}{AL},$$

Production function $f(k)$ must obey the properties:

$$y = f(k) \quad \begin{cases} f(0) = 0 \\ \lim_{k \rightarrow \infty} f'(k) = 0 \\ \lim_{k \rightarrow 0} f'(k) = \infty \end{cases}.$$

The first property states that nothing can be produced without capital, the second states that the marginal product of capital tends to zero as the quantity of capital grows unlimitedly, and the third property sets the condition that the marginal product of capital tends to infinity as the quantity of capital approaches zero. The latter two properties are known as the Inada conditions.

Dividing both sides of the output and expenditure equilibrium equation by the quantity of labor in efficiency units yields:

$$\frac{Y - C}{AL} = \frac{\dot{K}}{AL} + \frac{\delta K}{AL}.$$

Assuming that consumption is proportional to income, the previous expression transforms into:

$$sy = \frac{\dot{K}}{AL} + \delta k.$$

The derivative of k with respect to time yields:

$$\dot{k} = \frac{\dot{K}}{AL} - (g + n)k$$

Combining the latter two equations yields:

$$sy = \dot{k} + (g + n + \delta)k.$$

Dynamical System

The Solow exogenous growth model's dynamical system is given by the nonlinear first-order differential equation:

$$\dot{k} = sf(k) - (g + n + \delta)k. \quad (4.8)$$

In the steady state, $\dot{k} = 0$, equilibrium quantity of capital ($\bar{k}$) is obtained by solving the following equation:

$$sf(\bar{k}) = (g + n + \delta)\bar{k}.$$

The system has two points of equilibrium, $\bar{k} = 0$ and $\bar{k} = k^*$, as Fig. 4.1 shows. Assuming a solution different from zero, the previous equation may be written as:

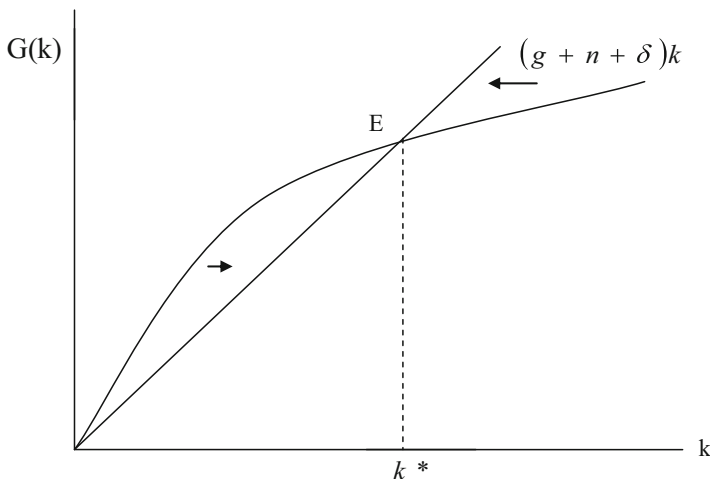


Fig. 4.1 Equilibrium of the Solow model

$$\frac{s}{k^*/f(k^*)} - \delta = g + n.$$

The ratio $k^*/f(k^*)$ is the equilibrium capital to output ratio. In the steady state, the natural growth rate of output is equal to the warranted growth rate of output: $\hat{Y}_n = \hat{Y}_w = \hat{Y}$. Therefore, there is no razor's edge in the Solow growth model. Where the two rates are different, the relative prices of the factors of production, capital, and labor adjust and the capital to output ratio changes so that the two rates will be equal in the steady state:

$$\hat{Y}_w = \frac{s}{v} - \delta = g + \delta = \hat{Y}_n. \quad (4.9)$$

Figures 4.1 and 4.2 show the model's dynamics. When the quantity of capital, measured in labor efficiency units, is out of equilibrium, whatever the initial position, the quantity of capital converges to k^* .

4.1.1 Predictions and Comparative Dynamics

The stable equilibrium of the Solow growth model leads to the following prediction:

Prediction 1 In the long term, the economy converges to k^* , regardless of the model's initial conditions.

Figure 4.3 shows the model's comparative dynamics when the saving rate increases ($s_1 > s_0$). The new equilibrium point corresponds to a higher capital to

Fig. 4.2 The phase diagram for k in the Solow model

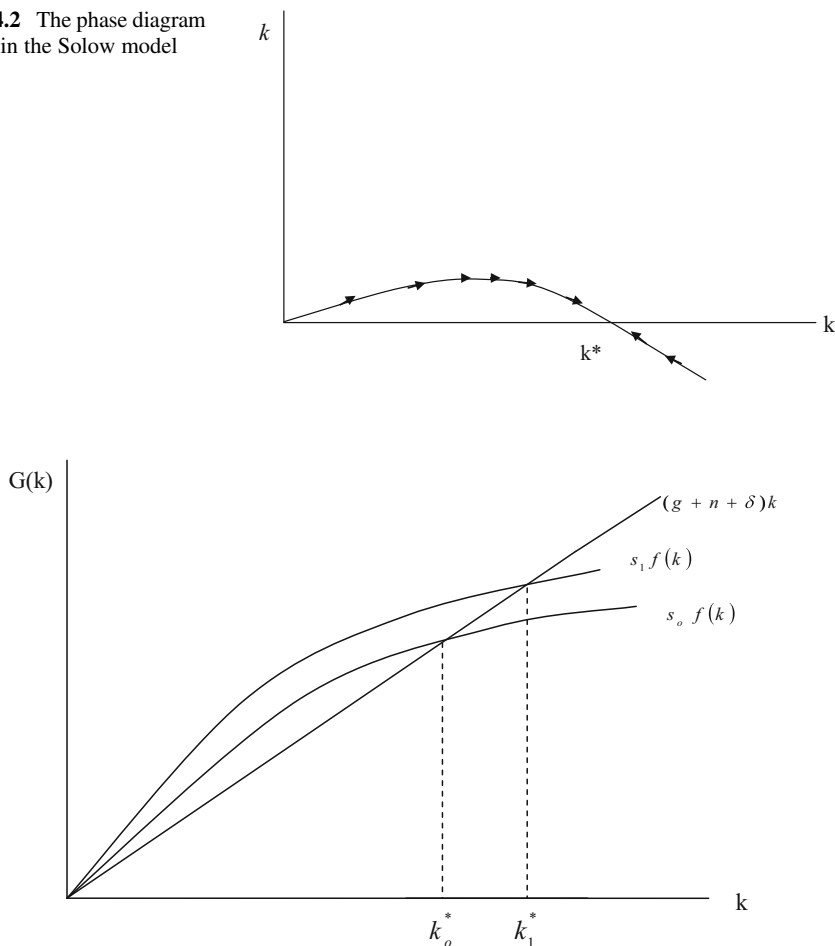


Fig. 4.3 The effects of an increase in the savings rate

labor ratio in efficiency units. Therefore, this economy's per-capita income will also be higher.

Prediction 2 The greater (lesser) the saving rate, the greater (lesser) the income per-capita in the long run term.

Figure 4.4 shows the comparative dynamics when the population growth rate increases ($n_1 > n_0$). At the new long-term equilibrium point, the quantity of capital, in labor efficiency units, decreases. This corresponds to a smaller income per-capita. This property leads to the following prediction:

Prediction 3 The greater the population growth rate, the smaller the income per-capita in the long term.

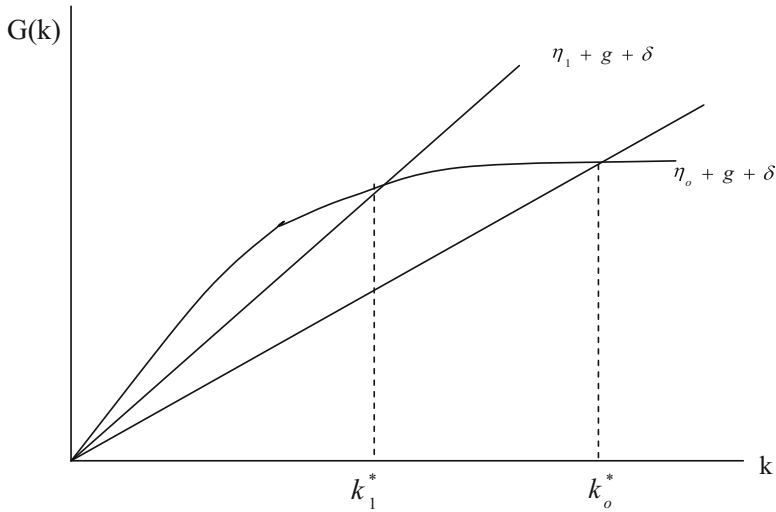


Fig. 4.4 The effects of an increase in the population growth rate s

The average productivity of labor at a stationary equilibrium equals the real output, measured in labor efficiency units, multiplied by the technological progress parameter:

$$\left(\frac{Y}{L}\right)^* = y^* A = y^* A_0 e^{gt}.$$

The growth rate of average productivity of labor is, therefore, equal to the growth rate of technological progress:

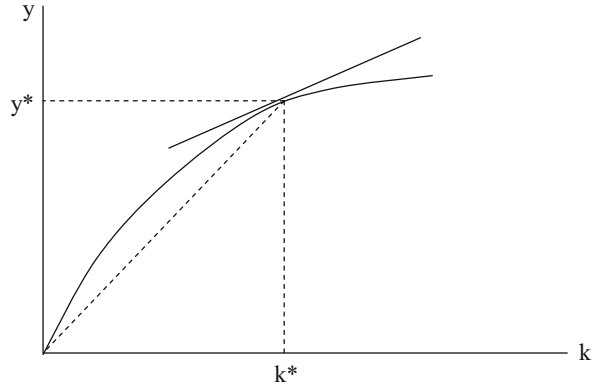
$$\left(\frac{\hat{Y}}{L}\right)^* = g. \quad (4.10)$$

Prediction 4 In the long run, the growth rate of income per-capita only depends on the growth rate of technological progress.

The capital to output ratio equals the capital to product ratio, both measured in terms of labor efficiency units. At the long-term equilibrium, the two variables are constant. Therefore, at equilibrium, the capital to output ratio is constant:

$$\frac{K^*}{Y^*} = \frac{(K/AL)^*}{(Y/AL)^*} = \frac{k^*}{y^*}. \quad (4.11)$$

Fig. 4.5 The interest rate is constant at the long run equilibrium



Prediction 5 In the long run the capital to output ratio is constant.

At the long-run term equilibrium, the interest rate is constant:

$$r^* = f'(k^*). \quad (4.12)$$

Figure 4.5 graphically depicts this ratio as the tangent of the production function curve. The wage per worker, measured in terms of efficiency, $\omega^* = (W/AL)^*$, is obtained by the following expression:

$$\omega^* = f(k^*) - f'(k^*)k^*.$$

It follows that the wage per worker grows at the same rate as technological progress:

$$\left(\frac{W}{L}\right)^* = \omega^* A_0 e^{gt}. \quad (4.13)$$

Prediction 6 In the long run term, the marginal product of capital is constant and the wage per worker grows at the same rate as technological progress.

Labor Productivity

The growth rate of labor productivity is equal to the sum of the rates of growth of technological progress and of output (by labor efficiency units), $\hat{Y} - \hat{L} = \hat{A} + \hat{y}$. Because $\hat{y} = \alpha_k \hat{k}$, where α_k is the share of capital in output, it follows that:

$$\hat{Y} - \hat{L} = \hat{A} + \alpha_k \left(\frac{sf(k)}{k} - (g + n + \delta) \right). \quad (4.14)$$

The growth rate of labor productivity has two components. The former is the long run-term growth rate, which is equal to the growth rate of technological

progress. The latter is a short run-term component that depends on the share of capital in output multiplied by the growth rate of capital per-capita measured in terms of labor efficiency. This growth rate of capital per-capita equals the difference between the warranted and natural rates. In the long run-term, this component is equal to zero.

4.1.2 The Golden Rule and Dynamic Inefficiency

In the Solow model's steady state, consumption is:

$$c^* = f(k^*) - (g + n + \delta)k^*.$$

The golden rule, which is based on the rule of conduct (do unto others as you would have them do unto you) is obtained by maximizing consumption relative to the capital stock. That is:

$$\frac{dc^*}{dk^*} = f'(k^*) - (g + n + \delta) = 0.$$

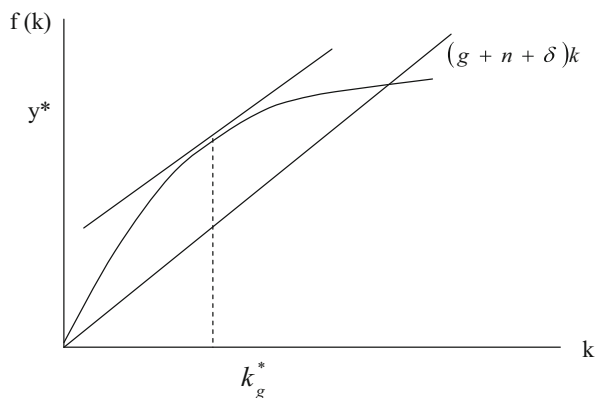
The interest rate that maximizes consumption equals the growth rate of the economy's real output:

$$f'(k_g^*) - \delta = g + n \quad (4.15)$$

where k_g^* is the capital stock, by labor efficiency units, that produces the golden rule.

Golden-rule consumption, in addition to being the maximum, must be such that savings would keep it constant over time. No generation would be in breach of the golden rule because they would all leave what they had to the coming generations. Figure 4.6 graphically depicts the golden rule. Consumption is maximum when the tangent of the production function is parallel with the line that runs through the origin and whose angular coefficient equals $g + n + \delta$. If the economy's capital stock

Fig. 4.6 The golden rule



is greater than the golden-rule capital stock, $k > k_g^*$, the real interest rate will be lower than the growth rate of real output, $f'(k) - \delta < g + n$, and the economy will be inefficient. Dynamic inefficiency is characterized by over-accumulation of capital. When an economy is dynamically inefficient, the welfare of everyone can be increased without harming anyone's welfare, since the economy is not Pareto-efficient. In an inefficient economy, society is saving too much, that is, a reduction in saving would increase consumption and the society's welfare.

The dynamic inefficiency condition may be written as follows:

$$\frac{f'(k)k}{y} < \frac{(g + n + \delta)k}{y}, \quad (4.16)$$

where both sides of the inequality are multiplied and divided by k and y , respectively. The left side of the inequality is the share of capital in output and the right side is the economy's steady-state investment rate:

$$\alpha_k = \frac{f'(k)k}{y}, \quad s = \frac{(g + n + \delta)k}{y}.$$

The economy will be dynamically inefficient when the share of capital in output is lower than the rate of investment: $\alpha_k < s$.

4.1.3 Convergence

The exogenous growth model can analyze two kinds of convergence: absolute and relative. Absolute convergence assumes that the parameters of two economies, or two regions, are exactly the same. Relative convergence assumes that this is not the case, that is, that the economies' parameters are different. To analyze both cases, one divides both sides of the Solow model's differential equation by k :

$$\frac{\dot{k}}{k} = s \frac{f(k)}{k} - (g + n + \delta). \quad (4.17)$$

The left side of this expression is the growth rate of k and the right side is the difference of two components. The former is proportional to the output to capital ratio and decreases as the capital stock by efficiency unit increases, while the second is constant, as Fig. 4.7 illustrates. Imagine two countries, or two regions, with different initial stocks of capital, as the figure shows. The country, or region, with the lower capital stock will show higher output growth. In the long term, the two will have the same income per-capita and the same growth rate of output.

Figure 4.8 illustrates the case of two countries, or two regions, where one has a higher rate of saving ($s_1 > s_2$). This is an example of relative convergence. The poorer

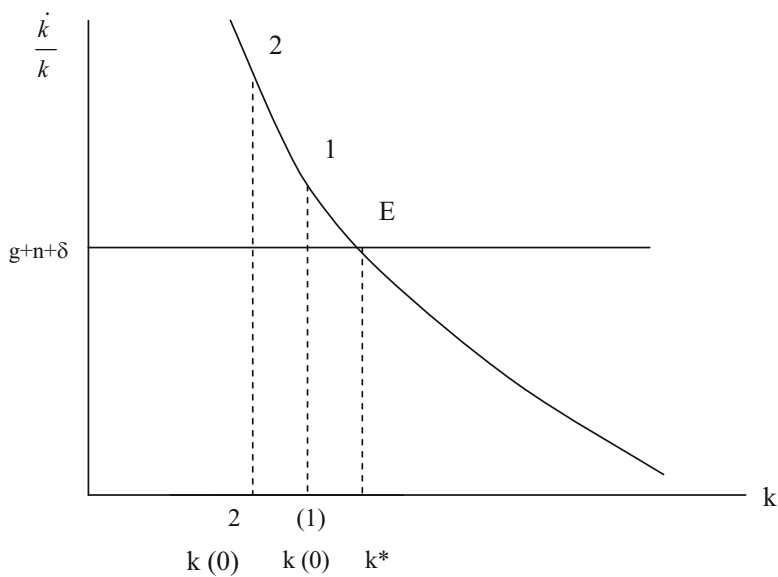


Fig. 4.7 Absolute convergence

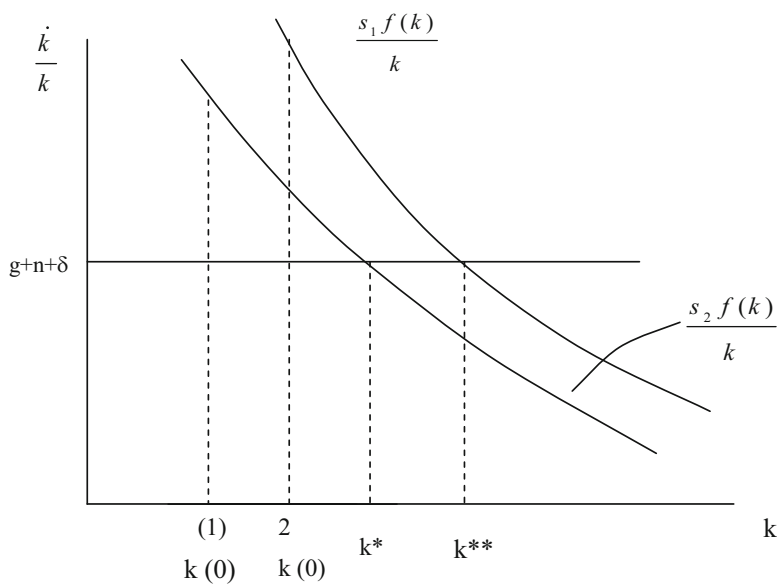


Fig. 4.8 Relative convergence

country, with a lower initial capital stock, will not necessarily have a higher growth rate than the rich country. In the long term, the two will show the same growth rate of output. But the richer country will remain richer because it will have a higher income per-capita.

Rate of Convergence

The Solow model's rate of convergence can be estimated using the Taylor expansion of the model's differential equation, which we reproduce below for convenience:

$$\dot{k} = sf(k) - (n + g + \delta)k.$$

The Taylor expansion disregards all terms of an order higher than the first:

$$\dot{k} = \left. \frac{\partial \dot{k}}{\partial k} \right|_{k=k^*} (k - k^*).$$

The derivative of $\dot{k}$ with respect to k equals:

$$\frac{\partial \dot{k}}{\partial k} = sf'(k) - (n + g + \delta).$$

Therefore:

$$\dot{k} = [sf'(k^*) - (n + g + \delta)](k - k^*).$$

Because at equilibrium $sf(k^*) = (n + g + \delta)k^*$, it follows that:

$$s = \frac{(n + g + \delta)k^*}{f(k^*)}.$$

The rate of convergence of the capital-labor ratio is, therefore, equal to:

$$\dot{k} = \left[\frac{(n + g + \delta)f'(k^*)k^*}{f(k^*)} - (n + g + \delta) \right] (k - k^*).$$

Placing $n + g + \delta$ in evidence yields:

$$\dot{k} = \left[\frac{f'(k^*)k^*}{f(k^*)} - 1 \right] (n + g + \delta)(k - k^*).$$

The share of capital in output equals:

$$\alpha_k = \frac{f'(k^*)k^*}{f(k^*)}.$$

Therefore:

$$\dot{k} = -(1 - \alpha_k)(n + g + \delta)(k - k^*).$$

Using λ to denote the expression that multiplies the deviation of k from its equilibrium value,

$$\lambda = (1 - \alpha_k)(n + g + \delta),$$

yields the expression:

$$\dot{k} = -\lambda(k - k^*). \quad (4.18)$$

The derivative of the production function $y = f(k)$ with respect to time is:

$$\frac{dy}{dt} = f'(k^*) \frac{dk}{dt}.$$

Taking into account the derivative of k with respect to time we get:

$$\dot{y} = -f'(k^*)\lambda(k - k^*).$$

A Taylor expansion of the production function $y = f(k)$, disregarding the terms higher than first order, yields:

$$f(k) = f(k^*) + f'(k^*)(k - k^*).$$

Combining the two equations above yields the equation for the rate of convergence of real output:

$$\dot{y} = -\lambda(y - y^*). \quad (4.19)$$

The rates of convergence of the capital stock, in labor efficiency units, and of real output are calculated using the following equations:

$$\begin{cases} \dot{k} = -\lambda(k - k^*) \\ \dot{y} = -\lambda(y - y^*) \end{cases} \quad (4.20)$$

They can be linearly approximated by:

$$\begin{aligned} k(t) - k^* &\cong e^{-\lambda t}(k(0) - k^*) \\ y(t) - y^* &\cong e^{-\lambda t}(y(0) - y^*) \end{aligned}$$

The Solow growth model's convergence is quite rapid. Indeed, consider an example where the economy's parameters are:

$$\begin{cases} \alpha_k = 1/3 \\ n = 1\% \\ g = 2\% \\ \delta = 3\% \end{cases}$$

The parameter λ of the rate of convergence is equal to 4%:

$$\lambda = (1 - \alpha_k)(n + g + \delta) = \frac{2}{3}(1 + 2 + 3) = 4\%.$$

Therefore, at this rate of convergence, the economy will take 17.5 years to cover half the distance: $e^{\lambda t} = e^{0.04t} = e^{0.04 \times 17.5} \cong 2$, as $e^{-\lambda t} = 1/2$.

4.1.4 Income Per-Capita: Differences Between Countries

The Solow growth model's stationary equilibrium is given by:

$$sy^* = (n + g + \delta)k^*.$$

Differentiating both sides of this equation yields:

$$\frac{dy^*}{y^*} + \frac{ds}{s} = \frac{dk^*}{k^*} + \frac{d(n + g + \delta)}{n + g + \delta}.$$

The differential of the production function $y^* = f(k^*)$, $dy^* = f'(k^*)dk^*$ yields:

$$\frac{dy^*}{y^*} = \frac{f'(k^*) \cdot k^*}{f(k^*)} \frac{dk^*}{k^*} = \alpha_k \frac{dk^*}{k^*},$$

where α_k is the share of capital in output $\alpha_k = f'(k^*)k^*/f(k^*)$. Substituting the value of dk^*/k^* in the differential of the equilibrium condition yields:

$$\frac{dy^*}{y^*} + \frac{ds}{s} = \frac{1}{\alpha_k} \frac{dy^*}{y^*} + \frac{d(n + g + \delta)}{n + g + \delta}.$$

Rearranging the terms of the expression we obtain:

$$\frac{dy^*}{y^*} = \frac{\alpha_k}{1 - \alpha_k} \left(\frac{ds}{s} - \frac{d(n + g + \delta)}{n + g + \delta} \right) \quad (4.21)$$

This equation's linear approximation is given by:

$$\log y^* = \frac{\alpha_k}{1 - \alpha_k} (\log s - \log(n + g + \delta)).$$

Therefore:

$$y^* = s^{\frac{\alpha_k}{1 - \alpha_k}} (n + g + \delta)^{-\frac{\alpha_k}{1 - \alpha_k}}. \quad (4.22)$$

Assume that country A's rate of saving is four times that of country B: s_A (country A) = 4 s_B (country B). Suppose that the share of capital in both countries is

one-third. Hence: $\alpha_k/(1 - \alpha_k = 1/2)$. The ratio between the two countries' incomes per-capita would then equal:

$$\frac{y_A^*}{y_B^*} = \left(\frac{s_A}{s_B} \right)^{\frac{1}{2}}, \quad (4.23)$$

assuming that all other parameters are the same for both countries. A country that saves four times as much as another would only have an income per-capita twice as big (in the steady state).

Consider another example, where country A's population growth is 1% and country B's is 3%: n_A (country A) = 1%, n_B (country B) = 3%; $g + \delta = 5\%$ per year, and in both countries the share of capital is the same as in the previous example. The difference in income per-capita between the two countries will be:

$$\frac{y_A^*}{y_B^*} = \left(\frac{n_A + g + \delta}{n_B + g + \delta} \right)^{-1/2} = \left(\frac{6}{8} \right)^{-1/2} = \left(\frac{8}{6} \right)^{-1/2} = 1.15. \quad (4.24)$$

We conclude from this exercise that the income per-capita of the country (A) with the lower population growth rate is 1.15 times that of the country (B) with the higher population growth rate.

Rate of Return on Capital and Income Per-Capita

The interest rate is equal to the marginal product of capital,

$$r = f'(k),$$

assuming, to simplify, a zero depreciation rate. This equation's differential is given by:

$$dr = f''(k)dk.$$

The differential of the production function $y = f(k)$, $dy = f'(k)dk$, when substituted in the previous equation, results in:

$$\frac{dr}{r} = \frac{f''(k)f}{[f'(k)]^2} \frac{dy}{y} = \frac{ff''}{(f')^2} \frac{dy}{y}.$$

Below, we will deduce the following equality from the definition of the elasticity of substitution σ between the factors of production:

$$\frac{f \cdot f''}{(f')^2} = -\frac{1 - \alpha_k}{\sigma \alpha_k}.$$

Substituting this result in the preceding expression we find that:

$$\frac{dr}{r} = - \left(\frac{1 - \alpha_k}{\sigma \alpha_k} \right) \frac{dy}{y}. \quad (4.25)$$

An approximation for this equation is:

$$\log r = - \left(\frac{1 - \alpha_k}{\sigma \alpha_k} \right) \log y.$$

The rates of return on capital and the incomes per-capita of two countries, A and B, are therefore related by means of the following equation:

$$\frac{r_A}{r_B} = \left(\frac{y_A}{y_B} \right)^{-\frac{1-\alpha_k}{\sigma \alpha_k}}. \quad (4.26)$$

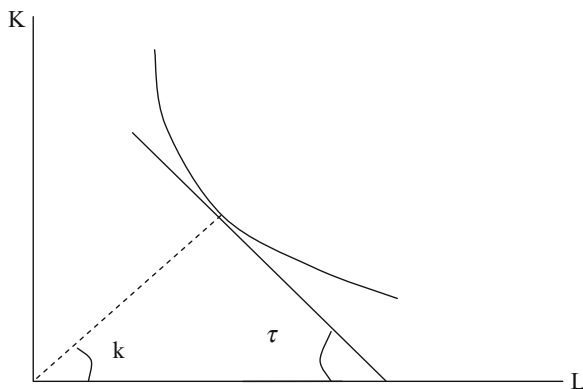
Assume that the production function is a Cobb-Douglas function. The elasticity of substitution is then equal to one: $\sigma = 1$. If the share of capital is equal to one-third, the absolute value of the exponent of the ratio of the incomes per-capita is equal to two. Therefore, a country with an income per-capita equal to half that of the other country would require a rate of return four times as big. This fact is not observed in the real world. If the share of capital were two-thirds, instead of one-third, the result would be more favorable.

Elasticity of Substitution

The elasticity of substitution measures the relative change in the proportion of factors when the marginal rate of substitution undergoes a percentage change along an isoquant, as Fig. 4.9 shows. The elasticity of substitution is thus defined by:

$$\sigma = \frac{\frac{\Delta k}{k}}{\frac{\Delta \tau}{\tau}}.$$

Fig. 4.9 The isoquant and the elasticity of substitution



When the variables are continuous, the elasticity of substitution is defined by the ratio of the marginal rate of substitution to the capital-labor ratio divided by the derivative of the marginal rate of substitution with respect to the capital-labor ratio:

$$\sigma = \frac{\tau/k}{d\tau/dk}. \quad (4.27)$$

The marginal rate of substitution equals the ratio between the marginal products of labor and capital. The production function may be written as the product of the quantity of labor multiplied by the function that depends on the capital-labor ratio. That is:

$$\tau = \frac{F_L}{F_K}, \quad Y = Lf(k).$$

The marginal products of labor and capital are respectively equal to:

$$F_L = \frac{\partial Y}{\partial L} = f(k) + Lf'(k) \left(-\frac{K}{L^2} \right) = f(k) - kf'(k),$$

$$F_K = \frac{\partial Y}{\partial K} = Lf'(k) \frac{1}{L} = f'(k).$$

Therefore, the marginal rate of substitution equals:

$$\tau = \frac{f(k) - kf'(k)}{f'(k)} = \frac{f(k)}{f'(k)} - k.$$

The derivative of the marginal rate of substitution with respect to k yields:

$$\frac{d\tau}{dk} = \frac{f'(k)f'(k) - f(k)f''(k)}{(f'(k))^2} - 1.$$

Once simplified, the expression may be written as:

$$\frac{d\tau}{dk} = -\frac{ff''}{(f')^2}.$$

The ratio of the marginal rate of substitution to k depends on the share of capital in output according to:

$$\frac{\tau}{k} = \frac{f(k)}{f'(k)k} - 1 = \frac{1}{\alpha_k} - 1 = \frac{1 - \alpha_k}{\alpha_k}.$$

The elasticity of substitution is obtained by dividing this expression by the previous one:

$$\sigma = \frac{(1 - \alpha_k)/\alpha_k}{\frac{-ff''}{(f')^2}} = -\left(\frac{1 - \alpha_k}{\alpha_k}\right) \frac{(f')^2}{ff''}. \quad (4.28)$$

The expression used in the text is an alternative way of writing the previous equation. That is:

$$\frac{ff''}{(f')^2} = -\left(\frac{1 - \alpha_k}{\sigma\alpha_k}\right)$$

Share of Capital in Output

The share of capital in output (α_k) is an important parameter for three relevant consequences of the Solow growth model: (i) differences in income per-capita as a result of rates of saving and population growth; (ii) speed of the convergence of output to the steady state; and (iii) differences in the rates of return on capital between poor and rich countries. In the three formulas, the share of capital is a parameter that affects the results, as can be easily seen by repeating the three expressions deduced earlier:

$$\begin{cases} \frac{dy^*}{y^*} = \frac{\alpha_k}{1 - \alpha_k} \left[\frac{ds}{s} - \frac{d(n + g + \delta)}{n + g + \delta} \right] \\ \dot{y} = (1 - \alpha_k)(n + g + \delta)(y - y^*) \\ \frac{dr}{r} = -\left(\frac{1 - \alpha_k}{\sigma\alpha_k}\right) \frac{dy}{y} \end{cases} \quad (4.29)$$

Based on these expressions, we conclude that the Solow growth model cannot explain: (i) the large differences in income per-capita observed around the world; (ii) the conditional rates of convergence that take place in practice; and (iii) the different rates of return on capital between poor and rich countries. One solution for these issues is to assume that the share of capital in output is far larger than that used in exogenous growth models. However, this assumption requires redefining capital. This topic will be introduced later, in the chapter 5 that presents the endogenous growth model.

4.2 The Solow Model with Human Capital

The Solow model may be generalized by introducing human capital as a factor of production. The production function thus depends on physical capital, human capital (H), and unskilled labor. The accumulation of physical capital plus depreciation is equal to saving, which is proportional to output. The accumulation of human capital plus its depreciation equals the saving diverted to this purpose. This saving also

absorbs a share of the output. Technological progress grows at a constant rate, and so does the population. The model is formed by the following equations:

$$Y = F(K, H, AL), \quad (4.30)$$

$$\dot{K} = s_K Y - \delta_K K, \quad (4.31)$$

$$\dot{H} = s_H Y - \delta_H H, \quad (4.32)$$

$$\dot{A} = gA, \quad (4.33)$$

$$\dot{L} = nL. \quad (4.34)$$

The production function, with constant returns to scale, and with the factors of production in units of labor efficiency, is:

$$y = \frac{Y}{AL} = F\left(\frac{K}{AL}, \frac{H}{AL}, 1\right) = f(k, h), \quad (4.35)$$

where $k = K/AL$, $h = H/AL$, and $f(k, h)$ is the intensive form of the production function.

Dynamical System

By taking the derivatives of k and h with respect to time and substituting the model's equations, using the same procedure as in the Solow model, yields the model's dynamical system for human capital, with two differential equations:

$$\dot{k} = s_K f(k, h) - (n + g + \delta_K)k, \quad (4.36)$$

$$\dot{h} = s_H f(k, h) - (n + g + \delta_H)h. \quad (4.37)$$

The differential equations system has a stable equilibrium point, as the determinant of its Jacobian matrix is positive and its trace is negative. Figure 4.10 shows the

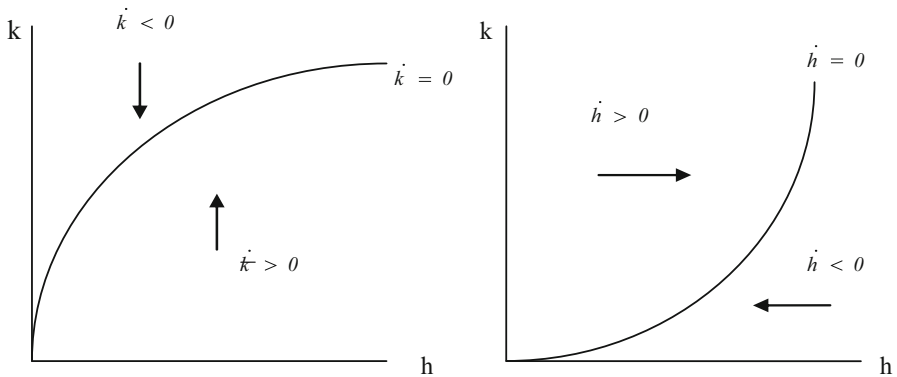


Fig. 4.10 The dynamics of k and h

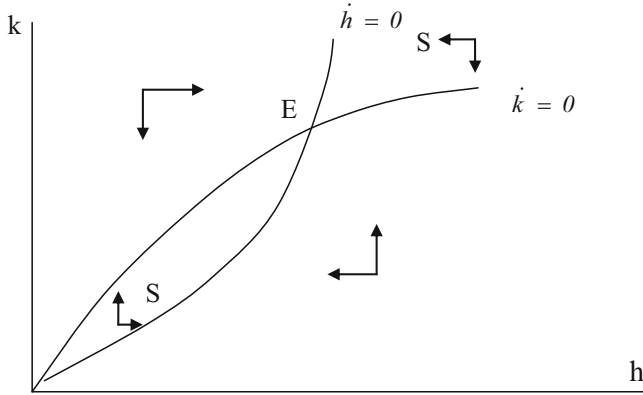


Fig. 4.11 The phase diagram of the Solow model with human capital

phase diagrams of each one of the model's differential equations. Figure 4.11 shows the model's phase diagram. Whatever the economy's initial point, it will converge to equilibrium point E . In the long term, the economy's growth rate grows at a rate equal to the sum of the rates of growth of technological progress and the population.

Labor Productivity

The growth rate of output measured in terms of labor efficiency is obtained by taking the derivative of the production function $f(k, h)$ with respect to time. That is:

$$\hat{y} = \alpha_k \hat{k} + \alpha_h \hat{h},$$

where α_k and α_h are, respectively, the shares of physical capital and human capital in output. Therefore, the growth rate of labor productivity equals:

$$\left(\frac{\hat{Y}}{\hat{L}}\right) = \hat{A} + \hat{y} = \hat{A} + \alpha_k \hat{k} + \alpha_h \hat{h}.$$

Substituting the values of $\hat{k}$ and $\hat{h}$ obtained from the model's two differential equations yields the following expression for the growth rate of labor productivity:

$$\left(\frac{\hat{Y}}{\hat{L}}\right) = \hat{A} + \alpha_k \left[\frac{s_k f(k, h)}{k} - (n + g + \delta_k) \right] + \alpha_h \left[\frac{s_h f(k, h)}{h} - (n + g + \delta_h) \right]. \quad (4.38)$$

In the long term, the growth rate of labor productivity equals the growth rate of technological progress. In the short term, the growth rate of labor productivity depends on three components: (i) the growth rate of technological progress; (ii) the product of the share of physical capital in output multiplied by the growth rate of per-capita capital, measured in terms of labor efficiency; (iii) the product of

the share of human capital in output multiplied by the growth rate of human capital, measured in terms of labor efficiency.

4.3 The Solow Model in the Small Open Economy

This section introduces the Solow model in a small open economy with perfect capital mobility. In the open economy, gross domestic product equals the sum of consumption (C), investment (I), and net exports (NX):

$$Y = C + I + NX. \quad (4.39)$$

The national product (Y_n) is obtained by adding all interest from abroad to the gross domestic product. That is:

$$Y_n = Y + rF, \quad (4.40)$$

where r is the real interest rate and F is the total of foreign assets, which will be positive where the country is a creditor and negative where it is a debtor. Therefore, the national product will exceed the domestic product for a creditor country and be lower than the domestic product for a debtor country.

The change in foreign assets – current account on the balance of payments – equals the sum of net exports and total interest:

$$\dot{F} = NX + rF. \quad (4.41)$$

Substituting the gross domestic product expression in the national product equation yields:

$$Y_n = C + I + NX + rF = C + I + \dot{F}.$$

Domestic savings equals the difference between national product and consumption. Therefore, in an open economy, the domestic savings finance investment and the purchase of new foreign assets:

$$S = Y_n - C = I + \dot{F}. \quad (4.42)$$

When the country is a debtor ($F < 0$), domestic and foreign savings finance investment.

In a small open economy with perfect capital mobility, the domestic real interest rate equals the real foreign interest rate by arbitrage:

$$r = r^*. \quad (4.43)$$

The Solow model assumes a production function with constant returns to scale and with labor-saving technological progress:

$$Y = G(K, AL), \quad (4.44)$$

where A is the parameter that represents technological progress, K is the capital stock, and L is the quantity of labor. Due to the constant returns to scale, this function can be written as follows:

$$y = \frac{Y}{AL} = G\left(\frac{K}{AL}, 1\right) = g(k), \quad (4.45)$$

where y and k represent output and the capital stock, respectively, measured in units of labor efficiency.

The Solow model's behavioral assumption is that savings is proportional to the economy's income according to:

$$S = s(y + rF), \quad (4.46)$$

where s is the rate of saving.

The Solow model further assumes a constant growth rate of technological progress (g) and population (n):

$$\frac{\dot{A}}{A} = g, \quad \frac{\dot{L}}{L} = n. \quad (4.47)$$

The addition to capital stock equals the capital investment minus depreciation:

$$\dot{K} = I - \delta K. \quad (4.48)$$

Gross Domestic Product Growth

In a small open economy, the domestic real interest rate equals the international interest rate. Therefore, the marginal product of capital equals the international interest rate:

$$g'(k) - \delta = r^*. \quad (4.49)$$

As a result, because the capital stock in units of labor efficiency is constant, the capital stock grows at the same rate as the sum of the rates of growth of technological progress and the population. Output in units of labor efficiency is also constant. It follows that this economy's gross domestic product grows at a rate equal to the sum of the rates of growth of technological progress and of the population. Any change in the international interest rate will not affect the economy's growth, but rather the level of product.

Unlike the closed-economy Solow growth model, there is no convergence process of the domestic product to a long-term equilibrium. The rate of saving also does not affect the domestic product's short-run growth rate and its long-run level. These conclusions are obtained with the implied, and implausible, assumption of instantaneous capital stock adjustment. Any change in the international interest rate instantaneously adjusts the capital stock.

4.3.1 Current Account on the Balance of Payments

The net change in capital stock equals capital investment minus depreciation. Investment equals savings minus the net change in foreign assets. In the Solow model, saving is proportional to national income. Therefore, the net change in capital is given by:

$$\dot{K} = S - \dot{F} - \delta K = s(Y + rF) - \dot{F} - \delta K$$

This equation may be rewritten as:

$$\frac{\dot{K} + \dot{F}}{AL} = s \frac{Y + rF}{AL} - \delta \frac{K}{AL},$$

where both sides are divided by the product of the technology coefficient multiplied by the quantity of labor. Defining $f = F/AL$ and taking into account the definition that $k = K/AL$, the previous equation becomes the following differential equation:

$$\dot{k} + (g + n)k + \dot{f} + (g + n)f = s(y + rf) - \delta k. \quad (4.50)$$

Given the international interest rate, the capital stock in efficiency units is constant, $k = \bar{k}$, and so its output is measured in the same unit, $y = \bar{y}$. Therefore $\dot{k} = 0$. It follows that the dynamics of the current account on the balance of payments, when measured in efficiency units, is given by the following differential equation:

$$\dot{f} = [s\bar{y} - (g + n + \delta)\bar{k}] - [(g + n) - sr^*]f. \quad (4.51)$$

The current account converges to stationary equilibrium:

$$\bar{f} = \frac{s\bar{y} - (g + n + \delta)\bar{k}}{g + n - sr^*}, \quad (4.52)$$

if the following inequality is satisfied:

$$g + n > sr^*. \quad (4.53)$$

At stationary equilibrium, the current account may be either positive or negative, depending on the expression's numerator, as the denominator is positive by assumption. To establish the conditions for one case or another to occur, the stationary equilibrium value may be written as follows:

$$\bar{f} = \frac{s\bar{k} \left[\frac{\bar{y}}{\bar{k}} - \frac{g+n+\delta}{s} \right]}{g + n - sr^*}.$$

In the closed economy, the Solow model has the following equation for the steady-state capital stock:

$$sg(k^c) = (g + n + \delta)k^c.$$

Therefore, the steady-state value of the current account is proportional to the difference between the steady-state output-capital ratio and the ratio that would be found in a closed economy:

$$\bar{f} = \frac{s\bar{k} \left[\frac{\bar{y}}{\bar{k}} - \frac{y^c}{\bar{k}^c} \right]}{g + n - sr^*}.$$

The production function has diminishing returns on capital. Therefore, the marginal product of capital decreases as the output to capital ratio decreases. Thus, if the international interest rate were equal to the interest rate of the closed economy, the current account on the balance of payments would be zero. On the other hand, if the open economy's output to capital ratio is greater than that which would prevail in a closed economy, then the international interest rate is greater than the interest rate that would apply in a closed economy, and the country is a creditor. For a debtor country, the international interest rate is lower than that in the closed economy and the current account is negative.

In the open-economy Solow model, the rate of saving does not affect the economy's growth rate, but rather the current account on the balance of payments. In a debtor country, a low rate of saving increases the equilibrium deficit, while in a creditor country, a low rate of saving decreases the surplus.

Gross National Product Dynamics

Unlike the gross domestic product, the gross national product measured in terms of efficiency units varies as the current account on the balance of payments converges to the steady state over time. The national product in efficiency units is:

$$y_n = \frac{Y_n}{AL} = \frac{Y}{AL} + r^* \frac{F}{AL} = y + r^* f.$$

As with the gross domestic product, given that the international interest rate is constant, it follows that the dynamic of national product is a function of the dynamics of the current account on the balance of payments according to:

$$\dot{y}_n = r^* \dot{f}$$

Substituting the differential equation of the current account on the balance of payments in this expression yields the differential equation of national product:

$$\dot{y}_n = r^* [s\bar{y} - (g + n + \delta)\bar{k}] - [(g + n) - sr^*]f.$$

A little algebra easily shows that this equation may be written as:

$$\dot{y}_n = -[(g + n) - sr^*](y_n - \bar{y}_n). \quad (4.54)$$

The stationary equilibrium national product is:

$$\bar{y}_n = \bar{y} + r^* \bar{f}.$$

4.3.2 Sustainability of the Foreign Debt

The current account on the balance of payments may be written dividing both sides of its expression by the product of the efficiency coefficient multiplied by the quantity of labor, according to:

$$\frac{\dot{F}}{AL} = \frac{NX}{AL} + r^* \frac{F}{AL}.$$

This equation yields the following differential equation:

$$\dot{f} = nx + [r^* - (g + n)]f.$$

At stationary equilibrium, the value of net exports (nx) is given by:

$$\bar{nx} = [r^* - (g + n)]\bar{f}.$$

The sustainability condition for foreign debt, in the case of a debtor country, is obtained by integrating the current account's differential equation. The following conditions are then obtained:

$$f = \int_t^\infty e^{-[r^* - (g+n)]\tau} nx \, d\tau \quad (4.55)$$

$$\lim_{T \rightarrow \infty} f(t+T)e^{-[r^* - (g+n)]T} = 0. \quad (4.56)$$

The former equation states that foreign debt must equal the present value of net exports discounted by the difference between the international interest rate and the growth rate of output. According to the latter equation, the foreign debt may grow over time, but must do so at a rate lower than the difference between the international interest rate and the economy's growth rate.

4.4 Growth Accounting

Economic growth accounting is a framework to identify the sources of growth. This method assigns a contribution to each factor of production and leaves a residual that cannot be identified as pertaining to a factor of production. The residual is called the total productivity of the factors of production. The accounting begins with a production function with constant returns to scale where output (Y) depends on the capital stock (K), the quantity of labor (L), and the parameter A measuring the total productivity of the factors of production. That is:

$$Y = AF(K, L). \quad (4.57)$$

The derivative of this production function with respect to time yields:

$$\frac{dY}{dt} = AF_K \frac{dK}{dt} + AF_L \frac{dL}{dt} + F(K, L) \frac{dA}{dt}. \quad (4.58)$$

A variable's index indicates the partial derivative of function F with respect to the represented variable. Dividing both sides of this expression by output and rearranging the terms so that the shares of each factor in the output are shown, assuming that both capital and labor are paid at their marginal products, the growth rate of output is given by:

$$\frac{1}{Y} \frac{dY}{dt} = \frac{AF_K K}{Y} \frac{dK}{dt} \frac{1}{K} + \frac{AF_L \cdot L}{Y} \frac{dL}{dt} \frac{1}{L} + \frac{F(K, L)}{Y} \frac{dA}{dt},$$

which may be written as:

$$\frac{1}{Y} \frac{dY}{dt} = \alpha_K \frac{dK}{dt} \frac{1}{K} + \alpha_L \frac{dL}{dt} \frac{1}{L} + \frac{1}{A} \frac{dA}{dt},$$

where α_K and α_L are the shares of capital and labor in output:

$$\alpha_K = \frac{AF_K K}{Y}; \quad \alpha_L = \frac{AF_L L}{Y}; \quad \alpha_K + \alpha_L = 1.$$

Using $\hat{X}$ to indicate the growth rate of variable X , the growth rate of output is equal to a weighted average of the rates of growth of capital stock and of the quantity of labor:

$$\hat{Y} = \alpha_K \hat{K} + \alpha_L \hat{L} + \hat{A}. \quad (4.59)$$

The difference between the growth rate of real output and the weighted average of the rates of growth of the two factors (capital and labor), represented by $\hat{A}$, is the growth rate of the total productivity of the factors of production. This rate is also known as the Solow residual.

Labor Productivity

The growth rate of labor productivity therefore depends on the growth rate of the capital-labor ratio and the growth rate of technological progress, according to:

$$\hat{Y} - \hat{L} = \alpha_K (\hat{K} - \hat{L}) + \hat{A}. \quad (4.60)$$

This expression takes into account the fact that the sum of the shares of factors in output is equal to unity.

Potential Output: Growth Rate

The growth accounting framework can be used to calculate the growth rate of the economy's potential output based on the following parameters: the shares of the factors in output, the rate of investment, the capital to output ratio, the depreciation

rate, the growth rate of labor, and the growth rate of the total productivity of factors. A growth rate of capital stock depends on investment and capital depreciation:

$$\hat{K} = \frac{dK}{dt} \frac{1}{K} = \frac{\dot{K}}{K} = \frac{I - \delta K}{K}.$$

This expression may be rewritten with respect to the rate of investment and the capital to output ratio:

$$\hat{K} = \frac{I}{K} - \delta = \frac{I/Y}{K/Y} - \delta$$

Substituting a growth rate of capital stock in the formula for the growth rate of potential output yields:

$$\hat{Y} = \alpha_K \left(\frac{I/Y}{K/Y} - \delta \right) + \alpha_L \hat{L} + \hat{A} \quad (4.61)$$

Table 4.1 exemplifies the application of the formula for the growth of potential output. The share of capital in output is 40%, labor has a share of 60%, the depreciation rate is 3.0%, the rate of investment is equal to 20.0%, and the capital to output ratio is 2.5. The growth rate of real output is therefore equal to 3.9% when the growth rate of technological progress is 1% and the quantity of labor increases by 1.5%.

Solow's classic growth accounting work (1956) concludes that a large portion of the United States' economic growth was due to technological progress. Since then, economic research has been attempting to unravel this work by identifying other contributing factors to economic growth. One candidate is education. The production function therefore depends on technological progress, on the quantity of capital, and on the quantity of labor multiplied by its schooling level (HL), where H measures the workforce's human capital. Symbolically:

$$Y = AF(K, H, L).$$

Table 4.1 The growth rate of potential output: parameters of the economy

α_K	0.40
α_L	0.60
δ	3.0%
I/Y	20.0%
K/Y	2.5%
$\hat{L}$	1.5
$\hat{A}$	1.0%
$\hat{Y}$	3.9%

The growth rate of real output is a weighted average of the rates of growth of capital and labor, added to two components, one measuring the contribution of the increase in human capital and another measuring the total increase in factor productivity:

$$\hat{Y} = \alpha_K \hat{K} + \alpha_L \hat{H} + \alpha_L \hat{L} + \hat{A}.$$

4.4.1 Labor Productivity

The increase in labor productivity depends on the increase in the capital-labor ratio, on the increase in human capital, and on the growth rate of the total productivity of the factors of production:

$$\hat{Y} - \hat{L} = \alpha_K (\hat{K} - \hat{L}) + \alpha_L \hat{H} + \hat{A}.$$

4.5 Exercises

1. Solve the Solow model for production functions given by the following specifications:
 - I. Cobb-Douglas: $Y = K^\alpha (AL)^{1-\alpha}$.
 - II. CES: $Y = \left[\delta K^{-\theta} + (1 - \delta)(AL)^{-\theta} \right]^{-\frac{1}{\theta}}$.
 - (a) What are the short- and long-term rates of growth of labor productivity in the case of the Cobb-Douglas function?
 - (b) What are the short- and long-term rates of growth of labor productivity in the case of the CES function?
 - (c) Does the value of the elasticity of substitution make a difference for the model's results?
2. Assume a Cobb-Douglas (intensive form) production function: $y = k^\alpha$, where α is the share of capital in output. In the Solow model, the economy is at the steady state:

$$sf(k) = (g + n + \delta)k.$$

Show that the log of labor productivity is given by:

$$\log \frac{Y}{L} = \log A_o + gt + \frac{\alpha}{1 - \alpha} \log s - \frac{\alpha}{1 - \alpha} \log (g + n + \delta).$$

3. Assume a Cobb-Douglas (intensive form) production function: $y = k^\alpha h^\beta$, where α is the share of capital in output and β is the share of human capital in output. The economy, in the model including human capital, is at steady state:

$$s_k f(k, h) = (n + g + \delta_k)k,$$

$$s_h f(k, h) = (n + g + \delta_h)h.$$

Show that the log of labor productivity is given by:

$$\log \frac{Y}{L} = \log A_o + gt + \frac{\alpha_k}{1 - \alpha_k - \alpha_n} \log s_k + \frac{\alpha_n}{1 - \alpha_k - \alpha_n} \log s_h - \frac{\alpha_k}{1 - \alpha_k - \alpha_n} \log(n + g + \delta_k) - \frac{\alpha_n}{1 - \alpha_k - \alpha_n} \log(n + g + \delta_n).$$

4. The differential equations of the exogenous growth model with human capital are given by:

$$\begin{cases} \dot{k} = s_k f(k, h) - (n + g + \delta_k)k \\ \dot{h} = s_h f(k, h) - (n + g + \delta_h)h \end{cases}$$

- (a) Deduce this system's Jacobian matrix.
 (b) Show that the determinant of this system's Jacobian matrix, at the stationary equilibrium point, is positive and that its trace is negative.
5. Consider the following model:

$$\text{Production function: } Y = AK + \gamma K^\alpha L^{1-\alpha},$$

$$\text{Investment = Saving: } \dot{K} - \delta K = sY,$$

$$\text{Population: } \dot{L} = nL.$$

- (a) What is this economy's short-term growth rate?
 (b) What is this economy's long-term growth rate?
6. The exogenous growth model with a government is specified by the following equations:

$$\text{Production function: } Y = F(K, AL),$$

$$\text{Saving: } S = s(Y - T),$$

$$\text{Investment Saving: } S = I = \dot{K} + \delta K,$$

$$\text{Government: } G = T,$$

$$\text{Technological progress: } \dot{A} = gA,$$

$$\text{Population: } \dot{L} = nL.$$

- (a) Deduce this model's differential equation for capital accumulation, measured in labor efficiency units.
 (b) Does the government affect the growth rate of output?
 (c) Does the government affect this economy's per-capita income?

7. The Solow model with money is specified by the following equations:

Production function: $y = f(k)$,

Assets: $a = m + k$,

Saving: $S = s(y + \tau - m\pi)$,

Investment Saving: $S = \dot{k} + \delta k + \dot{m}$,

Money Demand: $m = L(r)k$, $L' < 0$,

Real interest rate: $\rho = f'(k) - \delta$,

Monetary Policy: $m = m(\mu - \pi)$, $\mu = \frac{\dot{M}}{M} = \text{constant}$.

where $y = Y/L$, $k = K/L$, $m = M/PL$. Assume, for the sake of simplicity, that the population is constant.

- Use a phase diagram with k on the horizontal axis and m on the vertical axis to analyze this model's equilibrium and stability.
- Is the monetary policy neutral, that is, does the level of the money stock affect per-capita income?
- Is the monetary policy superneutral, that is, does the growth rate of money stock affect real output?

8. The Cobb-Douglas production function is given by:

$$Y = AK^\alpha L^{1-\alpha}.$$

The symbols have their conventional meanings.

(a) Show that:

$$\log \frac{Y}{L} = \frac{1}{1-\alpha} \log A + \frac{\alpha}{1-\alpha} \log \frac{K}{Y}$$

(b) What short- and long-term conclusions can be drawn from this equation?

Chapter 5

Economic Growth: Endogenous Savings and Growth



The first section in this chapter presents the Ramsey-Cass-Koopmans growth model, in which the consumption decision is based on the intertemporal allocation of resources. Section 5.2 presents the overlapping generations growth model, in which consumption is also an endogenous variable. Section 5.3 introduces endogenous growth models showing, in a very simple way, how each model solves the issue of endogenous growth. Section 5.4 introduces the AK model, in which the economy's growth rate is an endogenous variable arising from the fact that capital has no decreasing returns. Section 5.5 covers the Acemoglu-Ventura AK model of an open economy with no capital mobility but with goods and services mobility, in which a relative price – the terms of trade – affects the economy's growth rate. Section 5.6 introduces Lucas's human capital model. Section 5.7 presents Romer's model of innovation with new varieties of inputs, and Section 5.8 presents Aghion and Howit's Schumpeterian innovation model with creative destruction.

5.1 The Ramsey-Cass-Koopmans Model

The Solow growth model has two equations. The first equation is the law of motion of capital resulting from the goods and services market's equilibrium condition,

$$\dot{k} = f(k) - c - (g + n + \delta)k. \quad (5.1)$$

The second equation is a behavioral assumption that consumption is proportional to output:

$$c = (1 - s)f(k). \quad (5.2)$$

The Ramsey-Cass-Koopmans (RCK) model replaces this consumption equation with the Euler equation based on the consumer decision of its intertemporal resource allocation.

Euler Equation

Assume a consumer who has to choose between spending \$1 for consumption in period t or period $t+1$. Should they choose to consume in period t , their welfare shows an increase equal to the marginal utility of consumption in period t . Should they choose to consume in period $t+1$, they will invest \$1 in a financial asset yielding an interest rate equal to r , and spend the principal plus interest in the subsequent period. Their welfare will show an increase equal to the marginal utility of consumption in period $t+1$. To compare their welfare in period t with that in period $t+1$, they discount the latter at the rate of time preference ρ . In equilibrium, the consumer will be indifferent between consuming in period t or in period $t+1$. Therefore:

$$u'(C_t) = \frac{1+r}{1+\rho} u'(C_{t+1}).$$

The Taylor expansion of the marginal utility of consumption in period $t+1$ relative to consumption in period t is given by:

$$u'(C_{t+1}) = u'(C_t) + u''(C_t)(C_{t+1} - C_t),$$

where second-order terms have been disregarded. Substituting this expression in the Euler equation yields:

$$\frac{1+\rho}{1+r} = \frac{u'(C_{t+1})}{u'(C_t)} = 1 + \frac{u''(C_t)}{u'(C_t)}(C_{t+1} - C_t).$$

Taking the log of both sides of this equation yields:

$$\log(1+\rho) - \log(1+r) = \log \left[1 + \frac{u''(C_t)}{u'(C_t)}(C_{t+1} - C_t) \right].$$

Using the $\log(1+x) \cong x$ approximation:

$$\rho - r = \frac{u''(C_t)}{u'(C_t)}(C_{t+1} - C_t),$$

which may be written as:

$$C_{t+1} - C_t = \frac{u'(C_t)}{u''(C_t)}(\rho - r).$$

This expression, with continuous variables, transforms into:

$$\dot{C} = -\frac{u'(C)}{u''(C)}(r - \rho).$$

Dividing both sides of this equation by consumption yields:

$$\frac{\dot{C}}{C} = -\frac{u'(C)}{u''(C)C}(r - \rho).$$

Assuming a constant elasticity of substitution for the utility function:

$$u(C) = \frac{C^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}},$$

we obtain $u'(C) = C^{-\frac{1}{\sigma}}$ and $u''(C) = -\frac{1}{\sigma}C^{-\frac{1}{\sigma}-1}$. Therefore, the growth rate of consumption is:

$$\frac{\dot{C}}{C} = \sigma(r - \rho). \quad (5.3)$$

The growth rate of consumption is positive (negative) when the interest rate is higher (lower) than the rate of time preference. The consumer will therefore prefer to consume less (more) in the present and more (less) in the future if the interest rate is higher (lower) than the rate of time preference.

Consumption c , measured in units of labor efficiency, is given by:

$$c = \frac{C}{AL}. \quad (5.4)$$

The growth rate of consumption in units of labor efficiency equals the difference between the per-capita rates of consumption and of technological progress:

$$\frac{\dot{c}}{c} = \left(\frac{\widehat{C}}{L} \right) - \frac{\dot{A}}{A} = \left(\frac{\widehat{C}}{L} \right) - g.$$

The growth rate of consumption per-capita corresponds to the representative agent's Euler equation. Therefore, the differential equation for consumption in units of labor efficiency is:

$$\frac{\dot{c}}{c} = \sigma(r - \rho) - g,$$

which may be written as:

$$\frac{\dot{c}}{c} = \sigma \left(r - \rho - \frac{g}{\sigma} \right). \quad (5.5)$$

The first-order condition for maximizing the firm's profit implies that the interest rate is equal to the marginal product of capital,

$$r = f'(k) - \delta. \quad (5.6)$$

Therefore, the growth rate of consumption is:

$$\frac{\dot{c}}{c} = \sigma \left[f'(k) - \delta - \rho - \frac{g}{\sigma} \right]. \quad (5.7)$$

Dynamical System

The RCK model's dynamical system has two differential equations:

$$\begin{cases} \dot{k} = f(k) - c - (g + n + \delta)k \\ \dot{c} = \sigma \left[f'(k) - \delta - \rho - \frac{g}{\sigma} \right] \end{cases}. \quad (5.8)$$

The former is the capital accumulation equation, which is the same as in the Solow model. The latter is the Euler equation, in which the growth rate of consumption depends on the difference between the interest rate and the long-term equilibrium interest rate – the economy's natural rate, which is defined next.

At stationary equilibrium, $\dot{k} = \dot{c} = 0$. Therefore, equilibrium consumption and capital, both measured in labor efficiency units, are:

$$\begin{aligned} c^* &= f(k^*) - (g + n + \delta)k^*, \\ f'(k^*) - \delta &= \rho + \frac{1}{\sigma}g. \end{aligned}$$

In this model, the economy's starting capital $K(0)$ is given. This piece of information is not enough to solve the dynamical system of differential equations. An additional condition is needed. According to this condition, which is called the transversality condition, the limit of the present value of each agent's capital stock (K/L), evaluated at the marginal utility of consumption (λ), must equal zero as time tends to infinity. That is:

$$\lim_{t \rightarrow \infty} \lambda(t) \frac{K(t)}{L(t)} e^{-(\rho-n)t} = 0.$$

If this condition is not satisfied, the agent might increase their welfare by ceasing to invest and allocating their funds to consumption. The marginal utility of consumption, for the function with constant elasticity of substitution, equals: $\log \lambda = -(1/\sigma) \log (C/L)$. Consumption in the steady state grows at a rate g . Therefore, the marginal utility grows at a rate equal to $-g/\sigma$. The capital stock per

worker grows at a rate equal to g . Therefore, the transversality condition is satisfied when the following inequality is met:

$$\rho - n > g - \frac{g}{\sigma}$$

which is equivalent to:

$$\rho + \frac{g}{\sigma} > g + n \quad (5.9)$$

In this model there is no dynamic inefficiency because the interest rate ($\rho + g/\sigma$) will always exceed the growth rate of real output ($g + n$).

This dynamical system's Jacobian matrix, evaluated at the equilibrium point, is:

$$J = \begin{bmatrix} \frac{\partial \dot{c}}{\partial c} & \frac{\partial \dot{c}}{\partial k} \\ \frac{\partial \dot{k}}{\partial c} & \frac{\partial \dot{k}}{\partial k} \end{bmatrix} = \begin{bmatrix} 0 & \sigma f''(k^*)c^* \\ -1 & f'(k^*) - (g + n + \delta) \end{bmatrix}$$

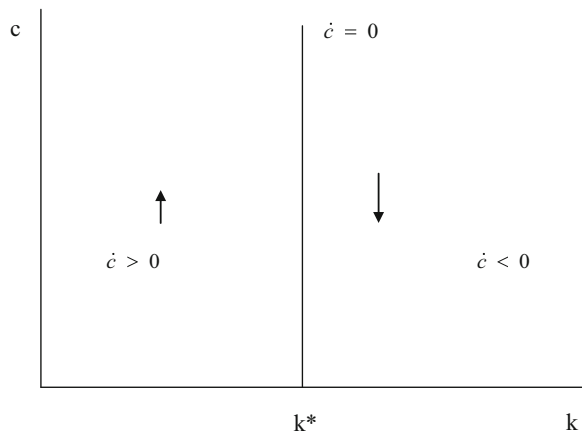
This matrix's determinant is negative,

$$|J| = \sigma f''(k^*)c^* < 0$$

because the marginal product of capital is decreasing as a result of the assumption of decreasing returns on each factor of production. Therefore, the system has a saddle point.

Figure 5.1 shows the phase diagram for the consumption equation. Consumption remains constant when capital corresponds to the economy's natural interest rate, which is equal to the sum of the rate of time preference and the ratio of the

Fig. 5.1 The dynamics of c



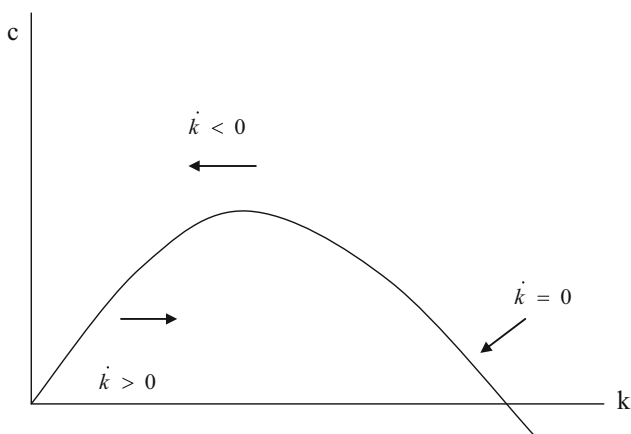


Fig. 5.2 The dynamics of k

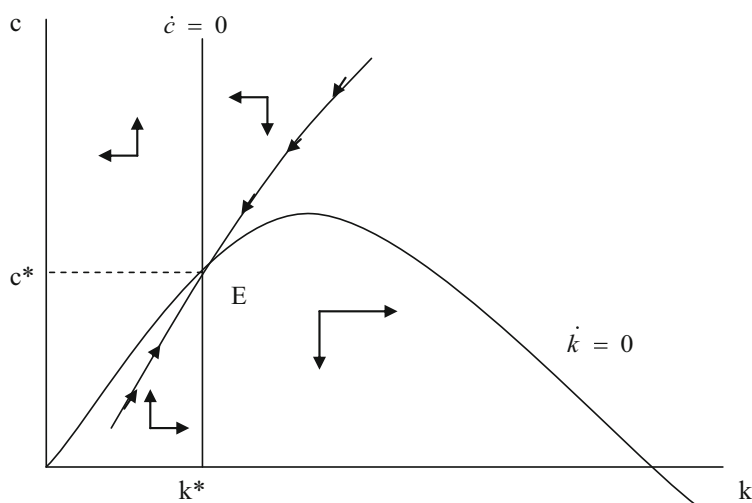


Fig. 5.3 The phase diagram for the c and k system

growth rate of technological progress to the elasticity of substitution between capital and labor.

Figure 5.2 shows the phase diagram of the capital equation. It cuts the horizontal axis at two points, at the origin and at a positive value of k . Below the curve, the quantity of capital increases, and above it, the quantity of capital decreases.

Figure 5.3 shows the phase diagram for the dynamical system of the model's two equations. Whatever the starting point of the capital-to-labor ratio, the economy converges to the equilibrium point on the saddle path represented in Fig. 5.3.

Labor Productivity

The growth rate of labor productivity in the RCK model also has two components, as in the Solow model. The difference between the two models lies in the short-term component because consumption is not proportional to income. Consumption and capital are related through the saddle path according to the function $c = c(k)$. The growth rate of labor productivity is therefore given by:

$$\hat{Y} - \hat{L} = \hat{A} + \alpha_k \left(\frac{f(k) - c(k)}{k} - (g + n + \delta) \right) \quad (5.10)$$

The two components of the growth rate of labor productivity are: (i) the growth rate of technological progress, and (ii) the product of the share of capital in output multiplied by the growth rate of the capital stock per capita, measured in units of labor efficiency.

Experiment

Figure 5.4 shows an unanticipated, permanent change in this economy's rate of time preference. The rate of time preference decreases from ρ_0 to ρ_1 . The economy's long-term interest rate decreases and the stock of capital increases as Fig. 5.5 shows. At first, consumption shows an instant drop and the economy follows a path on the model's new saddle. At the long-term equilibrium, consumption and capital will exceed their previous values, but the economy's growth rate remains the same as before in the long term.

Figure 5.6 shows an experiment in which the population growth rate decreases permanently to n_1 from n_0 .

The equation $\dot{c} = 0, f'(k^*) = \delta + \rho + \frac{1}{\sigma}g$ is independent from the population growth rate, and therefore remains unchanged, as Fig. 5.7 shows.

The equation $\dot{k} = 0, c = f(k) - (g + n + \delta)k$ depends on the population growth rate. Therefore, as the population growth rate decreases (increases), for a given quantity of capital, consumption increases (decreases). The equation $\dot{k} = 0$ shifts up, as Fig. 5.7 shows. The reduction in the population growth rate leads to an instant increase in consumption. The quantity of capital remains unchanged, as does the economy's real output. The portion of savings that kept the capital stock per worker constant now goes into consumption.



Fig. 5.4 An unanticipated permanent decrease in the rate of time preference

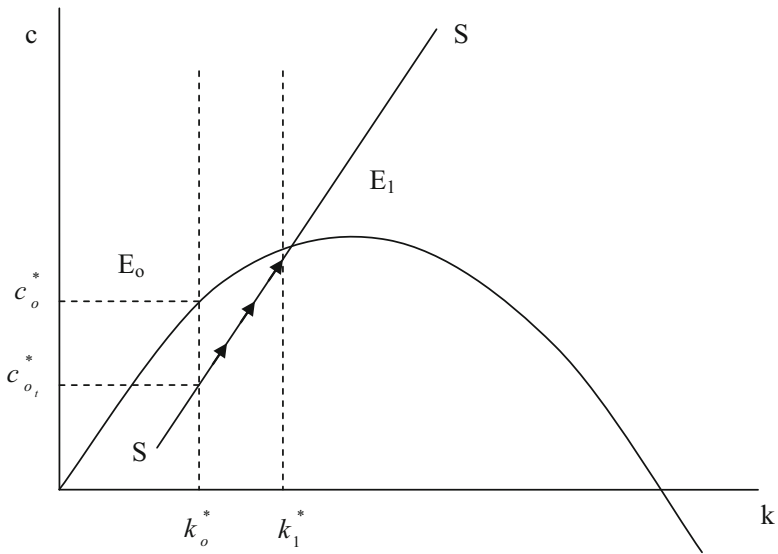


Fig. 5.5 The effects of an unanticipated permanent decrease in the rate of time preference

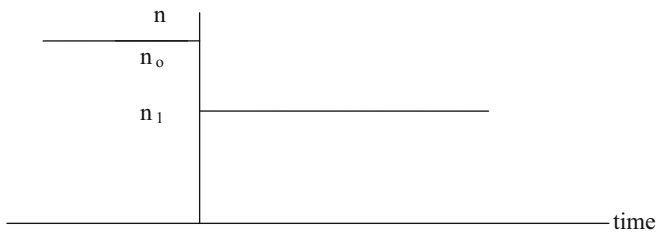


Fig. 5.6 An unanticipated permanent decrease in the population growth rate

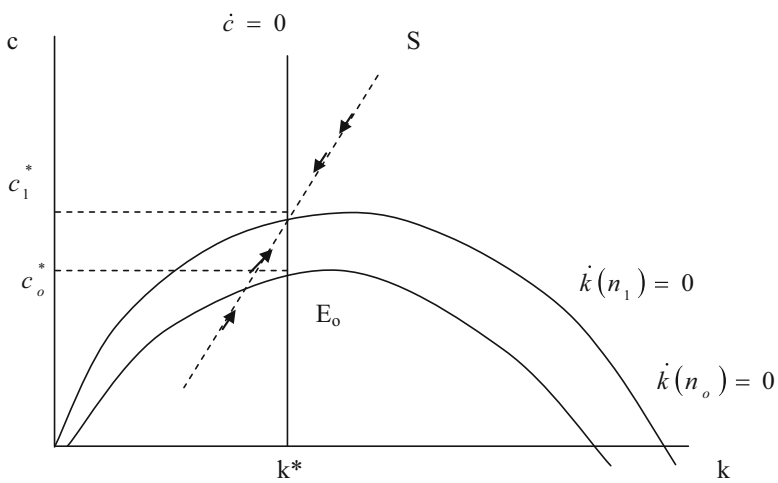


Fig. 5.7 The effects of an unanticipated permanent decrease in the population growth rate

5.2 Overlapping Generations Model

The infinite-life overlapping generations model (OLG) assumes that an individual born on date s has an infinite lifespan. A new generation is born at every moment. The population growth rate is equal to n . Each generation is born with no financial assets. On date t , the Euler equation for an agent born on date s equals the representative agent's Euler equation. That is:

$$\frac{\dot{C}(s, t)}{C(s, t)} = \sigma(r - \rho). \quad (5.11)$$

This economy's population on date t is equal to unity: $P(t) = 1$. The normalization simplifies the algebra. The population on date s was equal to: $P(s) = e^{-n(t-s)}$. To aggregate a variable $x(s, t)$ one must add together all generations existing on date t and born at some time in the past. At moment s the number of individuals born on s is equal to $nP(s)$. Therefore, the aggregate value X of variable $x(s, t)$ is given by:

$$X(t) = \int_{-\infty}^t nP(s)x(s, t)ds.$$

Substituting the value of $P(s)$ in this expression yields:

$$X(t) = \int_{-\infty}^t ne^{n(s-t)}x(s, t)ds. \quad (5.12)$$

This is the value per-capita of variable x because the population has been normalized at unity.

Consumption per-capita in this economy by individuals with infinite lives is therefore equal to:

$$C(t) = \int_{-\infty}^t ne^{n(s-t)}C(s, t)ds.$$

The derivative of consumption with respect to time is obtained via the Leibnitz rule:

$$\frac{dC}{dt} = ne^{n(t-t)}C(t, t) + \int_{-\infty}^t n \left[(-n)e^{n(s-t)}C(s, t) + e^{n(s-t)}\dot{C}(s, t) \right] ds,$$

which, taking into account the expression of $\dot{C}(s, t)$, may be written as:

$$\dot{C} = n \left(C(t, t) - C(t) \right) + \sigma(r - \rho)C. \quad (5.13)$$

This formula shows the consumption on date t of the generation born on t . Calculating this consumption requires using the individual's budget constraint.

The budget constraint, in terms of flows, of an individual born on date s at moment t has the change in wealth $\dot{a}(s, t)$ equal the difference between income from interest on financial assets and wages, and expenditures on consumer goods and services:

$$\dot{a}(s, t) = ra(s, t) + \omega(t) - C(s, \nu).$$

We assume the wage ω to be generation-independent. Resolving this differential equation yields the individual intertemporal constraint, assuming no Ponzi scheme. That is:

$$a(s, t) + \int_t^\infty e^{-r(v-t)} \omega(v) dv = \int_t^\infty e^{-r(v-t)} C(s, v) dv.$$

According to this constraint, the present value of an individual's expenditures equals the sum of the value of financial assets and the present value of wages, which we will represent by the letter h . Substituting the Euler equation for the rate of change of consumption in this constraint, we find that consumption is proportional to the individual's total wealth:

$$C(s, t) = \theta[a(s, t) + h(t)]. \quad (5.14)$$

Parameter θ is given by:

$$\theta = r + \sigma(\rho - r). \quad (5.15)$$

Consumption per-capita will also be proportional to total wealth:

$$C(t) = \theta[a(t) + h(t)].$$

The consumption of the generation born on date t , on date t , depends only on the present value of wages because it is assumed that $a(t, t) = 0$. Therefore:

$$C(t, t) = \theta[a(t, t) + h(t)] = \theta h(t).$$

It follows that:

$$C(t, t) - C(t) = -\theta a(t),$$

and the rate of change of consumption is:

$$\dot{C} = \sigma(r - \rho)C - n\theta a(t).$$

In this economy, financial assets are equal to the existing capital stock. Therefore, $a(t) = k$, and the rate of change of consumption depends on the capital-consumption ratio:

$$\frac{\dot{C}}{C} = \sigma(r - \rho) - n\theta \frac{k}{c}.$$

In the exogenous growth model, consumption c is measured in labor efficiency units. Its rate of change equals the difference between the rate of change of consumption per-capita and growth rate of technological progress:

$$\frac{\dot{c}}{c} = \widehat{C} - \widehat{A} = \sigma(r - \rho) - n\theta\frac{k}{c} - g.$$

Regrouping the first and third terms to the right of the second equals sign yields:

$$\frac{\dot{c}}{c} = \sigma\left(r - \rho - \frac{g}{\sigma}\right) - n\theta\frac{k}{c}.$$

The interest rate equals the net marginal product of capital, $r = f'(k) - \delta$. Therefore, the rate of change of consumption in units of labor efficiency is expressed as:

$$\frac{\dot{c}}{c} = \sigma\left(f'(k) - \delta - \rho - \frac{g}{\sigma}\right) - n\theta\frac{k}{c}. \quad (5.16)$$

Dynamical System

The infinite-life OLG exogenous growth model has the following differential equations system:

$$\begin{cases} \dot{k} = f(k) - c - (g + n + \delta)k \\ \frac{\dot{c}}{c} = \sigma\left(f'(k) - \delta - \rho - \frac{g}{\sigma}\right) - n\theta\frac{k}{c} \end{cases} \quad (5.17)$$

The first equation shows capital accumulation, which is the same as in other exogenous growth models. The second equation represents the consumption decision. The determinant of this system's Jacobian matrix is negative. The system therefore has a saddle point. The transversality condition of the optimization model for each generation's individual is that the present value of capital, valued at the marginal utility of consumption, must equal zero as time approaches infinity. That is:

$$\lim_{t \rightarrow \infty} \lambda(t)K(t)e^{-\rho t} = 0.$$

Capital per-capita in the steady state grows at a rate equal to g ; the marginal utility of consumption, also in the steady state, grows at a negative rate equal to g/σ . The transversality condition is therefore given by the following inequality:

$$\rho > g - \frac{g}{\sigma},$$

which is equivalent to:

$$\rho + \frac{g}{\sigma} > g. \quad (5.18)$$

At stationary equilibrium, the interest rate equals:

$$f'(\bar{k}) - \delta = \rho + \frac{g}{\sigma} + n\theta \frac{\bar{k}}{\bar{c}}.$$

The transversality condition does not ensure that this interest rate will be greater than the growth rate of real output ($g + n$). Therefore, under this model, one may have:

$$\rho + \frac{g}{\sigma} + n\theta \frac{\bar{k}}{\bar{c}} \leq g + n.$$

Should this inequality occur, the exogenous growth model with OLG is dynamically inefficient. This phenomenon of over-accumulation of capital makes the economy inefficient, as a reduction in the capital stock would increase the population's welfare.

Figure 5.8 shows the phase diagram of equation $\dot{c} = 0$:

$$c = \frac{\eta\theta k}{\sigma[f'(k) - \delta - \rho - \frac{g}{\sigma}]}.$$

In Fig. 5.8, this function's asymptote is the capital k^* corresponding to the interest rate:

$$f'(k^*) - \delta = \rho + \frac{g}{\sigma}.$$

The diagram was drawn assuming no dynamic inefficiency. Figure 5.9 shows the model's complete phase diagram. Whatever the economy's starting capital, the saddle path leads the economy to its steady state.

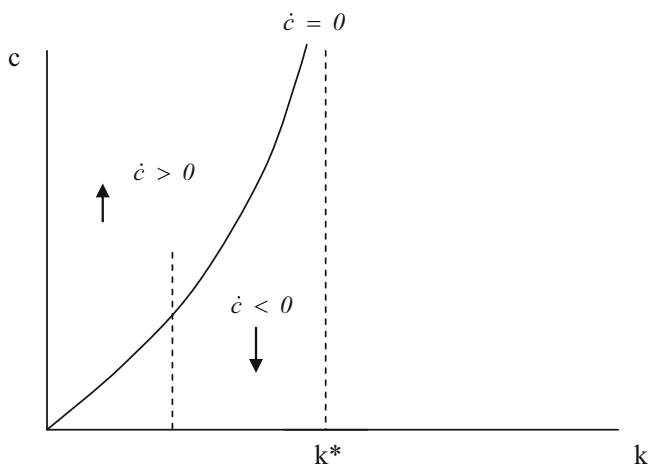


Fig. 5.8 The dynamics of c

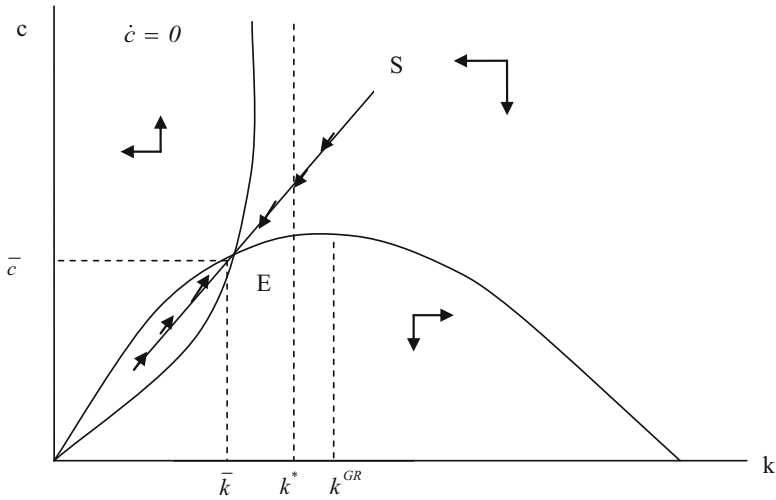


Fig. 5.9 $\rho + \frac{g}{\sigma} + n\theta\bar{k} > n + g$

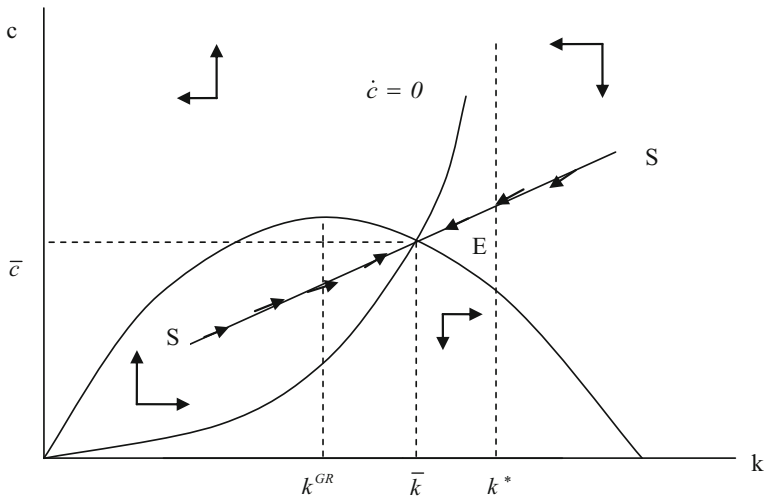


Fig. 5.10 $\rho + \frac{g}{\sigma} + n\theta\bar{k} < n + g$

Figure 5.10 shows the model's phase diagram in the presence of dynamic inefficiency. The interest rate is lower than the growth rate of real output. Once again, the saddle path brings the economy to the steady state, regardless of the starting capital.

Labor Productivity

The OLG model's growth rate of labor productivity has two components, as in the Solow and RCK models. The difference lies in the short-term component, as consumption is not proportional to income. Consumption and capital are related through the saddle path according to the function $c = c(k)$. The notation is the same as in the RCK model, but the functions are different, except where the population growth rate is equal to zero. The growth rate of labor productivity is therefore given by:

$$\hat{Y} - \hat{L} = \hat{A} + \alpha_k \left(\frac{f(k) - c(k)}{k} - (g + n + \delta) \right). \quad (5.19)$$

The two components of the growth rate of labor productivity are: (i) growth rate of technological progress; (ii) the product of the share of capital in output multiplied by the growth rate of capital stock per-capita, measured in labor efficiency units.

5.3 Endogenous Growth Models: An Introduction

The Solow, RCK, and OLG models assume competitive markets and that technological progress is exogenous. This assumption means that technological progress is a public service. Each business firm uses the technology without paying for it, and this does not affect its use by other firms. The factors of production – capital and labor – are paid according to their marginal products, and because the production function has constant returns to scale, payment for the factors absorbs the entire output. If business firms had to pay for their use of the technology, they would be unable to do so in this economic environment because payment for the factors of production exhausts total output.

Basic research in physics, chemistry, biology, and other scientific areas, which produces public goods, is generally carried out or financed by the state. Endogenous growth models assume that technology stems from Research and Development (R&D) investment by firms whose purpose is to make a profit. The patents system ensures that these innovations will not become public goods for a certain period of time. Firms obtain a temporary market power and cannot be treated as competitive. It follows that endogenous growth models must give up the assumption of competitive markets and embrace that of imperfectly competitive ones.

One possibility that first-generation economic growth models used was to assume that technology was transferred externally. In this case, firms benefit from the technology, but do not pay for its use. In such an economic environment, with competitive markets, the social decision differs from the private one, and it is the state's duty to correct the distortion.

The purpose of this introduction is to present the core of four endogenous growth models: the AK model, the human capital model, the varieties of inputs model, and the Schumpeterian creative destruction model. The following sections introduce

these models in greater detail. The central thread of this introduction is the Cobb-Douglas function:

$$Y = K^\alpha (EL)^{1-\alpha}, \quad (5.20)$$

where the notation is the same as in other chapters, except for the technological progress coefficient, which will be denoted by the letter E . The focus of attention for endogenous growth models is the modeling of the technological progress coefficient. Each of the models described above tells a different story for this coefficient.

The AK Model

One version of the AK model assumes that the technology coefficient is proportional to the quantity of capital per worker. That is:

$$E = \phi \frac{K}{L}. \quad (5.21)$$

Substituting this expression in the Cobb-Douglas function yields:

$$Y = K^\alpha \left(\phi \frac{K}{L} \right)^{1-\alpha} = \phi^{1-\alpha} K.$$

Therefore, in this model, output is a linear function of the capital stock:

$$Y = AK, A = \phi^{1-\alpha}. \quad (5.22)$$

The growth rate of output will be equal to the growth rate of capital stock.

Human Capital

In the human capital endogenous growth model, the technological progress coefficient is proportional to the worker's human capital. That is:

$$E = uh, \quad (5.23)$$

where the letter u stands for the time allocated to work. Investment in human capital is proportional to the stock of human capital and depends on the time spent acquiring this capital:

$$\dot{h} = \lambda(1 - u)h. \quad (5.24)$$

In the long term, the growth rate of labor productivity will be proportional to the time that the society allocates to building up its stock of human capital:

$$\frac{\dot{E}}{E} = \frac{\dot{h}}{h} = \lambda(1 - u). \quad (5.25)$$

Varieties of Inputs

In the varieties of inputs model, the production function depends on a variety of inputs according to:

$$Y = \left(\sum_{i=1}^N K_i^\alpha \right) L^{1-\alpha}, \quad (5.26)$$

where N is the number of inputs. Assume that the capital stock is evenly divided: $K_i = K/N$. Substituting this expression in the production function yields:

$$Y = K^\alpha (NL)^{1-\alpha}. \quad (5.27)$$

This model's technological progress coefficient is equal to the number of varieties. That is:

$$E = N. \quad (5.28)$$

The long-term growth rate of productivity depends on the growth rate of the number of varieties that results from R&D investment to create new varieties of inputs.

Schumpeterian Model

In the Schumpeterian model, each firm produces according to the following production function:

$$Y_{it} = A_{it} K_{it}^\alpha L_{it}^{1-\alpha}. \quad (5.29)$$

Total output is obtained by adding the output of every firm in the economy:

$$Y_t = \sum_{i=1}^N Y_{it}. \quad (5.30)$$

Assume that $K_{it} = K_t/N$ e $L_{it} = L_t/N$. It is easy to find that:

$$Y_t = \left(\frac{\sum_{i=1}^N A_{it}}{N} \right) K^\alpha L^{1-\alpha}.$$

The average coefficient of technological progress is:

$$A_t = \frac{\sum_{i=1}^N A_{it}}{N}. \quad (5.31)$$

The Schumpeterian model assumes that each firm invests in research to create new versions of existing products, so that the productivity coefficient of the new

versions is greater than that of old ones. The innovation flow is a random variable, but the average value of the rate of innovation depends on the probability μ that innovation will occur, according to:

$$\frac{\dot{A}_i}{A_i} = \mu(\gamma - 1), \gamma > 1. \quad (5.32)$$

The average value of the technology coefficient equals the average value of each firm. The probability of innovation depends on the resources R spent in research and development:

$$\mu = \lambda \frac{R}{A}. \quad (5.33)$$

Growth: Exogenous Versus Endogenous

Before introducing the endogenous models, it will be helpful to view the issue of exogenous versus endogenous growth from the point of view of the representative agent's Euler equation. That is, the growth rate of consumption per-capita is proportional to the difference between the interest rate and the rate of time preference:

$$\frac{\dot{c}}{c} = \sigma(r - \rho).$$

Using g to denote the growth rate of consumption per-capita, the real interest rate is:

$$r = \rho + \frac{1}{\sigma}g \quad (5.34)$$

In exogenous growth models, the growth rate of output per capita is equal to the growth rate consumption per-capita, an exogenous variable. Consequently, given g , the real interest rate is determined as Fig. 5.11a shows. In this figure, the positively sloped line EE is the Euler equation, and the vertical line CEx corresponds to exogenous growth.

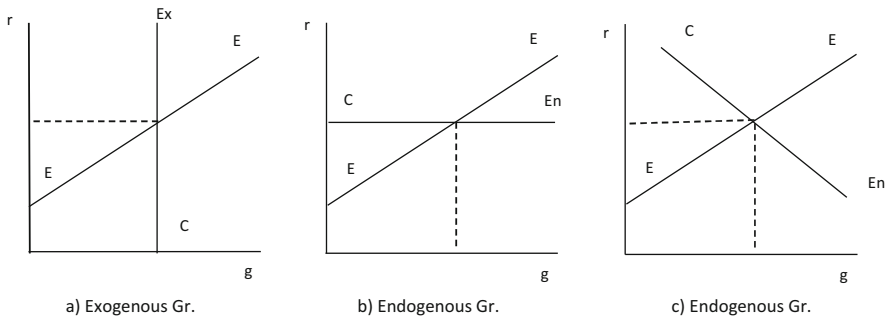


Fig. 5.11 (a) Exogenous growth. (b) Endogenous growth. (c) Endogenous growth

In the endogenous growth model, the real interest rate and the growth rate of output per capita are both endogenous variables. It should be pointed out that the technological innovations that increase the rate of return on capital, produce an increase in labor productivity, which benefits workers. In these models, capitalists and workers usually go hand-in-hand. In Fig. 5.11b the Euler equation is the same as in the previous case, and endogenous growth is given by the horizontal line C_{En} . The AK, open-economy AK, Lucas human capital model, and Romer varieties models, which we will discuss below, correspond to this case. In Fig. 5.11c the C_{En} endogenous growth line has a negative slope. This is precisely the shape of the Schumpeterian model discussed in the final section of this chapter.

5.4 The AK Model

In the Solow, RCK, and OLG exogenous growth models, the production function has constant returns to scale, but each factor's returns are decreasing. With capital accumulation, the marginal product of capital decreases and the economy converges to the steady state. In the exogenous economic growth model, the marginal product of labor does not decrease over time because of technological progress. Endogenous growth models make assumptions so that the factors' returns are not decreasing. This section introduces the AK model, where capital does not have decreasing returns.

In the AK endogenous growth model, the production function in the Solow model's differential equation, which we replicate for the sake of convenience,

$$\dot{k} = sf(k) - (n + \delta)k,$$

is replaced by the AK function, where capital is the only factor of production. The factor's symbol is the same, but its interpretation is different. It must include human capital besides physical capital. The new production function is specified by:

$$Y = AK.$$

The endogenous growth model's differential equation is therefore given by:

$$\dot{k} = sAk - (n + \delta)k \quad (5.35)$$

The rates of growth of the average productivity of labor and of the capital-labor ratio depend on the rate of saving s , on the technical coefficient A , on the population growth rate n , and on the depreciation rate δ , according to:

$$\frac{\dot{y}}{y} = \frac{\dot{k}}{k} = sA - (n + \delta).$$

This economy's growth rate of real output is:

$$\frac{\dot{Y}}{Y} = sA - \delta. \quad (5.36)$$

The growth rate of real output depends on the rate of saving (s), the marginal product of capital (A), and the capital depreciation rate (δ). The former two have a positive effect on the growth rate of output, while an increase (decrease) in the depreciation rate decreases (increases) the growth rate of real output.

The endogenous growth model does not differentiate between short and long term, that is, the economy does not converge to a certain income per-capita, as it would in the exogenous growth model. Differences in income per-capita and growth rate of output are persistent over time.

Labor Productivity

The AK model's growth rate of labor productivity is expressed by:

$$\hat{Y} - \hat{L} = sA - \delta - n \quad (5.37)$$

The growth rate of labor productivity depends on the saving rate (s), the technological coefficient (A), the depreciation rate (δ), and the population growth rate (n). Assume that $sA > \delta + n$. Under these circumstances, an increase in the saving rate permanently increases the growth rate of labor productivity.

In the representative agent model, the consumption Euler equation is:

$$\dot{c} = \sigma c(f'(k) - \delta - \rho).$$

The marginal product of capital equals the technical coefficient A :

$$f'(k) = A.$$

The AK model's dynamical system is therefore formed by the two differential equations:

$$\begin{cases} \dot{k} = Ak - c - (n + \delta)k \\ \dot{c} = \sigma c(A - \delta - \rho) \end{cases}. \quad (5.38)$$

Assume that $A - \delta - \rho > 0$, otherwise the growth rate of consumption would not be positive. In this system, the growth rate of consumption per-capita,

$$\frac{\dot{c}}{c} = \sigma(A - \delta - \rho),$$

must equal the rates of growth of the capital-labor ratio and of output per-capita:

$$\frac{\dot{k}}{k} = \frac{\dot{y}}{y} = \frac{\dot{c}}{c} = \sigma(A - \delta - \rho).$$

Therefore, it follows from the first differential equation that:

$$\frac{\dot{k}}{k} = A - \frac{c}{k} - (n + \delta)k = \sigma(A - \delta - \rho).$$

We conclude that:

$$\frac{c}{k} = A - (n + \delta) - \sigma(A - \delta) + \rho\sigma.$$

Rearranging the terms in this expression yields:

$$\frac{c}{k} = (A - \delta)(1 - \sigma) + \sigma\rho - n.$$

For the consumption-capital ratio to not be negative, the following inequality must be satisfied:

$$(A - \delta)(1 - \sigma) + \sigma\rho - n > 0.$$

Because $A - \rho - \delta > 0$, this and the preceding inequality produce the following constraints:

$$A > \rho + \delta > (\sigma - 1)(A - \delta - \rho) + n + \delta. \quad (5.39)$$

The growth rate of real output equals the previous one plus the population growth rate:

$$\hat{Y} = \frac{\dot{Y}}{Y} = \sigma(A - \delta - \rho) + n \quad (5.40)$$

In the AK growth model, the growth rate of real output increases (decreases) when: (i) the marginal product of capital increases (decreases); (ii) the capital depreciation rate decreases (increases); (iii) the rate of time preference decreases (increases); and (iv) the elasticity of substitution of consumption increases (decreases). The respective partial derivatives are as follows:

$$\begin{aligned} \frac{\partial \hat{Y}}{\partial A} &= \sigma > 0; \quad \frac{\partial \hat{Y}}{\partial \delta} = -\sigma < 0 \\ \frac{\partial \hat{Y}}{\partial \rho} &= -\sigma\rho < 0; \quad \frac{\partial \hat{Y}}{\partial \sigma} = A - \delta - \rho > 0 \end{aligned}$$

Labor Productivity

The AK endogenous growth model's growth rate of labor productivity is given by:

$$\hat{Y} - \hat{L} = \sigma(A - \delta - \rho) \quad (5.41)$$

The growth rate of labor productivity depends on four parameters, two (A, δ) represent technology and two (σ, ρ) consumer preferences. In the case of the latter two, the growth rate of labor productivity increases (decreases) when: (i) the elasticity of substitution increases (decreases), and (ii) the consumer's rate of time preference decreases (increases).

5.5 The Acemoglu-Ventura AK Model of an Open Economy

In the Acemoglu and Ventura AK model, the open economy produces two final goods/products: a consumer product and an investment product. These two final goods are not internationally traded. Nor do countries transact assets. There is a continuum of internationally traded intermediate goods made by competitive firms.

The representative agent maximizes:

$$\int_0^{\infty} e^{-\rho t} \log c \, dt, \quad (5.42)$$

subject to the following constraint:

$$\dot{k} = \frac{r}{p_I} k - \frac{p_c}{p_I} c. \quad (5.43)$$

In this economy, income derives solely and exclusively from capital, r is the rate of return on capital, the depreciation equals zero for the sake of simplicity, p_I is the price of the investment product, and p_c is the price of the consumer product. This problem's Hamiltonian is:

$$H = \log c + \lambda \left[\frac{r}{p_I} k - \frac{p_c}{p_I} c \right].$$

The first-order condition obtained by maximizing the Hamiltonian with respect to consumption is:

$$\frac{\partial H}{\partial c} = \frac{1}{c} - \lambda \frac{p_c}{p_I} = 0.$$

The shadow price of capital, the co-state variable, is obtained by:

$$\dot{\lambda} = \rho \lambda - \frac{\partial H}{\partial k} = \rho \lambda - \lambda \frac{r}{p_I}.$$

The transversality condition is:

$$\lim_{t \rightarrow \infty} \frac{p_I k}{p_c c} e^{-\rho t} = 0.$$

Combining the former two equations yields the Euler equation:

$$\frac{\dot{c}}{c} = \frac{r + \dot{p}_I}{p_I} - \frac{\dot{p}_c}{p_c} - \rho. \quad (5.44)$$

With a little algebra, we combine this equation with the budget stock constraint and the transversality condition to show that consumer spending is proportional to the value of wealth, according to:

$$p_c c = \rho p_I k. \quad (5.45)$$

Substituting this expression in the budget flow constraint yields the growth rate of capital stock:

$$\frac{\dot{k}}{k} = \frac{r}{p_I} - \rho. \quad (5.46)$$

Consumer and investment goods are produced using capital and intermediate inputs according to the production functions:

$$C = \chi K_C^{1-\tau} \left[\left(\int_0^M x(z)^{\frac{\varepsilon-1}{\varepsilon}} dz \right)^{\frac{\varepsilon}{\varepsilon-1}} \right]^\tau, \quad (5.47)$$

$$I = \xi^{-1} \chi K_I^{1-\tau} \left[\left(\int_0^M x(z)^{\frac{\varepsilon-1}{\varepsilon}} dz \right)^{\frac{\varepsilon}{\varepsilon-1}} \right]^\tau. \quad (5.48)$$

Production functions have constant returns to scale, ε is the elasticity of substitution between intermediate inputs, and the investment product's production function has the parameter ξ^{-1} to indicate that the economy is more efficient in the production of investment goods.

Production functions for two goods – consumer and investment – are of the Cobb-Douglas type. Therefore, the unit cost functions are also of the Cobb-Douglas type. That is:

$$B_C = r^{1-\tau} \left[\left(\int_0^M p(z)^{1-\varepsilon} dz \right)^{\frac{1}{1-\varepsilon}} \right]^\tau, \quad (5.49)$$

$$B_I = \varphi^{-1} r^{1-\tau} \left[\left(\int_0^M p(z)^{1-\varepsilon} dz \right)^{\frac{1}{1-\varepsilon}} \right]^\tau. \quad (5.50)$$

The price index of intermediate inputs will be adopted as numéraire. That is:

$$\left(\int_0^M p(z)^{1-\varepsilon} dz \right)^{\frac{1}{1-\varepsilon}} = 1. \quad (5.51)$$

The prices of the consumer and investment goods are equal to the respective unit production costs. It is easily seen that these prices are:

$$p_c = r^{1-\tau}, \quad (5.52)$$

$$p_I = \phi^{-1} r^{1-\tau}. \quad (5.53)$$

Intermediate goods are produced by competitive firms and one unit of capital produces one unit of the intermediate good. Therefore, the price p of the input equals the marginal cost of production, the cost of renting one unit of capital, or the interest rate. That is:

$$p = r.$$

This price is the price of this economy's intermediate input. The economy also imports a basket of inputs from other countries. Because this basket costs one unit, as a result of the normalized input price index, the price p measures the country's terms of trade.

In steady state, the economy's growth rate is equal to the growth rate of consumption and of capital, and equal to:

$$g = \frac{\dot{c}}{c} = \frac{\dot{k}}{k} = \phi p^\tau - \rho \quad (5.54)$$

The economy's growth rate in this open-economy model depends on the relative price p – the terms of trade – and on the rate of time preference. When the terms of trade increase (decrease), the economy's growth rate also increases (decreases). The rate of time preference, as in other endogenous growth models, is negatively correlated with the economy's growth rate.

5.6 The Lucas Human Capital Model

In the Lucas endogenous growth model with human capital, the production function depends on physical capital and on human capital according to:

$$Y = K^\alpha (u h L)^{1-\alpha}, \quad (5.55)$$

where the parameter that multiplies the quantity of labor now has a completely different specification. It equals the product of the share of time u that people allocate

to work multiplied by the quantity of human capital h that each worker has acquired in the process of investing part of their time, $1-u$, in human capital. The Lucas model's other assumption is that the growth rate of human capital is proportional to the time allocated to this investment:

$$\frac{\dot{h}}{h} = \lambda(1 - u). \quad (5.56)$$

The coefficient λ measures the productivity of education.

The production function may be written in the intensive form, dividing both sides by the quantity of labor:

$$y = k^\alpha (uh)^{1-\alpha}. \quad (5.57)$$

The representative agent maximizes the functional:

$$\int_0^\infty e^{-\rho t} u(c) dt, \quad (5.58)$$

subject to the following constraints:

$$\dot{k} = k^\alpha (uh)^{1-\alpha} - c, \quad (5.59)$$

$$\dot{h} = \lambda(1 - u)h. \quad (5.60)$$

The former corresponds to physical capital accumulation, and the latter to human capital. For simplicity, we assume that the rates of growth of the population and of capital depreciation are both zero. The starting values of physical capital and human capital are given.

This problem's current-value Hamiltonian is given by:

$$H = u(c) + \theta_1 \left[k^\alpha (uh)^{1-\alpha} - c \right] + \theta_2 [\lambda(1 - u)h].$$

The representative agent must choose the level of consumption and the number of working hours, maximizing the Hamiltonian. The problem's first-order conditions are:

$$\frac{\partial H}{\partial c} = u'(c) - \theta_1 = 0,$$

$$\frac{\partial H}{\partial u} = \theta_1 k^\alpha (1 - \alpha) (uh)^{-\alpha} h + \theta_2 \lambda h (-1) = 0.$$

The co-state variables θ_1 e θ_2 – respectively the shadow prices of physical capital and human capital – must attend the following conditions:

$$\begin{aligned}\dot{\theta}_1 &= \rho\theta_1 - \frac{\partial H}{\partial k} = \rho\theta_1 - \theta_1 \alpha k^{\alpha-1} (uh)^{1-\alpha}, \\ \dot{\theta}_2 &= \rho\theta_2 - \frac{\partial H}{\partial h} = \rho\theta_2 - \theta_1 k^\alpha (1-\alpha)(uh)^{-\alpha} u - \theta_2 \lambda (1-u).\end{aligned}$$

The first co-state variable equation may be written as:

$$\frac{\dot{\theta}_1}{\theta_1} = \rho - \alpha k^{\alpha-1} (uh)^{1-\alpha}.$$

The second co-state variable equation may be rewritten as:

$$\frac{\dot{\theta}_2}{\theta_2} = \rho - \frac{\theta_1}{\theta_2} k^\alpha (1-\alpha)(uh)^{-\alpha} u - \lambda (1-u).$$

The first-order condition for the number of working hours establishes the following relationship between the co-state variables:

$$\theta_1 (1-\alpha) k^\alpha (uh)^{-\alpha} = \theta_2 \lambda.$$

Substituting this expression in the previous one yields:

$$\frac{\dot{\theta}_2}{\theta_2} = \rho - \lambda.$$

The equation that relates the two co-state variables may also be written as:

$$\theta_1 (1-\alpha) y = \theta_2 \lambda u h.$$

In the long term, labor productivity grows at a rate equal to the growth rate of the stock of human capital. Therefore, the rates of growth of the co-state variables are the same:

$$\frac{\dot{\theta}_1}{\theta_1} = \frac{\dot{\theta}_2}{\theta_2} = \rho - \lambda.$$

Assume that the utility function has constant elasticity of substitution. That is:

$$u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}.$$

According to the first-order condition for consumption, the growth rate of consumption is equal to the co-state variable's rate of growth with a minus sign. That is:

$$\frac{\dot{c}}{c} = -\frac{\dot{\theta}_1}{\theta_1} = \sigma(\lambda - \rho). \quad (5.61)$$

Therefore, this model's real interest rate in stationary equilibrium equals the proportionality parameter of the equation for investment in human capital,

$$r^* = \lambda. \quad (5.62)$$

The Euler equation then enables the growth rate of labor productivity to be:

$$g = \sigma(\lambda - \rho). \quad (5.63)$$

The growth rate of labor productivity therefore depends on the intertemporal elasticity of substitution of consumption, on the productivity of education, and on the rate of time preference. It is positively correlated with the elasticity of substitution and with the productivity coefficient, and negatively correlated with the rate of time preference.

5.7 Romer's Varieties of Inputs Model

In Romer's varieties of inputs model, the consumer product is produced in a competitive market with technology given by the production function:

$$Y = \frac{1}{\alpha} \left(\sum_{i=1}^N X_i^\alpha \right) L^{1-\alpha}, \quad (5.64)$$

where X_i is the quantity of the input, L is the quantity of labor, N is the number of varieties of inputs, and the inverse of coefficient α (the first term in the production function) has the sole purpose of simplifying the formulas. The prices of inputs P_i and wage ω are given. The price of the final product is equal to unity, for normalization purposes. The firm selects the quantities of input in such a way as to maximize profit:

$$\Pi = Y - \sum_{i=1}^N P_i X_i - \omega L. \quad (5.65)$$

The problem's first-order conditions are:

$$(1 - \alpha) \frac{Y}{L} = \omega, \quad (5.66)$$

$$\left(\frac{X_i}{L} \right)^{\alpha-1} = P_i. \quad (5.67)$$

Inputs are produced by monopolist firms and each firm maximizes its profit. The i th firm maximizes:

$$\pi_i = P_i X_i - c X_i.$$

By taking into account the demand equation for the input it sells, and the fact that marginal revenue is equal to marginal cost c when profit is maximum, the price is given by:

$$P_i = \frac{c_i}{\alpha}.$$

The profit of the monopolist firm can be easily obtained with a little algebra. That is:

$$\pi_i = \kappa L,$$

where:

$$\kappa = \frac{1 - \alpha}{\alpha} \alpha^{\frac{1}{1-\alpha}} c^{\frac{-\alpha}{1-\alpha}}.$$

The present value of the monopolist firm's profit emerges from discounting the profits flow by the interest rate according to:

$$V_i = \int_0^{\infty} e^{-rt} \pi_i dt = \frac{\pi_i}{r} \quad (5.68)$$

Assume that the production cost of a new variety of input is equal to η . It follows that, in equilibrium, a new variety will be produced as long as the present value of profits is equal to the cost of innovation. That is:

$$V_i = \frac{\pi_i}{r} = \eta. \quad (5.69)$$

Because all monopolist firms are equal, we find that the equilibrium interest rate is given by:

$$r^* = \frac{\pi}{\eta} = \frac{\kappa L}{\eta}. \quad (5.70)$$

According to the representative agent's Euler equation, the growth rate of labor productivity is:

$$g = \sigma(r^* - \rho) = \sigma \left(\frac{\kappa L}{\eta} - \rho \right). \quad (5.71)$$

The growth rate of labor productivity depends on the intertemporal elasticity of substitution of consumption, on technology, on the market's size, on the cost of innovation, and on the rate of time preference. The growth rate of labor productivity is positively correlated with the elasticity of substitution of consumption and with the size of the market (L). It is negatively correlated with the cost of innovation and with the rate of time preference.

5.8 The Aghion and Howit's Schumpeterian Model

According to the Schumpeterian model, the consumer product is produced using machinery and labor according to the production function:

$$Y = \frac{1}{\alpha} \left(\int_0^1 \left(q(v,t)x(v,t) \right)^\alpha dv \right) L^{1-\alpha}, \quad (5.72)$$

where $q(v,t)$ is the quality of machine v at time t , $x(v,t)$ is the quantity of type- v machines at time t and of the quality indicated by q , L is the quantity of labor, and the first term in the expression is a normalization coefficient intended to simplify the algebra. A continuum of machines exists represented in the interval $[0,1]$.

In the Schumpeterian model, economic growth is due to the technological innovations that improve the quality of machines, rendering those that use old technology obsolete. The quality of each machine varies according to:

$$q(v,t) = \lambda^{n(v,t)} q(v,0), \quad (5.73)$$

where $n(v,t)$ is the number of innovations to the machine in time interval t , starting from the initial period, and the parameter $\lambda > 1$ measures the increase in the machine's quality by virtue of the technological innovation.

Machines are produced by monopolist firms that invest in research and development and hold the patent to their technological innovation. Demand for machines is easily deduced from the production function, which has the same shape as a Cobb-Douglas function. The demand for machines is:

$$x(v,t) = \left(\frac{q(v,t)}{p^x(v,t)} \right)^{\frac{1}{1-\alpha}} L. \quad (5.74)$$

Given this isoelastic demand equation, the monopolist firm maximizes its profit when marginal revenue is equal to the marginal cost of production ϕ . That is:

$$p^x(v,t) = \frac{\phi}{\alpha} q(v,t) = q(v,t).$$

The expression after the second equals sign was obtained by normalizing $\phi = \alpha$ to simplify the algebra. Combining the two previous equations, the quantity of machines produced equals the quantity of labor:

$$x(v,t) = L.$$

The monopolist firm's profit is therefore equal to:

$$\pi(v,t) = (1 - \alpha)q(v,t)L.$$

Substituting the quantity of machines in the production function, the economy's production of final goods is proportional to the quantity of labor:

$$Y = \frac{1}{\alpha}QL, \quad (5.75)$$

where Q measures the average quality of the machines,

$$Q = \int_0^1 q(v, t) dv.$$

To be attractive, the monopolist's business must produce a rate of return equal to the market interest rate. Therefore, by arbitrage:

$$r(t) = \frac{\pi(v, t)}{V(v, t)} + \frac{\dot{V}(v, t)}{V(v, t)} - z(v, t). \quad (5.76)$$

The capital V represents the market value of the monopolist firm. The first term in this expression is the rate of profit, the second is the capital gain or loss when the value of the firm changes over time. The third term represents the essence of the Schumpeterian model, and measures the rate of new innovations that took place in industry v in time t .

In the steady state, the value of the firm will be equal to its profit divided by the discount rate:

$$V(v, t) = \frac{(1 - \alpha)q(v, t)L}{r + z}.$$

This economy has no entry barriers against new machine-producing firms. Each machine's production cost is represented by the Greek letter η . Therefore, the following equilibrium condition must be fulfilled:

$$\eta V(v, t) = \lambda^{-1}q(v, t).$$

Combining the latter two expressions yields:

$$r + z = \lambda\eta(1 - \alpha)L.$$

This economy's output is proportional to the quantity of labor and the proportionality coefficient is the average quality of the machines. It follows that the growth rate of output is equal to the growth rate of the quality of the machines. That is:

$$\frac{\dot{Y}}{Y} = \frac{\dot{Q}}{Q}.$$

The growth rate of the quality of the machines is proportional to the rate of new innovations, according to:

$$\frac{dq(v, t)}{dt} \frac{1}{q(v, t)} = (\lambda - 1)z(v, t).$$

By taking the derivative of:

$$Q = \int_0^1 q(v, t) dv,$$

with respect to time, and taking the preceding equation into account, yields:

$$g = \frac{\dot{Y}}{Y} = (\lambda - 1)z.$$

Substituting the value of z given in this expression in the equation of the monopolist's discount rate ($r + z$) yields the equation for the economy's endogenous growth, establishing a relationship between the interest rate and the economy's growth rate:

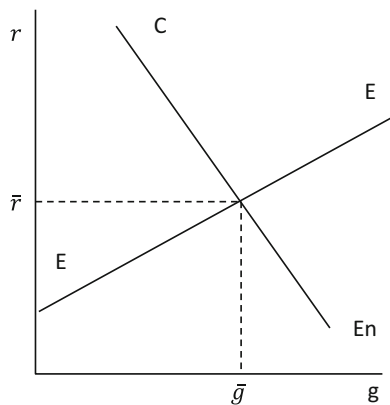
$$r = \lambda\eta(1 - \alpha)L - \frac{g}{\lambda - 1}. \quad (5.7)$$

The model's second equation is the Euler equation that establishes a relationship between the interest rate and the growth rate of labor productivity. That is:

$$r = \rho + \frac{1}{\sigma}g. \quad (5.78)$$

Figure 5.12, with the interest rate on the vertical axis and the growth rate of labor productivity on the horizontal axis, shows the Schumpeterian model's two curves. The positively sloped curve is the Euler equation and the negatively sloped one is the endogenous growth equation. The point where the two cross is the equilibrium point, with the interest rate and the growth rate of labor productivity given by:

Fig. 5.12 The Schumpeterian model cross diagram: the Euler equation curve and the endogenous growth equation curve



$$r = \rho + \frac{\lambda - 1}{\lambda - 1 + \sigma} [\lambda \eta (1 - \alpha) L - \rho], \quad (5.79)$$

$$g = \frac{\sigma(\lambda - 1)}{\lambda - 1 + \sigma} [\lambda \eta (1 - \alpha) L - \rho]. \quad (5.80)$$

5.9 Exercises

1. Show that, in the RCK model:

(a) The saving (investment) rate at stationary equilibrium is given by:

$$s^* = \left(\frac{I}{Y} \right)^* = \frac{\alpha(g + n + \delta)}{\rho + \delta + \frac{1}{\sigma}g},$$

where α is the coefficient of the Cobb-Douglas production function $f(k) = k^\alpha$ and the other symbols have the same meanings used throughout this chapter. The saving rate therefore depends on the parameters for technology (g , α , δ), demography (n), and preference (ρ , σ).

(b) the stationary equilibrium saving rate is lower than the share of capital in output: $s^* < \alpha$. [Hint: Use the model's transversality condition].

2. Show that, in the Solow, RCK, and OLG models, the saving (investment) rate s , the capital to output ratio (ν), the growth rate of output ($g + n$), and the depreciation rate (δ) are connected by the following equation in the long term:

$$\frac{s}{\nu} = g + n + \delta.$$

3. The representative agent maximizes:

$$\int_0^\infty e^{-(\rho-n)t} u(c, l) dt,$$

subject to the following constraints:

$$\dot{k} = k^\alpha (uh)^{1-\alpha} - c - (n + \delta)k,$$

$$\dot{h} = \lambda t_h h,$$

$$l + u + t_h = 1.$$

The symbols have the following meanings: l leisure, u time devoted to work, t_h time allocated to investment in human capital, h stock of human capital, c consumption, and k physical capital. The other symbols are positive parameters.

The initial stocks of physical and human capital are given. The utility function, which depends on consumption and leisure, is:

$$u(c, l) = \frac{c^{1-\frac{1}{\sigma}} + \phi l^{1-\frac{1}{\sigma}}}{1 - \frac{1}{\sigma}},$$

where σ is the intertemporal elasticity of consumption.

- (a) Write down the first-order conditions for this problem's solution.
- (b) What is the long-term growth rate of this economy's output?
- (c) What would the answer to the previous item be if the parameter ϕ equaled zero?

4. The representative agent maximizes:

$$\int_0^{\infty} e^{-(\rho-n)t} u(c) dt,$$

subject to the constraint:

$$\dot{k} = (k^\alpha + ak) - c - (n + \delta)k.$$

The symbols have the meanings used throughout this chapter and the utility function is:

$$u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1 - \frac{1}{\sigma}}.$$

- (a) Write down the first-order conditions for this problem's solution.
- (b) What is the long-term growth rate of this economy's output?
- (c) What would the answer to the previous item be if the parameter a equaled zero?
- (d) Calculate the share of labor in output for $k \rightarrow \infty$.

5. The representative agent maximizes:

$$\int_0^{\infty} e^{-\rho t} u(c) dt,$$

subject to the constraints:

$$\dot{k} = k^\alpha (uh)^{1-\alpha} \bar{h}^\gamma - c,$$

$$\dot{h} = \lambda(1 - u)h.$$

The first constraint corresponds to physical capital accumulation and the latter to human capital accumulation. Human capital produces an externality

represented by the average value of human capital $\bar{h}$ in the production function. The agent takes this value as a given and at equilibrium it equals the value of the economy. The starting values of physical capital and human capital are given.

- (a) Write down the problem's first-order conditions and the equations for the co-state variables.
- (b) What is this economy's growth rate of labor productivity?
- (c) What is the difference between the rates of growth of private and social labor productivity in this economy?
- (d) How might the government solve the previous item's distortion?

Part II

Sticky Price Models

Chapter 6

Keynesian Models: The IS and LM Curves, the Taylor Rule, and the Phillips Curve



This chapter presents three equations from Keynesian models: (i) the IS curve, which relates to the real interest rate and real output; (ii) the LM curve, which relates to the nominal interest rate and the quantity of money; (iii) and the Phillips curve, which relates to the unemployment rate (or output gap) and the inflation rate. Each of these equations is specified from two approaches: from the traditional Keynesian viewpoint, the equations are grounded on behavioral rules; from the new Keynesian approach, the specifications are based on microeconomics. The two approaches produce not only distinct specifications, but also different predictions that can be tested empirically.

The first two sections cover the IS curve. Section 6.1 introduces the traditional IS curve, while Sect. 6.2 presents the new Keynesian IS curve. Section 6.3 presents the concept of the natural interest rate according to the two IS curves. Sections 6.4 and 6.5 address the LM curve, the former using a behavioral model, the latter using microfounded ones. The LM curve stopped being used in short-term models in the 1990s for both Keynesian and new Keynesian models. It was replaced with a monetary policy rule—the Taylor rule—for the Central Bank’s nominal interest rate in the market for bank reserves. Section 6.6 of this chapter deals with this topic. Sections 6.7 and 6.8, respectively, show the traditional and microfounded Phillips curve models. In the former, the inflation rate is dependent on the past and is backward-looking. In the latter section, the inflation rate depends on the future and is forward-looking.

6.1 The Keynesian IS Curve

Expenditures in the goods and services market can be broken down into three components: consumption (c), investment (i), and government spending (g). Consumption depends on disposable income, which is obtained by subtracting total taxes (τ) from income (y). Marginal propensity to consume, $c' = dc/dy$, lies

between zero and one, $0 < c' < 1$. Investment depends on the internal rate of return, which is the marginal efficiency of capital, as Keynes puts it, and on the real interest rate expected by the market. The expected real interest rate equals the difference between the nominal interest rate (i) and the expected inflation rate (π^e). For a given internal rate of return, the higher (lower) the expected real interest rate, the lesser (greater) investment will be, that is, the derivative of investment with respect to the real interest rate ($\iota = di/dr$) is negative. Government spending is exogenous to the model. Expenditure in this economy is therefore equal to:

$$d = c(y - \tau) + \iota(i - \pi^e) + g. \quad (6.1)$$

The goods and services market is at equilibrium when expenditure equals output:

$$y = d. \quad (6.2)$$

Combining the two equations yields:

$$y = c(y - \tau) + \iota(r^e) + g. \quad (6.3)$$

The expected real interest rate, that is, the one predicted by the market, is defined by:

$$r^e = i - \pi^e. \quad (6.4)$$

The expected real interest rate is not an observable variable and some assumption has to be made as to how it relates to observable variables in the economy. Because this model has no uncertainty, since the variables are deterministic, let us assume that the prediction is perfect. That is, the predicted real interest rate equals the observed rate:

$$r^e = r. \quad (6.5)$$

The goods and services is in market equilibrium when:

$$y = c(y - \tau) + \iota(r) + g. \quad (6.6)$$

This equation corresponds to the IS curve. Figure 6.1 represents this curve on a plane where the horizontal axis measures real output and the vertical axis measures the real interest rate. The curve has a negative slope because if the real interest rate increases (decreases), then real output has to decrease (increase) to keep the goods and services market at equilibrium. When output equals full-employment output, the real interest rate is the long-term equilibrium real interest rate ($\bar{r}$), the natural interest rate. This rate depends on the government's fiscal policy and is affected by both spending and taxes.

The name of the IS curve comes from the fact that the goods and services market equilibrium is tantamount to equality between saving and investment. By subtracting

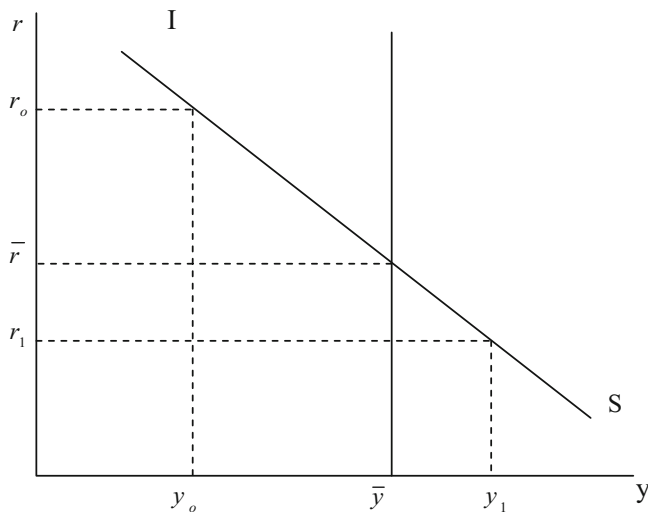


Fig. 6.1 The IS curve

the government's total tax revenue from both sides of the goods and services market equilibrium equation, this yields:

$$s = y - c(y - \tau) - \tau = \iota(r) + g - \tau,$$

or:

$$s(y - \tau) = \iota(r) + f. \quad (6.7)$$

When the economy is at full employment, saving has a constant value, as Fig. 6.2 shows. Investment varies in the opposite direction of the real interest rate. Investment plus public deficit corresponds to the IS curve in Fig. 6.2. The point where the vertical saving line intercepts the IS curve determines the real long-term interest rate, the natural interest rate.

Fiscal policy may vary with the cycle. When the economy is at full employment, the IS curve's equation is:

$$\bar{y} = c(\bar{y} - \bar{\tau}) + \iota(\bar{r}) + \bar{g}. \quad (6.8)$$

The variables with bars represent their values at full employment. The IS curve's equation may be written in terms of the variables' deviation from their full-employment values. Subtracting the previous equation from the IS curve equation yields:

$$y - \bar{y} = c(y - \tau) - c(\bar{y} - \bar{\tau}) + \iota(r) - \iota(\bar{r}) + g - \bar{g}.$$

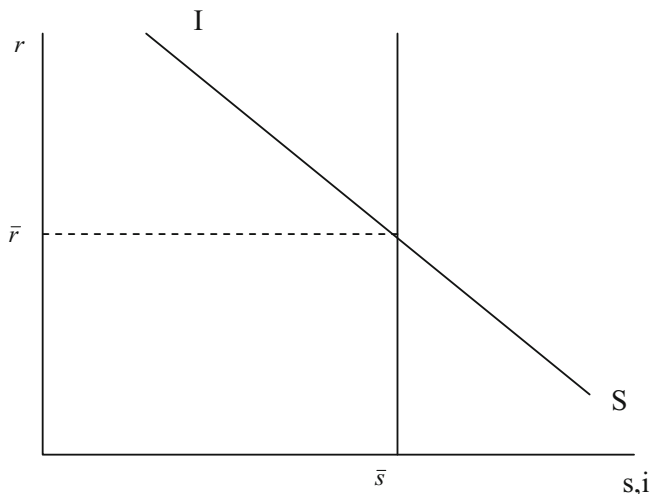


Fig. 6.2 The natural rate of interest

6.1.1 Algebra

The Taylor first-order expansions $[f(x) = f(x_0) + f'(x_0)(x - x_0)]$ of the consumption and investment functions around full employment are:

$$\begin{aligned} c(y - \tau) &= c(\bar{y} - \bar{\tau}) + c'[(y - \bar{y}) - (\tau - \bar{\tau})], \\ i(r) &= i(\bar{r}) + i'(r - \bar{r}). \end{aligned}$$

The derivatives of the consumption and investment functions, c' and i' , are valued at the full-employment point. Substituting these expressions in the IS curve yields:

$$y - \bar{y} = c'[(y - \bar{y}) - (\tau - \bar{\tau})] + i'(r - \bar{r}) + g - \bar{g}.$$

This equation may be written as:

$$y - \bar{y} = -\frac{c'}{1 - c'}(\tau - \bar{\tau}) + \frac{i'}{1 - c'}(r - \bar{r}) + \frac{1}{1 - c'}(g - \bar{g}).$$

The difference between real and potential output depends on the cyclical variations of taxes, the real interest rate, and government spending. The fiscal policy is represented by two variables – taxes and government spending – with distinct coefficients, because they have different effects on expenditure. One additional dollar of government spending initially increases expenditure by \$1, while \$1 less in taxes does not initially increase private-sector consumption by \$1 because this depends on the amount that the consumer will choose to save. In the extreme case where the entire tax reduction is saved, expenditure remains unchanged. The IS

curve may be written as function of public deficit, which is defined by subtracting total taxes from government spending:

$$f = g - \tau. \quad (6.9)$$

Full-employment public deficit is similarly defined:

$$\bar{f} = \bar{g} - \bar{\tau}. \quad (6.10)$$

The cyclical variation of public deficit is obtained by subtracting the full-employment public deficit from the current public deficit. That is:

$$f - \bar{f} = g - \bar{g} - (\tau - \bar{\tau}).$$

The IS curve, through simple algebraic manipulation, that is, adding and subtracting $c'(g - \bar{g})/(1 - c')$, may be written as:

$$y - \bar{y} = \frac{c'}{1 - c'} [(g - \tau) - \bar{g} - \bar{\tau}] + \frac{i'}{1 - c'} (r - \bar{r}) + \frac{1 - c'}{1 - c'} (g - \bar{g}),$$

or:

$$y - \bar{y} = \frac{c'}{1 - c'} (f - \bar{f}) + \frac{i'}{1 - c'} (r - \bar{r}) + g - \bar{g}.$$

A functional form that enables a more intuitive interpretation of the IS curve's parameters can be obtained by dividing both sides of the previous equation by potential output,

$$\frac{y - \bar{y}}{\bar{y}} = \frac{i'}{1 - c'} \left(\frac{1}{\bar{y}} \right) (r - \bar{r}) + \frac{c'}{1 - c'} \left(\frac{f}{\bar{y}} - \frac{\bar{f}}{\bar{y}} \right) + \frac{g}{\bar{y}} - \frac{\bar{g}}{\bar{y}}.$$

The fiscal variables are measured as shares of potential output. The left side of this expression is the output gap,

$$\frac{y - \bar{y}}{\bar{y}} \cong \log \left(1 + \frac{y - \bar{y}}{\bar{y}} \right) = \log \frac{y}{\bar{y}} = \log y - \log \bar{y}.$$

The interpretation of the equation's coefficients becomes simpler when the output gap is multiplied by 100, that is, $100 \log(y/\bar{y})$. The gap corresponds to a percentage. A similar procedure applies to all other variables in the equation.

The IS Curve Equation

Denoting the output gap as $x = y - \bar{y}$, the real interest rate's coefficient as $-\alpha$, and the public deficit's coefficient as β , the IS curve is specified as follows:

$$x = -\alpha(r - \bar{r}) + \beta(f - \bar{f}) + g - \bar{g}. \quad (6.11)$$

The variables in this IS curve use the same symbols as in its deduction, but the interpretation is different: (i) f is the public deficit as a share of potential output and $\bar{f}$ is the full-employment public deficit, also as a share of potential output; (ii) g is government spending and $\bar{g}$ is full-employment government spending, both as a share of potential output. The α parameter measures the effect on the output gap of sustained variation in the real interest rate relative to the natural interest rate. For example, if α equals two for each 1% increase in the real interest rate relative to the natural interest rate, the economy's idle capacity will increase by 2%. The β coefficient measures the effect of the variation of public deficit on the output gap. In the presence of Ricardian equivalence, this coefficient equals zero, as society reacts to the public deficit by increasing savings by the same amount in order to pay the taxes that will finance the deficit.

This IS curve specification enables simple and transparent analysis of the reasons that may lead the economy to show unemployment and idle capacity. The economy may find itself in this situation when at least one of the following occurs: (i) the real interest rate is different from the natural interest rate; (ii) public deficit is different to full-employment public deficit; and (iii) government spending differs from full-employment government spending. The latter two are caused by the fiscal policy. The real interest rate may differ from the natural interest rate for two reasons. One is a result of a monetary policy that may be contractionary (raising the interest rate) or expansionary (reducing the interest rate). The other is a change in the natural interest rate. This rate may change due to the government's fiscal policy or from the private sector's consumption and/or investment behavior.

6.2 The New Keynesian IS Curve

The new Keynesian IS curve is deduced from the problem of the intertemporal allocation of consumption for a representative agent. This section first characterizes consumers' preferences; then, it derives the first-order condition of the consumer optimization problem, the Euler equation.

6.2.1 Consumer Preferences

Figure 6.3 shows the consumer's utility curve, with consumption in period t measured on the horizontal axis and consumption in period $t + 1$ on the vertical axis. The marginal rate of substitution between consumption in the two periods is a point of tangency on the utility curve, or the derivative of consumption in period $t + 1$ with respect to consumption in period t , with its sign inverted, along a preference curve (at a constant utility level). This marginal rate (τ) equals the ratio of the two marginal utilities:

$$\tau = \frac{dc_{t+1}}{dc_t} = \frac{\partial u / \partial c_t}{\partial u / \partial c_{t+1}}. \quad (6.12)$$

Assume that the utility function has the following functional form:

$$u(c_t, c_{t+1}) = \frac{c_t^{1-\frac{1}{\sigma}} - 1}{1 - \frac{1}{\sigma}} + \frac{1}{1+\rho} \frac{c_{t+1}^{1-\frac{1}{\sigma}} - 1}{1 - \frac{1}{\sigma}}, \quad (6.13)$$

where σ is a parameter not equal to 1. The marginal utilities of consumption now (t) and tomorrow ($t+1$) are:

$$\frac{\partial u}{\partial c_t} = c_t^{-\frac{1}{\sigma}}; \quad \frac{\partial u}{\partial c_{t+1}} = \frac{1}{1+\rho} c_{t+1}^{-\frac{1}{\sigma}}$$

The marginal rate of substitution is then equal to:

$$\tau = \frac{c_t^{-\frac{1}{\sigma}}}{(1+\rho)^{-1} c_{t+1}^{-\frac{1}{\sigma}}} = \frac{1}{(1+\rho)^{-1}} \left(\frac{c_t}{c_{t+1}} \right)^{-\frac{1}{\sigma}} \quad (6.14)$$

When $c_t = c_{t+1} = c$, the marginal rate of substitution is equal to unity plus the ρ parameter:

$$\tau = 1 + \rho. \quad (6.15)$$

The rate of time preference is one of the parameters that characterize consumer preferences. This rate is the interest rate that would induce a consumer to maintain a constant level of consumption throughout life. That is, the rate of time preference is the rate of return on consumption. In Fig. 6.3, this corresponds to the tangent of the utility curve at the point where the 45° line from the origin intercepts the utility curve.

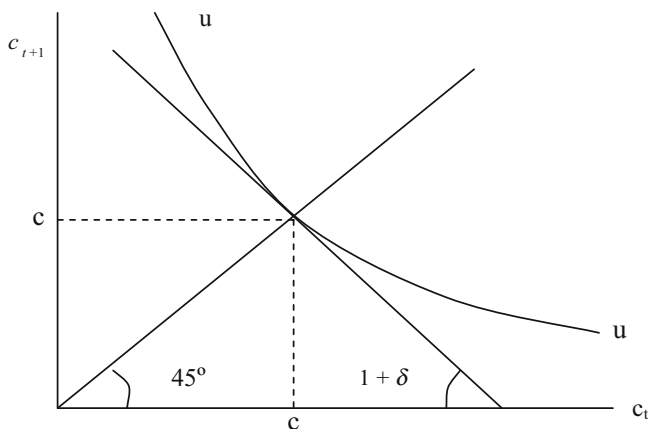


Fig. 6.3 The consumption choices and the rate of time preference

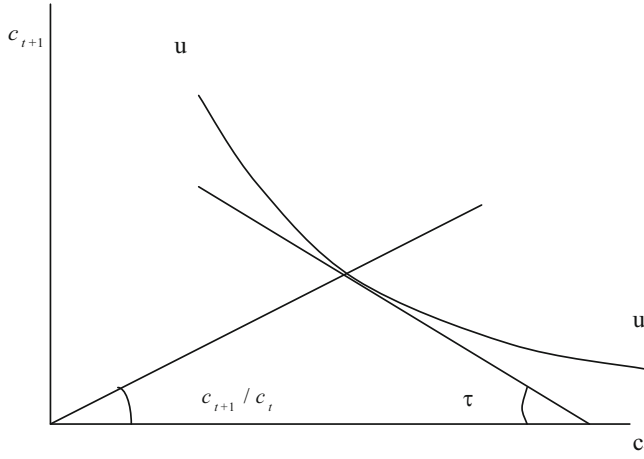


Fig. 6.4 The consumption choices and the elasticity of substitution

A second parameter that characterizes consumer preferences is the curvature of the utility function. The elasticity of substitution measures this curvature. Figure 6.4 illustrates its geometrical interpretation. The elasticity of substitution measures the response of the percentage change in the ratio of consumption tomorrow ($t + 1$) to consumption today (t) to a percentage change in the marginal rate of substitution. That is, the elasticity of substitution measures the relationship between a variation of the slope of the line connecting a point on the utility curve to the origin and the variation of the utility curve's tangent. Analytically, the elasticity of substitution is defined by:

$$\varepsilon_s = \frac{\Delta(c_{t+1}/c_t)}{\frac{c_{t+1}/c_t}{\Delta\tau/\tau}}. \quad (6.16)$$

Therefore, the elasticity of substitution in terms of its log derivative is:

$$\varepsilon_s = \frac{d \log(c_{t+1}/c_t)}{d \log \tau}.$$

The marginal rate of substitution for the previous utility function is:

$$\tau = (1 + \rho) \left(\frac{c_{t+1}}{c_t} \right)^{\frac{1}{\sigma}}.$$

This equation may be rewritten as:

$$\frac{c_{t+1}}{c_t} = \left(\frac{\tau}{1 + \rho} \right)^{\sigma}.$$

Taking the log of both sides of this expression yields:

$$\log(c_{t+1}/c_t) = \sigma \log \tau - \sigma \log(1 + \rho). \quad (6.17)$$

Therefore, the elasticity of substitution is equal to the σ parameter:

$$\varepsilon_s = \frac{d \log(c_{t+1}/c_t)}{d \log \tau} = \sigma. \quad (6.18)$$

6.2.2 Consumer Equilibrium: The Euler Equation

Imagine a consumer who must choose between spending \$1 in consumption today (t) or tomorrow ($t+1$). Should they choose immediate consumption, their welfare will show an increase equal to the marginal utility of consumption today. Should they choose to consume tomorrow, they will invest one dollar in a financial asset that will yield an interest rate equal to r , and spend the investment's principal and interest earned in a subsequent period. Their welfare tomorrow will show an increase equal to the marginal utility of consumption. However, in order to compare with today's welfare, they must discount tomorrow's welfare at the rate of time preference. At equilibrium, they will be indifferent between the two choices:

$$u'(c_t) = (1 + r_t) \frac{1}{1 + \rho} u'(c_{t+1}). \quad (6.19)$$

This equilibrium equation is named the Euler equation. The name comes from the first-order condition of the dynamical optimization problem for the calculus of variations. According to this condition, the consumer will invest their resources so that \$1 will have the same value in terms of welfare at every point along their lifetime. The basic rationale behind this equation is that the individual spaces out consumption along their life. They prefer to have a middle-class lifestyle throughout life than to live like a millionaire for a few days and spend the rest of their days in poverty.

When the utility function has the functional form,

$$u(c) = \frac{c^{1-\frac{1}{\sigma}} - 1}{1 - \frac{1}{\sigma}}, \quad (6.20)$$

marginal utility equals:

$$u'(c) = c^{-\frac{1}{\sigma}},$$

and the Euler equation is expressed by:

$$c_t^{-\frac{1}{\sigma}} = \frac{1 + r_t}{1 + \rho} c_{t+1}^{-\frac{1}{\sigma}}.$$

The ratio of consumption in the two periods is:

$$\frac{c_t}{c_{t+1}} = \left(\frac{1+r_t}{1+\rho} \right)^{-\sigma}.$$

Taking the log of both sides:

$$\log c_t - \log c_{t+1} = -\sigma[\log(1+r_t) - \log(1+\rho)].$$

The Euler equation is:

$$\log c_t = \log c_{t+1} - \sigma(r_t - \rho), \quad (6.21)$$

using the approximation $\log(1+x) \cong x$.

6.2.3 The New Keynesian IS Curve: Discrete Variables

The goods and services market is at equilibrium when expenditure, including consumption and government spending, equals the real output:

$$y_t = c_t + g_t. \quad (6.22)$$

The log-linear approximation [see derivation at the end of this section] of the goods and services market equilibrium around the steady state is:

$$\log y_t = \omega \log c_t + (1-\omega)\log g_t, \quad (6.23)$$

where ω is the consumption-to-income ratio at stationary equilibrium. Consumption is given by:

$$\log c_t = \log c_{t+1} - \sigma(r_t - \rho).$$

Substituting this equation in the equilibrium condition yields:

$$\log y_t = -\omega\sigma(r_t - \rho) + \omega \log c_{t+1} + (1-\omega)\log g_t.$$

Using the linear approximation of consumption for the following period:

$$\omega \log c_{t+1} = \log y_{t+1} - (1-\omega)\log g_{t+1},$$

one may then write the previous equation as:

$$\log y_t = -\omega\sigma(r_t - \rho) + \log y_{t+1} + (1-\omega)(\log g_t - \log g_{t+1}).$$

To simplify, assume that potential output is constant:

$$\bar{y}_t = \bar{y}_{t+1} = \bar{y}.$$

Subtracting the log of potential output from both sides of the real output equation yields the following IS curve:

$$\log y_t - \log \bar{y} = -\omega\sigma(r_t - \rho) + \log y_{t+1} - \log \bar{y} + (1 - \omega)(\log g_t - \log g_{t+1}),$$

The long-term equilibrium real interest rate – the natural interest rate – equals the rate of time preference ($\bar{r} = \rho$), assuming constant government spending.

The New Keynesian IS Curve Equation

Output gap x is defined by: $\log y_t - \log \bar{y} = x_t$. The new Keynesian IS curve becomes:

$$x_t = -\alpha(r_t - \bar{r}) + x_{t+1} + (1 - \omega)(g_t - g_{t+1}), \quad (6.24)$$

where $\alpha = \omega\sigma$ and the letter g now denotes the log of the variable. The effect of the real interest rate gap on the output gap is proportional to the size of the elasticity of substitution. When $g_t = g_{t+1}$, the IS curve simplifies as:

$$x_t = x_{t+1} - \alpha(r_t - \bar{r}). \quad (6.25)$$

Comparing the Keynesian and New Keynesian IS Curves

The previous expression written for the following period is:

$$x_{t+1} = x_{t+2} - \alpha(r_{t+1} - \bar{r}),$$

which, substituted in the previous equation, allows us to write the output gap as a function of the output gap two periods ahead and the gaps of the interest rates in periods t and $t+1$:

$$x_t = x_{t+2} - \alpha(r_t - \bar{r}) - \alpha(r_{t+1} - \bar{r}).$$

The output gap two periods ahead is:

$$x_{t+2} = x_{t+3} - \alpha(r_{t+2} - \bar{r}).$$

By means of this recursive forward substitution, the new Keynesian IS curve would depend on the entire future history of interest rate gaps according to:

$$x_t = -\alpha \sum_{i=0}^{\infty} (r_{t+i} - \bar{r}), \quad (6.26)$$

if the limit of the output gap in the remote future is equal to zero:

$$\lim_{T \rightarrow \infty} x_{t+T} = 0. \quad (6.27)$$

This assumption, however, is entirely arbitrary and unjustifiable, although it is used in every presentation of the new Keynesian model found in the literature. In fact, the forward substitution of the new Keynesian IS curve produces the following limit:

$$\lim_{T \rightarrow \infty} x_{t+T} = x_t + \alpha \sum_{i=0}^{\infty} (r_{t+i} - \bar{r}). \quad (6.28)$$

The traditional Keynesian IS curve is usually specified based on the past:

$$x_t = - \sum_{i=0}^n \alpha_i (r_{t-i} - \bar{r}). \quad (6.29)$$

The crucial difference between the Keynesian and new Keynesian IS curves is that, in the former, the real interest rate gap affects the output gap, whereas in the latter the real interest rate gap affects the rate of change of the output gap. It is inaccurate to state that the difference between the two curves is that in the Keynesian IS past interest rate gaps affect today's output gap, while in the new Keynesian IS curve it is the expected future interest rate gaps that affect today's output gap.

Log Approximation

Assume that z is a function of x and y according to:

$$z = f(x, y),$$

and that the variables with bars represent the solution's steady-state solution:

$$\bar{z} = f(\bar{x}, \bar{y}).$$

Differentiating the function f around stationary equilibrium yields:

$$dz = f_x(\bar{x}, \bar{y})dx + f_y(\bar{x}, \bar{y})dy.$$

Dividing both sides of this expression by $\bar{z}$, the differential of x by $\bar{x}$, and the differential of y by $\bar{y}$ yields:

$$\frac{dz}{\bar{z}} = \frac{f_x(\bar{x}, \bar{y})\bar{x}}{\bar{z}} \frac{dx}{\bar{x}} + \frac{f_y(\bar{x}, \bar{y})\bar{y}}{\bar{z}} \frac{dy}{\bar{y}}.$$

It follows that:

$$d \log z = \omega d \log x + (1 - \omega) d \log y,$$

where:

$$\omega = \frac{f_x(\bar{x}, \bar{y})\bar{x}}{\bar{z}}; 1 - \omega = \frac{f_y(\bar{x}, \bar{y})\bar{y}}{\bar{z}}.$$

Log-linear approximation around the stationary equilibrium point is:

$$\log z \cong \omega \log x + (1 - \omega) \log y.$$

6.2.4 New Keynesian IS Curve: Continuous Variables

The new Keynesian IS curve with continuous variables may be obtained from the IS curve with discrete variables,

$$x_t - x_{t+1} = -\alpha(r_t - \bar{r}).$$

The change in the output gap (x) is approximated using the derivative:

$$x_{t+1} - x_t \cong \dot{x} = \frac{dx}{dv}.$$

Therefore, the IS curve with continuous variables is:

$$\dot{x} = \alpha(r - \bar{r}). \quad (6.30)$$

The real interest rate gap affects the rate of change of the output gap and not its level, as would be the case in the Keynesian IS curve. The literature on the new Keynesian IS curve misinterprets the curve, arguing that consumer decision-making is forward-looking. Some algebra is needed to understand why this proposition is inaccurate. In terms of differentials, the IS curve may be written as:

$$dx = \alpha(r - \bar{r})dv.$$

Integrating both sides of the expression from today (t) until a period in the future (T) yields:

$$\int_t^T dx = \int_t^T \alpha(r - \bar{r})dv.$$

Therefore,

$$x(T) - x(t) = \int_t^T \alpha(r - \bar{r})dv.$$

The output gap in period t equals the output gap in future period T minus the component that depends on the interest rate gap in the relevant future period:

$$x(t) = x(T) - \int_t^T \alpha(r - \bar{r})dv. \quad (6.31)$$

If the output gap in period T is known, the future interest rate gaps would affect today's output gap. This assumption, however, is arbitrary and lacks economic grounds. The new Keynesian IS curve specifies how interest rate gaps affect the change in the output gap.

6.3 The Natural Interest Rate

In the new Keynesian model, the long-term equilibrium real interest rate – the natural interest rate – equals the consumer's rate of time preference when the economy's potential output is constant. When potential output varies over time, the IS curve is:

$$y_t - \bar{y}_t = y_{t+1} - \bar{y}_{t+1} - \bar{y}_t + \bar{y}_{t+1} - \sigma(r_t - \rho).$$

The letter y now represents the log of real output. This equation may also be written as:

$$y_t - \bar{y}_t = y_{t+1} - \bar{y}_{t+1} - \sigma \left[r_t - \rho - \frac{1}{\sigma} (\bar{y}_{t+1} - \bar{y}_t) \right].$$

The natural interest rate is therefore equal to the sum of two components. One is the consumer's rate of time preference. The other equals the product of the inverse of the elasticity of substitution multiplied by the growth rate of potential output. That is:

$$\bar{r} = \rho + \frac{1}{\sigma} (\bar{y}_{t+1} - \bar{y}_t). \quad (6.32)$$

In the new Keynesian model, the natural interest rate therefore depends on two parameters that characterize consumer preferences (ρ , σ) and on the growth of potential output.

In the Keynesian model, the long-term equilibrium real interest rate – the natural rate – depends on fiscal policy parameters. The natural rate is obtained from the IS curve's interception of the vertical line running through the abscissa of potential output, as Fig. 6.1 shows. Any movement of the IS curve affects the natural interest rate. Analytically, the natural interest rate therefore depends on public deficit and government spending. That is:

$$\bar{r} = \bar{r}(\bar{f}, \bar{g}, a). \quad (6.33)$$

An increase in public deficit or in public spending will increase the natural interest rate. The private sector, that is, the behavior of individuals in terms of consumption and their investment decisions, also affects the natural interest rate. The letter a in the expression above indicates that changes in the behavior of consumers, or of businesspersons in connection with investment, affect the natural interest rate. An autonomous increase (decrease) in consumption increases (decreases) the natural interest rate.

6.4 The LM Curve

The Central Bank’s main job is to buy and sell the currency it issues, and in which it is a monopolist. When the Central Bank sells its currency, it buys securities denominated in local or foreign currency. Local-currency denominated securities are, in general, government bonds. Foreign-currency denominated ones are also government bonds, albeit issued by foreign governments. When the Central Bank sells government bonds from its portfolio, it contracts the stock of the monetary base, as is also the case when it sells international reserves. The box below describes a typical Central Bank’s balance sheet. On the asset side lie international reserves and domestic government securities. On the liability side is the monetary base, which is the sum of currency held by the public and the banking system’s reserves held with the Central Bank. The banking reserves deposits is the one through which the economy’s entire payment system flows, and which holds reserve requirements on deposits that commercial banks are required to keep with the Central Bank.

The Central Bank has instruments to control the nominal stock of money by buying and selling securities. The mechanism by which it accomplishes this is quite simple, inducing the market to buy securities by means of the reduction of the price of such securities, and to sell them by means of price increases. The Central Bank’s sale of securities reduces the money stock, and its purchases of securities expand the money stock.

Central Bank Balance Sheet	
Assets	Liabilities
International Reserves (<i>IR</i>) Government Securities (<i>B^{CB}</i>)	Monetary Base (<i>M</i>) (a) Currency Held by the Public (<i>C</i>) (b) Reserve Requirements (<i>R</i>)
$IR + B^{CB} \equiv M \equiv C + R$	

The main characteristic of money that sets it apart from all other financial assets is its use as a means of payment. The price of money is the quantity of goods and services that can be bought with one unit of it, that is, the inverse of the price level (*1/P*). The opportunity cost of money is the nominal interest rate (*i*), the foregone earnings one would make investing in another financial asset, because money has no yield. The public’s demand for money is a demand for the quantity of goods and services that money can buy. The Central Bank controls the nominal stock of money and the public determines the real quantity of money it wishes to hold in its portfolio. This real quantity of money demanded depends on two variables: the transactions volume and the opportunity cost. The transactions volume can be measured by real output. The greater the real output, the greater the real quantity of money demanded and vice versa. When the opportunity cost of money – the interest rate – increases, the public is induced to decrease the quantity of money it wishes to hold. When the interest rate decreases, the real quantity of money demanded increases. The money demand equation is a function of real output and the nominal interest rate:

$$\frac{M^d}{P} = L(y, i). \quad (6.34)$$

The Central Bank controls the nominal stock of money and money supply is given by:

$$M^s = M. \quad (6.35)$$

The money market is at equilibrium when the quantity of money demanded equals the quantity supplied. That is:

$$M^d = M^s. \quad (6.36)$$

Therefore, the money market's equilibrium is:

$$\frac{M}{P} = L(y, i). \quad (6.37)$$

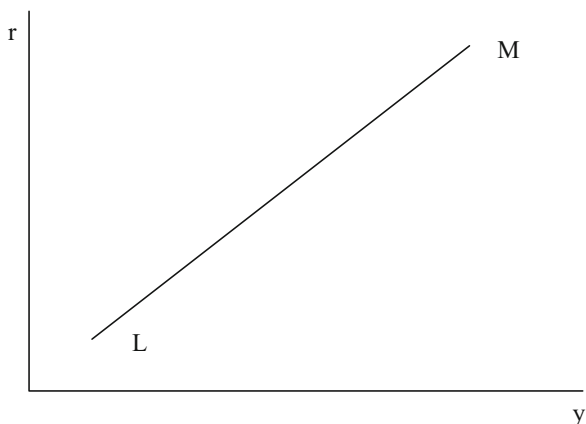
The LM curve in Fig. 6.5 describes the money market's equilibrium. The horizontal axis measures real output and the vertical axis measures the nominal interest rate. The LM curve has a positive slope because if the nominal interest rate increases (decreases), the real output must increase (decrease) to restore equilibrium in the money market. An expansion of the money supply or a reduction of the price level shifts the LM curve down and to the right. A reduction in the nominal stock of money or an increase in the price level shifts the LM curve up and to the left.

The LM curve equation in linear form is:

$$m - \alpha y - \beta i. \quad (6.38)$$

where m is the log of the real quantity of money, $m = \log(M/P)$. When the economy is at full employment, this equation becomes:

Fig. 6.5 The LM curve



$$\bar{m} = \alpha \bar{y} - \beta \bar{i}.$$

Variables with a bar on top are at full employment. Subtracting one equation from the other yields the LM curve in terms of the variables' deviation from their full-employment values:

$$m - \bar{m} = \alpha(y - \bar{y}) - \beta(i - \bar{i}). \quad (6.39)$$

6.5 The LM Curve: Microfoundations

Monetary theory deduces the money demand equation based on three approaches: (i) money in the utility function; (ii) cash-in-advance constraint; and (iii) transaction cost. These three approaches are presented next. The former two address the consumer's demand for money. The third approach – that of transaction cost – is applied to deducing the money demand from banks that uses it to transact in the payments system of a modern monetary economy.

6.5.1 Money in the Utility Function (MIU)

The money in the utility function approach assumes that people demand money for the service they render, similarly with demand for the services of durable goods. The utility function's variable z represents the service of money,

$$\cup(c, z), \quad (6.40)$$

and c represents the flow of consumer goods and services. The service of money is proportional to the stock of the real quantity of money according to:

$$z = k \frac{M}{P}, \quad k = 1. \quad (6.41)$$

It is helpful to select units such that the proportionality constant k equals one. Therefore, the service of money is the real stock of money. Another simplifying assumption is to make the utility function separable, where $u(c)$ is the utility of consumer goods and services and $v(m)$ is the utility of the services of money. That is:

$$\cup(c, m) = u(c) + v(m). \quad (6.42)$$

At the beginning of period t , the representative agent must decide whether to allocate their funds to the purchase of consumer goods and services or to hold those funds as money and spend them on the purchase of consumer goods in the following

period. The added welfare when they choose to purchase consumer goods equals the quantity of consumer goods multiplied by the marginal utility of consumption:

$$\frac{1}{P_t} u'(c_t).$$

When the agent prefers to hold money, the added welfare equals the sum of two items. One is the welfare provided by the service of money, and the other equals the welfare from consumer goods bought at the end of the period, when they convert the money into consumer goods:

$$\frac{1}{P_t} v'(m_t) + \frac{1}{1 + \rho} \frac{1}{P_{t+1}} u'(c_{t+1}).$$

At equilibrium, these choices must produce the same welfare. That is:

$$\frac{1}{P_t} u'(c_t) = \frac{1}{P_t} v'(m_t) + \frac{1}{1 + \rho} \frac{1}{P_{t+1}} u'(c_{t+1}). \quad (6.43)$$

This equation may be rewritten as follows:

$$u'(c_t) = v'(m_t) + \frac{1}{1 + \rho} \frac{1}{P_{t+1}/P_t} u'(c_{t+1}).$$

This model's Fisher equation is:

$$(1 + \rho) \frac{P_{t+1}}{P_t} = 1 + i_t \quad (6.44)$$

Assuming constant consumption,

$$c_t = c_{t+1} = c,$$

the equilibrium equation transforms into:

$$u'(c) \left[1 - \frac{1}{1 + i_t} \right] = v'(m_t).$$

The marginal rate of substitution between consumption and money equals the opportunity cost of holding money:

$$\frac{v'(m_t)}{u'(c)} = \frac{i_t}{1 + i_t}. \quad (6.45)$$

It is worth mentioning that the opportunity cost of holding money, in a discrete variables model, equals the present value of the interest rate, as the return on the financial asset is only paid at the end of the period. When the model is written with continuous variables, the equilibrium condition is:

$$\frac{v'(m)}{u'(c)} = i. \quad (6.46)$$

When the nominal interest rate increases (decreases), the real quantity of money decreases (increases) because the marginal utility of money must increase to restore equilibrium between the marginal rate of substitution of consumption and money, and the interest rate. This equation implicitly defines the money demand equation:

$$m = L(i, c). \quad (6.47)$$

6.5.2 Cash-in-Advance Constraint (CIA)

The cash-in-advance (CIA) constraint approach assumes that, in a monetary economy, goods are not exchanged for goods/products. Money buys goods, and goods buy money. To purchase consumer goods and services, an individual must have a sufficient quantity of money to pay for such goods and services. Analytically, this corresponds to the constraint:

$$M_{t-1} \geq P_t c_t, \quad (6.48)$$

where M_{t-1} is the accumulated money stock in period $t-1$, P_t is the price of the product, and c_t is the quantity of the consumer goods that will be purchased in period t . Money is essential in this economy because, without it, the consumer cannot buy the goods and services they desire. Because the marginal utility of goods and services is positive, the consumer's funds are not wasted. They will have a quantity of money identical to the value of their purchases, as the opportunity cost of money is the interest rate they forewent from financial investment. That is:

$$M_{t-1} = P_t c_t. \quad (6.49)$$

The nominal interest rate works as a tax on the purchase of goods and services. Indeed, for every dollar spent on the purchase of goods and services, interest is foregone, as this dollar must be in money form to pay for the consumer goods bought.

This approach may be extended to an economy with goods and services that may be bought on credit, rather than with money. In such an economy, the interest rate affects the relative price of the two types of goods – those that require money and those that may be bought on credit.

6.5.3 Transaction Cost

The central mechanisms of the payments system of modern economies (United States, Europe, Brazil, Switzerland, etc.) is a real-time gross settlement system (RTGS) that settles obligations in real time, transaction by transaction. The system settles payments for contracts in the money, foreign exchange, and capitals markets,

in addition to those operations made by the Central Bank and the Treasury. There is real-time monitoring of each bank's reserves. These reserves cannot have a negative balance at any time of the day. In addition, only the holder of an account may order debits to their own account, which assures full control of its balance. These procedures drastically reduce risks for participants in the payments system. On the other hand, there is an increased need for liquidity for managing each bank's reserves throughout the day.

As a result, if banks previously managed their balances to minimize the opportunity costs created by excess reserves while complying with required reserve, the RTGS gives rise to another reason for balance management: the need to settle payments in real time. This section introduces a reserves demand model in an RTGS environment. Assume that the transaction cost for the banking system to manage this system is:

$$c(t, T) = \frac{\alpha}{\beta} (t^\beta - 1) T^\delta, \quad (6.50)$$

where α and β are parameters of the cost function, t is the reserves turnover rate ($t = \frac{T}{R}$), T is the total amount of payments to be made during the period, and R is the total amount of banking reserves. This function assumes that the total amount of payments that the bank must make during the period affects the total cost of management of those payments. Two banks with the same turnover rate may show different costs, depending on their total payments. Assume that $\beta \geq 1$, $\alpha \geq 0$, and $c(1, T) = 0$. The δ parameter is unconstrained. The bank faces two costs, the loss of interest on reserves and the cost of managing the reserves account. The bank attempts to minimize its total costs and solves the following problem:

$$\min_t \left\{ iR + \frac{\alpha}{\beta} (t^\beta - 1) T^\delta \right\}. \quad (6.51)$$

The optimal turnover rate is:

$$t^* = \left(\frac{iT^{1-\delta}}{\alpha} \right)^{\frac{1}{\beta+1}}. \quad (6.52)$$

When $0 < \delta < 1$, economies of scale exist because the turnover rate increases with the volume of payments. The reserves demand equation is:

$$R = \alpha^{\frac{1}{1+\beta}} t^{-\frac{1}{1+\beta}} T^{\frac{\beta+\delta}{1+\beta}}. \quad (6.53)$$

The sum of the two elasticities does not equal 1, except where the δ parameter equals zero:

$$|\varepsilon_{R,i}| + |\varepsilon_{R,T}| = \frac{1}{1+\beta} + \frac{\beta+\delta}{1+\beta} = \frac{1+\beta+\delta}{1+\beta}. \quad (6.54)$$

6.6 The Taylor Rule

The Central Bank carries out monetary policy through the market for bank reserves. In this market, commercial banks exchange banking reserves, but the Central Bank controls the total amount of reserves, or the interest rate.

Figure 6.6 shows that if the Central Bank sets the interest rate at i_0 , the market will absorb reserves in a quantity equal to R_0 , and viceversa. However, the effects of these procedures are not equal. When, for any reason, the demand for reserves changes, the interest rate will change significantly if the Central Bank controls the quantity of reserves. On the other hand, if the Central Bank controls the interest rate, the level of reserves absorbs variations in the reserves demand curve, as Fig. 6.6 shows. Central Banks prefer a less volatile interest rate. In most cases, they will intervene in the market for bank reserves by setting the interest rate.

Should a Central Bank set the interest rate in a discretionary, ad hoc way, or by means of a rule that is common knowledge? Economics literature has long debated this issue. The argument for a discretionary policy is that the Central Bank would have a free hand to set the interest rate it deemed most appropriate at any given time. This kind of behavior, however, would create considerable unpredictability for businesspersons, workers, and consumers.

In the case of a monetary policy rule, society would have accurate information on the variables affecting the Central Bank's decisions. Furthermore, compliance with the rule would add to the Central Bank's credibility and reputation. Under these circumstances, merely announcing a policy would be enough for the economy's private sector to make decisions based on the announced policy. The argument that

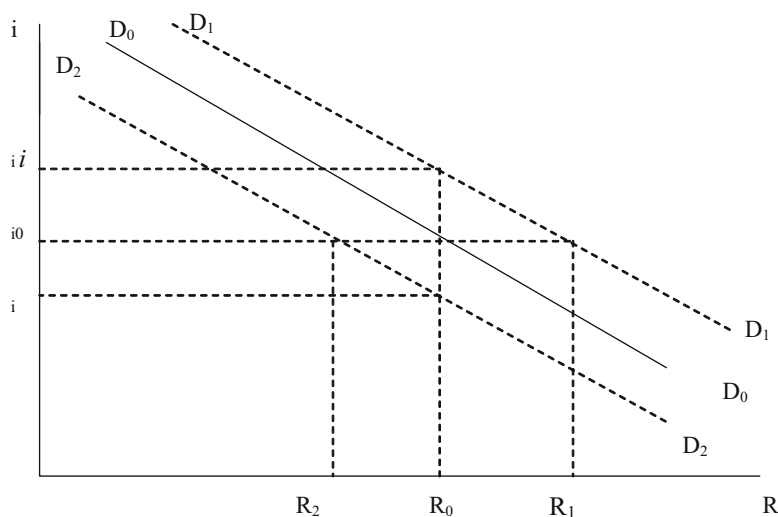


Fig. 6.6 The reserve market

adopting a rule might tie the hands of the Central Bank and prevent it from acting in exceptional situations is unfounded. The rule applies to ordinary circumstances. In the event of a tsunami, the rule is temporarily set aside, and its application would resume as soon as normality prevails. Society would understand the reasons for the temporary repeal of the rule and such an event would not mar the Central Bank's reputation and credibility.

Monetary Policy Rule

What variables determine the interest rate set by the Central Bank? The Taylor rule assumes that the Central Bank sets the banking reserves market's nominal interest rate based on: (i) the inflation gap, the difference between the inflation rate and its inflation target; (ii) the output gap; (iii) the natural interest rate; and (iv) the inflation rate, according to:

$$i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta(y - \bar{y}), \quad \phi > 0, \theta > 0. \quad (6.55)$$

Adding and subtracting inflation target $\bar{\pi}$ to and from this expression, the Taylor rule can be written as:

$$i = \bar{r} + \bar{\pi} + (1 + \phi)(\pi - \bar{\pi}) + \theta(y - \bar{y}). \quad (6.56)$$

This rule's main property is that whenever a deviation of 1% from the inflation target occurs, the Central Bank must raise the interest rate by more than 1% $[(1 + \phi)\%]$. When inflation is on target and the economy is at full employment, the nominal long-term interest rate will be equal to the sum of the natural interest rate and the inflation target.

6.7 The Phillips Curve

The Phillips curve can be derived from different assumptions regarding the goods and services market and the labor market. For price setting, assume that a firm has market power and sets its price by a markup on the marginal cost. The firm passes on any change in the marginal cost to their price. This cost depends on wages and the marginal productivity of labor. In the labor market, the wage depends on the conditions for the utilization of the economy's production capacity. These ingredients enable deducing the traditional Phillips curve.

Market Power and Price Setting

The firm faces a negatively sloped curve of demand for its product and has a marginal cost curve where the marginal cost increases with the quantity produced. Figure 6.7 illustrates the demand and marginal cost curves.

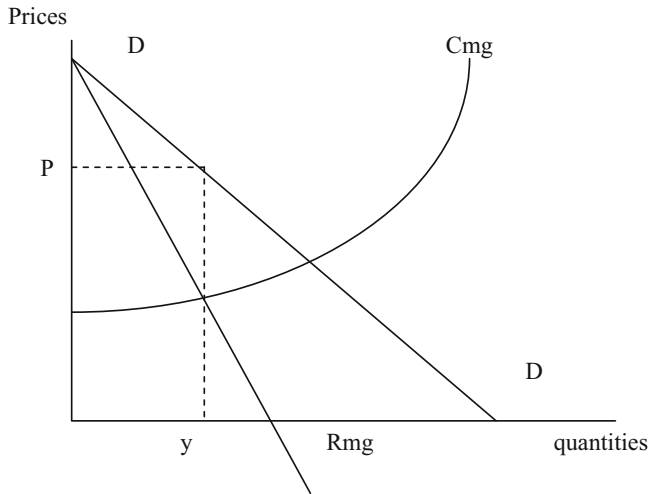


Fig. 6.7 The demand and marginal cost curves for a firm with market power

The firm's purpose is to maximize profit. Its revenues (R) equal the product of the price of the good/product (P) multiplied by the quantity of the goods sold (y). That is:

$$R = Py \quad (6.57)$$

The firm's production cost (C) depends on the quantity produced:

$$C = C(y)$$

Profit is the difference between revenue and production cost:

$$\text{Profit} = R - C$$

The first-order condition for profit maximization implies that the marginal revenue must equal the marginal cost:

$$\frac{dR}{dy} - \frac{dC}{dy} = 0 \Rightarrow Rmg = Cmg.$$

Marginal revenue depends on the price,

$$\frac{dR}{dy} = \frac{d}{dy}(Py) = P + y \frac{dP}{dy} = P \left(1 + \frac{y}{P} \frac{dP}{dy} \right),$$

and the elasticity of the quantity demanded relative to the product's price,

$$\varepsilon = \frac{dy}{dP} \frac{P}{y}; \quad |\varepsilon| = \frac{dy}{dP} \frac{P}{y}.$$

With a little algebra, the marginal revenue is a function of the product's price and the absolute value of the elasticity:

$$\text{Rmg} = P \left[1 + \frac{1}{\frac{Pdy}{y dP}} \right] = P \left(1 - \frac{1}{|\epsilon|} \right) = P \left(\frac{|\epsilon| - 1}{|\epsilon|} \right).$$

Marginal revenue may be written as a function of the k parameter, the firm's margin, and, at equilibrium, the marginal revenue is equal to the marginal cost. That is:

$$\text{Rmg} = \frac{P}{1 + k} = \text{Cmg}. \quad (6.58)$$

The parameter k depends on the price elasticity according to:

$$1 + k = \frac{|\epsilon|}{|\epsilon| - 1} \therefore k = \frac{|\epsilon|}{|\epsilon| - 1} - 1 = \frac{1}{|\epsilon| - 1}. \quad (6.59)$$

The price of a product sold by a firm that has market power is the mark-up of its marginal cost of production and the margin that depends on the price elasticity of the quantity demanded:

$$P = (1 + k)\text{Cmg}. \quad (6.60)$$

The firm changes the prices of its product when the margin changes, or when the marginal cost of production does.

Table 6.1 shows how the margin changes with the price elasticity of the quantity demanded. When absolute elasticity is equal to two, the firm's margin will equal 100%. When absolute elasticity is three, the margin will be 50%. Therefore, the margin decreases as elasticity increases, and, in the extreme case where elasticity is

Table 6.1 Price elasticity of the quantity demanded and the markup

$ \epsilon $	k
2	1 = 100%
3	$\frac{1}{2}$ = 50%
4	$\frac{1}{3}$ = 33%
5	$\frac{1}{4}$ = 25%
.	
.	
.	
11	$\frac{1}{10}$ = 10%
.	
.	
.	
∞	0 = 0%

infinite, the margin equals zero, the firm has no market power, and it operates as a firm under perfect competition.

Phillips Curve: Inflation and Unemployment

The marginal cost of production equals the additional cost of labor divided by the increased production achieved. That is:

$$Cmg = \frac{W\Delta L}{\Delta y} = \frac{W}{(\Delta y/\Delta L)}. \quad (6.61)$$

The marginal cost is therefore equal to nominal wages divided by the marginal productivity of labor. The price of the firm equals:

$$P = (1 + k) \frac{W}{(\Delta y/\Delta L)} = (1 + k) \frac{W}{Pmgl}, \quad (6.62)$$

where $Pmgl$ is the marginal productivity of labor. The price depends on three variables: (i) the firm's margin (k); (ii) the marginal productivity of labor ($Pmgl$); and (iii) the nominal wage (W) of the worker. Under constants productivity and margin, the inflation rate equals the rate of change of nominal wages:

$$\pi = \frac{\dot{P}}{P} = \frac{\dot{W}}{W}. \quad (6.63)$$

In the labor market, the rate of change of nominal wages depends on the expected inflation rate and on the conditions of the labor market. When the unemployment rate (u) is greater (lesser) than the natural unemployment rate ($\bar{u}$), wages tend to drop (rise). That is:

$$\frac{\dot{W}}{W} = \pi^e - a(u - \bar{u}). \quad (6.64)$$

Figure 6.8 illustrates this Phillips curve. The vertical axis measures the rate of change of wages, while the horizontal axis measures the unemployment rate. In the short term, there is a tradeoff between unemployment and inflation. This tradeoff does not exist in the long term and the curve is vertical.

Assuming a constant margin and a marginal productivity of labor for the firm, the inflation rate and the growth rate of wages will be equal. Therefore, the Phillips curve is:

$$\pi = \pi^e - a(u - \bar{u}). \quad (6.65)$$

Figure 6.8 also shows this Phillips curve. The vertical axis measures the inflation rate and the horizontal axis measures unemployment. In the long term, when the inflation rate is equal to the expected inflation rate, the unemployment rate will be equal to the natural unemployment rate.

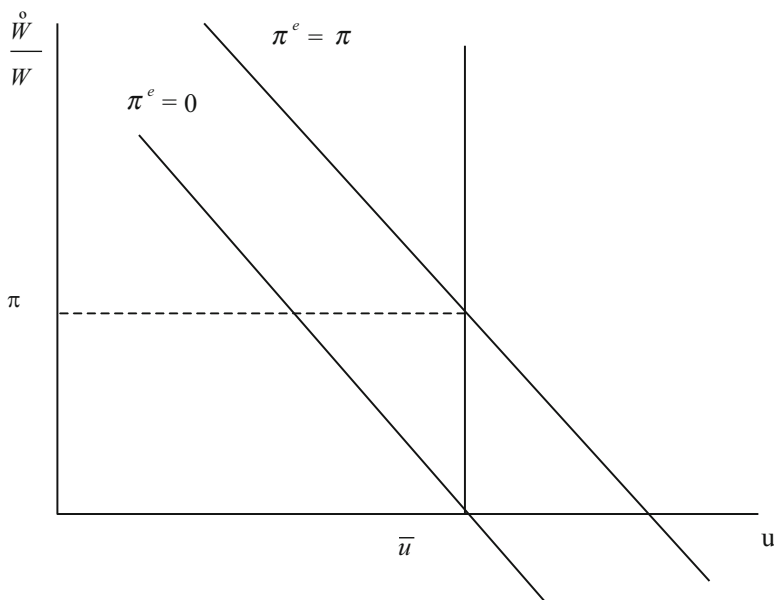


Fig. 6.8 The Phillips curve and the natural rate of unemployment

Okun's Law

Okun's Law relates to a deviation of the unemployment rate from the natural unemployment rate with the output gap. According to Okun's Law, each percentage point of increase of the unemployment rate relative to the natural unemployment rate, the economy's idle capacity increases by b percent, that is:

$$y - \bar{y} = -b(u - \bar{u}). \quad (6.66)$$

Estimates of the b parameter for the US economy produce values between two and three. This means that for each percentage point of unemployment, the economy's idle capacity would rise by 2–3%. This may appear inconsistent with the law of diminishing returns. However, when the natural unemployment rate varies, the rate of utilization of machinery and equipment, the number of working hours per worker, and the labor productivity also vary. Let us assume that the log of output (y) is a weighted average of the logs of the service of capital (k) and of the service of labor ($a + h + n$):

$$y = \alpha k + (1 - \alpha)(a + h + n),$$

where α is the elasticity of output with respect to capital, h is the log of the number of working hours, n is the log of employment, and a is the log of the coefficient measuring labor efficiency. This production function assumes constant returns to scale.

At full employment, the production function is:

$$\bar{y} = \alpha \bar{k} + (1 - \alpha)(\bar{a} + \bar{h} + n),$$

where a bar above a variable denotes its value at full employment.

Subtracting one equation from the other yields:

$$y - \bar{y} = \alpha(k - \bar{k}) + (1 - \alpha)(a - \bar{a} + h - \bar{h} + n - \bar{n}).$$

The output gap depends on the gaps of capital, labor productivity, the number of working hours, and employment.

Assume that total labor supply is given by n^* . The natural unemployment rate is defined by $u = n^* - n$ and the natural unemployment rate, with the economy at full employment, is equal to $\bar{u} = n^* - \bar{n}$. Therefore,

$$n - \bar{n} = -(u - \bar{u}).$$

Assume also that the gaps of capital, labor productivity, and the number of working hours are related to the unemployment gap according to the following equations:

$$k - \bar{k} = -\gamma(u - \bar{u}) \quad , \gamma > 0,$$

$$a - \bar{a} = -\lambda(u - \bar{u}) \quad , \lambda > 0,$$

$$h - \bar{h} = -\phi(u - \bar{u}) \quad , \phi > 0.$$

It follows that:

$$y - \bar{y} = -[\alpha\gamma + (1 - \alpha)(\lambda + \phi + 1)](u - \bar{u}).$$

The Okun's Law b coefficient therefore depends on α , γ , λ , and ϕ . That is:

$$b = \alpha\gamma + (1 - \alpha)(\lambda + \phi + 1).$$

In spite of the law of diminishing returns ($\alpha < 1$), the b parameter may be greater than unity, depending on the values of the model's other coefficients.

Phillips Curve: Inflation and Output Gap

The Phillips curve obtained by substituting the gap between the unemployment rate and the natural unemployment rate according to Okun's Law results in the following expression:

$$\frac{\dot{W}}{W} = \pi^e + \frac{a}{b}(y - \bar{y}). \quad (6.67)$$

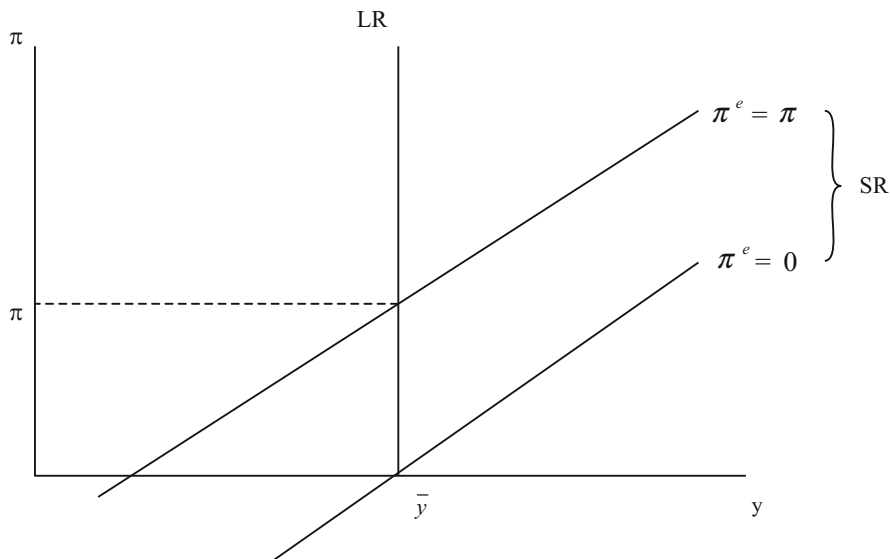


Fig. 6.9 The Phillips curve and the output gap

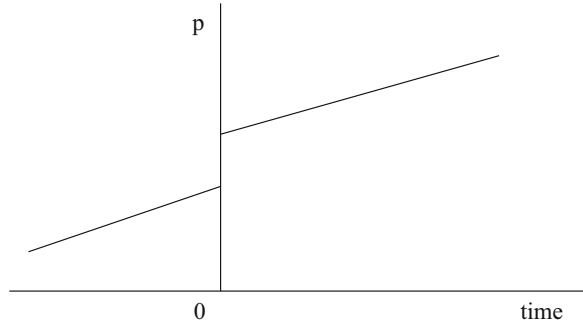
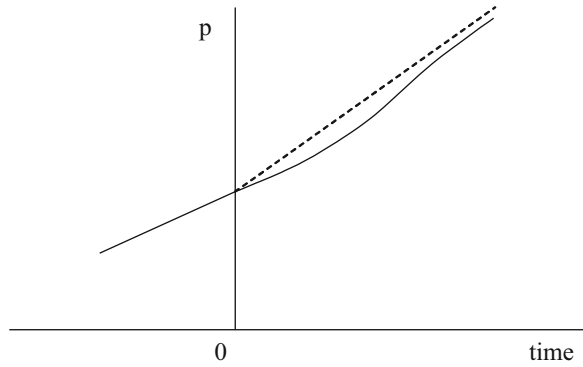
Assuming that the inflation rate equals the rate of change of wages yields the Phillips curve where the inflation rate depends on the expected inflation rate and the output gap ($y - \bar{y}$):

$$\pi = \pi^e + \varphi(y - \bar{y}), \quad \varphi = \frac{a}{b} > 0. \quad (6.68)$$

There is a short-term tradeoff between inflation and the output gap, but in the long term, where the inflation rate is equal to the expected inflation rate, the Phillips curve is vertical. Figure 6.9 shows the Phillips curve graph. The vertical axis measures the inflation rate and the horizontal axis measures real output. In the short term, for each given expected inflation rate, the Phillips curve has a positive slope. In the long term, the slope is vertical.

Phillips Curve: Inflation Inertia

The Phillips curve deduced above assumes that the price level is rigid in the short term. That is to say that the price level is a predetermined variable for the model and cannot instantly change values, as in Fig. 6.10. An additional assumption is that the inflation rate is also rigid in the short term, that is, there is inflation rate inertia, as in Fig. 6.11, where no abrupt change in the curve's tangent occurs. This means that, at the model's initial moment, both the price level and the inflation rate are predetermined variables: $p(0) = \log P(0)$ and $\pi(0)$ are given.

Fig. 6.10 Flexible prices**Fig. 6.11** Price rigidity and inflation inertia

Assume, then, that the expected inflation rate depends on the past inflation rate according to:

$$\pi_t^e = \pi(t - h), \quad (6.69)$$

where h indicates the relevant memory for the economic agent. Thus, the Phillips curve is:

$$\pi(t) = \pi(t - h) + \varphi(y_t - \bar{y}_t).$$

The inflation rate $\pi(t - h)$ is not known. Its value can be substituted in the previous equation by means of a Taylor expansion around inflation at point t . That is:

$$\pi(t - h) = \pi(t) + \dot{\pi}(t)[t - h - t],$$

or:

$$\pi(t - h) = \pi(t) - h\dot{\pi}(t).$$

Substituting this expression in the Phillips curve yields:

$$\pi(t) = \pi(t) - h\dot{\pi}(t) + \varphi(y_t - \bar{y}_t).$$

By removing equal terms that appear on both sides, the acceleration of inflation is a function of the output gap:

$$\ddot{\pi} = \frac{\varphi}{h} (y - \bar{y}) = \delta (y - \bar{y}). \quad (6.70)$$

The coefficient of the gap $\delta = \varphi/h$ depends on the memory represented by the letter h . That is, the greater the memory, the lower the output gap's coefficient. At the limit, if the memory ceases to exist, $\delta \rightarrow \infty$, there is no price rigidity. The prices system is therefore flexible and the economy is always at full employment.

Phillips Curve: Without Inflation Inertia

Another assumption in the model is that only the price level is rigid, as in Fig. 6.12. The inflation rate is not rigid and may change instantly, as is the case where $t = 0$ and the slope of the line changes in Fig. 6.12. This means that the price level, but not the inflation rate, is a predetermined model variable. Assume, therefore, that the expected inflation rate is equal to the future inflation rate,

$$\pi^e = \pi(t + h), \quad (6.71)$$

where h indicates the future horizon of relevance to the economic agent. The Phillips curve is then expressed as:

$$\pi(t) = \pi(t + h) + \varphi(y_t - \bar{y}_t).$$

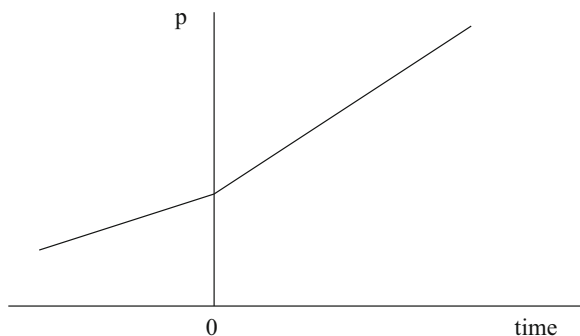
The future inflation rate is unknown. This problem may be solved by means of a Taylor expansion around the inflation rate at point t , that is:

$$\pi(t + h) = \pi(t) + \dot{\pi}(t)[t + h - t].$$

This expression, once simplified, becomes:

$$\pi(t + h) = \pi(t) + \dot{\pi}(t)h.$$

Fig. 6.12 Price rigidity and flexible inflation



Inserting this equation into the Phillips curve yields:

$$\pi(t) = \pi(t) + \dot{\pi}(t)h + \varphi(y - \bar{y}).$$

Removing the inflation rate on both sides, inflation acceleration is negatively related to the output gap:

$$\dot{\pi} = -\frac{\varphi}{h}(y - \bar{y}) = -\delta(y - \bar{y}). \quad (6.72)$$

This Phillips curve is the new Keynesian Phillips curve, as formalized by Calvo (1983) based on microfoundations (see exercise 20), to be deduced next in a model with discrete variables.

In this Phillips curve model, the agent's decisions are forward-looking. Solving this differential equation must take into account the fact that the integral's limits vary from today (t) to the future ($t + h$). That is:

$$\int_t^{t+h} d\pi = \int_t^{t+h} -\delta(y - \bar{y})dv.$$

The left-side integral is the acceleration of the inflation rate. Therefore:

$$\pi(t+h) - \pi(t) = -\int_t^{t+h} \delta(y - \bar{y})dv.$$

The inflation rate today (t) depends on the future inflation rate ($t + h$) and on the pressure of demand between today and the future:

$$\pi(t) = \pi(t+h) + \int_t^{t+h} \delta(y - \bar{y})dv. \quad (6.73)$$

In the backward-looking agent model, the differential equation's solution is given by:

$$\int_{t-h}^t d\pi = \int_{t-h}^t \delta(y - \bar{y})dv.$$

The integral's limits begin in the past ($t-h$) and extend into today (t). Therefore, the inflation rate depends on the past inflation rate and on the pressure of demand between the past and today. That is:

$$\pi(t) = \pi(t-h) + \int_{t-h}^t \delta(y - \bar{y})dv. \quad (6.74)$$

Comparison of the forward- and backward-looking Phillips curves shows that: (i) future output gaps affect the current inflation in the former case; and (ii) past output gaps affect the current inflation rate in the latter case.

Phillips Curve: Keynesian Versus New Keynesian

In the Keynesian model, there is an inertia component to inflation: the present depends on the past. In the continuous variables models, this property translates into the fact that the inflation rate is a predetermined model variable. Inflation is a function of the past's output gap. Therefore, the acceleration of inflation and the output gap are positively correlated.

In the new Keynesian model, inflation does not have an inertia component. The inflation rate at the model's initial instant is not a predetermined variable. It is a jump variable that can change values abruptly, according to the anticipated behavior of the future output gap. It may not be intuitive at first sight that the acceleration of inflation and the output gap are negatively correlated. The doubt fades, however, when the reasoning behind this Phillips curve is taken into account. For example, given a positive output gap, the inflation rate should rise at the initial instant and then decrease. When a negative gap occurs, the inflation should decrease and then rise. Therefore, the correlation between the acceleration of inflation and the output gap must be negative.

6.8 The New Keynesian Phillips Curve

A frequently used model to derive the Phillips curve assumes that each firm's price adjustment is not synchronous with those of the other firms. Each firm adjusts its price randomly when it receives a sign. The probability of receiving such a sign in the period is equal to λ . Therefore, the probability of the price adjustment taking place in j periods is given by:

$$P(X = j) = \lambda(1 - \lambda)^{j-1}, \quad j = 1, 2, 3, \dots \quad (6.75)$$

The firms' average price adjustment period equals the mathematical expectation of this geometrical distribution's random variable:

$$EX \sum_{j=1}^{\infty} jP(X = j) = \sum_{j=1}^{\infty} j\lambda(1 - \lambda)^{j-1} = \frac{1}{\lambda}. \quad (6.76)$$

When $\lambda = 0.25$, for example, and each period in the model is one-quarter, the average adjustment period will be four-quarters ($4 = 1/0.25$).

The fact that a firm does not adjust its price every period implies a loss. Assume that the expected value of this loss when the i^{th} firm adjusts its prices in period t is given by:

$$L = \frac{1}{2} E_t \sum_{j=0}^{\infty} \beta^j \left(p_{i,t} - p_{t+j}^* \right)^2, \quad (6.77)$$

where $P_{i,t}$ is the price set by the firm in t , P_{t+j}^* is the price it would charge in period $t+j$ if it were able to adjust its price, and $\beta = 1/(1 + \rho)$ is the discount factor that the firm uses.

The firm's objective is to set its price in period t , $p_{i,t}$ such that the expected value of L ,

$$\frac{1}{2} \sum_{j=0}^{\infty} (1 - \lambda)^j \beta^j E_t \left(p_{i,t} - p_{t+j}^* \right)^2, \quad (6.78)$$

is minimum. The first-order condition for a minimum implies that the partial derivative of this expression with respect to $p_{i,t}$ is equal to zero, which yields:

$$v_t = [1 - \beta(1 - \lambda)] \sum_{j=0}^{\infty} [\beta(1 - \lambda)]^j E_t p_{t+j}^*. \quad (6.79)$$

The letter v denotes the price of firms adjusting their prices in period t , as they share the same characteristics. This price is a weighted average of the expected optimum prices in the future ($E_t p_{t+j}^*$), based on the information available in the present. This equation may be written as (see exercise 13):

$$v_t = [1 - \beta(1 - \lambda)] p_t^* + \beta(1 - \lambda) E_t v_{t+1}. \quad (6.80)$$

The price index is the weighted average of the prices adjusted in period t and those that remained unchanged from the previous period. That is:

$$p_t = \lambda v_t + (1 - \lambda) p_{t-1}, \quad (6.81)$$

where λ is the share of firms that adjusted their prices in period t .

The new Keynesian Phillips curve is determined by a model formed by the two equations:

$$v_t = [1 - \beta(1 - \lambda)] p_t^* + \beta(1 - \lambda) E_t v_{t+1}, \quad (6.82)$$

$$p_t = \lambda v_t + (1 - \lambda) p_{t-1}. \quad (6.83)$$

Substituting the second equation's value of v in the first one yields:

$$\frac{p_t - (1 - \lambda) p_{t-1}}{\lambda} = [1 - \beta(1 - \lambda)] p_t^* + \beta(1 - \lambda) E_t \frac{p_{t+1} - (1 - \lambda) p_t}{\lambda}.$$

This equation, once simplified, produces the Phillips curve:

$$\pi_t = \beta E_t \pi_{t+1} + \frac{\lambda}{1 - \lambda} [1 - \beta(1 - \lambda)] (p_t^* - p_t). \quad (6.84)$$

The inflation rate $\pi_t = p_t - p_{t-1}$ depends on the subsequent period's expected inflation $E_t \pi_{t+1} = E_t p_{t+1} - p_t$ and on the difference between the price (p_t^*) that would

be optimum except for rigidity and the current price level (p_t). The optimum price is equal to the margin plus the marginal cost:

$$p_t^* = k + \text{cmg}_t.$$

Therefore,

$$p_t^* - p_t = k + \text{cmg}_t - p_t = k + \text{cmgr}_t$$

where $\text{cmgr}_t = \text{cmg}_t - p_t$ is the real marginal cost. The margin k equals the real long-term marginal cost with an inverted sign. Therefore,

$$p_t^* - p_t = \text{cmgr} - \bar{\text{cmgr}}.$$

The Taylor expansion of the real marginal cost around the full-employment output is:

$$\text{cmgr}_t = \bar{\text{cmgr}} + \text{cmgr}'(y_t - \bar{y}_t),$$

where cmgr' is the partial derivative of the marginal cost with respect to output, evaluated at full employment. The new Keynesian Phillips curve is:

$$\pi_t = \beta E_t \pi_{t+1} + \delta x. \quad (6.85)$$

The parameter δ of the output gap ($x = y - \bar{y}$) equals:

$$\delta = \text{cmgr}'\lambda[1 - \beta(1 - \lambda)]/(1 - \lambda).$$

In this Phillips curve, the price level is predetermined, but the inflation rate has no inertia because it does not depend on past inflation, but rather on the next period's expected inflation. Furthermore, in the long term, when the inflation rate and the expected inflation rate are equal, there is a tradeoff between inflation and output,

$$y_t = \bar{y}_t + \frac{1 - \beta}{\delta} \pi. \quad (6.86)$$

When β equals one, or the coefficient δ tends to infinity, the inflation rate does not affect real output in the long term. The fact that β is close to one does not mean that the long-term output gain from increased inflation is negligible. For example, assume β is equal to 0.99 and δ is equal to 0.2. Therefore, $(1 - \beta)/\delta = 0.05$. That is: $y = \bar{y} + 0.005\pi$. An inflation of 100% ($\pi = 1.0$) would lead to a 5% gain in real output.

The new Keynesian Phillips curve was deduced from a firm optimization problem that minimizes the expected value of losses from keeping the price fixed for a certain period. In theory, it would be immune to the Lucas critique because its parameters are independent from the economic policy rules in force. However, this is not the case of the λ parameter, which measures the probability of a firm receiving the sign

to raise its price. Calvo's ingenious formulation (1983) reproduces the stylized fact of price rigidity, but is certainly ad hoc because it is not microfounded.

Comparative analysis of the model in which the Phillips curve depends on past inflation with the model in which it is a function of future inflation becomes simpler with the use of continuous variables, instead of the discrete ones we have been using so far. The Phillips curve in both cases may be written as:

$$\dot{\pi} = \delta x. \quad (6.87)$$

When the parameter δ is positive, the price level and the inflation rate will be predetermined variables. In this case, inflation acceleration and the output gap are positively correlated. When δ is negative, the price level is predetermined, but the inflation can change instantaneously. In this case, there must be a negative correlation between inflation acceleration and the output gap. When $\delta \rightarrow \infty$, prices are flexible and output is at full employment.

6.9 Exercises

1. Suppose that investment depends on the real income level according to:

$$I = I(r - \pi^e, y).$$

Does the IS curve always have a negative slope?

2. Assume that consumption (c) depends on disposable income (y^d), $c = c(y^d)$, and that the disposable income is defined by $y^d = y - g$, where y is the real income and g is government spending.
 - (a) Why would you define disposable income in this way?
 - (b) Would a tax cut, for a given level of g , affect expenditures in this economy?
3. Assume that consumption depends on disposable income ($y^d = y - \tau$) and on the real quantity of money ($m = M/P$), $c = c(y - \tau, m)$.
 - (c) Is the IS curve independent from monetary policy?
 - (d) Is the full-employment real interest rate independent from monetary policy?
4. Consider the model:

$$IS : y = c(y - \tau) + I(i) + g,$$

$$LM : \frac{M}{P} = L(y, i),$$

$$MPR : i = \bar{i}.$$

When the Central Bank sets this economy's interest rate according to this monetary policy rule, is the economy's price level determined?

5. Assume that the utility function is:

$$u(c_t) = -\frac{1}{\alpha} e^{-\alpha c}, \quad \alpha > 0.$$

- (a) How do you interpret parameter α ?
 (b) Use the Euler equation to deduce the equation of the IS curve associated with this utility function.
6. The marginal utility of consumption in period $t + 1$ may be written as a function of the marginal utility in period t , of the derivative of the marginal utility of period t , and of the difference between consumption tomorrow and consumption today, according to the following Taylor expansion,

$$u'(c_{t+1}) \cong u'(c_t) + u''(c_t)(c_{t+1} - c_t).$$

This expansion disregards second-order terms. Show that the Euler equation with continuous variables is:

$$\dot{c} = -\frac{cu'(c)}{u''(c)}(r - \rho)$$

7. Suppose that the output gap depends on a lagged output gap and on the interest rate gap according to:

$$\begin{aligned} x_t &= \lambda x_{t-1} - \alpha(r_t - \bar{r}_t), \\ x_t &= \lambda x_{t+1} - \alpha(r_t - \bar{r}_t). \end{aligned}$$

The λ parameter lies between zero and unity and the α parameter is positive.

- (a) Write down the output gap, respectively, as:

$$\begin{aligned} x_t &= -\alpha \sum_{i=0}^{-\infty} \lambda^i (r_{t-i} - \bar{r}_{t-i}), \\ x_t &= -\alpha \sum_{i=0}^{\infty} \lambda^i (r_{t+i} - \bar{r}_{t+i}). \end{aligned}$$

- (b) What happens when the λ parameter is equal to unity? Can the output gap be written as a function of the past (future) interest rate gaps?
8. The consumer's utility function is:

$$u(c) = \frac{c^{1-\frac{1}{\sigma}} - 1}{1 - \frac{1}{\sigma}}.$$

Deduce the money demand equation when the utility function of money is specified by:

- (a) $v(m) = \frac{m^{1-\lambda}-1}{1-\lambda}$, $\lambda \neq 1$ and $v(m) = \log m$, $\lambda = 1$;
 (b) $v(m) = m(\alpha - \beta \log m)$, $\beta > 0$.

9. When uncertainty is introduced into the transaction cost model, the bank solves the following problem:

$$\min_t \{E[iR + c(t, T)]\}.$$

- (a) Show that this problem's first-order condition implies the following turn-over ratio:

$$t^* = \left(\frac{K}{\alpha E T^\delta} \right)^\gamma,$$

where $K = E\{i T\}$ and $\gamma = 1/(1 + \beta)$.

- (b) Assume, for simplicity, that T is log-normally distributed. For a normal X , it is known that:

$$E \exp(\tau X) = \exp(\mu \tau + 1/2 \sigma^2 \tau^2).$$

Show that this expression can be used to calculate the mathematical expectation of T^δ : $E T^\delta = E \exp(\delta \log T)$ and obtain:

$$\log t = \gamma \log \frac{K}{\alpha} - \gamma \delta E \log T - \frac{1}{2} \gamma \delta^2 \text{Var} \log T,$$

where Var represents variance.

- (c) In this model, does the volatility of payments (T) affect the turnover rate?

10. Consider the model:

Aggregate demand: $m + v = p + y$.

Aggregate supply: $p = p^e + \delta(y - \bar{y})$,

where m = nominal stock of money; v = income velocity of money; y = real income; p = price index; p^e = expected price index (all variables in logs).

Assuming rational expectations, what would be the expected value of the price index?

11. The income velocity of money is the ratio of nominal output to the nominal stock of money. That is:

$$V = \frac{Y}{M} = \frac{Py}{M}.$$

- (a) When the income elasticity of money is equal to unity, does its velocity depend on real output?
 - (b) Does the interest rate affect the velocity?
 - (c) Define $k = 1/V$. What is the unit of k ?
12. The former German Central Bank (the Bundesbank) carried out monetary programming based on the following identity:

$$M V \equiv P y.$$

Assume that potential output of the German economy grew at an annual rate of 2.5%. The Bundesbank's target was 2.5% annual inflation.

- (a) What information did the Bundesbank need to calculate the corresponding growth rate of M ?
 - (b) How would you obtain this information?
 - (c) Assume unstable income velocity. Would you adopt the same method?
13. Assume that the elasticity of money demand with respect to the interest rate is equal to minus infinity (liquidity trap).
- (a) Use the identity $M V \equiv P y$ to show why the monetary policy does not affect the economy's output.
 - (b) Some claim that the liquidity trap is a reasonable assumption when the nominal interest rate approaches zero. Others claim that, under these circumstances, the elasticity must equal zero. How might the issue be resolved?
14. Right, Wrong, or Maybe. Justify your answer.
- (c) The balanced budget multiplier (increased government spending = increased taxes) equals zero.
 - (d) In the short term, the inflation rate depends on monetary policy alone.
 - (e) When the monetary policy is expansionary, the economy's real liquidity decreases.
 - (f) Inflation inertia increases the social cost of fighting inflation.
 - (g) The real interest rate is independent from public deficit in the case of Ricardian equivalence.
 - (h) Increased government spending increases the economy's real output in both the short and the long term.
15. Suppose that an economic model can be represented by the following finite difference equation:

$$y_t = \alpha E(y_{t+1}/I_t) + \beta x_t, |\alpha| < 1.$$

Show how to obtain this model's fundamentals and bubble solutions, and apply the method to the following cases:

- (a) Arbitrage between fixed income and variable income (riskless):

$$\frac{E(P_{t+1}/I_t) - p_t + d_t}{p_t} = i.$$

- (b) Determination of the price level according to the Cagan model of money demand:

$$m_t - p_t = -\gamma[E(p_{t+1}/I_t) - p_t].$$

16. Consider the model:

$$\text{IS: } y = -\alpha i + u,$$

$$\text{LM: } m = -\beta i + \gamma y + v.$$

where u and v are random, non-correlated variables with averages equal to zero and the respective variances σ_u^2 and σ_v^2 . The Central Bank's loss function is given by:

$$L = y^2$$

The Central Bank may choose between the interest rate (i) or the quantity of money (m) as a policy instrument.

- (a) What value of m minimizes the expected value of the loss function?
 (b) What value of i minimizes the expected value of the loss function?
 (c) What instrument should the Central Bank choose?
17. The price level p_t is a weighted average of the price v_t and of the price in period $t-1$, according to:

$$p_t = \lambda v_t + (1 - \lambda)p_{t-1}.$$

- (a) Use recursive (back) substitution to show that:

$$p_t = \sum_{i=0}^{\infty} \lambda(1 - \lambda)^i v_{t-i}.$$

- (b) Solve the same exercise using the lag operator L , $Lz_t = z_{t-1}$, and the property. $\frac{1}{1-aL} = 1 + aL + a^2L^2 + \dots$
18. The optimum price v_t depends on price p_t^* and on the mathematical expectation $E_t v_{t+1}$, according to:

$$v_t = [1 - \beta(1 - \lambda)]p_t^* + \beta(1 - \lambda)E_t v_{t+1}.$$

- (a) Use recursive (forward) substitution to show that:

$$v_t = [1 - \beta(1 - \lambda)] \sum_{j=0}^{\infty} [\beta(1 - \lambda)]^j E_t p_{t+j}^*.$$

(b) Solve the same exercise using lag operator F , $Fz_t = z_{t+1}$, and the property:

$$\frac{1}{1 - aF} = 1 + aF + a^2F^2 + \dots$$

19. Consider the Phillips curve:

$$\pi_t - \pi^* = \beta(E_t \pi_{t+1} - \pi^*) + \delta(y_t - \bar{y}_t),$$

where π^* is the trend inflation. Is this curve vertical in the long-run?

20. Each firm sets the price of its product only when it receives a random, exponentially distributed sign. That is, the probability that the sign will be received in h periods from today is given by:

$$\delta e^{-\delta h}, \delta > 0.$$

The (log of) the price set by the firm in t , when it receives the sign, is:

$$v_t = \int_t^{\infty} (p_s + \alpha x_s) \delta e^{-\delta(s-t)} ds, \alpha > 0,$$

where p_s is the price level and x_s is excess demand, both in period s .

The (log of) price level (p) is defined by the formula:

$$p_t = \int_{-\infty}^t v_s \delta e^{-\delta(t-s)} ds.$$

(a) Show that the average of the random variable H is given by: $EH = 1/\delta$.

(b) Make explicit the arguments that justify the expressions of v_t and p_t .

(c) Take the derivative of v_t and p_t with respect to time, applying the Leibnitz rule, and show that:

$$\dot{v} = \delta(v - p - \alpha x),$$

$$\pi = \dot{p} = \delta(v - p).$$

(d) Take the derivative of π with respect to time and show that:

$$\dot{\pi} = -\alpha \delta^2 x.$$

(e) Suppose that v_t is given by:

$$v_t = \int_t^{\infty} e^{-\rho(s-t)} (p_s + \alpha x_s) \delta e^{-\delta(s-t)} ds, \alpha > 0,$$

where ρ is the discount rate. Show that:

$$\dot{\pi} = \rho \delta p + \rho \pi - \alpha \delta^2 x.$$

(f) Is the previous item's Phillips curve vertical in the long-run?

21. Consider the Phillips curve model,

$$\pi = \pi^e + \delta(y - \bar{y}).$$

Assume that the expected inflation rate follows the adaptive expectations model:

$$\dot{\pi}^e = \beta(\pi - \pi^e).$$

Show that the acceleration of the inflation rate is:

$$\dot{\pi} = \beta \delta (y - \bar{y}) + \delta \dot{y},$$

when the growth rate of potential output is equal to zero: $\dot{\bar{y}} = 0$.

22. The new Keynesian Phillips curve with perfect foresight is:

$$\pi_t = \beta \pi_{t+1} + \delta x.$$

Show that the new Keynesian Phillips curve with continuous variables is:

$$\dot{\pi} = \rho \pi - kx$$

where $\delta = (1 - \beta/\beta)$ and $k = -\delta/\beta$.

Chapter 7

Economic Fluctuation and Stabilization



This chapter introduces the equilibrium and dynamics of six sticky-price models. Section 7.1 features a model with an IS curve, a Phillips curve, a Taylor monetary policy rule, and inflation rate inertia. Section 7.2 presents a model that has the same equations as Section 7.1, but no inflation rate inertia in the Phillips curve. Section 7.3 is the new Keynesian model with no inflation rate inertia, an IS curve derived from the Euler equation, and a Calvo Phillips curve. Section 7.4 presents an encompassing specification that has the Keynesian and new Keynesian models as particular cases. Section 7.5 deals with a model in which the Central Bank controls the growth rate of money according to the Friedman rule, and there is both price-level and inflation-rate inertia. Although no Central Bank in the world adopts this rule, the model has properties that make it appealing for didactic purposes. Section 7.6 presents a Keynesian model for chronic inflation, which is characterized by a monetary policy rule whereas the Central Bank issues money to finance the public deficit.

7.1 Keynesian Model: Inflation Inertia

The model with sticky prices, inflation inertia, and an interest rate rule has three equations, an IS curve, a Phillips curve (PC), and the Taylor monetary policy rule (MPR). The simplifying assumption is that, under the fiscal policy, public deficit and government spending are constant and equal to their full-employment levels. The model is represented as follows:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(f - \bar{f}) + g - \bar{g}, \quad (7.1)$$

$$\text{PC: } \dot{\pi} = \delta x, \quad (7.2)$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \quad (7.3)$$

$$\text{IC: Given } p(0) = e\pi(0), \quad (7.4)$$

$$\text{Simplifying assumption: } f = \bar{f}, g = \bar{g}. \quad (7.5)$$

The symbol IC denotes the model's initial conditions. The parameters are positive, except for the inflation gap parameter in the monetary policy rule, which may be either positive or negative, depending on the Central Bank's behavior.

Algebra

The equation for the monetary policy rule allows for writing the difference between the real interest rate and the natural interest rate as a function of the inflation and output gaps:

$$r - \bar{r} = \phi(\pi - \bar{\pi}) + \theta x.$$

Substituting this expression in the IS curve, the output gap depends on the inflation gap according to:

$$x = -\frac{\alpha\phi}{1 + \alpha\theta}(\pi - \bar{\pi}).$$

Dynamical System

The model can be reduced to two equations: the Phillips curve and the aggregate demand equation resulting from the combination of the monetary policy rule and the IS curve:

$$\begin{cases} \dot{\pi} = \delta x \\ \pi = \bar{\pi} - \frac{(1 + \alpha\theta)}{\alpha\phi}x \end{cases} \quad (7.6)$$

Figure 7.1 shows the model's phase diagram. The horizontal axis measures the output gap and the vertical axis measures the inflation rate. When there is no inflation acceleration, output is equal to the potential output according to the Phillips curve. Inflation increases if the economy's real output is greater than its potential output and decreases when real output is lower than the potential output. The arrows in Fig. 7.1 illustrate these dynamics. The equation resulting from the combination of the monetary policy rule and the IS curve has a negative slope when the ϕ parameter is positive. In this case, the model is stable because, if the economy is anywhere other than the point of long-run equilibrium, its path will converge to the equilibrium point, as the arrows in Fig. 7.1 show.

Figure 7.2 shows the model's phase diagram when the monetary policy rule's ϕ parameter is negative. In this case, the equation resulting from the combination of the monetary policy rule and the IS curve has a positive slope. The phase diagram for the Phillips curve remains the same as in the previous case. Therefore, the model is unstable because the economy will not converge to the long-term equilibrium when it is off this equilibrium. The model's equilibrium point, to use the language of

Fig. 7.1 The phase diagram for the Keynesian model with inflation inertia: $\phi > 0$

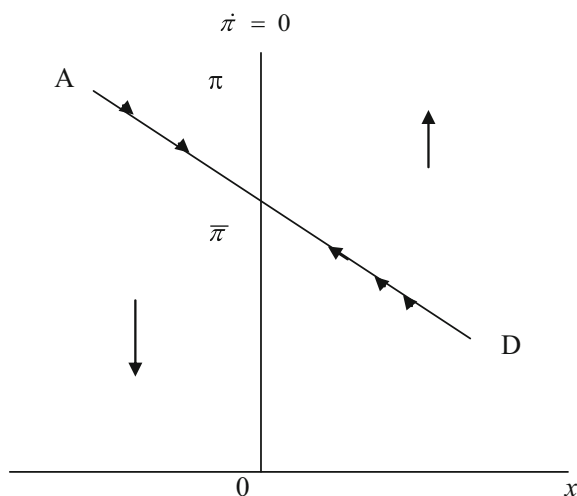
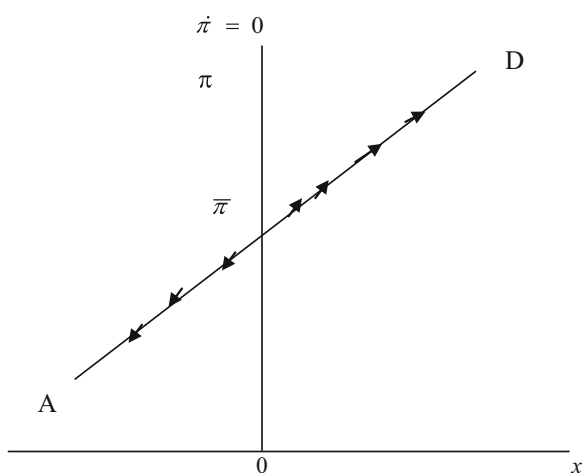


Fig. 7.2 The phase diagram for the Keynesian model with inflation inertia: $\phi < 0$



dynamical systems, is a repulsor rather than an attractor, as it was in the previous case. The model's instability is associated with the fact that the Central Bank does not respond properly in the presence of a deviation of the inflation rate from the inflation target. For each 1% that the inflation rate deviates from the target, the nominal interest rate increases by less than 1%. This kind of response from the Central Bank causes the economy's real interest rate to lie below the natural interest rate, heating up the economy and leading to an increased inflation rate.

Experiment

Figure 7.3 describes a monetary policy experiment in which the Central Bank, with no previous announcement, reduces the inflation target to $\bar{\pi}_1$ from $\bar{\pi}_0$.

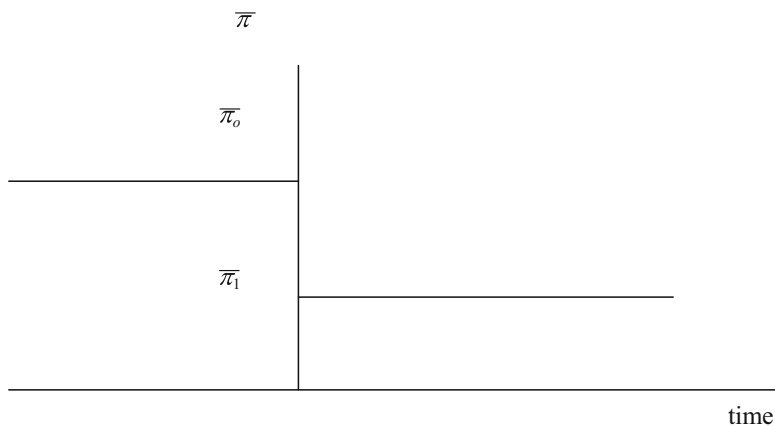


Fig. 7.3 Experiment: inflation target change

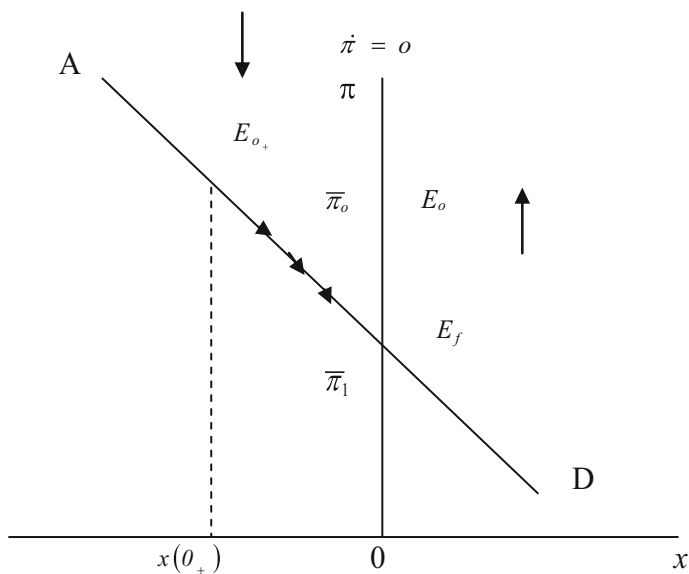


Fig. 7.4 The effects of an unanticipated permanent decrease in the inflation target

We analyze the experiment for the stable model, where the parameter ϕ is positive. The phase diagram in Fig. 7.4 is identical to the phase diagram in Fig. 7.2. The model's new long-run equilibrium is determined by the point where the ordinate of the model's new inflation target intercepts the abscissa of an output gap equal to zero, which corresponds to full-employment output. The point E_0 represents the model's initial equilibrium. The inflation rate is inertial in this model, which means that it does not change instantly. When the target changes,

the inflation rate remains equal to the previous target and real output drops because of the Central Bank's nominal interest rate increase. The economy dips into recession, the inflation rate starts to drop and the economy follows a path of convergence to the new long-run equilibrium, as the arrows in Fig. 7.4 show.

Demand Shock

The monetary policy rule of setting the interest rate assumes that the Central Bank knows the natural interest rate at every point in time. This is not always the case. When a demand shock occurs, due to changes in the private sector's behavior or in the fiscal policy, the natural interest rate changes.

Assume that the real long-run interest rate in the Central Bank's rule is not always equal to the natural interest rate. That is:

$$i = \bar{r}^{BC} + \pi + \phi(\pi - \bar{\pi}) + \phi x. \quad (7.7)$$

The aggregate demand curve, obtained by substituting the monetary policy rule's interest rate in the IS curve, is:

$$\pi = \bar{\pi} - \frac{(1 + \alpha\theta)}{\alpha\phi}x + \frac{1}{\phi}(\bar{r} - \bar{r}^{BC}). \quad (7.8)$$

When $\bar{r} = \bar{r}^{BC}$, we have the aggregate demand curve seen in Fig. 7.4.

Suppose that a positive demand shock occurs that causes $\bar{r} > \bar{r}^{BC}$. Figure 7.5 shows the consequences of such a demand shock. The aggregate demand curve shifts from A_0D_0 to A_1D_1 . Because of inflation rate inertia, the inflation rate remains

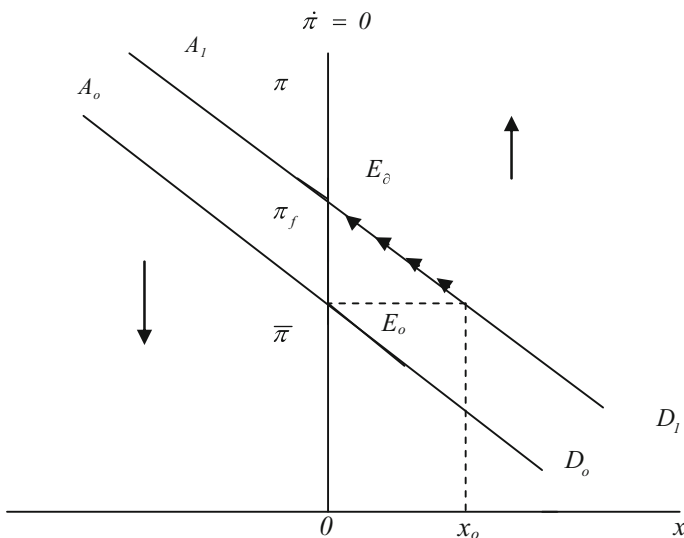


Fig. 7.5 The effects of a positive demand shock

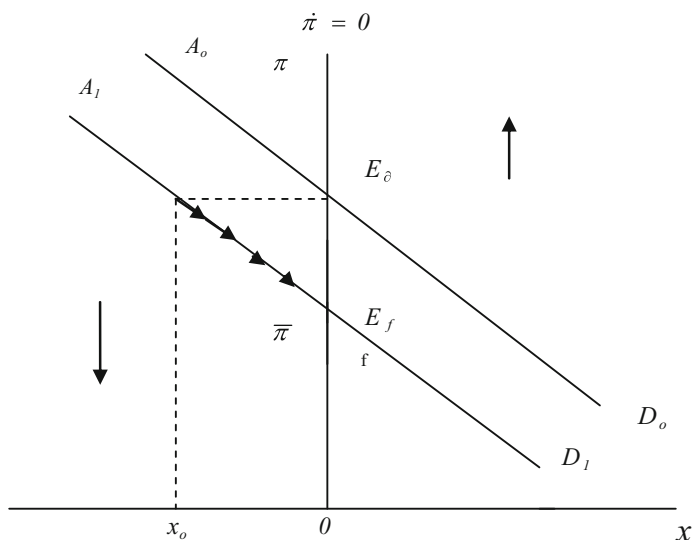


Fig. 7.6 The effects of a negative demand shock

unchanged at the time of the shock, but real output increases. The increase in real output brings about an inflation rate increase. The long-run equilibrium will occur at a higher inflation rate, above the target set by the Central Bank.

Figure 7.6 shows the dynamics when a negative demand shock occurs ($\bar{r} < \bar{r}^{BC}$) and the Central Bank does not adjust the nominal interest rate according to the new natural interest rate. Real output drops below potential output. The inflation rate then starts to decrease, converging to the new long-run equilibrium with an inflation rate below the Central Bank's target.

The conclusion is that the Central Bank must completely neutralize demand shocks to keep the economy at its full-employment equilibrium and with an inflation rate equal to the selected target. In practice, this is not always the case, as it assumes that the Central Bank knows the natural interest rate at every point in time.

Supply Shock

Supply shocks may be represented as changes in potential output. Figure 7.7 assumes a negative supply shock with reduced potential output. Such a shock reduces real output but increases the inflation rate, in a process known as stagflation – a combination of stagnation and inflation. The economy converges to a higher inflation rate, above the Central Bank's target rate. If the supply shock is permanent, the Central Bank will have to increase the interest rate to bring the economy back to the former inflation target. Otherwise, the target will be missed.

Figure 7.8 shows the dynamics of a supply shock that increases potential output. This shock increases real output and reduces the inflation rate. Inflation will converge to a rate below the inflation target. In this case, the Central Bank would have to reduce the nominal interest rate for the economy to return to the previous inflation target.

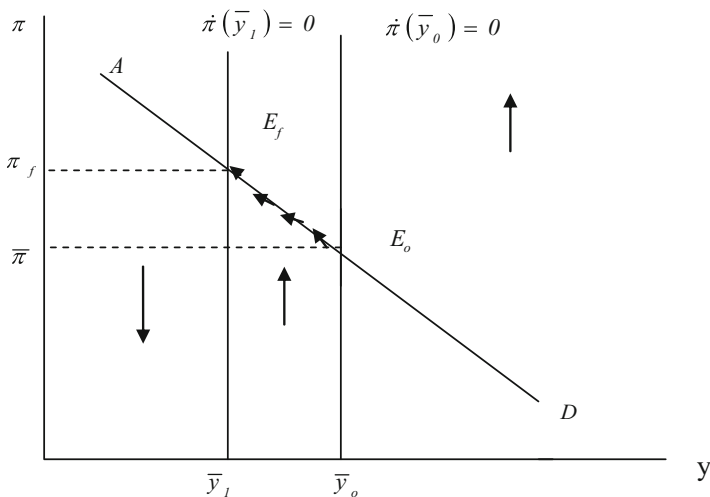


Fig. 7.7 The effects of a negative supply shock

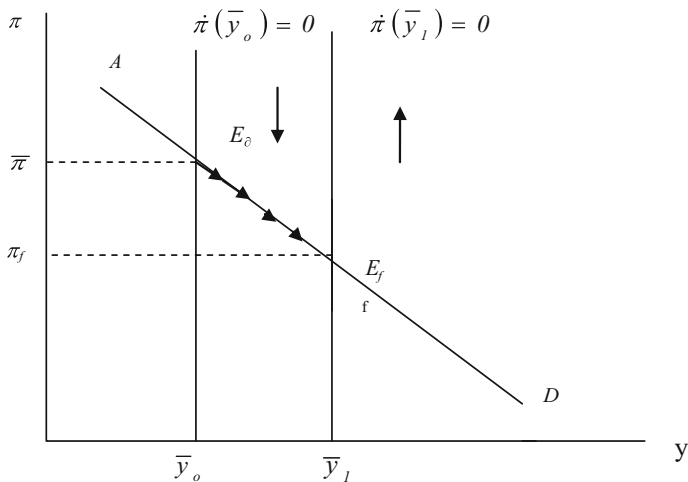


Fig. 7.8 The effects of a positive supply shock

Coordination of Monetary and Fiscal Policies

Consider a short-run model specified by an IS curve, a Phillips curve (PC), and two economic policy rules: one monetary (MPR) and one fiscal (FPR). That is:

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \phi > 0, \theta > 0, \quad (7.9)$$

$$\text{FPR: } f - \bar{f} = -\varphi x, \varphi > 0. \quad (7.10)$$

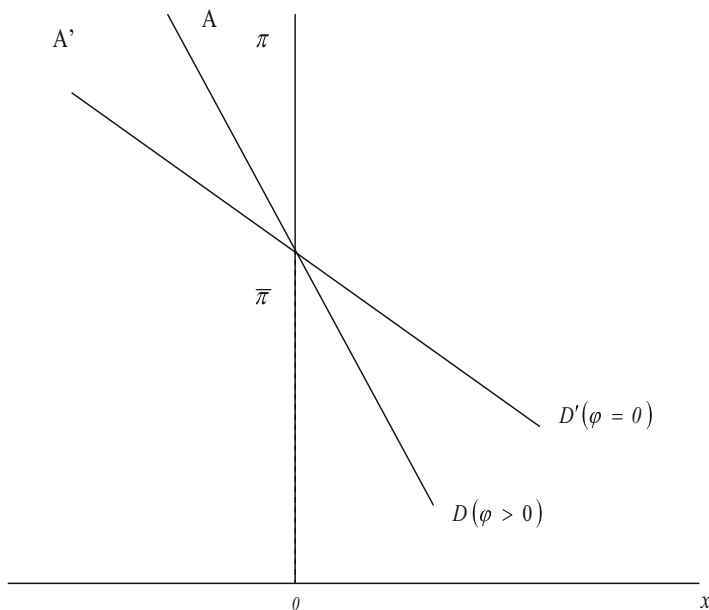


Fig. 7.9 Aggregate demand and the fiscal policy rule

The monetary policy rule is the Taylor rule. The fiscal policy rule is a counter-cyclical rule. When output exceeds full-employment output (the potential output), the public deficit will be below the full-employment deficit ($\bar{f}$) (the structural deficit). On the other hand, when the economy is in recession, with output below potential, the public deficit will exceed the structural deficit. The MPR's real interest rate gap and the FPR's public deficit gap can be substituted in the IS curve. That is:

$$x = -\alpha\phi(\pi - \bar{\pi}) - \alpha\theta x - \beta\phi x.$$

This equation may be rewritten as:

$$\pi = \bar{\pi} - \left(\frac{1 + \alpha\theta + \beta\phi}{\alpha\phi} \right) x. \quad (7.11)$$

Figure 7.9 shows the phase diagram. In the absence of a fiscal policy rule, the parameter ϕ equals zero. In this case, the curve is represented in Fig. 7.9 by the letters $A'D'$. In the presence of a countercyclical fiscal policy $\phi > 0$, the curve is represented by the letters AD . Note that the presence of a counter-cyclical fiscal policy makes the AD curve more sloped, with a greater angular coefficient than that found in the absence of a counter-cyclical fiscal policy.

Experiment

Consider a monetary policy experiment in which the Central Bank reduces the inflation target to $\bar{\pi}_1$ from $\bar{\pi}_0$, as Fig. 7.10 shows.

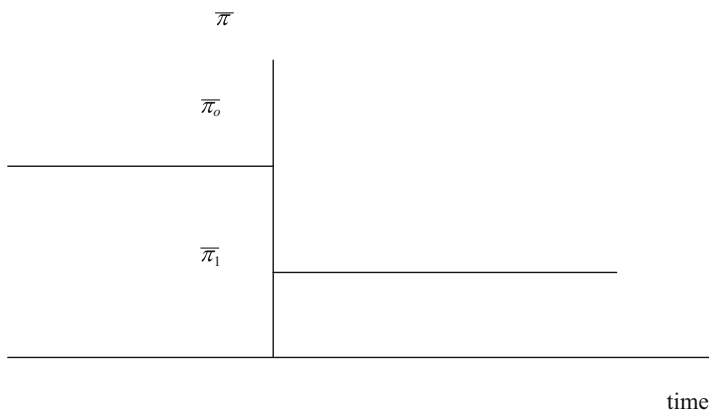


Fig. 7.10 An unanticipated permanent decrease in the inflation target

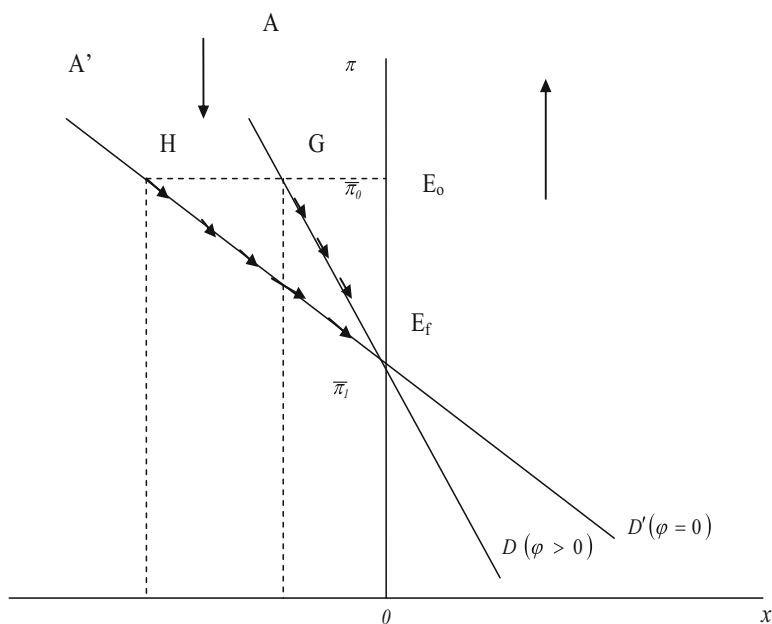


Fig. 7.11 The effects of an unanticipated permanent decrease in the inflation target

Figure 7.11 describes the dynamical adjustment process. The new equilibrium will be at point E_f . The initial equilibrium point is E_o . Assume inflation rate inertia, that is, the inflation rate does not change instantly (i.e., the inflation rate is not a jump variable). In the presence of a counter-cyclical fiscal policy, the economy experiences a drop in real output when the Central Bank raises the nominal interest rate, moving from E_o to G . Thereafter, the economy converges to full employment on the GE_f path indicated by the arrows.

When there is no coordination of the monetary and fiscal policies, the fiscal policy's coefficient φ equals zero, corresponding to the $A'D'$ curve in Fig. 7.11. In this case, at the time of the nominal interest rate change, interest rate inertia causes real output to decrease from E_o to H . The economy follows the convergence path HE_f , until full employment and the new inflation target are reached.

Figure 7.11 shows a simple visualization of the benefits of the fiscal policy rule: output loss is greatly reduced when a counter-cyclical fiscal policy rule is in place. The worst-case scenario would be that of a pro-cyclical fiscal policy rule. Under these circumstances, at some point the Central Bank and the Treasury would be pulling in opposite directions.

7.2 Keynesian Model: Without Inflation Inertia

The Keynesian model has rigid prices, no inflation rate inertia, and an interest rate rule. This model includes three equations: the IS curve, the Phillips curve, and the Taylor rule. The specification of the Phillips curve assumes a sticky price level, but that the inflation rate may change instantaneously, that is, no inflation rate inertia, but the price level is a predetermined variable. The three equations and the model's initial condition are as follows:

$$\text{IS: } x = -\alpha(r - \bar{r}), \quad (7.12)$$

$$\text{PC: } \dot{\pi} = -\delta x, \quad (7.13)$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \quad (7.14)$$

$$\text{IC: Given } p(0). \quad (7.15)$$

Dynamical System

The model may be reduced to two equations when the gap between the real interest rate and the monetary policy rule's natural rate is substituted in the IS curve equation. The model's two equations are:

$$\begin{cases} \dot{\pi} = -\delta x \\ \pi = \bar{\pi} - \frac{(1 + \alpha\theta)x}{\alpha\phi} \end{cases} \quad (7.16)$$

Figure 7.12 shows the model's phase diagram. The vertical axis measures the inflation rate and the horizontal axis measures the output gap. In the long-run, when inflation is constant ($\dot{\pi} = 0$), real output equals potential output ($x = 0$). When real output exceeds potential output, inflation acceleration is negative, and in the opposite case inflation acceleration is positive, as the arrows in Fig. 7.12 show. The aggregate demand curve has a negative slope, and for an output equal to the potential output, the inflation rate equals the Central Bank's inflation target.

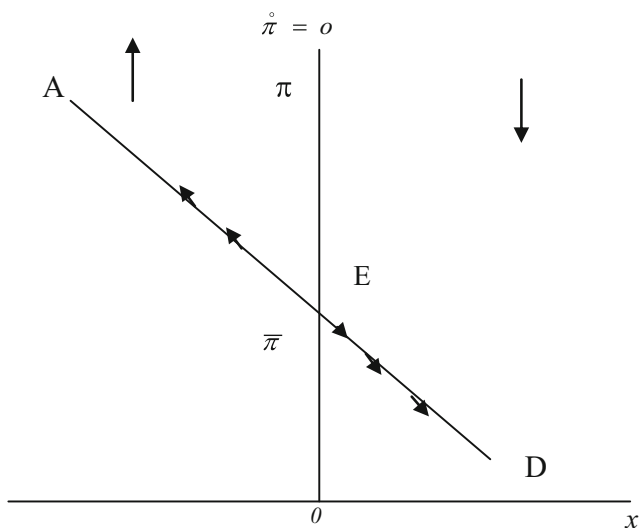


Fig. 7.12 The phase model for the Keynesian model without inflation inertia

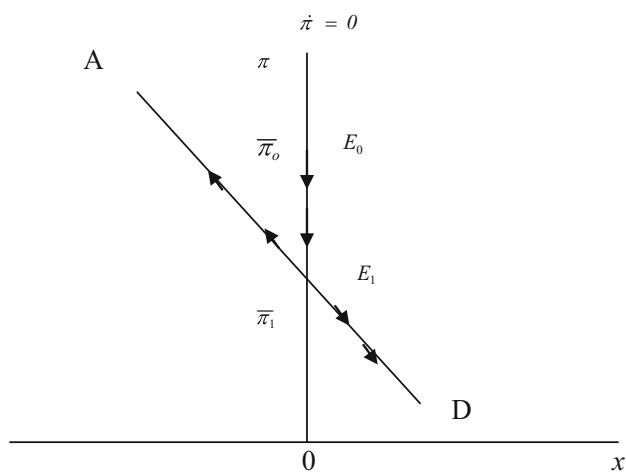


Fig. 7.13 The effects of a decrease in the inflation target in a Keynesian model without inflation inertia

Figure 7.12 shows that the model is unstable because if the economy is off its long-term equilibrium, it will not return to that equilibrium. How would this economy's dynamics operate should the Central Bank reduce the inflation target to $\bar{\pi}_1$ from $\bar{\pi}_0$, as in the experiment described in Fig. 7.10? Figure 7.13 answers this question, showing that the inflation rate would immediately adjust to the new target,

and real output would continue to equal potential output. Disinflation would be painless because no recession would ensue, as it would in the model presented in Sect. 1, where inflation rate inertia is present.

7.3 New Keynesian Model

The new Keynesian model was developed to replace the traditional model that relies on the IS and Phillips curves. The main argument justifying its specification is that it is a log-linear approximation of a microfounded model. Therefore, such a model would be immune to the Lucas critique that the parameters of traditional models change when there is change in economic policy rules.

The new Keynesian model, then, is specified by a Phillips curve, an IS curve, and a monetary policy rule. The Phillips curve in continuous time, is:

$$\dot{\pi} = -\delta x, \delta > 0.$$

This curve does not include a long-run tradeoff between inflation (π) and the output gap (x). The price level is predetermined and the price adjustments are not synchronized, with a small proportion of firms adjusting their prices at every moment.

The IS curve deduced from the Euler equation – the consumer intertemporal optimization – assumes that the acceleration of the output gap depends on the gap between the real interest rate ($i - \pi$) and the natural interest rate ($\bar{r}$). That is:

$$\dot{x} = \sigma(i - \pi - \bar{r}), \quad \sigma > 0,$$

where σ is the elasticity of intertemporal substitution.

The monetary policy rule sets the nominal interest rate by taking into account the natural interest rate, the inflation rate, the inflation gap between the inflation rate, the Central Bank's inflation target, and the output gap, according to,

$$i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x.$$

The parameter ϕ measures the nominal interest rate response to an inflation rate change. According to the Taylor rule, this parameter is positive, as the nominal interest rate increases by more than the change in the inflation rate. When this parameter is negative, the Central Bank's reaction while changing the nominal interest rate is less than the change in inflation. The sign of the parameter ϕ influences the model's stability, as seen in the previous section, and as will be demonstrated briefly for this section's model.

The new Keynesian model has three equations and the initial condition that the price level is given. That is:

$$\text{IS: } \dot{x} = \sigma(i - \pi - \bar{r}), \quad \sigma > 0, \quad (7.17)$$

$$\text{PC: } \dot{\pi} = -\delta x, \quad \delta > 0, \quad (7.18)$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \quad (7.19)$$

$$\text{IC: Given } p(0), \text{ free } \pi(0) \text{ and } x(0). \quad (7.20)$$

Algebra

Substituting the monetary policy rule in the IS curve yields the differential equation of the output gap:

$$\dot{x} = \sigma\phi(\pi - \bar{\pi}) + \sigma\theta x.$$

Dynamical System

The dynamical system contains this equation and the inflation rate acceleration equation:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & -\delta \\ \sigma\phi & \sigma\theta \end{bmatrix} \begin{bmatrix} \pi - \bar{\pi} \\ x \end{bmatrix}. \quad (7.21)$$

In this model, the output gap and the inflation rate are not predetermined variables. They are jump variables, and their initial values are free. This differential equations system has the following Jacobian matrix:

$$J = \begin{bmatrix} \frac{\partial \dot{\pi}}{\partial \pi} & \frac{\partial \dot{\pi}}{\partial x} \\ \frac{\partial \dot{x}}{\partial \pi} & \frac{\partial \dot{x}}{\partial x} \end{bmatrix} = \begin{bmatrix} 0 & -\delta \\ \sigma\phi & \sigma\theta \end{bmatrix}.$$

This matrix's determinant is equal to:

$$|J| = \sigma\phi\delta.$$

This determinant's sign may be either negative or positive, depending on the sign of the monetary policy rule's coefficient ϕ . Where ϕ is positive, so is the determinant, and the trace of the matrix J is positive:

$$\text{tr}J = \sigma\theta.$$

The system is therefore unstable in this case. The conclusion is that a Taylor monetary policy rule produces an unstable system. The model's solution, however, is unique. The phase diagram in Fig. 7.14 describes the system's dynamics.

When the parameter ϕ is negative, the differential equations system has a saddle point as shown by the phase diagram in Fig. 7.15. The line where the output gap remains constant has a positive slope and the one where the inflation rate does not

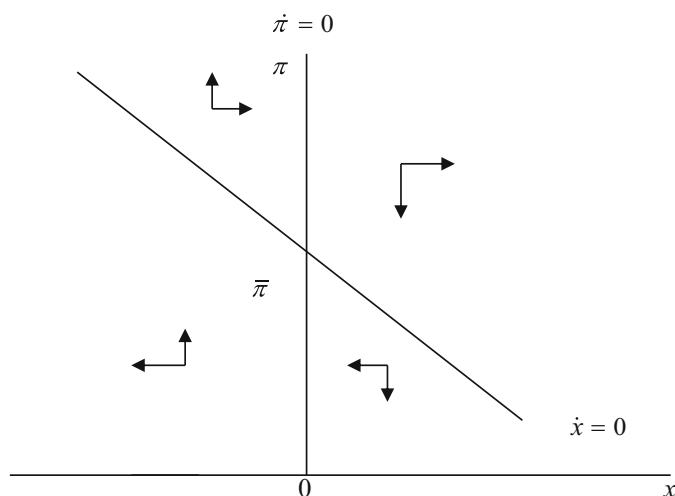


Fig. 7.14 The phase diagram for the New Keynesian model: $\phi > 0$

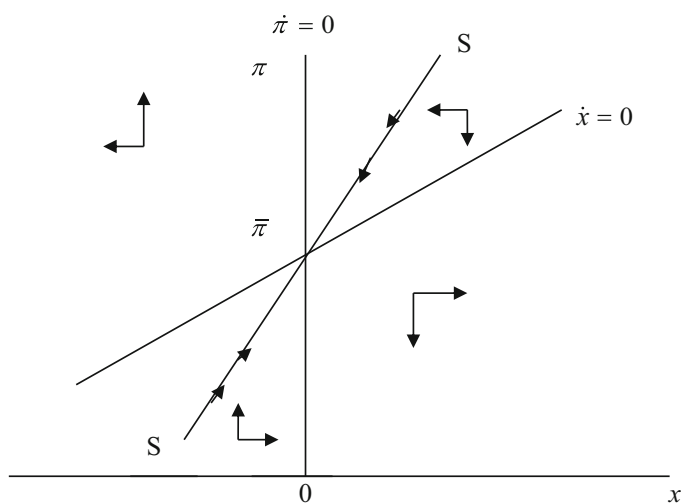


Fig. 7.15 The phase diagram for the New Keynesian model: $\phi < 0$

change is vertical, as in the previous case. The SS saddle path has a positive slope, and this is the only path of convergence to long-run equilibrium.

Experiment

Consider the monetary policy experiment described in Fig. 7.16, in which the Central Bank announces that at instant T , in the near future, the inflation target will be reduced to $\bar{\pi}_1$ from $\bar{\pi}_0$. There is no uncertainty in this model and, therefore, the announced policy will be implemented.

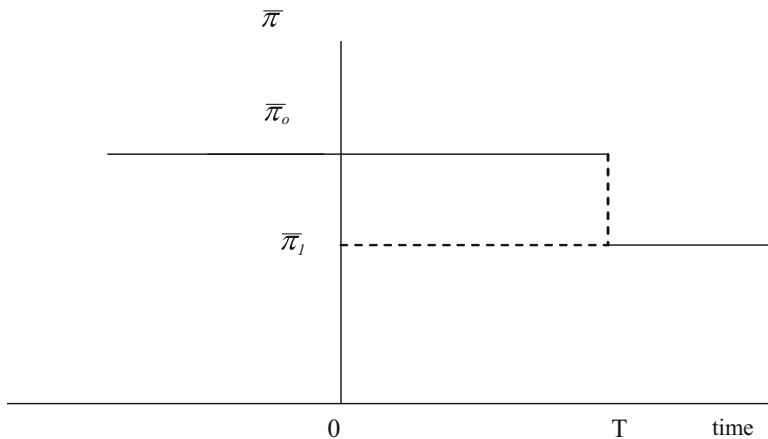


Fig. 7.16 An anticipated permanent decrease in the inflation target

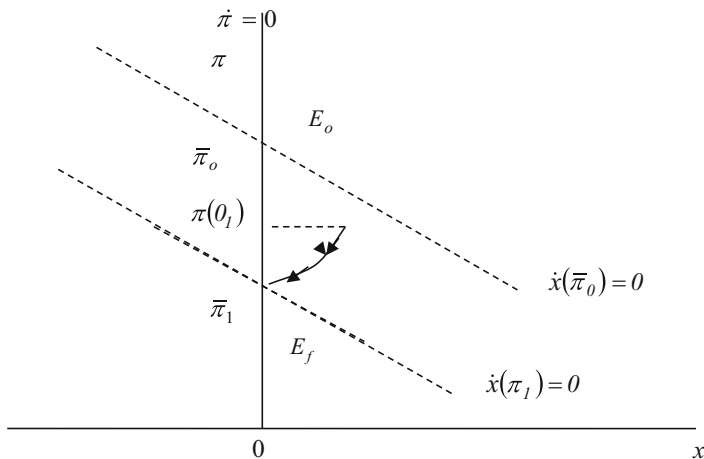


Fig. 7.17 The effects of an anticipated permanent decrease in the inflation target: $\phi > 0$

We will analyze the experiment using both the unstable model, where the parameter ϕ is positive, and the saddle-path model, with a negative parameter ϕ . At time zero, the monetary policy of inflation target reduction will be implemented instantaneously. This particular case corresponds to the experiment of a policy change that the public does not anticipate.

When the parameter ϕ is positive, the model is unstable. The phase diagram in Fig. 7.17 shows what happens in this economy as soon as the monetary policy is announced. In the new Keynesian model, the price level is a predetermined variable, but the inflation rate is a jump variable. Therefore, at the initial moment inflation decreases and real output increases. The economy then gradually converges to the new long-run equilibrium. When the change is implemented immediately, with no

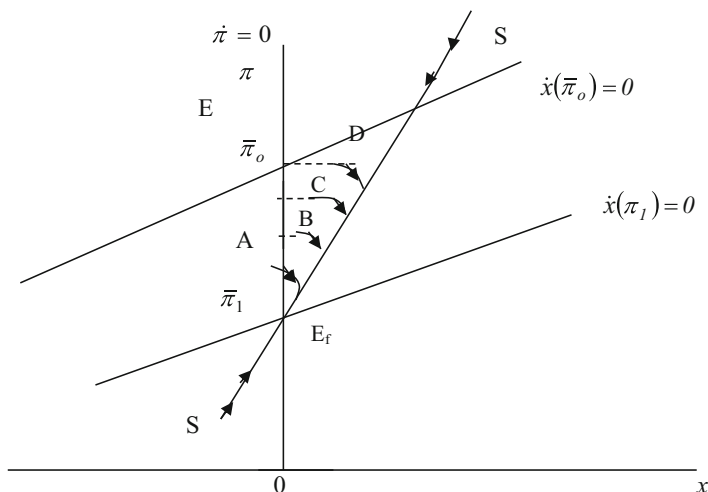


Fig. 7.18 The effects of an anticipated permanent decrease in the inflation target: $\phi < 0$

advance notice ($T = 0$), real output remains at the full-employment level and the inflation rate changes instantly to the Central Bank's new target.

When the monetary policy rule's ϕ parameter is negative, real output and the inflation rate change at the time of the announcement of the new policy, with inflation decreasing and real output increasing, as seen in point B of Fig. 7.18's phase diagram. At instant T , the economy's path finds the new saddle, with inflation and real output equal to $\pi(T)$ and $y(T)$, respectively, and then converging to the new long-run equilibrium. This phase diagram shows four solutions for the model (points A, B, C, and D). However, countless solutions exist, all with the same characteristics of those seen in Fig. 7.18. In this experiment, the previously announced program to combat inflation takes place at no social cost. Indeed, reducing the inflation rate heats up the economy, with a temporary real output gain.

As time interval T tends to zero, the period of announcement of the monetary policy change decreases. In the extreme case where T equals zero, the policy change is unanticipated by society. It is easily seen that, under these circumstances, the inflation rate instantly drops to the new level set by the Central Bank, even as the economy remains at full employment, with real output equal to potential output. The reduction in inflation is painless, and the Central Bank's announcement of the new target is enough to permanently affect the inflation rate.

In the new Keynesian model, a reduced Central Bank inflation target that is unanticipated by society produces an immediate reduction of the inflation rate to the new target, at no social cost, as real output remains at its full-employment level.

When the Central Bank announces an inflation target reduction in advance, the announcement causes the inflation rate to decrease and real output to increase. The economy gradually converges to the new target, with real output exceeding the

full-employment output, leading to a temporary social benefit. Countless solutions with these characteristics exist.

To what extent do such events take place in the real world? Empirical evidence from anti-inflation policies, whether announced in advance or not, in countries with traditionally low inflation rates, reject the new Keynesian model's prediction, as a reduction in inflation generally comes hand-in-hand with recession. On the other hand, for anti-hyperinflation programs, the facts do not contradict the new Keynesian model's predictions, as hyperinflations have ended at no cost, or even with real output gains. These two pieces of evidence would be consistent with a model in which the Phillips curve had both inertial and future inflation components. That is:

$$\pi(t) = \omega\pi(t-h) + (1-\omega)\pi(t+h) + \varphi x(t)],$$

where ω is the weight of inertia and h is the relevant time interval for economic decisions. The first-order expansion of the lagged inflation is given by $\pi(t-h) = \pi(t) + \dot{\pi}(t)(t-h-t)$ and the first-order expansion of the future inflation is equal to $\pi(t+h) = \pi(t) + \dot{\pi}(t)(t+h-t)$. Substituting these two expressions in the Phillips curve yields the differential equation of the inflation rate:

$$\dot{\pi} = \frac{\varphi}{(2\omega-1)h}x, \omega \neq 1/2, \quad (7.22)$$

and $x = 0$, when $\omega = 1/2$. Where the weight is .5, inflation acceleration is independent from the output gap. This weight is the critical value, because when it is over 50%, the coefficient of the Phillips curve will be positive, and when it is under 50%, the coefficient will be negative, as in the new Keynesian model. The theoretical question raised by this formulation would be to explain how the model's bifurcation occurs, that is, how this weight (ω) varies with the inflation rate itself.

The new Keynesian model has been used to analyze the monetary policy rules of countries with annual inflation rates below one digit. Although this model does have theoretical foundations, it is rejected by the empirical evidence of countries with traditionally low inflation rates. Therefore, because the model's logic lacks the factual backing, this kind of analysis must be taken with caution, or as just one part of the scientific process investigating the microfoundations of models to support the Central Banks' monetary policy decisions. One important issue to address as a part of this effort is to understand the mechanism by which the initial inflation rate ceases to be a predetermined model variable and becomes an endogenous variable.

7.4 Encompassing Keynesian Model

This section introduces a Keynesian model that encompasses the traditional Keynesian model (K), the new Keynesian model (NK), and the new Keynesian model with information rigidity (NKIR). Each Keynesian model is a particular case of the encompassing model, depending on the values of the parameters and on the

variables' initial conditions. The model has three equations: an IS curve, a Phillips curve (PC), and a monetary policy rule (MPR).

Keynesian IS Curve

The IS curve specifies the relationship between the output gap (x), the lagged output gap, and the gap between the real interest rate (r) and the natural interest rate ($\bar{r}$):

$$x_t = (1 + \gamma)x_{t-1} + \alpha(r_t - \bar{r}_t), \quad -1 < \gamma < 0, \alpha < 0.$$

This equation may be rewritten as:

$$\Delta x_t = \gamma x_{t-1} + \alpha(r_t - \bar{r}_t).$$

Therefore, the change in the output gap depends on the lagged output gap and on the interest rate gap. Figure 7.19 shows that the change in the output gap and the interest rate gap are negatively correlated. The continuous-variables version of the IS curve is given by:

$$\dot{x} = \gamma x + \alpha(r - \bar{r}).$$

New Keynesian IS Curve

The new Keynesian IS curve is derived from the Euler equation. According to this equation, the growth rate of consumption ($\dot{c}$) depends on the gap between the real interest rate and the rate of time preference (ρ):

$$\dot{c} = \sigma(r - \rho),$$

where σ is the intertemporal elasticity of substitution of consumption.

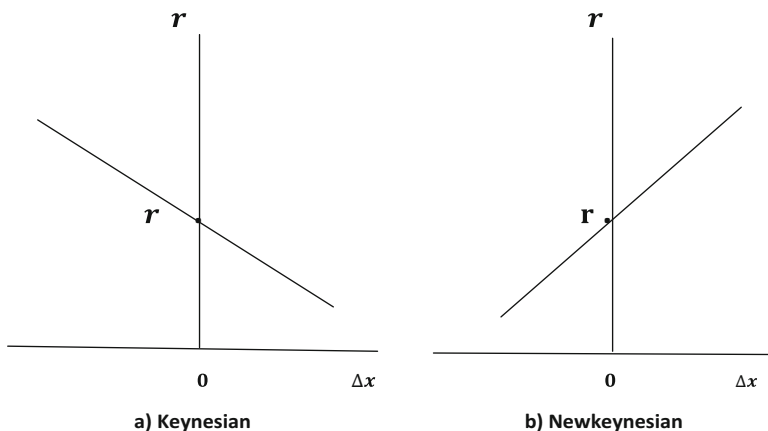


Fig. 7.19 The correlation between the change in output gap and the interest rate in the Keynesian and the New Keynesian models

The output gap equals the difference between output (y) and potential output ($\bar{y}$): $x = y - \bar{y}$. Taking the derivative of this equation with respect to time yields: $\dot{x} = \dot{y} - g, \dot{\bar{y}} = g$, where g is the growth rate of potential output. Assume a rather simple model where consumption and output are equal. Therefore, the rates of growth of the two variables are equal: $\dot{y} = \dot{c}$. Combining this expression with the definition of the output gap and the Euler equation yields the new Keynesian IS curve:

$$\dot{x} = \sigma(r - \rho) - g = \sigma\left(r - \rho - \frac{g}{\sigma}\right) = \sigma(r - \bar{r}), \bar{r} = \rho + \frac{g}{\sigma}.$$

The natural interest rate $\bar{r}$ depends on three parameters: the rate of time preference, the growth rate of potential output, and the intertemporal elasticity of substitution. The new Keynesian IS curve may be written as:

$$\dot{x} = \alpha(r - \bar{r}), \alpha = \sigma.$$

Figure 7.19 shows that, according to the new Keynesian IS curve, the change in output gap and the interest rate gap are positively correlated. The new Keynesian model's monetary transmission mechanism assumes that the nominal interest rate, as set by the Central Bank, changes the real interest rate in the short term and then affects the (expected) rate of growth of consumption. Therefore, a negative interest rate gap ($r < \bar{r}$) decreases the growth rate of the output gap, and a positive interest rate gap ($r > \bar{r}$) increases the growth rate of the output gap, as shown by Fig. 7.19.

Keynesian Phillips Curve

The accelerating Phillips curve assumes that inflation depends on the past, in a backward-looking process. The simpler model assumes that the acceleration of the inflation rate is proportional to the output gap. That is:

$$\Delta\pi_t = \kappa x_t.$$

The continuous-variables version of this equation is:

$$\dot{\pi} = \kappa x.$$

Figure 7.20 shows that, according to the Keynesian Phillips curve, changes in the inflation rate and the output gap are positively correlated.

New Keynesian Phillips Curve

The new Keynesian Phillips curve's inflation rate (π) depends on the expected inflation rate for the next period and on the output gap according to:

$$\pi_t = \beta\pi_{t+1} + \kappa_1 x_t = \beta(\pi_{t+1} - \pi_t) + \kappa_1 x_t + \beta\pi_t.$$

It thus follows that the acceleration of the inflation rate depends on the inflation rate and the output gap:

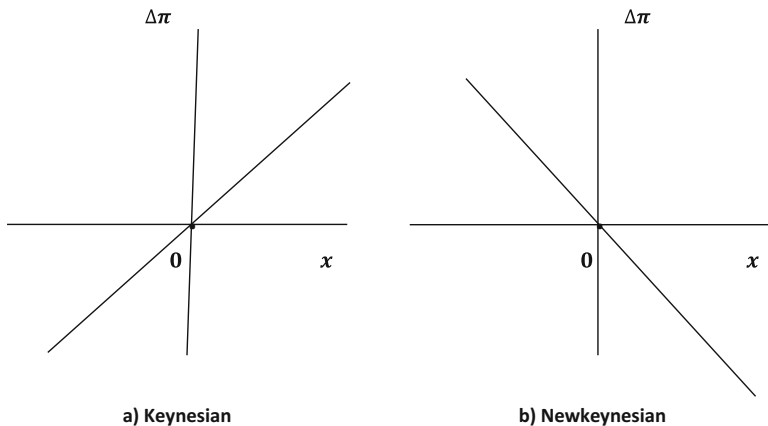


Fig. 7.20 The correlation between the change in the inflation rate and the output gap in the Keynesian and the New Keynesian model

$$\pi_{t+1} - \pi_t = \frac{1 - \beta}{\beta} \pi_t - \frac{\kappa_1}{\beta} x_t = \rho \pi_t + \kappa x_t, \kappa = -\frac{\kappa_1}{\beta} < 0,$$

where $\beta = 1/(1 + \rho)$ and ρ is the consumer's rate of time preference. This equation may be written in continuous time as follows:

$$\dot{\pi} = \rho \pi + \kappa x,$$

where $\dot{\pi} = d\pi/dt$. Note that this Phillips curve is not vertical in the long run if the rate of time preference is not zero. Figure 7.20 shows that, according to the new Keynesian Phillips curve, changes in the inflation rate and the output gap are negatively correlated.

New Keynesian Phillips Curve with Information Rigidity

The Phillips curve with information rigidity (NKIR) is specified as follows:

$$\pi_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j E_{t-1-j}(\pi_t + \alpha \Delta x_t) + \frac{\alpha \lambda}{1 - \lambda} x_t.$$

The notation is the same as in Mankiw and Reis (2002). The λ parameter is the probability that the firm will not, in period t , update its information set up until period $t - 1$, and α is the coefficient of the output gap in the equation that sets the firm's optimum price. The letter E represents the expected value of the variable, conditioned by the information available at the time that its index specifies. According to this specification, inflation depends on the output gap, on past expectations for the present inflation, and the change in the output gap. Let the lag operator be $LX_t = X_{t-1}$. One may then write $L^j E_{t-1} = E_{t-1-j}$. Using this operator, the Phillips curve with information rigidity is given by:

$$\pi_t = \lambda \sum_{j=0}^{\infty} [(1-\lambda)L]^j E_{t-1}(\pi_t + \alpha \Delta x_t) + \frac{\alpha \lambda}{1-\lambda} x_t.$$

It is straightforward to show that $\sum_{j=0}^{\infty} [(1-\lambda)L]^j = \frac{\lambda}{1-(1-\lambda)L}$. Therefore, taking this expression into account, one obtains the following equation:

$$\pi_t = \frac{\lambda}{1-(1-\lambda)L} E_{t-1}(\pi_t + \alpha \Delta x_t) + \frac{\alpha \lambda}{1-\lambda} x_t.$$

The rational expectations assumption implies that the expected value of a variable equals the variable's observed value minus a stochastic error uncorrelated with any information available at the time the forecast is made. Under this rational expectations assumption, the new Keynesian Philips curve with information rigidity may be expressed as:

$$\Delta \pi_t = \frac{\alpha \lambda^2}{(1-\lambda)^2} x_{t-1} + \frac{\alpha \lambda (2-\lambda)}{(1-\lambda)^2} \Delta x_t.$$

Inflation acceleration depends on the gap's output level and its rate of change. The continuous-variables specification of this Phillips curve is therefore given by:

$$\dot{\pi} = \kappa x + \phi \dot{x}.$$

Encompassing Keynesian Model

The three Keynesian models for a closed economy may be specified by the following equations system:

$$\text{IS: } \dot{x} = \gamma x + \alpha(r - \bar{r}), \quad (7.23)$$

$$\text{PC: } \dot{\pi} = \rho \pi + \kappa x + \phi \dot{x}, \quad (7.24)$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \quad (7.25)$$

where i is the nominal interest rate and $\bar{\pi}$ is the Central Bank's inflation target. The monetary rule is a Taylor rule. The monetary policy's parameters, ϕ for the inflation gap and θ for the output gap, are both positive.

Table 7.1 contains the parameters' values and the initial conditions for each model. All models assume sticky prices. The K and NKIR models further assume inflation inertia, which means that the initial inflation rate is given. On the other hand, the inflation rate is a jump variable in the NK model because it does not assume inflation inertia.

Dynamical System

Combining the monetary policy rule and the IS curve yields a linear differential equation for the output gap. Substituting the result in the Phillips curve produces a

Table 7.1 Encompassing Keynesian model

Models	Parameters					Initial
	γ	α	ρ	κ	φ	Conditions
Keynesian (K)	<0	<0	0	>0	0	$p(0), \pi(0)$
New Keynesian (NK)	0	>0	>0	<0	0	$p(0)$
New Keynesian Information Rigidity (NKIR)	0	>0	0	>0	>0	$p(0), \pi(0)$

Table 7.2 Stability of Keynesian models

Models	$ J $	$tr J$	Stability
Keynesian (K)	$-\alpha\phi\kappa > 0$	$\gamma + \alpha\theta < 0$	Stable
New Keynesian (NK)	$\rho\alpha\theta - \alpha\phi\kappa > 0$	$\rho + \alpha\theta > 0$	Unstable
New Keynesian Information Rigidity (NKIR)	$-\alpha\phi\kappa < 0$	$\alpha\theta + \alpha\varphi\phi > 0$	Saddle point

differential equation for the inflation rate. Thus, the liner system of first-order differential equations is:

$$\dot{\pi} = \rho\pi + (\kappa + \gamma\varphi + \alpha\varphi\theta)x + \alpha\varphi\phi(\pi - \bar{\pi}), \quad (7.26)$$

$$\dot{x} = (\gamma + \alpha\theta)x + \alpha\phi(\pi - \bar{\pi}). \quad (7.27)$$

This system's Jacobian matrix is:

$$J = \begin{bmatrix} \frac{\partial \dot{\pi}}{\partial \pi} & \frac{\partial \dot{\pi}}{\partial x} \\ \frac{\partial \dot{x}}{\partial \pi} & \frac{\partial \dot{x}}{\partial x} \end{bmatrix} = \begin{bmatrix} \rho + \alpha\varphi\phi & \kappa + \gamma\varphi + \alpha\varphi\theta \\ \alpha\phi & \gamma + \alpha\theta \end{bmatrix}.$$

The matrix's determinant and trace are equal to:

$$|J| = \rho\gamma + \rho\alpha\theta - \alpha\phi\kappa,$$

$$tr J = \rho + \alpha\varphi\phi + \gamma + \alpha\theta.$$

Table 7.2 shows the three models' stability. Model K is stable because the determinant is positive and the trace is negative. Model NK is unstable because both the determinant and the trace are positive. The NKIR model has a saddle point because the determinant is negative.

Figure 7.21 shows the K model's phase diagram, with the inflation rate on the vertical axis and the output gap on the horizontal axis. In the steady state, the inflation rate is on target and the output gap equals zero. The points where $\dot{\pi} = 0$ are vertical and those where $\dot{x} = 0$ are negatively sloped. The arrows show how the inflation rate and the output gap move over time. The inflation rate is a predetermined variable because inertia is present; that is, the initial inflation rate is a given for the problem.

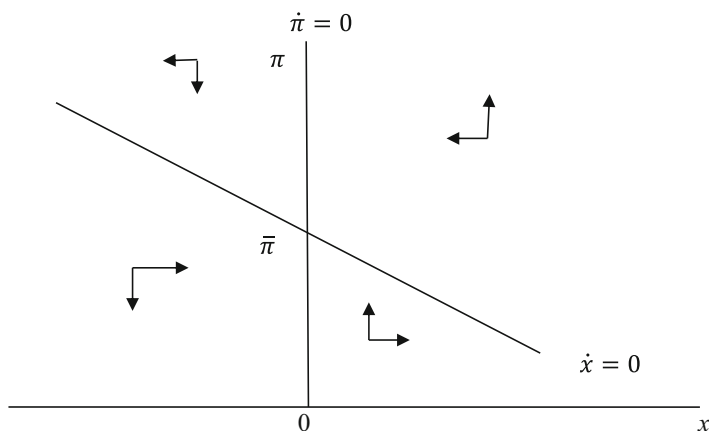


Fig. 7.21 The phase diagram of the Keynesian model

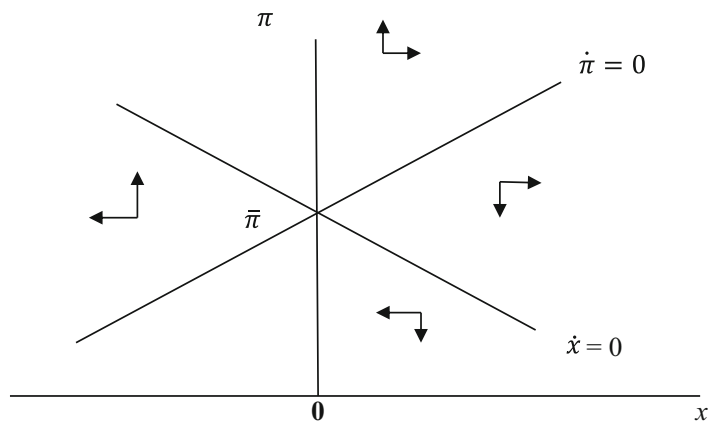


Fig. 7.22 The phase diagram of the New Keynesian model

Figure 7.22 shows the NK model's phase diagram. In this case, the Phillips curve is vertical in the long run. The line $\dot{\pi} = 0$ is positively sloped and $\dot{x} = 0$ is negatively sloped. In the steady state, the inflation rate is equal to the inflation target and the output gap equals zero. The arrows indicate the model's dynamics. The system is unstable and the inflation rate is a jump variable.

Figure 7.23 shows the NKIR model's phase diagram. The slope of the $\dot{\pi} = 0$ line is greater than that of $\dot{x} = 0$. Both are negatively sloped. At equilibrium, the inflation rate is equal to the inflation target and the output gap equals zero. The arrows in Fig. 7.23 show how the inflation rate and the output gap move over time, when the economy is in disequilibrium. The SS – the convergent saddle arm – is negatively sloped. The inflation rate is not a jump variable in this model because inertia is present.

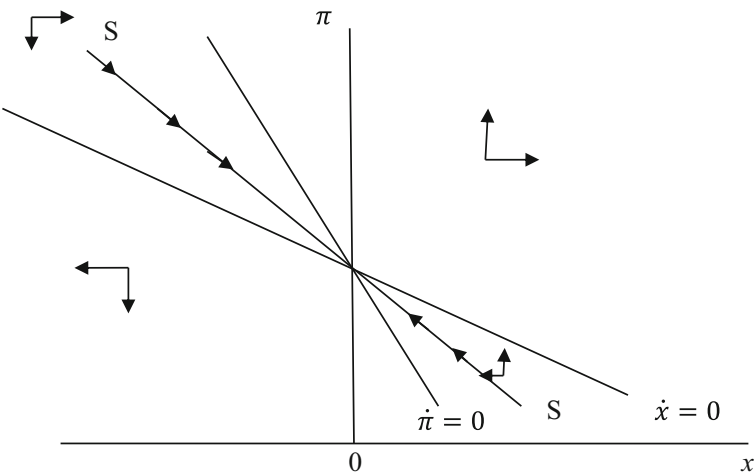
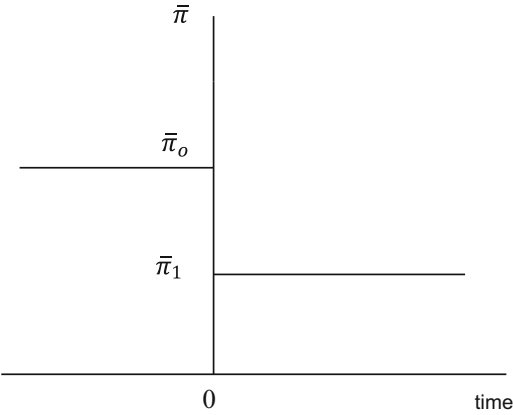


Fig. 7.23 The phase diagram of the New Keynesian model with information rigidity

Fig. 7.24 An unanticipated permanent decrease in the inflation target



Experiment

Consider the following experiment for the three Keynesian models: at time zero, an unanticipated, permanent change reduces the inflation target rate to $\bar{\pi}_1$ from $\bar{\pi}_0$, as Fig. 7.24 illustrates.

Figures 7.25, 7.26, and 7.27 show the adjustment of the economy to the changed inflation target in each model. The K model (Fig. 7.25) predicts a recession and a slow convergence to the new inflation target. Inflation instantaneously adjusts to the new target in the NK model (Fig. 7.26) without output loss. In the NKIR model, the economy jumps to the new saddle (Fig. 7.27), the output gap becomes negative, and inflation converges to the new target.

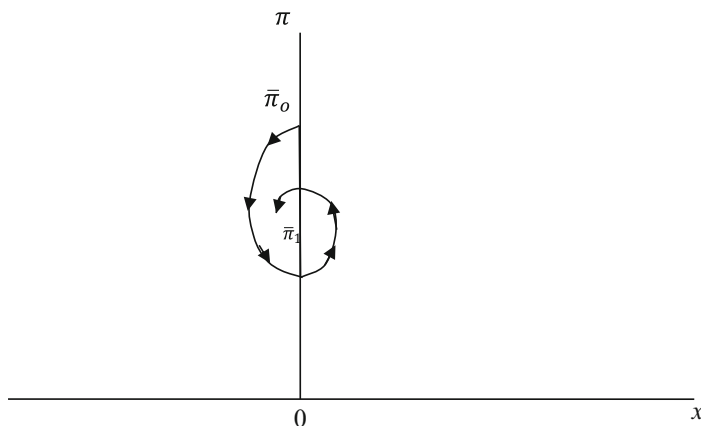
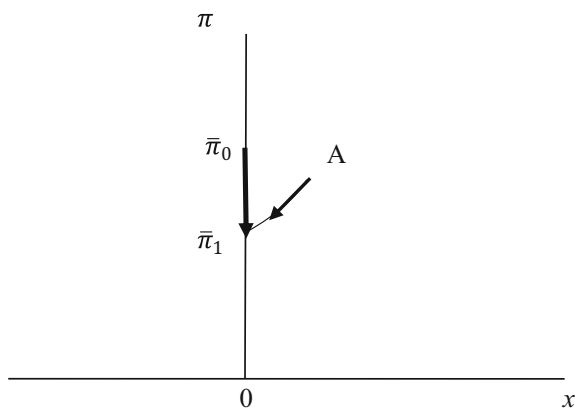


Fig. 7.25 The effects of an unanticipated permanent decrease in the inflation target in the Keynesian model

Fig. 7.26 The effects of an unanticipated permanent decrease in the inflation target in the new Keynesian model



In a different experiment: Assume that a new inflation target is announced at moment $t = 0$ to be implemented at moment $t = T$. The inflation target will be reduced as in the previous experiment, and will remain at the new level permanently. In the interval between the announcement and the change in the inflation target, nothing happens in the K model. Starting from implementation of the new target, the output gap becomes negative and the inflation rate starts converging to the new target. In the NK model, the inflation rate changes instantaneously, jumping to point A in Fig. 7.26, while output increases so that, at instant T , the economy will be at a new equilibrium. The NKIR model predicts a recession at the time of the announcement (Fig. 7.27, point A). The economy will reach the saddle path at instant T and the recession will persist while the economy converges to the new inflation target.

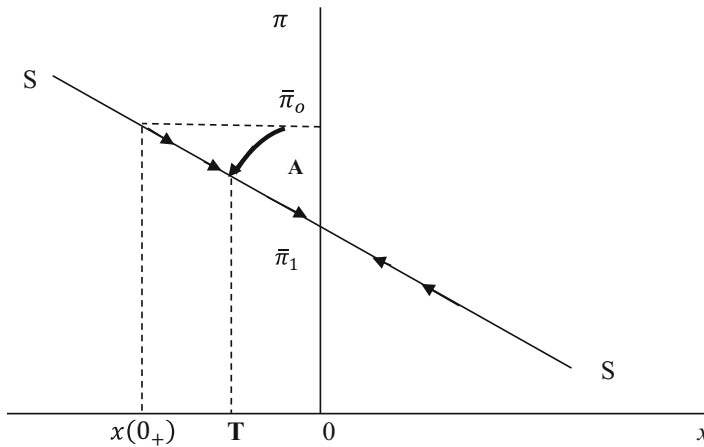


Fig. 7.27 The effects of an unanticipated permanent decrease in the inflation target in the New Keynesian model with information rigidity

Conclusions

The Keynesian IS curve predicts that the output gap and the change in the output gap are negatively correlated with the interest rate gap. The new Keynesian IS curve, on the other hand, predicts that the change in the output gap and the interest rate gap are positively correlated, but no correlation exists between the output gap and the interest rate gap.

The Keynesian and information rigidity Phillips curves predict a positive correlation between changes in the inflation rate and the output gap are. The new Keynesian Phillips curve predicts that changes in the inflation rate and the output gap are negatively correlated.

Therefore, the encompassing specification of Keynesian models may be used to test them. Furthermore, the encompassing specification allows for analyzing the dynamic of the Keynesian models. Anticipated disinflation heats up the economy in the new Keynesian model, and causes a recession in the traditional and information rigidity models. The IS curve has no influence on this outcome. This property is due to inflation rate inertia in the Phillips curves of the traditional Keynesian and information rigidity models.

7.5 Friedman's Model

This section's model has four equations: an IS curve, a Phillips curve, an LM curve, and a monetary policy rule (MPR), and one simplifying fiscal-policy assumption. The Phillips curve assumes that both the output gap and its rate of change affect the inflation rate. That is, if the growth rate of the economy's real output is greater than

the growth rate of potential output, the inflation rate increases, and vice-versa. The Friedman rule assumes that the Central Bank increases the monetary base at a constant rate equal to μ , regardless of the economy's situation. Friedman's model is specified as follows:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(f - \bar{f}) + g - \bar{g}, \quad (7.28)$$

$$\text{PC: } \dot{\pi} = \delta x + \phi \dot{x}, \quad (7.29)$$

$$\text{LM: } m - \bar{m} = \lambda x - \theta(i - \bar{i}), \quad (7.30)$$

$$\text{MPR: } \mu = \frac{d \log M}{dt} = \text{constants}, \quad (7.31)$$

$$\text{IC: Given } p(0) \text{ and } \pi(0), \quad (7.32)$$

$$\text{Simplifying assumptions: } f = \bar{f}, \quad g = \bar{g}. \quad (7.33)$$

Algebra

The Fisher equation allows the gap between the short- and long-run nominal interest rates to be written as:

$$i - \bar{i} = r - \bar{r} + \pi - \bar{\pi}.$$

Substituting this expression in the LM equation results in:

$$m - \bar{m} = \lambda x - \theta(r - \bar{r}) - \theta(\pi - \bar{\pi}).$$

This equation may be written as:

$$r - \bar{r} = -(\pi - \bar{\pi}) + \frac{\lambda}{\theta}x - \frac{1}{\theta}(m - \bar{m}).$$

Substituting the gap between short- and long-run real interest rates into the IS curve equation yields:

$$x = \alpha(\pi - \bar{\pi}) - \frac{\alpha\lambda}{\theta}x + \frac{\alpha}{\theta}(m - \bar{m}).$$

The output gap is therefore given by:

$$x = \frac{\alpha}{1 + \frac{\alpha\lambda}{\theta}}(\pi - \bar{\pi}) + \frac{\alpha/\theta}{1 + \frac{\alpha\lambda}{\theta}}(m - \bar{m}).$$

Differentiating this expression with respect to time yields the second equation in the differential equations system below, taking into account the fact that the derivative with respect to time of the log of the real stock of money equals the difference between the growth rate of the nominal stock of money (μ) and the inflation rate (π).

The first equation in the differential equations system is the Phillips curve equation. That is:

$$\begin{aligned}\dot{\pi} &= \delta x + \phi \dot{x} \\ \dot{x} &= \frac{\alpha}{1 + \frac{\alpha\lambda}{\theta}} \dot{\pi} + \frac{\alpha/\theta}{1 + \frac{\alpha\lambda}{\theta}} (\mu - \pi).\end{aligned}$$

The differential equations system may be written in matrix notation as follows:

$$\begin{bmatrix} 1 & -\phi \\ -\frac{\alpha}{1 + \frac{\alpha\lambda}{\theta}} & 1 \end{bmatrix} \begin{bmatrix} \dot{\pi} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \delta x \\ \frac{\alpha/\theta}{1 + \frac{\alpha\lambda}{\theta}} (\mu - \pi) \end{bmatrix}.$$

The solution to this linear equations system is:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \end{bmatrix} = \frac{1}{1 - \frac{\alpha\phi}{1 + \frac{\alpha\lambda}{\theta}}} \begin{bmatrix} 1 & \phi \\ \frac{\alpha}{1 + \frac{\alpha\lambda}{\theta}} & 1 \end{bmatrix} \begin{bmatrix} \delta x \\ \frac{\alpha/\theta}{1 + \frac{\alpha\lambda}{\theta}} (\mu - \pi) \end{bmatrix}.$$

Dynamical System

The model's two differential equations – for the rate of change of the output gap and for inflation acceleration – are therefore given by:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \end{bmatrix} = \frac{1}{\theta + \alpha(\lambda - \phi\theta)} \begin{bmatrix} -\phi\alpha & \delta(\theta + \alpha\lambda) \\ -\alpha & \alpha\theta\delta \end{bmatrix} \begin{bmatrix} \pi - \mu \\ x \end{bmatrix}. \quad (7.34)$$

The Jacobian matrix is:

$$J = \begin{bmatrix} \frac{\partial \dot{\pi}}{\partial \pi} & \frac{\partial \dot{\pi}}{\partial x} \\ \frac{\partial \dot{x}}{\partial \pi} & \frac{\partial \dot{x}}{\partial x} \end{bmatrix} = \frac{1}{\theta + \alpha(\lambda - \phi\theta)} \begin{bmatrix} -\phi\alpha & \delta(\theta + \alpha\lambda) \\ -\alpha & \alpha\theta\delta \end{bmatrix}.$$

This matrix's determinant and trace are equal to:

$$|J| = \frac{-\alpha\theta\delta\phi\alpha + \alpha\delta(\theta + \alpha\lambda)}{[\theta + \alpha(\lambda - \phi\theta)]^2}, \quad trJ = \frac{\alpha\theta\delta - \phi\alpha}{\theta + \alpha(\lambda - \phi\theta)}.$$

The model is stable when the determinant is positive and the trace is negative. The determinant is positive when the parameters satisfy the following restriction:

$$|J| > 0 \Rightarrow \alpha\delta(\theta + \alpha\lambda) > \alpha\theta\delta\phi\alpha$$

$$\theta + \alpha\lambda > \theta\alpha\phi$$

$$\theta + \alpha(\lambda - \phi\theta) > 0$$

The trace is negative when the model's parameters satisfy the inequality:

$$trJ < 0 \Rightarrow \alpha\theta\delta - \phi\alpha < 0$$

$$\theta\delta < \phi$$

Combining the determinant and trace constraints for model stability implies that the parameters must obey the following:

$$\theta\delta < \phi < \frac{\theta + \alpha\lambda}{\theta\alpha}.$$

It is worth emphasizing that if $\phi = 0 \Rightarrow |J| > 0$, $trJ > 0$. Therefore, the model is unstable in this case.

Figure 7.28 shows the phase diagram of the output gap equation. The line where the output gap remains constant has a positive slope. Output increases in the region below and to the right of this line, and decreases above and to the left of it, as the arrows show.

Figure 7.29 corresponds to the phase diagram of the differential equation for the inflation rate. It has a positive slope. The inflation rate decreases in the region above and to the left of the line, and increases below and to the right of it, as the arrows show. The output gap and inflation rate lines have positive slopes. The inequalities

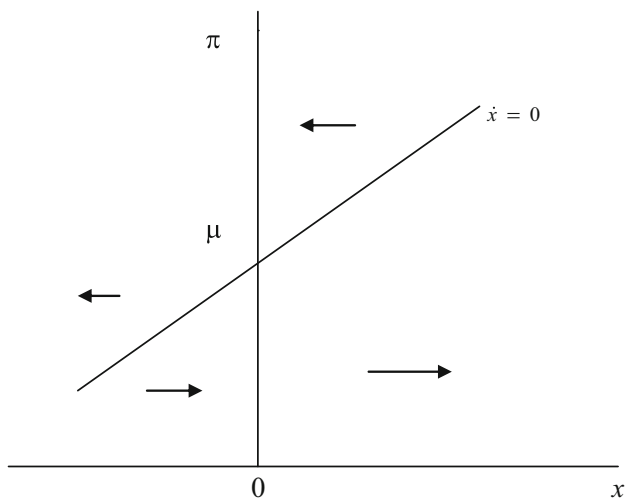
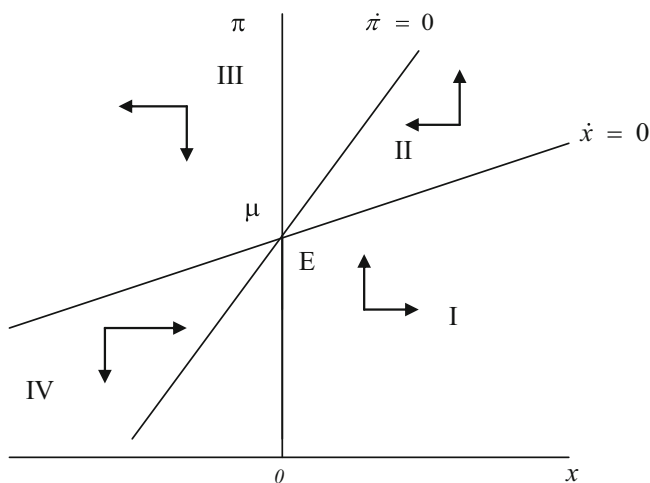
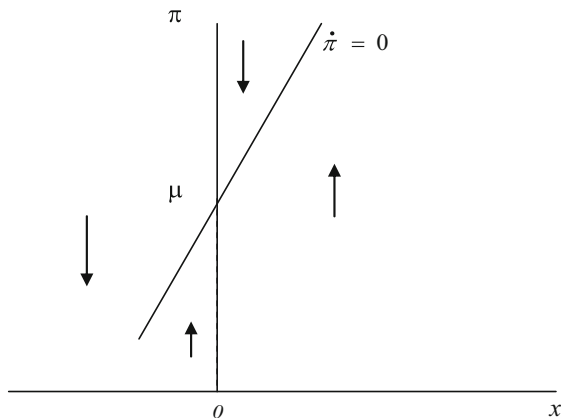


Fig. 7.28 Phase diagram: $\dot{x} = 0$

Fig. 7.29 Phase diagram: $\dot{\pi} = 0$ **Fig. 7.30** The model's phase diagram

that the model parameters must obey imply that the angular coefficient of the line where the inflation rate is constant is greater than that of the line where the output gap is constant. That is:

$$\frac{\delta(\theta + \alpha\lambda)}{\phi\alpha} \div \theta\delta = \frac{\delta(\theta + \alpha\lambda)}{\phi\alpha\theta\delta} > 1.$$

Figure 7.30 shows the model's phase diagram, with four distinct regions. In region I the economy moves northeast, in region II the inflation rate and real output move northwest, in region III the economy's path is southwest, and in region IV the two variables move southeast.

Experiment

Figure 7.31 describes a monetary policy experiment in which the Central Bank increases the monetary base expansion rate to μ_1 from μ_0 . In this model the price level and the inflation rate do not change instantaneously because of the rigidity and inertia assumptions. At the moment of the monetary policy change, the economy is at equilibrium at point E_0 , as Fig. 7.32 shows. To prevent overloading the graph, the figure does not show the two curves of the dynamical system that led to this equilibrium. At the time of the monetary policy change, the dynamical system's equations shift up and cross the new long-run equilibrium point, corresponding to inflation rate π_1 and to the economy's full-employment output level.

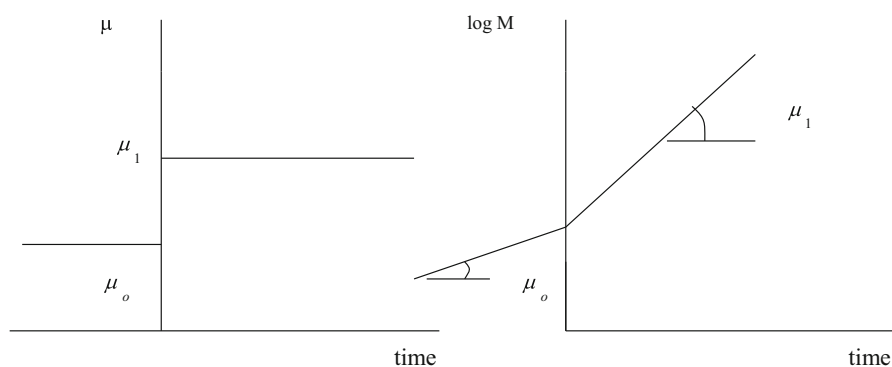


Fig. 7.31 Monetary policy experiment

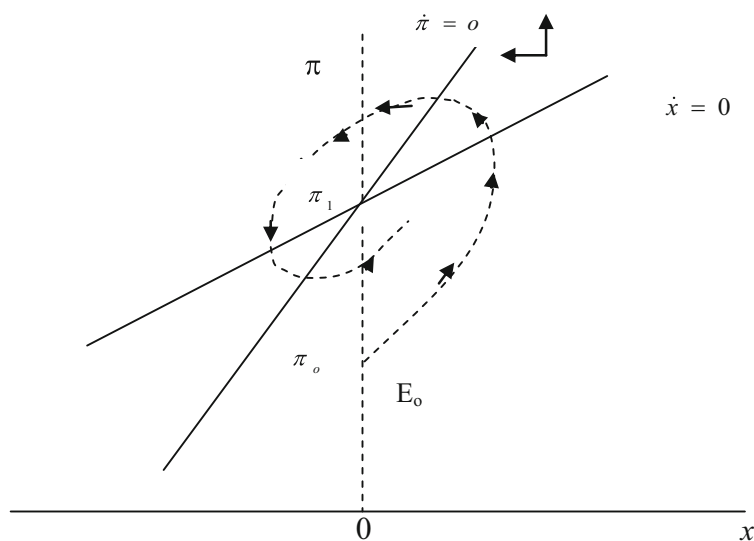


Fig. 7.32 Model dynamics: overshooting

The increased growth rate of the monetary base pushes down the nominal interest rate, and the same occurs with the real interest rate due to inflation rate rigidity. The economy's real output enters an expansion path and inflation gradually increases. In this process, the inflation rate overshoots the new long-run equilibrium inflation rate ($\pi_1 = \mu_1$) and continues to rise for a certain period of time until it reaches the line $\dot{\pi} = 0$. From then on, the inflation rate begins to drop and converges to the new equilibrium in an oscillating movement, according to the assumptions based on which the graph was drawn.

Overshooting occurs because, in the long-run, the nominal interest rate rises to a new level and the real quantity of money demanded by the public decreases. Because in the beginning of the adjustment process the inflation rate is lower than the new expansion rate of the monetary base, the real quantity of money initially increases. For it to drop to its new long-run equilibrium, the inflation rate must rise at a greater rate than the growth rate of the monetary base for a certain period of time.

The phase diagram in Fig. 7.32 shows the complexity of the economy's adjustment to a monetary shock. It shows a period in which both the inflation rate and the output gap increase. This is followed by a period in which the output gap begins to drop, but the inflation rate continues to rise, and then a period in which both the output gap and the inflation rate decrease. The dynamical adjustment process continues for a period in which the output gap increases and the inflation rate decreases. In the long-run, the inflation rate equals the growth rate of the monetary base, but not so in the short-run.

7.6 Chronic Inflation

Chronic inflation is a phenomenon that occurs when the Central Bank issues money to finance the public deficit on a permanent basis. The traditional Keynesian model with an IS curve, an LM curve, and a Phillips curve can be used to model chronic inflation. In this model, the monetary policy rule states that the Central Bank finances the public deficit f according to:

$$\frac{\dot{M}}{P} = f. \quad (7.35)$$

Real cash balance is defined by: $m = M/P$. By taking the derivative of this expression with respect to time, we obtain the rule of monetary policy (MPR): $\dot{m} = f - m\pi$.

The chronic inflation model has four equations and two initial conditions:

$$\text{IS: } x = -\alpha(r - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{LM: } \log m - \log \bar{m} = \lambda x - \theta(i - \bar{i}),$$

$$\text{MPR: } \dot{m} = f - m\pi,$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

Algebra

The aggregate demand curve is obtained by combining the IS and LM curves and it can be written as:

$$x = -a + b \log m + c \pi^e,$$

where the coefficients a , b , and c are given by: $a = \alpha(\log \bar{m} + \theta \bar{\pi})/(\theta + \alpha\lambda)$; $b = \alpha/(\theta + \alpha\lambda)$, and $c = \alpha\theta/(\theta + \alpha\lambda)$. These coefficients are positive. The coefficient b is the Keynes effect, or the liquidity effect. The coefficient c is the Fisher effect, or the expected inflation effect.

We assume perfect foresight: $\pi^e = \pi$. By taking into account this hypothesis, the aggregate demand and the Phillips curves, when combined, yield the following equation:

$$\dot{\pi} = -\kappa + \beta \log m + \gamma m,$$

where $\kappa = \delta a$, $\beta = \delta b$, and $\gamma = \delta c$.

Dynamical System

The chronic inflation model has two nonlinear differential equations:

$$\dot{m} = f - m\pi, \quad (7.36)$$

$$\dot{\pi} = -k + \beta \log m + \gamma \pi. \quad (7.37)$$

Figure 7.33 shows the phase diagram of the model, assuming two steady-state rates of inflation: point A is the high inflation rate and point B is the low inflation rate. The arrows indicate the direction of motion when the economy is not in equilibrium (Fig. 7.33).

Local Stability of the Equilibrium Points

The equilibrium points of the model are points for which $\dot{m} = 0$ and $\dot{\pi} = 0$. Thus, the equilibrium values for the real quantity of money are obtained from the following equation:

$$g(m) = \beta \log m + \frac{\gamma f}{m} = k.$$

Figure 7.35 shows the graph of this function assuming that there are two equilibrium points. For this to happen, the parameters of the model must satisfy the inequality:

$$k > \beta \left(1 + \log \frac{\gamma f}{\beta} \right)$$

The two values of the equilibrium for the real quantity of money are such that (Fig. 7.34):

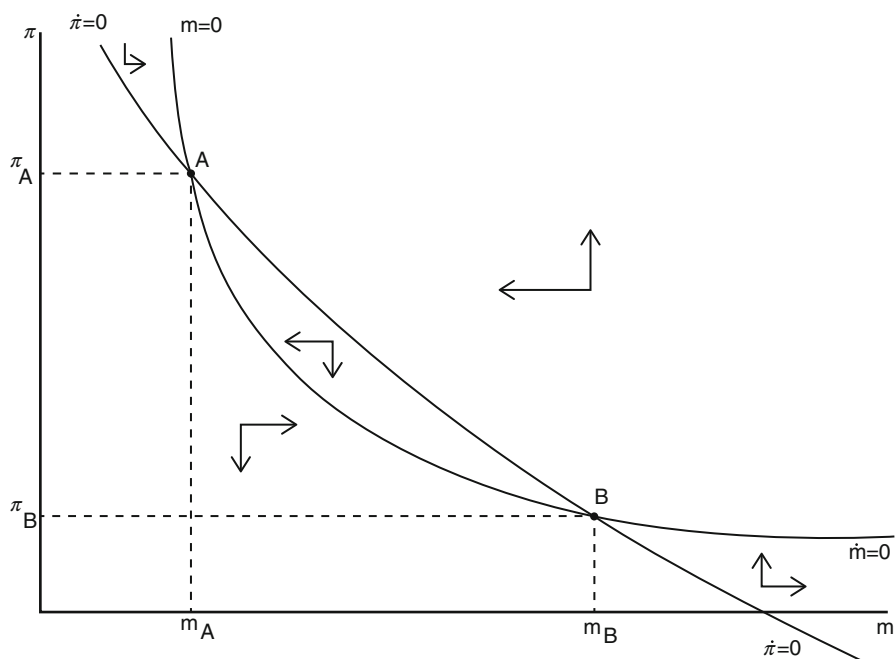


Fig. 7.33 The phase diagram of the chronic inflation model

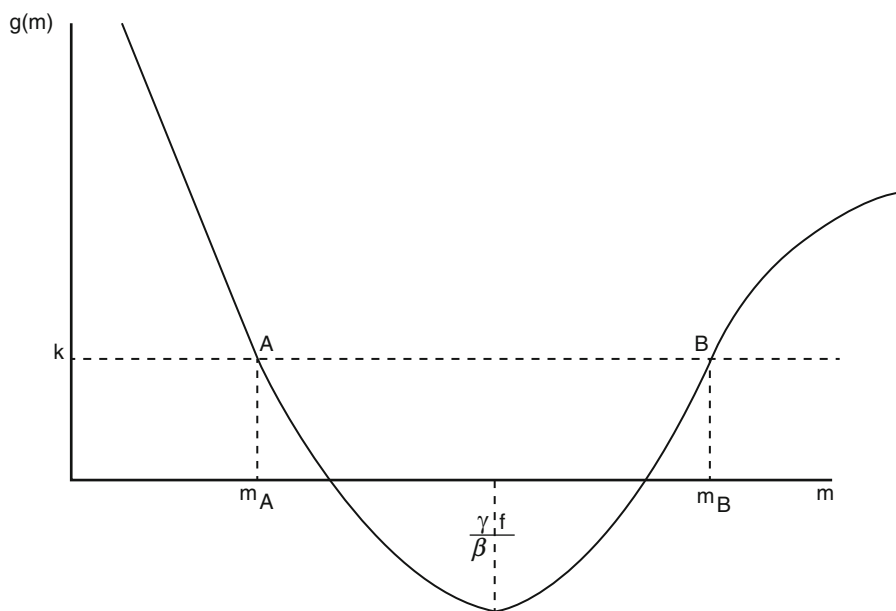


Fig. 7.34 The chronic inflation model: multiple equilibria

$$m_A < \frac{\gamma f}{\beta} < m_B \quad (7.38)$$

The Jacobian of the dynamical system for any point $(\bar{m}, \bar{\pi})$ is given by:

$$J = \begin{bmatrix} -\bar{\pi} & -\bar{m} \\ \frac{\beta}{\bar{m}} & \gamma \end{bmatrix}$$

The determinant and the trace of this matrix are:

$$|J| = \beta - \gamma \bar{\pi}$$

$$trJ = \gamma - \bar{\pi}$$

The characteristic roots of the matrix J obtained from the equation $|J - \lambda I| = 0$ are given by:

$$\lambda = \frac{trJ}{2} \pm \frac{1}{2} \sqrt{\Delta}$$

where:

$$\Delta = (\gamma - \bar{\pi})^2 - 4(\beta - \bar{\pi}\gamma).$$

The equilibrium point A is a saddle point. This follows from the fact that $m_A < \gamma f / \beta$. Thus $|J_A| = \beta - \gamma \bar{\pi}_A = \beta - \gamma f / m_A < 0$ and there are two real distinct roots of opposite signs.

The determinant of matrix J evaluated at point B is positive because $m_B > \gamma f / \beta$. Thus, $|J_B| = \beta - \gamma \bar{\pi}_B = \beta - \gamma f / m_B > 0$. The equilibrium point B can be either stable or unstable depending on the sign of the trace of matrix J evaluated at this point. The sign of this trace depends on the values of the parameters of the model. When $trJ_B = \gamma - \bar{\pi}_B > 0$, point B is a source (locally unstable). On the other hand, if $trJ_B = \gamma - \bar{\pi}_B < 0$, point B is a sink (locally stable). The characteristic roots can be real if $\Delta > 0$ and complexes if $\Delta < 0$. If the roots are complex and the trace is equal to zero, $trJ_B = \gamma - \bar{\pi}_B = 0$, the roots are pure imaginary.

Periodical Orbits and Hopf Bifurcation

The qualitative analysis of the dynamical equations of the chronic inflation model when we increase the fiscal deficit (f), holding constant the parameters k, γ, β , shows that the stability of the low inflation equilibrium point changes. It is easy to verify that the low inflation equilibrium (point B) increases when the public deficit increases. Thus we use $\pi_B(f)$ in our analysis as the shifting parameter, and this function has the following properties:

$$0 < \pi_B(f) < \frac{\beta}{\gamma}, \quad \text{and} \quad \pi'_B(f) > 0. \quad (7.39)$$

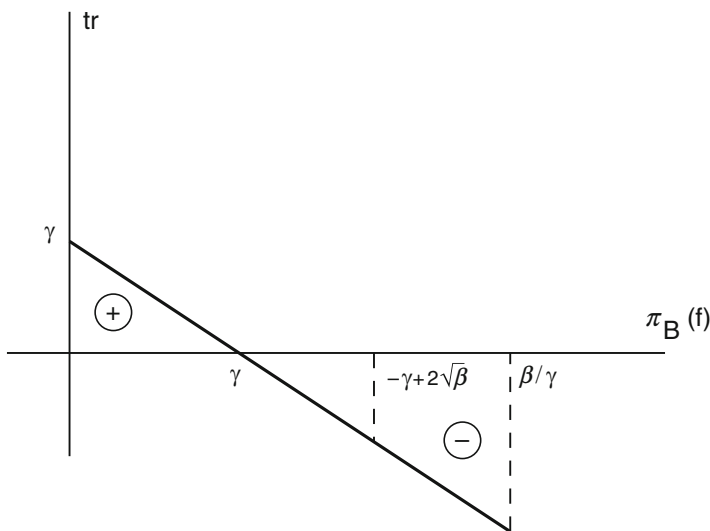


Fig. 7.35 Hopf bifurcation at the low inflation point

The trace of the Jacobian matrix evaluated at point B is linear in the rate of inflation ($trJ_B = \gamma - \pi_B(f)$) as shown in Fig. 7.35. For $\pi_B(f) < \pi_B(f_0)$, the system is unstable since the trace is positive. For $\pi_B(f) > \pi_B(f_0)$, the system is stable since the trace is negative. As f crosses f_0 , the system becomes stable. The real part of the complex characteristic root (R) is equal to half of the trace, and its derivative, with respect to the fiscal deficit, is given by:

$$\frac{dR}{df} = \frac{d(\gamma - \pi_B(f))/2}{df} = -\frac{1}{2} \pi'_B(f) < 0. \quad (7.40)$$

The fact that this derivative is different from zero allows us to conclude that for $f < f_0$, the system has a stable limit cycle. Thus, $f = f_0$ is a Hopf bifurcation point because there is a change in stability accompanied by the creation of a limit cycle.¹

Figure 7.36 shows some possibilities for the dynamics of the economy in points close to $f = f_0$. Figure 7.36a supposes that for the public deficit $f < f_0$, there is one limit cycle. In the dotted and dashed areas of Fig. 7.36a there is no hyperinflation, with point B being unstable. However, the curve H-H is a hyperinflation path. Figure 7.36b shows the dynamics of the economy when $f \geq f_0$. In this case, point B is stable and in the dotted area there is no hyperinflation. However, the curve H-H is a hyperinflation path.

Table 7.3 provides the information necessary to analyze the stability of equilibrium at point B, the low rate of inflation equilibrium. The point can be either stable or

¹Hopf bifurcation theory requires the computation of the third-order derivative of a function and its sign according to Marsden and McCracken [(1976), p. 65]. This computation was carried out and the sign of this derivative is negative.

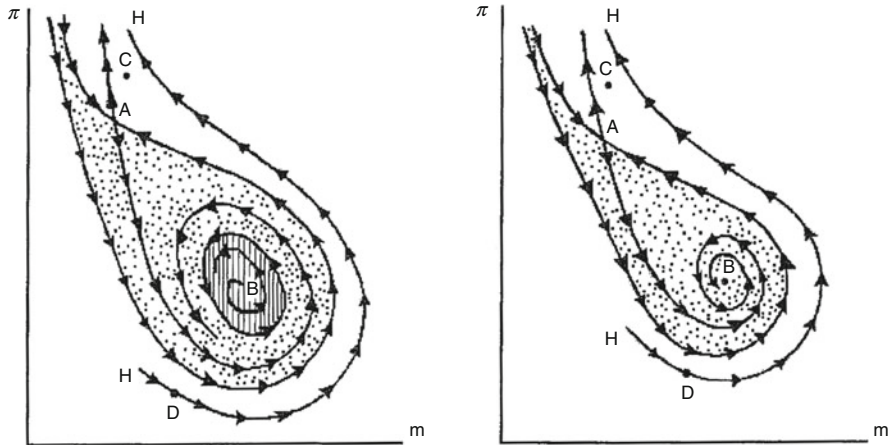


Fig. 7.36 The dynamics of the economy under chronic inflation

Table 7.3 Stability of equilibrium at the low inflation point

$\pi_B(f)$	$\text{tr}J_B$	Δ_B	Stability
$0 < \pi_B(f) < \gamma$	>0	<0	Unstable (complex roots)
$\pi_B(f) = \gamma$	$=0$	<0	Hopf Bifurcation
$\gamma < \pi_B(f) < -\gamma + 2\sqrt{\beta}$	<0	<0	Stable (complex roots)
$-\gamma + 2\sqrt{\beta} < \pi_B(f) < \beta/\lambda$	<0	>0	Stable (real roots)

unstable, depending on the parameter values of the model. When the equilibrium inflation rate is greater than zero and less than γ , the roots are complex and point B is stable. When the inflation rate is equal to γ , there is a Hopf bifurcation. When the inflation rate is greater than γ and less than $-\gamma + 2\sqrt{\beta}$, the model is stable at point B and the roots are complex. When the inflation rate is greater than $-\gamma + 2\sqrt{\beta}$ and less than β/γ , the model is stable at point B and the roots are real.

The change in the Phillips curve parameter affects γ but does not affect the ratio β/γ . Thus, when the average period of the contracts or of the indexations mechanism increases, the parameter γ also increases and the region of instability at point B becomes larger.

7.7 Exercises

1. Consider the model:

$$\text{IS: } \dot{x} = \sigma(i - \pi - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x,$$

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; π = inflation rate; $\bar{\pi}$ = inflation rate target; $\dot{\pi} = d\pi/dt$; i = nominal interest rate; $\alpha, \delta, \sigma, \phi, \theta$ are positive parameters.

Analyze the consequences of an unanticipated, permanent change in the inflation target.

2. Consider the model:

$$\text{IS: } \dot{x} = -\lambda x - \alpha(r - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x,$$

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: x = output gap; $\dot{x} = dx/dt$; r = real interest rate; $\bar{r}$ = natural interest rate; π = inflation rate; $\dot{\pi} = d\pi/dt$; i = nominal interest rate; $\lambda, \alpha, \delta, \phi$ and θ are positive parameters.

- Analyze the model's equilibrium and dynamics on a phase diagram, with the inflation rate (π) on the vertical axis and the output gap (x) on the horizontal axis.
- Use item (a)'s phase diagram to show what happens in this model when the Central Bank raises the inflation rate target.
- What happens in this model if the parameter ϕ is negative? How do you interpret the monetary policy rule in this case?

3. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(f - \bar{f}) + g - \bar{g},$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x,$$

IC: Given $p(0)$ and $\pi(0)$.

$$\text{Fiscal policy rule: } f - \bar{f} = -\varphi_1 x, g - \bar{g} = -\varphi_2 x.$$

The parameters $\alpha, \beta, \delta, \phi, \theta, \varphi_1, \varphi_2$ are positive and the other symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; f = public deficit; $\bar{f}$ = full-employment public deficit; g = government spending; $\bar{g}$ = full-employment government spending; π = inflation rate; $\bar{\pi}$ = inflation target; $\dot{\pi} = d\pi/dt$.

- How would you interpret the monetary and fiscal policy rules?
- Analyze the model's equilibrium and dynamics on a diagram with π on the vertical axis and x on the horizontal axis.
- What would happen in this model if the parameter ϕ was negative?
- What would happen in this economy if the government raised the inflation rate target to $\bar{\pi}_1 > \bar{\pi}_0$ from $\bar{\pi}_0$?

4. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}), \quad \alpha > 0,$$

$$\text{PC: } \dot{\pi} = -\delta x, \quad \delta > 0,$$

$$\text{MPR: } i = \bar{r}^{BC} + \pi + \phi(\pi - \bar{\pi}) + \theta x,$$

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; π = inflation rate; $\dot{\pi} = d\pi/dt$; i = nominal interest rate; $\bar{r}^{BC}$ = real Central Bank interest rate; α, δ, ϕ and θ are positive parameters.

(a) What are the short- and long run consequences of $\bar{r} \neq \bar{r}^{BC}$?

(b) How would you interpret $\bar{r} \neq \bar{r}^{BC}$?

5. Consider the model:

$$\text{IS: } \dot{x} = -\alpha(r - \bar{r}), \quad \alpha > 0,$$

$$\text{PC: } \dot{\pi} = \delta x, \quad \delta > 0,$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \quad \phi > 0, \theta > 0,$$

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: $\dot{x} = dx/dt$; x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; i = nominal interest rate; π = inflation rate; $\bar{\pi}$ = inflation rate target; $\dot{\pi} = d\pi/dt$; α, δ, ϕ and θ are positive parameters.

(a) Discuss the specification of each of the model's equations;

(b) Analyze the model's equilibrium and dynamics on a phase diagram with π on the vertical axis and x on the horizontal axis.

(c) Use the previous item's phase diagram to show this model's dynamics when the Central Bank reduces the inflation target.

(d) What would happen in this model if the parameter ϕ was negative?

6. Consider the model:

$$\text{IS: } u - \bar{u} = \alpha(r - \bar{r}), \quad \alpha > 0,$$

$$\text{PC: } \dot{\pi} = -\delta(u - \bar{u}), \quad \delta > 0,$$

$$\text{MPR: } \dot{i} = \phi(\pi - \bar{\pi}) - \theta \dot{u},$$

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: u = natural unemployment rate; $\bar{u}$ = natural unemployment rate; r = real interest rate; $\bar{r}$ = natural interest rate; i = nominal interest rate; $\dot{i} = di/dt$; π = inflation rate; $\bar{\pi}$ = inflation rate target; $\dot{u} = du/dt$; α, δ, ϕ and θ are positive parameters.

(a) How would you obtain this model's IS curve?

(b) Does the monetary policy rule need information on a non-observable variable?

- (c) Analyze the model's equilibrium and dynamics on a phase diagram with inflation on the horizontal axis and the unemployment rate on the horizontal axis. Is this model stable?
- (d) Assume that the model's initial inflation is an endogenous variable and that the parameter δ is negative. Analyze the model's equilibrium and dynamics under these circumstances.
- (e) Would you recommend using this monetary policy rule?

7. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x.$$

- (a) In this model, is the long-run inflation a monetary phenomenon?
- (b) Assume that the LM curve is specified by:

$$LM : m - \bar{m} = \lambda x - \beta(i - \bar{i}),$$

where m is the real quantity of money $m = M/P$. Does the inflation rate target equal the growth rate of the monetary base in the long term?

- (c) Does the monetary policy rule that sets the interest rate imply that inflation is not a monetary phenomenon in the long-run?
- (d) Based on this model, might one argue that if the society's objective is to reduce the interest rate, the interest rate must rise initially?

8. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{MPR: } i^s = \bar{r}^s + \pi + \theta(\pi - \bar{\pi}) + \phi x,$$

$$\text{Credit: } i = i^s + \bar{s}p + \beta(sp - \bar{s}p),$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

$$\text{Definitions: } i = r - \pi, \quad \bar{r} = \bar{r}^s + \bar{s}p$$

The symbol sp represents the spread of the interest rate on the credit market, and $\bar{s}p$ is the long-run equilibrium spread.

- (a) Show that this model's aggregate demand equation is given by:

$$\pi = \bar{\pi} - \frac{(1 + \alpha\phi)}{\alpha\theta}x - \frac{\beta}{\theta}(sp - \bar{s}p).$$

- (b) Use a phase diagram (π on the vertical axis and x on the horizontal axis) to show what happens to the output gap and the inflation rate when a credit market shock causes $sp - \bar{s}p > 0$.

9. Consider the model:

IS: $\dot{x} = -\alpha(r - \bar{r}), \alpha > 0,$

PC: $\dot{\pi} = -\gamma(\pi - \bar{\pi}) + \delta x, \delta > 0,$

MPR: $i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \phi > 0, \theta > 0,$

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: $\dot{x} = dx/dt$; x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; i = nominal interest rate; π = inflation rate; $\bar{\pi}$ = inflation rate target; $\dot{\pi} = d\pi/dt$; α, δ, ϕ and θ are positive parameters.

- Analyze the model's equilibrium and dynamics on a phase diagram with π on the vertical axis and x on the horizontal axis.
- Use the previous item's phase diagram to show this model's dynamics when the Central Bank reduces the inflation target.
- What would happen in this model if the parameter ϕ was negative?

10. Consider the model:

IS: $\dot{x} = -\alpha(r - \bar{r}), \alpha > 0,$

PC: $\dot{\pi} = \delta x, \delta > 0,$

MPR: $i = \bar{i},$

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: $\dot{x} = dx/dt$; x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; i = nominal interest rate; π = inflation rate; $\bar{\pi}$ = inflation rate target; $\dot{\pi} = d\pi/dt$; α, δ are positive parameters.

- Analyze the model's equilibrium and dynamics on a phase diagram with π on the vertical axis and x on the horizontal axis.
- Use the previous item's phase diagram to show the model's dynamics when the Central Bank reduces the inflation target.

11. Consider the model:

IS: $\dot{x} = \gamma x + \sigma(r - \bar{r}),$

PC: $\dot{\pi} = \delta x,$

MPR: $i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x,$

IC: Given $p(0)$.

- Analyze the model's equilibrium and dynamics.
- What are the model's properties when $\gamma = 0$ and $\delta < 0$ (new Keynesian model)?
- What are the model's properties when $\gamma < 0, \delta > 0$ and $\sigma < 0$ (Keynesian model)?

12. Consider the model:

$$\text{IS: } \dot{x} = -\alpha(r - \bar{r}), \alpha > 0,$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{MPR: } \frac{di}{dt} = \phi(\pi - \bar{\pi}) + \theta x.$$

- (a) What condition must the parameter θ meet for the model to be stable?
- (b) Assume that $\alpha < 0$ and $\delta < 0$. Does the model have multiple equilibria?
- (c) Would you recommend that the Central Bank should use this monetary policy?

13. Consider the new Keynesian model:

$$\text{IS: } \dot{x} = \sigma(r - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \rho\pi - \kappa x,$$

$$\text{MPR: } r - \bar{r} = \phi(\pi - \bar{\pi}) + \theta(x - \bar{x}).$$

The inflation target $\bar{\pi}$ and the $\bar{x}$ gap are such that $\bar{x} = \rho\bar{\pi}/\kappa$.

- (a) Does this model have a unique equilibrium?
- (b) Is the monetary policy long-run neutral in the new Keynesian model?

Chapter 8

Open Economy Macroeconomics



This chapter presents the basic concepts of open economy macroeconomics. Sections 8.1 and 8.2 discuss arbitrage-pricing models for goods and services in international trade and for interest rates in the capital flow between countries. Section 8.3 introduces the Marshall-Lerner condition, which sets a condition for a positive correlation between the terms of trade and the current account on the balance of payments. Section 8.4 addresses the specification of the IS curve in an open economy; showing the relationship between real output, real interest rate, and real exchange rate. The specifications of the IS curves are shown for the traditional and the new Keynesian models. Section 8.5 analyzes the determination of the long-run equilibrium real exchange rate, the natural exchange rate. Section 8.6 discusses the specification of the Taylor Rule in an open economy. Section 8.7 covers the specification of the Phillips curve in an open economy in the Keynesian and new Keynesian models.

8.1 Goods and Services Arbitrage

An open economy features mobility of goods and services, capital, and labor between countries. Immigration laws constrain labor mobility and, therefore, our open-economy models will not take this kind of mobility into consideration. Goods/products and services mobility occurs via international trade. The exporter and the importer are agents whose main business is to arbitrage pricing between countries.

8.1.1 *Absolute Purchasing Power Parity*

The exchange rate (E) is the quantity of domestic money needed to purchase one unit of the foreign currency. Assume that the price of the domestic product is P and that

the price of the same imported product in the currency of its country of origin is P^* . The price of the imported product, once converted into the domestic currency, is therefore equal to EP^* . If this price is lower than that of the domestic product, the imported product will be bought. Otherwise, the domestic product will. According to the law of one price, a product must be sold for the same price:

$$EP^* = P. \quad (8.1)$$

The nominal exchange rate may be written as the ratio of the inverse of the foreign price index and the inverse of the domestic price index,

$$E = \frac{1/P^*}{1/P}. \quad (8.2)$$

This expression's numerator is the foreign currency's purchasing power, and its denominator is the domestic currency's purchasing power. This equation shows that the arbitrage pricing of goods and services establishes the purchasing power parity between currencies.

The real exchange rate is the ratio of the two countries' price levels measured in the same currency:

$$Q = \frac{EP^*}{P}. \quad (8.3)$$

Q is the real exchange rate, E is the nominal exchange rate, P^* is the foreign price index, and P is the domestic price index. The real exchange rate is the relative price of the two countries' goods and services, measuring the quantity of domestic goods and services needed to buy one physical unit of the foreign product. When the goods are homogeneous, the absolute purchasing power parity implies that the real exchange rate must be equal to unity:

$$Q = 1. \quad (8.4)$$

8.1.2 Relative Purchasing Power Parity

Assume that computation of each country's price index involves two goods, but that the price of each product is not the same in the two countries. The ratio of expenditure on product two in the domestic index is α and the same ratio in the foreign price index is equal to β . The geometrical price indices of each country are:

$$P = P_1^{1-\alpha} P_2^\alpha = P_1 (P_2/P_1)^\alpha, \quad (8.5)$$

$$P^* = (P_1^*)^{1-\beta} (P_2^*)^\beta = P_1^* \left(\frac{P_2^*}{P_1^*} \right)^\beta. \quad (8.6)$$

The real exchange rate, based on these two indices, is:

$$Q = \frac{EP^*}{P} = \frac{EP_1^* p^\beta}{P_1 p^\alpha}, \quad (8.7)$$

where p is the relative price of the two goods. International trade arbitrage implies the law of one price:

$$EP_1^* = P_1; \quad EP_2^* = P_2. \quad (8.8)$$

Therefore, the real exchange rate is:

$$Q = p^{\beta-\alpha}. \quad (8.9)$$

When $\alpha = \beta$, $Q = 1$. If $\alpha \neq \beta$ and $p = \text{constant}$, then the relative purchasing power parity is obtained:

$$Q = \text{constant}. \quad (8.10)$$

Relative purchasing power parity is equivalent to the proposition that the rate of change of the nominal foreign exchange rate equals the difference between the domestic and the foreign inflation rates:

$$\widehat{E} = \widehat{P} - \widehat{P}^*. \quad (8.11)$$

The hat on top of the variable denotes a rate of change.

8.1.3 Tradable and Non-Tradable Goods

A number of goods and services exist in each country that are not internationally traded. These goods and services are called non-tradable (N). Each country's price index is a geometric average of the prices of the two goods:

$$P = P_T^{1-\omega} P_N^\omega = P_T \left(P_N / P_T \right)^\omega, \quad (8.12)$$

$$P^* = P_T^{*1-\omega} P_N^{*\omega} = P_T^* \left(\frac{P_N^*}{P_T^*} \right)^\omega. \quad (8.13)$$

It is assumed, for simplicity, that each product has the same weight in both countries. For tradable goods (T), the law of one price implies:

$$EP_T^* = P_T. \quad (8.14)$$

The real exchange rate therefore depends on the ratio of the two countries' relative prices:

$$Q = \left(\frac{p^*}{p} \right)^\omega. \quad (8.15)$$

In this case, neither the relative nor the absolute purchasing power holds and the rate of change of the real exchange rate is proportional to the difference between the rates of change of the two relative prices:

$$\hat{Q} = \omega(\hat{p}^* - \hat{p}). \quad (8.16)$$

8.1.4 Terms of Trade and Real Exchange Rate

The terms of trade represent the relative price of a country's imported and exported goods, that is, the terms of trade are defined by the ratio of these two prices. Some analysts choose to use the price of exports as the numerator and that of imports as the denominator. This book adopts the opposite approach, defining the terms of trade (S) as the ratio of the price of imported goods (P_M) to that of exported goods (P_X):

$$S = \frac{P_M}{P_X}. \quad (8.17)$$

This definition enables consistency with the definition of the real exchange rate that this book adopts. However, if the preferred definition is to use the price of exports as the numerator, it will be defined as the inverse ($1/S$) and all this requires is to use the inverse of the definition of the real exchange rate ($1/Q$) as well.

When both exportable and importable goods exist, the previous sub-section's real exchange rate may be written as:

$$Q = \frac{EP_T^*}{P_T} \frac{(p^*)^{\omega^*}}{p^{\omega}} = Q_T \frac{(p^*)^{\omega^*}}{p^{\omega}}. \quad (8.18)$$

This expression assumes that the weights of non-tradable goods in the price indices are not the same in the two countries. The real exchange rate has three components: (i) the relative price of tradable goods; (ii) the relative price of non-tradable goods in the foreign country; and (iii) the relative price of non-tradable goods in the domestic country.

The relative price of tradable goods depends on the prices of exportable and importable goods, according to the following price indices:

$$P_T^* = (P_X^*)^{1-\alpha^*} (P_M^*)^{\alpha^*} = P_X^* \left(\frac{P_M^*}{P_X^*} \right)^{\alpha^*},$$

$$P_T = P_X^{1-\alpha} P_M^\alpha = P_X \left(\frac{P_M}{P_X} \right)^\alpha.$$

It follows that the real exchange rate can be written as a function of the price of tradable goods as:

$$Q_T = \frac{EP_X^* (P_M^*/P_X^*)^{\alpha^*}}{P_X (P_M/P_X)^\alpha}.$$

The foreign country's exported goods are the domestic country's imported goods, and the domestic country's exported goods are the foreign country's imported goods. Therefore, the following relationships exist between them:

$$EP_X^* = P_M, \quad (8.19)$$

$$EP_M^* = P_X. \quad (8.20)$$

Substituting these expressions in the real exchange rate yields, with a little algebra, the relationship between the real exchange rate and the terms of trade:

$$Q_T = S^{1-\alpha-\alpha^*}. \quad (8.21)$$

Based on this, it is easily seen that the real exchange rate, the terms of trade, and the relative prices of non-tradable goods in the two countries satisfy the following equation:

$$q = (1 - \alpha - \alpha^*)s + \omega^* p^* - \omega p. \quad (8.22)$$

Lower-case letters indicate the logs of the respective variables. The real exchange rate therefore depends on the terms of trade, the relative price of non-tradable goods in the foreign country, and the relative price of non-tradable goods in the domestic country. The terms of trade and the real exchange rate are positively correlated when the sign of $1 - \alpha - \alpha^*$ is positive. Otherwise, the correlation will be negative. When the sum of the weights is equal to one ($\alpha + \alpha^* = 1$), the terms of trade do not affect the real exchange rate.

8.2 Interest Rate Arbitrage

Capital mobility interconnects the financial markets of the various countries, enabling interest rate arbitrage of both fixed- and variable-income securities issued in different currencies. This section assumes perfect capital mobility, that the assets are perfect substitutes, and that the agents are risk-neutral.

8.2.1 *Uncovered Interest Rate Parity*

Consider an investor that may invest their funds domestically or abroad. Should they invest in their own country, upon redemption, they will receive the principal plus the interest accrued:

$$1 + i_t. \quad (8.23)$$

Should they choose to invest abroad, they will have to convert their domestic currency on the foreign exchange market to obtain $1/E$ in foreign currency, invest it at the foreign interest rate i^* , and, upon redemption, convert the principal and interest at the exchange rate E_{t+1} :

$$\frac{1}{E_t} (1 + i_t^*) E_{t+1}^e. \quad (8.24)$$

The index e is added to the exchange rate in period $t + 1$ to indicate the fact that, at time t of the investment, this rate is unknown. For a risk-neutral investor, the two choices are equivalent. Therefore, according to the law of one price:

$$1 + i_t = (1 + i_t^*) \frac{E_{t+1}^*}{E_t}. \quad (8.25)$$

Taking the natural log of both sides of the expression and the $\log(1 + x) \cong x$ approximation, one finds that the anticipated change in the exchange rate equals the differential of the interest rates:

$$e_{t+1}^e - e_t = i_t - i_t^*. \quad (8.26)$$

8.2.2 *Exchange Rate Determination*

The exchange rate – the price of a financial asset – is determined based on the arbitrage of domestic and foreign assets. The uncovered interest rate arbitrage equation may be written as:

$$i_t + e_t = i_t^* + e_{t+1}^e. \quad (8.27)$$

Figure 8.1 graphs this equation with the interest rate on the horizontal axis and the exchange rate on the vertical axis, assuming that the foreign interest rate and the future exchange rate are known. For a domestic interest rate equal to i_0 , the exchange rate determined by arbitrage is equal to e_0 . Today's exchange rate depends on tomorrow's predicted exchange rate. If a change in the future exchange rate is anticipated, the exchange rate changes immediately, as Fig. 8.1 shows. The AA curve shifts to $A'A'$ and the exchange rate increases to e_1 from e_0 .

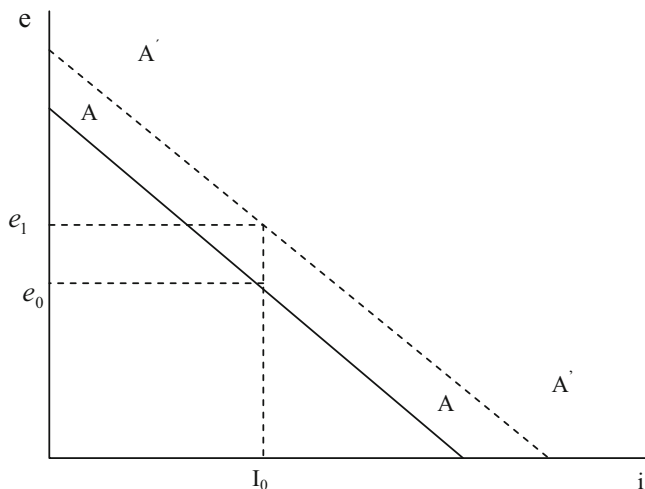


Fig. 8.1 Uncovered interest rate arbitrage

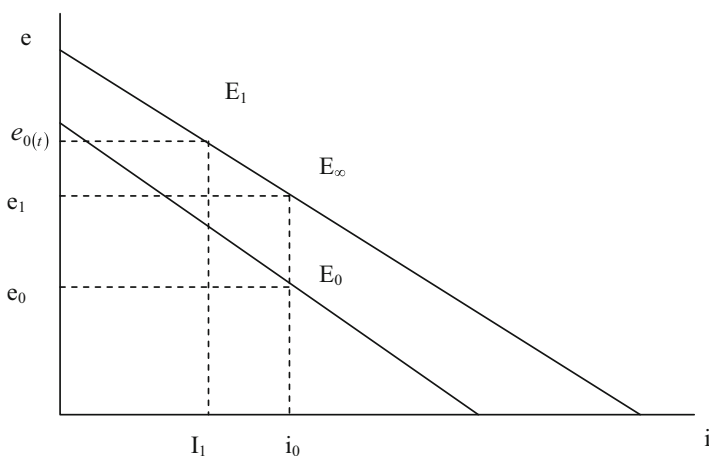


Fig. 8.2 The effects of an unanticipated decrease in the domestic interest rate on the exchange rate

The exchange rate, like the price of any other asset, may show a bubble, that is, its price may vary without a change to the fundamentals. All this requires is for the people active in the market to believe that prices will rise in the future. This belief may lead to a self-fulfilling prophecy. The absence of fundamentals to sustain the price rise will sooner or later prevail, and the bubble will dissipate.

The exchange rate is a jump variable, like the price of any other asset. The prices of goods and services, however, may be sticky and do not change values abruptly in the short run; that is, the prices of goods and services may be predetermined variables. Assume that this is the case. Further assume that the Central Bank reduces the interest rate to i_1 from i_0 , as Fig. 8.2 shows. The interest rate reduction increases the

quantity of money and in the long term the price level will increase proportionally with the increase in the money stock. The future exchange rate will increase because there has been no change in the real exchange rate. In the long term, the interest rate returns to its initial value because of the price level increase, which keeps the economy's real liquidity unchanged. In the long run, the exchange rate will be equal to e_1 . In the short run, the exchange rate will be higher to prevent arbitrage gains, as the exchange rate must appreciate after its initial jump. This phenomenon is called overshooting, where in the short run the exchange rate exceeds its long-run value.

8.2.3 Covered Interest Rate Parity

Now consider an investor who may invest their funds in the domestic market at interest rate i . Upon redemption, their investment will have the following value in the domestic currency:

$$1 + i_t. \quad (8.28)$$

The same investor may choose to go to the spot exchange market and purchase $1/E$ in foreign currency to invest at interest rate i^* . At the time of the investment, they know the exact value in foreign currency to be received upon redemption. The investor may sell this foreign currency in the future or forward market at the time of the investment at the price F on the future market or the forward market for delivery of the foreign currency in period $t + 1$. Upon redemption of the investment, its value will be:

$$\frac{1}{E_t} (1 + i_t^*) F_t. \quad (8.29)$$

The law of one price implies that the returns of the two assets must be the same because the assets are identical. That is:

$$1 + i_t = (1 + i_t^*) \frac{F_t}{E_t}. \quad (8.30)$$

In this case, there is no need to assume risk neutrality because the investment is riskless. Taking the natural log of both sides of the expression, with the same approximation used before, we find that the spread between the future or forward market and the spot market is equal to the interest rate spread:

$$f_t - e_t = i_t - i_t^*. \quad (8.31)$$

Comparing this expression with its equivalent for the uncovered interest rate parity, one can easily conclude that the future market's exchange rate equals the expected exchange rate:

$$e_{t+1}^e = f_t. \quad (8.32)$$

8.2.4 Uncovered Real Interest Rate Parity

The uncovered interest rate parity has the differential between the domestic and foreign interest rates equal the anticipated rate of change of the exchange rate:

$$1 + i_t = (1 + i_t^*) \frac{E_{t+1}}{E_t}. \quad (8.33)$$

This expression remains unchanged when the two sides of the equation are multiplied and divided by the ratio of domestic to foreign price indices in periods t and $t + 1$:

$$\frac{1 + i_t}{P_{t+1}/P_t} = \frac{1 + i_t^*}{P_{t+1}^*/P_t^*} \frac{\frac{E_{t+1}P_{t+1}^*}{P_{t+1}}}{\frac{E_tP_t^*}{P_t}}.$$

The left side of this equation is equal to one plus the domestic real interest rate. The first fraction on the right side is equal to one plus the foreign real interest rate. The second fraction on the right side equals the ratio of the real exchange rate in period $t + 1$ to the real exchange rate in period t . The uncovered real interest rate parity's expression is similar to that of the nominal interest rate parity. That is:

$$1 + r_t = (1 + r_t^*) \frac{Q_{t+1}}{Q_t}. \quad (8.34)$$

Taking the log of both sides of the equation, we conclude that the real interest rate differential equals the expected value of the change in the real exchange rate:

$$r_t = r_t^* + q_{t+1} - q_t, \quad (8.35)$$

where $q_t = \log Q_t$. This expression may be written as continuous variables, that is, at each point in time the difference between the real interest rates equals the derivative of the real exchange rate with respect to time:

$$r = r^* + \dot{q}. \quad (8.36)$$

In the long run, when the real exchange rate is at equilibrium $\dot{q} = 0$, the real domestic interest rate equals the real foreign interest rate: $\bar{r} = r^*$.

8.3 The Marshall-Lerner Condition

The gross domestic product equals the sum of consumption (C), investment (I), government spending (G), and non-factor goods and services exports (X), minus total non-factor goods and services imports (Z). That is:

$$Y = C + I + G + X - Z. \quad (8.37)$$

Subtracting the net income transferred abroad from the gross domestic product yields the gross national product. Net non-factor goods and services exports minus net income transferred abroad is equal to the current account on the balance of payments. Therefore, the meaning of the symbol Y , whether gross domestic product or gross national product, must be clear from the context:

$$Y = \text{GDP} \Rightarrow X - Z = \text{net goods and services exports};$$

$$Y = \text{GNP} \Rightarrow X - Z = \text{current account on the balance of payments}.$$

Dividing the expression of nominal gross national product by the price index of domestic goods yields:

$$\frac{Y}{P} = \frac{C}{P} + \frac{I}{P} + \frac{G}{P} + \frac{X}{P} - \frac{EP^*}{P}z.$$

The nominal value of imports is obtained by multiplying the nominal exchange rate (E) by the price index (P^*) of imported goods and services multiplied by the quantity of imported goods and services (z). Therefore, the gross national product in real terms is:

$$y = c + i + g + x - Sz. \quad (8.38)$$

S is the terms of trade. The current account on the balance of payments equals the net exports of factor and non-factor goods and services. That is:

$$cc = x - Sz. \quad (8.39)$$

Goods and services exports vary in the same direction as the terms of trade, that is, when the terms of trade increase (decrease) net exports will increase (decrease):

$$x = x(S, \dots), \quad \frac{\partial x}{\partial S} > 0. \quad (8.40)$$

Goods and services imports, in turn, vary in the opposite direction of the terms of trade. The partial derivative of imports with respect to the terms of trade is negative:

$$z = z(S, \dots), \quad \frac{\partial z}{\partial S} < 0. \quad (8.41)$$

Algebra

The derivative of the current account with respect to the terms of trade is:

$$\frac{\partial cc}{\partial S} = \frac{\partial x}{\partial S} - \left(z + S \frac{\partial z}{\partial S} \right).$$

This expression may be written as:

$$\frac{\partial cc}{\partial S} = \frac{x}{S} \frac{\partial x}{\partial S} \frac{S}{x} - z \left(1 + \frac{S}{z} \frac{\partial z}{\partial S} \right).$$

The elasticities of exports and imports with respect to the terms of trade are defined by:

$$\eta_{x,s} = \frac{\partial x}{\partial S} \frac{S}{x}, \quad \eta_{z,s} = \frac{\partial z}{\partial S} \frac{S}{z}.$$

The derivative of the current account with respect to the terms of trade may be written as a function of the two elasticities, according to:

$$\frac{\partial cc}{\partial S} = \frac{x}{S} \eta_{x,s} - z(1 + \eta_{z,s}).$$

If $x = S z$, that is, if the current account is initially in equilibrium, we may place z , or x/S , in evidence, and the derivative of the current account with respect to the terms of trade is given by:

$$\frac{\partial cc}{\partial S} = \frac{x}{S} (\eta_{x,s} - \eta_{z,s} - 1).$$

Marshall-Lerner

The elasticity of imports with respect to the terms of trade is negative. Its absolute value is therefore equal to the value of the elasticity of imports with an inverted sign, and the derivative of the current account with respect to the terms of trade depends on the sum of the elasticities according to:

$$\frac{\partial cc}{\partial S} = \frac{x}{S} (\eta_{x,Q} + |\eta_{z,Q}| - 1). \quad (8.42)$$

The Marshall-Lerner condition sets the restriction that must be met for the terms of trade and the current account to vary in the same direction. When the sum of the absolute values of the two elasticities is greater than unity, the current account increases (decreases) when the terms of trade increase (decrease):

$$\eta_{x,s} + |\eta_{z,s}| - 1 > 0 \Rightarrow \frac{\partial cc}{\partial S} > 0. \quad (8.43)$$

8.4 IS Curve in an Open Economy

The Keynesian IS curve in an open economy must take into account certain characteristics of such an environment. First, the presence of two goods, a domestic one that is exported and a foreign one that is imported; second, the possible effects of the relative prices of these goods on the domestic product's absorption; and third, the possible effects of a surplus (deficit) of the current account on the balance of payments on the domestic product's absorption.

8.4.1 Keynesian IS Curve

In an open economy, the gross national product is the sum of consumption, investment, government spending, and the current account on the balance of payments:

$$y = c + \iota + g + cc. \quad (8.44)$$

Consumption depends on disposable income, investment depends on the real interest rate, and the current account is a function of the terms of trade:

$$y = c(y - \tau) + \iota(r) + g + cc(S).$$

At full-employment equilibrium, this equation becomes:

$$\bar{y} = c(\bar{y} - \bar{\tau}) + \iota(\bar{r}) + \bar{g} + cc(\bar{S}).$$

Algebra

The expansions of the consumption, investment, and current account functions around the point that corresponds to full employment, disregarding second-order terms, are:

$$c(y - \tau) = c(\bar{y} - \bar{\tau}) + c'(y - \bar{y} - (\tau - \bar{\tau})),$$

$$\iota(r) = \iota(\bar{r}) + \iota'(r - \bar{r}),$$

$$cc(S) = cc(\bar{S}) + cc'(S - \bar{S}).$$

Subtracting full-employment output from the output and substituting the above expressions yields:

$$y - \bar{y} = c''[(y - \bar{y}) - (\tau - \bar{\tau})] + \iota'(r - \bar{r}) + g - \bar{g} + cc'(S - \bar{S}).$$

This equation may be written as:

$$y - \bar{y} = -\frac{c'}{1 - c'}(\tau - \bar{\tau}) + \frac{\iota'}{1 - c'}(r - \bar{r}) + \frac{1}{1 - c'}(g - \bar{g}) + \frac{cc'}{1 - c'}(S - \bar{S}).$$

Public deficit may replace one of the fiscal policy variables in this equation. The government's tax revenue equals the difference between government spending and public deficit, $\tau = g - f$. Therefore, $\tau - \bar{\tau} = g - \bar{g} - (f - \bar{f})$ and:

$$y - \bar{y} = \frac{t'}{1 - c'}(r - \bar{r}) + \frac{cc'}{1 - c'}(S - \bar{S}) + \frac{c'}{1 - c'}(f - \bar{f}) + g - \bar{g}.$$

This equation, like the closed-economy IS curve, may be written with the output gap on the left side. All that is required is to divide both sides of the equation by potential output. The term that contains the deviation of the terms of trade from the full-employment terms of trade must be multiplied and divided by the full-employment terms of trade. In this manner, the deviation of the log of the terms of trade from the log of the long-run terms of trade will be one of the IS curve's arguments. That is:

$$\frac{S - \bar{S}}{\bar{S}} \cong \log\left(\frac{1 + S - \bar{S}}{\bar{S}}\right) = \log S - \log \bar{S} = s - \bar{s}.$$

IS Curve Equation

In an open economy, it is important to make a distinction between the consumer price index (P_c) and the price index of domestically produced goods (P). The consumer price index is a weighted average of the prices of domestic and imported goods, with the weight ω being equal to the share of consumer expenditures that goes to imported goods and services. That is:

$$P_c = P^{1-\omega}(EP^*)^\omega = P\left(\frac{EP^*}{P}\right)^\omega. \quad (8.45)$$

The real exchange rate equals the ratio of the consumer, foreign, and domestic price indices, all expressed in the same currency:

$$Q = \frac{EP^*}{P_c}. \quad (8.46)$$

Substituting the consumer price index in this expression shows that the real exchange rate and the terms of trade are connected according to:

$$Q = S^{1-\omega}. \quad (8.47)$$

The log of the real exchange rate is therefore proportional to the log of the terms of trade:

$$q = \log Q = (1 - \omega)\log S = (1 - \omega)s.$$

Therefore, the real exchange rate gap is proportional to the terms of the trade gap:

$$q - \bar{q} = (1 - \omega)(s - \bar{s}). \quad (8.48)$$

Instead of using the terms of the trade gap in the IS curve, one may use the real exchange rate gap. The only difference lies in the gap's coefficient, but there is no substantive specification difference. Adopting one or the other will be a matter of algebraic convenience, involving an appropriate interpretation of the model's variables, as we will see below.

In the open-economy IS curve equation, the output gap (x) therefore depends on the gaps of the real interest rate, the real exchange rate (or the terms of trade), the public deficit, and government spending according to:

$$x = -\alpha(r - \bar{r}) + \beta(q - \bar{q}) + \gamma(f - \bar{f}) + g - \bar{g}. \quad (8.49)$$

The coefficients α , β , γ are positive. In the event of Ricardian equivalence, the parameter γ equals zero because the public deficit does not affect the output gap.

8.4.2 New Keynesian IS Curve

A small open economy's new Keynesian IS curve has two versions: one based on the representative agent model and another based on the overlapping generations model. Section 2.5 of Chap. 2 deduces the representative agent model's IS curve [Eq. (2.48)], the continuous-variables specification of which is:

$$\dot{x} = \omega\sigma(i_t - E_t\pi_{H,t+1} - \bar{r}_t). \quad (8.50)$$

The rate of change of the output gap is proportional to the real interest rate gap. This proportionality coefficient is the product of the intertemporal elasticity of consumption σ multiplied by parameter $\omega = 1 + \gamma(2 - \gamma)(\eta/\sigma) - 1$. The parameter ω depends on the economy's openness degree (γ), intertemporal elasticity of substitution of consumption (σ), and intratemporal elasticity of consumption (η). The parameter ω is equal to one in two situations: (i) in a closed economy ($\gamma = 0$), and (ii) when the intertemporal and intratemporal elasticities of substitution are equal ($\sigma = \eta$).

The natural interest rate in this new Keynesian model of a small open economy is a weighted average of the small open economy's natural interest rate if it were a closed economy and the real international interest rate. That is:

$$\bar{r}_t = \frac{1}{\omega}\bar{r}_t^{CE} + \frac{\omega - 1}{\omega}r_t^*. \quad (8.51)$$

When ω is equal to one, the economy is closed and the natural interest rate corresponds to that of a closed economy. When $\omega \rightarrow \infty$ ($\eta \rightarrow \infty$ or $\sigma \rightarrow 0$), the small

open economy's natural interest rate will be equal to the real international interest rate. It is worth noting that, even in this case, the interest rate of the small open economy is not equal to the natural world interest rate. In the small open economy, the natural interest rate varies according to the real international interest rate.

The overlapping generations model's new Keynesian IS curve, deduced in Sect. 3.4 of Chap. 3 [Eq. (3.22)], is given by:

$$\dot{x} = \beta_x x + \beta_r (r - \bar{r}) - \beta_a (a - \bar{a}) - \beta_s (s - \bar{s}). \quad (8.52)$$

The rate of change of the output gap depends on its own level, and on the interest rate, wealth, and terms of trade gaps. Therefore, the total of net foreign assets plays an explicit role in this IS curve. This curve's parameters are related to the model's structural parameters according to:

$$\begin{aligned} \beta_x &= \frac{n\psi\bar{a}}{\bar{c}_y}; \beta_r = \omega_1\sigma; \\ \beta_a &= \frac{\omega_1 n\psi}{\bar{c}_y}; \beta_s = \frac{n\psi\bar{a}\omega_2\xi}{\bar{c}_y}. \end{aligned}$$

The coefficients β_r and β_a are positive. On the other hand, the coefficients β_x and β_s may be either positive or negative. They are positive if the country is a net creditor ($\bar{a} > 0$), and negative if the country is a net debtor ($\bar{a} < 0$). When $n = 0$, it is the particular case of the representative agent model's IS curve because $\beta_x = \beta_a = \beta_s = 0$.

In this model, in the long term, the natural interest rate will be equal to the international interest rate:

$$\bar{r} = \rho + \frac{n\psi}{\sigma} \frac{\bar{a}}{\bar{c}_y} = r^*. \quad (8.53)$$

In the long run, the ratio of financial wealth to consumption will adjust to make it equal to the difference between the international rate of interest and the rate of time preference, which may be positive or negative, depending on the degree of the impatience of the economy compared with the international interest rate.

8.5 Natural Exchange Rate

The IS curve represents the equilibrium condition between saving and investment. In an open economy the deficit of the current account on the balance of payments equals foreign savings. In the presence of a current-account surplus, domestic savings finance the purchase of foreign assets and, if a deficit occurs, foreign savings purchase domestic assets. Subtracting taxes and consumption from the gross national product yields the domestic savings:

$$y - \tau - c(y - \tau) = i(r) + g - \tau + cc(q).$$

This IS curve is specified with the real exchange rate gap instead of the terms of trade gap because the purpose of this section is to determine the natural exchange rate. In the Keynesian model, the real exchange rate is proportional to the terms of trade. This is not the case in other models.

Domestic savings finance investment, public deficit, and the surplus of the current account on the balance of payments, as follows:

$$s(y - \tau) = i(r) + g - \tau + cc(q).$$

When the economy is at full employment, this equality may be written as:

$$s(\bar{y} - \bar{\tau}) - cc(\bar{q}) = i(\bar{r}) + \bar{g} - \bar{\tau}. \quad (8.54)$$

Domestic and foreign savings therefore finance investment and public deficit:

$$s(\bar{y} - \bar{\tau}) + s_e(\bar{q}) = i(\bar{r}) + \bar{f}.$$

At the long run -term equilibrium of an open economy with perfect capital mobility, the real long-term interest rate equals the foreign interest rate. Therefore, the fiscal policy affects the long-run equilibrium real exchange rate, as Fig. 8.3 illustrates. The figure's vertical axis measures the real interest rate and its horizontal axis measures saving and investment.

The IS curve is the sum of investment and public deficit. It has a negative slope because the real interest rate and investment are negatively correlated. If the economy were a closed economy, the natural rate of interest would be given by the IS curve's interception of the SS vertical curve of saving. In a small open economy, the

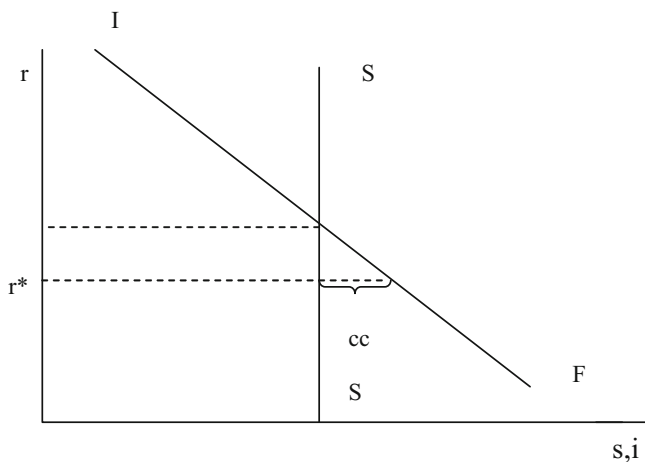


Fig. 8.3 The real international interest rate and the current account deficit

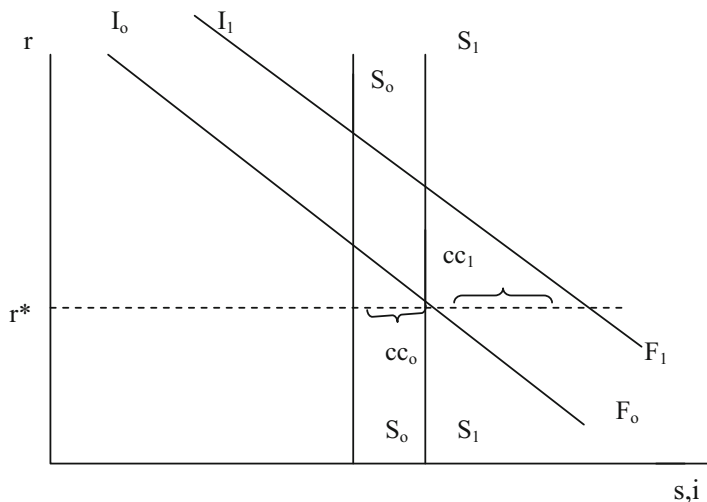


Fig. 8.4 The effects of an increase in the public deficit on the current account: the twin deficits

real long-run interest rate, the natural rate of interest, is equal to the real foreign interest rate (r^*). Figure 8.3 shows that, in this case, the balance of payments will show a deficit, and this deficit determines the long-term real exchange rate.

Figure 8.4 shows what happens in this economy when the full-employment public deficit increases. In a closed economy, the rising deficit would lead to an increase in the real interest rate. In a small open economy with perfect capital mobility, an increase in the public deficit causes the real exchange rate to appreciate, crowding out exports, producing a deficit in the balance of payments current account, a phenomenon known as the twin deficits.

The open-economy Keynesian (Mundell-Fleming) model has an inconsistency that can be easily grasped by analyzing Fig. 8.3. In a small open economy, the foreign interest rate does not generally produce a balanced current account. In Fig. 8.3, the interest rate r^* corresponds to a current account deficit. This deficit increases the country's stock of foreign debt. For as long as the foreign interest rate lies under the rate that would bring a current account deficit, the foreign debt will continue to grow. This model lacks a mechanism to correct this disequilibrium.

Wealth in the IS Curve

One way to address this anomaly would be to assume that domestic consumption depends not only on disposable income, but also on the country's net wealth (a):

$$c = c(y - \tau, a), \quad \frac{\partial c}{\partial a} > 0. \quad (8.55)$$

Saving thus depends on the disposable income and on net wealth according to:

$$s = s(y - \tau, a), \quad \frac{\partial s}{\partial a} < 0. \quad (8.56)$$

When a country's foreign debt increases, its wealth decreases. Consumption then decreases and saving increases. The saving curve (SS) in Fig. 8.3 shifts to the right to the point where the current account equals zero. Therefore, in the long run, the wealth effect adjusts the current account on the balance of payments.

It is straightforward to show that the open economy's IS curve, with the wealth effect on the consumption function, has an additional term for the wealth gap. That is:

$$x = -\alpha(r - \bar{r}) + \beta(q - \bar{q}) + \gamma(f - \bar{f}) + g - \bar{g} + \psi(a - \bar{a}) \quad (8.57)$$

where $\bar{a}$ is the long-run equilibrium net wealth measured as a share of potential output.

8.6 Taylor Rule in an Open Economy

In small open economies, the exchange rate may play an important role in the monetary authority's decision-making process. The Taylor rule then undergoes three important changes. Firstly, the long-run equilibrium real interest rate – the natural rate – becomes dependent on the real international interest rate and varies over time. Therefore, the assumption that it is constant must not be used as is generally the case in empirical studies of the Taylor rule in a large economy assumed to be closed in the sense that external events do not affect its interest rate. Secondly, in a small open economy the exchange rate is an important variable in the Central Bank's decision-making process while setting the interest rate. Therefore, the Taylor rule gains an additional term in small open economies, usually some kind of transformation of the exchange rate. This term must have a zero mean in the long run so that, at equilibrium, the interest rate will equal the natural rate plus inflation, as in a closed economy. Some empirical studies fail to comply with this property, causing a specification error.

The Taylor rule for a small open economy may be specified by the following equation, for example:

$$i_t = \bar{r}_t + \pi_t + \phi\tilde{\pi}_t + \theta\tilde{y}_t + \gamma_1\tilde{q}_t + \gamma_2\Delta\tilde{q}_t, \quad (8.58)$$

where $\bar{r}_t$ is the natural interest rate with an index to indicate that it varies over time, q_t is the real exchange rate, the tilde represents the difference between the current value and the long-run equilibrium value, and Δ is the first difference of the indicated variable. The natural interest rate in a small open economy with perfect capital mobility equals the international interest rate. Note that this international rate is the

current rate of the international financial market, not the natural international interest rate.

The third change in the Taylor rule for an open economy concerns the choice of an appropriate price index. A natural candidate is the inflation rate as measured by the consumer price index. In the Keynesian model, for example, this index is a geometric average of the domestic price and the imported product's price. The inflation rate as measured by the consumer price index is therefore equal to the inflation rate as measured by the domestic product plus a percentage of the rate of change of the terms of trade. In the long run, assuming constant terms of trade, the two measures of the inflation rate will coincide. Although the inflation rate, as measured by the consumer price index, is a natural candidate for use with the Taylor rule, there is nothing to prevent using the inflation rate for domestic goods prices. Using the relationship between the two rates allows checking the equivalence of Taylor rules using different inflation rate specifications.

8.7 Phillips Curve in an Open Economy

The open-economy Phillips curve assumes that the prices of domestic goods are sticky in the short run. The Keynesian Phillips curve also assumes inflation rate rigidity, that is, the inflation rate on domestic goods has an inertia component. In the new Keynesian Phillips curve, the inflation rate is not a predetermined variable, and may change values abruptly.

8.7.1 Keynesian Phillips Curve

In an open economy, as mentioned in the previous section, the consumer price index (P_c) is a weighted average of the prices of domestic and imported goods, with the weight ω being equal to the share of consumer expenditures on imported goods and services. That is:

$$P_c = P^{1-\omega}(EP^*)^\omega = P\left(\frac{EP^*}{P}\right)^\omega. \quad (8.59)$$

The consumer price index may be written as a function of the domestic price index and of the terms of trade S :

$$P_c = PS^\omega. \quad (8.60)$$

Therefore, inflation rate, as measured by the consumer index, depends on the inflation rate of domestic goods and on the rate of change of the terms of trade:

$$\pi_c = \pi + \omega \dot{s}. \quad (8.61)$$

The adjustment rate of wages depends on the expected inflation rate, as measured by the inflation rate of consumer prices, on the conditions of the labor market as measured by the output gap (x), and on the growth of labor productivity. That is:

$$\frac{\dot{W}}{W} = \pi_c^e + \delta^* x + \frac{(\dot{y}/L)}{y/L}. \quad (8.62)$$

Firms adjust the prices of domestic goods by adding a margin to the unit production cost:

$$P = (1 + k) \frac{WL}{y} = (1 + k) \frac{W}{y/L}. \quad (8.63)$$

The inflation rate of domestic goods and services equals the difference between the growth rate of wages and the growth rate of labor productivity:

$$\pi = \frac{\dot{W}}{W} - \frac{(\dot{y}/L)}{y/L}. \quad (8.64)$$

Substituting the growth rate of wages in this expression yields the following Phillips curve:

$$\pi = \pi^e + \omega \dot{s} + \delta^* x. \quad (8.65)$$

The difference between this Phillips curve and the one deduced for a closed economy lies in the inclusion of the rate of change of the terms of trade as one of the arguments of the inflation rate of domestic goods and services.

When inflation has an inertia component, the expected inflation rate depends on the past inflation rate:

$$\pi^e = \pi(t - h). \quad (8.66)$$

The Phillips curve is then expressed as:

$$\pi = \pi(t - h) + \omega \dot{s} + \delta^* x.$$

The first-order expansion of the past inflation rate around the point that corresponds to the current inflation rate yields,

$$\pi(t - h) = \pi(t) + \dot{\pi}(t - (t - h)).$$

Therefore, the Phillips curve expresses inflation acceleration as a function of the rate of change of the terms of trade and of the output gap:

$$\dot{\pi} = \gamma \dot{s} + \delta x, \quad (8.67)$$

where $\gamma = \omega/h$ and $\delta = \delta^*/h$.

When the expected inflation rate is based on future inflation,

$$\pi^e = \pi(t+h), \quad (8.68)$$

the first-order expansion of this function around point t is given by:

$$\pi(t+h) = \pi(t) + \dot{\pi}(t)(t+h-t).$$

Substituting this expression in the wages equation yields, with a little algebra, the following Phillips curve:

$$\dot{\pi} = -\gamma\dot{s} - \delta x. \quad (8.69)$$

8.7.2 New Keynesian Phillips Curve

The new Keynesian Phillips curve for an open economy assumes that the prices of domestic goods are rigid and uses the same Calvo mechanism for a closed economy. The new Keynesian Phillips curve is therefore given by:

$$\pi_{H,t} = \beta E_t \pi_{H,t+1} + \delta(p_{H,t}^* - p_{H,t}), \quad (8.70)$$

where the index H denotes the domestic product, that is, the inflation rate of domestic goods, and $p_{H,t}^*$ would be the (log of) the optimum price in the absence of rigidity, while $p_{H,t}$ would be the (log of) the effective market price. This section's notation is the same used in the deduction of the new Keynesian IS curve in Chap. 2 [Sect. 2.5]. The δ coefficient depends on the discount rate (β) and on the price rigidity parameter (λ).

The optimum price equals the marginal cost of production plus a margin according to the pricing at a firm in monopolistic competition. The marginal cost equals wage divided by the marginal product of labor. Assuming a linear production function, $Y = AN$, the marginal cost in real terms, deflated at the price index or domestic goods, is given by:

$$Cm_{gr} = \frac{W}{AP_H}.$$

Using lower-case letters to denote the logs of the capital letters, this expression may be written as:

$$cm_{gr} = w - p + p - p_H - a.$$

The consumer price index has been added and subtracted in this expression. The marginal cost breaks down into three components: (i) real wages; (ii) the price of domestic goods relative to the consumer price index; and (iii) the marginal product of labor.

In the new Keynesian model, the real wage equals the marginal rate of substitution between consumption and leisure. Assume that the utility function has the following functional form:

$$u(C, N) = \frac{C^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} - \frac{N^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}}.$$

It follows that real wage is:

$$\frac{W}{P} = \frac{-\frac{\partial u}{\partial N}}{\frac{\partial u}{\partial C}} = \frac{N^{\frac{1}{\varphi}}}{C^{-\frac{1}{\sigma}}}.$$

Taking the log of both sides of the expression, it transforms into:

$$w - p = \frac{1}{\varphi} n + \frac{1}{\sigma} c.$$

Equations (2.44), (2.45) and (2.47), from Chap. 2, which we replicate below for convenience, establish the following relationships:

$$y = y^* + \frac{\sigma\omega}{1-\gamma}q, \quad (8.71)$$

$$c = y^* + \sigma q, \quad (8.72)$$

$$p = p_H + \frac{\gamma}{1-\gamma}q. \quad (8.73)$$

Using Eq. (8.73) and the equation of real wages, the real marginal cost is expressed as:

$$\text{cmgr} = \frac{1}{\varphi}n + \frac{1}{\sigma}c + \frac{\gamma}{1-\gamma}q - a.$$

The production function may be used to eliminate the quantity of labor from this equation, and consumption can be eliminated using Eq. (8.72), yielding the following expression:

$$\text{cmgr} = \frac{1}{\varphi}y + \frac{1}{\sigma}y^* + \frac{1}{1-\gamma}q - \left(1 + \frac{1}{\varphi}\right)a.$$

The real exchange rate from Eq. (8.71), when substituted in this equation, produces the following real marginal cost:

$$\text{cmgr} = \left(\frac{1}{\varphi} + \frac{1}{\sigma\omega}\right)y + \frac{1}{\sigma}\left(\frac{\omega-1}{\omega}\right)y^* - \left(1 + \frac{1}{\varphi}\right)a.$$

The relative price according to the Phillips curve is therefore equal to the marginal cost plus the profit margin (μ). That is:

$$p_{H,T}^* - p_{H,t} = \mu + \text{cmgr}_t.$$

In the long run, when this relative price is equal to zero, the margin equals the long-run marginal cost with an inverted sign. It follows that this relative price equals the difference between the short- and long-run marginal costs. The previous real marginal cost expression allows writing:

$$p_{H,T}^* - p_{H,t} = \left(\frac{1}{\varphi} + \frac{1}{\sigma\omega} \right) (y_t - \bar{y}_t),$$

where potential output is represented by $\bar{y}$. The new Keynesian Phillips curve is given by:

$$\pi_{H,t} = \beta E_t \pi_{H,t+1} + \kappa x_t. \quad (8.74)$$

The coefficient of the output gap depends on the model's structural parameters according to:

$$\kappa = \delta \left(\frac{1}{\varphi} + \frac{1}{\sigma\omega} \right).$$

Taking into account the ratio of the price index of domestic goods to the consumer price index, the Phillips curve for the inflation rate as measured by this index is:

$$\pi_t = \beta E_t \pi_{t+1} + \frac{\gamma}{1-\gamma} (q_t - \beta E_t q_{t+1}) + \kappa x_t. \quad (8.75)$$

A little algebra enables easy deduction of the new Keynesian Phillips curves with continuous variables for the inflation rates of domestic goods and the consumer goods. They are given by the following expressions, respectively:

$$\dot{\pi}_H = \rho \pi_H + \kappa x, \quad (8.76)$$

$$\dot{\pi} = \rho \pi + \frac{\gamma}{1-\gamma} (\dot{q} - \rho q) + \kappa x. \quad (8.77)$$

The rate of time preference ρ and the discount rate β are related by means of $\rho = (1 - \beta)/\beta$. In the long run, as in a closed economy, the new Keynesian Phillips curve is not vertical, with a trade-off between inflation and output.

Despite the rigorous microfounded deduction of the new Keynesian Phillips curve, it is worth emphasizing that at least one of the ingredients used in the deduction is not palatable. Equation 8.72 may be interpreted as a “consumption function” in which consumption depends on world income and the real exchange rate. Empirical evidence rejects this assumption.

8.8 Exercises

1. Consider the following model of a small open economy (monetary approach of the balance of payments with a fixed exchange rate):

$$\begin{aligned}M^s &= C + R \\M^d &= PL(y, i) \\i &= i^* \\y &= \bar{y} \\P &= EP^*, E = \bar{E} = \text{constant}\end{aligned}$$

The symbols have the following meanings: M^s = quantity of money supplied; C = net domestic credit; R = international reserves; M^d = quantity of money demanded; P = domestic price level; i = domestic interest rate; i^* = international interest rate; E = nominal exchange rate; P^* = international price level.

- What is the effect of a foreign exchange devaluation on the balance of payments?
 - What is the effect of an increase in net domestic credit on the balance of payments?
2. Consider the following model of a small open economy (monetary approach of the balance of payments with a flexible exchange rate):
- Equilibrium in country A's money market: $\frac{M}{P} = L(y, i)$.
- Equilibrium in country B's money market: $\frac{M^*}{P^*} = L(y^*, i^*)$.
- Exchange rate: $E = \frac{P}{P^*}$.
- Comment on the following:
- The exchange rate depreciates when the country grows more quickly than others.
 - The exchange rate appreciates when the money stock grows more quickly than that of other countries.
3. Consider the following model of a small open economy (monetary approach of the balance of payments with a flexible exchange rate):
- Equilibrium in country A's money market: $m - p = \alpha y - \beta i$,
- Equilibrium in country B's money market: $m^* - p^* = \alpha y^* - \beta i^*$,
- Exchange rate: $e = p - p^*$,
- Uncovered interest parity: $i = i^* + \dot{e}$.
- Deduce the differential equation for exchange rate determination.
 - Does the solution to this equation include a bubble component?
4. Consider the regression:

$$e_{t+1} - e_t = a_0 + a_1(f_t - e_t) + \epsilon_t,$$

or

$$e_{t+1} - e_t = a_0 + a_1(i_t - i_t^*) + \epsilon_t.$$

The symbols have the following meanings: e is the log of the exchange rate on the spot market, f is the log of the exchange rate on the forward market, i_t is the domestic interest rate, and i_t^* is the foreign interest rate.

- (a) Is the forward market's exchange rate, or the differential of the interest rate, a good predictor of the future exchange rate on the forward market?
 - (b) Several empirical studies have obtained negative values for the a_1 parameter. How would you interpret this result?
5. (Harberger-Laursen-Metzler (HLM) Effect). In an open economy, the national product (Y) equals the sum of absorption (A) and the balance of the current account on the balance of payments ($X-Z$):

$$Y = A + X - Z.$$

The price index of absorption is a geometric average of the price of the domestic product (P) and the price of the foreign product, converted into the domestic currency at the exchange rate (SP^*):

$$P_a = P^{1-\alpha}(EP^*)^\alpha = P\left(\frac{EP^*}{P}\right)^\alpha = PS^\alpha.$$

where α is the share of the imported good in absorption and $S = EP^*/P$ is the terms of trade. The product may be written in real terms as:

$$y = d + x - Sz,$$

where $d = \frac{P_a a}{P}$, $x = \frac{X}{P}$ and $z = \frac{Z}{SP^*}$. Real absorption (a) depends on the real income as defined by: $y_a = \frac{P_y}{P_a}$.

- (a) Show that the elasticity of expenditures with respect to the terms of trade ($\eta_{d,S}$) is:

$$\eta_{d,S} = \alpha(1 - \eta_{a,y_a}),$$

where η_{a,y_a} is the elasticity of absorption with respect to the real income.

- (b) Show that saving ($s \cdot y - d - \tau$, where τ represents taxes) varies with the terms of trade:

$$\frac{\partial s}{\partial S} = \frac{\alpha d}{S} (\eta_{a,y_a} - 1).$$

- (c) According to the HLM effect, worsening (improving) terms of trade decrease (increase) the economy's real income, reducing (increasing) saving. For a given level of investments, the reduction (increase) in saving deteriorates (improves) the current account on the balance of payments. What happens to the current account on the balance of payments if $\eta_{a,y_a} < 1$ and the country's terms of trade improve?

6. Consider the model:

$$\begin{cases} M = m(i, i^* + \dot{e})W \\ B = b(i, i^* + \dot{e})W \\ EF = f(i, i^* + \dot{e})W \\ W = M + B + EF \\ \dot{F} = \varphi(EP^*/P) + i^*F \end{cases}$$

The symbols have the following meanings: M = money stock; B = stock of domestic securities; F = stock of foreign currency denominated securities; E = exchange rate; e = log of exchange rate; i = domestic interest rate; i^* = foreign interest rate; P = domestic price level (exogenous); P^* = foreign price level (exogenous).

- (a) Discuss the specification of each of the model's equations and analyze its equilibrium.
 (b) Show what happens to s and F in each of the following circumstances: (i) an increase in M ; (ii) an increase in B .

7. Consider the following model of a flexible exchange rate portfolio:

$$\begin{aligned} E &= g(F, M, B, i^*), \quad \frac{\partial E}{\partial F} < 0 \\ \dot{F} &= X\left(\frac{E}{P}\right) + i^*F \end{aligned}$$

where E is the exchange rate, F is total foreign currency denominated securities, M is the money stock, B is the securities stock, i^* is the international interest rate, P is the domestic price level. $\dot{F} = dF/dt$, and $X(E/P)$ represents net exports. Based on this model, comment on the following proposition. "A country with a deficit on the current account on the balance of payments tends to depreciate the exchange rate, while one with a surplus tends to appreciate the exchange rate."

8. In a small open economy, the Central Bank sets the nominal interest rate. That is:

$$i = \bar{i}.$$

- (a) Is this economy's nominal exchange rate determined? (Hint: use the uncovered interest rate parity).
 (b) Compare your response with what would happen in a closed economy if the Central Bank used the same monetary policy rule.

9. The Taylor rule adopted by a small open economy is:

$$i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x + \lambda q.$$

The coefficients ϕ , θ , and λ are positive and assume, for simplicity, that the long-run real foreign exchange rate is equal to zero: $\bar{q} = 0$.

The foreign country's Taylor rule is:

$$i^* = \bar{r}^* + \pi^* + \phi(\pi^* - \bar{\pi}^*) + \theta x^*.$$

The asterisk (*) represents the foreign variables and the two rules' parameters are equal to simplify the algebra.

The two countries' real interest rates are related by means of the real interest rate parity:

$$r - r^* = \dot{q}.$$

(a) Show that the real exchange rate follows the differential equation:

$$\dot{q} = -f + \lambda q,$$

and point out the model's fundamentals (f).

(b) Show the solution to this differential equation with the two components, fundamentals and bubble.

Chapter 9

Economic Fluctuation and Stabilization in an Open Economy



This chapter presents economic fluctuation and stabilization models in a small open economy. Section 9.1 addresses the Mundell-Fleming-Dornbusch model under a fixed exchange rate regime. Section 9.2 adds the balance-of-payments equation to this model, since wealth is an argument in the consumption function. Section 9.3 introduces the fixed exchange rate regime in a new Keynesian model. Section 9.4 deals with the Mundell-Fleming-Dornbusch model under a flexible exchange rate regime. Section 9.5 analyzes the extended Mundell-Fleming-Dornbusch model for a flexible exchange rate. Section 9.6 presents the flexible exchange rate for the new Keynesian model.

9.1 Mundell-Fleming-Dornbusch Model: Fixed Exchange Rate

Assume that the Keynesian model, also known as the Mundell-Fleming-Dornbusch model, for a small open economy has the following equations: the IS curve, the Phillips curve, the uncovered interest rate parity, and the monetary policy rule of the exchange rate. The equations are:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(s - \bar{s}), \quad (9.1)$$

$$\text{PC: } \dot{\pi} = \gamma\dot{s} + \delta x, \quad (9.2)$$

$$\text{UIP: } r = \bar{r} + \dot{s}, \bar{r} = r^*, \quad (9.3)$$

$$\text{MPR: } e = \bar{e}, \quad (9.4)$$

$$\text{IC: Given } p(0) \text{ and } \pi(0). \quad (9.5)$$

The real interest rate in the uncovered interest rate parity equation is measured by deflating the nominal interest rate by the price index of domestic goods, rather than at

the consumer price index. For simplicity, the IS curve assumes public deficit and government spending at their full-employment values. The model has five variables: the output gap, the real interest rate, the terms of trade, the inflation rate, and the nominal exchange rate. The model's fifth equation is the natural log definition of the terms of trade, expressed as:

$$s = e + p^* - p. \quad (9.6)$$

Algebra

This is a dynamical model that can be solved in different ways. The solution shown next will reduce it to a dynamical system with two variables – the inflation rate and the real output. To this end, we start by taking the derivative of both sides of the terms of trade expression with respect to time. That is:

$$\dot{s} = \dot{e} + \pi^* - \pi.$$

The monetary policy rule assumes a fixed nominal exchange rate ($\dot{e} = 0$). Therefore, the rate of change of the terms of trade is equal to the gap between the foreign and the domestic inflation rates:

$$\dot{s} = \dot{\pi} - \pi.$$

According to the uncovered interest rate parity, the difference between the domestic and foreign real interest rates equals the variation in the terms of trade. Therefore, it follows from the last equation that:

$$r - \bar{r} = \dot{s} = \dot{\pi} - \pi.$$

Substituting this result in the Phillips curve, we find that inflation acceleration depends on the gap between the foreign and domestic inflation rates and on the output gap:

$$\dot{\pi} = \gamma(\pi^* - \pi) + \delta x.$$

To obtain the model's second differential equation, let us begin by taking the derivative of the IS curve with respect to time:

$$\dot{x} = -\alpha \dot{r} + \beta \dot{s}.$$

Bearing in mind that the variation of the terms of trade equals the difference between the foreign and domestic inflation rates, it follows that:

$$\dot{x} = -\alpha \dot{r} + \beta(\pi^* - \pi).$$

Because the difference between the domestic and foreign real interest rates equals the derivative of the terms of trade with respect to time, which equals the difference between the foreign and domestic inflation rates, it is easily deduced that:

$$\dot{r} = \ddot{s} = \dot{\pi}^* - \dot{\pi} = -\dot{\pi}.$$

This equation assumes a constant foreign inflation rate ($\dot{\pi}^* = 0$). Substituting this expression in the equation of the derivative of real output with respect to time yields:

$$\dot{x} = \alpha\dot{\pi} + \beta(\pi^* - \pi).$$

Substituting the expression of inflation acceleration in this equation results in:

$$\dot{x} = \alpha\gamma(x^* - \pi) + \alpha\delta x + \beta(\pi^* - \pi).$$

Rearranging the terms of this equation, we may write that the rate of change of the output gap depends on the gap between the foreign and domestic inflation rates and on the output gap according to:

$$\dot{x} = -(\alpha\gamma + \beta)(\pi - \pi^*) + \alpha\delta x.$$

Dynamical System

The model's dynamical system with a fixed exchange rate regime has two differential equations, one for the inflation rate and another for the output gap:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} -\gamma & \delta \\ -(\alpha\gamma + \beta) & \alpha\delta \end{bmatrix} \begin{bmatrix} \pi - \pi^* \\ x \end{bmatrix}. \quad (9.7)$$

This system's Jacobian matrix is:

$$J = \begin{bmatrix} \frac{\partial \dot{\pi}}{\partial \pi} & \frac{\partial \dot{\pi}}{\partial x} \\ \frac{\partial \dot{x}}{\partial \pi} & \frac{\partial \dot{x}}{\partial x} \end{bmatrix} = \begin{bmatrix} -\gamma & \delta \\ -(\alpha\gamma + \beta) & \alpha\delta \end{bmatrix}.$$

This matrix's determinant is positive,

$$|J| = -\alpha\delta\gamma + \delta(\alpha\gamma + \beta) = \delta\beta > 0,$$

and its trace may be either positive or negative:

$$trJ = \alpha\delta - \gamma.$$

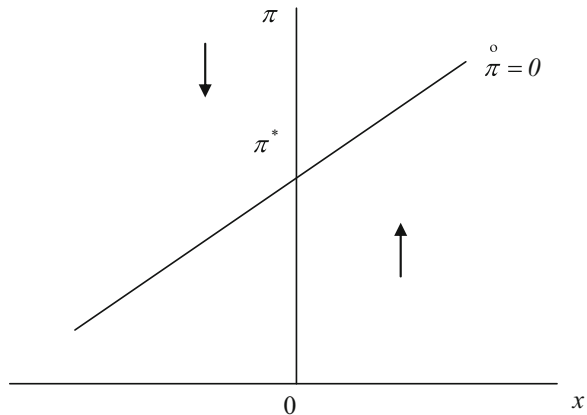
For the dynamical system to be stable, assume that the following inequality is valid:

$$\alpha\delta - \gamma < 0.$$

In this case, the trace of the dynamical system's Jacobian matrix is negative.

The points in the phase diagram, with inflation on the vertical axis and the output gap on the horizontal axis, along which the value of inflation does not change, correspond to:

Fig. 9.1 The dynamics of the inflation rate



$$\dot{\pi} = 0 \Rightarrow \pi = \pi^* + \frac{\delta}{\gamma}x.$$

Figure 9.1 graphs the inflation rate and the arrows indicate what happens with the inflation rate's dynamics. Inflation decreases below the line and increases above it.

To draw the phase diagram of the differential equation of the output gap, the equation in which the output gap's value does not change is given by:

$$\dot{x} = 0 \Rightarrow \pi = \pi^* + \frac{\alpha\delta}{\alpha\gamma + \beta}x.$$

This equation may be written as:

$$\pi = \pi^* + \frac{\delta}{\gamma\left(1 + \frac{\beta}{\alpha\gamma}\right)}x.$$

This curve's angular coefficient is less than that of the line where the value of the inflation rate remains constant. Figure 9.2 graphs the line and the direction of the output gap. The output gap decreases above the line and increases below it.

Figure 9.3 shows the model's phase diagram with four regions. The economy moves northeast in region I; northwest in region II; southwest in region III, when the economy is at disequilibrium; and southeast in region IV. At the equilibrium point, the inflation rate equals the foreign inflation rate and output equals the potential output.

Experiment

Figure 9.4 describes an experiment in which the foreign inflation rate rises to π^*_1 from π^*_0 . At the new long-run equilibrium, the inflation rate equals the new foreign inflation rate. In the short-run, the inflation rate rises slowly because of inflation inertia. The economy is in region I and real output starts to increase as Fig. 9.5 shows. After some time, the inflation rate overshoots the foreign inflation rate and

Fig. 9.2 The dynamics of the output gap

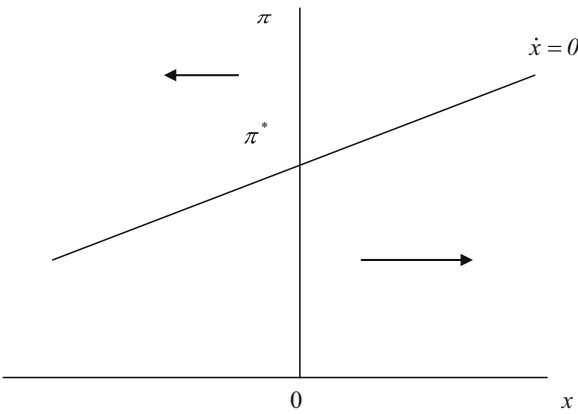


Fig. 9.3 The phase diagram for the inflation rate and the output gap

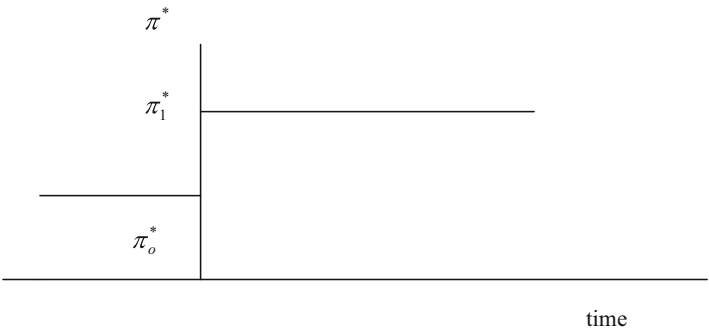
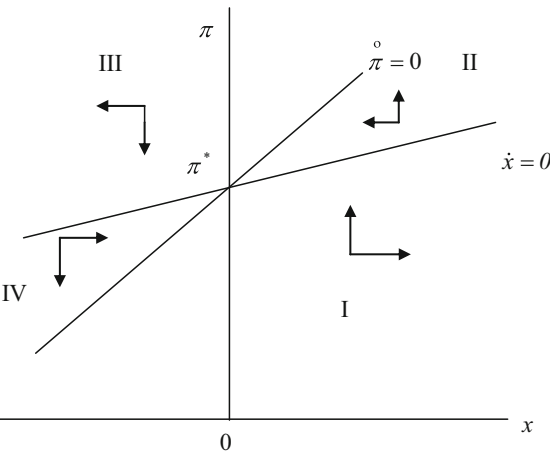


Fig. 9.4 An unanticipated permanent increase in the foreign inflation rate

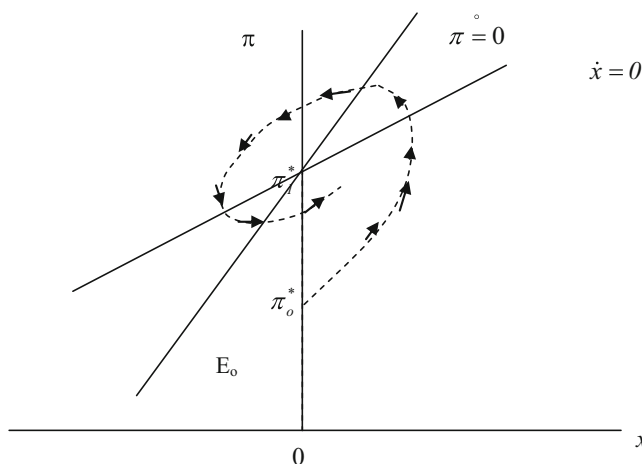
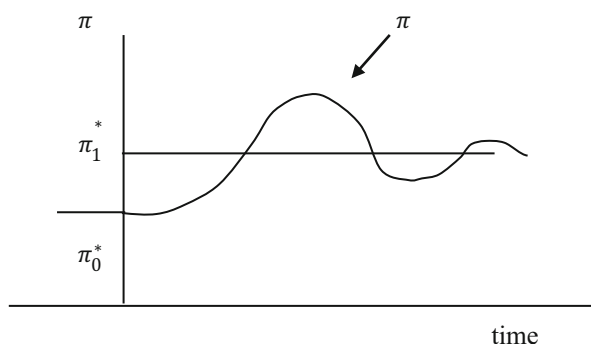


Fig. 9.5 The effects of an unanticipated permanent increase in the foreign inflation rate

Fig. 9.6 The overshooting of the domestic inflation rate



output continues to increase. The reason for the overshooting is that the real exchange rate increases in the beginning of the adjustment process but returns to its initial value in the long-run. For this to occur, the domestic inflation rate must grow faster than the foreign inflation rate for some time.

Figure 9.6 describes the domestic inflation's path of adjustment to the change in the foreign inflation rate, assuming that the model's parameters produce an oscillatory damping adjustment.

9.2 Extended Mundell-Fleming-Dornbusch Model: Fixed Exchange Rate

The balance of payments surplus (deficit) equals the sum of interest revenues (spending) and net exports:

$$\frac{\dot{A}}{\bar{y}} = r \frac{A}{\bar{y}} + \frac{EX}{\bar{y}}. \quad (9.8)$$

The variables have been defined proportionally to potential output. The letter a indicates the ratio of wealth to potential output $a = A/\bar{y}$, and the ratio of net exports to output is defined as: $ex = EX/\bar{y}$. It follows that:

$$\dot{a} = ra + ex.$$

For simplicity, we assume that the growth rate of potential output is zero. At stationary equilibrium ($\dot{a} = 0$):

$$\bar{r}\bar{a} + \bar{e}\bar{x} = 0.$$

Therefore,

$$\dot{a} = ra - \bar{r}\bar{a} + ex - \bar{e}\bar{x}.$$

The first-order expansion of the function $f(a, r) = ra$ is given by:

$$ra = \bar{r}\bar{a} + \bar{r}(a - \bar{a}) + \bar{a}(r - \bar{r}).$$

The $\bar{a}$ coefficient will be positive if the country is a net creditor and negative if it is a net debtor. The linear approximation of net exports is expressed by:

$$ex = \bar{e}\bar{x} + \Phi(s - \bar{s}). \quad (9.9)$$

The balance of payments surplus (deficit) can then be written as follows:

$$\dot{a} = \bar{r}(a - \bar{a}) + \bar{a}(r - \bar{r}) + \Phi(s - \bar{s}). \quad (9.10)$$

Model

The Keynesian model of a small open economy under the fixed exchange rate regime, with wealth effect on consumption, has: an IS curve that depends on the gaps of the interest rate, the terms of trade, and of wealth; a Phillips curve whose arguments are the rate of change of the terms of trade and the output gap; an interest rate parity equation; a current account balance of payments equation; and the monetary policy rule according to which the exchange rate is fixed by the Central Bank. That is:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(s - \bar{s}) + \Psi(a - \bar{a}), \quad (9.11)$$

$$\text{PC: } \dot{x} = \gamma\dot{s} + \delta x, \quad (9.12)$$

$$\text{UIP: } \dot{s} = r - \bar{r}, \quad (9.13)$$

$$\text{BP: } \dot{a} = \bar{r}(a - \bar{a}) + \bar{a}(r - \bar{r}) + \Phi(s - \bar{s}), \quad (9.14)$$

$$\text{MPR: } e = ct, \quad (9.15)$$

$$\text{IC: Given, } p(0), \pi(0), s(0) \text{ and } a(0). \quad (9.16)$$

Algebra

The terms of trade are defined by:

$$s = e + p^* - p. \quad (9.17)$$

By taking the derivative of both sides of the expression with respect to time and taking into account that $\dot{e} = 0$ as a result of the monetary policy rule:

$$\dot{s} = \pi^* - \pi = r - \bar{r},$$

where $\pi^* = dp^*/dt$ is the foreign inflation rate. Substituting the real interest rate gap in the IS curve, uncovered interest rate parity, and balance-of-payments equations yields:

$$\begin{aligned} x &= \alpha(\pi - \pi^*) + \beta(s - \bar{s}) + \Psi(a - \bar{a}), \\ \dot{s} &= -(\pi - \pi^*), \\ \dot{a} &= -\bar{a}(\pi - \pi^*) + \Phi(s - \bar{s}) + \bar{r}(a - \bar{a}). \end{aligned}$$

Substituting the interest rate gap and the output gap in the Phillips curve yields:

$$\dot{\pi} = (-\gamma + \alpha\delta)(\pi - \pi^*) + \beta\delta(s - \bar{s}) + \delta\Psi(a - \bar{a}).$$

Dynamical System

The dynamical system of a small open economy Keynesian model with wealth effect on consumption has three differential equations: one for the inflation rate, one for the terms of trade, and one for the balance of payments current account. That is:

$$\begin{bmatrix} \dot{\pi} \\ \dot{s} \\ \dot{a} \end{bmatrix} = \begin{bmatrix} -\gamma + \alpha\delta & \beta\delta & \delta\Psi \\ -1 & 0 & 0 \\ -\bar{a} & \Phi & \bar{r} \end{bmatrix} \begin{bmatrix} \pi - \pi^* \\ s - \bar{s} \\ a - \bar{a} \end{bmatrix}. \quad (9.18)$$

This system's Jacobian matrix is:

$$J = \begin{bmatrix} -\gamma + \alpha\delta & \beta\delta & \delta\Psi \\ -1 & 0 & 0 \\ -\bar{a} & \Phi & \bar{r} \end{bmatrix}.$$

This matrix's determinant and trace are respectively equal to:

$$\begin{aligned} |J| &= -\Phi\delta\Psi + \beta\delta\bar{r}, \\ \text{tr}J &= -\gamma + \alpha\delta + \bar{r}. \end{aligned}$$

For the dynamical system to be stable, the three characteristic roots should be negative. Thus, the determinant and trace must be negative. For this to occur, the following inequality has to be fulfilled:

$$\bar{r} < \min\left(\frac{\Phi\Psi}{\beta}, \gamma - \alpha\delta\right).$$

The dynamical system's solution is given by:

$$\begin{aligned}\pi &= \pi^* + c_1 v_{11} e^{\mu_1 t} + c_2 v_{21} e^{\mu_2 t} + c_3 v_{31} e^{\mu_3 t} \\ s &= \bar{s} + c_1 v_{12} e^{\mu_1 t} + c_2 v_{22} e^{\mu_2 t} + c_3 v_{32} e^{\mu_3 t}, \\ a &= \bar{a} + c_1 v_{13} e^{\mu_1 t} + c_2 v_{23} e^{\mu_2 t} + c_3 v_{33} e^{\mu_3 t},\end{aligned}$$

where $\mu_1 > \mu_2 > \mu_3$ are the characteristic roots obtained from the characteristic equation:

$$\begin{vmatrix} -\gamma + \alpha\delta & \beta\delta & \delta\Psi \\ -1 & -\mu & 0 \\ -\bar{a} & \Phi & \bar{r} - \mu \end{vmatrix} = 0.$$

The constants c_1 , c_2 , and c_3 are determined based on the initial conditions, and v_{ij} are the components of the characteristic vectors determined by the solutions of linear system:

$$\begin{bmatrix} -\gamma + \alpha\delta - \mu & \beta\delta & \delta\Psi \\ -1 & -\mu & 0 \\ -\bar{a} & \Phi & \bar{r} - \mu \end{bmatrix} \begin{bmatrix} v_{i1} \\ v_{i2} \\ v_{i3} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

This system satisfies the following relationships:

$$\begin{aligned}(-\gamma + \alpha\delta - \mu)v_{i1} + \beta\delta v_{i2} + \delta\Psi v_{i3} &= 0 \\ -v_{i1} - \mu v_{i2} &= 0 \\ -\bar{a}v_{i1} + \Phi v_{i2} + (\bar{r} - \mu)v_{i3} &= 0\end{aligned}$$

According to the second equation, the ratio of the first and second elements of the i th vector is positive, as the characteristic root is negative. That is:

$$\frac{v_{i1}}{v_{i2}} = -\mu > 0.$$

This property will be helpful below to deduce an empirical implication of this model.

Predictions

The limit, as time increases indefinitely, of the ratio of the inflation gap to the terms of trade gap is given by:

$$\lim_{t \rightarrow \infty} \frac{\pi - \pi^*}{s - \bar{s}} = \lim_{t \rightarrow \infty} \frac{c_1 v_{11} e^{\mu_1 t} \left[1 + \frac{c_1 v_{21}}{c_1 v_{11}} e^{(\mu_2 - \mu_1)t} + \frac{c_3 v_{31}}{c_1 v_{11}} e^{(\mu_3 - \mu_1)t} \right]}{c_1 v_{12} e^{\mu_1 t} \left[1 + \frac{c_2 v_{22}}{c_1 v_{12}} e^{(\mu_2 - \mu_1)t} + \frac{c_3 v_{32}}{c_1 v_{12}} e^{(\mu_3 - \mu_1)t} \right]}. \quad (9.19)$$

Assume that $\mu_1 > \mu_2 > \mu_3$. Therefore, $\mu_2 - \mu_1 < 0$ and $\mu_3 - \mu_1 < 0$. When $t \rightarrow \infty$, the terms in brackets in the numerator and denominator approach one. Therefore, at the limit,

$$\lim_{t \rightarrow \infty} \frac{\pi - \pi^*}{s - \bar{s}} = \frac{v_{11}}{v_{12}} > 0. \quad (9.20)$$

This means that if the inflation rate is higher than the world inflation rate as the former converges to the latter, the terms of trade will also exceed the long-term terms of trade. Figure 9.7 shows this property. The figure's phase diagram is built from the differential equation resulting from the combination of interest-rate parity with the fixed exchange rate monetary rule. That is:

$$\dot{s} = -(\pi - \pi^*),$$

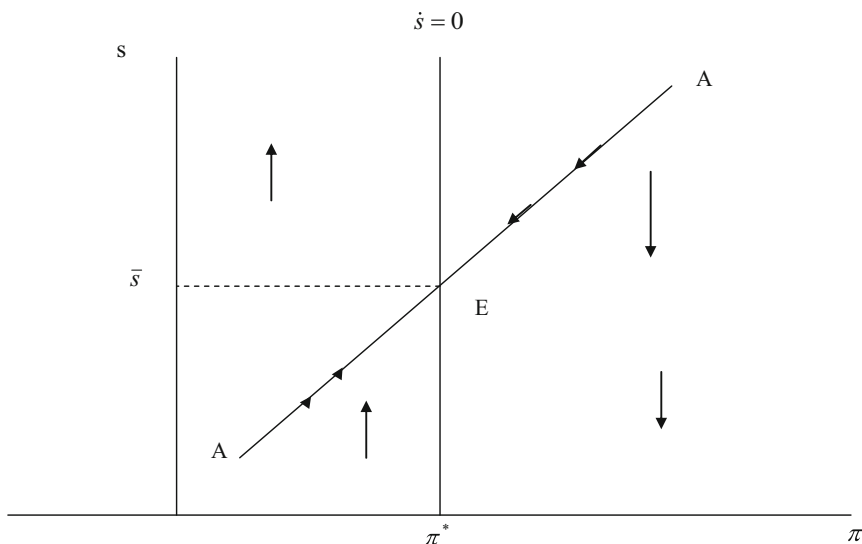


Fig. 9.7 The phase diagram for the terms of trade and the inflation rate

when $\pi > \pi^*$, $\dot{s} < 0$ and if $\pi < \pi^*$, $\dot{s} > 0$, as the arrows in Fig. 9.7 show. The AA line is the model's dominant eigenvector (characteristic vector) ray.

Another important property this model shows is the relationship between the terms of trade and the country's net foreign wealth (debt). To deduce this property, one calculates the limit of the ratio of the terms of trade gap to the wealth gap. That is:

$$\lim_{t \rightarrow \infty} \frac{s - \bar{s}}{a - \bar{a}} = \lim_{t \rightarrow \infty} \frac{c_1 v_{12} e^{\mu_1 t} \left[1 + \frac{c_1 v_{22}}{c_1 v_{12}} e^{(\mu_2 - \mu_1)t} + \frac{c_3 v_{32}}{c_1 v_{12}} e^{(\mu_3 - \mu_1)t} \right]}{c_1 v_{13} e^{\mu_1 t} \left[1 + \frac{c_2 v_{23}}{c_1 v_{12}} e^{(\mu_2 - \mu_1)t} + \frac{c_3 v_{33}}{c_1 v_{13}} e^{(\mu_3 - \mu_1)t} \right]}. \quad (9.21)$$

This limit, for the same reasons applied to the previous one, equals:

$$\lim_{t \rightarrow \infty} \frac{s - \bar{s}}{a - \bar{a}} = \frac{v_{12}}{v_{13}} < 0. \quad (9.22)$$

The sign of the ratio of v_{12} to v_{13} can be obtained as follows. The combination of the characteristic vector's first two equations implies:

$$\frac{v_{12}}{v_{13}} = \frac{-\delta \Psi}{[-(-\gamma + \alpha \delta - \mu)\mu + \beta \delta]}.$$

The model's characteristic equation is given by:

$$-(-\gamma + \alpha \delta - \mu)\mu(\bar{r} - \mu) - \Phi \delta \Psi - \bar{a} \mu \delta \Psi + \beta \delta (\bar{r} - \mu) = 0,$$

which may be rewritten as:

$$[-(-\gamma + \alpha \delta - \mu)\mu + \beta \delta](\bar{r} - \mu) = \Phi \delta \Psi + \bar{a} \mu \delta \Psi.$$

Assuming that $\bar{a} < 0$ (debtor country), $\bar{a} \mu > 0$. Therefore, the right side of this expression is positive. Since $\mu < 0$, $\bar{r} - \mu > 0$. Therefore,

$$-(-\gamma + \alpha \delta - \mu)\mu + \beta \delta > 0.$$

When the country is a debtor, $a = -d$ and $\bar{a} = -\bar{d}$, where d is the country's net foreign debt. Therefore:

$$\lim_{t \rightarrow \infty} \frac{s - \bar{s}}{d - \bar{d}} > 0. \quad (9.23)$$

This model implies that if the country's foreign debt exceeds the long-term equilibrium level, the terms of trade will exceed its long-term equilibrium for the debt to be reduced and reach its stationary equilibrium level.

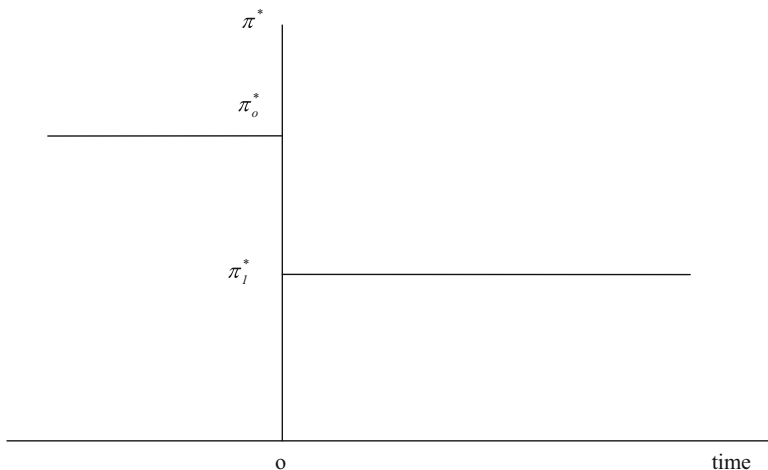


Fig. 9.8 An unanticipated permanent decrease in the foreign inflation rate

Experiment

Consider the experiment in Fig. 9.8, which assumes a permanent reduction of the world inflation to π_l^* from π_o^* .

Figure 9.9 reproduces the phase diagram in Fig. 9.7 to analyze the experiment. Point E_0 is the initial equilibrium point when world inflation – which equals domestic inflation – is equal to π_o^* and the terms of trade equals $\bar{s}$. The ending equilibrium point is point E_f when the inflation rate equals π_l^* . The long-term equilibrium terms of trade remain unchanged because they are independent from the economy's nominal variables. The adjustment path implies a process of foreign exchange appreciation and rising foreign debt. The terms of trade only begin to recover when the inflation rate drops below its long-term equilibrium value, or undershoots, as Fig. 9.9 shows.

9.3 New Keynesian Model: Fixed Exchange Rate

A fundamental ingredient of the new Keynesian model is the IS curve deduced from the problem of intertemporal allocation of consumption. This curve's specification is the first topic in this section. The second topic is the *BP* curve, which shows the evolution of the balance of payments current account. The third and final topic is the model's specification and analysis.

IS Curve

In the overlapping generations model of a small open economy, the IS curve is given by:

$$\dot{x} = \beta_x x + \beta_r (r - \bar{r}) - \beta_a (a - \bar{a}) - \beta_s (s - \bar{s}). \quad (9.24)$$

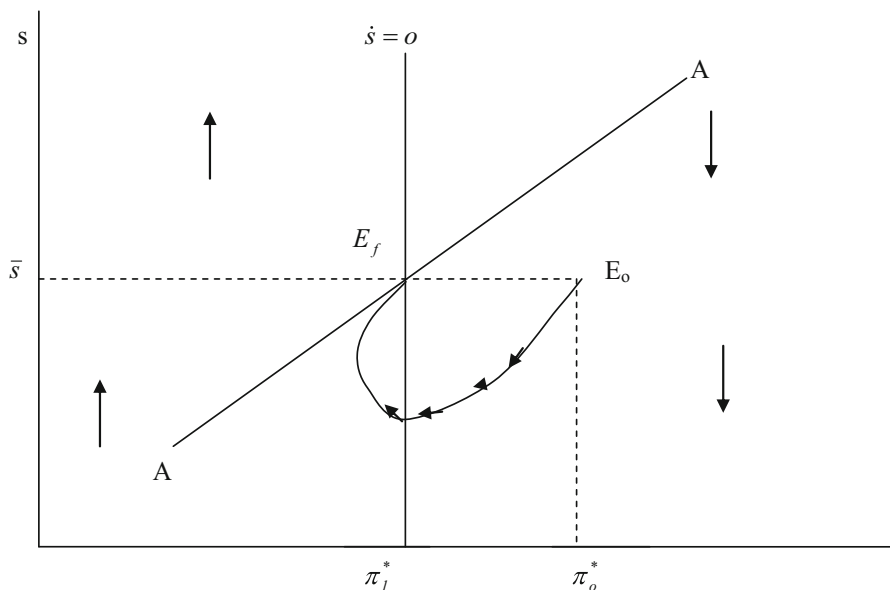


Fig. 9.9 The effects of an unanticipated permanent decrease in the foreign inflation rate

The rate of change of the output gap depends on the gap itself and on the real interest rate, wealth, and terms of trade gaps. The parameters of this IS curve are defined by the following expressions:

$$\beta_x = \frac{n\theta\bar{a}}{\bar{c}_y}; \beta_r = \omega_1\sigma + \omega_2\xi;$$

$$\beta_\alpha = \frac{\omega_1 n\theta}{\bar{c}_y}; \beta_s = \frac{n\theta\bar{a}\omega_2\xi}{\bar{c}_y}.$$

The coefficients β_r and β_a are positive. The coefficients β_x and β_s may be either positive or negative. They will be positive if the country is a net foreign creditor ($\bar{a} > 0$) and negative if it is a debtor ($\bar{a} < 0$).

BP Curve

In the small open economy overlapping generations model, the equation for the balance of payments current account is given by:

$$\dot{A} = (r - n)A + Ex. \quad (9.25)$$

This equation, when normalized at the economy's potential output, may be written as:

$$\dot{a} = (r - n)a + \frac{Ex}{\bar{y}}.$$

At stationary equilibrium ($\dot{a} = 0$):

$$0 = \frac{(\bar{r} - n)\bar{a} + \bar{E}x}{\bar{y}}.$$

Therefore,

$$\dot{a} = (r - n)a - (\bar{r} - n)\bar{a} + \frac{Ex - \bar{E}x}{\bar{y}}.$$

A first-order expansion of the ra term and of net exports yields:

$$\begin{aligned} ra &= \bar{r}\bar{a} + \bar{r}(a - \bar{a}) + \bar{a}(r - \bar{r}), \\ \frac{Ex}{\bar{y}} &= \frac{\bar{E}x}{\bar{y}} + \frac{\bar{S}Ex'}{\bar{y}} \left(\frac{S - \bar{S}}{\bar{S}} \right) = \frac{\bar{E}x}{\bar{y}} + \varepsilon(s - \bar{s}). \end{aligned}$$

Substituting these expressions in the $\dot{a}$ equation yields:

$$\dot{a} = \alpha_a(a - \bar{a}) + \alpha_r(r - \bar{r}) + \alpha_s(s - \bar{s}).$$

This equation's parameters are defined by:

$$\alpha_a = \bar{\rho} - n\alpha_r = \bar{\alpha}, \alpha_s = \varepsilon. \quad (9.26)$$

The coefficients α_a and α_s are positive. The sign of the coefficient α_r depends on whether the country is a creditor ($\bar{a} > 0$) or a debtor ($\bar{a} < 0$).

Model

The IS curve of a new Keynesian model of a small open economy under a fixed exchange rate regime is deduced from the Euler equation, a Calvo Phillips curve, the uncovered interest rate parity, a monetary policy rule where the Central Bank sets the exchange rate, and the equation for the balance of payments current account. The model's initial conditions assume that the price level, the terms of trade, and net wealth are predetermined variables. All other variables are free. The model is specified as follows:

$$\text{IS: } \dot{x} = \beta_x x + \beta_r(r - \bar{r}) - \beta_a(a - \bar{a}) - \beta_s(s - \bar{s}), \quad (9.27)$$

$$\text{PC: } \dot{\pi} = -\delta x, \quad (9.28)$$

$$\text{UIP: } \dot{s} = r - \bar{r}, \quad (9.29)$$

$$\text{MPR: } e = ct, \quad (9.30)$$

$$\text{BP: } \dot{a} = \alpha_a(a - \bar{a}) + \alpha_r(r - \bar{r}) + \alpha_s(s - \bar{s}), \quad (9.31)$$

$$\text{IC: } \text{Given } p(0), s(0) \text{ and } a(0). \quad (9.32)$$

Algebra

The terms of trade are defined by:

$$s = e + p^* - p. \quad (9.33)$$

By taking the derivative of both sides of this expression with respect to time and bearing in mind that the nominal exchange rate is constant ($\dot{e} = 0$), we get:

$$\dot{s} = \dot{e} + \pi^* - \pi = \pi^* - \pi = r - \bar{r}.$$

Replacing the real interest rate gap with the inflation rate gap in the IS curve, interest rate parity, and balance of payments equations yield:

$$\begin{aligned} \dot{x} &= \beta_x x - \beta_r (\pi - \pi^*) - \beta_a (a - \bar{a}) - \beta_s (s - \bar{s}), \\ \dot{s} &= -(\pi - \pi^*), \\ \dot{a} &= -\alpha_r (\pi - \pi^*) + \alpha_s (s - \bar{s}) + \alpha_a (a - \bar{a}). \end{aligned}$$

Dynamical System

The dynamical system of the new Keynesian model of a small open economy under a fixed exchange rate regime has four differential equations: inflation rate, output gap, terms of trade, and net wealth. That is:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \\ \dot{s} \\ \dot{a} \end{bmatrix} = \begin{bmatrix} 0 & -\delta & 0 & 0 \\ -\beta_r & -\beta_x & -\beta_s & -\beta_a \\ -1 & 0 & 0 & 0 \\ -\alpha_r & 0 & \alpha_s & \alpha_a \end{bmatrix} \begin{bmatrix} \pi - \pi^* \\ x \\ s - \bar{s} \\ a - \bar{a} \end{bmatrix}. \quad (9.34)$$

This system's Jacobian matrix is:

$$J = \begin{bmatrix} 0 & -\delta & 0 & 0 \\ -\beta_r & -\beta_x & -\beta_s & -\beta_a \\ -1 & 0 & 0 & 0 \\ -\alpha_r & 0 & \alpha_s & \alpha_a \end{bmatrix}.$$

This matrix's determinant and trace are respectively given by:

$$\begin{aligned} |J| &= \delta[\alpha_s \beta_a - \alpha_a \beta_s], \\ \text{tr} J &= \beta_x + \alpha_a. \end{aligned}$$

For this model to have a saddle point, with two positive and two negative roots, the determinant must be positive. The determinant will also be positive if all four roots have the same sign (whether positive or negative). Therefore, we must prove that all roots have the same sign. For this to be the case, the coefficient of the μ^2 term of the characteristic equation must be negative [Sydsaeter, Strom, and Berck (2000), p.8]:

$$\sum_{i < j} \mu_i \mu_j = \text{coefficient of } \mu^2.$$

Note that if all characteristic roots have the same sign, the μ^2 coefficient will be positive. The dynamical system's characteristic equation is given by:

$$\begin{vmatrix} -\mu & -\delta & 0 & 0 \\ -\beta_r & -\beta_x - \mu & -\beta_s & -\beta_a \\ -1 & 0 & -\mu & 0 \\ -\alpha_r & 0 & \alpha_s & \alpha_s - \mu \end{vmatrix} = 0.$$

Resolving this determinant by means of the expansion of the first line's cofactors yields the characteristic equation:

$$\begin{aligned} \mu^4 - (\alpha_a + \beta_x)^3 + (\alpha_a \beta_x - \delta \beta_r) \mu^2 + \\ \delta(\alpha_a \beta_r + \alpha_r \beta_a + \beta_s) \mu + \delta(\alpha_s \beta_a \alpha_a \beta_s) = 0. \end{aligned}$$

The coefficient of μ^2 must be negative. For this to occur, the following inequality must be satisfied:

$$\beta_r > \frac{\beta_x \alpha_a}{\delta}.$$

The dynamical system's solution, assuming two negative characteristic roots $\mu_1 > \mu_2$, is:

$$\pi = \pi^* + c_1 v_{11} e^{\mu_1 t} + c_2 v_{21} e^{\mu_2 t}, \quad (9.35)$$

$$x = c_1 v_{12} e^{\mu_1 t} + c_2 v_{22} e^{\mu_2 t}, \quad (9.36)$$

$$s = \bar{s} + c_1 v_{13} e^{\mu_1 t} + c_2 v_{23} e^{\mu_2 t}, \quad (9.37)$$

$$a = \bar{a} + c_1 v_{14} e^{\mu_1 t} + c_2 v_{24} e^{\mu_2 t}. \quad (9.38)$$

The c_1 and c_2 constants are obtained from the problem's initial conditions and the elements of the v_{ij} vectors emerge from the solution of the linear system:

$$\begin{bmatrix} -\mu & -\delta & 0 & 0 \\ -\beta_r & \beta_x - \mu & -\beta_s & -\beta_a \\ -1 & 0 & -\mu & 0 \\ -\alpha_r & 0 & \alpha_s & \alpha_a - \mu \end{bmatrix} \begin{bmatrix} v_{i1} \\ v_{i2} \\ v_{i3} \\ v_{i4} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

This system's first and third equations produce the following inequalities:

$$\begin{aligned} \frac{v_{i1}}{v_{i2}} &= -\frac{\delta}{\mu} > 0, \\ \frac{v_{i1}}{v_{i3}} &= -\mu > 0. \end{aligned}$$

The fourth equation, combined with the third, yields the inequality:

$$\frac{v_{i3}}{v_{i4}} = -\frac{\alpha_a - \mu}{\alpha_r \mu + \alpha_s} < 0.$$

Predictions

Dividing the inflation rate gap by the interest rate gap and taking the limit of this ratio as time tends to infinity yields:

$$\lim_{t \rightarrow \infty} \frac{\pi - \pi^*}{x} = \lim_{t \rightarrow \infty} \frac{c_1 v_{11} e^{\mu_1 t} \left[1 + \frac{c_2 v_{21}}{c_1 v_{11}} e^{(\mu_2 - \mu_1)t} \right]}{c_1 v_{12} e^{\mu_1 t} \left[1 + \frac{c_2 v_{22}}{c_1 v_{12}} e^{(\mu_2 - \mu_1)t} \right]}. \quad (9.39)$$

When $t \rightarrow \infty$, as $\mu_2 - \mu_1 < 0$, the terms in brackets in the numerator and the denominator converge to one. Therefore,

$$\lim_{t \rightarrow \infty} \frac{\pi - \pi^*}{x} = \frac{v_{11}}{v_{12}} > 0 \quad (9.40)$$

In this model, the ratio of the inflation rate gap to the output gap converge to stationary equilibrium, approaching the dominant ray (of the eigenvalue) in such a manner that the ratio remains positive along this path. This means, for example, that if the domestic inflation rate is higher than the international inflation rate, the output gap will be positive.

The same procedure can be used to prove that:

$$\lim_{t \rightarrow \infty} \frac{\pi - \pi^*}{s - \bar{s}} = \frac{v_{11}}{v_{13}} > 0, \quad (9.41)$$

$$\lim_{t \rightarrow \infty} \frac{s - \bar{s}}{a - \bar{a}} = \frac{v_{13}}{v_{14}} > 0. \quad (9.42)$$

The ratio of the inflation gap to the terms of trade gap approaches the dominant ray with a positive value. On the other hand, the ratio of the terms of trade gap to the wealth gap has a negative value when it approaches the model's dominant ray.

The real interest rate gap equals the derivative of the real exchange rate with respect to time. Therefore, the interest rate gap equation is obtained by taking the derivative of the terms of trade equation with respect to time. That is:

$$r - \bar{r} = \dot{s} = c_1 v_{13} \mu_1 e^{\mu_1 t} + c_2 v_{23} \mu_2 e^{\mu_2 t}.$$

Using the same limit calculation technique for the ratio of the inflation gap to the output gap, we find that:

$$\lim_{t \rightarrow \infty} \frac{r - \bar{r}}{s - \bar{s}} = \mu_1 < 0. \quad (9.43)$$

This proposition states that the real interest rate and the terms of trade, when off the long-run equilibrium, move in opposite directions on the path that approaches the model's dominant ray.

9.4 Mundell-Fleming-Dornbusch Model: Flexible Exchange Rate

The IS and Phillips curves and the uncovered interest rate parity are specified in the same way as in the Keynesian model with a fixed exchange rate regime. The monetary policy rule is the Taylor rule. The model therefore has the following equations:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(s - \bar{s}), \quad (9.44)$$

$$\text{PC: } \dot{\pi} = \gamma\dot{s} + \delta x, \quad (9.45)$$

$$\text{UIP: } r = \bar{r} + \dot{s}, \bar{r} = r^*, \quad (9.46)$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \quad (9.47)$$

$$\text{IC: Given } p(0) \text{ and } \pi(0). \quad (9.48)$$

The model can be solved in different ways. We first present the solution in which it is reduced to a dynamical system of the differential equations for the inflation rate and the output gap. The model's solution is then obtained from a dynamical system of differential equations for the interest rate and the terms of trade.

Algebra

Combining the Phillips curve with the uncovered interest rate parity allows writing the acceleration of the inflation rate as a function of the gaps of the real interest rate and output:

$$\dot{\pi} = \gamma(r - \bar{r}) + \delta x.$$

According to the monetary policy rule, the real interest rate gap is:

$$r - \bar{r} = \phi(\pi - \bar{\pi}) + \theta x.$$

Substituting this equation in the previous expression yields the differential equation of inflation rate as a function of the inflation rate gap and the output gap:

$$\dot{\pi} = \gamma\phi(\pi - \bar{\pi}) + (\gamma\theta + \delta)x.$$

To obtain the differential equation of real output, we begin by taking the derivatives of the IS curve and the monetary policy rule with respect to time:

$$\begin{aligned}\dot{x} &= -\alpha\dot{r} + \beta\dot{s} \\ \dot{r} &= \phi\dot{\pi} + \theta\dot{x}.\end{aligned}$$

Substituting the second equation in the first equation results in:

$$\dot{x} = -\alpha\phi\dot{\pi} - \alpha\theta\dot{x} + \beta(r - \bar{r}).$$

By taking the monetary policy rule into account to substitute for the real interest rate gap and transferring the term that contains the derivative of the output gap with respect to time to the left side, we get:

$$(1 + \alpha\theta)\dot{x} = -\alpha\phi\dot{\pi} + \beta\phi(\pi - \bar{\pi}) + \beta\theta x.$$

Replacing the derivative of inflation with respect to time with the expression of its differential equation yields:

$$(1 + \alpha\theta)\dot{x} = -\alpha\phi\gamma\phi(\pi - \bar{\pi}) - \alpha\phi(\gamma\theta + \delta)x + \beta\phi(\pi - \bar{\pi}) + \beta\theta x.$$

This expression may be rewritten as:

$$(1 + \alpha\theta)\dot{x} = \phi(\beta - \alpha\phi\gamma)(\pi - \bar{\pi}) + [\beta\theta - \alpha\phi(\gamma\theta + \delta)]x.$$

Dynamical System

The small open economy model with a flexible exchange rate regime may be solved as a dynamical system of two differential equations: for the inflation rate and for real output gap:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \gamma\phi & \gamma\theta + \delta \\ \frac{\phi(\beta - \alpha\phi\gamma)}{1 + \alpha\theta} & \frac{\beta\theta - \alpha\phi(\gamma\theta + \delta)}{1 + \alpha\theta} \end{bmatrix} \begin{bmatrix} \pi - \bar{\pi} \\ x \end{bmatrix}. \quad (9.49)$$

This system's Jacobian matrix is:

$$J = \begin{bmatrix} \frac{\partial \circ \pi}{\partial \pi} & \frac{\partial \circ \pi}{\partial x} \\ \frac{\partial \circ x}{\partial \pi} & \frac{\partial \circ x}{\partial x} \end{bmatrix} = \begin{bmatrix} \gamma\phi & \gamma\theta + \delta \\ \frac{\phi(\beta - \alpha\phi\gamma)}{1 + \alpha\theta} & \frac{\beta\theta - \alpha\phi(\gamma\theta + \delta)}{1 + \alpha\theta} \end{bmatrix}.$$

This matrix's determinant is negative and the system has a saddle point. That is:

$$|J| = \frac{\gamma\phi\beta\theta - \gamma\phi\alpha\phi(\gamma\theta + \delta) - (\gamma\theta + \delta)(\phi\beta - \alpha\phi\phi\gamma)}{1 + \alpha\theta} = -\frac{\delta\phi\beta}{1 + \alpha\theta},$$

$$|J| < 0 \Rightarrow \text{saddle}.$$

Fig. 9.10 The dynamics of the inflation rate

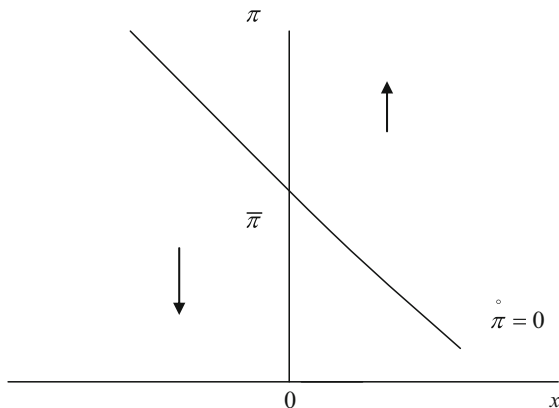


Figure 9.10 shows the phase diagram of the inflation rate's differential equation. The vertical axis measures the inflation rate and the horizontal axis measures the output gap. The equation describing the points at which the inflation rate does not change is a negatively sloped straight line. That is:

$$\dot{\pi} = 0 \Rightarrow \pi = \bar{\pi} - \frac{\gamma\theta + \delta}{1 + \alpha\theta}x.$$

The inflation rate increases at points above the line and decreases below it.

The phase diagram of the output gap's differential equation depends on the values of the model's parameters. The equation $\dot{x} = 0$ is a straight line, but its slope depends on the signs of the parameters, as can be seen by examining the output gap's coefficient in:

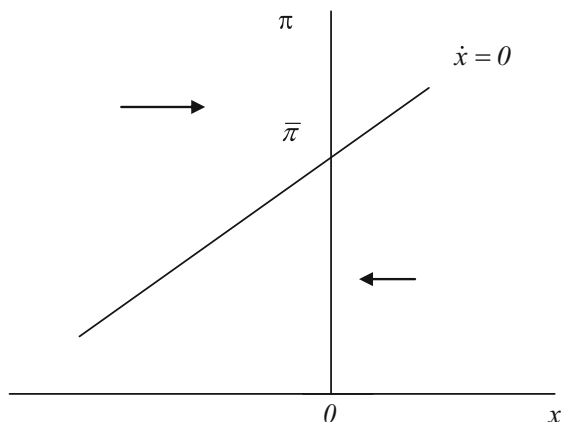
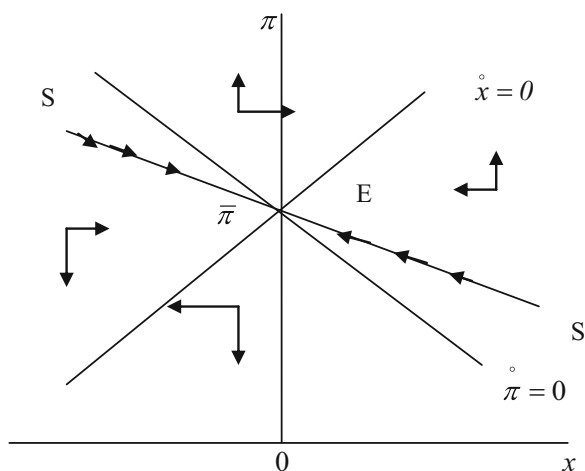
$$\dot{x} = 0 \Rightarrow \pi = \bar{\pi} - \frac{[\beta\theta - \alpha\phi(\gamma\theta + \delta)]}{\phi(\beta - \alpha\phi\gamma)}x.$$

The previous equation may be written as:

$$\pi = \bar{\pi} - \frac{\theta(\beta - \alpha\phi\gamma) - \alpha\phi\delta}{\phi(\beta - \alpha\phi\gamma)}x.$$

If $\beta - \alpha\phi\gamma < 0$ the slope will be negative. If $\beta - \alpha\phi\gamma > 0$ the lines corresponding to $\dot{x} = 0$ and $\dot{\pi} = 0$ will be negatively sloped. To find out which slope is greater, we check the difference between the absolute values of the two angular coefficients. That is, the absolute value of the angular coefficient of $\dot{x} = 0$ is greater than that of $\dot{\pi} = 0$:

$$\frac{\theta(\beta - \alpha\phi\gamma) - \alpha\phi\delta}{\phi(\beta - \alpha\phi\gamma)} - \frac{\gamma\theta + \delta}{\gamma\phi} = \frac{-\beta\delta}{\gamma\phi(\beta - \alpha\phi\gamma)} > 0.$$

Fig. 9.11 The dynamics of the output gap**Fig. 9.12** The phase diagram for the inflation rate and the output gap

If the model's parameters satisfy the inequality $\beta - \alpha\phi\gamma > 0$, two possibilities exist. If $\theta(\beta - \alpha\phi\gamma) > \alpha\phi\delta$ then $\dot{x} = 0$ has a negative slope and the difference between the coefficients (see the above expression) is negative, that is, the absolute value of this line's angular coefficient is less than that of the $\dot{\pi} = 0$ line. If $\theta(\beta - \alpha\phi\gamma) < \alpha\phi\delta$ then the $\dot{x} = 0$ line will have a positive slope. Figure 9.11 shows the phase diagram for this case. The output gap decreases at points below the line and increases above it.

Figure 9.12 shows the dynamical system's phase diagram, with a positively sloped $\dot{x} = 0$ line. The converging *SS* saddle path has a negative slope.

Experiment

This model can be used to analyze the economic policy experiment that consists of the Central Bank changing its inflation target. Assume that the Central Bank reduces the inflation target to $\bar{\pi}_1$ from $\bar{\pi}_0$, as Fig. 9.13 shows. The inflation rate does not

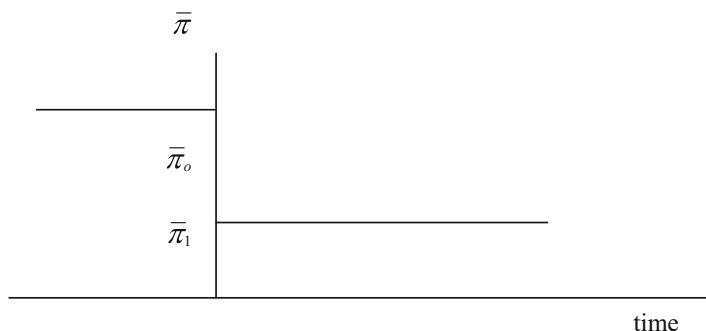
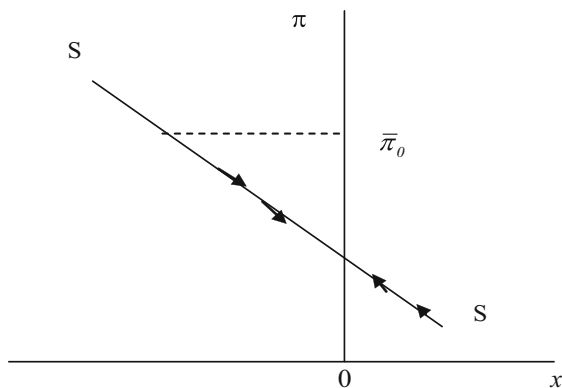


Fig. 9.13 An unanticipated permanent decrease in the inflation target

Fig. 9.14 The effects of an unanticipated permanent decrease in the inflation target



change instantaneously in this model because it is a predetermined variable. Figure 9.14 graphs the economy's adjustment dynamics. The output gap drops at the time of the announcement of the new inflation target and the economy enters a recession. The inflation rate starts to gradually drop, following the SS saddle path until it reaches the new inflation target and the economy returns to full-employment real output.

Algebra

The model of this section can also be solved using a system of differential equations for the real exchange rate and real interest rate variables. The real exchange rate's differential equation is the uncovered interest rate parity equation:

$$\dot{s} = r - \bar{r}.$$

To obtain the real interest rate's differential equation, we begin by taking derivatives of the monetary policy rule and IS curve equations with respect to time. That is:

$$\begin{aligned}\dot{r} &= \phi\dot{\pi} + \theta\dot{x}, \\ \dot{x} &= -\alpha\dot{r} + \beta\dot{s}.\end{aligned}$$

Substituting the value of $\dot{x}$ in the equation for $\dot{r}$ yields:

$$\dot{r} = \phi\dot{\pi} - \alpha\theta\dot{r} + \beta\theta\dot{s}.$$

Inflation acceleration depends on the variation of the terms of trade and on the output gap. Therefore:

$$(1 + \alpha\theta)\dot{r} = \phi[\gamma\dot{s} + \delta x] + \beta\theta\dot{s}.$$

This equation may be written as:

$$(1 + \alpha\theta)\dot{r} = \phi\delta x + (\beta\theta + \phi\gamma)\dot{s}.$$

The output gap depends on the real interest rate and terms of trade gaps according to the IS curve. The previous expression thus becomes:

$$(1 + \alpha\theta)\dot{r} = \phi\delta[-\alpha(r - \bar{r}) + \beta(s - \bar{s})] + (\beta\theta + \phi\gamma)(r - \bar{r})$$

Adding the terms corresponding to the interest rate and terms of trade gaps yields the real interest rate's differential equation:

$$(1 + \alpha\theta)\dot{r} = (\beta\theta + \phi\gamma - \alpha\phi\delta)(r - \bar{r}) + \beta\phi\delta(s - \bar{s}).$$

Dynamical System

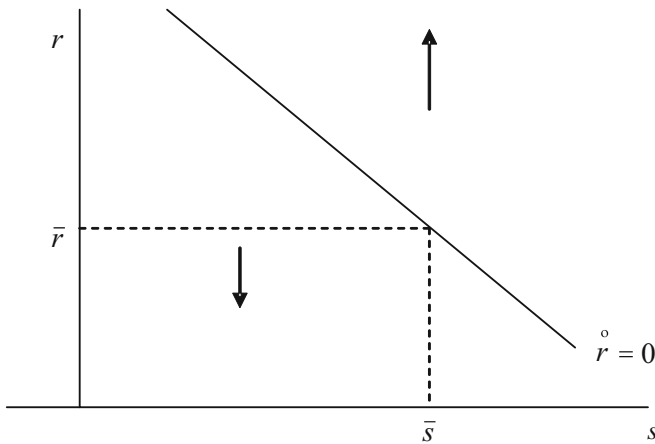
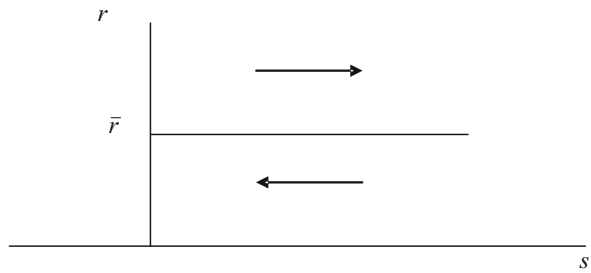
The model's dynamical system, in a flexible exchange rate regime, has two differential equations: one for the real interest rate and another for the terms of trade:

$$\begin{bmatrix} \dot{r} \\ \dot{s} \end{bmatrix} = \begin{bmatrix} \frac{\beta\theta + \phi\gamma - \alpha\phi\delta}{1 + \alpha\theta} & \frac{\beta\phi\delta}{1 + \alpha\theta} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} r - \bar{r} \\ s - \bar{s} \end{bmatrix}. \quad (9.50)$$

In this dynamical system, the two variables – the real interest rate and the terms of trade – are not predetermined, but its initial values are free. These values, however, must simultaneously satisfy the IS curve and the monetary policy rule, as we will show below.

This system's Jacobian matrix is:

$$J = \begin{bmatrix} \frac{\partial \dot{r}}{\partial r} & \frac{\partial \dot{r}}{\partial s} \\ \frac{\partial \dot{s}}{\partial r} & \frac{\partial \dot{s}}{\partial s} \end{bmatrix} = \begin{bmatrix} \frac{\beta\theta + \phi\gamma - \alpha\phi\delta}{1 + \alpha\theta} & \frac{\beta\phi\delta}{1 + \alpha\theta} \\ 1 & 0 \end{bmatrix}.$$

Fig. 9.15 The dynamics of the terms of trade**Fig. 9.16** The dynamics of the real interest rate

This matrix's determinant is negative:

$$|J| = -\frac{\beta\phi\delta}{1 + \alpha\theta} < 0 \Rightarrow \text{saddle.}$$

Figure 9.15 shows the phase diagram for the terms of trade differential equation, with the real interest rate on the vertical axis and the terms of trade on the horizontal axis. When the terms of trade remain unchanged, real interest rate equals the real foreign interest rate:

$$\dot{s} = 0 \Rightarrow r = \bar{r}.$$

The terms of trade increase at points above the real foreign interest rate and decrease at points below it, as the arrows show.

Figure 9.16 shows the phase diagram for the real interest rate differential equation. The line corresponding to points where the real interest rate remains constant may be either negatively or positively sloped, depending on the model's parameters:

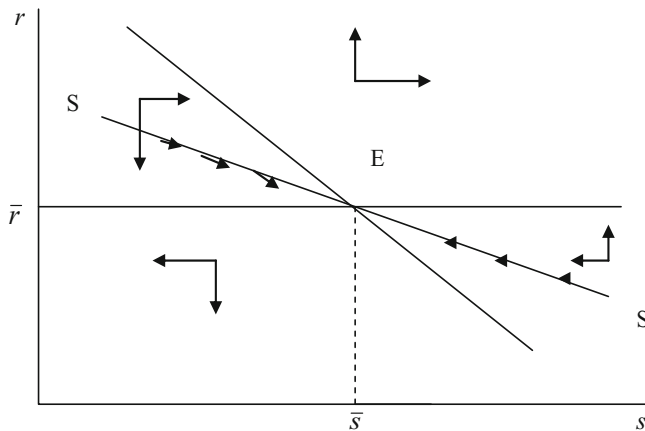


Fig. 9.17 The phase diagram for the real rate of interest and the terms of trade

$$\dot{r} = 0 \Rightarrow r = \bar{r} - \frac{\beta\phi\delta}{\beta\theta + \phi\gamma - \alpha\phi\delta}(s - \bar{s}).$$

Figure 9.16 assumes that the $\dot{r} = 0$ line has a negative slope. This assumption has no effect on the model's qualitative conclusions. The real interest rate increases at points above this line and decreases below it.

Figure 9.17 shows the dynamical system's phase diagram for the real interest rate and terms of trade differential equations. The SS saddle path has a negative slope. The main conclusion is that the real interest rate and the terms of trade move in opposite directions, that is, the real interest rate and the terms of trade must be negatively correlated. Analytically, on the saddle path, the terms of trade gap and the interest rate gap would be related by an equation such as:

$$s - \bar{s} = -\kappa(r - \bar{r}), \quad (9.51)$$

where the κ coefficient is positive and depends on the model's parameters. When substituted in the IS curve equation, this equation results in:

$$x = -\alpha(r - \bar{r}) + \beta(s - \bar{s}) = -(\alpha + \beta\kappa)(r - \bar{r}). \quad (9.52)$$

The effect of the monetary policy on the output gap is greater in an open economy because the interest rate affects the terms of trade, which in their turn affect the current account of the balance of payments.

Experiment

Figure 9.18's economic policy experiment, in which the Central Bank reduces the inflation target to $\bar{\pi}_1$ from $\bar{\pi}_0$, may be analyzed based on the phase diagram in Fig. 9.18. The two lines, $\dot{r} = 0$ and $\dot{s} = 0$, remain unchanged after the monetary

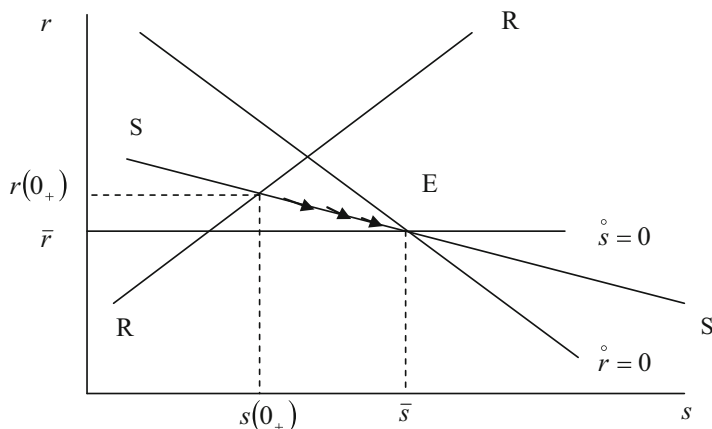


Fig. 9.18 The effects of an unanticipated permanent decrease in the inflation target

policy change. The economy, however, does not remain at equilibrium point E , which is the long-run equilibrium point. At the time of the announcement of the new inflation target, the Central Bank increases the nominal interest rate. Because inflation is a predetermined variable, the real interest rate also increases, and the economy's output is affected. On the other hand, the nominal interest rate increase reduces the nominal exchange rate as funds flow into the foreign exchange market. Therefore, the terms of trade appreciate. The two variables – the real interest rate and the terms of trade – change instantaneously. How do we determine the point on the SS saddle path to which the variables jump from their initial equilibrium? They must at all times satisfy the IS curve and the monetary policy rule. Eliminating the output gap from the two equations yields:

$$r - \bar{r} = \frac{\phi}{1 + \alpha\theta} (\pi - \bar{\pi}) + \frac{\beta\theta}{1 + \alpha\theta} (s - \bar{s}).$$

Figure 9.18 graphs this line, represented by the letters RR , at the time of the monetary policy change. At this instant, the inflation rate gap is measured as the difference between the old and the new inflation targets. The point where the RR line intercepts the SS saddle path is the initial point of the interest rate and the terms of trade immediately after the announcement of the new inflation target. The economy then starts to move in the direction of its long-run equilibrium (point E), with a reduction of the real interest rate and an increase in the terms of trade along the SS saddle path.

9.5 Extended Mundell-Fleming-Dornbusch Model: Flexible Exchange Rate

The Keynesian model of a small open economy under a flexible exchange rate regime and with the wealth effect on consumption has the same specification as in the fixed exchange rate regime, except for the monetary policy rule. Under a flexible exchange rate regime, the Central Bank controls the nominal interest rate and the markets determine the exchange rate. The monetary policy rule is the Taylor rule. To simplify the algebra, this rule does not include the output gap term. The model is specified as follows:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(s - \bar{s}) + \Psi(a - \bar{a}), \quad (9.53)$$

$$\text{PC: } \dot{\pi} = \gamma\dot{s} + \delta x, \quad (9.54)$$

$$\text{UIP: } \dot{s} = r - \bar{r}, \quad (9.55)$$

$$\text{BP: } \dot{a} = \bar{r}(a - \bar{a}) + \bar{a}(r - \bar{r}) + \Phi(s - \bar{s}), \quad (9.56)$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}), \quad (9.57)$$

$$\text{IC: Given } p(0), \pi(0) \text{ and } a(0).$$

Algebra

The monetary policy rule may be written as:

$$r - \bar{r} = \phi(\pi - \bar{\pi}).$$

Substituting this value of the real interest rate gap on the IS curve and on the current account of the balance of payments curve yields:

$$x = -\alpha\phi(\pi - \bar{\pi}) + \beta(s - \bar{s}) + \rho(a - \bar{a}),$$

$$\dot{a} = \phi\bar{a}(\pi - \bar{\pi}) + \Phi(s - \bar{s}) + \bar{r}(a - \bar{a}).$$

The Phillips curve, after substituting the real interest rate gap and the output gap, results in the following differential equation:

$$\dot{\pi} = (\gamma - \alpha\delta\phi)(\pi - \bar{\pi}) + \beta\delta(s - \bar{s}) + \delta\Psi(a - \bar{a}).$$

Dynamical System

This model's dynamical system has three differential equations: inflation rate, terms of trade, and foreign wealth. That is:

$$\begin{bmatrix} \dot{\pi} \\ \dot{s} \\ \dot{a} \end{bmatrix} = \begin{bmatrix} \gamma - \alpha\delta\phi & \beta\delta & \delta\Psi \\ \phi & 0 & 0 \\ \phi\bar{a} & \Phi & \bar{r} \end{bmatrix} \begin{bmatrix} \pi - \bar{\pi} \\ s - \bar{s} \\ a - \bar{a} \end{bmatrix}. \quad (9.58)$$

This system's Jacobian matrix is:

$$J = \begin{bmatrix} \gamma - \alpha\delta\phi & \beta\delta & \delta\Psi \\ \phi & 0 & 0 \\ \phi\bar{a} & \Phi & \bar{r} \end{bmatrix}.$$

The matrix's determinant and trace are respectively equal to:

$$\begin{aligned} |J| &= \phi\Phi\delta\Psi - \phi\beta\delta\bar{r}, \\ \text{tr}J &= \gamma - \alpha\delta\phi + \bar{r}. \end{aligned}$$

For this dynamical system to have a saddle point with two negative and one positive characteristic root, the Jacobian matrix's determinant must be positive and its trace must be negative. For the saddle point to exist, the real long-term interest rate must satisfy the following inequality:

$$\bar{r} < \min\left(\frac{\Phi\Psi}{\beta}, \alpha\delta\phi - \gamma\right).$$

The dynamical system's solution is given by the following equations:

$$\begin{aligned} \pi &= \bar{\pi} + c_1 v_{11} e^{\mu_1 t} + c_2 v_{21} e^{\mu_2 t}, \\ s &= \bar{s} + c_1 v_{12} e^{\mu_1 t} + c_2 v_{22} e^{\mu_2 t}, \\ a &= \bar{a} + c_1 v_{13} e^{\mu_1 t} + c_2 v_{23} e^{\mu_2 t}, \end{aligned}$$

where μ_1, μ_2 are the negative characteristic roots, $\mu_1 > \mu_2$, c_1 and c_2 are constants determined by the initial conditions, and v_{ij} are the elements of the characteristic vectors associated with each root, obtained by solving the equations system:

$$\begin{bmatrix} \gamma - \alpha\delta - \mu & \beta\delta & \delta\Psi \\ \phi & -\mu & 0 \\ \phi\bar{a} & \Phi & \bar{r} - \mu \end{bmatrix} \begin{bmatrix} v_{i1} \\ v_{i2} \\ v_{i3} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The system's second equation implies that:

$$\frac{v_{i1}}{v_{i2}} = \frac{\mu}{\phi} < 0.$$

The inequality sign stems from the fact that the characteristic root is negative and the parameter ϕ is positive.

Predictions

As time increases indefinitely, the limit of the ratio of the inflation gap and the terms of trade gap is:

$$\lim_{t \rightarrow \infty} \frac{\pi - \bar{\pi}}{s - \bar{s}} = \lim_{t \rightarrow \infty} \frac{c_1 v_{11} e^{\mu_1 t} \left[1 + \frac{c_2 v_{21}}{c_1 v_{11}} e^{(\mu_2 - \mu_1)t} \right]}{c_1 v_{12} e^{\mu_1 t} \left[1 + \frac{c_2 v_{22}}{c_1 v_{12}} e^{(\mu_2 - \mu_1)t} \right]}. \quad (9.59)$$

Because $\mu_2 - \mu_1 < 0$, the terms in brackets in the numerator and the denominator converge to one. Therefore, this limit is negative:

$$\lim_{t \rightarrow \infty} \frac{r - \bar{r}}{s - \bar{s}} < 0 \quad (9.60)$$

This means that if the real interest rate is high, the terms of trade (the real exchange rate) will be appreciated and vice-versa.

The limit of the ratio of the terms of trade gap and the foreign wealth gap is obtained by means of the following expression:

$$\lim_{t \rightarrow \infty} \frac{s - \bar{s}}{a - \bar{a}} = \lim_{t \rightarrow \infty} \frac{c_1 v_{12} e^{\mu_1 t} \left[1 + \frac{c_1 v_{22}}{c_1 v_{12}} e^{(\mu_2 - \mu_1)t} \right]}{c_1 v_{13} e^{\mu_1 t} \left[1 + \frac{c_2 v_{23}}{c_1 v_{13}} e^{(\mu_2 - \mu_1)t} \right]}. \quad (9.61)$$

Using the same properties as in the calculation of the previous limit, we find that:

$$\lim_{t \rightarrow \infty} \frac{s - \bar{s}}{a - \bar{a}} = \frac{v_{12}}{v_{13}} < 0. \quad (9.62)$$

When the terms of trade are above its long-term equilibrium value, net foreign wealth will be below its long-term value. This inequality's sign can be obtained as follows: Combining the first two equations from the system that produces the characteristic vectors:

$$\frac{v_{12}}{v_{13}} = -\frac{\delta \Psi \phi}{(\gamma - \alpha \delta - \mu)\mu + \beta \delta \phi} < 0.$$

The differential equations system's characteristic equation is:

$$-(\gamma - \alpha \delta - \mu)\mu(\bar{r} - \mu) + \phi \Phi \delta \Psi + \phi \bar{a} \mu \delta \Psi - \phi \beta \delta (\bar{r} - \mu) = 0.$$

Rearranging the terms, we get:

$$-(\bar{r} - \mu)[(\gamma - \alpha \delta - \mu)\mu + \beta \delta \phi] = -\phi \Phi \delta \Psi - \phi \bar{a} \mu \delta \Psi.$$

Therefore, because $\mu < 0$ and if $\bar{a} < 0$:

$$(\gamma - \alpha \delta - \mu)\mu + \beta \delta \phi > 0.$$

The output gap equation can be obtained from the Phillips curve, for example:

$$x = \frac{\dot{\pi} - \gamma \dot{s}}{\delta}$$

A little algebra shows that the limit of the ratio of the output to the inflation rate gap is negative:

$$\lim_{t \rightarrow \infty} \frac{\pi - \bar{\pi}}{x} < 0. \quad (9.63)$$

In the process of convergence to stationary equilibrium, if the inflation is above its target, for example, the economy will be in recession (negative output gap).

9.6 New Keynesian Model: Flexible Exchange Rate

The new Keynesian model for a flexible exchange rate regime has the same equations for the IS and Phillips curves, the uncovered interest rate parity and the balance of payments as the one for a fixed exchange rate regime. The monetary policy rule assumes that the Central Bank controls the nominal interest rate. To simplify the algebra, the Taylor rule does not include the output gap term. The initial conditions of the flexible exchange rate regime model are different from those of the fixed exchange rate regime model. The initial conditions assume that the price level and net foreign wealth are predetermined variables. All other variables are free. The model is specified as follows:

$$\text{IS: } \dot{x} = \beta_x x + \beta_r (r - \bar{r}) - \beta_a (a - \bar{a}) - \beta_s (s - \bar{s}), \quad (9.64)$$

$$\text{PC: } \dot{\pi} = -\delta x, \quad (9.65)$$

$$\text{IRP: } \dot{s} = r - \bar{r}, \quad (9.66)$$

$$\text{BP: } \dot{a} = \rho_a (a - \bar{a}) + \alpha_r (r - \bar{r}) + \alpha_s (s - \bar{s}), \quad (9.67)$$

$$\text{MPR: } i = \bar{r} + \pi + \phi(\pi - \bar{\pi}), \quad (9.68)$$

$$\text{IC: Given } p(0) \text{ and } a(0). \quad (9.69)$$

Algebra

The monetary policy rule may be written as:

$$r - \bar{r} = \phi(\pi - \bar{\pi})$$

Substituting the value of this real interest rate gap in the IS curve's equations for interest rate parity and the balance of payments yields:

$$\begin{aligned}
\dot{x} &= \beta_x x + \beta_r \phi(\pi - \bar{\pi}) - \beta_a(a - \bar{a}) - \beta_s(s - \bar{s}), \\
\dot{s} &= \phi(\pi - \bar{\pi}), \\
\dot{a} &= \alpha_a(a - \bar{a}) + \alpha_r \phi(\pi - \bar{\pi}) + \alpha_s(s - \bar{s}).
\end{aligned}$$

Dynamical System

The dynamical system for a new Keynesian model in a flexible exchange rate regime consists of the differential equations for the inflation rate, the output gap, the terms of trade, and the current account of the balance of payments. That is:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \\ \dot{s} \\ \dot{a} \end{bmatrix} = \begin{bmatrix} 0 & -\delta & 0 & 0 \\ \beta_r \phi & \beta_x & -\beta_s & -\beta_a \\ \phi & 0 & 0 & 0 \\ \alpha_r \phi & 0 & \alpha_s & \alpha_a \end{bmatrix} \begin{bmatrix} \pi - \bar{\pi} \\ x \\ s - \bar{s} \\ a - \bar{a} \end{bmatrix}. \quad (9.70)$$

This system's Jacobian matrix is:

$$J = \begin{bmatrix} 0 & -\delta & 0 & 0 \\ \beta_r \phi & \beta_x & -\beta_s & -\beta_a \\ \phi & 0 & 0 & 0 \\ \alpha_r \phi & 0 & \alpha_s & \alpha_a \end{bmatrix}.$$

Calculation of this matrix's determinant by means of the expansion rule of the first line's cofactor equals:

$$|J| = (-\delta)(-1)^{1+2} \begin{vmatrix} \beta_r \phi & \beta_s & -\beta_a \\ \phi & 0 & 0 \\ \alpha_r \phi & \alpha_s & \alpha_a \end{vmatrix}.$$

Therefore,

$$|J| = -\delta \phi (\alpha_s \beta_a + \beta_s \alpha_a).$$

The Jacobian matrix's trace is:

$$trJ = \beta_x + \alpha_a.$$

Assume that the parameters are such that the determinant is negative: $|J| < 0$. The dynamical system then has one negative and three positive characteristic roots, or else three negative and one positive characteristic roots. To find out whether a solution exists with three positive roots, one must analyze the model's characteristic equation. This characteristic equation is obtained by solving the following determinant:

$$\begin{vmatrix} -\mu & -\delta & 0 & 0 \\ \beta_r\phi & \beta_x - \mu & \beta_s & -\beta_a \\ \phi & 0 & -\mu & 0 \\ \alpha_r\phi & 0 & \alpha_s & \alpha_a - \mu \end{vmatrix} = 0.$$

Using the cofactor expansion rule for the first line, for example, and with a little algebra, yields the following equation:

$$\mu^4 - (\alpha_a + \beta_x)\mu^3 + (\alpha_a\beta_x + \beta_r\delta\phi)\mu^2 - \delta\phi(\alpha_a\beta_r + \alpha_r\beta_a - \beta_s)\mu + \phi(\alpha_s\beta_a + \alpha_a\beta_s) = 0.$$

If the number of sign changes in the sequence of coefficients [Sydsaeter, Strom and Berck (2000), p.8] is equal to three, there will be three positive roots:

$$\begin{aligned} a_4 &= 1, a_3 = -(\alpha_a + \beta_x), a_2 = (\alpha_a\beta_x + \delta\phi), \\ a_1 &= \delta\phi(\alpha_a\beta_r + \alpha_r\beta_a - \beta_s), a_0 = \phi(\alpha_s\beta_r + \alpha\beta_s). \end{aligned}$$

Assume that the coefficients are such that this condition is met.

The dynamical system's solution is obtained through the following equations:

$$\pi = \bar{\pi} + c_1 v_{11} e^{\mu_1 t},$$

$$x = c_1 v_{12} e^{\mu_1 t},$$

$$s = \bar{s} + c_1 v_{13} e^{\mu_1 t},$$

$$a = \bar{a} + c_1 v_{14} e^{\mu_1 t},$$

where the c_1 constant is obtained from the initial value of predetermined variable $a(0)$, μ_1 is the negative characteristic root and the v_i elements are obtained from the solution of the following linear system:

$$\begin{bmatrix} -\mu & -\delta & 0 & 0 \\ \beta_r\phi & \beta_x - \mu & -\beta_s & -\beta_a \\ \phi & 0 & -\mu & 0 \\ \alpha_r\phi & 0 & \alpha_s & \alpha_a - \mu \end{bmatrix} \begin{bmatrix} v_{i1} \\ v_{i2} \\ v_{i3} \\ v_{i4} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

The first, third, and fourth equations imply the following properties:

$$\frac{v_{11}}{v_{12}} = -\frac{\delta}{\mu} > 0,$$

$$\frac{v_{11}}{v_{13}} = \frac{\mu}{\phi} < 0,$$

$$\frac{v_{13}}{v_{14}} = \frac{\phi(\alpha_a - \mu)}{\alpha_r\phi_\mu + \alpha_s\phi} < 0.$$

The inequalities' signs stem from the fact that the characteristic root μ is negative.

Predictions

Dividing the inflation rate gap by the output gap yields:

$$\frac{\pi - \bar{\pi}}{x} = \frac{v_{11}}{v_{12}} > 0 \quad (9.71)$$

The inflation gap and the output gap must therefore move in the same direction on the equilibrium path.

Dividing the inflation rate gap by the terms of trade gap, yields:

$$\frac{\pi - \bar{\pi}}{s - \bar{s}} = \frac{v_{11}}{v_{13}} < 0. \quad (9.72)$$

The inflation rate and terms of trade gaps must move in opposite directions. Since the real interest rate gap is proportional to the inflation rate gap because of the monetary policy rule, the real interest rate and terms of trade gaps move in opposite directions: the interest upward (downward) and the exchange rate downward (upward). This proposition can also be deduced by using the fact that $r - \bar{r} = \dot{s}$. It is easily concluded that:

$$r - \bar{r} = c_1 v_{13} \mu e^{\mu_1 t} = \mu_1 (s - \bar{s}), \mu_1 < 0.$$

Dividing the wealth gap by the terms of trade gap yields:

$$\frac{a - \bar{a}}{s - \bar{s}} = \frac{v_{14}}{v_{13}} < 0. \quad (9.73)$$

Therefore, when the terms of trade lie above (below) the long-term value, the country's net wealth will lie below (above) its long-term equilibrium value.

9.7 Exercises

1. Consider the following model of a small open economy (Mundell-Fleming-Dornbusch model):

$$\begin{aligned} \dot{p} &= \delta(d - y) \\ d &= k + \alpha(e + p^* - p) - \beta i \\ m - p &= -\gamma i + \phi y \\ i &= i^* + \dot{e}^e \\ \dot{e}^e &= \dot{e} \end{aligned}$$

The symbols have the following meanings: p = price index; d = expenditure; y = real output; e = nominal exchange rate; p^* = foreign price index; i = interest rate; m = money stock; i^* = foreign interest rate; $\delta, k, \alpha, \beta, \gamma, \phi$ are positive parameters and the e index indicates the variable's expected value. The model's variables are the respective logs, except for the interest rate.

- (a) What are the short- and long run effects of a public deficit increase on the exchange rate?
- (b) What are the short- and long run effects of an increase in the price index of goods and services produced abroad on the exchange rate?

2. Consider the model:

$$\begin{aligned}\dot{p} &= \phi(y - \bar{y}) \\ y &= \alpha_0 + \alpha_1(e + p^* - p) - \alpha_2 i \\ m - p &= \beta_0 + \beta_1 y - \beta_2 i \\ i &= i^* + \dot{e}\end{aligned}$$

The symbols have the following meanings: y = real output, and $=$ nominal exchange rate, p^* = foreign price index, p = domestic price index, i = interest rate, m = nominal quantity of money, i^* = international interest rate, $\bar{y}$ = potential output. The model's variables are the respective logs, except for the interest rate.

- (a) What is the effect of an increase in the quantity of money on the exchange rate?
- (b) Compare your response to the previous item with the one that you would obtain replacing the $\dot{p}$ equation with: $\dot{p} = \phi(d - y)$ with the second equation becoming the expenditure equation: $d = \alpha_0 + \alpha_1(e + p^* - p) - k_2 i$. Is there a chance that a monetary expansion may cause undershooting instead of overshooting?

3. Consider the following model of a small open economy:

$$\begin{aligned}\text{IS: } x &= -\alpha(r - \bar{r}) + \beta(s - \bar{s}), \\ \text{PC: } \dot{\pi} &= \gamma \dot{s} + \delta x, \\ \text{UIP: } r &= \bar{r} + \dot{s}, \bar{r} = r^*, \\ \text{MPR: } i &= \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta(s - \bar{s}), \\ \text{IC: } &\text{Given } p(0) \text{ and } \pi(0).\end{aligned}$$

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; s = terms of trade; $\bar{s}$ = natural terms of trade; π = inflation rate; $\dot{\pi} = d\pi/dt$; $\dot{s} = ds/dt$; r^* = real foreign interest rate; i = nominal interest rate; $\alpha, \beta, \gamma, \delta, \phi$ and θ are positive parameters.

- (a) Analyze the model's equilibrium and dynamics on a phase diagram with the real interest rate (r) on the vertical axis and the terms of trade (s) on the horizontal axis.
- (b) Are the real interest rate and the terms of trade negatively correlated regardless of the values of the model's parameters?
- (c) Use the phase diagram from item (a) to show what happens in this model when public deficit increases.
- (d) Use the phase diagram from item (a) to show what happens in this model when the real foreign interest rate increases.

4. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(s - \bar{s}),$$

$$\text{PC: } \dot{\pi} = \phi\dot{s} + \theta x,$$

$$\text{UIP: } r = \bar{r} + \dot{s}\bar{r} = r^*,$$

$$\text{MPR: } e = \bar{e},$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; e = terms of trade; $\bar{s}$ = natural terms of trade; i = nominal interest rate; π = inflation rate; $\dot{\pi} = d\pi/dt$; $\dot{s} = ds/dt$; r^* = real foreign interest rate. The α, β, ϕ and θ parameters are positive. Assume a constant foreign inflation rate ($\pi^* = dp^*/dt = \text{constant}$).

Use a phase diagram with r on the vertical axis and s on the horizontal axis to show this model's equilibrium and dynamics.

5. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(s - \bar{s}),$$

$$\text{PC: } \dot{\pi} = \gamma\dot{s} + \delta x,$$

$$\text{UIP: } r = \bar{r} + \dot{s},$$

$$\text{MPR: } i = \bar{r} + \pi + \theta(\pi - \bar{\pi}) + \phi\dot{s},$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; s = terms of trade; $\bar{s}$ = natural terms of trade; π = inflation rate; $\dot{\pi} = d\pi/dt$; $\bar{\pi}$ = inflation rate target; $\dot{s} = ds/dt$; i = nominal interest rate.

- (a) Analyze the model's equilibrium and dynamics on a phase diagram with the inflation rate on the vertical axis and the terms of trade on the horizontal axis;
- (b) Use the previous item's phase diagram to show what happens when the Central Bank lowers the inflation target to $\bar{\pi}_1 (< \bar{\pi}_0)$ from $\bar{\pi}_0$.

6. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(q - \bar{q}),$$

$$\text{PC: } \dot{\pi} = \gamma\dot{q} + \delta x,$$

$$\text{UIP: } r = \bar{r} + \dot{q},$$

$$\text{MPR: } i = \bar{r} + \pi_C + \theta(\pi_C - \bar{\pi}_C),$$

$$\text{CPI: } \pi_C = \pi + \chi\dot{q}, \chi = \omega/(1 - \omega),$$

$$\text{FE: } i = r + \pi_C,$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; q = real exchange rate; $\bar{q}$ = natural exchange rate; π = inflation rate; $\dot{\pi} = d\pi/dt$; $\bar{\pi}_C$ = inflation rate target; $\dot{q} = dp/dt$; i = nominal interest rate. The real exchange rate is defined by: $q = e + p^* - p_C$ and the consumer price index equals: $p_C = (1 - \omega)p + \omega(e + p^*)$. FE = Fisher Equation.

- Analyze the model's equilibrium and dynamics on a phase diagram with the real interest rate (on the vertical axis) and the real exchange rate (on the horizontal axis);
- Use the previous item's phase diagram to show what happens in this economy when a real interest rate increases to $r_1^* > r_0^*$ from r_0^* .
- Analyze the model's equilibrium and dynamics with the inflation rate on the vertical axis and real output on the horizontal axis.
- Use the previous item's phase diagram to show what happens when the Central Bank lowers the inflation target to $\bar{\pi}_1 (< \bar{\pi}_0)$ from $\bar{\pi}_0$.

7. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(q - \bar{q}),$$

$$\text{PC: } \dot{\pi} = \gamma\dot{q} + \delta x,$$

$$\text{UIP: } r = \bar{r} + \dot{q},$$

$$\text{MPR: } e = \bar{e},$$

$$\text{CPI: } \pi_c = \pi + \omega\dot{q},$$

$$\text{FE: } i = r + \pi_C,$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; q = real exchange rate; $\bar{q}$ = natural exchange rate; π = inflation rate of domestic goods; π_C = inflation rate of consumer prices; $\dot{\pi} = d\pi/dt$; $\dot{q} = dq/dt$; i = nominal interest rate, e = nominal exchange rate. The

real exchange rate is defined by: $q = e + p^* - p_C$ and the consumer price index equals: $p_C = (1 - \omega)p + \omega(e + p^*)$. FE = Fisher Equation.

- (a) Analyze the model's equilibrium and dynamics on a phase diagram with the real interest rate (on the vertical axis) and the real exchange rate (on the horizontal axis);
- (b) Use the previous item's phase diagram to show what happens in this economy when the real interest rate increases to $r_1^* > r_0^*$ from r_0^* .
- (c) Analyze the model's equilibrium and dynamics with the inflation rate on the vertical axis and real output on the horizontal axis.
- (d) Use the previous item's phase diagram to show what happens when the international inflation rate decreases to $\pi_1^* (< \pi_0^*)$ from π_0^* .

8. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}) + \beta(q - \bar{q}),$$

$$\text{PC: } \dot{\pi} = \gamma\dot{q} + \delta x,$$

$$\text{UIP: } r = \bar{r} + \dot{q},$$

$$\text{MPR: } \dot{e} = \pi - \pi^*,$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

The symbols have the following meanings: x = output gap; r = real interest rate; $\bar{r}$ = natural interest rate; q = real exchange rate; $\bar{q}$ = natural exchange rate; π = inflation rate; e = nominal exchange rate.

- (a) Analyze the model's equilibrium and dynamics.
 - (b) Show what happens in this economy when the natural interest rate increases.
9. Assume that imports and labor are used as inputs in the production of a domestic product. The economy's real output is then equal to the sum of consumption and exports:

$$y_t = \omega_1 c_t + \omega_2 ex_t.$$

The variables are in log form and the weight ω_i is the respective variable's proportion in the steady state. The demand for export equation is the demand equation for an input that depends on the world output and the relative price:

$$ex_t = y_t^* + \eta s_t + \kappa.$$

The η parameter is the elasticity of substitution between the input and labor, κ is a constant and s is the terms of trade as defined by: $s_t = e_t + p_t^* - p_t$.

The Euler equation is given by:

$$c_t = -\sigma(r_t - \rho) + c_{t+1}.$$

- (a) Show that the IS curve for this open economy model is the following:

$$x_t = x_{t+1} - \omega_1 \sigma(r_t - \bar{r}_t) - \omega_2 \eta \Delta(s_{t+1} - \bar{s}_{t+1}).$$

- (b) Show that the natural interest rate is given by the expression:

$$\bar{r}_t = \rho + \frac{\Delta \bar{y}_{t+1}}{\omega_1 \sigma} - \frac{\omega_2 \Delta y_{t+1}^*}{\omega_1 \sigma} - \frac{\omega_2 \eta \Delta \bar{s}_{t+1}}{\omega_1 \sigma}.$$

- (c) Show that this expression of the natural rate can be simplified, using a little algebra, to obtain the same formula as in a closed economy:

$$\bar{r}_t = \rho + \frac{1}{\sigma} \Delta \bar{y}_{t+1}.$$

- (d) What is your verdict regarding this IS curve model of a small open economy? (Hint: Compare the previous item's natural interest rate with the one obtained based on the uncovered interest rate parity).

Part III
Monetary and Fiscal Policy Models

Chapter 10

Government Budget Constraint



This chapter presents the government budget constraint and analyzes several issues based on this accounting framework. Section 10.1 shows that the government budget constraint results from the consolidation of the Treasury and Central Bank balance sheets. Section 10.2 presents the conditions for the public debt to be sustainable. Section 10.3 addresses the inflation tax and the various alternative ways to calculate its social cost. Section 10.4 covers various hyperinflation models. Section 10.5 presents the Ricardian equivalence theorem. Section 10.6 deals with the fiscal theory of the price level. Section 10.7 analyzes the conditions for a monetary regime to be sustainable.

10.1 Consolidating the Treasury and the Central Bank Balance Sheets

Table 10.1 shows a Central Bank's balance sheet. The assets include foreign-currency denominated securities under International Reserves (IR), and domestic-currency denominated securities – usually Treasury Bills and Bonds (B^{BC}). The Central Bank's liabilities consist of the Monetary Base (M) and Treasury Deposits (TD). Treasury Deposits are not necessarily held with the Central Bank, depending on each country's laws. The monetary base is made up of currency held by the public and bank reserves. The bank reserves are deposits that banks hold with the Central Bank, and include their required reserves, among other deposits. This account is the conduit for all of the Central Bank's transactions with the banks, and for all operations under the payments system, which the Central Bank supervises.

The Central Bank has the monopoly on issuing money, and its business consists of selling and buying the money it issues. It buys domestic or foreign securities when it wants to sell money. By purchasing foreign securities, the Central Bank buys foreign currency on the spot market for immediate delivery, or on the forward market

Table 10.1 Central Bank balance sheet

Assets	Liabilities
1. International Reserves (IR) 2. Government Securities (B^{BC})	1. Monetary Base (M) (a) Currency held by the Public (b) Bank Reserves 2. Treasury Deposits (TD)
$IR + B^{BC} \equiv M + DT$	

for future delivery. While purchasing domestic securities, it may carry out definitive purchases, or temporary ones through repurchase deals. When the Central Bank wishes to buy money, it sells foreign or domestic securities from its portfolio. When selling or buying foreign securities, it operates on the foreign exchange market and when selling or purchasing domestic securities, it operates on the secondary market for government bills and bonds. In both cases, the operations are conducted by the Central Bank's operations desk.

The analysis below makes the simplifying assumption that international reserves and treasury deposits are equal to zero, $IR = TD = 0$. Under these hypotheses, the Central Bank's profits will be equal to the interest on the government bills and bonds in its portfolio, disregarding operational costs. Hence:

$$L^{BC} = i B^{BC}. \quad (10.1)$$

The change of the monetary base equals the change of the stock of government securities in its portfolio:

$$\dot{M} = \dot{B}^{BC}. \quad (10.2)$$

Treasury finances government spending with tax revenues, the Central Bank's profits, and the issuing of public debt:

$$G - T + iB^T - L^{BC} \equiv \dot{B}^T. \quad (10.3)$$

The change of the stock of public debt equals that of the stock held by the public and the change of the stock held by the Central Bank:

$$\dot{B}^T = \dot{B} + \dot{B}^{BC} \quad (10.4)$$

Public debt issued by the Treasury is held either by the public or by the Central Bank:

$$B^T = B + B^{BC} \quad (10.5)$$

Substituting the Central Bank's profit and the public debt's breakdown in the Treasury constraint yields:

$$G - T + iB^T - iB^{BC} \equiv \dot{B} + \dot{B}^{BC}.$$

Treasury finances a deficit by selling government bonds directly to the private sector and indirectly to the Central Bank, which purchases Treasury securities on the secondary market.

Consolidating the Treasury and Central Bank balance sheets yields the government budget constraint:

$$G - T + iB \equiv \dot{B} + \dot{M}. \quad (10.6)$$

A more appropriate name for this constraint might be government financing sources, because government spending on consumption and investment (G) and on the public debt service is financed with taxes, increased public debt, or the printing of money. This financing equation is easily understood when measured in terms of the economy's nominal gross domestic product. That is:

$$\frac{G - T}{Y} + i\frac{B}{Y} \equiv \frac{\dot{B}}{Y} + \frac{\dot{M}}{Y}. \quad (10.7)$$

The primary surplus is defined as the difference between tax revenues and government spending except for interest payments on the public debt:

$$\frac{G - T}{Y} = -f_s. \quad (10.8)$$

The symbol f_s represents the primary surplus. The letter b represents the debt-to-GDP ratio ($b = B/Y$). The derivative of the debt-to GDP ratio with respect to time yields:

$$\dot{b} = \frac{\dot{B}}{Y} - \frac{B\dot{Y}}{Y^2} = \frac{\dot{B}}{Y} - \frac{B}{Y} \frac{\dot{Y}}{Y}.$$

The economy's nominal output equals the price index multiplied by the real output index, $Y = Py$. Therefore, the growth rate of nominal output equals the sum of the inflation rate and the growth rate (n) of real output:

$$\frac{\dot{Y}}{Y} = \frac{\dot{P}}{P} + \frac{\dot{y}}{y} = \pi + n.$$

The increase in public debt as a share of the gross domestic product can then be written as follows:

$$\frac{\dot{B}}{Y} = \dot{b} + b(\pi + n).$$

Using the letter m to denote the ratio of the monetary base to the economy's nominal output, taking the derivative of m with respect to time and proceeding in the same way as with the debt-to-GDP ratio yields:

$$\frac{\dot{M}}{Y} = \dot{m} + m(\pi + n).$$

Substituting the previous equations in the government budget constraint yields the following budget constraint:

$$-f_s + ib \equiv \dot{b} + (\pi + n)b + \dot{m} + (\pi + n)m,$$

or:

$$-f_s + (i - \pi - n)b - (\pi + n)m \equiv \dot{b} + \dot{m} \equiv f.$$

Fiscal deficit f is defined as the change in the government's liabilities, either in the form of government securities or money. Hence:

$$f = \dot{b} + \dot{m}. \quad (10.9)$$

10.2 Public Debt Sustainability

This section analyzes public debt sustainability based on the assumption of a constant stock of money:

$$\dot{M} = 0. \quad (10.10)$$

The government budget constraint simplifies to:

$$\dot{b} = (i - \pi - n)b - f_s.$$

Because the spread between the nominal interest rate and the inflation rate equals the real interest rate, the previous equation becomes:

$$\dot{b} = (r - n)b - f_s. \quad (10.11)$$

10.2.1 Constant Primary Deficit (Surplus)

When the real interest rate is lower than the growth rate of real output, $r - n < 0$, the differential equation that describes the public debt's dynamics has a negative slope, as Fig. 10.1 shows. When there is a primary deficit, $f_d = -f_s > 0$ the debt-to-GDP ratio converges to an equilibrium value. When there is a primary surplus, if Treasury is initially indebted at first, it will redeem its bonds and start purchasing assets over time. The assets-to-GDP ratio converges to a fixed value, as the left panel of Fig. 10.1 shows.

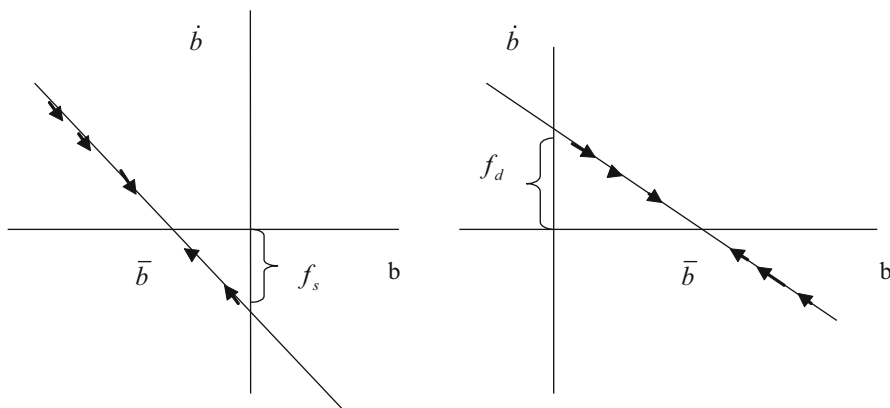


Fig. 10.1 $r - n < 0$

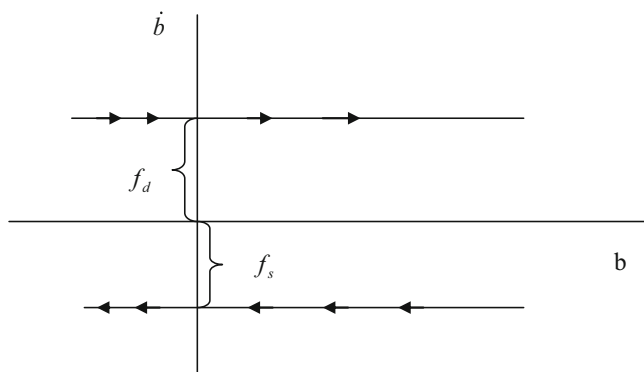


Fig. 10.2 $r - n = 0$

Whatever the initial value of the debt-to-GDP ratio, the ratio converges to its equilibrium value:

$$\bar{b} = \frac{f_s}{r - n}. \quad (10.12)$$

This value is positive in the presence of a primary deficit, and negative in the case of a primary surplus.

When the real interest rate is equal to the growth rate of real output ($r = n$) and there is a fiscal deficit financed with government borrowing, the path of public debt is explosive, as the phase diagram in Fig. 10.2 shows. Whatever the initial value of the debt-to-GDP ratio, it will grow indefinitely. When Treasury has a primary surplus, it will build up assets and their stock will grow indefinitely as well.

When the real interest rate exceeds the growth rate of the economy's real output, $r - n > 0$, the differential equation describing the evolution of the debt-to-GDP ratio

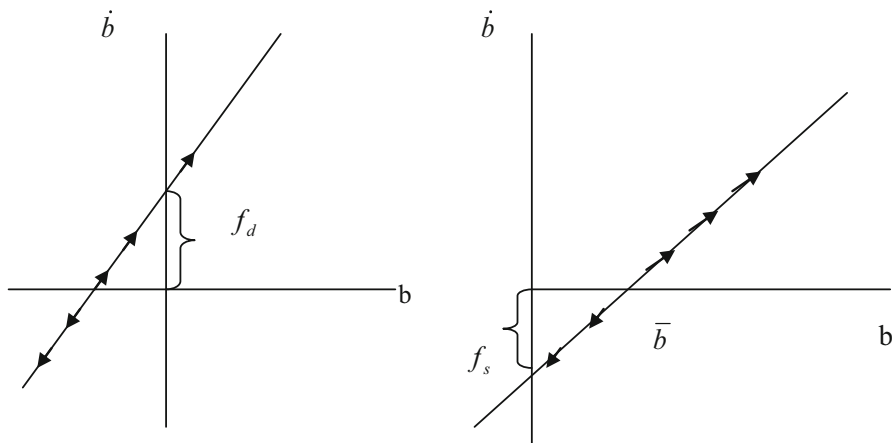


Fig. 10.3 $r - n > 0$

has a positive slope, as Fig. 10.3 shows. When there is a primary surplus, an equilibrium value exists for the debt-to-GDP ratio. If this ratio's initial value is other than the equilibrium value, public debt will be unsustainable because the model is unstable. If Treasury has a primary deficit, it must have a stock of assets for the fiscal policy to be sustainable. The left panel of Fig. 10.3 shows the phase diagram corresponding to this case.

10.2.2 Variable Primary Deficit (Surplus)

When the primary surplus (deficit) is variable over time, the f_s term in differential equation

$$\dot{b} - (r - n)b = -f_s, \quad (10.13)$$

which describes the law of motion of the debt-to-GDP ratio, is no longer a parameter, but rather a function of time.

Algebra

To determine the solution to this equation, let us begin by multiplying both sides by the natural number raised to the power of the difference between the real interest rate and the real output growth rate multiplied by time with a negative sign. That is:

$$e^{-(r-n)v}[b - (r - n)b] = -f_s e^{-(r-n)v}.$$

It is easily seen that:

$$\frac{d}{dv} b e^{-(r-n)v} = -f_s e^{-(r-n)v}.$$

Therefore,

$$d b e^{-(r-n)v} = -f_s e^{-(r-n)v} dv.$$

Integrating this expression from today (t) to the future (T) yields:

$$\int_t^T d b e^{-(r-n)v} = - \int_t^T f_s e^{-(r-n)v} dv.$$

Calculating the integral of the left side of this expression yields:

$$b(T)e^{-(r-n)T} - b(t)e^{-(r-n)t} = - \int_t^T f_s e^{-(r-n)v} dv.$$

This equation may be written as:

$$b(t) = b(T)e^{-(r-n)(T-t)} + \int_t^T f_s e^{-(r-n)(v-t)} dv. \quad (10.14)$$

Debt Sustainability

A debt becomes unsustainable when its issuer not only rolls the principal, but also fully finances the interest with new debt issues. This behavior is known as the Ponzi game. The condition for the absence of a Ponzi game is for the following limit to be equal to zero:

$$\lim_{T \rightarrow \infty} b(T)e^{-(r-n)(T-t)} = 0. \quad (10.15)$$

According to this limit, public debt is sustainable when public debt as a share of output grows at a rate lower than the real interest rate minus the rate of growth of real output, assuming that $r > n$. This constraint has no implication for the level of the public debt, only for its growth rate. Under the no-Ponzi game hypothesis, the existing debt-to-GDP ratio must equal the present value of future primary surpluses discounted at the real interest rate net of the economy's growth rate:

$$b(t) = \int_t^\infty f_s e^{-(r-n)(v-t)} dv \quad (10.16)$$

This equation corresponds to the government's intertemporal budget constraint, that is, the present value of the taxes is equal to the present value of spending plus the value of public debt held by the public:

$$b(t) + \int_t^\infty g e^{-(r-n)(v-t)} dv = \int_t^\infty \tau e^{-(r-n)(v-t)} dv. \quad (10.17)$$

It is worth mentioning that if the primary surplus is constant, the ratio of the primary surplus to public debt must equal the net real interest rate: $r - n = \bar{f}_s/b$.

When the real interest rate is lower than the growth rate of real output, $r < n$, the solution of the differential equation of public debt may be written as:

$$b(T) = b(t)e^{-(n-r)(T-t)} + \int_t^T f_d e^{-(n-r)(T-v)} dv.$$

Because:

$$\lim_{T \rightarrow \infty} b(t)e^{-(n-r)(T-t)} = 0,$$

it follows that:

$$\lim_{T \rightarrow \infty} b(T) = \int_t^\infty f_d e^{-(n-r)(T-v)} dv. \quad (10.18)$$

When the primary deficit, f_d , is constant, the public debt converges to a finite value equal to $f_d/(\eta - r)$. When the primary deficit varies with time, the limit of the public debt as a share of gross domestic product must be finite. However, sustainability analysis does not indicate the value of this limit.

10.3 Inflation Tax

The government budget constraint shows a revenue source that is equal to the product of the inflation rate multiplied by the real quantity of money. This revenue is the inflation tax:

$$\tau = \pi m. \quad (10.19)$$

Inflation rate π is the rate of the tax, and m is its base. The quantity of money demanded depends on the level of real output and on the nominal interest rate, the opportunity cost of holding money. Assume, for simplicity, that the elasticity of the real quantity of money with respect to real output is equal to one. The money demand equation may be written with the nominal interest rate as a function of the real quantity of money:

$$m = \frac{M}{P_y} = L(i), i = L^{-1}(m) = r(m). \quad (10.20)$$

Figure 10.4 shows the money demand curve. The inflation tax is the hashed area. This area measures the loss of purchasing power, as a result of inflation, of holders of money stock. To recover the real stock of money, the public demand an additional quantity of money, the Central Bank issues this money and appropriates the corresponding real resources to finance government spending. Figure 10.5 shows the shape of the tax curve in two particular cases. The vertical axis measures the inflation tax and the horizontal axis measures the inflation rate – the tax rate. In Fig. 10.5a the tax has a maximum and, in Fig. 10.5b, the tax increases as its rate increases.

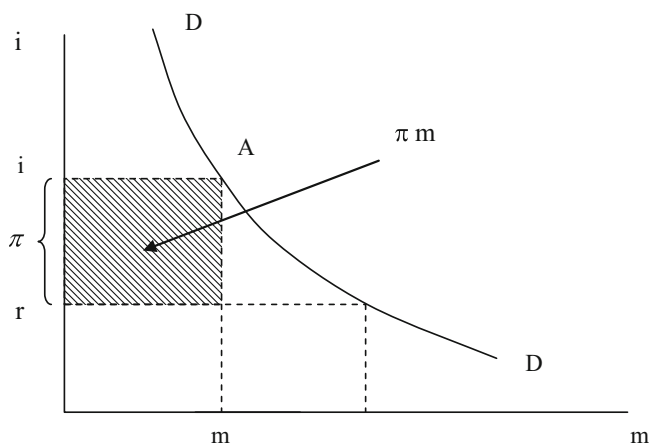


Fig. 10.4 The money demand curve and the inflation tax

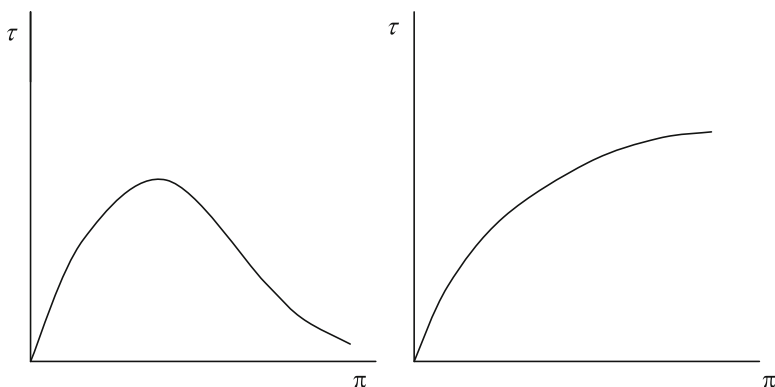
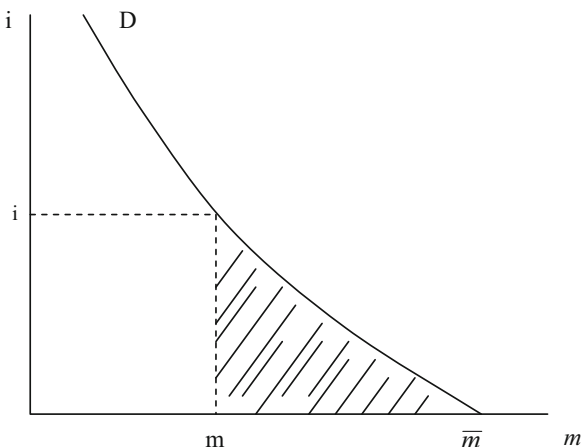


Fig. 10.5 The inflation tax curves

The Social Cost of the Inflation Tax: Consumer Surplus

Like any other tax, the inflation tax has a social cost, or welfare cost, because it creates a wedge between the value for money holders and value for the issuer of money. The hashed area in Fig. 10.6 shows the social cost of inflation, disregarding the Central Bank's operational costs and taking into account that the cost of production of money is quite small and can be disregarded too. The area corresponds to the following integral:

$$\omega(i) = \int_{m(i)}^{m(0)} i(m) dm = \int_0^i m(v) dv - im \quad (10.21)$$

Fig. 10.6 The social cost of inflation

Calculation of the social cost from the first integral uses the inverse function of the money demand equation, while the formula following the second equal sign uses the money demand equation. Analysis of Fig. 10.6 clearly shows the equivalence of the two formulas.

For a money demand equation with constant semi-elasticity, specified by $\log m = -\alpha i$, the social cost of inflation is:

$$\omega(i) = \int_0^r e^{-av} dv - ie^{-ai} = \frac{1}{\alpha}(1 - e^{-ai}) - ie^{-ai}. \quad (10.22)$$

When $i \rightarrow \infty$, the social cost of hyperinflation is finite and is equal to the inverse of the parameter α . This is to say that the social cost is proportional to the quantity of money that society wants to hold when the interest rate equals zero. The proportionality coefficient is equal to the inverse of the semi-elasticity.

When the money demand equation has constant elasticity specified by $\log m = -\alpha \log i$, $\alpha < 1$, the social cost of inflation is given by:

$$\omega(i) = \int_0^i v^{-\alpha} dv - ii^{-\alpha} = \frac{i^{1-\alpha}}{1-\alpha} - i^{1-\alpha} = \frac{\alpha}{1-\alpha} i^{1-\alpha}. \quad (10.23)$$

It is straightforward to show from this equation that the social cost of hyperinflation in this case is infinite because:

$$\lim_{i \rightarrow \infty} \omega(i) = \infty. \quad (10.24)$$

Analysis of the two cases – constant semi-elasticity and constant elasticity – shows that the social cost of hyperinflation is finite if money is not essential, but infinite if money is essential. Therefore, the functional form of the money demand equation produces completely different results in terms of the social cost of hyperinflation.

Social Cost of the Inflation Tax: Representative Agent

The social cost, or welfare cost, of inflation calculated in the previous equations is based on the consumer surplus from partial equilibrium analysis. In a microfounded monetary model, consumption (c) and the services of money (m) are arguments in the utility function:

$$u = u(c, m).$$

One dollar, allocated as money, produces an increase in welfare equal to the marginal utility of money ($\partial u / \partial m$). The opportunity cost of money equals the nominal interest not earned in an alternative financial asset. Therefore, when \$1 is held as money, the loss of welfare equals the product of interest rate i multiplied by the marginal utility of consumption ($\partial u / \partial c$). Hence, at equilibrium:

$$i = \frac{\partial u / \partial m}{\partial u / \partial c} = \frac{u_m(c, m)}{u_c(c, m)}. \quad (10.25)$$

The representative agent's total utility is maximum when the marginal utility of money is zero. Therefore, the optimum quantity of money is obtained when the nominal interest rate is equal to zero. To calculate the social cost of inflation from this framework, it is convenient to normalize consumption, making it equal to one. Therefore,

$$i = \frac{u_m(1, m)}{u_c(1, m)}.$$

This expression implicitly defines the demand for money equation: $m = m(i)$.

The welfare cost of inflation is the percentage of the income that a consumer must receive to become indifferent between two situations: one in which the interest rate is equal to i and another in which the interest rate is zero. The welfare cost of inflation $\omega(i)$ is therefore defined as:

$$u(1 + \omega(i), m(i)) = u(1, m(0)). \quad (10.26)$$

The approximate value of $\omega(i)$ can be obtained by expanding this function in a Taylor series around the point $i = 0$, disregarding third-order terms. That is:

$$\omega(i) \cong \omega(0) + \omega'(0)(i - 0) + \frac{\omega''(0)}{2}(i - 0)^2.$$

The social cost of $i = 0$ is zero: $\omega(0) = 0$. To obtain the expressions of $\omega'(0)$ and $\omega''(0)$, one takes the derivative of the equation that defines the social cost.

$$u_c \omega'(i) + u_m m'(i) = 0.$$

Because $i u_c = u_m$ at equilibrium, it follows that:

$$\omega'(i) = -im'(i).$$

The second derivative of the social cost with respect to the interest rate is therefore given by:

$$\omega''(i) = -(m'(i) + im''(i)).$$

Clearly, $\omega'(0) = 0$ and $\omega''(0) = -m'(0)$. The welfare cost of inflation is then approximated by:

$$\omega(i) \cong -\frac{m'(0)}{2}i^2. \quad (10.27)$$

The social cost of inflation is proportional to the square of the interest rate. For small interest rate values, the social cost is small as well. For high values, however, the social cost increases at the square of the interest rate. According to the specification of the semi-log money demand equation, $\log m = -\alpha i$, $m'(i) = -\alpha m(i)$, the social cost depends on the initial stock of the real quantity of money and on the semi-elasticity α of the demand equation, according to:

$$\omega(i) = \frac{1}{2}m(0)\alpha i^2 \quad (10.28)$$

10.4 Hyperinflation

A theoretical framework for explaining hyperinflation must reproduce some of the facts observed in the process. These facts are common to all experiences and are documented in several classical studies of the subject. The stylized facts of hyperinflation are as follows: (i) the real quantity of money approaches zero; (ii) the inflation rate increases explosively; (iii) public deficit is financed by printing money; (iv) the duration of hyperinflation is variable, depending on each country's experience; (v) the local currency experiences competition from a foreign currency that first takes over the store of value and unit of account functions, and then replaces the local currency as a means of payment; and (vi) hyperinflation is stopped overnight by means of a stabilization program that changes the monetary and fiscal policy regimes.

There are four hypotheses to explain this phenomenon: (i) bubble; (ii) multiple equilibria; (iii) fiscal crisis and adaptive expectations or partial adjustment; and (iv) fiscal crisis and rational expectations. Not all of these hypotheses can explain the stylized facts listed above.

The hyperinflation models analyzed in this section assume that the public deficit is financed with money only. For simplicity, public debt and its rate of change are zero:

$$B = \dot{B} = 0. \quad (10.29)$$

Inflation Dynamics

The government budget constraint in terms of flows is given by:

$$\dot{m} = f - \pi m. \quad (10.30)$$

Adding and subtracting the real interest rate multiplied by the real quantity of money, the constraint remains unchanged:

$$\dot{m} = f - rm - \pi m + rm.$$

Using the Fisher equation, according to which the nominal interest rate equals the sum of the real interest rate and the inflation rate, we obtain a differential equation for the real quantity of money:

$$\dot{m} = f + rm - im. \quad (10.31)$$

The value of the services of money im is a function of the real quantity of money, as Fig. 10.7 shows. Therefore, this equation may be written as:

$$\dot{m} - rm = f - s(m). \quad (10.32)$$

The function that represents the services of money may have different shapes, depending on the elasticity of the real quantity of money with respect to the inflation rate. Figure 10.8 illustrates two cases. In the bell curve (inverted “U”), money is not essential because the value of the services of money tends to zero when the real quantity of money approaches zero. In the second curve in Fig. 10.8, the value of the services of money converges to a finite value when the real quantity of money tends to zero, and money in this case is essential.

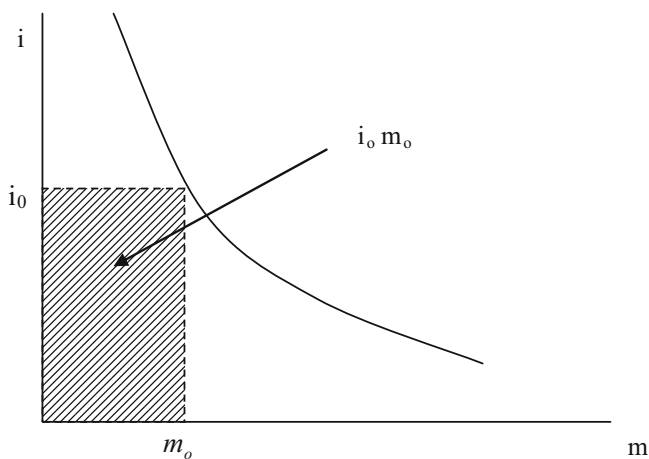


Fig. 10.7 The demand for money and the value of the services of money

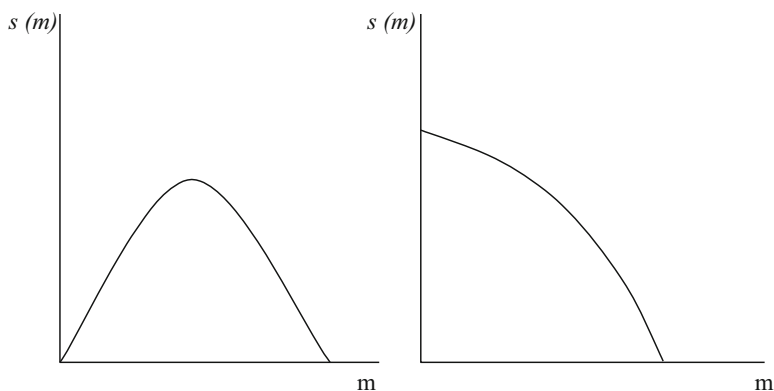


Fig. 10.8 The value of the services of money curves

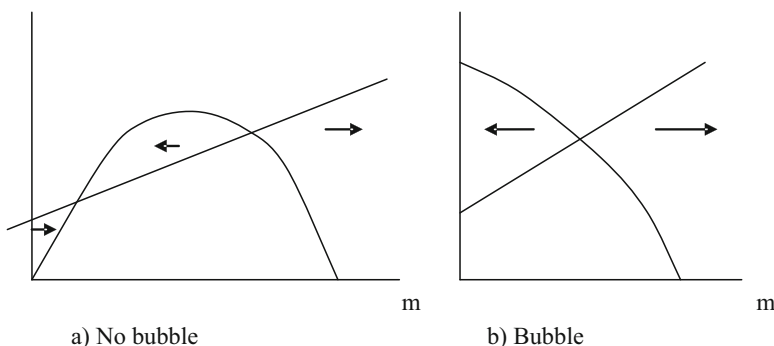


Fig. 10.9 (a) No bubble (b) Bubble

10.4.1 Bubble

The price of money, as for any other financial asset, may eventually have a bubble. The price of money has two components: one corresponding to fundamentals and another to the bubble. Figure 10.9 shows both cases. In the left panel, the services of money curve is bell-shaped, and two points of equilibrium exist. The high inflation equilibrium point, where the real quantity of money is lower, is a stable equilibrium. In this case, the possibility of bubble hyperinflation does not exist because the real quantity of money does not converge to zero.

In the right panel of Fig. 10.9, the curve of the money services has a negative slope and a single unstable equilibrium point. In this case, a bubble path may exist on which the real quantity of money converges to zero. This shape of the services of money curve assumes that the elasticity of the real quantity of money with respect to the inflation rate is, in absolute values, less than one. That is, a bubble path may exist as long as the money demand is inelastic.

10.4.2 Multiple Equilibria

The multiple equilibria hypothesis to explain hyperinflation does not actually explain hyperinflation, but a high inflation process. Figure 10.10 shows the various components of the differential equation:

$$\dot{m} = f + rm - s(m). \quad (10.33)$$

The $s(m)$ curve is bell-shaped and the positively sloped line $f + rm$ crosses the curve at two points. Point m_B corresponds to the low inflation point and point m_A is the high inflation point. In the region between the two points: $\dot{m}$ is negative, the real quantity of money decreases, and the inflation rate rises. At points above the low inflation point: $\dot{m}$ is positive, the real quantity of money increases, and the inflation rate decreases. At points between m_A and the origin, $\dot{m}$ is positive, the real quantity of money increases, and the inflation decreases. Therefore, the low inflation point is unstable and the high inflation point is stable.

The multiple equilibria hypothesis assumes that, for some reason, the economy will enter a path starting at some point between A and B and ending at A , on which the real quantity of money drops and the inflation rate rises. This path is in fact a transitory disequilibrium path because there has been no change to the economy's fundamentals; that is, the money-financed public deficit remains exactly the same. The inflation rate reaches a new equilibrium level, but no hyperinflation occurs.

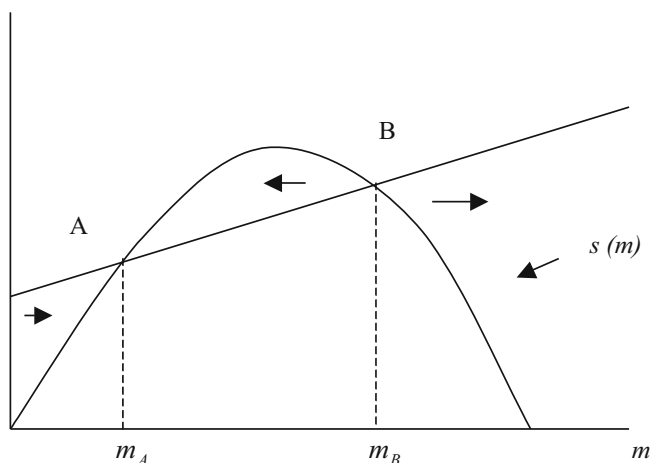


Fig. 10.10 Multiple equilibria

10.4.3 Fiscal Crisis and Rigidity

The model of hyperinflation caused by a fiscal crisis with adaptive expectations has four equations:

$$\dot{m} = f - m\pi \quad (10.34)$$

$$\log m = k - \alpha\pi^e \quad (10.35)$$

$$\dot{\pi}^e = \beta(\pi - \pi^e) \quad (10.36)$$

$$f = f(t) \quad (10.37)$$

The first equation assumes that the public deficit f is financed with money; the second one is the money demand equation, where the log of the real quantity of money $m(=M/P)$ depends on the expected inflation rate π^e and α is the semi-elasticity of money demand; the third equation is the adaptive expectations mechanism, with β as the parameter measuring the speed of adjustment of expectations; and the fourth equation represents public deficit evolution over time.

Algebra

Taking the derivative of the second equation with respect to time yields the following expression, after taking into account the expectations mechanism:

$$\frac{\dot{m}}{m} = -\alpha\dot{\pi}^e = -\alpha\beta(\pi - \pi^e) = -\alpha\beta\pi + \alpha\beta\pi^e.$$

This equation becomes:

$$\frac{\dot{m}}{m} = -\alpha\beta\pi + \beta k - \beta \log m,$$

by substituting the value of expected inflation rate π^e in the money demand equation. The inflation rate is obtained by combining this expression with the model's first equation. That is:

$$\pi = \frac{1}{1 - \alpha\beta} \left(\frac{f}{m} - \beta k + \beta \log m \right). \quad (10.38)$$

The inflation tax $\tau(m)$ equals the product of the inflation rate (π) and the real quantity of money (m). Thus, it follows from the previous expression that:

$$\tau(m) = \frac{1}{1 - \alpha\beta} (f - \beta km + \beta m \log m). \quad (10.39)$$

The derivatives of $\tau(m)$ with respect to the real quantity of money m are given by:

$$\frac{d\tau(m)}{dm} = \frac{1}{1 - \alpha\beta} (-\beta k + \beta(\log m + 1)), \quad \frac{d^2\tau(m)}{dm^2} = \frac{\beta}{(1 - \alpha\beta)m} > 0.$$

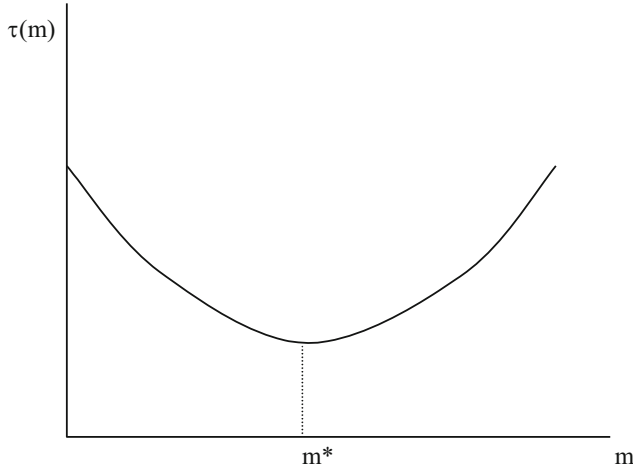


Fig. 10.11 Inflation tax curve for the fiscal crisis and rigidity hyperinflation model

The inequality assumes that $1 - \alpha\beta > 0$, otherwise the model would not produce hyperinflation. Therefore, the inflation tax function is convex, with a minimum at point $m^* = \exp. (k-1)$, and the tax increases as m approaches zero. Figure 10.11 shows that the inflation tax curve is U-shaped. This function is completely different from the bell shape of the Laffer curve, which corresponds to Cagan's money demand equation when the expected and observed inflation rates are equal. The absolute value of the elasticity of the quantity of money demanded is greater than one for m , greater than the value that corresponds to the minimum tax ($m = m^*$), and less than one, in absolute values, for values of m between zero and the real quantity of money that minimizes the inflation tax. The limit of the inflation tax, as m approaches zero, is positive:

$$\lim_{m \rightarrow 0} \tau(m) = \frac{f}{1 - \alpha\beta} > 0. \quad (10.40)$$

Money is essential in this model because the inflation tax does not approach zero when the real quantity of money tends to zero, despite the fact that the functional form of the money demand equation assumes that the absolute value of the elasticity with respect to the expected inflation rate tends to infinity.

Hyperinflation Dynamics

Combining the inflation tax equation with the model's first equation yields the non-linear differential equation:

$$\dot{m} = -\frac{a\beta}{1 - \alpha\beta}f(t) + \frac{\beta km}{1 - \alpha\beta} - \frac{\beta}{1 - \alpha\beta}m \log m \quad (10.41)$$

Fig. 10.12 The phase diagram of the dynamics of hyperinflation in the fiscal crisis and rigidity model

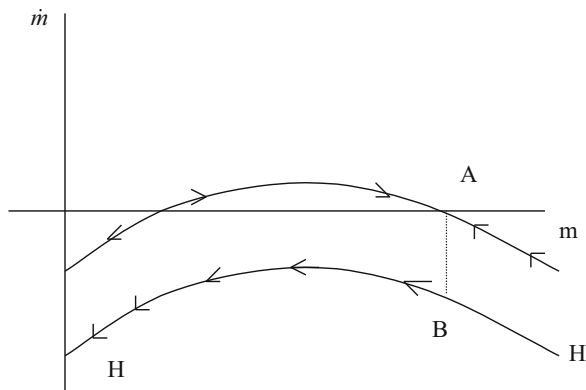
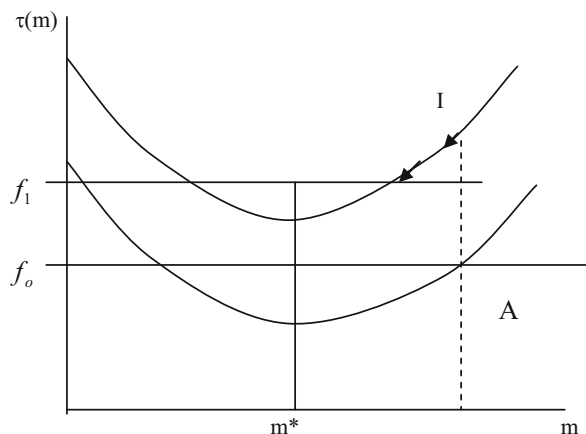


Fig. 10.13 The inflation tax curve shifts upward in the fiscal crisis



This equation describes the model's dynamics. Hyperinflation occurs when $f(t)$ is a step function,

$$f(t) = \begin{cases} f_o < f^*, & \text{if } t < 0 \\ \bar{f} > f^* = \frac{e^{k-1}}{\alpha}, & \text{if } t \geq 0 \end{cases} \quad (10.42)$$

The phase diagram of Fig. 10.12, with the real quantity of money on the horizontal axis and its rate of change over time on the vertical axis, shows the dynamics of hyperinflation. At the moment before the fiscal crisis, the economy is at equilibrium at point A. When the money-financed public deficit increases to $\bar{f}$ from f_o , the economy moves to point B and the hyperinflation path begins. The real quantity of money does not jump but starts to drop until it reaches a value equal to zero. The inflation tax curve shifts upward, as Fig. 10.13 shows. The inflation tax increases immediately (to point I from point A), and then starts to drop, reaches a minimum, and later increases until it reaches a maximum when it touches the vertical

axis. The correlation between the inflation tax and the inflation rate cannot be inferred from the shape of the inflation tax curve because it has both downward and upward slopes. The model's dynamics are more complex and not always recognized in the literature on this model.

10.4.4 *Intertemporal Approach: Fiscal Crisis and Rational Expectations*

The hypothesis of hyperinflation caused by a fiscal crisis and rational expectations makes two different assumptions for the money demand equation. According to the first one, money is inelastic and therefore essential. This assumption corresponds to strong hyperinflation, or hyperinflation proper. According to the second assumption, money is non-inelastic. In this case the fiscal crisis causes weak hyperinflation because the real quantity of money does not converge to zero and the inflation rate does not tend to infinity. In both cases, the model can be summarized as two equations: one for the dynamics of the real quantity of money and another for the fiscal crisis:

$$\dot{m} = f + rm - s(m) \quad (10.43)$$

$$f = f(t), \dot{f} > 0, f < \bar{f} \quad (10.44)$$

Specification of the fiscal crisis assumes that the money-financed public deficit increases over time, but is lower than a predetermined ceiling ($\bar{f}$). This ceiling is greater than what this economy could generate as inflation tax revenue in a permanent basis. The model produces a non-autonomous differential equation.

Integrating the previous differential equation from moment t to future time T yields the government's intertemporal constraint:

$$m(t) = m(T)e^{-r(T-t)} + \int_t^T (s(m) - f)e^{-r(v-t)} dv.$$

The condition for the absence of hyperdeflation is for the following limit to be equal to zero:

$$\lim_{T \rightarrow \infty} m(T)e^{-r(T-t)} = 0.$$

The government's intertemporal constraint is therefore given by the following expression:

$$m(t) = \int_t^\infty [s(m) - f]e^{-r(v-t)} dv. \quad (10.45)$$

According to this constraint, the real quantity of money held by the public equals the present value of the services of money minus the money-financed public deficit. Another way to read this constraint is as follows: The real quantity of money held by the public (the government's monetary liability) plus the present value of the public deficit must equal the present value of the services of money. If this constraint is not satisfied, the policy of financing the public deficit by issuing money will be unsustainable. That is, assume that the money-financed public deficit increases over time in such a manner that at a certain point it exceeds the value of the services of money. There is, then, a time t_h when the real quantity of money will be equal to zero:

$$m(t_h) = \int_t^{t_h} [s(m) - f] e^{-r(v-t)} dv = 0. \quad (10.46)$$

The $t_h - t$ interval is the maximum duration of the hyperinflation given the monetary regime of financing the public deficit issuing money. According to this intertemporal approach, hyperinflation is analyzed in the same way as a fiscal regime producing a public debt crisis, that is, the government's intertemporal budget constraint ceases to be satisfied. Analytically, the model's solution requires solving a non-autonomous differential equation because the public deficit changes over time. The model can, however, be solved graphically, as we will show next.

Strong Hyperinflation

Figure 10.14 describes the solution of the differential equation for the inelastic money case. At the initial moment, the money-financed public deficit is equal to $f(0)$. The real quantity of money at this point is equal to $m(0)$ and the value of $\dot{m}$ is negative. The real quantity of money starts to decrease and the inflation rate increases on a hyperinflationary path. Society knows beforehand that the budget constraint is not sustainable and that the economy will stabilize at some point in time.

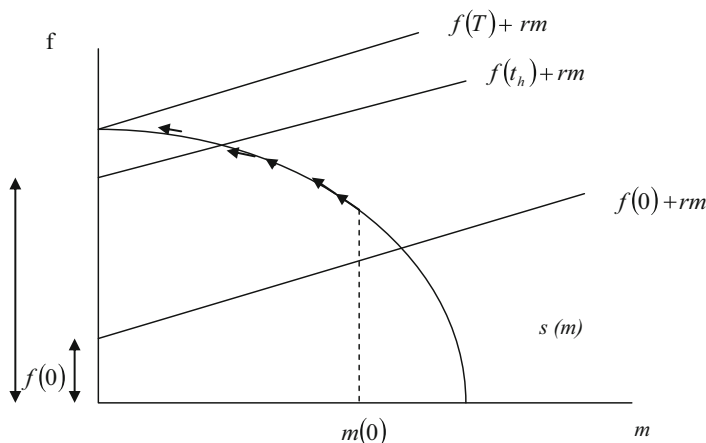


Fig. 10.14 Strong hyperinflation

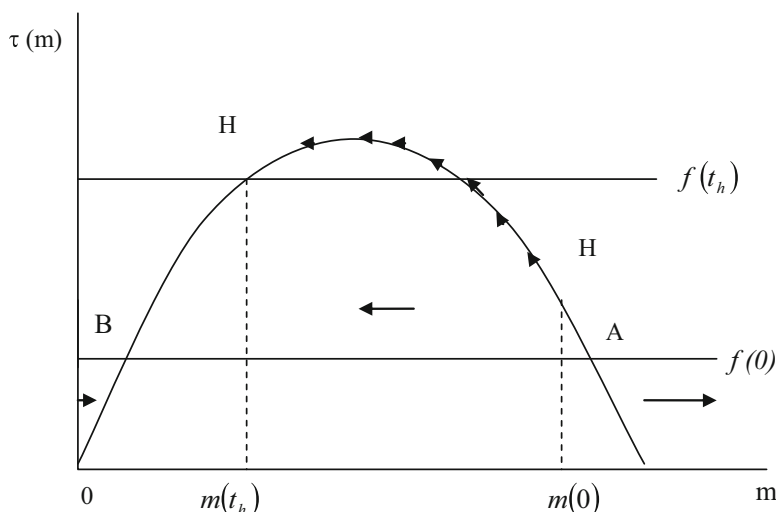


Fig. 10.15 Weak hyperinflation

The period of time t_h indicates the longest possible duration of the hyperinflation. The economy may stabilize at any moment before this period ends. However, despite being aware that the situation is unsustainable, society continues to use money because it is essential for economic transactions.

Weak Hyperinflation

Figure 10.15 shows the model's solution when money is non-inelastic. At the initial moment when public deficit is $f(0)$, the real quantity of money is equal to $m(0)$ and $\dot{m}(0) < 0$. That is, the real quantity of money starts to decrease as the inflation rate rises. The dynamics of the process may eventually take the economy to the "wrong side" of the Laffer curve. This does not stem from irrational government behavior in the sense that it might make greater revenues with less inflation. It arises from the irrational fact that the money financed fiscal crisis is an unsustainable policy because the government's intertemporal constraint is not satisfied.

10.5 Ricardian Equivalence

According to the Ricardian equivalence theorem, public debt and taxes are equivalent for a given government spending path. The theorem makes the following assumptions: (i) the representative agent's horizon is infinite; (ii) the representative agent borrows and lends at the same real interest rate as the government; (iii) future taxes are predictable; and (iv) taxes have no allocation effects and are of the lump sum type. The theorem's demonstration is quite simple, and merely requires consolidating the private sector and the government budget constraints.

Budget Constraint: Private Sector

The private sector's budget constraint shows how the private funds are used. Income Y plus interest on government bonds held iB finance consumer spending C , tax payments T , the change in money stock $\dot{M}$, the change in the stock of government bonds $\dot{B}$:

$$Y + iB = C + T + \dot{M} + \dot{B}. \quad (10.47)$$

Dividing both sides of this expression by the price level yields:

$$\frac{Y}{P} + i\frac{B}{P} = \frac{C}{P} + \frac{T}{P} + \frac{\dot{M}}{P} + \frac{\dot{B}}{P}.$$

Using lower-case letters to denote the variables' real values, the constraint may be written as:

$$y + ib = c + \tau + \dot{m} + \pi m + \dot{b} + \pi b.$$

Adding and subtracting the product of the real interest rate and the real quantity of money does not change the previous expression. That is:

$$y + (i - \pi)b = c + \tau + \dot{m} + \dot{b} + mr - mr + \pi m.$$

The services of money depend on the real quantity of money, $im = s(m)$. Therefore, the private sector's budget constraint shows that income and financial asset yields are used to buy goods and services, pay taxes, consume the services of money, and change the real stocks of money and public debt:

$$y + r(m + b) = c + \tau + s(m) + \dot{m} + \dot{b}.$$

The private sector's total financial assets equal the sum of the money stock and the stock of bonds: $a = m + b$. It follows that the private sector's budget constraint in terms of flows is described by the differential equation:

$$\dot{a} = ra + y - c - \tau - s(m).$$

Integrating this differential equation between the dates t and T produces the private sector's budget constraint in terms of stock:

$$a(t) = a(T)e^{-r(T-t)} + \int_t^T [c + \tau + s(m) - y]e^{-r(v-t)} dv. \quad (10.48)$$

For a Ponzi game to not occur, the following limit must be satisfied:

$$\lim_{T \rightarrow \infty} a(T)e^{-r(T-t)} = 0. \quad (10.49)$$

According to this condition, the agent cannot refinance the debt and interest with additional debt. This is equivalent to the statement that the present value of their spending on consumption, taxes, and the services of money must equal the present value of their income plus their current stock of financial assets:

$$a(t) + \int_t^\infty ye^{-r(v-t)} dv = \int_t^\infty [c + \tau + s(m)]e^{-r(v-t)} dv. \quad (10.50)$$

Budget Constraint: Public Sector

According to the government budget constraint, the government finances its spending with taxes, printing money, and issuing debt:

$$\frac{G}{P} + i\frac{B}{P} - \frac{T}{P} = \frac{\dot{M}}{P} + \frac{\dot{B}}{P}. \quad (10.51)$$

The values have been deflated by the price index. Using lower-case letters to denote the real values of the variables, and bearing in mind that $\dot{M}/P = \dot{m} + \pi m$ and $\dot{B}/P = \dot{b} + \pi b$, the previous constraint becomes:

$$g - \tau + i b = \dot{m} + \pi m + \dot{b} + \pi b.$$

Adding and subtracting rm yields:

$$g - \tau + (i - \pi)b = \dot{m} + \dot{b} + rm - rm + \pi m.$$

This equation is equivalent to:

$$g - \tau + r(m + b) = \dot{m} + \dot{b} + im.$$

The government budget constraint in terms of flows is given by the following differential equation:

$$\dot{a} = ra + g - \tau - s(m). \quad (10.52)$$

Integrating both sides of this expression yields the government budget constraint in terms of stocks:

$$a(t) = \int_t^\infty [\tau + s(m) - g]e^{-r(v-t)} dv. \quad (10.53)$$

This expression assumes that the condition for the absence of a Ponzi game is satisfied:

$$\lim_{T \rightarrow \infty} a(T)e^{-r(T-t)} = 0. \quad (10.54)$$

Consolidating the Private and Public Sectors

Substituting the government budget constraint's value of $a(t)$ in the private sector's budget constraint, we find that the present value of consumption spending is equal to the present value of the private sector's income flow minus government spending:

$$\int_t^{\infty} c e^{-r(v-t)} dv = \int_t^{\infty} (y - g) e^{-r(v-t)} dv. \quad (10.55)$$

Public debt and taxes are irrelevant to the representative agent's consumer spending decisions. The basic reason for this is that public debt is not regarded as part of the private sector's wealth because the value of public debt corresponds to a liability equal to the present value of future taxes needed to repay the debt.

10.6 Fiscal Theory of the Price Level

The fiscal theory of the price level assumes that the Central Bank sets the nominal interest rate

$$i = \bar{i} \quad (10.56)$$

and that the treasury sets the primary surplus

$$f_s = \bar{f}_s. \quad (10.57)$$

The fiscal policy is a non-Ricardian policy because the government can freely choose its primary surplus and is not bound to that which would be consistent with the government budget constraint.

According to the fiscal theory of the price level, the government budget constraint is not an identity, but an equilibrium condition for the model. The combination of monetary and fiscal policies would solve the problem of the indeterminacy of the price level when the Central Bank sets the nominal interest rate. In Fig. 10.16 the real quantity of money is determined by the nominal interest rate, but the price level and the nominal quantity of money are indeterminate. The fiscal theory solves this problem by determining the price level on the fiscal side such that the public debt is solvent.

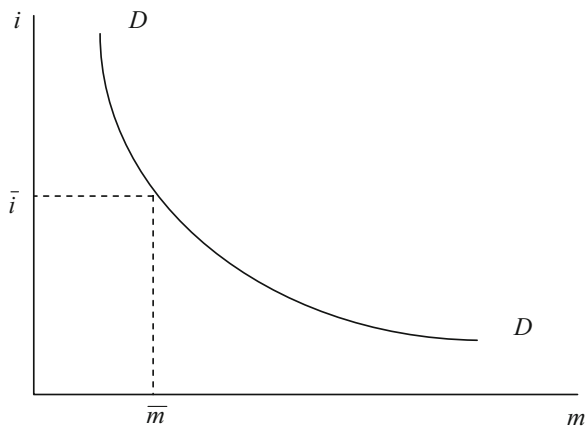
Goods and services market equilibrium occurs when expenditure equals output:

$$y = c + g, y = \bar{y}. \quad (10.58)$$

In this economy, real output is assumed to be fixed, that is, real output is equal to $\bar{y}$.

Government debt has two components: the real quantity of money held by the public and the real stock of public debt, which by assumption is not indexed to any price index. Hence:

Fig. 10.16 The central bank pegs the nominal interest rate and the demand for money gives the real quantity of money



$$a(t) = m(t) + b(t) = \bar{m} + b(t). \quad (10.59)$$

The government's debt will be sustainable when it is equal to the present value of primary surplus flows:

$$a(t) = \int_t^{\infty} \bar{f}_s e^{-r(v-t)} dv. \quad (10.60)$$

This solvency condition, combined with the definition of the debt, allows us to write:

$$\bar{m} + b(t) = \int_t^{\infty} \bar{f}_s e^{-r(v-t)} dv.$$

Because the real stock of debt equals the ratio of the nominal stock to the price index, it follows that:

$$\frac{B(t)}{P(t)} = \int_t^{\infty} \bar{f}_s e^{-r(v-t)} dv - \bar{m}. \quad (10.61)$$

The price level is therefore given by the ratio of the nominal stock of the public debt to the present value of primary surpluses minus the real quantity of money:

$$P(t) = \frac{B(t)}{\int_t^{\infty} \bar{f}_s e^{-r(v-t)} dv - \bar{m}}. \quad (10.62)$$

The mechanism that lies behind the determination of the price level is a wealth effect stemming from the non-Ricardian fiscal policy. When the government sets the primary surplus, the public's wealth increases and this wealth increase causes an increase in the purchase of goods and services. As a consequence of this, spending and price level increases, thus re-establishing the system's logic by making the government debt equal to the present value of primary surplus flows.

10.7 Sustainable Monetary Regime

The Central Bank may choose the monetary regime by setting the foreign exchange rate or the interest rate. Under a fixed exchange rate regime, the Central Bank buys and sells foreign currency at a fixed price. In such a regime, the inflation rate will be equal in the long term to the inflation rate of the country to which the exchange rate is pegged. In a flexible exchange rate regime, the Central Bank controls the interbank market's interest rate. The foreign exchange rate is given by the foreign exchange market. Under such a regime, the Central Bank may freely choose the country's long-run inflation rate.

In either case, the long-run inflation rate – that is, the inflation tax rate – is chosen without taking the government budget constraint into consideration. Are these monetary regimes sustainable in the long-run? To answer this question, one must analyze the government's intertemporal budget constraint.

The government budget constraint may be written in terms of flows as:

$$\dot{b} + \dot{m} = (r - n)(b + m) - f_s - (r + \pi)m. \quad (10.63)$$

The notation is the same that has been used throughout this chapter. Because $a = b + m$ and $s(m) = (r + \pi)m$, the constraint is equivalent to:

$$\dot{a} = (r - n)a - f_s - s(m). \quad (10.64)$$

The government's intertemporal constraint, assuming no Ponzi game, is therefore given by:

$$a(t) = \int_t^\infty [f_s + s(m)]e^{-(r-n)(v-t)} dv. \quad (10.65)$$

In the long-run, whatever the monetary regime, the value of the services of money depends on the inflation rate and is fixed. That is:

$$s(m) = (r + \bar{\pi})\bar{m} = \bar{s}. \quad (10.66)$$

where the long-run inflation rate ($\bar{\pi}$) equals the foreign inflation rate ($\bar{\pi} = \pi^*$) when the exchange rate is fixed, or the implied or explicit inflation target under a flexible exchange rate regime. The nominal interest rate ($r + \bar{\pi}$) determines the real quantity of money ($\bar{m}$) and, therefore, the value of the services of money ($\bar{s}$).

Because $a(t) = b(t) + \bar{m}$, it follows that the government's intertemporal budget constraint may be written as:

$$b(t) = \int_t^\infty f_s e^{-(r-n)(v-t)} dv + \frac{(n + \bar{\pi})\bar{m}}{r - n}. \quad (10.67)$$

We conclude, based on this intertemporal constraint, that the sustainability of any monetary regime – be it fixed or floating exchange rate – depends on the fiscal regime. If the fiscal regime is unsustainable, the monetary regime will also be unsustainable.

10.8 Exercises

1. The government finances its deficit by issuing bonds and money according to:

$$g - \tau + ib = \dot{b} + (n + \pi)b + \mu m,$$

where g , τ , b , and m are spending, taxes, public debt, and the monetary base as shares of nominal output, i is the nominal interest rate, n is the growth rate of real output, π is the inflation rate, μ is the growth rate of the monetary base, and $\dot{b} = db/dt$. Assume that the fiscal policy follows the rule:

$$f = g - \tau + ib = a + \alpha b > 0.$$

The letters a and α are parameters and $0 \leq \alpha \leq r$. Analyze the sustainability of the public debt under this fiscal policy rule.

2. Consider the model:

Bonds and money financed public deficit: $g - \tau + rb = \dot{b} + \mu m$,

Tax dependent on public debt: $\tau = \tau(b)$, $\tau' > 0$,

Money demand: $m = f(i)$, $f' < 0$,

Monetary policy rule: $\dot{m} = m(\mu - \pi)$ constant

Constant real interest rate.

Analyze the model's equilibrium and dynamics.

3. Consider the model:

Money demand: $m = \alpha - \beta\pi^e$, $\alpha > 0$, $\beta > 0$, $m \leq \bar{m}$,

Money financed public deficit: $f = \frac{dM}{dt} \frac{1}{P}$, $f = \text{constant}$,

Expected inflation rate: $\dot{\pi}^e = \theta(\pi - \pi^e)$.

The symbols have the following meanings: m is the real quantity of money ($m = M/P$), π^e is the expected inflation rate, M is the nominal stock of money, P is the price index, and π is the inflation rate.

(a) Analyze the model's dynamics and equilibrium.

(b) Repeat item (a) for $\theta \rightarrow \infty$.

(c) What inflation rate maximizes the inflation tax revenue?

4. Assume an economy described by:

$$\dot{\pi} = F(\pi, m, \alpha),$$

where $\pi = d \log P/dt$ is the inflation rate, P is the price index, m is the real liquidity level ($m = M/P$, where M is the money stock), and α is a parameter vector for the economy.

Assume two economic policy regimes:

MPR (Monetary regime, active monetary policy): $\dot{m} = m(\mu - \pi)$.

MPR (Fiscal regime, passive monetary policy): $\dot{m} = f - m\pi$.

The symbols have the following meanings: μ is the expansion rate of the money stock ($\mu = d \log M/dt$) and f is the real public deficit.

- (a) Specify an F function for the $\dot{\pi}$ equation, indicating the assumptions you made for your specification.
- (b) Analyze the economy's dynamics for each economic policy regime.
- (c) What happens when the real deficit (f) decreases?
- (d) What happens when the monetary expansion rate (μ) decreases?

5. Consider the model:

$$\text{Aggregate demand: } y = k + \alpha(m - p) + \beta\pi,$$

$$\text{PC: } \pi = \pi^e + \delta(y - \bar{y}),$$

$$\text{Expectation: } \pi^e = \mu,$$

$$\text{MPR: } \frac{\dot{M}}{M} = \mu = \text{constant}.$$

The symbols have the following meanings: y = log of real output; m = log of the nominal stock of money; p = log of the price index; $\dot{p} = \pi$.

- (a) In this economy, can the output differ from full-employment output ($\bar{y}$)?
- (b) Might one say that the model's stability is independent from the values of the parameters?
- (c) Would you be able to solve this model when the monetary policy rule is replaced by the following rule: $\frac{\dot{M}}{P} = f = \text{constant}$ and answer the previous item's question?

6. Consider the following government budget constraint:

$$f + rb = \dot{b} + \frac{\dot{M}}{P}$$

Suppose that the government adopts the following monetary/fiscal policy rule:

$$\frac{\dot{M}}{P} = \alpha f.$$

- (a) What value must parameter α have for the public debt to be sustainable?
- (b) Suppose that a financial innovation occurs that shifts the money demand curve. Might this economy show hyperinflation?

7. The government budget constraint is:

$$G_t + i_{t-1}B_{t-1} = T_t + B_t - B_{t-1}.$$

$$\text{Define: } d_t = \frac{d_{t-1}}{1 + r_{t-1}} = 1, 2, \dots$$

$$d_o = 1$$

(a) Show that: $\sum_{t=1}^{\infty} d_t T_t = \sum_{t=1}^{\infty} d_t G_t + B_0 - \lim_{T \rightarrow \infty} d_T B_T$.

(b) If $\lim_{T \rightarrow \infty} d_T B_T > 0$, would you buy government bonds?

The symbols have the following meanings: B = public debt; G = government spending; T = taxes; i = interest rate.

8. The government finances public debt by printing money according to:

$$G_t - T_t = M_t - M_{t-1}.$$

The symbols have the following meanings: G = (nominal) government spending, T = (nominal) tax revenue, M_t = monetary base stock.

How would you measure the model's inflation tax with discrete variables?

9. The balance of the current account on the balance of payments ($\dot{B}$) is given by the following expression:

$$\dot{B} = -TB + i^* B.$$

TB is the balance of exports and imports of goods and services, r^* is the interest rate on foreign debt B (denominated in a foreign currency). Divide both sides of this expression by the gross domestic product Y (in the foreign currency)

of the internationally traded goods and services. That is: $\frac{\dot{B}}{Y} = -\frac{TB}{Y} + i^* \frac{B}{Y}$.

Analyze the sustainability of this country's foreign debt.

10. Assume that the government defaults on its public debt. Assume, also, that the Ricardian equivalence theorem is valid. What are the consequences of such a default?

11. A political party's government program lists the following objectives:

- (a) To keep the tax burden stable;
- (b) To increase the ratio of government spending to output;
- (c) To reduce the debt service;
- (d) To keep the inflation rate stable.

Analyze the program for consistency.

12. The government budget constraint is given by:

$$\frac{G - T}{Y} + i \frac{B}{Y} \equiv \frac{\dot{B}}{Y} + \frac{\dot{M}}{Y}.$$

The symbols have the following meanings: G = government spending; T = tax revenue; Y = nominal output; i = nominal interest rate; B = public debt stock; M = monetary base stock; $\dot{B} = dB/dt$; $\dot{M} = dM/dt$. Assume that the growth of the monetary base can be disregarded ($\dot{M} = 0$).

- (a) Suppose that the government has a nominal deficit of 3% as an objective. What would be the value of the primary surplus needed to achieve this objective?
- (b) Suppose that the government sets its primary surplus according to the following rule:
 $\frac{T-G}{Y} = \alpha b$. The real interest rate is $r = i - \pi$ and α is a positive coefficient. What constraint must coefficient α satisfy for the public debt to be sustainable?
- (c) Suppose that the primary surplus rule was: $\frac{T-G}{Y} = \alpha b$. Is the public debt sustainable?
13. Consider the model (Perverse Monetarist Arithmetic):
 Bonds and money financed public deficit: $\dot{b} = f + rb - \mu m$,
 Monetary policy: $\dot{m} = m(\mu - \pi)$, $\mu = \text{constant}$,
 Money demand: $m = m(i)$, $m' < 0$,
 Fisher equation: $i = r + \pi$.
 Assume a constant public deficit f .
- (a) Analyze the model's dynamics and equilibrium on a phase diagram with b on the vertical axis and m on the horizontal axis.
- (b) Assume that the Central Bank decides to reduce the monetary expansion rate, but that there is a public debt ceiling equal to $b(T)$. What happens to the inflation rate at the time of the monetary policy change?
14. Consider the limit,

$$\lim_{T \rightarrow \infty} m(T)e^{-r(T-t)} = Ce^{\rho t},$$

where: $C = \lim_{T \rightarrow \infty} m(T)e^{-rT}$. When $C \neq 0$, what happens to the solution of differential equation:

$$\dot{m} = rm + f - s(m).$$

15. Take the derivative of $b(t)$ with respect to time,

$$b(t) = \int_t^\infty f_s e^{-(r-n)(v-t)} dv,$$

to obtain the differential equation:

$$\dot{b} = (r - n)b - f_s.$$

Solving this exercise requires application of the Leibnitz rule of the derivative of $V(r)$ with respect to r :

$$V(r) = \int_{\alpha(r)}^{\beta(r)} f(x, r) dx.$$

The Leibnitz rule is as follows:

$$\frac{dV(r)}{dr} = f(\beta(r), r) \frac{d\beta(r)}{dr} - f(\alpha(r), r) \frac{d\alpha(r)}{dr} + \int_{\alpha(r)}^{\beta(r)} \frac{\partial f(x, r)}{\partial r} dx.$$

16. The derivative of $m(t)$ with respect to time:

$$m(t) = \int_t^{\infty} [s(m) - f] e^{-r(v-t)} dv,$$

is equal to the differential equation:

$$\dot{m} = rm + f - s(m).$$

17. The derivative of $a(t)$ with respect to time:

$$a(t) = \int_t^{\infty} [\tau + s(m) - g] e^{-r(v-t)} dv,$$

is equal to the differential equation:

$$\dot{a} = ra + g - \tau - s(m).$$

18. Given the budget constraint:

$$-f_s + (r - n)b - (\pi + n)m = \dot{b} + \dot{m}.$$

(a) Show that it may be written as:

$$a(t) = \int_t^{\infty} [f_s + s(m)] e^{-(r-n)(v-t)} dv,$$

where $a = b + m$.

(b) Show that:

$$\begin{aligned} \int_t^{\infty} s(m) e^{-(r-n)(v-t)} dv - m(t) &= \int_t^{\infty} (r - n)[m(v) - m(t)] e^{-(r-n)(v-t)} dv + \\ &\quad \int_t^{\infty} (\pi + n)m(v) e^{-(r-n)(v-t)} dv \end{aligned}$$

(c) How do you interpret the two components on the right side of the above equation?

19. The public deficit is financed with Treasury sales of bonds denominated in local and in foreign currency, according to:

$$G - T + iB + i^*SB^* \equiv \dot{B} + \dot{SB}^*,$$

where G = government spending, T = taxes, i = nominal interest rate, B = stock of local currency denominated bonds, S = exchange rate, B^* = stock of foreign currency denominated bonds. $\dot{B} = dB/dt$; $\dot{B}^* = dB^*/dt$. Let Y be the economy's nominal output: $Y = P_y$, P is the price index and y is real output. Define: $f = (G - T)/T$; $b = B/Y$, $b^* = SB^*/Y$.

- (a) Show that the government budget constraint may be written as:

$$f + (i - \pi - n)b + (i^* - n - \pi + \dot{s})b^* = \dot{b} + \dot{b}^*,$$

where $\pi = \dot{P}/P$, $\dot{s} = \dot{s}/s$, $n = \dot{y}/y$.

- (b) Assuming that the uncovered interest rate arbitrage, $\dot{q} = r - r^*$, show that the budget constraint transforms into:

$$f + (r - n)(b + b^*) = \dot{b} + \dot{b}^*.$$

20. The European Union's Maastricht Treaty sets forth that, in order to adopt the Euro, countries had to meet the following conditions: (i) public debt cannot exceed 60% of GDP and (ii) public deficit cannot exceed 3% of GDP.

- (a) Show that for the debt-to-GDP ratio to remain stable, $\dot{b} = 0$, the following equation must be satisfied:

$$f = b(n + \pi).$$

- (b) For $b = 60\%$, $n = 3\%$, and $\pi = 2\%$, does the 3% fiscal deficit keep the debt-to-GDP ratio stable?

21. Consider the model:

Production function: $y = f(k)$,

Government budget constraint: $g + rb = \tau(y + rb)$,

Gross interest rate: $r = f'(k)$,

Net interest rate: $\rho = (1 - \tau)r$.

- (a) Show that:

$$\frac{dk}{db} = \left[\frac{\tau}{1 - \tau} + \frac{ff''}{(f')^2} \right]^{-1}.$$

- (b) Assume that $y = k^\alpha$, $\tau = 1/3$, and $\alpha = 1/3$. Show that: $\frac{dk}{db} = -\frac{2}{3}$.

Chapter 11

Monetary Theory and Policy



This chapter introduces several monetary theory and policy issues. Section 11.1 covers the determination of the price of money as a financial asset, with its two components of fundamentals and bubbles; shows the possibility of multiple equilibria in a monetary economy, where money has no value at one equilibrium; and addresses the indeterminacy of the price of money when the Central Bank pegs the nominal interest rate. Section 11.2 presents the optimum quantity of money in a flexible price economy. Section 11.3 analyzes the liquidity trap in its modern version, with a zero-lower bound nominal interest rate. Section 11.4 deals with the problem of dynamic inconsistency in the presence of incentives for decisions made in the present for the future not to be carried out. Section 11.5 covers the smoothing of the interest rate by central bankers who prefer not to change the interest rate abruptly, but gradually, producing some inertia in the behavior of the interbank interest rate. Section 11.6 covers inflation targeting, a system adopted by several Central Banks around the world since the 1990s. Section 11.7 analyzes the operational procedures of monetary policy in the bank reserves market, where the Central Bank plays a dominant role. Section 11.8 shows how the term structure of interest rates can be introduced in short-term macroeconomic models. This framework allows for the analysis of the effect of Central Bank announcements of the interest rate on the output gap and the inflation rate.

11.1 Price of Money

The price of money equals the quantity of goods and services that can be bought with one monetary unit. That is, the price of money (q) equals the inverse of the price level (P):

$$q = \frac{1}{P} = \frac{m}{M}. \quad (11.1)$$

The term after the second equality sign shows that the price of money equals the ratio of the real quantity (m) to the nominal stock (M) of money. The traditional view of the value of money involves precisely the analysis of the variables affecting the real and nominal stocks of money. Individuals decide on the real quantity of money they wish to hold in their portfolio and the Central Bank has instruments that enable it to control the nominal stock.

11.1.1 Bubbles $\times$ Fundamentals

A central problem in monetary theory is the microfoundations of the money demand equation. The literature has used four approaches to this end: (i) the money in the utility function (MIU); (ii) the cash in advance constraint (CIA); (iii) transaction costs (TC); and (iv) the overlapping generations models (OLG). The first three were introduced in Chap. 6 and the last one was abandoned because it fails to take into account the main attribute of money: its role as a means of payment. Any of the above approaches produces an equation for the real quantity of money demanded,

$$i = i(m), i' < 0. \quad (11.2)$$

The nominal interest rate i is the opportunity cost of holding money. The money demand equation also depends on a scale variable, such as income or consumption. The analysis next assumes that this variable remains constant.

The liquidity services of money can be measured as the opportunity cost of resources used in money form. That is:

$$s = im = s(m). \quad (11.3)$$

The value of the services corresponds to the hashed area in Fig. 9.1. The services of money per money unit can be obtained by dividing the services flow (s) by the nominal stock of money: $s(m)/M$ (Fig. 11.1).

Examples of the $s(m)$ function correspond to the specifications of the money demand equation in its semi-log and double-log functional forms. In the semi-log form $\log m = -\alpha i$, $\alpha > 0$, $s(m) = -(m \log m)/\alpha$, $s'(m) = 0$ when $m = \exp(-1)$ and $s''(m) < 0$. The double-log functional form is a particular case of $\log m = -\epsilon \log(\beta + i)$, $\epsilon > 0$, when $\beta = 0$. The services of money function in this case is $s(m) = m(m^{-1/\epsilon} - \beta)$, $s'(m) < 0$ and $s''(m) \geq 0$ when $\epsilon \leq 1$. The graphs in Fig. 11.2 correspond to the particular cases of the semi-log and double-log equations where $\epsilon = 1$. In Fig. 11.2a the elasticity of the real quantity of money demanded with respect to the interest rate ranges from zero to minus infinity. In Fig. 11.2b the elasticity is below one in absolute values. Money is defined as essential in this case.

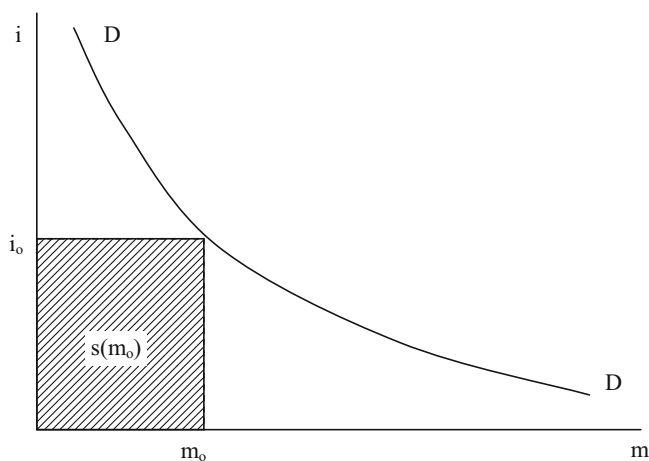


Fig. 11.1 The demand for money and the value of the services of money

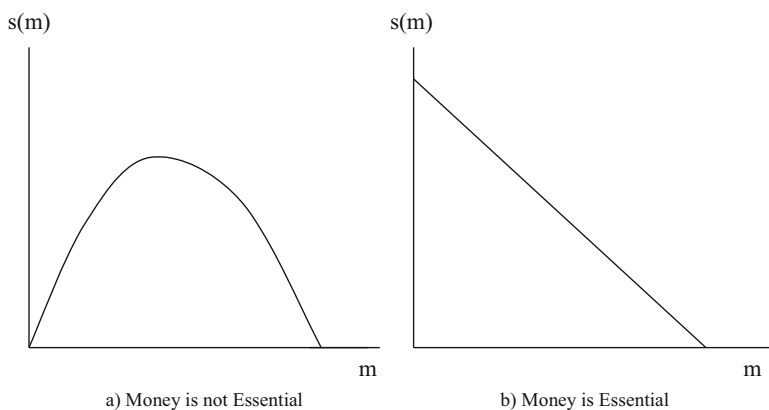


Fig. 11.2 (a) Money is not essential. (b) Money is essential.

Money is essential when it cannot be easily substituted by other financial assets in its function as a means of payment. Money is a social convention that depends both on the legal frame work that establishes the legal conditions for the financial settlement of contracts and on the degree to which society is forced to comply with the law in force. Therefore, the essentiality of money is determined not only by the technology of economic transactions but also by each country's institutions.

The equilibrium price of any asset, in the presence of certainty and perfect capital markets, is such that its rate of return equals the interest rate. Otherwise, an opportunity for arbitrage exists. Therefore, if r is the real interest rate, the price of money satisfies the following arbitrage condition:

$$r = \frac{\dot{q} + \frac{s(m)}{M}}{q}, \quad (11.4)$$

where $\dot{q} = dq/dt$ is the capital gain (loss) when the price of money increases (decreases).

To deduce this equation, just combine $s(m) = im$ and the Fisher equation $i = r + \pi = r - \dot{q}/q$, that is, $s(m) = (r - \dot{q}/q)Mq$, taking into account the fact that $m = Mq$. The arbitrage equation is:

$$\dot{q} = rq - s(Mq)/M.$$

Solving this differential equation requires specifying the function that represents the services of money $s(m)$ and the process that generates the nominal stock (M) of money. However, whatever the solution, it has two components: one for fundamentals and another for the bubble:

$$q(t) = q_f(t) + q_b(t). \quad (11.5)$$

The bubble solution q_b is:

$$q_b(t) = Ce^{rt}. \quad (11.6)$$

It is straightforward to show that this expression is a solution of the arbitrage equation. Indeed, taking the derivative of both sides of it, while bearing in mind the two components of the solution and substituting in the arbitrage equation, yields:

$$\dot{q}_f + rCe^{rt} = rq_f + rCe^{rt} - s(m)/M.$$

Simplifying:

$$\dot{q}_f = rq_f - s(m)/M. \quad (11.7)$$

Therefore, the solution of the arbitrage equation has two components. The constant C in the bubble solution may be negative because in the case of fiat money, there is nothing to prevent its price from being equal to zero, unlike a real asset for which no bubble could bring the price down to zero. The fundamentals solution (q_f) equals the present value of the flows of the services of money. That is:

$$q_f(t) = \int_t^\infty \frac{s(m)}{M} e^{-r(v-t)} dv. \quad (11.8)$$

The statement that money is a financial asset used as a means of payment and that its price equals the present value of the flows of its services must be qualified. It is based on the following theorem: the price of money is given by the present value of the flows of its services if, and only if, $i = i(m)$ and $i = r + \pi$, where $i(\cdot)$ is the inverse function of the money demand equation and $i = r + \pi$ is the Fisher equation. Proof

that this condition is sufficient has already been made for the foregoing expression. To show that this condition is necessary, all one has to do is take the derivative of the same expression with respect to time and use the definition of the $s(m)$ function.

11.1.2 Multiple Equilibria

To analyze the existence of multiple equilibria, assume that the nominal stock of money is constant and equal to one: $M = 1$. This simplifies the problem without any loss of generality. The money arbitrage equation then becomes:

$$\dot{q} = rq - s(q). \quad (11.9)$$

The phase diagram in Fig. 11.3, with the price of money on the horizontal axis and its derivative with respect to time on the vertical axis, shows that this differential equation may have two stationary equilibria or a single one, depending on the limit of the $s(q)$ function, when the price of money approaches zero,

$$\lim_{q \rightarrow 0_+} s(q) \begin{cases} = 0 \\ > 0 \end{cases}. \quad (11.10)$$

When money is essential, this limit is positive and a single equilibrium exists. When money is not essential, the limit equals zero and two equilibria exist. At one, money has no value and hyperinflationary paths exist that bring the value of money down to zero. The paths leading to the equilibrium where the price of money equals zero are called bubbles. They do not, however, correspond to proper bubble solutions because the value of money at equilibrium is zero because the flow of the services of money is zero in this case.

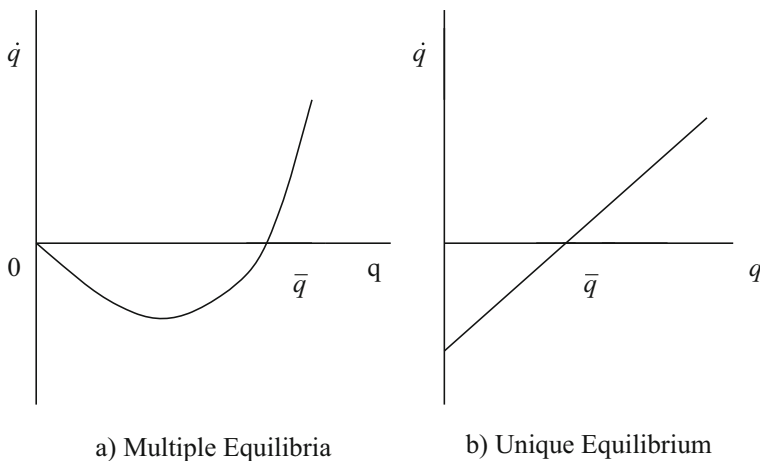


Fig. 11.3 (a) Multiple equilibria. (b) Unique equilibrium.

How does one empirically test which of the two situations is of relevance in practice? The answer is simple because if money is essential, the elasticity of the real quantity of money with respect to the inflation rate will be equal to or less than one in absolute values. This will not be the case when money is non-essential. Therefore, in principle, data from hyperinflation experiences can be used to test this hypothesis.

11.1.3 Indeterminacy

The price of money is indeterminate when the Central Bank sets the nominal interest rate $i = \bar{i}$. Under these circumstances, the flow of services is constant, $s(m) = \bar{s}$, but the nominal quantity of money is undetermined. Therefore, the differential equation of arbitrage is:

$$\dot{q} = rq - \frac{\bar{s}}{M}. \quad (11.11)$$

The phase diagram in Fig. 11.4 shows that for each value of M , an equilibrium value exists for the price of money. Indeterminacy can be solved if the Central Bank changes the interest-rate setting rule to one of the following types:

$$i = \bar{i} + \alpha(\bar{q} - q), \quad \alpha > 0,$$

where $\bar{q}$ is the Central Bank's target price of money. At equilibrium, when $i = \bar{i}$, $q = \bar{q}$, the price is determined.

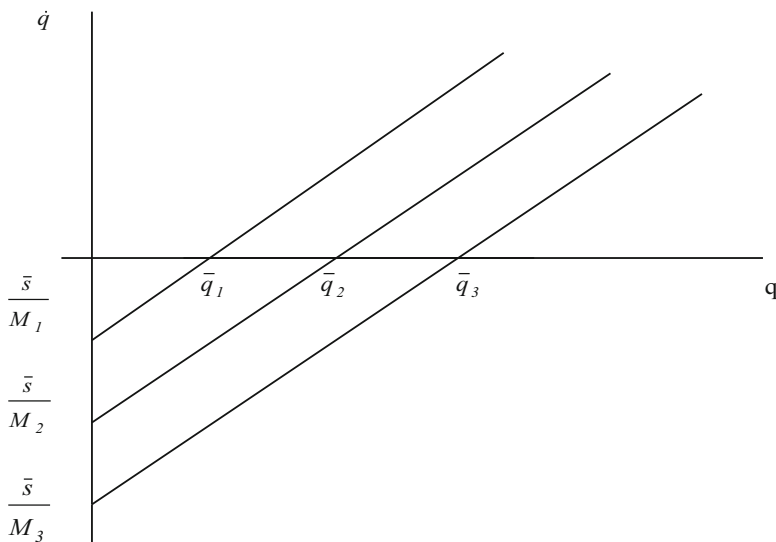


Fig. 11.4 The phase diagram for the price of money

11.2 Optimum Quantity of Money

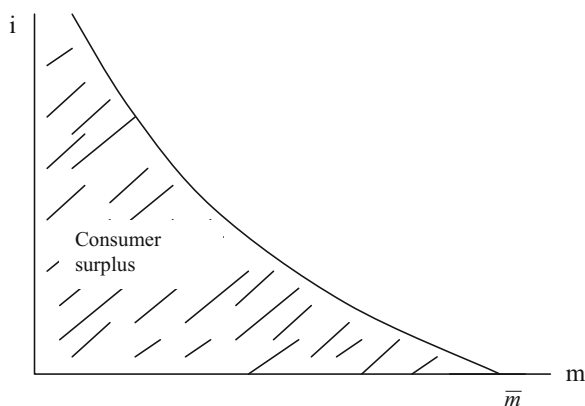
The private sector's opportunity cost of holding money is the nominal interest rate. In a fiat-money economy, the social cost of money production is zero. Figure 11.5 shows that the consumer surplus is maximum when the nominal interest rate is zero and the real quantity of money demanded equals $\bar{m}$. The value of the services of money is zero in this case: $s(\bar{m}) = 0$. When the value of the services of money equals zero, the arbitrage differential equation is:

$$\dot{q} = rq. \quad (11.12)$$

The price of money grows at a rate equal to the real interest rate. According to the fundamentals, the price of money equals zero. The price of money, however, is given by the bubble component. This solution may appear strange, but it is not a typical bubble solution for intertemporal optimization models because of the presence of a finite real quantity of money when the nominal interest rate equals zero. Furthermore, the more appropriate interpretation in this case would not be by means of a bubble, but the price of an asset that does not produce a cash flow, such as gold, whose price at equilibrium, in a world with no uncertainty and with perfect markets, should increase at a rate equal to the real interest rate. The analogy in this case is with the familiar rule for non-renewable natural resources. According to this rule, the shadow price of a non-renewable resource must rise at a rate equal to the interest rate. When the marginal cost of extraction of the non-renewable resource is zero, the price of this asset will rise at a rate equal to the interest rate.

The Friedman rule of a nominal interest rate equal to zero, which determines the optimum quantity of money, suffers from the indeterminacy problem analyzed in the previous section if the Central Bank decides to implement it by setting the interest rate. At the initial instant, the price of money is undefined because the present value of the flows of the services of money is zero. This condition cannot therefore be used to calculate the initial price of money. The Central Bank may thus choose any initial

Fig. 11.5 The optimum quantity of money



value, according to the modified interest rate rule as a function of the gap of the purchasing power of money. This means that there are an infinite number of paths for the price of money.

The price of money must rise at a rate equal to the real interest rate, whatever its initial value. Instead of setting the interest rate, the Central Bank may implement the optimum monetary policy by setting the path of the quantity of money, with its nominal stock dropping at a rate equal to the real interest rate. The two forms of implementing the optimum policy are equivalent, as long as the initial price chosen for the interest rate rule corresponds to the initial stock of the path of the nominal stock of money.

11.3 Zero Lower Bound Nominal Interest Rate

The lower bound of the nominal interest rate set by the Central Bank is zero because a negative rate makes no sense in the presence of paper money. No one would be willing to invest \$100 to get paid nothing upon redemption. Therefore, if i^* is the interest rate calculated by means of a monetary policy rule, such as the Taylor rule, the Central Bank's monetary policy rule would be as follows:

$$i = \begin{cases} i^*, & \text{if } i^* > 0 \\ 0, & \text{if } i^* \leq 0 \end{cases} \quad (11.13)$$

The interest rate i^* is:

$$i^* = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta(y - \bar{y}). \quad (11.14)$$

When this rate is zero or less, the Central Bank sets the nominal interest rate of the interbank reserves market at zero. This case corresponds to the modern version of the liquidity trap. What would happen to the equilibrium and dynamic of an economy operating under such circumstances? To analyze the problem, assume a model that has a traditional IS curve, a Phillips curve with rigid prices and inflation rate inertia, and a monetary policy rule with zero nominal interest rate. That is:

$$\text{IS: } y - \bar{y} = -\alpha(i - \pi - \bar{r}), \quad (11.15)$$

$$\text{PC: } \dot{\pi} = \delta(y - \bar{y}), \quad (11.16)$$

$$\text{MPR: } i = 0, \quad (11.17)$$

$$\text{IC: Given } p(0) \text{ and } \pi(0). \quad (11.18)$$

Substituting the monetary policy rule for the interest rate in the IS curve yields:

$$y - \bar{y} = \alpha(\bar{r} + \pi). \quad (11.19)$$

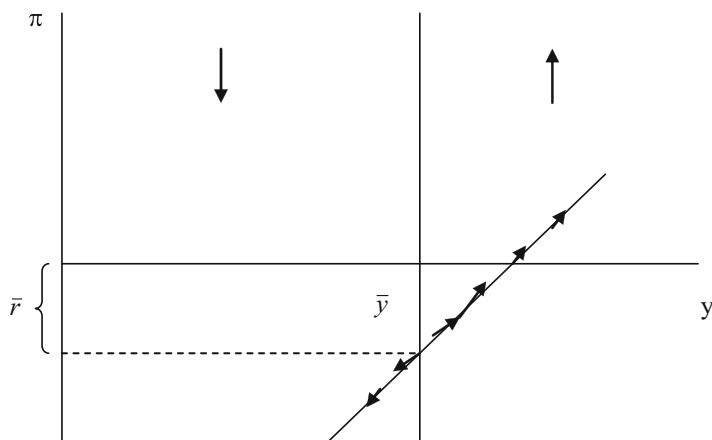


Fig. 11.6 $\delta > 0$

This equation, combined with the Phillips curve, can be analyzed on a phase diagram with the inflation rate on the vertical axis and real output on the horizontal axis, as Fig. 11.6 shows. The equilibrium inflation rate is negative; that is, a deflation equal to the real interest rate. At equilibrium, real output equals the economy's potential output. The model's equilibrium is unstable, as the arrows of the positively sloped line in Fig. 11.6 show. What would happen in this model if the natural interest rate were to change? In this case, because of inflation rate inertia, the economy would enter a recession and not return to full employment if the natural interest rate dropped. In the opposite case, if the natural interest rate were to rise, real output would increase without a mechanism to bring the economy back to full employment.

The model's other possibility is for the Phillips curve's δ parameter to be negative. This occurs when the price level is rigid and the inflation rate is not inertial and may jump instantaneously. The model is stable, as the phase diagram in Fig. 11.7 shows. However, multiple solutions exist. A change in the natural interest rate is consistent with an endless number of solutions because the inflation rate is a jump variable that can change values abruptly.

11.4 Dynamic Inconsistency

Dynamic inconsistency is the phenomenon that occurs when there is no way to force an economic agent to fulfill tomorrow's promises made today. Assume that the central banker announces an inflation target at $\bar{\pi}$ and that their objective function depends on the square of the gap between the observed inflation rate and the inflation rate target, and on the square of the gap between real output and the real output target y^* . That is:

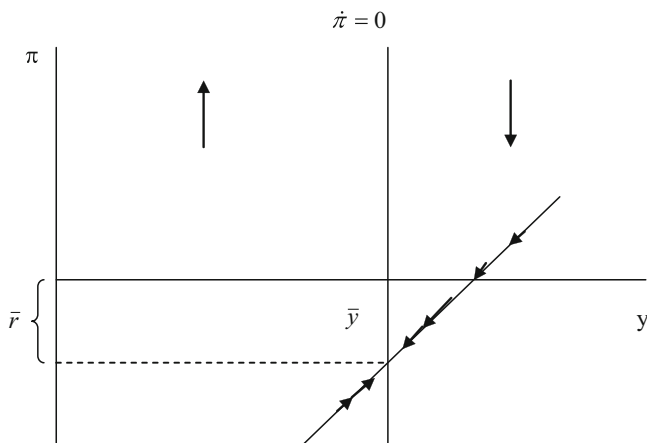


Fig. 11.7 $\delta < 0$

$$L = \frac{\varphi}{2}(\pi - \bar{\pi})^2 + \frac{1}{2}(y - y^*)^2. \quad (11.20)$$

This objective function's coefficient φ indicates the relative weight that the Central Bank assigns to the inflation rate relative to output. When φ equals zero the Central Bank is only concerned with real output. On the other hand, when the parameter tends to infinity, the Central Bank is concerned with inflation only.

The real output target is equal to potential output plus a constant k :

$$y^* = \bar{y} + k, k > 0. \quad (11.21)$$

Substituting this in the Central Bank's objective function – the L loss function – yields:

$$L = \frac{\varphi}{2}(\pi - \bar{\pi})^2 + \frac{1}{2}(y - \bar{y} - k)^2.$$

The Phillips curve will be simplified to avoid including parameters that are irrelevant to the qualitative conclusions of this section. The inflation rate equals the expected inflation rate and the output gap coefficient is equal to unity:

$$\pi = \pi^e + y - \bar{y}. \quad (11.22)$$

Substituting the Phillips curve's output gap in the loss function yields:

$$L = \frac{\varphi}{2}(\pi - \bar{\pi})^2 + \frac{1}{2}(\pi - \pi^e - k)^2. \quad (11.23)$$

Despite announcing an inflation target, the Central Bank chooses the inflation rate by minimizing the loss function with respect to the inflation rate. The partial derivative of L with respect to inflation rate π is:

$$\frac{\partial L}{\partial \pi} = \varphi(\pi - \bar{\pi}) + (\pi - \pi^e - k) = 0.$$

The solution to this first-order condition is the following inflation rate:

$$\pi = \frac{\varphi\bar{\pi} + \pi^e + k}{1 + \varphi}. \quad (11.24)$$

This economy's agents have information on the Central Bank's criteria and rational expectations that, in this case, are equivalent to perfect foresight. Therefore, the expected and observed inflation rates are equal:

$$\pi^e = \pi. \quad (11.25)$$

Substituting the observed for the expected inflation rate in the first-order condition's solution, the inflation rate will be equal to the inflation target promised by the Central Bank plus a positive term that depends on the constant k and the parameter φ of the Central Bank's loss function. That is:

$$\pi = \bar{\pi} + \frac{k}{\varphi}. \quad (11.26)$$

The monetary policy shows dynamic inconsistency because the announced inflation target was not met. How relevant is this model, which is quite popular in the literature, to explaining a central banker's behavior? The crucial assumption for the model's result is that the Central Bank aims to reach a higher target for real output than the potential output. This assumption is very implausible. It would be appropriate in a country under a populist administration that wanted to use the Central Bank to reach unattainable objectives. The notion that the problem of the Central Bank's inflationary bias (k/φ) could be solved by appointing a conservative central banker with a rather high φ parameter that is different to that of the population does not appear appropriate in a democratic society. In a democratic society, institutions that fail to represent the population's preferences are unlikely to survive. Furthermore, a conservative central banker's objective function would not include as an argument a real output target other than the economy's potential output.

11.5 Interest Rate Smoothing

A stylized fact in the behavior of Central Banks is the gradual adjustment of the interest rate. They avoid abrupt changes. The consequence is to make the interest rate less volatile. The literature refers to this phenomenon as interest rate smoothing. The

interest rate then depends on its own past, as described, for example, in the following equation:

$$i_t = \omega i_t^* + (1 - \omega)i_{t-1} + \varepsilon_t, 0 < \omega \leq 1. \quad (11.27)$$

The interest rate in period t is a function of the interest rate in period $t - 1$ and the desired interest rate i^* , ω is the weight of the desired interest rate i^* , which must be specified according to the Central Bank's monetary policy rule, and ε represents stochastic shocks. Some empirical studies report that, for quarterly data, the ω parameter lies between .1 and .2, suggesting a very slow adjustment and highly significant interest rate inertia.

The reasons that dictate this kind of behavior are not yet well known to monetary theory. Two are worth mentioning here: one is a concern with preserving the financial industry's health and the other is to affect the long-term interest rate, which depends on expectations of the Central Bank's future interest rate. Whatever the motivation, interest rate inertia on the part of Central Banks is a well-documented empirical fact. The purpose of this section is to analyze the conditions that the smoothing parameter must satisfy for the monetary policy rule to be stable. Two models will be analyzed: the Keynesian and the new Keynesian models.

11.5.1 Keynesian Model

Interest rate smoothing in the continuous-time model is described by a partial adjustment mechanism where the change in the interest rate is proportional to the gap between desired and current interest rates. That is:

$$\dot{i} = \lambda(i^* - i), \lambda > 0. \quad (11.28)$$

This equation assumes the existence of an adjustment cost that prevents the Central Bank from immediately setting the nominal interest rate at the desired level. When the parameter $\lambda \rightarrow \infty$, the adjustment is instantaneous. Otherwise, the interest rate adjusts gradually to its equilibrium position. The desired interest rate follows the Taylor rule:

$$i^* = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x. \quad (11.29)$$

The model for analyzing the economy's equilibrium conditions with interest rate smoothing has an IS curve and a Phillips curve. The IS curve assumes that the output gap depends on the gap between the short- (r) and long-term ($\bar{r}$) interest rates, according to:

$$x = -\alpha(r - \bar{r}), \alpha > 0, \quad (11.30)$$

where x is the output gap.

The Phillips curve assumes that inflation acceleration is proportional to the output gap:

$$\dot{\pi} = \delta x, \delta > 0. \quad (11.31)$$

The model is thus specified according to:

$$\text{IS: } x = -\alpha(r - \bar{r}), \alpha > 0, \quad (11.32)$$

$$\text{PC: } \dot{\pi} = \delta x, \quad \delta > 0, \quad (11.33)$$

$$\text{MPR: } \dot{i} = \lambda(i^* - i), \quad i^* = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x, \quad (11.34)$$

$$\text{IC: Given } p(0) \text{ and } \pi(0). \quad (11.35)$$

Differentiating the smoothing equation together with the monetary policy rule, and after a little algebra, we get the following differential equation for the real interest rate:

$$\dot{r} = \phi\lambda(\pi - \bar{\pi}) - [(1 + \alpha\theta)\lambda - \alpha\delta](r - \bar{r}).$$

The model's second differential equation is obtained by combining the IS and Phillips curves. That is:

$$\dot{\pi} = -\alpha\delta(r - \bar{r}).$$

Dynamical System

The model has two differential equations: one for the inflation rate and the other for the real interest rate:

$$\begin{bmatrix} \dot{\pi} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} 0 & -\alpha\delta \\ \phi\lambda & -[(1 + \alpha\theta)\lambda - \alpha\delta] \end{bmatrix} \begin{bmatrix} \pi - \bar{\pi} \\ r - \bar{r} \end{bmatrix}. \quad (11.36)$$

The differential equations system has the following Jacobian matrix:

$$J = \begin{bmatrix} \frac{\partial \dot{\pi}}{\partial \pi} & \frac{\partial \dot{\pi}}{\partial r} \\ \frac{\partial \dot{r}}{\partial \pi} & \frac{\partial \dot{r}}{\partial r} \end{bmatrix} = \begin{bmatrix} 0 & -\alpha\delta \\ \phi\lambda & -[(1 + \alpha\theta)\lambda - \alpha\delta] \end{bmatrix}.$$

This matrix's determinant is positive:

$$|J| = \alpha\delta\phi\lambda > 0$$

The trace of the J matrix may be either positive or negative. For the dynamical system to have a stable equilibrium, the trace must be negative. This means that the interest rate adjustment parameter λ must satisfy the restriction:

Table 11.1 Upper bound of the inertial coefficient $\alpha = 1.0$

δ	θ		
	.25	.50	1.00
.25	.82	.85	.88
.50	.67	.72	.78
1.00	.45	.51	.61

Table 11.2 Upper bound of the inertia coefficient $\alpha = 2.0$

δ	θ		
	.25	.50	1.00
.25	.72	.78	.85
.50	.51	.61	.72
1.00	.26	.37	.51

$$\text{tr } J < 0 \text{ if } \lambda > \lambda_- = \frac{\alpha\delta}{1 + \alpha\theta}.$$

We conclude from this restriction that the Central Bank must not be too slow to adjust the interest rate when the desired interest rate changes. The model's stability sets a lower bound for the interest rate adjustment parameter λ , which depends on the IS curve's parameter α , on the Phillips curve's parameter δ , and on the monetary policy rule's parameter θ . The greater the former two, the lesser the interest rate inertia. The greater the response of the monetary policy to the output gap, the lower the lower bound of the interest rate adjustment coefficient.

The presence of a lower bound for the coefficient λ means that the specification of the smoothing equation has an upper bound for the degree of interest rate smoothing by the Central Bank, that is, inertia. The parameters of the economy and of the monetary policy rule impose a restriction on the Central Bank's interest rate smoothing behavior. Tables 11.1 and 11.2 show some upper bound values for the inertia coefficient as a function of the values of the parameters α , δ , and θ .

Table 11.1 assumes that the coefficient α equals one, whereas Table 11.2 calculates the inertia coefficient assuming α equals two. The two values are representative of values used in calibration exercises, as well as being close to the econometric estimates. Tables 11.1 and 11.2 use the same values for the parameters δ and θ , which take on the values of .25, .50, and 1.00. The first rows of Tables 11.1 and 11.2 show that inertia coefficients between .8 and .9, as observed in the behavior of Central Banks, are consistent with some values for the model's parameters.

The phase diagram in Fig. 11.8 shows the model's dynamics, with four regions with different inflation and real interest rates movements, where the economy is not at the long-run equilibrium point E .

Experiment

Figure 11.9 describes the economy's adjustment dynamics when the Central Bank decides to change the inflation target, in a monetary policy experiment where the target is lowered to $\bar{\pi}_1$ from $\bar{\pi}_0$. The inflation rate begins to gradually drop, while the real interest rate rises until it reaches its maximum level at the point where the

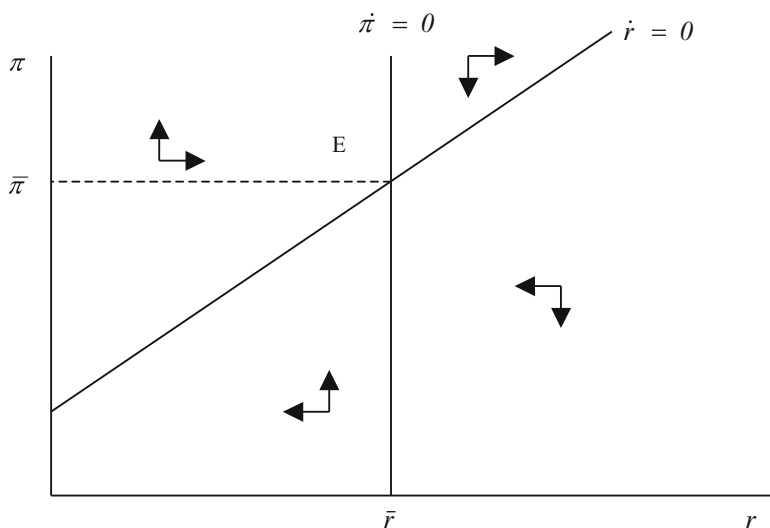


Fig. 11.8 The phase diagram for the inflation rate and the real interest rate

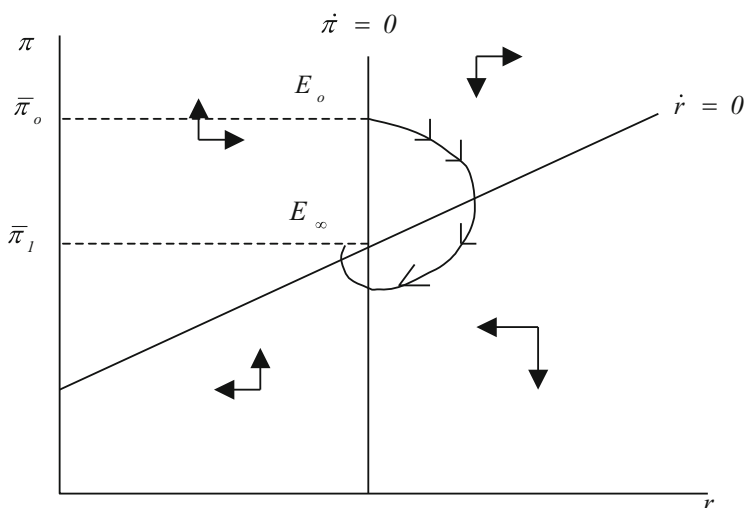


Fig. 11.9 The effects of a permanent decrease in the inflation target

economy's path intercepts the line where $\dot{r} = 0$. The inflation rate continues to drop, reaching a lower value than that of the long-run equilibrium (undershooting) and climbing back up until it converges to the new inflation target. The economy shows a recessive path from the monetary policy change until the inflation rate reaches its lowest values, at which point a period of increased activity begins with real output

exceeding the potential output, because the real interest rate lies below its long-run equilibrium value for a certain period of time.

11.5.2 New Keynesian Model

In the new Keynesian model with interest rate smoothing, the Central Bank does not immediately adjust the interest rate to the desired level, but does so gradually in a smoothing process. The model has a Calvo Phillips curve, the IS curve is an Euler equation, the interest equation assumes a gradual adjustment of the nominal interest rate, and the monetary policy follows a Taylor rule. The model is specified as follows:

$$\text{IS: } \dot{x} = \sigma(i - \pi - \bar{r}),$$

$$\text{PC: } \dot{\pi} = -\delta x,$$

$$\text{SI: } \dot{i} = \lambda(i^* - i),$$

$$\text{MPR: } i^* = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x,$$

$$\text{IC: } p(0) \text{ and } r(0) \text{ given, } \pi(0) \text{ and } x(0) \text{ free.}$$

Algebra

The IS curve may be written as a function of the nominal interest rate and inflation rate gaps, according to:

$$\dot{x} = \sigma(i - \pi + \bar{\pi} - \bar{\pi} - \bar{r}) = \sigma(i - \bar{i}) - \sigma(\pi - \bar{\pi}),$$

where $\bar{i} = \bar{r} + \bar{\pi}$.

Substituting the desired nominal interest rate (i^*) in the nominal interest rate's differential equation yields:

$$\dot{i} = \lambda[\bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x - i].$$

Adding and subtracting the inflation target ($\bar{\pi}$) on the right side of the expression and rearranging its terms we obtain:

$$\dot{i} = \lambda(1 + \phi)(\pi - \bar{\pi}) + \lambda\theta x - \lambda(i - \bar{i}).$$

Dynamical System

The new Keynesian model is a dynamical system of three differential equations: for the inflation rate, the output gap, and the nominal interest rate. That is:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} 0 & -\delta & 0 \\ -\sigma & 0 & \sigma \\ \lambda(1 + \phi) & \lambda\theta & -\lambda \end{bmatrix} \begin{bmatrix} \pi - \bar{\pi} \\ x \\ i - \bar{i} \end{bmatrix}. \quad (11.37)$$

This system's Jacobian matrix equals:

$$J = \begin{bmatrix} 0 & -\delta & 0 \\ -\sigma & 0 & \sigma \\ \lambda(1 + \phi) & \lambda\theta & -\lambda \end{bmatrix}.$$

The matrix's determinant and trace are:

$$\begin{aligned} |J| &= -\delta\sigma\lambda(1 + \phi) + \sigma\delta\lambda = -\lambda\delta\sigma\phi < 0 \\ \text{tr } J &= -\lambda < 0 \end{aligned}$$

This dynamical system may have three negative characteristic roots or a single negative root. To determine which, the following theorem is used: the necessary and sufficient condition for a system of three first-order linear differential equations to have an equilibrium is that $|A| < 0$, $\text{tr } A < 0$ and that the following determinant be negative:

$$\begin{bmatrix} a_{22} + a_{33} & -a_{12} & -a_{13} \\ -a_{21} & a_{11} + a_{33} & -a_{23} \\ -a_{31} & -a_{32} & a_{11} + a_{22} \end{bmatrix} < 0.$$

In the new Keynesian model this determinant is:

$$\begin{vmatrix} -\lambda & \delta & 0 \\ \sigma & -\lambda & -\sigma \\ -\lambda(1 + \phi) & -\lambda\theta & 0 \end{vmatrix} = \delta\sigma\lambda(1 + \phi) + \lambda\sigma\lambda\theta > 0$$

Therefore, the determinant is positive and the three characteristic roots are not negative. The model thus has a saddle point with one negative and two positive roots.

Model Solution

The model's solution is obtained by making the constants of the positive roots equal to zero. That is:

$$\begin{aligned} \pi &= \bar{\pi} + c_1 v_{11} e^{\mu_1 t}, \\ x &= c_1 v_{12} e^{\mu_1 t}, \\ i &= \bar{i} + c_1 v_{13} e^{\mu_1 t}, \end{aligned} \tag{11.38}$$

where $\mu_1 < 0$ is the negative characteristic root, c_1 is a constant to be determined, and the vector $[v_{11} \ v_{12} \ v_{13}]$ is the vector associated with this root, according to:

$$\begin{bmatrix} -\mu & -\delta & 0 \\ -\sigma & -\mu & \gamma \\ \lambda(1 + \phi) & \lambda\theta & -\lambda - \mu \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \\ v_{13} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

It follows that:

$$\begin{aligned} -\mu v_{11} - \delta v_{12} &= 0, \\ -\sigma v_{11} - \mu v_{12} + \gamma v_{13} &= 0, \\ \lambda(I + \phi)v_{11} + \lambda\theta v_{12} - (\lambda + \mu)v_{13} &= 0. \end{aligned}$$

One of the characteristic vector's components can be normalized. For convenience, let $v_{13} = 1$. The ratio of the former to the second component of the characteristic vector is positive because the μ root is negative:

$$\frac{v_{11}}{v_{12}} = -\frac{\delta}{\mu} > 0.$$

The two components of vector v may be determined by the first two equations from the previous system:

$$\begin{bmatrix} \mu & \delta \\ \sigma & \mu \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ \delta \end{bmatrix}.$$

This system's solution is:

$$v_{11} = -\frac{\delta\sigma}{\mu^2 - \sigma\delta} < 0; \quad v_{12} = -\frac{\mu\sigma}{\mu^2 - \sigma\delta} < 0.$$

To check the sign of each of these elements, we will use the differential equations system's characteristic equation:

$$\begin{vmatrix} -\mu & -\delta & 0 \\ -\sigma & -\mu & \sigma \\ \lambda(1 + \phi) & \lambda\theta & -(\lambda + \mu) \end{vmatrix} = 0.$$

The characteristic equation is:

$$-\mu(\mu^2 - \sigma\delta) = \delta\sigma\lambda\phi + \lambda\mu^2 - \mu\sigma\lambda\theta.$$

Since $\mu < 0$, it follows that:

$$-\mu(\mu^2 - \sigma\delta) > 0.$$

Therefore $\mu^2 - \sigma\delta > 0$, and the characteristic vector's two elements are negative: $v_{11} < 0$ and $v_{12} < 0$.

Prediction

Dividing the inflation gap equation by the output gap equation yields:

$$\frac{\pi - \bar{\pi}}{x} = \frac{c_1 v_{11} e^{\mu_1 t}}{c_1 v_{12} e^{\mu_2 t}} = \frac{v_{11}}{v_{12}} > 0 \quad (11.39)$$

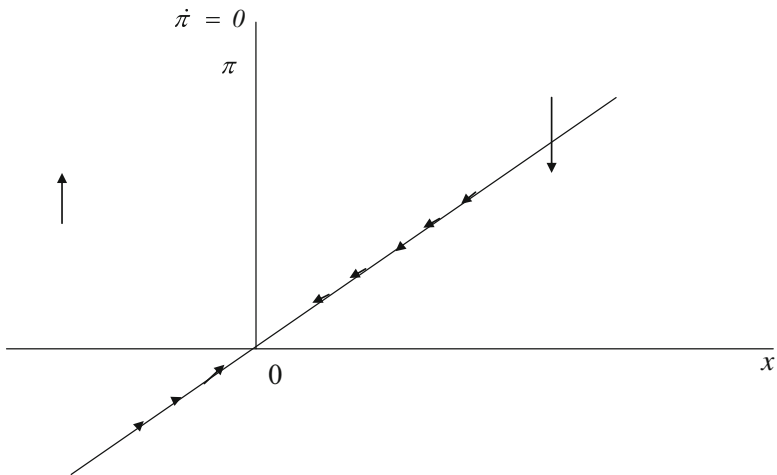


Fig. 11.10 The dynamics of the inflation rate and the output gap

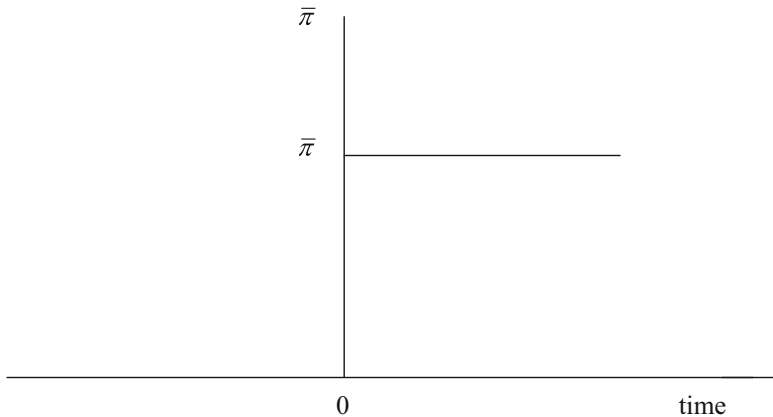


Fig. 11.11 An unanticipated permanent increase in the inflation target

The conclusion is that the inflation and output gaps converge in the same direction to stationary equilibrium. Figure 11.10 illustrates this. The figure’s arrows stem from the model’s Phillips curve. Figure 11.10 assumes that the inflation target equals zero ($\bar{\pi} = 0$).

Experiment

Figure 11.11 describes an experiment in which the inflation target increases to $\bar{\pi}$ from zero in a permanent unanticipated change.

Figure 11.12 shows that the inflation jumps at the initial moment, overshooting its equilibrium value. The output gap increases to $x(0)$ from zero. How do we determine

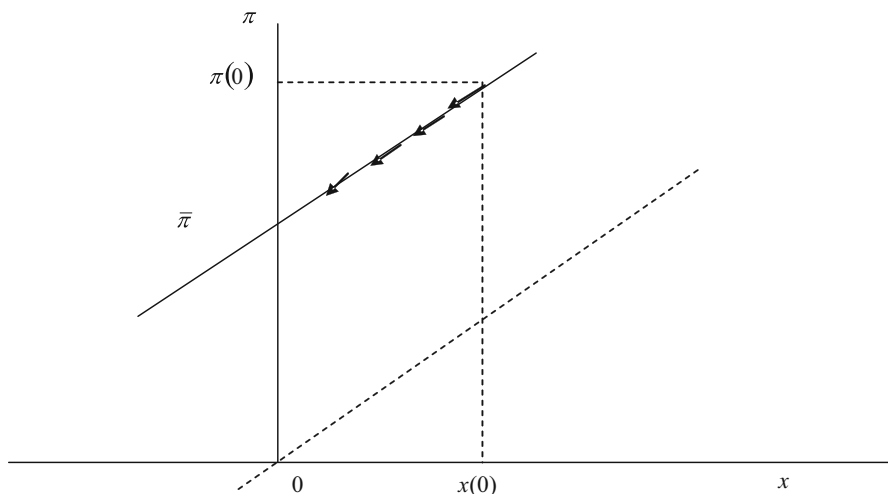


Fig. 11.12 The effects of an unanticipated permanent increase in the inflation target

the values of $\pi(0)$ and $x(0)$ in this model? Firstly, one must determine the value of the c_1 constant that appears in the model's solution. When $t = 0$, $i = i(0)$. It follows that:

$$i(0) - \bar{i} = c_1.$$

Therefore, the initial values of inflation and of the output gap are:

$$\begin{aligned}\pi(0) &= \bar{\pi} + v_{11}[i(0) - \bar{i}], \\ x(0) &= v_{12}[i(0) - \bar{i}].\end{aligned}$$

Raising the inflation target causes the long-run interest rate ($\bar{i}$) to increase. Because the nominal interest rate is a predetermined variable, $i(0) - \bar{i} < 0$. Therefore, since v_{11} and v_{12} are both negative, it follows that the initial inflation rate is greater than the new target rate and the output gap at the initial moment becomes positive, as Fig. 11.12 shows.

11.6 Inflation Targeting

In the latter half of the 1960s, theoretical arguments showed that there was no long-run tradeoff between inflation and unemployment. The long-run Phillips curve is vertical. Subsequent empirical evidence did not reject this hypothesis. The practical consequence of this piece of evidence was that central bankers became convinced that the monetary policy is responsible for the inflation trend, that is, the Central Bank does not control the inflation rate at each point in time, but rather its trend.

A practical issue that emerged among central bankers was the means of implementing the monetary policy to reach a certain inflation rate. In the early 1990s, the Central Bank of New Zealand introduced inflation targeting, an innovation that was eventually copied by several other Central Banks. Under such a program, the Central Bank announces the inflation target for a certain horizon and calibrates the interbank reserves market's interest rate to hit this target. The predicted inflation rate for the specified horizon becomes a crucial variable in this process. The Central Bank raises the interest rate when the predicted inflation rate is above target and lowers the interbank reserves market's interest rate when the predicted inflation rate is below target. The proposition may be deduced from a rather simple model that includes a two-period lag for the interest rate to affect the inflation rate.

The model has three equations: an IS curve, a Phillips curve, and a Central Bank loss function. The equations are:

$$y_{t+1} - \bar{y} = \lambda(y_t - \bar{y}) - \alpha(i_t - \pi_t - \bar{r}) + \varepsilon_{t+1}, \quad (11.40)$$

$$\pi_{t+1} = \pi_t + \delta(y_t - \bar{y}) + u_{t+1}, \quad (11.41)$$

$$L = \beta^j \frac{1}{2} (\pi_{t+j} - \bar{\pi})^2. \quad (11.42)$$

In this model, the output gap has a lagged effect on the inflation rate, and the interest rate has a lagged effect on the output gap. Therefore, the interest rate affects the inflation rate after two periods. For a model using annual data, this means that it takes the Central Bank 2 years to reach the inflation rate. The symbols u and ε are the shocks to the Phillips and IS curves. These shocks are zero-average, have constant variance, and have no serial correlation. The β coefficient is the Central Bank's discount rate.

The inflation rate two periods ahead is:

$$\pi_{t+2} = (1 + \alpha\delta)\pi_t + \delta(1 + \lambda)(y_t - \bar{y}) - \alpha\delta(i_t - \bar{r}) + u_{t+1} + \delta\varepsilon_{t+1} + u_{t+2}.$$

The Central Bank's problem is to choose a nominal interest rate such that the expected value of the loss 2 years from now is as low as possible. That is:

$$\min_{r_t} E_t \frac{\beta^2}{2} (\pi_{t+2} - \bar{\pi})^2. \quad (11.43)$$

This problem's first-order condition establishes that the forecasted inflation two periods from now must be equal to the inflation rate target:

$$E_t \pi_{t+2} = \bar{\pi}. \quad (11.44)$$

The intermediate objective of inflation targeting therefore is the forecast of the inflation rate. This prediction allows for the calibration of the interbank reserves market's nominal interest rate to attain the final objective, that is, to hit the inflation

target. Indeed, the equation for the inflation rate two periods ahead allows for the calculation of the predicted inflation rate:

$$E_t \pi_{t+2} = (1 + \alpha\delta)\pi_t + \delta(1 + \lambda)(y_t - \bar{y}) - \alpha\delta(i_t - \bar{r}) = \bar{\pi}.$$

The nominal interest rate needed to reach the inflation rate target is obtained from this expression by writing the interest rate as a function of the other variables, which are known in period t . That is:

$$i_t = \bar{r} + \pi_t + \frac{1}{\alpha\delta}(\pi_t - \bar{\pi}) + \frac{\delta(1 + \lambda)}{\alpha\delta}(y_t - \bar{y}). \quad (11.45)$$

The inflation rate in period $t+2$ will be equal to its value as predicted in period t , plus the shocks had over the period:

$$\pi_{t+2} = E_t \pi_{t+2} + u_{t+1} + \delta \varepsilon_{t+1} + u_{t+2}. \quad (11.46)$$

11.7 Monetary Policy Operational Procedures

The main monetary policy instrument for the majority of the world's Central Banks is the bank reserves market's interest rate. This rate is called the SELIC rate in Brazil, the FED Funds rate in the United States, the cash rate in New Zealand, the bank rate in the United Kingdom, the call rate in Japan. The Central Bank's monetary policy committee, or its Board of Governors, decides the level of this rate and tells the respective trade desks to implement the decision.

An operating procedure that is becoming commonplace is for the Central Bank to set an interest rate corridor with an upper and a lower limit. If a bank happens to have excess reserves, it may invest its reserves with the Central Bank, receiving a return equal to the lower interest-rate bound. Should the bank lack reserves, it may borrow from the Central Bank, which will charge an interest rate equal to the upper interest-rate bound. Figure 11.13 describes the mechanism. The interest rate i^c is the one set by the Central Bank's monetary policy committee. The interest rate i^s is one that the Central Bank charges from reserves it lends out (marginal lending facility) and the interest rate i^d is the one that the Central Bank pays on commercial banks' reserves (marginal deposit facility). This procedure prevents the bank reserves market's interest rate from overshooting the upper bound or undershooting the lower bound.

The Central Bank's trading desk keeps the bank reserves market's interest rate at the level selected by the monetary policy committee by buying and selling securities, usually under repurchase agreements ("repos"). The model with the two equations below describes the bank reserves market in a stylized manner. The first equation in the model is the trading desk's reaction function. When the interest rate lies below the one set by the committee, the desk raises the interest rate by removing reserves from the market. On the other hand, when the rate lies above the one set by the



Fig. 11.13 The interest rate corridor

monetary policy committee, the desk injects reserves into the market to bring down the rate. The parameter ψ measures the trading desk's reaction speed in connection with the interest rate spread. The second model in the equation is the bank reserves demand equation, where the quantity of reserves demanded increases (decreases) as the interest rate decreases (increases). The interest rate has an upper bound (i^s) and a lower bound (i^i). The model has the following specification:

$$\begin{cases} \dot{i} = \psi(i^c - i), \psi > 0 \\ i = a - bR \\ i^i < i^c < i^s \end{cases} \quad (11.47)$$

Figure 11.14 shows the model's phase diagram, with the interest rate on the vertical axis and the quantity of reserves on the horizontal axis. The diagram shows the interest-rate corridor, with its upper and lower bounds. The model is stable, indicating that, by injecting and removing reserves, the desk can implement the Central Bank's monetary policy committee's decision.

11.8 Term Structure of Interest Rates

The term structure of interest rates is the relationship between short- and long-term rates. Central Banks control the short-term interest rate in the interbank market, while long-run rates are determined in the financial market. Figure 11.15 shows three types of yield curves that have been observed in practice. The horizontal axis measures a security's maturity and the vertical axis measures the corresponding

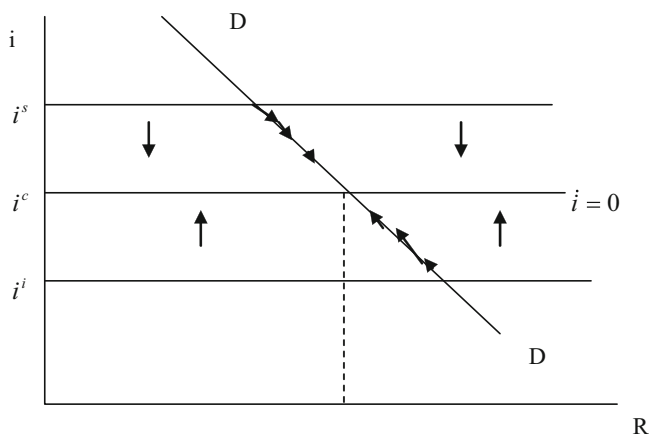


Fig. 11.14 The dynamics of the interest rate and the quantity of reserves

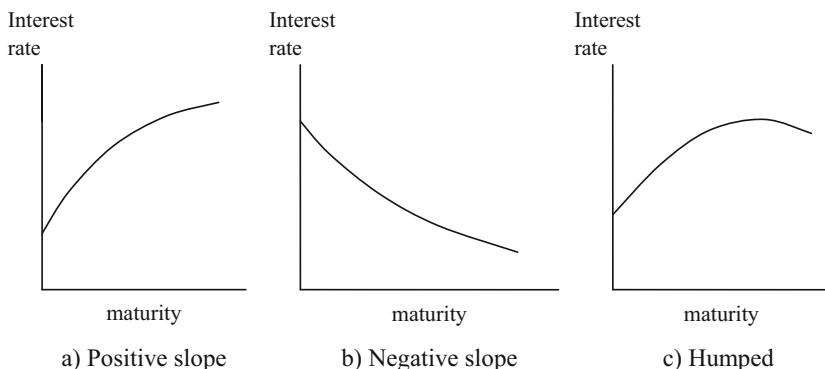


Fig. 11.15 The term structure of interest rates

interest rate. The yield curve in Fig. 11.15a has a positive slope, that is, the long-run rate is higher than the short-run rate. The yield curve in Fig. 11.15b has a negative slope, that is, the long-run rate is lower than the short-run rate. The yield curve in Fig. 11.15c is humped. What kind of theoretical framework might explain these facts? This section presents the expectations hypothesis of the term structure of interest rates.

Assume that someone wants to invest their funds for two periods, from period t to period $t+2$, as Fig. 11.16 shows. They may invest in period t , buying a security maturing in one period and upon maturity reinvesting the principal and interest in another one-period security. The value of the investment at the end of the second period is:

$$(1 + i_{t,t+1})(1 + i_{t+1,t+2}), \quad (11.48)$$

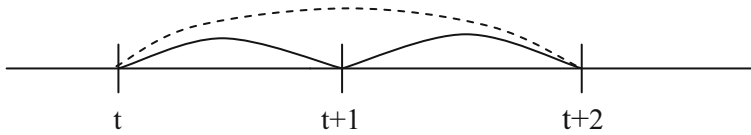


Fig. 11.16 Securities maturing in one and two periods

where $i_{t, t+1}$ is the interest rate at t for one period and $i_{t+1, t+2}$ will be the interest rate at $t+1$ for a security maturing on $t+2$. Alternatively, the investor may buy a security maturing in two periods with an interest rate per period $i_{t, t+2}$. The value of their investment upon maturity is:

$$(1 + i_{t, t+2})^2. \quad (11.49)$$

What is the best alternative? The expectations hypothesis of the term structure of interest rates assumes rational economic agents and that, by arbitrage, the two alternatives are exactly the same. Otherwise, there would be an opportunity for gains. That is:

$$(1 + i_{t, t+2})^2 = (1 + i_{t, t+1})(1 + i_{t+1, t+2}) \quad (11.50)$$

Taking the log of both sides of this expression and using the approximation $\log(1 + x) \cong x$ yields:

$$i_{t, t+2} = \frac{i_{t, t+1} + i_{t+1, t+2}}{2} \quad (11.51)$$

Therefore, we conclude that the longer-term rate $i_{t, t+2}$ is an average of the shorter-term rates ($i_{t, t+1}$ and $i_{t+1, t+2}$). It is easily seen that:

- (i) if $i_{t, t+2} > i_{t, t+1} \Rightarrow i_{t+1, t+2} > i_{t, t+1}$
- (ii) if $i_{t, t+2} < i_{t, t+1} \Rightarrow i_{t+1, t+2} < i_{t, t+1}$.

The expectations hypothesis's first conclusion, then, is that if the long-term rate is higher (lower) than the short-term rate, then the latter must rise (drop) in the future.

Arbitrage between the long-term rate and the short-term ones may be written in the continuous capitalization system. Let i_L be the long-term rate for maturity T and let i_S be the short-term interest rate at each point in time. By arbitrage:

$$e^{i_L(T-t)} = e^{\int_t^T i_S(v)dv}. \quad (11.52)$$

Therefore, the long-term rate is an average of the short-term rates:

$$i_L = \int_t^T i_S(v)dv / (T - t). \quad (11.53)$$

The derivative with respect to time t yields:

$$\dot{i}_L = \frac{di_L}{dt} = \frac{1}{T-t}(i_L - i_S). \quad (11.54)$$

The second conclusion of the expectations hypothesis of the term structure of interest rates may be easily deduced from the above expression, when $i_L > i_S$, $\dot{i}_L > 0$. That is, according to the second conclusion of the expectations hypothesis of the term structure, if the long-term rate is higher than the short-term rate, the long-term rate must rise, causing a capital loss for holders of long-term securities.

Central Banks control the interbank market's interest rate, the one that exchanges reserves between banks (usually maturing in 1 day). The long-term interest rate is affected by Central Bank announcements regarding the future path of the short-term rate. Therefore, even if the short-term rate does not change at the present time, the mere announcement of future changes may affect the real economy.

Model with Term Structure of Interest Rates

Assume the following model:

$$\text{IS: } x = -\alpha(i_L - \pi - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{TSIR: } \dot{i}_L = \lambda(i_L - i_S),$$

$$\text{MPR: } i_S = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x,$$

C.I.: $p(0)$ and $\pi(0)$ given. The equation of the term structure of interest rates (TSIR) assumes that the long-term rate is determined by the expectations hypothesis. The IS curve assumes that the long-term rate (i_L) affects the spending decisions of economic agents. According to the monetary policy rule (MPR), the Central Bank determines the economy's short-term interest rate (i_S).

Algebra

The derivative of the IS curve with respect to time, substituting the Phillips curve, and the term structure of interest rates equations, and using the value of i_L from the IS curve yields, with a little algebra, the following differential equation for the output gap:

$$\dot{x} = (\lambda + \alpha\lambda\theta + \alpha\delta)x + \alpha\lambda\phi(\pi - \bar{\pi}).$$

Dynamical System

The differential equations system has this equation and the Phillips curve:

$$\begin{bmatrix} \dot{\pi} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & \delta \\ \alpha\lambda\phi & \lambda + \alpha\lambda\theta + \alpha\delta \end{bmatrix} \begin{bmatrix} \pi - \bar{\pi} \\ x \end{bmatrix} \quad (11.55)$$

The system has the following Jacobian matrix:

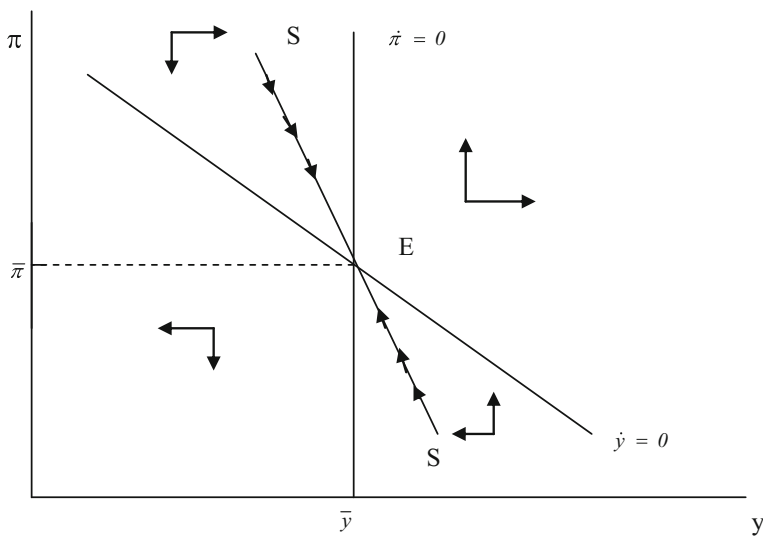


Fig. 11.17 The phase diagram for the inflation rate and real output

$$J = \begin{bmatrix} \frac{\partial \dot{\pi}}{\partial \pi} & \frac{\partial \dot{\pi}}{\partial x} \\ \frac{\partial \dot{y}}{\partial \pi} & \frac{\partial \dot{y}}{\partial x} \end{bmatrix} = \begin{bmatrix} 0 & \delta \\ \alpha\lambda\phi & \lambda + \alpha\lambda\theta + \alpha\delta \end{bmatrix}.$$

This matrix's determinant is negative: $|J| = -\alpha\lambda\phi\delta < 0$. Therefore, the system has a saddle point. Figure 11.17 is the model's phase diagram, with the inflation rate (π) on the vertical axis and real output (y) on the horizontal axis. The $\dot{\pi} = 0$ line is vertical. The $\dot{y} = 0$ curve has a negative slope, as does the SS saddle path. Given the model's initial inflation rate, the economy converges on the saddle path to the inflation target and full-employment output.

Experiment

Figure 11.18 describes a monetary policy experiment in which the Central Bank announces today that at instant T , in the near future, the inflation target will be lowered to $\bar{\pi}_1 < \bar{\pi}_0$ from $\bar{\pi}_0$.

This announcement means that at instant T the Central Bank will increase the short-term interest rate. According to the expectations hypothesis of the term structure of interest rates, the long-term interest rate (i_L) increases at the instant of the announcement. The economy's expenditure will then drop and the inflation rate will start to decrease before the Central Bank actually reduces the short-term interest rate, as Fig. 11.19 shows. At the instant of the announcement, the economy's real output contracts because of the higher long-term real interest rate, and the economy

11.9 Exercises

1. Consider the model:

$$\text{IS: } x = -\alpha(r - \bar{r}), \alpha > 0,$$

$$\text{PC: } \dot{\pi} = \delta x, \delta > 0,$$

$$\text{MPR: } \dot{i} = \lambda(i^* - i), \lambda > 0, i^* = \bar{r} + \pi + \phi(\pi - \bar{\pi}),$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

The symbols have the following meanings: x = output gap; r = real interest rate; π = inflation rate; i = nominal interest rate; $\bar{\pi}$ = inflation rate target; $\dot{\pi} = d\pi/dt$.

- Analyze the model's equilibrium and dynamics on a phase diagram with π on the vertical axis and r on the horizontal axis.
- Show what happens when the inflation target is lowered to $\bar{\pi}_1$ from $\bar{\pi}_0$.

2. Consider the model:

$$\text{IS: } x = -\alpha(i - \pi - \bar{r}),$$

$$\text{PC: } \dot{\pi} = \delta x,$$

$$\text{MPR: } i = \bar{i},$$

$$\text{IC: Given } p(0) \text{ and } \pi(0).$$

The symbols have the following meanings: x = output gap; $\bar{r}$ = real long-term interest rate; π = inflation rate; $\dot{\pi} = d\pi/dt$; i = nominal interest rate; $\bar{i}$ = nominal interest rate set by the Central Bank; α, δ are parameters.

- Analyze the model's equilibrium and dynamics.
- Assume that the parameter δ is negative. Analyze the model's equilibrium and dynamics under this assumption.
- Would you recommend using this monetary policy rule?

3. The Central Bank's loss function is:

$$L = \frac{\alpha}{2}\pi^2 - y + \bar{y}, \quad \alpha > 0.$$

The symbols have the following meanings: π is the inflation rate, $\bar{y}$ is the potential output, y is the real output, and α is a parameter. This economy's Phillips curve is given by:

$$\pi = \pi^e + \beta(y - \bar{y}), \quad \beta > 0.$$

The symbol π^e is the expected inflation rate. Assume that the private sector's expected inflation is set before the Central Bank makes a decision about the inflation rate.

- (a) What is this economy's inflation rate if the Central Bank acts in an ad-hoc manner?
- (b) Would a monetary policy rule that could be enforced be better for this economy?
- (c) Would a conservative central banker produce better results than an ad-hoc one at the Central Bank?
- (d) Would a conservative central banker's loss function be the one specified in this exercise?

4. Consider the model:

$$\text{IS: } y_t = \bar{y} - \alpha(r_t - \bar{r}) + \epsilon_t,$$

$$\text{PC: } \pi_t = \pi_{t-1} + \beta(y_t - \bar{y}) + v_t.$$

The symbols have the traditional meanings, and ϵ_t and v_t are uncorrelated zero-average, constant variance, random variables.

The Central Bank's loss function is:

$$L = \gamma(\pi_t - \bar{\pi})^2 + (y_t - \bar{y})^2.$$

What is the monetary policy rule for the real interest rate when the Central Bank's objective is to minimize the expected value of L ?

5. Consider the following model of the bank reserves market:

$$R^d = R_0 - \alpha i$$

$$R^s = \text{BR} + \text{NBR}$$

$$\text{BR} = \beta(i - i^d)$$

$$R^d = R^s$$

The symbols have the following meanings: R^d is the total demand for bank reserves, R^s is the total supply of bank reserves, i is the interest rate, BR is the share of reserves obtained at the discount rate, NBR is the share of reserves over which the Central Bank has effective control, i^d is the discount rate (non-punitive, $i > i^d$).

Consider three Central Bank operational procedures: (i) setting $i = \bar{i}$; (ii) setting $\text{BR} = \bar{\text{BR}}$; (iii) setting $\text{NBR} = \bar{\text{NBR}}$. Analyze what would happen in the bank reserves market for each of these procedures when a change occurs in the demand for bank reserves.

6. Consider the following model of the bank reserves market:

$$R_t^d = \alpha - \beta r_t + \delta i_{t+1}^e,$$

$$R_t^s = \bar{R}$$

$$R_t^d = R_t^s,$$

The symbols have the following meanings: R^s = volume of reserves offered by the Central Bank; i = interest rate; α , β and δ are positive parameters; i_{t+1}^e = interest rate anticipated at t for period $t + 1$.

Graphically show what happens today when the market expects the Central Bank to raise the interest rate tomorrow.

7. Assume that the long-term security is a perpetuity paying \$1 per period. This security's price P is the inverse of the long-term interest rate: $P = 1/i_L$.

(a) Show why the short-term interest rate must satisfy the equation:

$$i_s = \frac{1 + \dot{P}}{P}.$$

(b) Show that: $\frac{\dot{i}_L}{i_L} = i_L - i_s$.

8. Consider the following model:

IS: $x = -\alpha(r - \bar{r})$, $\alpha > 0$,

PC: $\dot{\pi} = \delta x$, $\delta > 0$,

TSIR: $i_s = \beta(r - r^s)$, $\beta > 0$, $r^s = i_s - \pi$,

MPR: $i_s = \bar{r} + \pi + \phi(\pi - \bar{\pi}) + \theta x$,

IC: Given $p(0)$ and $\pi(0)$.

The symbols have the following meanings: x = output gap; r = real long-term interest rate; r^s = real short-term interest rate; π = inflation rate; i_s = nominal short-term interest rate; $\bar{\pi}$ = target inflation rate; $\dot{\pi} = d\pi/dt$.

- How do you interpret the TSIR (term structure of interest rates) equation?
- Analyze the model's equilibrium and dynamics on a phase diagram with π on the vertical axis and x on the horizontal axis.
- Show what happens when the inflation target is lowered to $\bar{\pi}_1$ from $\bar{\pi}_0$ in the following situations: (i) unanticipated reduction and (ii) anticipated reduction.

9. Consider the model:

IS: $x = -\alpha(R - \pi - \bar{r})$,

PC: $\dot{\pi} = -\delta x$,

TSIR: $R = i + \lambda \dot{i}$,

MPR: $i = \bar{r} + \pi + \phi(\pi - \bar{\pi})$,

IC: Given $p(0)$.

- Show what conditions the model's parameters must satisfy for a unique equilibrium to exist.
- Does the model have a unique equilibrium when $\delta < 0$?

Appendix A: Differential Equations

This appendix presents some basic results of linear differential equations that are widely used throughout the text. Section A.1 covers the first-order linear differential equation. Section A.2 analyzes the second-order linear differential equation. Section A.3 deals with the linear system of first-order differential equations. Section A.4 addresses the linear system of n first-order differential equations. Section A.5 studies the determination of the differential equations systems' initial and terminal conditions. Section A.6 analyzes the phenomenon of hysteresis.

A.1 First-Order Linear Differential Equation

A homogeneous first-order linear differential equation is defined by:

$$\dot{x} + ax = 0, \quad (\text{A.1})$$

where a is a time independent coefficient. If $x = e^{rt}$ is a solution, $\dot{x} = re^{rt}$, it must satisfy this equation:

$$re^{rt} + ae^{rt} = 0.$$

Therefore, $r = -a$. The equation's solution is therefore given by:

$$x = Ce^{-at}, \quad (\text{A.2})$$

where C is a constant to be determined. This equation may show the following situations: (i) the a coefficient is positive, $a > 0$, and the equation's solution is stable, as it converges to zero; (ii) the a coefficient is negative, $a < 0$, and the equation's solution is unstable because the value of x grows indefinitely.

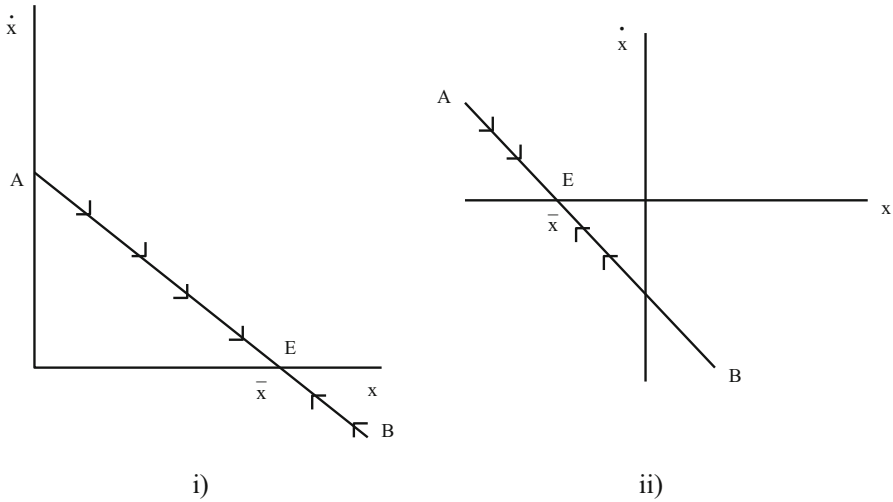


Fig. A.1 Stable solution. (i) $a > 0, k > 0$. (ii) $a > 0, k < 0$

Now consider the non-homogeneous first-order differential equation:

$$\dot{x} + ax = k, \quad (\text{A.3})$$

where k is parameter. The solution to this equation is given by:

$$x = \bar{x} + Ce^{-at}, \quad (\text{A.4})$$

where $\bar{x} = k/a$. If $a > 0$, the solution is stable, and if $a < 0$, the solution is unstable. It is worth mentioning that if the value of constant C equals zero, there is a stationary unstable solution.

Stability analysis for this equation can be done with a phase diagram with the value of $\dot{x}$ on the vertical axis and the value of x on the horizontal axis, as Figs. A.1 and A.2 show. The AB line represents the equation $\dot{x} + ax = k$. The arrows in Fig. A.1 show that the value of x converges to $\bar{x}$ because for all values to the left of $\bar{x}$, $\dot{x} > 0$ and, therefore, x is increasing. On the other hand, for values of x to the right of $\bar{x}$, $\dot{x} < 0$, that is x is decreasing. The arrows in Fig. A.2 show that for values of x different from $\bar{x}$, the path of x diverges from point E. The solution is therefore unstable.

A.2 Second-Order Linear Differential Equation

The second-order linear differential equation is defined by:

$$\ddot{x} + a\dot{x} + bx = 0, \quad (\text{A.5})$$

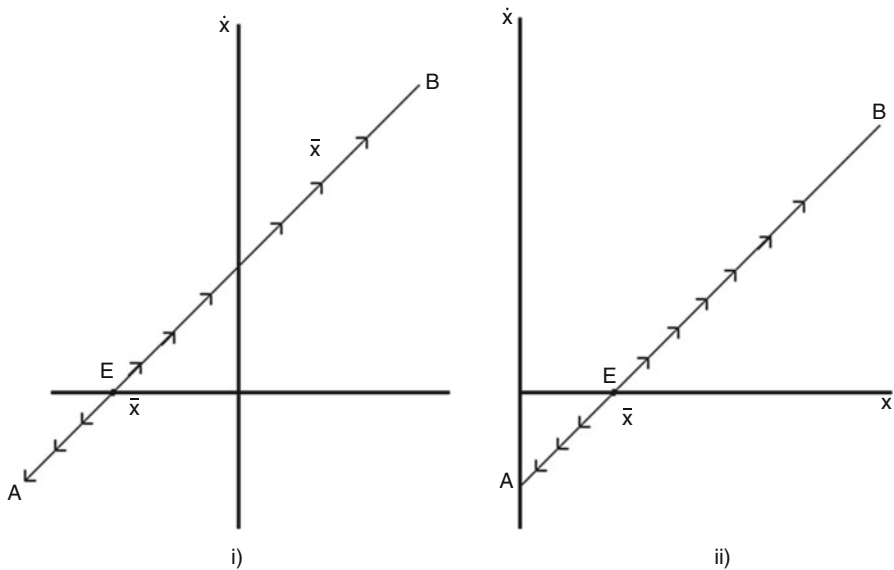


Fig. A.2 Unstable solution. (i) $a < 0, k > 0$. (ii) $a < 0, k < 0$

where a and b are time-independent coefficients. If $x = e^{rt}$ is a solution to this equation:

$$\dot{x} = re^{rt}, \quad \ddot{x} = r^2 e^{rt}.$$

Substituting the values of x , $\dot{x}$ and $\ddot{x}$ in the differential equation yields the equation:

$$r^2 + ar + b = 0, \quad (\text{A.6})$$

whose roots are:

$$r_1 = -\frac{a}{2} + \left(\frac{a^2}{4} - b\right)^{1/2},$$

$$r_2 = -\frac{a}{2} - \left(\frac{a^2}{4} - b\right)^{1/2}.$$

The sum of the roots is equal to minus a and their product is equal to b . Hence:

$$r_1 + r_2 = -a, \quad (\text{A.7})$$

$$r_1 r_2 = b. \quad (\text{A.8})$$

Distinct Real Roots

The roots are real and distinct when:

$$\frac{a^2}{4} - b > 0.$$

The differential equation solution is given by:

$$x = C_1 e^{r_1 t} + C_2 e^{r_2 t}, \quad (\text{A.9})$$

where C_1 and C_2 are two constants to be determined. The following cases are possible: (a) Both roots are negative: $r_1 < r_2 < 0$. In this case the solution is stable because the value of x converges to zero; (b) Both roots are positive: $r_1 > r_2 > 0$. In this case the value of x grows without limits, and the solution is unstable. (c) One root is positive and the other is negative: $r_1 > 0 > r_2$. If the constant C_1 is different from zero, the value of x grows indefinitely over time. If the value of the constant C_1 is zero and the constant C_2 is different from zero, the solution is stable because x converges to zero. This case corresponds to a saddle point. (d) One root is zero and the other is negative: $r_1 = 0$ and $r_2 < 0$, the solution is stable. (e) One root is zero and the other is positive: $r_1 = 0$ and $r_2 > 0$, the solution is unstable.

Equal Real Roots

The roots are real and equal when:

$$\frac{a^2}{4} - b = 0.$$

The differential equation solution is given by:

$$x = (C_1 + C_2 t) e^{-\frac{1}{2} a t}, \quad (\text{A.10})$$

because $r_1 = r_2 = -\frac{a}{2}$. Two cases are possible: (a) when $a > 0$, the solution is stable and (b) when $a < 0$, the solution is unstable.

Complex Conjugate Roots

The roots are complex and conjugate when the following inequality is satisfied:

$$\frac{a^2}{4} - b < 0.$$

Let $r_1 = \alpha + \beta i$ and $r_2 = \beta i$ be the two complex roots where $i^2 = -1$ and $r_2 = \alpha - \beta i$

$$\alpha = -\frac{a}{2}; \quad \beta = \left(b - \frac{a^2}{4}\right)^{1/2}.$$

We write:

$$\begin{aligned} e^{r_1 t} &= e^{(\alpha + \beta i)t} = e^{\alpha t} e^{\beta i t} = e^{\alpha t} (\cos \beta t + i \sin \beta t), \\ e^{r_2 t} &= e^{(\alpha - \beta i)t} = e^{\alpha t} e^{-\beta i t} = e^{\alpha t} (\cos \beta t - i \sin \beta t), \end{aligned}$$

because $e^{z i} = \cos z + i \sin z$. We need the following result: Let $u(t)$ and $v(t)$ be two solutions for the differential equation $\ddot{x} + a\dot{x} + bx = 0$. The linear combination $\theta u(t) + \phi v(t)$, where θ and ϕ are parameters, is a solution of this differential equation:

$$(\theta \ddot{u} + \phi \ddot{v}) + a(\theta \dot{u} + \phi \dot{v}) + b(\theta u + \phi v) = \theta \cdot 0 + \phi \cdot 0 = 0.$$

Therefore, $\theta u(t) + \phi v(t)$ is also a solution for $\ddot{x} + a\dot{x} + bx = 0$. So if $e^{r_1 t}$ and $e^{r_2 t}$ are solutions for these equations, it is straightforward to show that the linear combinations:

$$\begin{aligned} \frac{1}{2} e^{r_1 t} + \frac{1}{2} e^{r_2 t} &= e^{\alpha t} \cos \beta t, \\ \frac{1}{2i} e^{r_1 t} - \frac{1}{2} e^{r_2 t} &= e^{\alpha t} \sin \beta t, \end{aligned}$$

are solutions for $\ddot{x} + a\dot{x} + b = 0$. We thus find that:

$$x = e^{\alpha t} (C_1 \cos \beta t + C_2 \sin \beta t), \quad (\text{A.11})$$

is a solution for the second-order linear differential equation, as it stems from a linear combination of the two previous solutions. The following situations are possible: (a) The roots are purely imaginary: $\alpha = a = 0$. In this case the solution oscillates within fixed bounds. (b) The real part of the complex number is negative: $\alpha < 0$, that is, $a > 0$. The solution oscillates and converges to zero. (c) The real part of the complex number is positive: $\alpha > 0$, that is, $a < 0$. The solution oscillates and grows indefinitely.

We can now establish the following theorem: the second-order differential equation $\ddot{x} + a\dot{x} + b = 0$ is stable only if $a > 0$ and $b > 0$.

Second-Order Non-homogeneous Differential Equation

The second-order non-homogeneous differential equation is defined by:

$$\ddot{x} + a\dot{x} + bx = k. \quad (\text{A.12})$$

This equation solution is given by:

$$x = \bar{x} + C_1 e^{r_1 t} + C_2 e^{r_2 t}, \quad (\text{A.13})$$

where $\bar{x} = k/b$ is the stationary equilibrium solution and r_1 and r_2 are roots of $r^2 + ar + b = 0$. The results for homogeneous equations apply in this case.

A.3 First-Order Linear Differential Equations System

The first-order differential equations system is defined by:

$$\begin{cases} \dot{x} = a_{11}x + a_{12}y + b_1 \\ \dot{y} = a_{21}x + a_{22}y + b_2 \end{cases}, \quad (\text{A.14})$$

where the a_{ij} and b_i coefficients are time independent. Let the matrix be made up of the values of a_{ij} defined by A :

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad (\text{A.15})$$

whose trace and determinant are given by:

$$\text{tr } A = a_{11} + a_{22}, \quad (\text{A.16})$$

$$|A| = a_{11}a_{22} - a_{21}a_{12}. \quad (\text{A.17})$$

The first-order differential equations system is equivalent to two second-order differential equations for variables x and y . This result can be obtained as follows. First, take the derivative of the equation for $\dot{x}$ with respect to time:

$$\ddot{x} = a_{11}\dot{x} + a_{12}\dot{y}.$$

Substituting the expression of $\dot{y}$ in this equation yields:

$$\ddot{x} = a_{11}\dot{x} + a_{12}(a_{21}x + a_{22}y + b_2).$$

The differential equation for $\dot{x}$ allows us to write: $y = (\dot{x} - a_{11}x - b_1)/a_{12}$. Substituting this value of y in the previous expression yields:

$$\ddot{x} = (a_{11} + a_{22})\dot{x} - (a_{11}a_{22} - a_{12}a_{21})x + a_{12}b_2 - a_{22}b_1.$$

This is the second-order differential equation for variable x associated with the first-order linear differential equations system. The y variable's second-order differential equation is obtained in a similar manner. The two differential equations can be written as:

$$\begin{cases} \ddot{x} - (\text{tr } A)\dot{x} + |A|x = a_{12}b_2 - a_{22}b_1 \\ \ddot{y} - (\text{tr } A)\dot{y} + |A|y = a_{21}b_1 - a_{11}b_2 \end{cases}$$

The previous results for the stability of the second-order differential equation apply. That is, the necessary and sufficient condition for the linear system of

differential equations to be stable is that the determinant of the matrix is positive and its trace is negative:

$$|A| > 0, \quad (\text{A.18})$$

$$\text{tr } A < 0. \quad (\text{A.19})$$

The $\bar{x}$ and $\bar{y}$ stationary equilibrium solution for the differential equations system is obtained by solving the system of two equations and two unknowns,

$$a_{11}\bar{x} + a_{12}\bar{y} + b_1 = 0$$

$$a_{21}\bar{x} + a_{22}\bar{y} + b_2 = 0$$

The solution is easily obtained from the previous system by means of $\ddot{x} = \dot{x} = \dot{y} = \ddot{y} = 0$. Hence:

$$\bar{x} = \frac{a_{12}b_2 - a_{22}b_1}{|A|},$$

$$\bar{y} = \frac{a_{21}b_1 - a_{11}b_2}{|A|}.$$

The first-order differential equations system can alternatively be written by subtracting the system's equilibrium solution equations $(\bar{x}, \bar{y})$ from the equations for $\dot{x}$ and $\dot{y}$. The first-order differential equations system is then expressed as:

$$\dot{x} = a_{11}(x - \bar{x}) + a_{12}(y - \bar{y}),$$

$$\dot{y} = a_{21}(x - \bar{x}) + a_{22}(y - \bar{y}).$$

When $x = \bar{x}$ and $y = \bar{y}$, $\dot{x} = \dot{y} = 0$. On the other hand, when at least one variable is not equal to its stationary equilibrium value, the dynamical system moves toward the equilibrium point if it is an attractor, or away from it if the point is a repulsor. Analysis of the system's stability can be done with the phase diagram with y on the vertical axis and x on the horizontal axis.

Stable System

Assume that $a_{11} < 0$, $a_{22} < 0$, $a_{21} > 0$, and $a_{12} < 0$. It is easily seen that the system is stable with these values, as $\text{tr } A < 0$ and $|A| > 0$. When $\dot{x} = 0$, the values of y and x that correspond to this geometric locus are given by:

$$y = -\frac{a_{11}}{a_{12}}x - \frac{b_1}{a_{12}}.$$

When $\dot{x} > 0$, the following inequality must be satisfied:

$$y < -\frac{a_{11}}{a_{12}}x - \frac{b_1}{a_{12}}.$$

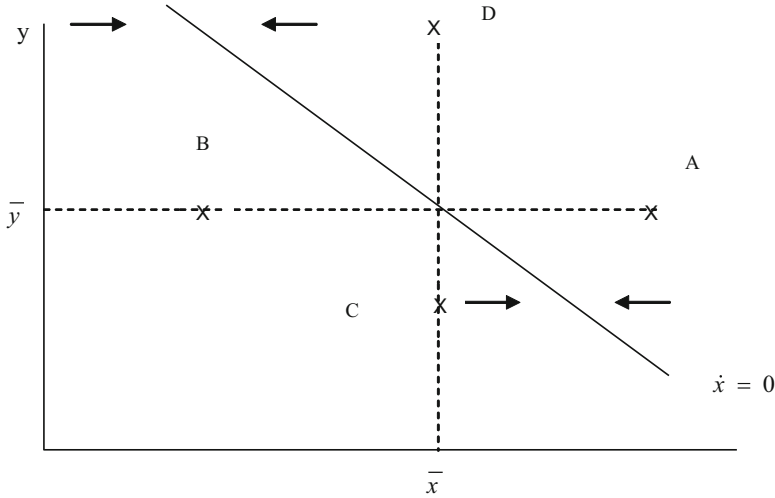


Fig. A.3 Phase diagram: Equation $\dot{x} = 0$

On the other hand, when $\dot{x} < 0$:

$$y > -\frac{a_{11}}{a_{12}}x - \frac{b_1}{a_{12}}.$$

The arrows in Fig. A.3 show what happens in terms of movement of the points outside the $\dot{x} = 0$ line. Variable x 's movement can be easily obtained by examining what happens at points A, B, C, or D in Fig. A.3. For example, at point A the value of y is equal to $\bar{y}$, but x is greater than $\bar{x}$. Therefore, because $a_{11} < 0$ it follows that $\dot{x} < 0$. At point C the abscissa x is equal to $\bar{x}$, but at this point $y < \bar{y}$ and $a_{12} < 0$; it follows that $\dot{x} > 0$. A similar reasoning can be applied to points B and D.

When $\dot{y} = 0$, the values of y and x corresponding to this geometric locus are given by:

$$y = -\frac{a_{21}}{a_{22}}x - \frac{b_2}{a_{22}}.$$

If $\dot{y} > 0$, the following inequality occurs:

$$y < -\frac{a_{21}}{a_{22}}x - \frac{b_2}{a_{22}}.$$

On the other hand, when $\dot{y} < 0$, the inequality is as follows:

$$y > -\frac{a_{21}}{a_{22}}x - \frac{b_2}{a_{22}}.$$

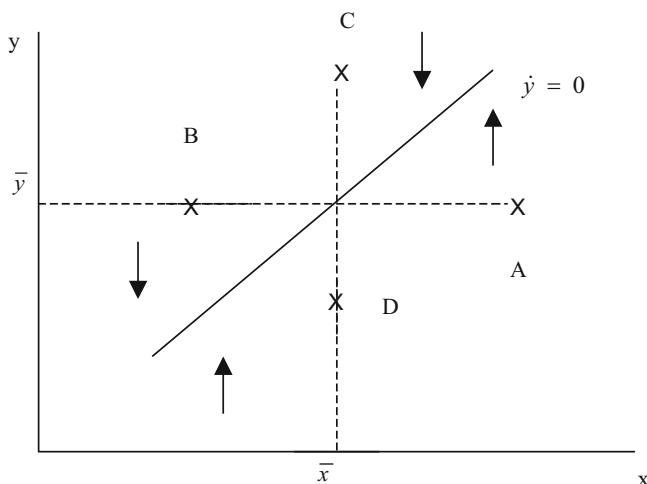


Fig. A.4 Phase diagram: Equation $\dot{y} = 0$

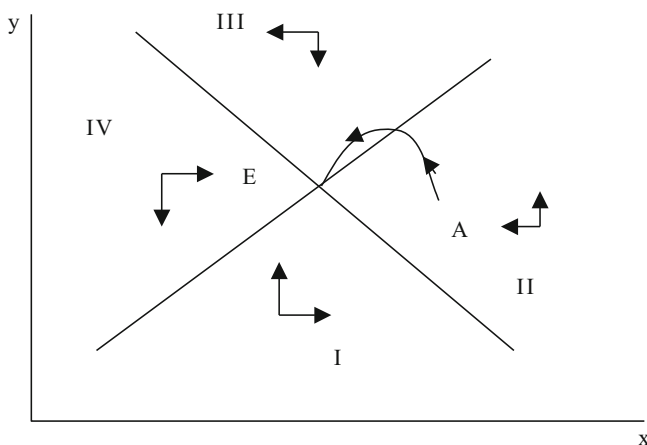


Fig. A.5 Phase diagram and stability analysis

The arrows in Fig. A.4 show what happens with the movement of points off the $\dot{y} = 0$ line. The arrows' directions, as in the previous case, may be obtained by analyzing one of the points A, B, C, or D in Fig. A.4. At point B the ordinate is equal to $\bar{y}$ and the abscissa x is lower than $\bar{x}$. Because the coefficient $a_{21} > 0$ it follows that $\dot{y} < 0$. At point C the abscissa is equal to $\bar{x}$ and the ordinate is greater than $\bar{y}$. Since $a_{22} < 0$ it follows that $\dot{y} < 0$. A similar reasoning can be applied to points A and D.

The graphs in Figs. A.3 and A.4 can be combined to analyze the equations system's stability, as Fig. A.5 shows. The arrows in Fig. A.5 show that, starting from any point, such as point A, the system converges to point E, which is a steady state, or an equilibrium point.

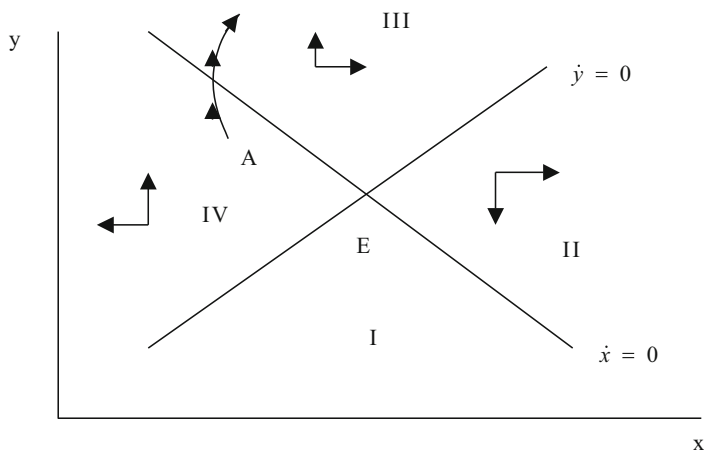


Fig. A.6 Phase diagram: unstable system

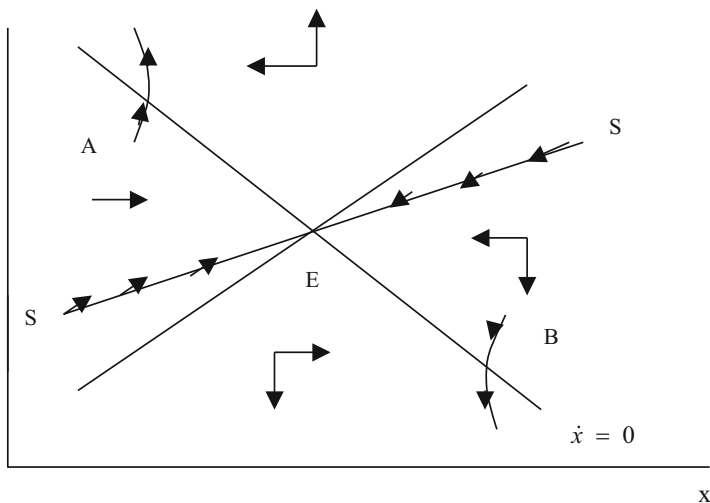


Fig. A.7 Phase diagram: saddle point

Unstable System

Assume that $a_{11} > 0$, $a_{12} > 0$, $a_{21} < 0$, and $a_{22} > 0$. The system is unstable because $\text{tr}A > 0$. Figure A.6 shows the phase diagram for this case. The system is unstable because starting from any point, such as point A, it moves away from equilibrium point E.

Unstable System: Saddle Point

Now assume that $a_{11} < 0$, $a_{12} < 0$, $a_{21} < 0$, and $a_{22} > 0$. The system is unstable because $|A| < 0$. The sign of matrix A's trace is indeterminate a priori $\text{tr}A \gtrless 0$, and has

no importance at all in this case. Figure A.7 shows the phase diagram for this example. The system is unstable because, starting from points such as A and B , it moves away from the equilibrium position (point E).

Note that, in this case, one of the roots of $\ddot{x} - (\text{tr}A)\dot{x} + |A|x = 0$ is positive while the other is negative because $r_1 r_2 = |A| < 0$. Therefore, the value of the constant associated with the positive root is zero, the system converges to equilibrium point E . Under these circumstances, the equilibrium is called a saddle point, so that there is a path, represented in Fig. A.7 by the SS line, the saddle path, that brings the system to equilibrium.

A.4 Linear System of n First-Order Differential Equations

A linear system of n homogeneous first-order differential equations is defined by:

$$\dot{z} = Az, \quad (\text{A.20})$$

where vector z has n components and matrix $A = [a_{ij}]$ is a nonsingular matrix ($|A| \neq 0$) $n \times n$. In the case of the previous section, for two differential equations, the z vector and the A matrix are given by:

$$Z = \begin{bmatrix} y \\ x \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Let $z(t)$ be a solution vector of the system of n first-order differential equations,

$$Z(t) = ve^{\mu t}, \quad (\text{A.21})$$

where v is a vector $n \times 1$ and μ is a scalar. The derivative of $Z(t)$ with respect to time is:

$$\dot{Z}(t) = \mu ve^{\mu t} = Ave^{\mu t}.$$

The expression to the right of the second equality sign takes into account the fact that $\dot{z}(t) = Az(t)$. Therefore, for $z(t)$ to be a solution, the following condition must be satisfied:

$$Av = \mu v.$$

This equation may be written as:

$$(A - \mu I)v = 0.$$

The μ parameter is the characteristic root – also known as eigenvalue – and v is the characteristic vector – or eigenvector – of matrix A . For this equations system

to have a solution other than the trivial $v = 0$, the linear system's determinant must be zero:

$$|A - \mu I| = 0. \quad (\text{A.22})$$

In general, this characteristic equation has n roots, μ_i , $i = 1, \dots, n$. One characteristic vector is associated with each root.

The solution of a system of n linear homogeneous differential equations of the first-order is therefore given by:

$$z = \sum_{i=1}^n C_i v_i e^{\mu_i t}, \quad (\text{A.23})$$

where the v_i vector corresponds to the μ_i root and C_i is the constant to be determined by the dynamical system's initial and terminal conditions.

The linear system of n first-order non-homogeneous differential equations is defined by:

$$\dot{z} = Az + b, \quad (\text{A.24})$$

where b is a time-independent vector. In this case, the system is autonomous. This particular system's solution is obtained by having $\dot{z} = 0$. That is:

$$\dot{z} = A\bar{z} + b = 0.$$

The particular solution is therefore given by:

$$\bar{z} = -A^{-1}b.$$

The general solution of a linear system of n non-homogeneous first-order differential equations is obtained by adding the particular solution and the solution of the homogeneous part of the linear system. Hence:

$$z = \bar{z} + \sum_{i=1}^n C_i v_i e^{\mu_i t}. \quad (\text{A.25})$$

A.5 Initial and Terminal Conditions of the Differential Equations Solution

The solution of a linear system of n first-order differential equations includes n constants to be determined, all of which depend on the system's initial and/or terminal initial conditions. Such a differential equations system has two types of variables: predetermined and free or jump variables. Predetermined variables are those whose values at the initial moment are given. Free variables, as the name

suggests, may take on any value at the initial instant and must be determined based on the problem.

We analyze three cases, where the linear system has: (i) n predetermined variables; (ii) n free variables; and (iii) n_1 predetermined variables and n_2 free variables ($n = n_1 + n_2$).

When a linear system has n predetermined variables, it has a solution that converges to stationary equilibrium if all characteristic roots are negative. In this case, the constants are determined by solving the following equations system:

$$z(0) = \bar{z} + \sum_{i=1}^n C_i v_i e^{0 \cdot t}.$$

It follows that:

$$\sum_{i=1}^n C_i v_i = z(0) - \bar{z}. \quad (\text{A.26})$$

When a linear system has n jump variables, the system has a unique solution if all characteristic roots are positive. In this case, all constants must be zero:

$$C_i = 0, \quad i = 1, \dots, n, \quad (\text{A.27})$$

and the solution of the linear system of differential equations is the stationary equilibrium. Hence:

$$z(0) = \bar{z}.$$

When a linear system has n_1 predetermined variables and n_2 jump variables, the system's solution is unique if the number of negative roots is equal to the number of predetermined variables. The constants of the jump variables, which correspond to positive characteristic roots, are equal to zero:

$$C_i = 0, \quad i = n_1 + 1, \dots, n. \quad (\text{A.28})$$

The predetermined variables' constants are determined by the solution of the system:

$$\sum_{i=1}^{n_1} C_i v_i^* = y(0) - \bar{y}, \quad (\text{A.29})$$

where the z and v_i vectors have been partitioned according to $z = [yx]^1$ and $v_i = [v_i^*, v_i^{**}]^1$, y is the vector for predetermined variables and x is that for jump variables, and v_i^* has n_1 elements.

A.6 Hysteresis

When the determinant of the A matrix of a linear system of differential equations is equal to zero,

$$|A| = a_{11}a_{22} - a_{21}a_{12} = 0, \quad (\text{A.30})$$

one of the roots will be equal to zero (bear in mind that the product of the roots is equal to the determinant). Assume that the differential equations system has an infinite number of solutions. Therefore, the following condition must be satisfied:

$$\frac{b_1}{b_2} = \frac{a_{12}}{a_{22}} = \frac{a_{11}}{a_{21}}.$$

In this case the differential equations system produces a phenomenon called hysteresis, where the system's solution depends on history, that is, on the model's initial conditions. As such, the two equations for $\dot{x} = 0$ and for $\dot{y} = 0$ are identical:

$$a_{11}\bar{x} + a_{12}\bar{y} + b_1 = 0,$$

$$a_{21}\bar{x} + a_{22}\bar{y} + b_2 = 0.$$

Stable System

The equations system is stable when the trace of the A matrix is negative:

$$\text{tr } A = a_{11} + a_{22} < 0.$$

For the trace to be negative, the coefficients of the A matrix must satisfy the following constraints:

$$a_{11} < 0, \quad a_{22} < 0, \quad a_{12} > 0, \quad a_{21} > 0.$$

When $\dot{x} = 0$ and $\dot{y} = 0$, the points corresponding to abscissa x and ordinate y belong to the line:

$$y = -\frac{b_1}{a_{12}} - \frac{a_{11}}{a_{12}}x.$$

Figure A.8's diagrams show the system's dynamics for points off this line. Analysis of points A , B , C , or D allows easy determination of the direction of the arrows indicating the movement of variables x and y . The phase diagram in Fig. A.9 combines the two diagrams and describes the differential equations system's dynamics. Assume that the system's initial point is point A . The system's equilibrium converges to point B , which is a stable solution for the system.

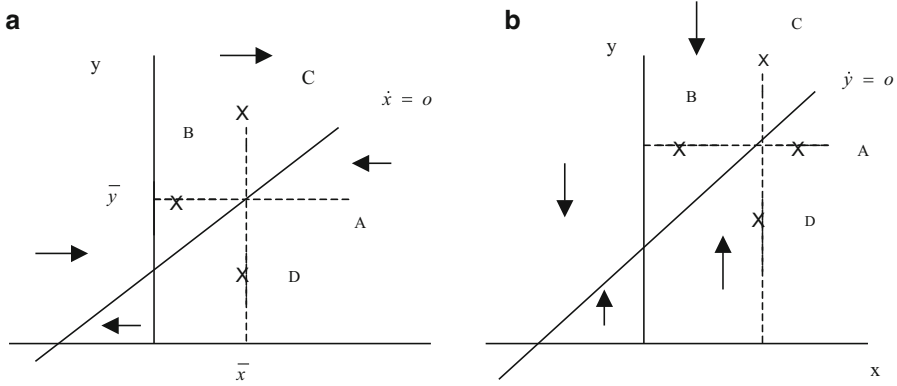


Fig. A.8 Phase diagram. (a) Equation $\dot{x} = 0$. (b) Equation $\dot{y} = 0$

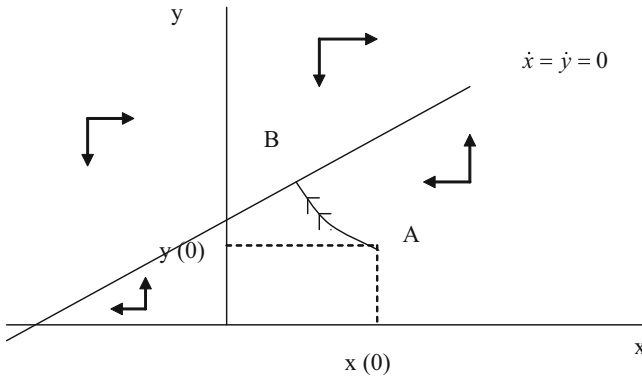


Fig. A.9 Phase diagram: stable system

Unstable Solution

The differential equations system's solution is unstable when the trace of the A matrix is positive,

$$\text{tr}A = a_{11} + a_{22} > 0.$$

For the trace to be positive, the coefficients of matrix A must satisfy the following inequalities:

$$a_{11} > 0, \quad a_{22} > 0, \quad a_{12} < 0, \quad a_{21} < 0.$$

Figure A.10's phase diagrams indicate the direction of the movement of the system's variables when in disequilibrium. Figure A.11 combines the two diagrams to show that the variables do not converge to equilibrium if the system is not at stationary equilibrium.

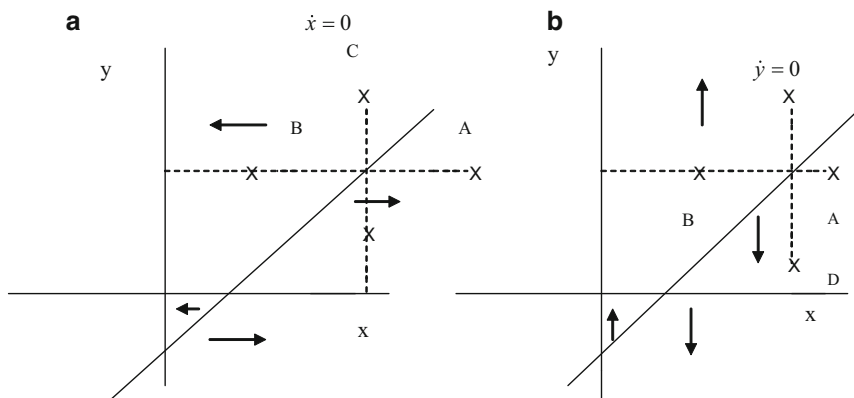


Fig. A.10 Phase diagram. (a) Equation $\dot{x} = 0$. (b) Equation $\dot{y} = 0$

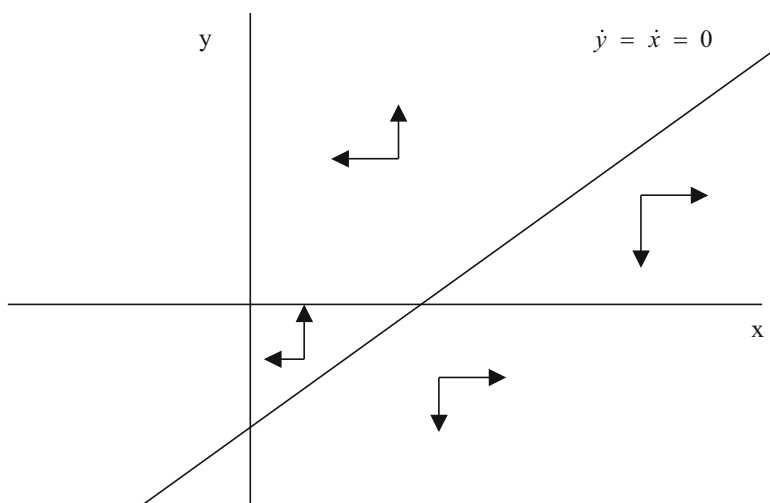


Fig. A.11 Phase diagram: unstable system

A.7 Exercises

1. With help from operator $Dx = \dot{x}$, the differential equations system:

$$\begin{cases} \dot{x} = a_{11}x + a_{12}y + b_1 \\ \dot{y} = a_{21}x + a_{22}y + b_2 \end{cases}$$

may be written as:

$$\begin{cases} (D - a_{11})x - a_{12}y = b_1 \\ -a_{21}x + (D - a_{22})y = b_2 \end{cases} \quad \text{or yet as :} \quad \begin{bmatrix} D - a_{11} & -a_{12} \\ -a_{21} & D - a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

(a) Show that this system solution is given by:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{(D - a_{11})(D - a_{22}) - a_{21}a_{12}} \begin{bmatrix} D - a_{22} & a_{12} \\ a_{21} & D - a_{11} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

(b) Show that the properties of operator D , $Dx = \dot{x}$, and $D^2x = \ddot{x}$, allow for the writing of the system as:

$$\begin{cases} \ddot{x} - (\text{tr } A)\dot{x} + |A|x &= a_{12}b_2 - a_{22}b_1 \\ \ddot{y} - (\text{tr } A)\dot{y} + |A|y &= a_{21}b_1 - a_{11}b_2 \end{cases}$$

2. A goods's market model is specified by the equations:

$$\text{demand: } q^d = \alpha - \beta p,$$

$$\text{supply: } q^s = \gamma + \delta p,$$

$$\text{adjustment: } \dot{p} = \frac{dp}{dt} = \phi(q^d - q^s), \phi > 0.$$

(a) What is the goods's equilibrium price?

(b) Given an initial price of $[p(0) = p_0]$, what is the price's path?

(c) Use a phase diagram [$\dot{p}$ on the vertical axis and p on the horizontal axis] to illustrate your answers to items (a) and (b).

3. *Harrod-Domar Economic Growth Model* – Savings is proportional to gross domestic product,

$$S = sY.$$

Investment is proportional to the rate of change of output,

$$I = v\dot{Y}.$$

The s and v parameters are positive. The goods and services market is at equilibrium when saving equals investment:

$$I = S.$$

(a) What is the path of this economy's output, assuming that the value of the output at the initial instant is Y_0 ?

(b) Use a phase diagram [$\dot{Y}$ on the vertical axis and Y on the horizontal axis] to show the path of this economy's output.

4. *Asset-Pricing Model* – An economy has two assets: one fixed income, the other variable income. The fixed income asset's interest rate is r . The variable asset is riskless and pays dividends of v per unit of time. By arbitrage, the two assets' rates of return must be equal. That is:

$$r = \frac{v + \dot{p}^e}{p},$$

where $\dot{p}^e$ is the expected change of the price p of the variable income asset. Assume perfect foresight of the asset's price in the sense that:

$$\dot{p}^e = \dot{p}.$$

- (a) What is this asset's equilibrium price?
 - (b) Can the market price and the equilibrium price differ?
 - (c) Would you be able to set up this model with discrete variables?
5. An economy's model is specified by the equations:

goods and services market adjustment: $\dot{y} = \alpha(d - y)$,

expenditure: $d = c + i + g$,

consumption: $c = \beta y$,

investment: $i = \bar{i}$,

fiscal policy rule: $\dot{g} = -\gamma(y - \bar{y})$.

The symbols have the following meanings: y = real output, $\dot{y} = dy/dt$, d = expenditure, c = consumption, i = investment, g = government spending, $\bar{y}$ = full-employment output; α , β , and γ are positive parameters.

- (a) What is the value of the real equilibrium output?
 - (b) Does this economy's fiscal policy rule lead its real output to full-employment output?
6. An economy's model is specified by the equations:

goods and services market adjustment: $\dot{y} = \alpha(d - y)$,

expenditure: $d = c + i$,

consumption: $c = \beta y$,

money market adjustment: $\dot{r} = \gamma(m^d - m)$,

demand for money: $m^d = \delta y - \lambda r$.

The symbols have the following meanings: y = real output, $\dot{y} = dy/dt$, d = expenditure, c = consumption, i = investment, r = interest rate, m^d = quantity of money demanded, m = quantity of money supplied, α , β , γ , δ , and λ are positive parameters.

- (a) Analyze the model's equilibrium and dynamics when investment is constant, $i = \bar{i}$.
- (b) Analyze the model's equilibrium and dynamics when investment depends on the interest rate according to $i = \tau - \theta r$, where θ is a positive parameter.

7. Consider the model:

$$\begin{aligned}\dot{P} &= \alpha(d - y), \alpha > 0 \\ \dot{r} &= \beta \left(L(y, r) - \frac{M}{P} \right), \beta > 0 \\ d &= d(y, r), \frac{\partial d}{\partial y} > 0, \frac{\partial d}{\partial r} < 0 \\ y &= \bar{y}\end{aligned}$$

where P is the price index, d is the level of expenditure, y is real output, r is the interest rate, M is the nominal stock of money, $\bar{y}$ is potential output.

Analyze this model's equilibrium and stability.

8. Consider the model:

$$\begin{aligned}\dot{p} &= \beta \left(\frac{M}{p} - L(y, r) \right), \beta > 0 \\ \dot{r} &= \alpha(i + g - t - s), \alpha > 0 \\ i &= i(r), \frac{\partial i}{\partial r} < 0 \\ s &= s(y), \frac{\partial s}{\partial y} > 0 \\ y &= \bar{y}\end{aligned}$$

where p is the price index, M is the nominal stock of money, y is real output, r is the interest rate, i is the investment level, g is government spending, t is the tax revenue, s is the level of saving, $\bar{y}$ is potential output. Analyze the model's equilibrium and stability.

9. Consider the model:

$$\begin{aligned}\dot{y} &= \alpha(d - y), \alpha > 0 \\ \frac{M}{P} &= L(y, r) \\ d &= d(y, r), \frac{\partial d}{\partial y} > 0, \frac{\partial d}{\partial r} < 0 \\ M &= \bar{M} \quad , \quad P = \bar{P}\end{aligned}$$

where y is real output, d is the expenditure level, M is the nominal stock of money, p is the price index, r is the interest rate. Analyze the model's equilibrium and stability.

10. Consider the model:

$$\begin{aligned}\dot{y} &= \phi(d - y) \\ \pi &= \pi^e + \delta(y - \bar{y}) \\ \dot{\pi}^e &= \theta(\pi - \pi^e) \\ d &= d(y, \pi^e), \frac{\partial d}{\partial y} > 0, \frac{\partial d}{\partial \pi^e} > 0\end{aligned}$$

where y is the real output level, $\bar{y}$ is the potential output level, d is the level of expenditure, π is the inflation rate, π^e is the expected inflation rate, ϕ , δ , and θ are positive parameters. Analyze the model's equilibrium and dynamics.

11. Consider the model:

$$\begin{aligned}\pi &= \pi^e + \phi(d - y) \\ \dot{y} &= \psi(\bar{y} - y) \\ \dot{\pi}^e &= \theta(\pi - \pi^e) \\ d &= d(y, \pi^e), \frac{\partial d}{\partial y} > 0, \frac{\partial d}{\partial \pi^e} > 0\end{aligned}$$

where the symbols have the same meanings as in the previous exercise, and ϕ , ψ , and θ are positive parameters. Analyze the model's equilibrium and dynamics.

12. Consider the model:

$$\begin{aligned}d &= \alpha_0 + \alpha_1 y - \alpha_2(R - \pi^e) + \alpha_3 f \\ \dot{y} &= \phi(d - y) \\ m - p &= \beta_0 + \beta_1 y - \beta_2 r \\ \dot{R} &= R - r\end{aligned}$$

where: d = real expenditure; y = real output; R = long-term interest rate; π^e = expected inflation rate; f = fiscal policy variable (government spending); m = log of the nominal quantity of money; p = log of the price index; r = short-term interest rate.

- (a) What is the effect of an increase in government spending on real output?
- (b) What is the effect of an increase in the quantity of money on real output?

13. In a disequilibrium situation, the market interest rate (r) differs from the natural interest rate (r^*), and the surplus of investment (I) over saving (S) is financed with the issuing of money. That is:

$$I - S = -\alpha(r - r^*) = \frac{dM}{dt}, \quad \alpha > 0,$$

The price change $\left(\dot{p} = \frac{dp}{dt}\right)$ depends on excess demand:

$$\dot{p} = \beta(I - S) \quad , \quad \beta > 0.$$

The Central Bank changes the interest rate $\left(\dot{r} = \frac{dr}{dt}\right)$ according to the equation:

$$\dot{r} = \gamma(p - \bar{p}) + \delta\dot{p}, \quad \gamma > 0, \quad \delta > 0,$$

where $\bar{p}$ is the price-level target. Analyze the model's equilibrium and dynamics.

14. Consider the model:

aggregate demand: $y = k + \alpha \log \frac{M}{P} + \beta \pi^e + \gamma f$,

Phillips curve: $\pi = \pi^e + \delta(y - \bar{y})$,

perfect prediction: $\pi^e = \pi$,

where the symbols have the following meanings: y is real output, M is the nominal stock of money, P is the price index, π^e is the expected inflation rate, f represents a fiscal variable, π is the inflation rate ($\pi = d \log P / dt$), $\bar{y}$ is the economy's potential output.

(a) Analyze the effects of a change in the government's fiscal policy in the following situations:

- (i) permanent, unanticipated;
- (ii) permanent, anticipated;
- (iii) transitory, unanticipated;
- (iv) transitory, anticipated.

(b) Analyze the effect of a change in the growth rate of money stock $\left(\mu = \frac{d \log M}{dt}\right)$ in the following situations:

- (i) permanent, unanticipated;
- (ii) permanent, anticipated;
- (iii) transitory, unanticipated;
- (iv) transitory, anticipated.

15. Assume that the explicit or implicit rent (A) of a house is given by:

$$A = (\delta + r - \pi + \tau)P - \dot{P},$$

where P is the house's price, δ is the depreciation rate, r is the nominal interest rate, π is the inflation rate, τ is the property tax on the house, and $\dot{P} = dP/dt$ is the change in the house's price. The house's inverse demand equation depends on the quantity of pieces of house (H) according to,

$$A = A(H), \frac{dA}{dH} < 0.$$

The change in the stock of houses depends on the supply of new houses and on the depreciation rate. That is:

$$\dot{H} = S(P) - \delta H,$$

where $S(P)$ is the new houses supply curve and $dS/dP > 0$.

- (a) Analyze the model's equilibrium and dynamics on the H (horizontal axis) and P (vertical axis) plane.
- (b) Analyze the consequences for the house market (P and H) of an increase in the property tax rate when the rate increase is:
 - (i) permanent, unanticipated;
 - (ii) permanent, anticipated;
 - (iii) transitory, unanticipated;
 - (iv) transitory, anticipated.

16. Consider the model:

$$\dot{q} = (r + \delta)q - F'(K), \quad F''(K) < 0,$$

$$\dot{K} = I(q) - \delta K, \quad I'(q) > 0,$$

where the symbols have the following meanings: q = Tobin's q ; r = interest rate; δ = depreciation rate; $F'(K)$ = marginal product of capital; K = capital stock, $\dot{q} = dq/dt$, $\dot{K} = dK/dt$.

- (a) What is the economic interpretation of the model's equations?
- (b) On the saddle path, are q and K negatively correlated?
- (c) What are the short- and long-term effects of an interest-rate increase that is:
 - (i) unanticipated, permanent;
 - (ii) permanent, anticipated;
 - (iii) transitory, unanticipated;
 - (iv) transitory, anticipated.

Appendix B: Optimal Control Theory

This appendix deals with optimal control theory. Section B.1 presents the basic optimal control problem. Section B.2 introduces the Hamiltonian and the transversality condition. Section B.3 covers the optimal control problem with a discount rate and an infinite horizon. Section B.4 introduces linear optimal control. Section B.5 analyzes the comparative dynamics of the solution of the optimal control problem.

B.1 Optimal Control: Basic Problem

The optimal control theory deals with the optimization of dynamical systems. The objective to be maximized is a function that associates the paths of the state (x) and control (u) variables to a real number. That is:

$$\max \int_{t_0}^{t_1} f(t, x(t), u(t)) dt, \quad (\text{B.1})$$

subject to the following constraints:

$$\dot{x} = g(t, x(t), u(t)), \quad (\text{B.2})$$

$$x(t_0) = x_0, \quad (\text{B.3})$$

$$x(t_1) \text{ free}, \quad (\text{B.4})$$

$$t_0, x_0 \text{ and } t_1 \text{ fixed}. \quad (\text{B.5})$$

The symbol t represents time. The problem's initial t_0 and final t_1 times are fixed. The initial value of the state variable is given, but its final value is unknown. The first-order differential equation for the state variable is the state, or transition, equation. Given the control variable's path, the state equation allows for the calculation of the state variable's path.

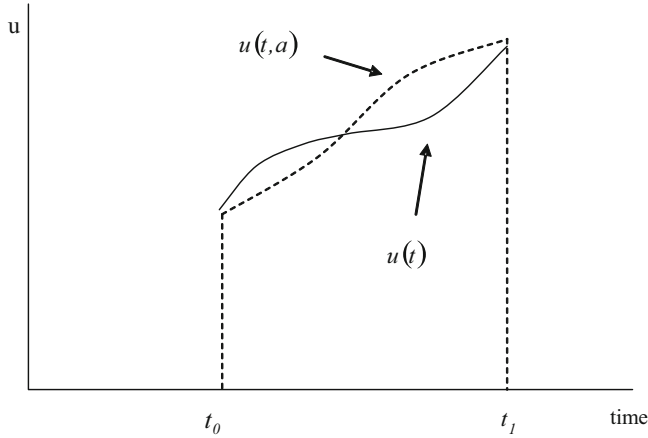


Fig. B.1 Small disturbances in the neighbourhood of the solution path

Assume that $u(t)$ is a solution for this problem and consider small disturbances in the region of this path, according to Fig. B.1. Analytically, the path in the region of optimal control is given by:

$$u(t, a) = u(t) + ah(t),$$

where a is a parameter and $h(t)$ is a fixed arbitrary function. The path close to optimal control corresponds to a state variable path, $u(t, a) \rightarrow x(t, a)$. When $a = 0$, we have the state variable's optimal path, $x(t, 0) = x(t)$, and the state variable's fixed initial value, $x(t_0, a) = x_0$.

The state equation may be written as:

$$g(t, x(t, a), u(t, a)) - \dot{x}(t, a) \equiv 0, \forall t \in [t_0, t_1].$$

It follows that:

$$\int_{t_0}^{t_1} \mu(t) [g(t, x(t, a), u(t, a)) - \dot{x}(t, a)] dt \equiv 0,$$

where $\mu(t)$ is a costate variable, a function of time, but unknown.

Using this constraint, the first-order conditions of the optimal control problem can be deduced from the function F of parameter a . That is:

$$F(a) = \int_{t_0}^{t_1} [f(t, x(t, a), u(t, a)) + \mu(t) [g(t, x(t, a), u(t, a)) - \dot{x}(t, a)]] dt.$$

The integration by parts, $\int u dv = uv - \int v du$, of the term in the expression that contains the derivative of the state variable with respect to time results in:

$$\int_{t_0}^{t_1} \mu(t) \dot{x}(t, a) dt = \mu(t) x(t, a) \Big|_{t_0}^{t_1} - \int_{t_0}^{t_1} \dot{\mu} x(t, a) dt,$$

or:

$$\int_{t_0}^{t_1} \mu(t) \dot{x}(t, a) dt = \mu(t_1) x(t_1, a) - \mu(t_0) x(t_0, a) - \int_{t_0}^{t_1} \dot{\mu} x(t, a) dt.$$

Substituting this expression in the $F(a)$ function yields:

$$F(a) = \int_{t_0}^{t_1} [f(t, x(t, a), u(t, a)) + \mu(t) g(t, x(t, a), u(t, a)) + \dot{\mu} x(t, a)] dt - \mu(t_1) x(t_1, a) + \mu(t_0) x(t_0, a).$$

$F(a)$ is a function of the parameter a and $F(0) \geq F(a)$. Therefore: $F'(0) = 0$:

$$F'(a) = \int_{t_0}^{t_1} (f_x + \mu g_x + \dot{\mu}) \frac{\partial x(t, a)}{\partial a} dt + \int_{t_0}^{t_1} (f_u + \mu g_u) h(t) dt - \mu(t_1) \frac{\partial x(t_1, a)}{\partial a} + \mu(t_0) \frac{\partial x(t_0, a)}{\partial a} = 0$$

The partial derivatives are defined by:

$$f_x = \frac{\partial f(t, x, u)}{\partial x}; f_u = \frac{\partial f(t, x, u)}{\partial u}; g_x = \frac{\partial g(t, x, u)}{\partial x}; g_u = \frac{\partial g(t, x, u)}{\partial u}.$$

The first-order conditions for a maximum of $F(a)$, $F'(0) = 0$, are therefore given by:

$$f_x + \mu g_x + \dot{\mu} = 0, \quad (\text{B.6})$$

$$f_u + \mu g_u = 0, \quad (\text{B.7})$$

$$\mu(t)_1 = 0. \quad (\text{B.8})$$

When the state variable's initial condition is unknown, that is, when $x(t_0)$ is not a given, but rather an endogenous variable, the costate variable at the initial point must equal zero: $\mu(t_0) = 0$. This condition stems from the fact that, under these circumstances, $\partial x(t_0, a) / \partial a \neq 0$.

B.2 Hamiltonian and Transversality Condition

The optimal control problem's first-order conditions can be written with the help from the Hamiltonian H defined by:

$$H = H(t, x, u, \mu) = f(t, x, u) + \mu g(t, x, u). \quad (\text{B.9})$$

The first-order condition implies that the Hamiltonian must be maximized with respect to the control variable:

$$\frac{\partial H}{\partial u} = f_u + \mu g_u = 0. \quad (\text{B.10})$$

The Hamiltonian's partial derivative with respect to the state variable equals the derivative of the variable with respect to time with a minus sign:

$$\frac{\partial H}{\partial x} = f_x + \mu g_x = -\dot{\mu}. \quad (\text{B.11})$$

The Hamiltonian's partial derivative with respect to the costate variable results in the transition equation:

$$\frac{\partial H}{\partial \mu} = g(t, x, u) = \dot{x}. \quad (\text{B.12})$$

The problem's initial condition is given by:

$$x(t_0) = x_0. \quad (\text{B.13})$$

The other condition for determining the problem's solution is the terminal boundary condition, also known as the transversality condition, given by:

$$\mu(t_1) = 0. \quad (\text{B.14})$$

Transversality Condition: Geometric Interpretation

Let $V = \int_{t_0}^{t_1} f(t, x, u) dt$ the maximum value of the optimal control problem. Assume that final instant t_1 and value $x(t_1)$ are unknown. The optimal value then depends on the following variables:

$$V = V(t_1, x(t_1)) = V(t_1, x_1).$$

Assume, also, that t_1 and $x(t_1)$ must satisfy the constraint:

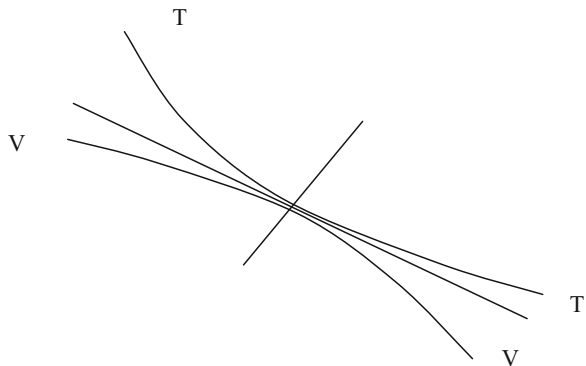
$$T = T(t_1, x(t_1)) = T(t_1, x_1) = 0$$

The terminal conditions are therefore obtained by maximizing:

$$V(t_1, x_1) \quad (\text{B.15})$$

with the condition that the following constraint is satisfied:

$$T(t_1, x_1) = 0. \quad (\text{B.16})$$

Fig. B.2 The transversality condition

The problem's Lagrangian is given by:

$$\ell = V(t_1, x_1) + \varphi T(t_1, x_1) \quad (\text{B.17})$$

where φ is the Lagrange multiplier. This problem's first-order condition implies that:

$$\frac{\frac{\partial V}{\partial x_1}}{\frac{\partial V}{\partial t_1}} = \frac{\frac{\partial T}{\partial x_1}}{\frac{\partial T}{\partial t_1}} \quad (\text{B.18})$$

The term transversality condition stems from the following geometric interpretation of this equation: the gradient of curve V intercepts curve T at a right angle at the optimal terminal point, as Fig. B.2 shows. When time t_1 is fixed, the T function is given by:

$$T(t_1, x_1) = t_1 - \bar{t}_1 = 0. \quad (\text{B.19})$$

Therefore, $\partial T / \partial x_1 = 0$ and

$$\frac{\partial V}{\partial x_1} = \mu(t_1) = 0. \quad (\text{B.20})$$

Dynamical System

The first-order conditions for determination of optimal control, the maximum principle, plus the state variable's initial condition and the transversality condition are as follows:

$$H_u = 0 \quad (\text{B.21})$$

$$\dot{\mu} = -H_x \quad (\text{B.22})$$

$$\dot{x} = H_\mu \quad (\text{B.23})$$

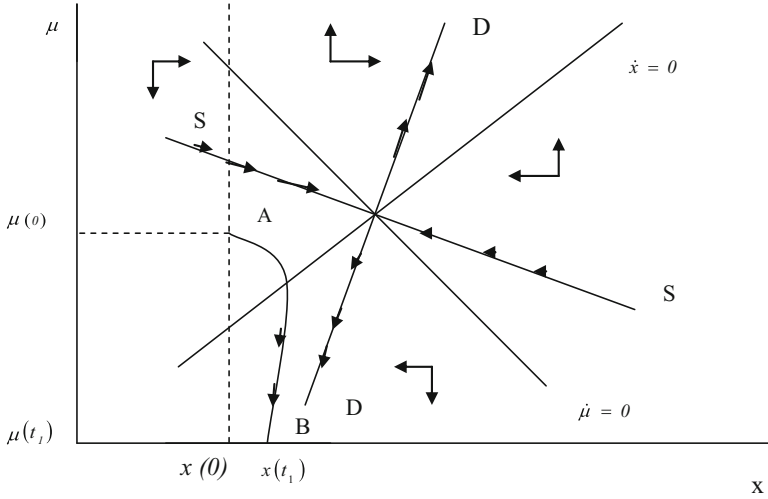


Fig. B.3 The phase diagram for the costate and the state variables

$$x(0) = x_0 \quad (\text{B.24})$$

$$\mu(t_1) = 0. \quad (\text{B.25})$$

Note that this problem's solution transforms into a system of two differential equations, when the control variable obtained from the Hamiltonian maximization first-order condition as a function of the state and costate variables is substituted in the differential equations of these variables. The state variable's initial condition and the costate variable's transversality condition therefore determine these variables' paths. That is:

$$\dot{\mu} = M(\mu, x, \alpha) \quad (\text{B.26})$$

$$\dot{x} = X(\mu, x, \alpha) \quad (\text{B.27})$$

$$x(0) = x_0 \quad (\text{B.28})$$

$$\mu(t_1) = 0 \quad (\text{B.29})$$

where α is a vector of parameters affecting the dynamical system. The system assumes that time is not an argument of the differential equations. Thus, the system is autonomous. Figure B.3 shows this dynamical system's phase diagram, assuming that there is a saddle point. SS is the convergent saddle path, and DD is the divergent saddle. The AB path is the control problem's optimal solution. Point A 's abscissa is the initial value ($x(t_0)$) of the state variable, and point B 's ordinate is the costate variable's terminal value ($\mu(t_1)$). The AB path is such that the time interval to cover it is equal to t_1 .

Economic Interpretation

An economic interpretation of the maximum principle can be obtained by multiplying the Hamiltonian by time interval dt :

$$Hdt = f(t, x, u)dt + \mu g(t, x, u)dt = f(t, x, u)dt + \mu \dot{x} dt,$$

where the transition equation is taken into account. Since $\dot{x} dt = dx$, it follows that:

$$Hdt = f(t, x, u)dt + \mu dx.$$

This expression's first component is the direct contribution to the problem's objective, when the state variable equals x and the u control is applied. Its second component measures the indirect contribution to the problem's objective of the change in state variable x when the control variable is equal to u in time interval dt . The μ coefficient can be interpreted as the shadow price of the state variable.

B.3 Discounted Infinite Horizon Optimal Control

In the majority of optimal control problems in economics, the objective to maximize is discounted by a factor, the rate of time preference in the case of consumers, and the interest rate in the business firms' cash flow. Let ρ be the discount rate and $f(x, u)$ the function representing the agent's criterion. The optimal control problem therefore consists of:

$$\max \int_0^T e^{-\rho t} f(x, u) dt, \quad (\text{B.30})$$

subject to the following constraints:

$$\dot{x} = g(x, u), \quad (\text{B.31})$$

$$x(0) = x_0, \text{ given.} \quad (\text{B.32})$$

The present-value Hamiltonian H^* is expressed by:

$$H^* = e^{-\rho t} f(x, u) + \mu g(x, u). \quad (\text{B.33})$$

The current-value Hamiltonian, H , is defined by:

$$H = H^* e^{\rho t}. \quad (\text{B.34})$$

Substituting the equation for H^* in the previous equation yields:

$$H = f(x, u) + \lambda g(x, u) \quad (\text{B.35})$$

where the new costate variable λ equals:

$$\lambda = \mu e^{\rho t}.$$

The problem's first-order conditions are:

$$\begin{aligned}\frac{\partial H^*}{\partial u} &= e^{-\rho t} f_u + \mu g_u = 0, \\ \frac{\partial H^*}{\partial x} &= e^{-\rho t} f_x + \mu g_x = -\dot{\mu}, \\ \frac{\partial H^*}{\partial u} &= g(x, u) = \dot{x}.\end{aligned}$$

The first equation implies that:

$$f_u + \mu e^{\rho t} g_u = f_u + \lambda g_u = 0,$$

where the definition of λ has been taken into account. Therefore, the first-order condition to maximize the present-value Hamiltonian equals the following condition for the maximization of the current-value Hamiltonian:

$$\frac{\partial H}{\partial u} = f_u + \lambda g_u = 0. \quad (\text{B.36})$$

The second first-order condition equation for the optimal control problem may be written as:

$$f_x + \mu e^{\rho t} g_x = -e^{\rho t} \dot{\mu}.$$

The derivative of costate variable λ with respect to time is:

$$\frac{d\lambda}{dt} = \dot{\mu} e^{\rho t} + \rho \mu e^{\rho t}.$$

Therefore:

$$e^{\rho t} \dot{\mu} = \dot{\lambda} - \rho \lambda.$$

Substituting this expression in the first-order condition yields:

$$f_x + \lambda g_x = -(\dot{\lambda} - \rho \lambda).$$

This equation may be written as:

$$\dot{\lambda} = \rho \lambda - (f_x + g_x) = \rho \lambda - H_x. \quad (\text{B.37})$$

It is straightforward to show that:

$$\frac{\partial H}{\partial \lambda} = g(x, u) = \dot{x}. \quad (\text{B.38})$$

The state variable's initial condition is given. That is:

$$x(t_0) = x_0.$$

The transversality condition becomes:

$$\mu(T) = \lambda(T)e^{-\rho T} = 0. \quad (\text{B.39})$$

The optimal control problem with a discount rate may be solved both with help from the present-value and the current-value Hamiltonians. It is a matter of preference to use one or the other. The problem's solution is precisely the same, except for the interpretation of the respective costate variable. The first-order conditions, the initial condition, and the transversality condition are therefore given by:

$$\begin{cases} H_u = f_u + \lambda g_u = 0 \\ \dot{\lambda} = \rho\lambda - H_x = \rho\lambda - (f_x + \lambda g_x) \\ H_\lambda = \dot{x} = g(x, u) \\ x(t_0) = x_0 \\ \lambda(T) = 0 \end{cases} \quad (\text{B.40})$$

When $T \rightarrow \infty$, the infinite horizon case occurs. Obviously, the integral to optimize:

$$\int_0^\infty e^{-\rho t} f(x, u) dt, \quad (\text{B.41})$$

must be a proper integral, that is, it must exist. The transversality condition becomes:

$$\lim_{T \rightarrow \infty} \lambda(T)e^{-\rho T} = 0. \quad (\text{B.42})$$

When the state variable must satisfy the non-negativity condition in the final period, $x(T) \geq 0$, the transversality condition is given by:

$$\lim_{T \rightarrow \infty} \lambda(T)x(T)e^{-\rho T} = 0.$$

The costate variable's differential equation,

$$\dot{\lambda} = \rho\lambda - H_x,$$

may be written as an arbitrage equation for the determination of price λ , that is:

$$\frac{\dot{\lambda} + H_x}{\lambda} = \rho. \quad (\text{B.43})$$

The rate of return on capital obtained by dividing the sum of the profit from an additional unit of capital with the capital gain by its price should, in equilibrium, be equal to the interest rate. This equation's solution determines the price of capital λ , according to:

$$\lambda(t) = \int_t^{\infty} e^{-\rho(\tau-t)} H_x d\tau. \quad (\text{B.44})$$

B.4 Linear Optimal Control

In some optimal control problems, the Hamiltonian is linear in the control variable. That is:

$$H = f(x, u) + \lambda g(x, u) = F(x, \lambda) + \sigma(x, \lambda)u, \quad (\text{B.45})$$

where F and σ are functions that do not depend on the control variable. Assume that the control variable has both upper ($\bar{u}$) and lower ($\underline{u}$) bounds, according to:

$$\underline{u} \leq u \leq \bar{u}. \quad (\text{B.46})$$

The $\sigma(x, \lambda)$ function is called a switching function. If $\sigma > 0$, the maximum of H occurs when $u = \bar{u}$. If $\sigma < 0$, the Hamiltonian maximum corresponds to the inferior bound. This kind of optimal control is called “bang-bang control” because the control variable oscillates between these two values, according to the sign of σ .

When values of x and λ exist such that $\sigma(x, \lambda) = 0$, a solution may exist for the optimal control problem that is not obtained by means of the maximum principle. In this case, the optimal control is called a singular control.

B.5 Comparative Dynamics

The solution of the infinite-horizon optimal control problem may be obtained from an autonomous system of differential equations for the costate (λ) and state (x) variables, when the optimal control value (u) in the $H_u = 0$ equation is substituted in the other first-order condition equations. The system is therefore given by:

$$\dot{\lambda} = \Gamma(\lambda, x, \alpha) \quad (\text{B.47})$$

$$\dot{x} = G(\lambda, x, \alpha) \quad (\text{B.48})$$

$$x(0) = x_0 \quad (\text{B.49})$$

$$\lim_{T \rightarrow \infty} \lambda(T)x(T)e^{-\rho T} = 0 \quad (\text{B.50})$$

where α is a vector of parameters, the state variable's initial value is given, and the transversality condition must be satisfied. In the steady state, $\dot{\lambda} = \dot{x} = 0$, the costate and state variables depend on vector α . That is:

$$\bar{\lambda} = \bar{\lambda}(\alpha) \quad (\text{B.51})$$

$$\bar{x} = \bar{x}(\alpha). \quad (\text{B.52})$$

Comparative dynamics analyzes how the problem's optimal solution changes with the value of the parameter α . Four cases must be considered: (i) permanent, unanticipated; (ii) permanent, anticipated; (iii) transitory, unanticipated; (iv) transitory, anticipated.

Permanent Change: Unanticipated Versus Anticipated

Figure B.4 shows an unanticipated change in the α parameter to α_1 from α_0 . The phase diagram in Fig. B.4 assumes that the steady-state values of the costate and state variables will rise when this happens. This assumption will also be used in the other cases that this section analyzes. The state variable is predetermined at the initial moment, but the costate variable is free at the initial instant and may change values abruptly.

Figure B.5 shows the system's dynamics when the parameter α changes. At the initial moment, when the change occurs, the costate variable jumps to the new saddle path, the costate and state variables then converge to the new steady state.

Figure B.6 shows a permanent, anticipated change in parameter α . At the initial moment, that is, when $t = 0$, an announcement is made that the parameter will change to α_1 from α_0 in T periods.

Figure B.7 illustrates the dynamical system's behavior after the announcement of the change in the parameter α . At the time of the announcement, the costate variable jumps, that is, changes values suddenly to $\lambda(0_+)$. The state variable does not change values at the time of the announcement because it is a predetermined variable. At

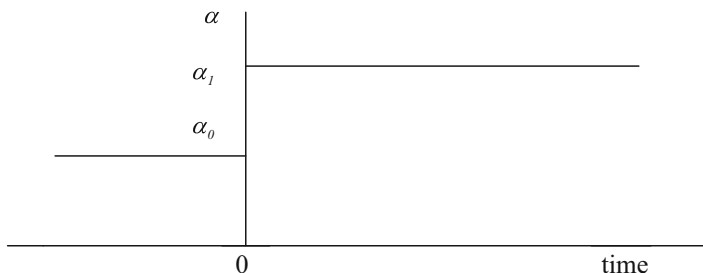


Fig. B.4 An unanticipated permanent increase in parameter α

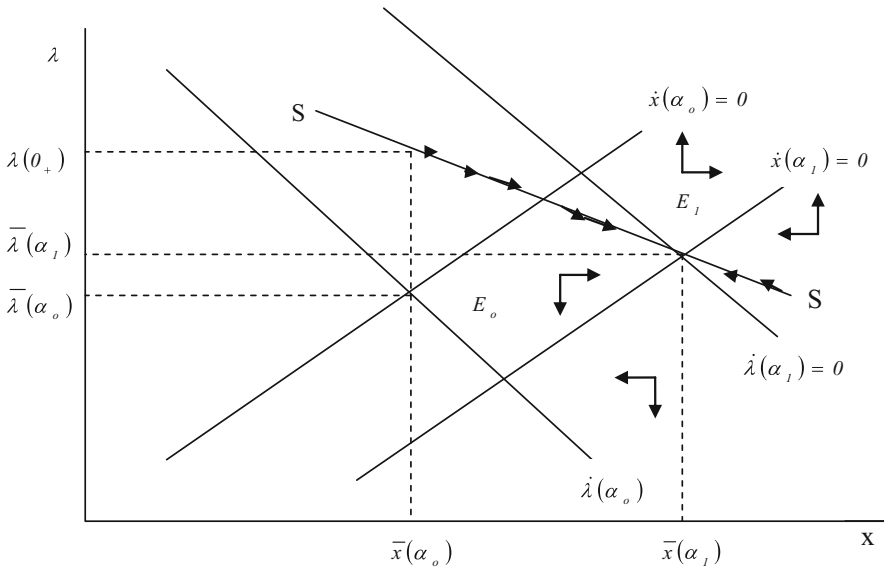


Fig. B.5 The effects of an unanticipated permanent increase in parameter α

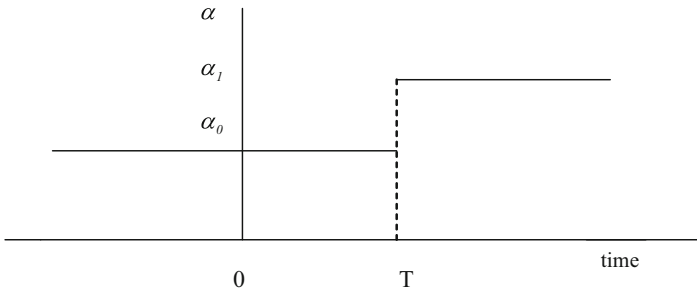


Fig. B.6 An anticipated permanent increase in parameter α

moment T , the differential equations system changes and finds its new equilibrium point at E_1 . The costate variable's jump at the initial moment must be such that the economy is at the new saddle path at time T , or else it would not converge to the new equilibrium point. The path in the time interval before the parameter changes follows the dynamics of the dashed arrows, due northeast. From moment T onward, the optimal path follows the saddle path until it reaches the new equilibrium point at E_1 .

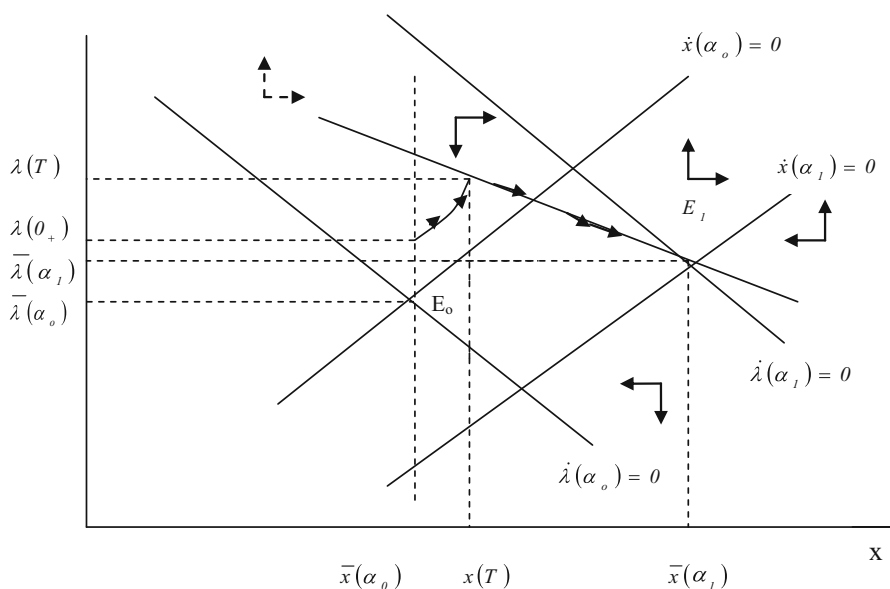


Fig. B.7 The effects of an anticipated permanent increase in parameter α

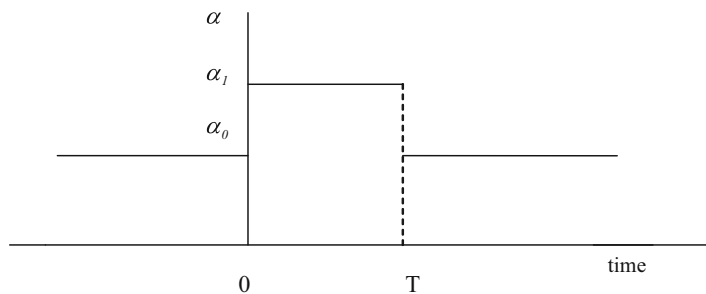


Fig. B.8 An anticipated transitory increase in parameter α

Transitory Change: Unanticipated Versus Anticipated

Figure B.8 shows a transitory, unanticipated change in parameter α to α_1 from α_0 at moment $t = 0$, but returning to the initial value after a time interval T . At the initial moment, the differential equations system shifts, passing through the new equilibrium point that corresponds to α_1 , as Fig. B.9 shows. The system's dynamics are determined by the figure's arrows. At moment T , the differential equations return to their original positions and the saddle path is represented by SS . The costate variable jumps to a new value at the initial moment, such that at moment T , the economy lies on the convergent saddle path SS . In the long term, the economy returns to its initial equilibrium.

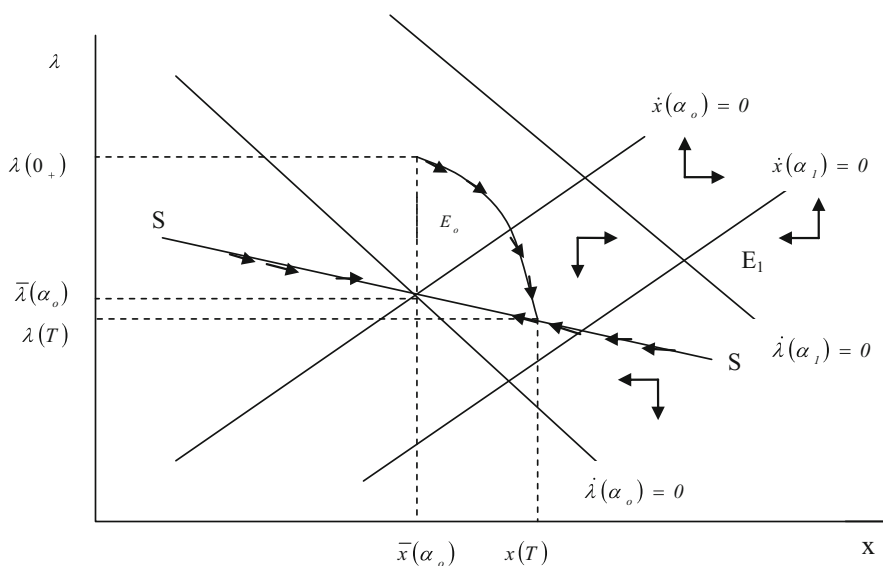


Fig. B.9 The effects of an unanticipated transitory increase in parameter α

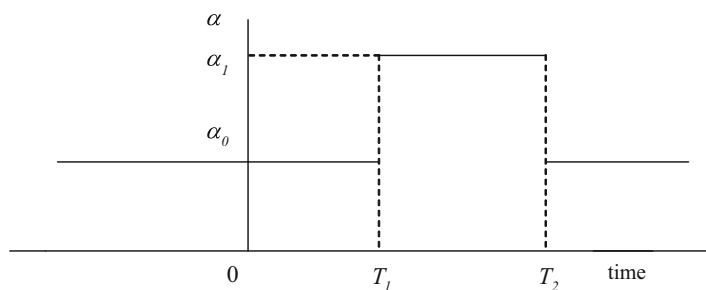


Fig. B.10 An anticipated transitory increase in parameter α

Figure B.10 describes an anticipated change in parameter α , to α_1 from α_0 within T_1 periods, returning to its initial value after a time interval T_2 .

At the time of the announcement, the costate variable changes values and jumps to $\lambda(0_+)$, as Fig. B.11 shows. At moment T_1 , when the parameter changes values, the system's dynamics will be governed by the new configuration of differential equations (orthogonal arrows). Up until then, the dynamics are given by the initial configuration (dashed orthogonal arrows). Therefore, after the costate variable's jump, the system follows a northeast path. At moment T_1 , when the parameter changes, the dynamics of the differential equations system is given by the new system, whose stationary equilibrium point is at E_1 , and the path heads southeast. At moment T_2 , the parameter α returns to its previous value. At this moment, the economy's path must meet the system's saddle path so that it may converge to its long-term stationary equilibrium.

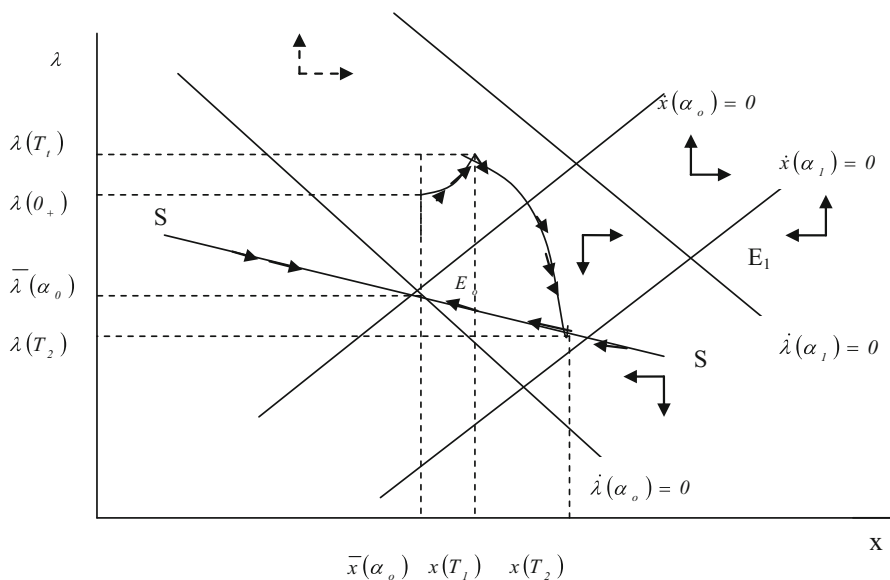


Fig. B.11 The effects of an anticipated transitory increase in parameter α

B.6 Exercises

1. (Calculus of Variations) Consider the problem:

$$\max \int_{t_0}^{t_1} F(t, x, \dot{x}) dt,$$

$$x(t_0) = x_0, \text{ given,}$$

$$x(t_1) \text{ free.}$$

- (a) Transform this problem into an optimal control problem by defining the control variable $u = \dot{x}$.
 (b) Deduce the necessary condition for solving the problem, the Euler equation:

$$\frac{d}{dt} \left(\frac{\partial F}{\partial \dot{x}} \right) = \frac{\partial F}{\partial x},$$

or, in its alternative form:

$$\frac{\partial^2 F}{\partial \dot{x} \partial \dot{x}} \ddot{x} + \frac{\partial^2 F}{\partial x \partial \dot{x}} \dot{x} + \frac{\partial^2 F}{\partial t \partial \dot{x}} - \frac{\partial F}{\partial x} = 0.$$

(c) Show that the transversality condition is given by:

$$\left. \frac{\partial F}{\partial \dot{x}} \right|_{t=t_1} = 0.$$

2. The representative agent maximizes:

$$\int_0^{\infty} e^{-\rho t} u(c) dt,$$

subject to the following constraints:

$$\begin{aligned} \dot{k} &= f(k) - c - \delta k \\ k(0) &= k_0 \end{aligned}$$

where c is consumption, $u(c)$ is the utility function, ρ is the rate of time preference, k is capital stock, $f(k)$ is the production function, and δ is the capital depreciation rate. The utility and output functions show their traditional properties. Use λ to denote the problem's costate variable.

- (a) Use a phase diagram with λ on the vertical axis and k on the horizontal axis to analyze this model's equilibrium and dynamics.
 - (b) Use calculus of variations (exercise 1) to solve this problem.
3. An individual, when at a disequilibrium situation, has an adjustment cost (C) with two components. One is the cost of not being in equilibrium ($x - \bar{x}$) and the other is the cost of change ($\dot{x}$), according to the following function:

$$C = \frac{\alpha}{2} (x - \bar{x})^2 + \frac{1}{2} \dot{x}^2.$$

The individual solves the following problem:

$$\begin{aligned} \min \int_0^{\infty} e^{-\rho t} \left[\frac{\alpha}{2} (x - \bar{x})^2 + \frac{1}{2} \dot{x}^2 \right] dt, \\ x(0) = x_0, \text{ given.} \end{aligned}$$

Apply the Euler equation to show that the path of variable x is given by the differential equation:

$$\dot{x} = k(\bar{x} - x).$$

4. Consider the model:

$$\begin{aligned} \text{ISy} - \bar{y} &= -\alpha(r - \bar{r}) \\ \text{PC} : \dot{\pi} &= \beta(y - \bar{y}). \end{aligned}$$

The Central Bank's loss function is given by:

$$L = \frac{1}{2} (\pi - \bar{\pi})^2 + \frac{\varphi}{2} (y - \bar{y})^2.$$

The Central Bank determines the nominal interest rate by solving the following problem:

$$\min \int_0^\infty e^{-\delta t} \left[\frac{1}{2} (\pi - \bar{\pi})^2 + \frac{\varphi}{2} (y - \bar{y})^2 \right] dt,$$

subject to the following constraint:

$$\dot{\pi} = \beta(y - \bar{y}),$$

$$p(0) \text{ and } \pi(0) \text{ given,}$$

where δ is the Central Bank's discount rate and the initial price level and inflation rate are both given.

Deduce the formula of the nominal interest rate set by the Central Bank.

5. A consumer decides the time structure of their consumption by solving the following problem:

$$\max \int_0^\infty e^{-\rho t} u(c) dt,$$

subject to the following constraints:

$$\dot{a} = ra + y - c,$$

$$a(0) = a_0, \quad \text{given,}$$

where ρ is the rate of time preference, $u(c)$ is the consumption function, $u'(c) > 0$, $u''(c) < 0$, a is the total of financial assets, y is the income level, and c is consumption.

- (a) Assume stationary equilibrium. Analyze the model's equilibrium and dynamics on a phase diagram with c on the vertical axis and a on the horizontal axis.
- (b) In the previous item's case, use the phase diagram to show what happens when income undergoes the following changes:
 - (i) permanent, unanticipated;
 - (ii) permanent, anticipated;
 - (iii) transitory, unanticipated;
 - (iv) transitory, anticipated.
- (c) In the previous item's case, do transitory changes have permanent effects (unit root problem)?

(d) Assume no stationary equilibrium. Deduce the consumption equation when:

$$u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}, \quad \sigma \neq 1,$$

$$u(c) = \log c, \quad \sigma = 1.$$

6. (Tobin's q) A business firm's cash flow is obtained by subtracting the cost of labor (ωL), investment (I), and the cost of installation ($C(I)$) from its revenues. The firm then solves the following problem:

$$\max \int_0^\infty e^{-\rho t} [pQ(K, L) - \omega L - I - C(I)] dt,$$

subject to the following constraints:

$$\begin{aligned} \dot{K} &= I - \delta K, \\ K(0) &= K_0, \quad \text{given,} \end{aligned}$$

$Q = Q(K, L)$ is the production function with constant returns to scale, which depends on capital (K) and labor (L), p is the price of the product, ω is wages, I is investment, δ is the capital depreciation rate. The installation cost function is given by:

$$C(I) = \alpha \frac{I^2}{K}.$$

- (a) Use the letter q (Tobin's q) to represent the costate variable and analyze the model's equilibrium and dynamics on a phase diagram with q on the vertical axis and K on the horizontal axis.
 (b) Show that:

$$q(0)K_0 = V_0,$$

where V_0 is the value of the firm. Interpret this result.

- (c) What happens when the parameter α equals zero?

7. The objective of monetary policy is to maximize the present value of seigniorage,

$$\frac{\dot{M}}{P} = \frac{\dot{M}}{M} \frac{M}{P} = \mu m,$$

where $\mu = \dot{M}/M$, $m = M/P$, M is the monetary base, and P is the price level. That is:

$$\max \int_0^\infty e^{-\rho t} \mu m dt,$$

subject to the constraint:

$$\dot{m} = \mu m - \tau(m),$$

where $\tau(m) = \pi m$ is the inflation tax. Assume rational expectations, that the initial price level $P(0)$ is an endogenous variable, and that the monetary base at the initial instant, $M(0)$, is given.

(a) Show that the first-order condition implies that:

$$\tau'(m) = -\rho,$$

rather than $\tau'(m) = 0$, which maximizes the inflation tax.

(b) Show that this policy is inconsistent because the Central Bank is under an incentive to change it, increasing the monetary expansion rate to a level compatible with the maximization of the inflation tax.

8. Using the constraint:

$$\dot{m} = \mu m - \tau(m),$$

show that:

$$\int_0^\infty e^{-\rho t} \mu m dt = \int_0^\infty e^{-\rho t} \tau(m) dt + \infty \int_0^\infty e^{-\rho t} \rho (m - m(0)) dt,$$

assuming $\lim_{t \rightarrow \infty} m e^{-\rho t} = 0$

9. ('Honest' Government Policy). Assume that the objective of the monetary policy is to maximize the present value of seigniorage, but that the price level at the initial moment $P(0)$ is kept constant by means of variations in the initial stock of money. Under these circumstances, the government is honest because it allows no capital gains or losses for money holders.

The Central Bank's objective therefore consists of maximizing,

$$\int_0^\infty e^{-\rho t} \mu m dt + m(0) - m(0_-),$$

where $m(0) - m(0_-)$ is the variation of the real stock of money at the initial moment.

Show that no dynamic inconsistency occurs in this case because the Central Bank is under no incentive to change the monetary policy.

10. The objective of the monetary policy is to maximize the present value of the seigniorage,

$$\int_0^\infty e^{-\rho t} \mu m \, dt,$$

subject to the constraint,

$$\dot{m} = -\frac{\alpha\beta}{1-\alpha\beta}\mu m - \frac{\beta}{1-\alpha\beta}m \log m,$$

and initial condition,

$$m(0) = m_0,$$

(a) Show that the state equation is obtained from the money demand equation,

$$\log m = -\alpha\pi^e.$$

and the adaptive expectations mechanism,

$$\dot{\pi}^e = \beta(\pi - \pi^e),$$

where α and β are positive parameters and π^e is the expected inflation rate.

(a) Show that no optimal control exists for this problem unless $m_0 = \bar{m}$, where $\bar{m}$ is the stationary solution of linear control.

(b) Suppose that the Central Bank may inject or remove money at the initial moment. Is there an optimal monetary policy in this case?

11. (Dynamic Inconsistency) The economy produces a consumer product (c) whose output is affected by tax (x) according to:

$$\begin{aligned} c &= f(x), f(x) > 0, \underline{x} < x < \bar{x}, \underline{x} < 0, \bar{x} > 0, \\ f(\underline{x}) &= f(\bar{x}) = 0, f''(x) < 0, f(0) > 0, f'(0) = 0. \end{aligned}$$

The real quantity of money demanded ($m = M/P$) depends on the expected inflation rate:

$$\log m^d = -\alpha\pi^e, \alpha > 0.$$

Expectations are rational and the money market is at equilibrium:

$$\begin{aligned} \pi^e &= \pi \\ m^d &= m^s \end{aligned}$$

The government budget constraint is expressed by:

$$\frac{\dot{M}}{P} = -x.$$

The representative agent maximizes their welfare. The utility function is additive:

$$U(c, m) = u(c) + v(m),$$

with the $u(c)$ and $v(m)$ functions following the standard assumptions.

- (a) Show that the representative agent's problem consists of maximizing:

$$\int_0^{\infty} e^{-\rho t} [u(c(f(x))) + v(m)] dt,$$

subject to the constraint:

$$\dot{m} = m \frac{\log m}{\alpha} - x.$$

- (b) Analyze the model's equilibrium and dynamics on a phase diagram with the costate variable on the vertical axis and the real quantity of money on the horizontal axis.
- (c) The initial real quantity of money (m_0) is not given in this problem. It is an endogenous model variable. The transversality condition requires the costate variable to be zero at the initial moment, $\lambda(0) = 0$. What happens under these circumstances?

12. The Central Bank controls the growth rate of the monetary base μ and sets its value by minimizing:

$$\int_0^{\infty} e^{-\rho t} \left[\frac{\varphi}{2} x^2 + \frac{1}{2} \pi^2 \right] dt,$$

where ρ is the real interest rate, x is the output gap, π is the inflation rate, and φ is a positive parameter, subject to the following constraints:

$$\dot{\pi} = \alpha(\pi - \mu) + \beta x$$

$$\dot{x} = \gamma(\pi - \mu) + \delta x$$

$$\pi(0) = \mu_0$$

$$x(0) = 0$$

- (a) The Hamiltonian (H) is linear for control variable μ . Determine a solution for the model such that H is independent from the control variable.
- (b) Analyze the solution's equilibrium and dynamics on a phase diagram with π on the vertical axis and x on the horizontal axis.
13. The society has a non-renewable resource whose current stock equals S . Consumption (c) of this resource must satisfy the condition:

$$\int_0^T c(t)dt = S,$$

where T is the horizon of time over which the resource will be fully consumed. The remaining stock (x) of the non-renewable resource at time t is therefore defined by:

$$x(t) = S - \int_0^t c(\tau)d\tau.$$

The consumer maximizes:

$$\int_0^T e^{-\rho t} u(c)dt,$$

subject to the following conditions:

$$\dot{x}(t) = -c(t)$$

$$x(0) = S$$

$$x(T) = 0$$

- (a) Determine the optimal consumption path.
- (b) What happens to the path when $\rho = 0$?

14. A firm extracts a non-renewable resource that has a stock equal to S :

$$\int_0^T q(t)dt = S.$$

The remaining stock (x) in period t is therefore given by:

$$x(t) = S - \int_0^t q(\tau)d\tau.$$

The resource's cost of extraction depends on the quantity extracted (q) and the remaining stock according to:

$$C = C(q, x), C_q > 0; C_x < 0,$$

$$C_{qq} < 0; C_{xx} > 0.$$

The firm maximizes its cash flow,

$$\int_0^T e^{-\rho t} [pq - C(q, x)]dt,$$

subject to the following constraints:

$$\begin{aligned}\dot{x} &= -q \\ x(0) &= S \\ 0 &< q < S\end{aligned}$$

- (a) When $C(q, x) = 0$, what is the problem's solution? Is the interest rate relevant to the firm's decision?
- (b) Assume that $C(q, x) = c(x)q$, with $c'(x) < 0$, $c''(x) > 0$. What is the optimal control solution in this case?

15. The stock of fish (x) evolves according to the transition equation:

$$\dot{x} = f(x) - q,$$

where q is the quantity fished and the reproduction function $f(x)$ has the following properties: $f(x) > 0$, $f'(x) > 0$, $f(0) = f(S) = 0$.

A fishery solves the following problem:

$$\max \int_0^{\infty} e^{-\rho t} [p - c(x)] q dt,$$

subject to the constraints:

$$\begin{aligned}\dot{x} &= f(x) - q \\ x(0) &= 0 \\ 0 &\leq q \leq s\end{aligned}$$

where $c(x)$, $c'(x) < 0$ is the unit cost of fishing.

- (a) Determine this firm's optimal control.
- (b) Repeat the previous item when the $f(x)$ function is given by:

$$f(x) = \alpha x \left(1 - \frac{x}{s} \right).$$

Appendix C: Difference Equations

This appendix presents the stability conditions for first-order linear difference equations. The results are applied to study the stability of the Keynesian and the new Keynesian models.

C.1 First-Order Linear Difference Equations

The endogenous variable, in the first-order difference equation we analyze here, depends on its expected value, according to:

$$y_t = \alpha E_t y_{t+1} + \beta x_t, y_0 \text{ free}, \quad (\text{C.1})$$

where y and x are scalars, and x is a bounded and well-behaved stochastic process. We examine three cases for the solution of this equation, namely: (i) $|\alpha| < 1$, (ii) $|\alpha| > 1$, and (iii) $|\alpha| = 1$.

When $|\alpha| < 1$, we use the forward operator F ($Fx_t = x_{t+1}$) to write:

$$E_t[(1 - \alpha F)y_t - \beta x_t] = 0.$$

After inverting the F polynomial, $(1 - \alpha F)^{-1} = 1 + \alpha F + \alpha^2 F^2 + \dots$ we get the solution:

$$y_t = E_t \sum_{i=0}^{\infty} \beta \alpha^i x_{t+i}. \quad (\text{C.2})$$

When $|\alpha| > 1$, the difference Eq. (C.1) can be written as:

$$y_t = \frac{1}{\alpha} y_{t-1} - \frac{\beta}{\alpha} x_{t-1} + \varepsilon_t, \quad (\text{C.3})$$

where ε is a stochastic variable with expected value zero, $E_t \varepsilon_{t+1} = 0$. Therefore, when $|\alpha| > 1$, the model is indeterminate because the initial values y_0 and ε are arbitrary. We can use the lag operator L ($Lx_t = x_{t-1}$) to write:

$$\left(1 - \frac{1}{\alpha}L\right)y_t = -\frac{\beta}{\alpha}\left(x_{t-1} - \frac{\alpha}{\beta}\varepsilon_t\right).$$

Dividing both sides by the L polynomial and using $(1 - \alpha^{-1}L)^{-1} = 1 + \alpha^{-1}L + \alpha^{-2}L^2 + \dots$ yields the solution:

$$y_t = -\frac{\beta}{\alpha} \sum_{i=0}^{\infty} \left(\frac{1}{\alpha}\right)^i \left(x_{t-1-i} - \frac{\alpha}{\beta}\varepsilon_{t-i}\right) \quad (\text{C.4})$$

When $|\alpha| < 1$, the solution of Eq. (C.1) is forward looking and when $|\alpha| > 1$, the solution of Eq. (C.1) is backward looking. When $|\alpha| = 1$, Eq. (C.1) has two solutions, one forward and the other backward looking. The forward solution is given by:

$$y_t = E_t \sum_{i=0}^{\infty} \beta x_{t+i}, \quad (\text{C.5})$$

and the backward solution is given by:

$$y_t = -\sum_{i=0}^{\infty} (\beta x_{t-i} - \varepsilon_{t-i}), \quad (\text{C.6})$$

where ε_t is an arbitrary stochastic process, such that $E_t \varepsilon_{t+1} = 0$.

Linear System

Now consider the linear model:

$$y_t = AE_t y_{t+1} + \varepsilon_t, \quad (\text{C.7})$$

where y is a vector with two components and the matrix A is given by:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

The determinant and trace of matrix A are:

$$\begin{aligned} \det A &= a_{11}a_{22} - a_{21}a_{12}, \\ \text{tr} A &= a_{11} + a_{22}. \end{aligned}$$

The characteristic roots of matrix A are obtained by solving the characteristic equation:

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0.$$

This equation is equivalent to:

$$\lambda^2 - (\text{tr}A)\lambda + \det A = 0.$$

Thus, the equation's roots are related to the trace and the determinant according to:

$$\text{tr}A = \lambda_1 + \lambda_2$$

$$\det A = \lambda_1 \lambda_2$$

We would like to establish the necessary and sufficient conditions for the roots to lie within the $(-1, 1)$ interval, outside of this interval, and also, for a root to be less than one in absolute value and for the other to be greater than one in absolute value. Let the polynomial be:

$$p(\lambda) = \lambda^2 - (\text{tr}A)\lambda + \det A = (\lambda - \lambda_1)(\lambda - \lambda_2). \quad (\text{C.8})$$

Hence,

$$p(1) = (1 - \lambda_1)(1 - \lambda_2) = 1 - \text{tr}A + \det A, \quad (\text{C.9})$$

$$p(-1) = (1 + \lambda_1)(1 + \lambda_2) = 1 + \text{tr}A + \det A. \quad (\text{C.10})$$

- I. Let $p(1) = 1 - \text{tr}A + \det A > 0$ and $p(-1) = 1 + \text{tr}A + \det A > 0$. Then, if $|\det A| > 1$, the roots will be greater than one in absolute values: $|\lambda_1| > 1$ and $|\lambda_2| > 1$. On the other hand, if $|\det A| < 1$, the roots will be less than one in absolute values: $|\lambda_1| < 1$ and $|\lambda_2| < 1$.
- II. Let $p(1) = 1 - \text{tr}A + \det A < 0$ and $p(-1) = 1 + \text{tr}A + \det A < 0$. Then the roots will be greater than one in absolute values: $|\lambda_1| > 1$ and $|\lambda_2| > 1$.
- III. Let $p(1)p(-1) = (1 - \text{tr}A + \det A)(1 + \text{tr}A + \det A) < 0$. Then one root will be greater and the other will be less than one in absolute values: $|\lambda_1| < 1$ and $|\lambda_2| > 1$.

C.2 Forward- and Backward-looking Models

The forward-looking model is:

$$y_t = AE_t y_{t+1} + Bx_t, y_0 \text{ free.} \quad (\text{C.11})$$

Assume that the characteristic roots of the matrix A lie within the unit circle, Λ is the diagonal matrix of the eigenvalues of A , and Q is the matrix of the eigenvectors of A . Hence,

$$A = Q\Lambda Q^{-1}.$$

Using this property, the difference Eq. (C.11) may be written as:

$$Q^{-1}y_t = \Lambda Q^{-1}E_t y_{t+1} + Bx_t.$$

This equation is equivalent to:

$$z_t = \Lambda E_t z_{t+1} + Q^{-1}Bx_t.$$

Using the linear transformation: $Q^{-1}y_t = z_t$ we obtain:

$$y_t = E_t \sum_{i=0}^{\infty} \Lambda^i Bx_{t+1}, \quad (\text{C.12})$$

by taking into account:

$$(I - \Lambda F)^{-1} = I + \Lambda F + \Lambda^2 F^2 + \dots$$

and that the powers of A are given by:

$$A^i = Q\Lambda^i Q^{-1}.$$

When the roots of A lie outside the unit circle, the roots of the inverse matrix lie within the unit circle. In this case, the solution of the Eq. (C.11) is expressed by:

$$y_t = A^{-1}y_{t-1} - A^{-1}Bx_{t-1} + \varepsilon_t, \quad (\text{C.13})$$

where ε_t is a stochastic variable with expected value zero: $E_t \varepsilon_{t+1} = 0$. The initial value and ε are arbitrary. Therefore, Eq. (C.11) has an infinite number of solutions. To verify that the above equation is a solution for Eq. (C.11), we write:

$$y_{t+1} = A^{-1}y_t - A^{-1}Bx_t + \varepsilon_{t+1}.$$

The expected value with respect to the information available at t is equal to:

$$E_t y_{t+1} = A^{-1}y_t - A^{-1}Bx_t.$$

Substituting this in Eq. (C.11) yields:

$$y_t = A(A^{-1}y_t - A^{-1}Bx_t) + Bx_t = y_t.$$

The backward-looking model is:

$$y_t = Ay_{t-1} + Bx_t, y_0 \text{ given} \quad (\text{C.14})$$

Assume that the roots of the matrix A lie within the unit circle. Therefore:

$$Q^{-1}y_t = \Lambda Q^{-1}y_{t-1} + Q^{-1}Bx_t.$$

Using the linear transformation $Q^{-1}y_t = z_t$, this equation may be written as:

$$z_t = \Lambda z_{t-1} + Q^{-1}Bx_t.$$

It is easily seen that:

$$y_t = \sum_{i=0}^{\infty} A^i Bx_{t-i}, \quad (\text{C.15})$$

by taking into account the inverse of the following matrix:

$$(I - \Lambda L)^{-1} = I + \Lambda L + \Lambda^2 L^2 + \dots$$

C.3 Keynesian Model

The Keynesian model includes an IS curve, a Phillips curve, and a monetary policy rule according to the specification:

$$\text{IS: } x_t = \lambda x_{t-1} - \alpha(r_t - \bar{r}_t) \quad (\text{C.16})$$

$$\text{PC: } \pi_t = \pi_{t-1} + \kappa x_t, \quad (\text{C.17})$$

$$\text{MPR: } r_t - \bar{r}_t = \phi(\pi_t - \bar{\pi}) + \theta x_t \quad (\text{C.18})$$

The notation is the following: π is the inflation rate, x is the output gap, r is the real interest rate, and all other symbols are positive parameters.

Combining the IS curve and the monetary policy rule yields:

$$x_t = \lambda x_{t-1} - \alpha\phi(\pi_t - \bar{\pi}) - \alpha\theta x_t.$$

This equation and the Phillips curve may be written as a difference equation system. That is:

$$\begin{bmatrix} 1 & -\kappa \\ \alpha\phi & 1 + \alpha\theta \end{bmatrix} \begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} \pi_{t-1} \\ x_{t-1} \end{bmatrix}.$$

Inverting the matrix on the left side of the system yields:

$$\begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = A \begin{bmatrix} \pi_{t-1} \\ x_{t-1} \end{bmatrix}. \quad (\text{C.19})$$

The matrix A is given by:

$$A = \frac{1}{1 + \alpha\theta + \alpha\phi\kappa} \begin{bmatrix} 1 + \alpha\theta & \kappa\lambda \\ -\alpha\phi & \lambda \end{bmatrix}. \quad (\text{C.20})$$

The determinant and trace of the A matrix are equal to:

$$\det A = \frac{\lambda}{1 + \alpha\theta + \alpha\phi\kappa},$$

$$\text{tr } A = \frac{1 + \alpha\theta + \lambda}{1 + \alpha\theta + \alpha\phi\kappa}.$$

The determinant and trace of the A matrix are both positive. Therefore, $1 + \text{tr}A + \det A > 0$ and

$$1 - \text{tr}A + \det A = \frac{\alpha\phi\kappa}{1 + \alpha\theta + \alpha\phi\kappa}.$$

It follows that if $\phi > 0$, the model's equilibrium is determinate.

C.4 New Keynesian Model

The new Keynesian model has two equations: a Phillips curve and an IS curve:

$$\text{PC: } \pi_t = \beta E_t \pi_{t+1} + \kappa x_t, \quad (\text{C.21})$$

$$\text{IS} x_t = E_t x_{t+1} - \sigma(r_t - \bar{r}_t) \quad (\text{C.22})$$

The notation is the same as in the previous model, E is the expectations operator, and all other symbols are positive parameters.

Central Bank Sets the Real Interest Rate

Assume that the Central Bank sets the nominal interest rate so that it controls the real interest rate gap. A model with such a monetary policy rule can be reduced to the following system of difference equation:

$$\begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = A \begin{bmatrix} E_t \pi_{t+1} \\ E_t x_{t+1} \end{bmatrix} + \begin{bmatrix} 0 \\ \sigma(r_t - \bar{r}_t) \end{bmatrix}. \quad (\text{C.23})$$

The model has a unique equilibrium only if both of the eigenvalues of matrix A lie within the unit circle. The A matrix is given by:

$$A = \begin{bmatrix} \beta & \kappa \\ 0 & 1 \end{bmatrix}. \quad (\text{C.24})$$

The determinant and trace of this matrix are:

$$\det A = \beta, \quad \text{tr}A = 1 + \beta.$$

Both of matrix A 's roots lie within the unit circle only if the conditions in case I, for stability analysis of the linear system, are satisfied. It is easily seen that one root is equal to and the other is less than one. Therefore, the model's equilibrium is not unique.

According to the well-known Sargent and Wallace proposition, an interest rate rule that specifies an exogenous path for the nominal interest rate produces an indeterminate equilibrium. This proposition is therefore also valid for an exogenous real interest rate path.

The Phillips curve and the IS curve can be written as:

$$\pi_t = E_t \sum_{i=0}^{\infty} \kappa \beta^i x_{t+i}, \quad (\text{C.25})$$

$$\Delta x_t = \sigma(r_{t-1} - \bar{r}_{t-1}). \quad (\text{C.26})$$

Combining the two expressions, the acceleration of the inflation rate depends on the expected value of the interest gaps:

$$\Delta \pi_t = \kappa \sigma E_t \sum_{i=0}^{\infty} \beta^i (r_{t+i-1} - \bar{r}_{t+i-1}). \quad (\text{C.27})$$

The conclusion from this model is that the change in the output gap is backward-looking, while inflation rate acceleration is forward-looking.

Central Bank Uses the Taylor Rule

The new Keynesian model is specified as follows:

$$\text{PC: } \pi_t = \beta E_t \pi_{t+1} + \kappa x_t, \quad (\text{C.28})$$

$$\text{IS: } x_t = E_t x_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \bar{r}_t), \quad (\text{C.29})$$

$$\text{MPR: } i_t = \bar{r}_t + (1 + \phi)\pi_t + \theta x_t, \quad (\text{C.30})$$

where i is the nominal interest rate. Combining the IS curve and the monetary policy rule yields:

$$(1 + \sigma\theta)x_t + \sigma(1 + \phi)\pi_t = E_t x_{t+1} + \sigma E_t \pi_{t+1}.$$

This equation and the Phillips curve can be written as the following system of difference equations:

$$\begin{bmatrix} 1 & -\kappa \\ \sigma(1 + \phi) & 1 + \sigma\theta \end{bmatrix} \begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = \begin{bmatrix} \beta & 0 \\ \sigma & 1 \end{bmatrix} \begin{bmatrix} E_t \pi_{t+1} \\ E_t x_{t+1} \end{bmatrix}.$$

Inverting the matrix on the left side of the equation yields the following system:

$$\begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = A \begin{bmatrix} E_t \pi_{t+1} \\ E_t x_{t+1} \end{bmatrix}, \quad (\text{C.31})$$

where the matrix A is:

$$A = \frac{1}{1 + \sigma\theta + \sigma(1 + \phi)\kappa} \begin{bmatrix} (1 + \sigma\theta)\beta + \sigma\kappa & \kappa \\ -\sigma(1 + \phi)\beta + \sigma & 1 \end{bmatrix}. \quad (\text{C.32})$$

The determinant and trace of this matrix are equal to:

$$\det A = \frac{\beta}{1 + \sigma\theta + \sigma(1 + \phi)\kappa},$$

$$\text{tr} A = \frac{(1 + \sigma\theta)\beta + \sigma\kappa + 1}{1 + \sigma\theta + \sigma(1 + \phi)\kappa}.$$

The determinant and trace are both positive and $1 + \text{tr} A + \det A > 0$. Therefore, if:

$$\theta(1 - \beta) + \phi\kappa > 0,$$

the model has a unique equilibrium because:

$$1 - \text{tr} A + \det A = \frac{\sigma(\theta(1 - \beta) + \phi\kappa)}{1 + \sigma\theta + \sigma(1 + \phi)\kappa} > 0.$$

C.5 Encompassing Keynesian Model

The encompassing specification of the Keynesian model, with discrete variables, is given by:

$$\text{IS: } \Delta x_t = \gamma x_{t-1} + \alpha_0(r_t - \bar{r}_t) + \alpha_1(r_{t-1} - \bar{r}_{t-1}) + \varepsilon_t, \quad (\text{C.33})$$

$$\text{PC: } \Delta \pi_t = \rho \pi_{t-1} + \kappa_0 x_t + \kappa_1 x_{t-1} + \mu, \quad (\text{C.34})$$

where ε_t and μ_t are stochastic variables. Table C.1 contains the signs of the parameters and the initial conditions for each Keynesian model: Keynesian (K), New Keynesian (NK), New Keynesian with Information Rigidity (NKIR) (see Chapter 7).

Table C.1 Keynesian models: old and new

Models	Parameters						Initial conditions
	γ	α_0	α_1	κ_0	κ_1	ρ	
Keynesian (K)	<0	<0	0	>0	0	0	$p(0), \pi(0)$
New Keynesian (NK)	0	0	>0	0	<0	>0	$p(0)$
New Keynesian Information Rigidity (NKIR)	0	0	>0	>0	<0 ($\kappa_0 + \kappa_1 > 0$)	0	$p(0), \pi(0)$

The monetary policy rule is the Taylor rule:

$$r_t - \bar{r} = \phi(\pi_t - \bar{\pi}) + \theta x_t. \quad (\text{C.35})$$

Combining this rule and the IS curve yields the following equation:

$$(1 - \alpha_0\theta)x_t = (1 + \gamma + \alpha_1\theta)x_{t-1} + \alpha_0\phi\pi_t + \alpha_1\phi\pi_{t-1}.$$

We have assumed, for simplicity, that the inflation target equals zero, $\bar{\pi} = 0$, and that the stochastic terms, too, are zero. This equation and the Phillips curve forms the following system of difference equations:

$$\begin{bmatrix} 1 & -\kappa_0 \\ -\alpha_0\phi & 1 - \alpha_0\theta \end{bmatrix} \begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = \begin{bmatrix} 1 + \rho & \kappa_1 \\ \alpha_1\phi & 1 + \gamma + \alpha_1\theta \end{bmatrix} \begin{bmatrix} \pi_{t-1} \\ x_{t-1} \end{bmatrix}.$$

Therefore, inverting the matrix on the left of the equation yields the system of difference equations for the inflation rate and the output gap:

$$\begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = A \begin{bmatrix} \pi_{t-1} \\ x_{t-1} \end{bmatrix}. \quad (\text{C.36})$$

The matrix A is:

$$A = \frac{1}{1 - \alpha_0(\theta + \phi\kappa_0)} \begin{bmatrix} (1 - \alpha_0\theta)(1 + \rho) + \kappa_0\alpha_1\phi & (1 - \alpha_0\theta)\kappa_1 + \kappa_0(1 + \gamma + \alpha_1\theta) \\ \alpha_0(1 + \rho) + \alpha_1\phi & \alpha_0\phi\kappa_1 + 1 + \gamma + \alpha_1\theta \end{bmatrix}. \quad (\text{C.37})$$

The determinant and trace of this matrix are equal to:

$$\det A = \frac{(1 + \gamma + \alpha_1\theta)(1 + \rho) - \alpha_1\phi\kappa_1}{1 - \alpha_0(\theta + \phi\kappa_0)},$$

$$\text{tr} A = \frac{(1 - \alpha_0\theta)(1 + \rho) + \phi(\alpha_0\kappa_1 + \alpha_1\kappa_0) + 1 + \gamma + \alpha_1\theta}{1 - \alpha_0(\theta + \phi\kappa_0)}.$$

To analyze the stability of the K, NK, and NKIR models, we must calculate the signs of $1 + \text{tr} A + \det A$ and $1 - \text{tr} A + \det A$. Table C.2 contains this information. For

Table C.2 Stability of the Keynesian models

Models	$1 + \text{tr} A + \det A$	$1 - \text{tr} A + \det A$	$ A $	Roots
Keynesian (K)	>0	>0	<1	$ \lambda_1 < 1$ $ \lambda_2 < 1$
New Keynesian (NK)	>0	<0	>1	$ \lambda_1 > 1$ $ \lambda_2 > 1$
New Keynesian Information Rigidity (NKIR)	>0	>0	X	$ \lambda_1 > 1$ $ \lambda_2 < 1$

the Keynesian model, $1 + \text{tr}A + \det A > 0$, $1 - \text{tr}A + \det A > 0$ and the absolute value of the determinant is less than one, the two characteristic roots are less than one, in absolute values, and the model's equilibrium is determinate. For the new Keynesian model, $1 + \text{tr}A + \det A > 0$, $1 - \text{tr}A + \det A > 0$ and the determinant of A is greater than one, in absolute values, the two characteristic roots are greater than one, and the model's equilibrium is determinate. For the rigid information model, $1 + \text{tr}A + \det A > 0$ and $1 - \text{tr}A + \det A < 0$. Therefore, one characteristic root is greater and the other is less than one. Therefore, the equilibrium point is a saddle point.

C.6 Hybrid Models

Consider the model:

$$y_t = A_t E_t y_{t+1} + A_b y_{t-1} + Cx_t, \quad y_0 \text{ free and } y_{-1} \text{ given.} \quad (\text{C.38})$$

If $A_f = 0$ and the roots of A_b lie within the unit circle, the model is backward-looking.

If $A_b = 0$ and the roots of A_f lie within the unit circle, the model is forward looking.

The z and ν vectors are defined by:

$$z_t = \begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix}, \quad \nu_t = \begin{bmatrix} -Cx_t \\ 0 \end{bmatrix}.$$

The model can be written in matrix form according to:

$$\begin{bmatrix} A_f & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} E_t y_{t+1} \\ E_t y_t \end{bmatrix} = \begin{bmatrix} I & -A_b \\ I & 0 \end{bmatrix} \begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} -Cx_t \\ 0 \end{bmatrix}.$$

Inverting the matrix on the left side of the expression and using the previous definitions, the system may be rewritten as:

$$E_t z_{t+1} = A z_t + \nu_t. \quad (\text{C.39})$$

The matrix A is:

$$A = \begin{bmatrix} A_f^{-1} & -A_f^{-1}A_b \\ I & 0 \end{bmatrix}. \quad (\text{C.40})$$

This dynamical system admits a unique solution only if the characteristic roots of the characteristic equation:

$$|A - \lambda I| = 0,$$

lie outside the unit circle. This characteristic equation may be rewritten as:

$$\det \begin{bmatrix} A_f^{-1} - \lambda I & -A_f^{-1}A_b \\ I & -\lambda I \end{bmatrix} = 0.$$

The determinant of the partitioned matrix:

$$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix},$$

is equal to [see, Graybill (1969), p.164–166]:

$$|B| = |B_{22}| |B_{11} - B_{12}B_{22}^{-1}B_{21}|.$$

Applying this formula to the previous characteristic equation yields:

$$|A - \lambda I| = |-\lambda I| |A_f^{-1} - \lambda I - A_f^{-1}A_b\lambda^{-1}|.$$

Therefore, the necessary and sufficient condition for the rational expectations equilibrium to be unique is that n roots of the characteristic polynomial:

$$p(\lambda) = |A_f^{-1} - \lambda I - A_f^{-1}A_b\lambda^{-1}|$$

lie outside the unit circle and n roots lie within it.

Particular Case: One Equation

Consider a model in which the endogenous variable depends on its expected value in the following period, on its lagged value from the previous period, and on the exogenous process of $\{x_t\}$, according to:

$$y_t = a_t E_t y_{t+1} + a_b y_{t-1} + b x_t, y_{-1} \text{ given}, y_0 \quad (\text{C.41})$$

Let:

$$z_t = \begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix}, \quad \nu_t = \begin{bmatrix} -c x_t \\ 0 \end{bmatrix}.$$

The model can be written in matrix form as:

$$\begin{bmatrix} a_f & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} E_t y_{t+1} \\ E_t y_t \end{bmatrix} = \begin{bmatrix} 1 & -a_b \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} -c x_t \\ 0 \end{bmatrix}.$$

Inverting the matrix on the left side of the expression, this equation may be written as:

$$E_t z_{t+1} = A z_t + \nu_t. \quad (\text{C.42})$$

The matrix A is:

$$A = \begin{bmatrix} \frac{1}{a_f} & -\frac{a_b}{a_f} \\ 1 & 0 \end{bmatrix}. \quad (\text{C.43})$$

The eigenvalue polynomial associated with this matrix is given by:

$$p(\lambda) = \lambda^2 - \frac{1}{a_f}\lambda + \frac{a_b}{a_f}.$$

The sum and the product of the roots are:

$$\lambda_1 + \lambda_2 = \frac{1}{a_f}; \quad \lambda_1 \lambda_2 = \frac{a_b}{a_f}$$

The model has a unique equilibrium only if one root is outside and the other is within the unit circle. Therefore, the following condition must be satisfied:

$$a_b + a_f < 1.$$

The model is indeterminate, that is, it has multiple solutions when the characteristic polynomial is such that $a_b + a_f > 1$, the roots are outside the unit circle if $a_f < a_b$, and are within if $a_f > a_b$.

Equation (C.41) may be written forward- or backward-looking depending on the characteristic roots of the polynomial. Equation (C.41) can be written as follows:

$$E_t \left[a_f \left(y_{t+1} + \frac{a_b}{a_f} y_{t-1} - \frac{1}{a_f} y_t \right) + b x_t \right] = 0$$

Using forward operator F and lag operator L , this expression may be rewritten as:

$$E_t \left[a_f L \left(F^2 - \frac{1}{a_f} F + \frac{a_b}{a_f} \right) y_t + b x_t \right] = 0$$

The polynomial F can be decomposed as the product of two monomials:

$$F^2 - \frac{1}{a_f} F + \frac{a_b}{a_f} = (F - \lambda_1)(F - \lambda_2) = \frac{a_b}{a_f} \left(1 - \frac{1}{\lambda_1} F \right) \left(1 - \frac{1}{\lambda_2} F \right)$$

This expression assumes that the two roots are greater than one. If one root is less than one, the following transformation is used:

$$1 - \frac{1}{\lambda_1} F = -\frac{1}{\lambda_1 L} (1 - \lambda_1 L).$$

When a unique rational expectations equilibrium exists, one root will be within and the other outside the unit circle. Therefore, Eq. (C.41) may be written with backward- and forward-looking components:

$$E_t \left[-\frac{a_b}{\lambda_1} (1 - \lambda_1 L) \left(1 - \frac{1}{\lambda_2} F \right) y_t + b x_t \right] = 0$$

This expression may be rewritten as:

$$y_t = \frac{b\lambda_1}{a_b} E_t \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \lambda_1^j \left(\frac{1}{\lambda_2} \right)^i x_{t+i-j}, \quad (\text{C.44})$$

and this expression shows that the endogenous variable depends on past, present, and future expected values of the exogenous variable. Alternatively, the following solutions are obtained:

$$y_t = E_t \frac{1}{\lambda_2} y_{t+1} + \frac{\lambda_1 b}{a_b} \sum_{i=0}^{\infty} \lambda_1^i x_{t-i} \quad (\text{C.45})$$

$$y_t = \lambda_1 y_{t-1} + \frac{\lambda_1 b}{a_b} E_t \sum_{i=0}^{\infty} \left(\frac{1}{\lambda_2} \right)^i x_{t+i} \quad (\text{C.46})$$

In the former equation, the endogenous variable depends on its own expected value in the next period and on the past values of the exogenous variable. In the latter equation, the endogenous variable depends on its own value in the past period and on the discounted value of the exogenous variable's expected future values.

When $a_b + a_f = 1$, at least one unit root exists; if $a_f < a_b$, one unit root and one root outside the unit circle exist; if $a_f > a_b$, there is one unit root and one root outside the unit circle; and two unit roots exist if $a_f = a_b$. Therefore, if the sum of the two coefficients is equal to one, Eq. (C.41) may be written as:

$$\Delta y_t = \frac{a_f}{a_b} E_t \Delta y_{t+1} + \frac{b}{a_b} x_t. \quad (\text{C.47})$$

When $a_f < a_b$, the unique rational expectations equilibrium is given by:

$$\Delta y_t = \frac{c}{a_b} E_t \sum_{i=0}^{\infty} \left(\frac{a_f}{a_b} \right)^i x_{t+i} \quad (\text{C.48})$$

When $a_f > a_b$, Eq. (C.48) may be rewritten as:

$$E_t \Delta y_{t+1} = \frac{a_b}{a_f} \Delta y_t - \frac{c}{a_f} x_t \quad (\text{C.49})$$

It is easily seen that multiple rational expectations equilibria exist for an arbitrary sequence of shocks ε according to:

$$\Delta y_{t+1} = \frac{a_b}{a_f} \Delta y_t - \frac{c}{a_f} x_t + \varepsilon_{t+1} \quad (\text{C.50})$$

This equation may be written in backward-looking form:

$$\Delta y_t = -\frac{c}{a_f} \sum_{i=0}^{\infty} \left(\frac{a_b}{a_f} \right)^i \left(x_{t-i} \frac{a_f}{c} \varepsilon_{t-i} \right) \quad (\text{C.51})$$

When $a_b = 0$ in Eq. (C.41), the model has a unique solution only if $|a_f| < 1$. When $|a_f| \geq 1$, the model has multiple solutions. When $a_f = 0$, the model has a unique solution only if $|a_b| < 1$. Otherwise, the model has multiple solutions.

A Phillips Curve Example

The Phillips curve with indexation as specified in Woodford [(2003), p. 215, Eq. (3.6)] is:

$$\pi_t = \beta E_t(\pi_{t+1} - \gamma \pi_t) + \gamma \pi_{t-1} + \kappa x_t, \quad \pi_0 \text{ free}, \quad \pi_{-1} \text{ given} \quad (\text{C.52})$$

This equation may be rewritten as:

$$\pi_t = \frac{\beta}{1 + \beta\gamma} E_t \pi_{t+1} + \frac{\gamma}{1 + \beta\gamma} \pi_{t-1} + \frac{1}{1 + \beta\gamma} x_t. \quad (\text{C.53})$$

Therefore,

$$a_f = \frac{\beta}{1 + \beta\gamma}, \quad a_b = \frac{\gamma}{1 + \beta\gamma}, \quad b = \frac{1}{1 + \beta\gamma}. \quad (\text{C.54})$$

Since $a_f + a_b < 1$, the model has a unique rational expectations equilibrium and the inflation rate depends on the past, present, and future expected output gaps, according to:

$$\pi_t = \frac{b\lambda_1}{a_b} E_t \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \lambda_1^j \left(\frac{1}{\lambda_2} \right)^i x_{t+i-j}, \quad (\text{C.55})$$

and:

$$\lambda_1 + \lambda_2 = \frac{1}{a_f}; \quad \lambda_1 \lambda_2 = \frac{a_b}{a_f}$$

C.7 Hybrid New Keynesian Model

The hybrid new Keynesian model (HNKM) has three equations: a Phillips curve (PC), an IS curve, and a monetary policy rule for the interest rate (MPR):

$$\text{PC: } \pi_t = \omega_\pi E_t \pi_{t+1} + (1 - \omega_\pi) \pi_{t-1} + \kappa x_t, \quad (\text{C.56})$$

$$\text{IS: } x_t = \omega_x E_t x_{t+1} + (1 - \omega_x) x_{t-1} - \sigma(i_t - E_t \pi_{t+1} - \bar{r}_t), \quad (\text{C.57})$$

$$\text{MPR: } i_t = \rho + \phi_\pi \pi_t + \phi_x x_t. \quad (\text{C.58})$$

The weights ω_π and ω_x are the forward-looking coefficients of the Phillips and IS curves, respectively, and they lie within the zero-to-one range. Using the interest rate rule to eliminate the nominal interest rate from the IS curve yields:

$$\begin{aligned} x_t = & \frac{\omega_x}{1 + \sigma\phi_x} E_t x_{t+1} + \frac{\sigma}{1 + \sigma\phi_x} E_t \pi_{t+1} - \frac{\sigma\phi_\pi}{1 + \sigma\phi_x} \pi_t \\ & + \frac{1 - \omega_x}{1 + \sigma\phi_x} x_{t-1} + \frac{\sigma}{1 + \sigma\phi_x} (\bar{r}_t - \rho). \end{aligned}$$

This equation and the Phillips curve may be written as the following equation system:

$$\begin{aligned} \begin{bmatrix} 1 & -\kappa \\ \frac{\sigma\phi_\pi}{1 + \sigma\phi_x} & 1 \end{bmatrix} \begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = & \begin{bmatrix} \frac{\omega_\pi}{1 + \sigma\phi_x} & \frac{0}{1 + \sigma\phi_x} \\ \frac{\sigma}{1 + \sigma\phi_x} & \frac{0}{1 + \sigma\phi_x} \end{bmatrix} \begin{bmatrix} E_t \pi_{t+1} \\ E_t x_{t+1} \end{bmatrix} \\ & + \begin{bmatrix} 1 - \omega_\pi & 0 \\ 0 & \frac{1 - \omega_x}{1 + \sigma\phi_x} \end{bmatrix} \begin{bmatrix} \pi_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\sigma(\bar{r}_t - \rho)}{1 + \sigma\phi_x} \end{bmatrix}. \end{aligned}$$

After inverting the matrix on the left side, this system transforms into:

$$\begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = A_f \begin{bmatrix} E_t \pi_{t+1} \\ E_t x_{t+1} \end{bmatrix} + A_b \begin{bmatrix} \pi_{t-1} \\ x_{t-1} \end{bmatrix} + \nu_t, \quad (\text{C.59})$$

where:

$$\begin{aligned} A = \eta \begin{bmatrix} \omega_\pi + \frac{\sigma}{1 + \sigma\phi_x} & \frac{\kappa\omega_x}{1 + \sigma\phi_x} \\ \frac{-\sigma\phi_\pi\omega_\pi + \sigma}{1 + \sigma\phi_x} & \frac{\omega_x}{1 + \sigma\phi_x} \end{bmatrix}, A_b = \eta \begin{bmatrix} 1 - \omega_\pi & \frac{\kappa(1 - \omega_x)}{1 + \sigma\phi_x} \\ \frac{-\sigma\phi_x(1 - \omega_\pi)}{1 + \sigma\phi_x} & \frac{1 - \omega_x}{1 + \sigma\phi_x} \end{bmatrix} \\ \eta = \frac{1 + \sigma(\phi_x + \kappa\phi_\pi)}{1 + \sigma\phi_x}, \nu_t = \begin{bmatrix} \frac{\kappa\sigma(\bar{r}_t - \rho)}{1 + \sigma\phi_x} \\ \frac{\sigma(\bar{r}_t - \rho)}{1 + \sigma\phi_x} \end{bmatrix} \end{aligned} \quad (\text{C.60})$$

Central Bank Sets the Real Interest Rate

When the Central Bank controls the real interest rate gap, the hybrid model (HNKM) may be written as the following equation system:

$$\begin{bmatrix} \pi_t \\ x_t \end{bmatrix} = \begin{bmatrix} \omega_\pi & \kappa\omega_x \\ 0 & \omega_x \end{bmatrix} \begin{bmatrix} E_t\pi_{t+1} \\ E_tx_{t+1} \end{bmatrix} + \begin{bmatrix} 1 - \omega_\pi & \kappa\omega_x \\ 0 & 1 - \omega_x \end{bmatrix} \begin{bmatrix} \pi_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} -\kappa\sigma(r_t - \bar{r}_t) \\ -\sigma(r_t - \bar{r}_t) \end{bmatrix}$$

This model is a particular case of the model to be introduced next, and its equilibrium will be analyzed below.

Central Bank Uses the Taylor Rule

When the Central Bank has an inflation target, the hybrid model (HNKM) has a hybrid Phillips curve, an IS curve, and the monetary policy rule:

$$i_t = \rho + \phi E_t\pi_{t+1}.$$

It is assumed, for simplicity, that the Central Bank adopts a strict interest-rate rule and that the inflation target is zero. Therefore, the interest rate rule excludes the output gap. Using this monetary policy rule to eliminate the nominal interest rate from the IS curve yields the following difference equation system:

$$y_t = A_f E_t y_{t+1} + A_b y_{t-1} + C z_t, \quad y_{-1} \text{ given}, \quad y_0 \text{ free}, \quad (\text{C.61})$$

where:

$$y_t = \begin{bmatrix} \pi_t \\ x_t \end{bmatrix}; A_f = \begin{bmatrix} \omega_\pi - \kappa\sigma(\phi - 1) & \kappa\omega_x \\ -\sigma(\phi - 1) & \omega_x \end{bmatrix}; A_b = \begin{bmatrix} 1 - \omega_\pi & \kappa(1 - \omega_x) \\ 0 & 1 - \omega_x \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & \kappa \\ 0 & 1 \end{bmatrix}; z_t = \begin{bmatrix} 0 \\ \sigma(\bar{r}_t - \rho) \end{bmatrix}.$$

The characteristic polynomial of this dynamical system is:

$$p(\lambda) = a_4 \lambda^4 + a_3 \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0,$$

where:

$$a_4 = 1; a_3 = -[\omega_x + \omega_\pi - \kappa\sigma(\phi - 1)]/\omega_x\omega_\pi;$$

$$a_2 = [1 + \omega_\pi(1 - \omega_x) + \omega_x(1 - \omega_\pi)]/\omega_x\omega_\pi$$

$$a_1 = -[1 - \omega_x + 1 - \omega_\pi]/\omega_x\omega_\pi; a_0 = (1 - \omega_x)(1 - \omega_\pi)/\omega_x\omega_\pi.$$

Given that the inflation rate and output gap in period $t - 1$ are predetermined variables and those in period t are free, the hybrid model (HNKM) has a determinate rational expectations equilibrium only if two roots lie outside and two lie inside the unit circle.

The $p(\lambda)$ characteristic polynomial has the following properties:

$$\begin{aligned} p(1) &= \kappa\sigma(\phi - 1)/(\omega_x\omega_\pi) > 0 \text{ if } \phi > 1 \\ p(0) &= (1 - \omega_x)(1 - \omega_\pi)/(\omega_x\omega_\pi) > 0 \\ p(-1) &= [4 - \kappa\sigma(\phi - 1)]/(\omega_x\omega_\pi) > 0 \text{ if } \phi < 1 + \frac{4}{\kappa\sigma} \\ \lim_{\lambda \rightarrow \infty} p(\lambda) &= \lim_{\lambda \rightarrow -\infty} p(\lambda) = \infty \end{aligned}$$

When $\phi = 1$, $p(1) = 0$. Therefore, the model has at least one unit root. In fact, there are two unit roots and the other two roots are positive and may be greater or less than one, according to,

$$\lambda_1 = \lambda_2 = 1; \quad \lambda_3 = \frac{1 - \omega_x}{\omega_x}; \quad \lambda_4 = \frac{1 - \omega_\pi}{\omega_\pi}$$

The two roots are greater than one if the forward-looking coefficients are not dominant: $\omega_x < 1/2$ and $\omega_\pi < 1/2$. In this case, the model has a high level of inertia and can be shown to be determinate. Otherwise, the model is indeterminate. Sargent and Wallace (1975) show that interest-rate rules specifying an exogenous path for the nominal interest rate produce an indeterminate equilibrium. For the hybrid model (HNKM), the Sargent and Wallace proposition is not valid for an exogenous real interest rate path when the backward-looking coefficients are dominant.

According to Descartes' rule of signs, the number of positive roots is equal to four or two if $a_3 < 0$. For this to occur, the monetary policy coefficient must satisfy the following upper bound:

$$\phi < 1 + \frac{\omega_x + \omega_\pi}{\kappa\sigma}.$$

This condition is tantamount to saying that the sum of the roots is positive. Because $p(-1) = (1 + \lambda_1)(1 + \lambda_2)(1 + \lambda_3)(1 + \lambda_4) > 0$ and at least two roots are positive, if the other two are negative they will both lie inside or outside the unit circle. The final property of the characteristic polynomial states that the polynomial is always positive for high values of the λ roots, be they positive or negative.

Based on the characteristic polynomial's signs and properties, one cannot determine how many roots lie inside or outside the unit circle. Because of this analytical difficulty, numerical techniques are used. The hybrid model's parameters, κ , σ , and ϕ , will be calibrated according to the values used in the literature. These are: $\kappa = 0.01, 0.03, 0.05$; $\sigma = 1.0$, and $\phi = 1.5$. The forward-looking coefficients of the Phillips and IS curves take on the following values: $\omega_\pi, \omega_x = 0.1, 0.2, \dots, 0.8, 0.9$. Calibration involves 243 HNKM. Only a few of these –less than 10%– have determinate rational expectations equilibria because the condition that two roots lie outside and two lie inside the unit circle is not satisfied. Models with a rational expectations equilibrium correspond to parameters along the diagonal in Fig. C.1, for $\kappa = 0, 01$ where the forward- and backward-looking coefficients are equal, or along or near the diagonal as in Fig. C.2, for $\kappa = 0, 03$ and $\kappa = 0, 05$. Therefore, we

Fig. C.1 Rational expectations equilibria ($\kappa = 0.01$)

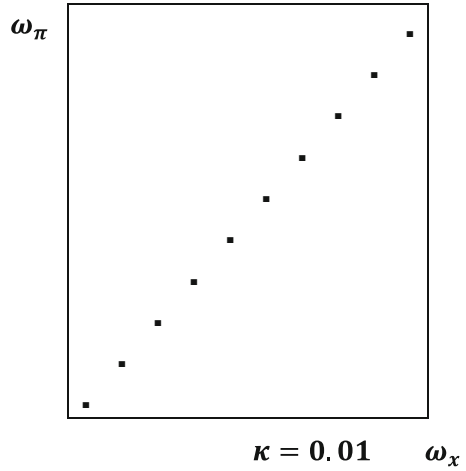
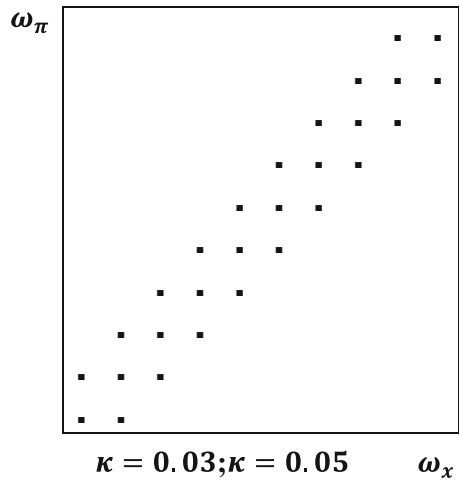


Fig. C.2 Rational expectations equilibria ($\kappa = 0.03$ and $\kappa = 0.05$)



may conclude that the new Keynesian hybrid model (HNKM) has few rational expectations equilibria.

C.8 Exercises

1. Consider the model:

$$\text{IS: } x_t = E_t x_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \bar{r}_t)$$

$$\text{PC: } \pi_t = \beta E_t \pi_{t+1} + \kappa x_t$$

$$\text{MPR: } i_t = \bar{r}_t$$

- (a) Analyze this model's equilibrium and stability.
- (b) Is the solution unique?

2. Consider the new Keynesian model:

$$\text{PC: } \pi_t = \beta E_t \pi_{t+1} + \kappa x_t$$

$$\text{IS: } x_t = E_t x_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \bar{r}_t)$$

$$\text{MPR: } i_t = \bar{r}_t + \pi_t + \varphi \pi_t + \theta x_t$$

- (a) What condition must the parameters satisfy for the model to have a unique equilibrium?
- (b) Does this condition imply that the ϕ parameter must always be positive?

3. Consider the new Keynesian model:

$$\text{PC: } \pi_t = \beta E_t \pi_{t+1} + \kappa x_t$$

$$\text{IS: } x_t = E_t x_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \bar{r}_t)$$

$$\text{MPR: } i_t = \bar{r}_t + \pi_t + \phi E_t \pi_{t+1} + \theta x_t$$

- (a) What condition must the parameters satisfy for the model to have a unique equilibrium?
- (b) Does this condition imply that the ϕ parameter must always be positive?
- (c) Answer the items (a) and (b) when the monetary policy rule is given by $i_t = \bar{r}_t + \pi_t + \phi E_t \pi_{t+1} + \theta E_t x_{t+1}$.

4. Consider the new Keynesian IS curve $x_t = E_t x_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \bar{r}_t)$.

- (a) What are the solutions of this difference equation?
- (b) Is the forward solution consistent with the consumption theory this curve was derived from?

5. Consider the new Keynesian Phillips curve $\pi_t = E_t \pi_{t+1} + \kappa x_t$.

- (a) What are the solutions of this difference equation?
- (b) Are there any criteria to choose one solution?

6. Consider the new Keynesian model:

$$\text{IS: } x_t = E_t x_{t+1} - \sigma(i_t - E_t \pi_{t+1} - \bar{r}_t)$$

$$\text{PC: } \pi_t - \bar{\pi} = \beta E_t (\pi_{t+1} - \bar{\pi}) + \kappa x_t$$

$$\text{MPR: } i_t = \bar{r}_t + \pi_t + \phi E_t (\pi_{t+1} - \bar{\pi}) + \theta x_t$$

where $\bar{\pi}$ is the inflation target.

- (a) Is this Phillips curve vertical in the long run?
- (b) What condition must the parameters satisfy for the model to have a unique equilibrium?